basics - math

basics

Nov 03, 2026

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This material is part of the **basics-books project**. It is also available as a .pdf document.

Contents.

- **Linear Algebra.** …, matrix factorization,…; basics of lots of numerical methods; **Vector and Tensor Algebra** provides the mathematical framework for manipulating vectors and tensors, the mathematical objects meant to represent **absolute quantities** - invariant under coordinate transformations - fundamental in geometry and physics.
- **Multivariable Calculus** provides the tools for working with continuous functions of many variables; **Differential Geometry** studies smooth curves, surfaces, volumes (and manifolds in general), within the framework of a Riemann structure, providing a metric to define distances, angles, curvatures and other geometric properties; **Vector and Tensor Calculus** studies vector and tensor fields defined on a manifold, along with their differentiation and integration, and thus being essential for a mature approach to geometry and physics;
- **Functional analysis** and **Complex Calculus** (complex analysis, transforms,…)
- ODEs
- PDEs
- Calculus of Variations: theoretical background; Lagrange multiplier method for constraints; sensitivty; gradient-based methods,…
- Optimization

Part I

Linear Algebra

CHAPTER

ONE

INTRODUCTION TO LINEAR ALGEBRA

- Matrices:
  - properties
  - representation of linear transformations
  - factorizations
  - linear systems

todo:

- existence of solution of equilibria of LTI and their stability, in *LTI:Equilibria* and in *LTI:Stability*

CHAPTER

TWO

MATRICES

$\mathbf{A} \in \mathbb{K}^{m,n}$ with usually $\mathbb{K}^{m,n} = \mathbb{R}^{m,n}$ or $\mathbb{C}^{m,n}$

Hermitian matrix. The Hermitian matrix $\mathbf{A}^*$ of a matrix $\mathbf{A}$ is the transpose and complex conjugate matrix (if $\mathbb{K} = \mathbb{C}$),

$$[\mathbf{A}^*]_{ij} = A^*_{ji} \ ,$$

with the notation of a^* for the complex conjugate of a numerical quantity.

2.1 Subspaces

2.1.1 Range, Image

$$R(\mathbf{A}) = \{\mathbf{y} \in \mathbb{K}^m \mid \exists \mathbf{x} \in \mathbb{K}^m \ , \text{ s.t. } \mathbf{A}\mathbf{x} = \mathbf{y}\}$$

The range of a matrix $\mathbf{A}$ is the linear space built on the columns of $\mathbf{A}$, since the operation $\mathbf{A}\mathbf{x}$ represents nothing but a linear combination of the columns of matrix.

2.1.2 Null, Kernel

$$K(\mathbf{A}) = \{\mathbf{x} \in \mathbb{K}^n \mid \mathbf{A}\mathbf{x} = \mathbf{0}\}$$

2.2 Theorem

2.2.1 Orthogonality of $R(\mathbf{A})$ and $K(\mathbf{A}^*)$

The following holds,

$$R(\mathbf{A}) \perp K(\mathbf{A}^*) \ ,$$

meaning that $\forall \mathbf{u} \in R(\mathbf{A})$ and $\forall \mathbf{v} \in K(\mathbf{A}^*)$, $\mathbf{u}^*\mathbf{v} = 0$.

Proof.

$$\mathbf{u} = \mathbf{A}\mathbf{x}$$
$$\mathbf{0} = \mathbf{A}^*\mathbf{v}$$

and thus, premultiplication by $\mathbf{x}^*$ of the second relation gives

$$0 = \mathbf{x}^*\mathbf{0} = \underbrace{\mathbf{x}^*\mathbf{A}^*}_{=(\mathbf{A}\mathbf{x})^*=\mathbf{u}^*}\mathbf{v} = \mathbf{u}^*\mathbf{v}\,.$$

This theorem becomes quite useful, e.g. for constrained linear systems and projections… (e.g. N-S, or other constrained linear systems…)

todo add links

CHAPTER

THREE

MATRIX FACTORIZATIONS

- **Singular Value Decomposition (SVD)**
- **Spectral decomposition** Eigenvalues, eigenvectors; Jordan canonical formula…

3.1 Gershgorin circle theorem

Let **A** a complex n, n matrix. Let $R_i = \sum_{j \neq i} |a_{ij}|$, and $D(a_{ii}, R) \subseteq \mathbb{C}$ the closed disc centered at a_{ii} with radius R_i in the complex plane.

Every eigenvalue of **A** lies within at least one of the Gershgorin discs $D(a_{ii}, R_i)$.

Proof

Let λ an eigenvalue of **A**. Thus, it exsits a vector **v** s.t. $\mathbf{Av} = \lambda\mathbf{v}$, or

$$\sum_j a_{ij} v_j = \lambda v_i \ .$$

Moving a_{ii} from the LHS to the RHS, it follows

$$\sum_{j \neq i} a_{ij} v_j = (\lambda - a_{ii})\, v_i \ .$$

Selecting i s.t. $|v_i| \geq |v_j|$, $\forall j$ it follows that

$$|\lambda - a_{ii}| = \left| \sum_{j \neq i} a_{ij} \frac{v_j}{v_i} \right| \leq \sum_{j \neq i} \left| a_{ij} \frac{v_j}{v_i} \right| \leq \sum_{j \neq i} |a_{ij}| \leq R_i \ .$$

3.2 Spectral radius

Spectral radius of a matrix **A** is defined as the maximum of the absolute value of its eigenvalues

$$\rho(\mathbf{A}) = \max |\lambda(\mathbf{A})| \ .$$

- **QR**
- **LU**
- **Schur**

- **Cholesky** Symmetric positive definite matrices have Choleski decomposition,

$$\mathbf{M} = \mathbf{L}\mathbf{L}^* ,$$

with $\mathbf{L}$ lower triangular matrix. And thus quite easy to "invert", for solving linear systems.

3.3 Schur decomposition

Any square matrix $\mathbf{A} \in \mathbb{C}^{n,n}$ can be written as

$$\mathbf{A} = \mathbf{Q}\mathbf{U}\mathbf{Q}^* ,$$

with $\mathbf{U}$ upper triangular and $\mathbf{Q}$ unitary, s.t. $\mathbf{Q}^{-1} = \mathbf{Q}^*$.

Proof by induction

This proof uses induction[1] on the dimension of the matrix $\mathbf{A} \in \mathbb{C}^{n,n}$. For $n = 1$, the transformation is trivial

$$A = 1 \cdot A \cdot 1 .$$

Now, let's prove the relation for n, assuming that it's true for $n - 1$. A vector $\mathbf{v}_1 \in \mathbb{C}$ s.t. $\mathbf{A}\mathbf{v}_1 = s_1\mathbf{v}_1$ exists for some $s_1 \in \mathbb{C}$. Then it's possible to build a orthonormal basis of \mathbb{C}^{n}, complementing $\mathbf{v}_1$,

$$\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{bmatrix} .$$

Evaluating the product $\mathbf{V}^*\mathbf{A}\mathbf{V}$,

$$\begin{aligned}
\mathbf{V}^*\mathbf{A}\mathbf{V} &= \begin{bmatrix} \mathbf{v}_1^* \\ \mathbf{v}_2^* \\ \dots \\ \mathbf{v}_n^* \end{bmatrix} \mathbf{A} \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{bmatrix} = \\
&= \begin{bmatrix} \mathbf{v}_1^* \\ \mathbf{v}_2^* \\ \dots \\ \mathbf{v}_n^* \end{bmatrix} \begin{bmatrix} \mathbf{A}\mathbf{v}_1 & \mathbf{A}\mathbf{v}_2 & \dots & \mathbf{A}\mathbf{v}_n \end{bmatrix} = \\
&= \begin{bmatrix} \mathbf{v}_1^* \\ \mathbf{v}_2^* \\ \dots \\ \mathbf{v}_n^* \end{bmatrix} \begin{bmatrix} s_1\mathbf{v}_1 & \mathbf{A}\mathbf{v}_2 & \dots & \mathbf{A}\mathbf{v}_n \end{bmatrix} = \\
&= \begin{bmatrix} s_1 & \mathbf{v}_1^*\mathbf{A}\mathbf{v}_2 & \dots & \mathbf{v}_1^*\mathbf{A}\mathbf{v}_n \\ 0 & \mathbf{v}_2^*\mathbf{A}\mathbf{v}_2 & \dots & \mathbf{v}_2^*\mathbf{A}\mathbf{v}_n \\ \dots & \dots & \dots & \dots \\ 0 & \mathbf{v}_n^*\mathbf{A}\mathbf{v}_2 & \dots & \mathbf{v}_n^*\mathbf{A}\mathbf{v}_n \end{bmatrix} = \\
&= \begin{bmatrix} s_1 & \mathbf{a}_1^* \\ \mathbf{0} & \mathbf{A}_1 \end{bmatrix} = \mathbf{T}
\end{aligned}$$

[1] **todo** Add a reference to induction in math notes, in sections about logics, either for high-school or in this personal notes. Find something in the old .pdf material.

Thus $\mathbf{A} = \mathbf{VTV}^*$. Induction assumes that $\mathbf{A}_1 = \mathbf{Q}_1\mathbf{U}_1\mathbf{Q}_1^*$, so that

$$\begin{aligned}
\mathbf{A} &= \mathbf{VTV}^* = \\
&= \mathbf{V}\begin{bmatrix} s_1 & \mathbf{a}_1^* \\ \mathbf{0} & \mathbf{A}_1 \end{bmatrix}\mathbf{V}^* = \\
&= \mathbf{V}\begin{bmatrix} s_1 & \mathbf{a}_1^* \\ \mathbf{0} & \mathbf{Q}_1\mathbf{U}_1\mathbf{Q}_1^* \end{bmatrix}\mathbf{V}^* = \\
&= \underbrace{\mathbf{V}\begin{bmatrix} 1 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{Q}_1 \end{bmatrix}}_{=\mathbf{Q}}\underbrace{\begin{bmatrix} s_1 & \mathbf{a}_1^* \\ \mathbf{0} & \mathbf{U}_1 \end{bmatrix}}_{=\mathbf{U}}\underbrace{\begin{bmatrix} 1 & \mathbf{0}^* \\ \mathbf{0} & \mathbf{Q}_1^* \end{bmatrix}\mathbf{V}^*}_{=\mathbf{Q}^*} = \\
&= \mathbf{QUQ}^* .
\end{aligned}$$

As the existence of Schur decomposition for $\mathbf{A}_1 \in \mathbb{C}^{n-1,n-1}$ implies the existence of Schur decomposition for $\mathbf{A} \in \mathbb{C}^{n,n}$, the proof is complete.

Lemma

For every matrix $\mathbf{A} \in \mathbb{C}^{n,n}$, there is a vector $\mathbf{v} \in \mathbb{C}^n$, $\mathbf{v} \neq \mathbf{0}$, so that $\mathbf{Av} = a\mathbf{v}$, for some value of $a \in \mathbb{C}$. This relation holds for non-trivial vectors $\mathbf{v}$ if

$$(\mathbf{A} - s\mathbf{I})\,\mathbf{v} = \mathbf{0} ,$$

i.e. $\mathbf{v} \in \text{Ker}\,(\mathbf{A} - s\mathbf{I})$, i.e. $\text{Ker}\,(\mathbf{A} - s\mathbf{I})$ is not empty. This is true if

$$0 = |\mathbf{A} - s\mathbf{I}| .$$

This is a polynomial equation of degree n and thus, for the *fundamental theorem of algebra*, it has at least one complex solution, $s = \overline{s}$. For $s = \overline{s}$, a vector $\mathbf{v}$ satisfying $\mathbf{Av} = \overline{s}\mathbf{v}$.

3.4 Cholesky decomposition

For a semi-positive definite matrix $\mathbf{M}$

$$\mathbf{M} = \mathbf{LL}^* = \mathbf{U}^*\mathbf{U}$$

with $\mathbf{L}$ lower triangular matrix.

3.5 Jordan normal form

$$\mathbf{A} = \mathbf{VJV}^{-1} ,$$

with $\mathbf{J} = ...$

- link to solution of system of linear ODEs, and stability conditions

3.6 Singular Value Decomposition

Singular value decomposition of a matrix $\mathbf{A} \in \mathbb{C}^{m,n}$

$$\mathbf{A}_{(m,n)} = (\text{SVD}) = \mathbf{U}_{(m,m)}\mathbf{S}_{(m,n)}\mathbf{V}^*_{(n,n)}$$

with $\mathbf{U}^*\mathbf{U} = \mathbf{I}_{(m,m)}$, and $\mathbf{V}^*\mathbf{V} = \mathbf{I}_{(n,n)}$, $\mathbf{S} = \text{diag}\{\sigma_i\}$, $\sigma_i \geq 0$.

Exploiting the definition of matrix product, the SVD of matrix A can be written as

$$\mathbf{A} = \sum_{i=\min(m,n)} \sigma_i \mathbf{u}_i \mathbf{v}_i^* \ ,$$

see also economic decomposition below.

3.6.1 Properties

Relation with *range* and *kernel* the matrix.

$$\begin{aligned} \text{R}(\mathbf{A}) &= \{\mathbf{u}_i | \sigma_i > 0\} \\ \text{K}(\mathbf{A}) &= \{\mathbf{v}_i | \sigma_i = 0\} \\ \text{R}(\mathbf{A}^*) &= \{\mathbf{v}_i | \sigma_i > 0\} \\ \text{K}(\mathbf{A}^*) &= \{\mathbf{u}_i | \sigma_i = 0\} \end{aligned}$$

It's immediate to prove $R(\mathbf{A}) \perp K(\mathbf{A}^*)$.

Full or economic decomposition. In general the $\mathbf{S}$ of the full decompsition in not square.

$$\mathbf{A} = (\text{SVD}) = \mathbf{U}_{(m,m)}\mathbf{S}_{(m,n)}\mathbf{V}^*_{(n,n)} = \mathbf{U}^e_{(m,k)}\mathbf{S}^e_{(k,k)}\mathbf{V}^{e*}_{(k,n)} \ ,$$

with $k = \min(m, n)$.

3.6.2 Applications

Solution of under-determined linear systems

Norms and optimization

If L^2-norm is used for vector norms, see *Example 3.6.1*

$$\text{Find} \max_{||\mathbf{x}||=1} ||\mathbf{A}\mathbf{x}||^2$$

$$\mathbf{x}^*\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{x}^*\mathbf{V}\mathbf{S}^* \underbrace{\mathbf{U}^*\mathbf{U}}_{\mathbf{I}} \mathbf{S} \underbrace{\mathbf{V}^*\mathbf{x}}_{\mathbf{z}}$$

and $\mathbf{z} = \mathbf{V}^*\mathbf{x}$ is unitary as well (since $1 = \mathbf{x}^*\mathbf{x} = \mathbf{z}^*\mathbf{V}^*\mathbf{V}\mathbf{z} = \mathbf{z}^*\mathbf{z}$).

After defining $\mathbf{S}_2 := \mathbf{S}^*\mathbf{S}$, the problem thus becomes

$$\text{Find} \max_{||\mathbf{z}||=1} \mathbf{z}^*\mathbf{S}_2\mathbf{z}$$

Manipulating the objective function as $\sum_i \sigma_i^2 |z_i|^2$, the constraint optimization problem has global maximum σ_1^2 (sorted singular values from the largest to the smalles) when $\mathbf{z}_1 = (1, 0, 0, \dots, 0)^T$. Going back to the original variable, optimal condition

- is achieved for $\mathbf{x}_1 = \mathbf{v}_1$;
- has value $\max_{||\mathbf{x}||=1} ||\mathbf{Ax}|| = \sigma_1^2$
- and the response of the system is $\mathbf{y}_1 = \sigma_1 \mathbf{u}_1$ as

$$\mathbf{y}_1 := \mathbf{Ax}_1 = \mathbf{USV}^* \mathbf{v}_1 = \sum_k \left(\sigma_k \mathbf{u}_k \mathbf{v}_k^*\right) \mathbf{v}_1 = \sigma_1 \mathbf{u}_1 \ .$$

Example 3.6.1 (Other norms - variations of the L^2-norm)

This kind of problem may results as the discrete counterpart of a continuous problem, as an example from *finite element methods*, where **x**, **y** contain the coefficients of the basis functions. In this case, the discrete counterpart of the continous norm-measure of the continuous fields may involve a "mass matrix" (symmetric, definite positive - and thus with Choloeski factorization…),

$$\int_{\Omega_x} |x(\vec{r})|^2 d\vec{r} \simeq \mathbf{x}^* \mathbf{M}_x \mathbf{x}$$
$$\int_{\Omega_y} |y(\vec{r})|^2 d\vec{r} \simeq \mathbf{y}^* \mathbf{M}_y \mathbf{y}$$

Continuous and discrete optimization problems are

$$\text{Find} \max_{|x(\vec{r})|_{L^2(\Omega_x)}=1} |y|^2_{L^2(\Omega_y)}$$

$$\text{Find} \max_{\mathbf{x}^* \mathbf{M}_x \mathbf{x}=1} \mathbf{x}^* \mathbf{A}^* \mathbf{M}_y \mathbf{A} \mathbf{x}$$

with the relation $\mathbf{y} = \mathbf{Ax}$ between the discrete input and output.

This problem can be easily (and efficiently?) recast to the standard form of the problem, using *Cholesky decomposition* of matrix $\mathbf{M}_x = \mathbf{L}_x \mathbf{L}_x^*$, with the definition of the variable $\mathbf{z} = \mathbf{L}_x^* \mathbf{x}$

$$\text{Find} \max_{||\mathbf{z}||=1} \mathbf{z}^* \mathbf{L}_x^{-1} \mathbf{A}^* \mathbf{L}_y \mathbf{L}_y^* \mathbf{A} \mathbf{L}_x^{-*} \mathbf{z}$$

This problem can be efficiently solved with iterative algorithms to compute the SVD of the matrix $\widetilde{\mathbf{A}} := \mathbf{L}_y^* \mathbf{A} \mathbf{L}_x^{-*}$, that doesn't need the expensive full inversion of a matrix but only its action on a vector (instead of evaluating the inverse, a linear system - here triangular! Easier to solve - can be efficiently solved). Algorithms like **Arnoldi algorithm** evaluates the largest eigenvalues or singular values(if no options to set other goals) and the corresponding eigenvectors and singular vectors, alternating **power iterations** and **orthogonalization**. Here power iteration to evaluate the action of the matrix $\widetilde{\mathbf{A}} = \mathbf{L}_y^* \mathbf{A} \mathbf{L}_x^{-*}$ on a generic vector **z** is made of the following steps:

1. solution of the linear system $\mathbf{L}_x^* \mathbf{a} = \mathbf{z} \rightarrow \ \mathbf{a} = ...$
2. matrix-vector multiplication $\mathbf{b} = \mathbf{Aa}$
3. matrix-vector multiplication $\mathbf{c} = \mathbf{L}_y^* \mathbf{b}$

Once the SVD is solved, with $\mathbf{z}_1 = \mathbf{v}_1$

$$\mathbf{x}_1 = \mathbf{L}_x^{-*} \mathbf{v}_1$$
$$\mathbf{y}_1 = \mathbf{Ax}_1 = \sigma_1 \mathbf{L}_y^{-*} \mathbf{u}_1$$

CHAPTER

FOUR

LINEAR SYSTEMS

The linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

with $\mathbf{A} \in \mathbb{R}^{m,n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$ has solution if there exists (at least) a vector $\tilde{\mathbf{x}}$ whose product with $\mathbf{A}$ gives $\mathbf{b}$.

Condition for the existence of a solution. A solution exists if $\mathbf{b}$ belongs to the *range of* $\mathbf{A}$.

Uniqueness of a solution. If a solution $\tilde{\mathbf{x}}$ exists, it's unique if the *kernel of* $\mathbf{A}$ is empty, $K(\mathbf{A}) = \emptyset$. If the kernel is not empty,

$$\mathbf{b} = \mathbf{A}(\tilde{\mathbf{x}} + \mathbf{u}) = \mathbf{A}\tilde{\mathbf{x}} + \underbrace{\mathbf{A}\mathbf{u}}_{=\mathbf{0}} ,$$

for $\forall \mathbf{u} \in K(\mathbf{A})$, and thus an infinite number of solutions exists. Given a vector basis of the kernel $\mathbf{K}(\mathbf{A})$, where $\dim\left(K(\mathbf{A})\right) = n_K$, $\{\mathbf{u}_1, \dots, \mathbf{u}_{n_K}\}$, the general solution has n_K "degrees of arbitrarieness", since the general solution of the problem is

$$\tilde{\mathbf{x}} + \sum_{i=1}^{n_K} a_i \mathbf{u}_i = \tilde{\mathbf{x}} + \mathbf{U}\mathbf{a} .$$

todo treat under-, det-, over-determined lin sys

CHAPTER

FIVE

SPECTRAL DECOMPOSITION

- **Introduction.** What's spectral (or eigenvalue) decomposition? Arbitrarieness. When is it possible? Special matrices. Properties and theorems (*Cayley-Hamilton*)

$$\mathbf{A}\mathbf{u}_i = s_i \mathbf{u}_i$$

- **Generalized spectral decompositions.** Generalized spectral decomposition of first order systems; generalized spectral deocmposition of second order systems[1]

$$\mathbf{A}\mathbf{u}_i = s_i \mathbf{B}\mathbf{u}_i$$

$$\left[s_i^2 \mathbf{M} + s_i \mathbf{C} + \mathbf{K} \right] \mathbf{u}_i = \mathbf{0} \, .$$

- **Sensitivity of spectral decomposition.** ...

5.1 Spectral decomposition of symmetric matrices

In this section the spectral decomposition of symmetric matrices (or Hermitian, if defined on the complex field), $\mathbf{A} = \mathbf{A}^*$, is discussed.

Theorem 5.1.1 (Spectral theorem of a symmetric matrix)

A symmetric matrix has a spectral decomposition, with a complete set of unit orthogonal vectors and real eigenvalues,

$$\mathbf{A} = \mathbf{R}\mathbf{S}\mathbf{R}^* \, ,$$

with $\mathbf{R} = [\, \mathbf{r}_1 \mid \ldots \mid \mathbf{r}_n \,]$, and $\mathbf{r}_i^* \mathbf{r}_k = \delta_{ik}$, and $\mathbf{S} = \text{diag}\{s_i\}$, $s_i \in \mathbb{R}$.

[1] It's likely that mainly underdamped systems with simultaneously diagonalizable matrices will be treated, as this kind of systems often arises in structural dynamics of elastic structures with small structural damping.

Proof

todo *By construction…*

$$\begin{aligned}
\mathbf{A} &= \mathbf{R}\mathbf{S}\mathbf{R}^* = \\
&= \begin{bmatrix}\mathbf{r}_1 & \dots & \mathbf{r}_n\end{bmatrix} \operatorname{diag}\{s_i\} \begin{bmatrix}\mathbf{r}_1^* \\ \dots \\ \mathbf{r}_n^*\end{bmatrix} = \\
&= \begin{bmatrix}s_1\mathbf{r}_1 & \dots & s_n\mathbf{r}_n\end{bmatrix} \begin{bmatrix}\mathbf{r}_1^* \\ \dots \\ \mathbf{r}_n^*\end{bmatrix} = \\
&= s_1\mathbf{r}_1\mathbf{r}_1^* + \dots + s_n\mathbf{r}_n\mathbf{r}_n^* \ .
\end{aligned}$$

Properties.

- Left and right eigenvectors are the same. This immediately follows from the symmetry of the matrix and the definition of the left and right eigenvectors

$$\begin{aligned}
\mathbf{A}\mathbf{r}_k &= s_k\mathbf{r}_k \\
\mathbf{l}_j^*\mathbf{A} &= s_j^*\mathbf{l}_i^* \quad \rightarrow \quad \mathbf{A}\mathbf{l}_j = s_j\mathbf{l}_j \ ,
\end{aligned}$$

- Eigenvalues are real, as

$$0 = \mathbf{l}_k^* \left(\mathbf{A}\mathbf{r}_k\right) - \left(\mathbf{l}_k^*\mathbf{A}\right)\mathbf{r}_k = \left(s_k - s_k^*\right)\mathbf{l}_k^*\mathbf{r}_k = 2\,\mathrm{im}\{s_k\}\underbrace{\mathbf{l}_k^*\mathbf{r}_k}_{=|\mathbf{r}_k|^2} \ ,$$

  and thus $\mathrm{im}\{s_k\} = 0$.

- Eigenvectors with different eigenvalues are orthogonal, as

$$0 = \mathbf{l}_j^* \left(\mathbf{A}\mathbf{r}_k\right) - \left(\mathbf{l}_j^*\mathbf{A}\right)\mathbf{r}_k = \left(s_k - s_j^*\right)\mathbf{l}_j^*\mathbf{r}_k \ ,$$

  and thus $\mathbf{l}_j^*\mathbf{r}_k = 0$ for $j \neq k$, if $s_j \neq s_k$. Left and right eigenvectors are orthogonal w.r.t. matrix $\mathbf{A}$ as well.

5.2 Sensitivity of spectral decomposition

See also

Sensitivity of the spectral decomposition of a second-order system is used in Continuum Mechanics: Structural Mechanics: Structural Damping to discuss the nature of structural damping in linear structures.

Matrices involved in a eigenvalue problem can be function of some parameters p. Sensitivity analysis evaluates first-order changes[1] of eigenvalues and eigenvectors following a increment of the parameter $p = \overline{p} + \Delta p$,

$$\begin{aligned}
s_i(p) &= s_i(\overline{p}) + \Delta p\, s_{i/p}(\overline{p}) + o(\Delta p) \\
\mathbf{u}_i(p) &= \mathbf{u}_i(\overline{p}) + \Delta p\, \mathbf{u}_{i/p}(\overline{p}) + o(\Delta p) \ .
\end{aligned}$$

Here the terms $s_{i/p}(\overline{p})$ and $\mathbf{u}_{i/p}(\overline{p})$ are the **sensitivity** of the ith eigenvalue and eigenvector respectively w.r.t. to an increment Δp of the variable p, starting from the reference configuration determined by the value $\overline{p}$ of the parameter.

Rank of sensitivity. Sensitivity of eigenvalue to a scalar parameter is a scalar quantity. Sensitivity of eivenvector to a scalar parameter is a vector quantity. If the parameter $\mathbf{p}$ is a vector quantity, sensitivity of eigenvalue w.r.t. $\mathbf{p}$ is a vector quantity, and sensitivity of eigenvector w.r.t. $\mathbf{p}$ is a matrix (or tensor, sometimes…) quantity.

[1] First order changes should be meant as derivatives w.r.t. the parameter p, evaluated for the reference value of $\overline{p}$ that appear in a Taylor series of the quantity of interest, $s(p_i) = s(\overline{p}_i) + \Delta p_k\, \partial_{p_k} s(\overline{p}_i) + o(\Delta p_i)$.

5.2.1 Generalized eigenvalue problem (first order)

Sensitivity of eigenvalue and right eigenvector to a parameter p of matrices $\mathbf{A}(p)$, $\mathbf{B}(p)$ read

$$s_{i/p} = \mathbf{w}_i^* \left(\mathbf{A}_{/p} - s_i \mathbf{B}_{/p} \right) \mathbf{u}_i \tag{5.1}$$

$$\mathbf{u}_{i/p} = \sum_{a \notin \{i\}} \mathbf{w}_a^* \left(-\mathbf{A}_{/p} + s_i \mathbf{B} \right) \mathbf{u}_i \frac{1}{s_a - s_i} \mathbf{u}_a \ ,$$

having exploited normalization condition $\mathbf{w}_i^* \mathbf{B} \mathbf{u}_i = 1$, being $\mathbf{u}_i$ and $\mathbf{w}_i$ the[2] right and left eigenvectors associated with eigenvalue s_i.

Right and left eigenvalue problem

Right and left eigenvalue problems are defined as

$$\begin{aligned} \mathbf{A}\mathbf{u}_i &= s_i \mathbf{B}\mathbf{u}_i \\ \mathbf{v}_j^* \mathbf{A} &= s_j \mathbf{v}_j^* \mathbf{B} \end{aligned} \tag{5.2}$$

Property 5.2.1 (Eigenvalues of the right and left spectral problems)

Left and right eigenvalue problems have the same set of eigenvalues.

Proof

todo

Property 5.2.2 (Orthogonality conditions of right and left spectral problems)

Left and right eigenvectors associated with different eigenvalues are orthogonal w.r.t. the matrices of the system, i.e.

$$\begin{aligned} \mathbf{v}_j^* \mathbf{B} \mathbf{u}_i &= b^{(i)} \delta_{ij} \\ \mathbf{v}_j^* \mathbf{A} \mathbf{u}_i &= a^{(i)} \delta_{ij} \ , \end{aligned}$$

with $a^{(i)} = s_i b^{(i)}$. Parameters $a^{(i)}$, $b^{(i)}$ are not uniquely determined, and one of them (or their combination) can be used in a **normalization condition** removing arbitrarieness of eigenvectors to a multiplicative factor.

Proof

Starting from the left and right eigenvalue problems (5.2) for two different eigenvalues $s_i \neq s_j$, and multiplying (on the left) the right eigenvalue problem by $\mathbf{v}_j^*$ and multiplying (on the right) the left eigenvalue problem by $\mathbf{u}_i$

$$\begin{aligned} \mathbf{v}_j^* \mathbf{A} \mathbf{u}_i &= s_i \mathbf{v}_j^* \mathbf{B} \mathbf{u}_i \\ \mathbf{v}_j^* \mathbf{A} \mathbf{u}_i &= s_j \mathbf{v}_j^* \mathbf{B} \mathbf{u}_i \end{aligned} \tag{5.3}$$

[2] Eigenvalues with algebraic and geometric multiplicity larger than one, $m_i^{(a)} = m_i^{(g)} > 1$, have the associated eigenvectors that spans a subspace of dimension m_i. In this subspace, it's always possible to define a set of orthogonal vectors. **todo** write it better; refer to introduction to spectral decomposition… **todo** treat sensitivity to parameters of eigenvalues and eigenvectors with multiplicity larger than one…

and subracting the two equations

$$0 = (s_i - s_j)\mathbf{v}_j^* \mathbf{B}\mathbf{u}_i$$

$$\rightarrow \qquad 0 = \mathbf{v}_j^* \mathbf{B}\mathbf{u}_i \quad \text{if } s_i \neq s_j \,, \tag{5.4}$$

as the left-hand-side terms are the same, and $s_i \neq s_j$. As (5.4) holds, from (5.3) it also follows that

$$\rightarrow \qquad 0 = \mathbf{v}_j^* \mathbf{A}\mathbf{u}_i \quad \text{if } s_i \neq s_j \,. \tag{5.5}$$

These relations don't hold for left and right eigenvectors associated with the same eigenvalues, and thus these conditions can be used as normalization conditions.

Sensitivity of eigenvalue

The derivative w.r.t. parameter p of the right eigenvalue problem

$$\mathbf{A}(p)\mathbf{u}_i(p) = s_i(p)\mathbf{B}(p)\mathbf{u}_i(p) \,,$$

reads

$$0 = (\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p})\mathbf{u}_i + (\mathbf{A} - s_i\mathbf{B})\mathbf{u}_{i/p} \,.$$

Multiplying on the left by the left eigenvector $\mathbf{w}_i$, and recalling that the left eigenvalue problem reads $0 = \mathbf{w}_i^* \, (\mathbf{A} - s_i\mathbf{B})$, the last term is identically zero and

$$0 = \mathbf{w}_i^* (\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p})\mathbf{u}_i \,,$$

so that it's possible to find the expression of the eigenvalue sensitivity as

$$s_{i/p} = \frac{\mathbf{w}_i \left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p}\right) \mathbf{u}_i}{\mathbf{w}_i^* \mathbf{B}\mathbf{u}_i} = \mathbf{w}_i \left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p}\right) \mathbf{u}_i \,, \tag{5.6}$$

where the last step exploits the normalization condition $\mathbf{w}_i^* \mathbf{B}\mathbf{u}_i = b^{(i)} = 1$.

Sensitivity of eigenvector

The sensitivity of the eigenvector $\mathbf{u}_{i/p}$ is the solution of the linear system that can be obtained from the derivative of the eigenvalue problem,

$$(\mathbf{A} - s_i\mathbf{B})\mathbf{u}_{i/p} = \underbrace{-(\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p})\mathbf{u}_i}_{=:\mathbf{b}_i} \,,$$

once the sensitivity of the eigenvector $s_{i/p}$ is known from expression (5.6).

Property 5.2.3 (Linear system is singular, $\mathbf{K}(A) = \{\mathbf{u}_i\}$)

Linear system is singular as s_i is an eigenvalue of the problem, $|\mathbf{A} - s_i\mathbf{B}| = 0$, and the kernel of the matrix is the space spanned by the right eigenvectors associated with s_i, as

$$(\mathbf{A} - s_i\mathbf{B}) \, \mathbf{u}_i = \mathbf{0} \,.$$

As the matrix of the linear system is singular, the linear system has **no solution if** the right-hand side is not in the range of the matrix of the system, i.e. $\mathbf{b}_i \notin \mathrm{R}(\mathbf{A} - s_i\mathbf{B})$.

Fortunately, the RHS belongs to the range of the matrix, $\mathbf{b}_i \in \mathrm{R}(\mathbf{A} - s_i\mathbf{B})$, as it's proved in the box below.

Property 5.2.4 ($\mathbf{b}_i \in \mathbf{R}(\mathbf{A} - s_i\mathbf{B})$)

If $\mathbf{b}_i$ belongs to $\mathrm{R}(\mathbf{A} - s_i\mathbf{B})$, it's orthogonal to $\mathrm{K}(\mathbf{A}^* - s_i^*\mathbf{B}^*)$ and viceversa, by theorem … **todo** add link

Kernel of $\mathbf{A}^* - s_i^*\mathbf{B}^*$ is spanned by the left eigenvectors $\mathbf{v}_i$. Thus, using expression of the eigenvalue sensitivity in evaluating the product

$$\begin{aligned}\mathbf{v}_i^*\mathbf{b}_i &= -\mathbf{v}_i^*\left(\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i\mathbf{B}_{/p})\mathbf{u}_i\right) = \\ &= -\mathbf{v}_i^*\left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p})\mathbf{u}_i\right) + \mathbf{v}_i^*\left(\mathbf{A}_{/p} - s_i\mathbf{B}_{/p}\right)\mathbf{u}_i\,\underbrace{\mathbf{v}_i^*\mathbf{B}\mathbf{u}_i}_{=1} = \\ &= 0\,,\end{aligned}$$

it's readily proved that $\mathbf{v}_i^*\mathbf{b}_i = 0$. **todo** treat the case where eigenvalue multiplicity is larger than 1.

As the singular linear system has at least a solution, it has infinite solution as the kernel of the matrix is not empty. If $\widetilde{\mathbf{u}_{/p}}$ is a solution of the system, adding a linear combination of the vectors of $\mathrm{K}(\mathbf{A} - s_i\mathbf{B}) = \{\mathbf{u}_i\}$ produces a solution as well,

$$\widetilde{\mathbf{u}_{i/p}} + \mathbf{U}_i\beta\,,$$

as $(\mathbf{A} - s_i\mathbf{B})\mathbf{U}_i = \mathbf{0}$, being $\mathbf{U}_i$ the matrix whose columns are the right eigenvectors $\mathbf{u}_i$ associated with eigenvalue s_i, spanning the kernel of the matrix.

In order to remove this arbitrariness, it's possible to introduce the orthogonality condition $\mathbf{V}_i^*\mathbf{B}\widetilde{\mathbf{u}_{i/p}} = \mathbf{0}$, setting the coefficients of the linear combination of the vectors of the kernel $\beta = \mathbf{0}$.

Property 5.2.5 (Augmented linear system)

$$\begin{bmatrix}\mathbf{A} - s_i\mathbf{B} & \mathbf{U}_i \\ \mathbf{V}_i^*\mathbf{B} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\widetilde{\mathbf{u}_{i/p}} \\ \beta\end{bmatrix} = \begin{bmatrix}\mathbf{b}_i \\ \mathbf{0}\end{bmatrix}$$

If everything goes right, the coefficients β must be zero.

Property 5.2.6 (Decomposition of the soluton using the modal basis)

Writing the solution as a linear combination of the right eigenvectors,

$$\widetilde{\mathbf{u}_{i/p}} = \mathbf{U}_{\notin i}\alpha + \mathbf{U}_i\beta\,,$$

it's possible to solve for α and for β and then retrieve the solution of the underdetermined linear system. Projecting the linear system on $\mathbf{V}_{\notin i}$, and prescibing the orthogonality condition $\mathbf{V}_i^*\mathbf{B}\widetilde{\mathbf{u}_{i/p}} = \mathbf{0}$ on the solution, two decoupled sets of equations for α and β appears,

$$\begin{cases}\mathbf{V}_{\notin i}^*\left(\mathbf{A} - s_i\mathbf{B}\right)\mathbf{U}_{\notin i}\alpha = \mathbf{V}_{\notin i}^*\mathbf{b}_i \\ \mathbf{V}_i^*\mathbf{B}\mathbf{U}_i\beta = \mathbf{0}\,,\end{cases}$$

as $\left(\mathbf{A} - s_i\mathbf{B}\right)\mathbf{U}_i = \mathbf{0}$ and $\mathbf{V}_i^*\mathbf{B}\mathbf{U}_{\notin i} = \mathbf{0}$. Exploiting diagonalization properties, the system can be written as two uncoupled diagonal systems

$$\begin{cases}\mathrm{diag}\left\{a^{(j)\notin i} - s_i b^{(j)\notin i}\right\}\alpha = \mathbf{V}_{\notin i}^*\mathbf{b}_i \\ \mathrm{diag}\left\{b^{(i)}\right\}\beta = \mathbf{0}\,,\end{cases}$$

or, exploiting normalization condition $\mathbf{v}_i^* \mathbf{B} \mathbf{u}_i = 1$, and thus $b^{(j)} = 1$ and $a^{(j)} = s_j$,

$$\begin{cases} \text{diag}\left\{s_{j \notin i} - s_i\right\} \alpha = \mathbf{V}_{\notin i}^* \mathbf{b}_i \\ \mathbf{I}\, \beta = \mathbf{0}\,, \end{cases}$$

The solution reads

$$\begin{cases} \alpha = \text{diag}\left\{ \dfrac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* \mathbf{b}_i \\ \beta = \mathbf{0}\,, \end{cases}$$

and thus

$$\begin{aligned} \tilde{\mathbf{u}}_{i/p} &= \mathbf{U}_{\notin i} \text{diag}\left\{ \frac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* \mathbf{b}_i = \\ &= \mathbf{U}_{\notin i} \text{diag}\left\{ \frac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* (-(\mathbf{A}_{/p} - s_{i/p}\mathbf{B} - s_i \mathbf{B}_{/p}) \mathbf{u}_i) = \\ &= \mathbf{U}_{\notin i} \text{diag}\left\{ \frac{1}{s_{j \notin i} - s_i} \right\} \mathbf{V}_{\notin i}^* (-(\mathbf{A}_{/p} - s_i \mathbf{B}_{/p}) \mathbf{u}_i) = \\ &= \sum_{j \notin i} \frac{\mathbf{v}_j^* (-\mathbf{A}_{/p} + s_i \mathbf{B}_{/p}) \mathbf{u}_i}{s_j - s_i} \mathbf{u}_j\,. \end{aligned}$$

Some algebra with components

$$\begin{aligned} u_{a/p}^i &= \sum_{b \neq i} u_a^{(b)} \delta_{bc} D^{(c)} v_d^{(c)\,*} b_d^{(i)} = \\ &= \sum_{b \neq i} u_a^{(b)} \delta_{bc} D^{(c)} v_d^{(c)\,*} M_{de} u_e^{(i)} = \\ &= \sum_{b \neq i} u_a^{(b)} D^{(b)} v_d^{(b)\,*} M_{de} u_e^{(i)} = \\ &= \sum_{b \neq i} C^{b,i} u_a^{(b)}\,, \end{aligned}$$

with $C^{b,i} = D^{(b)} v_d^{(b)\,*} M_{de/p} u_e^{(i)} = \frac{\mathbf{v}_b^* \mathbf{M} \mathbf{u}_i}{s_b - s_i}$, and $\mathbf{M} = -\mathbf{A}_{/p} + s_i \mathbf{B}_{/p}$.

5.2.2 Generalized eigenvalue problem (second order)

See as an example, sensitivity of the spectral decomposition of a second-order system is used in Continuum Mechanics: Structural Mechanics: Structural Damping to discuss the nature of structural damping in linear structures.

Right and left eigenvalue problem

CHAPTER

SIX

CAYLEY-HAMILTON THEOREM

Every square matrix $\mathbf{A}$ satisfies its own characteristic equation,

$$\mathbf{A}^n + a_{n-1}\mathbf{A}^{n-1} + \cdots + a_1\mathbf{A} + a_0\mathbf{I} = \mathbf{0} \, .$$

todo Proof using the adjugate matrix and the identity (**prove it!**) $\det(\mathbf{M})\,\mathbf{I} = \mathbf{M}\,\mathrm{adj}(\mathbf{M})$, and the telescopic sum. This identity comes from the properties of the determinant. Need to write a section about it.

Part II

Multivariable Calculus

CHAPTER

SEVEN

INTRODUCTION TO MULTI-VARIABLE CALCULUS

7.1 Function

7.2 Limit

7.3 Derivatives

7.4 Integrals

7.5 Theorems

7.5.1 Green's lemma

$$\int_S \frac{\partial F}{\partial y} dxdy = -\oint_{\partial S} F dx$$
$$\int_S \frac{\partial G}{\partial x} dxdy = \oint_{\partial S} G dy$$

Proof for simple domains.

In a simple domain in x, so that the closed contour ∂S is delimited by the curves $y = Y_1(x)$, $y = Y_2(x) > Y_1(x)$, for $x \in [x_1, x_2]$,

$$\begin{aligned}
\int_S \frac{\partial F}{\partial y} dxdy &= \int_{x=x_1}^{x_2} \int_{y=Y_1(x)}^{Y_2(x)} \frac{\partial F}{\partial y} dy\, dx = \\
&= \int_{x=x_1}^{x_2} \left[F(x, Y_2(x)) - F(x, Y_1(x))\right] dx = \\
&= -\int_{x=x_1}^{x_2} F(x, Y_1(x)) - \int_{x=x_2}^{x_1} F(x, Y_2(x)) dx = \\
&= -\oint_{\partial S} F(x, y) dx
\end{aligned}$$

Proof for generic domain

todo

In a simple domain in y, so that the closed contour ∂S is delimited by the curves $x = X_1(y)$, $x = X_2(y) > X_1(y)$ for $y \in [y_1, y_2]$,

$$\begin{aligned}
\int_S \frac{\partial G}{\partial x} dx dy &= \int_{y=y_1}^{y_2} \int_{x=X_1(y)}^{X_2(y)} \frac{\partial G}{\partial x} dx\, dy = \\
&= \int_{y=y_1}^{y_2} \left[G(X_2(y), y) - G(X_1(y), y) \right] dy = \\
&= \int_{y=y_1}^{y_2} G(X_1(y), y) dy + \int_{y=y_2}^{y_1} G(X_2(y), y) dy = \\
&= \oint_{\partial S} G(x, y) dy
\end{aligned}$$

CHAPTER

EIGHT

INTERCHANGING OPERATORS

- $I \subseteq \mathbb{R}$ interval
- Y set with accumulation point y_0
- $f : I \times Y \to \mathbb{R}$

Let the pointwise limit

$$g(x) := \lim_{y \to y_0} f(x, y)$$

exists for $\forall x \in I$, i.e. analogously $\forall \varepsilon > 0$, $\forall x \in I$, $\exists \delta > 0$ so that $|f(x, y) - g(x)| < \delta$ for $\forall y \in U_{y_0, \delta}$.

8.1 Interchange limits - Moore-Osgood theorem

Let x_0 a limit point of I. Suppose

1. for $\forall y \in V_{y_0}$ (punctured neighborhood of y_0), $\varphi(y) := \lim_{x \to x_0} f(x, y)$ exists. (*Here y is taken as fixed. If y is taken as a parameter, these are common limits of function of one variable, for all the possible parameters y; can it be interpreted as pointwise convergence as $x \to x_0$ for all the points $y \in V_{y_0}$?*)
2. $f(x, y) \to g(x)$ as $y \to y_0$ uniformly for $x \in U_{x_0}$ (punctured)

Then

- $\lim_{y \to y_0} \varphi(y)$ exists and

$$\lim_{y \to y_0} \varphi(y) = \lim_{x \to x_0} g(x) = \lim_{(x,y) \to (x_0, y_0)} f(x, y) \, .$$

Proof

Step 1. By 2. (see definition of uniform convergence), for $\forall \varepsilon > 0$, $\exists \delta_1 > 0$ s.t. $|f(x, y) - g(x)| < \frac{\varepsilon}{3}$ for $\forall y \in V_{y_0, \delta_1}$, $\forall x \in U_{x_0}$. Take $y_1, y_2 \in V_{y_0, \delta_1}$, then

$$|f(x, y_1) - f(x, y_2)| \leq |f(x, y_1) - g(x)| + |g(x) - f(x, y_2)| < \frac{2}{3}\varepsilon \, .$$

As $x \to x_0$, from 1.,

$$\lim_{x \to x_0} |f(x, y_1) - f(x, y_2)| = |\varphi(y_1) - \varphi(y_2)| \leq \frac{2}{3}\varepsilon \, .$$

Thus, as $\varphi(y)$ satisfies the Cauchy condition, the limit $\lim_{y \to y_0} \varphi(y) =: L$ exists.

Step 2. Shrinking V_{y_0,δ_1} to V_{y_0,δ_2} so that $|\varphi(y) - L| < \frac{\varepsilon}{3}$ for $\forall y \in V_{y_0,\delta_2}$. From the definition of $\varphi(y)$, then $\exists \delta_x > 0$ s.t.

$$|f(x,y) - \varphi(y)| < \frac{\varepsilon}{3} \quad , \quad \forall 0 < |x - x_0| < \delta_x \, .$$

Thus

$$|g(x) - L| \leq |g(x) - f(x,y)| + |f(x,y) - \varphi(y)| + |\varphi(y) - L| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \, ,$$

i.e. $g(x) \to L$ as $x \to x_0$.

Counter-example. $f(x,y) = \frac{x^2}{x^2+y^2}$ on $(x,y) \neq (0,0)$ for $x_0 = y_0 = 0$. This function is not uniformly convergent as $y \to y_0$.

…

Pointwise and uniform convergence

$$\lim_{y \to y_0} f(x,y) = g(x) \, .$$

Pointwise. For $\forall \varepsilon > 0$, $\forall x \in X$, $\exists \delta > 0$ s.t. $|f(x,y) - g(x)| < \varepsilon$ for $\forall y \in Y_{y_0,\delta}$. Here $\delta(\varepsilon, x)$.

Uniform. For $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $|f(x,y) - g(x)| < \varepsilon$ for $\forall y \in Y_{y_0,\delta}$, $\forall x \in X$, . Here $\delta(\varepsilon)$. This can be written in terms of $\sup |f(x,y) - g(x)|$ as well, as it must hold for $\forall x \in X$.

Cauchy criterion

Let $I \subseteq \mathbb{R}$ be an interval containing a limit point x_0, and let $f : I \setminus \{x_0\} \to \mathbb{R}$.

The limit $\lim_{x \to x_0} f(x)$ exists in $\mathbb{R}$ if and only if for every $\varepsilon > 0$, there exists a deleted neighborhood $U = (x_0 - \delta, x_0 + \delta) \setminus \{x_0\}$ such that for all $x_1, x_2 \in U$,

$$|f(x_1) - f(x_2)| < \varepsilon \, .$$

Proof.

Direction ($\Rightarrow$): if the limit exists, the Cauchy condition holds. Assume $\lim_{x \to x_0} f(x) = L$ exists. Fix $\varepsilon > 0$. By definition of a functional limit, there exists $\delta > 0$ such that for all $x \in I$ with $0 < |x - x_0| < \delta$:

$$|f(x) - L| < \frac{\varepsilon}{2}$$

Take any $x_1, x_2 \in I$ such that $0 < |x_1 - x_0| < \delta$ and $0 < |x_2 - x_0| < \delta$. By the triangle inequality:

$$|f(x_1) - f(x_2)| = |(f(x_1) - L) - (f(x_2) - L)| \leq |f(x_1) - L| + |f(x_2) - L|$$

Substitution gives $|f(x_1) - f(x_2)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Thus, the condition holds. □

Direction ($\Leftarrow$): if the Cauchy condition holds, then the limit exists.

…**todo**…

Bolzano-Weierstrass theorem

Every bounded sequence of real numbers has a convergence subsequence, i.e. let $(x_n)_{x=1}^{+\infty}$ be a sequence in $\mathbb{R}$. If $\exists a, b \in \mathbb{R}$ s.t. $a \leq x_n \leq b$, then $(x_n)_{n=1}^{+\infty}$ contains a convergent subsequence $(x_{n_k})_{k=1}^{+\infty}$

Proof. Take the original interval $I_0 = [a, b]$, and split it with its midpoint $c_0 = \frac{a+b}{2}$. (At least) one of the two sub-intervals contains infinitey many terms of the original sequence. Choose this sub-interval I_1, and repeat the process. Select the sub-sequence with these rules: pick $x_{n_1} \in I_1$; pick $x_{n_2} \in I_2$ s.t. $n_2 > n_1$ (always possible, as there's a infinite number of terms); pick $x_{n_3} \in I_3$ s.t. $n_3 > n_2$;... The sequence of the left endpoints (a_k) is non-decreasing and bounded above by b, so by monotone convergence theorem, then $\lim_{k\to+\infty} a_k = L$, for some $L \in \mathbb{R}$. The series of right endpoints satisfies $b_k = a_k + \frac{b-a}{2^k}$ and thus $\lim_{k\to+\infty} b_k = L$ as well. Thus, as $a_k \leq x_{n_k} \leq b_k$, it follows (squeeze theorem) that $\lim_{k\to+\infty} x_{n_k} = L$. ■

From $<$ to $\leq$ in limits

Let $f(x) < C$ for $x \in U_{x_0}$. Then $L := \lim_{x\to x_0} f(x) \leq C$.

Proof. By contradiction, let's assume $L > C$. 1) Define $\varepsilon := L - C > 0$; 2) apply the definition of limit. For $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $|f(x) - L| < \varepsilon$ for $\forall y \in U_{x_0,\delta}$; 3) this implies $L - \varepsilon < f(x) < L + \varepsilon$; 4) replacing $\varepsilon = L - C$, it follows $C < f(x) < -C$, and this contains a contradiction with the assumption of the theorem $f(x) < C$. It follows that the premise $L > C$ is false, and thus $L \leq C$. ■

8.2 Interchange limit and integral

8.2.1 Riemann integral

If

- $[a, b]$ compact
- $f(\cdot, y)$ Riemann integrable on $[a, b]$ for each y
- $f(x, y) \to g(x)$ uniformly as $y \to y_0$ uniformly for $x \in [a, b]$

Then

- g is Riemann integrable and

$$\lim_{y\to y_0} \int_{x=a}^{b} f(x, y)\, dx = \int_{x=a}^{b} g(x)\, dx\ .$$

Proof

Integrability of $g(x)$. Given $\varepsilon > 0$, there exists a neighborhood of y_0 so that

$$\sup_{x\in[a,b]} |f(x, y) - g(x)| < \frac{\varepsilon}{4(b-a)}\ .$$

Since $f(\cdot, y)$ is Riemann integrable, it's possible to choose a partition P so that $U(f(\cdot, y), P) - L(f(\cdot, y), P) < \frac{\varepsilon}{2}$, with

$$L(f(x), P) = \sum_n \inf_{x\in I_n} f(x)(x_n - x_{n-1})$$
$$U(f(x), P) = \sum_n \sup_{x\in I_n} f(x)(x_n - x_{n-1})$$

As $\sup g = \sup(f+g-f) \le \sup f + \sup|f-g|$, $\inf g = \inf(f+g-f) \le \inf f - \sup|f-g|$,

$$U(g,P) - L(g,P) \le U(f,P) - U(f,P) + 2\sup|f-g|(b-a) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \, .$$

For Riemann crieterion, g is Riemann integrable. Then

$$\left| \int_a^b f(x,y)\,dx - \int_a^b g(x)\,dx \right| \le \int_a^b |f(x,y) - g(x)|\,dx \le (b-a) \sup_{x\in[a,b]} |f(x,y) - g(x)| < \frac{varepsilon}{4} \, ,$$

and thus

$$\lim_{y\to y_0} \int_a^b f(x,y)\,dx = \int_a^b \lim_{y\to y_0} f(x,y)\,dx \, .$$

Counter-example. $f(x,y) = y^2 x(1-x)^y$, for $x \in [0,1]$, for $y \to +\infty$.

Details

As $y \to +\infty$, $f(x,y) \to 0$ for all $x \in [0,1]$, and thus $\int_0^1 \lim_{y\to+\infty} f(x,y)\,dx = 0$. On the other hand,

$$\begin{aligned}
\int_0^1 y^2 x(1-x)^y\,dx &= (u = 1-x,\ du = -dx) \\
&= \int_0^1 y^2(1-u)u^y du = \\
&= y^2 \left(\frac{1}{y+1} u^{y+1} - \frac{1}{y+2} u^{y+2} \right) \Big|_{u=0}^{1} = \\
&= \frac{y^2}{(y+1)(y+2)} \, ,
\end{aligned}$$

and thus $\lim_{y\to+\infty} \int_0^1 f(x,y)\,dx = 1$.

This occurs because $f(x,y)$ doesn't converge uniformly as $y \to +\infty$. This can be easily proved as:

- the function $f(x,y)$ converges pointwise to 0 as $y \to +\infty$
- but

$$\sup_{x\in[0,1]} |f(x,y) - 0| = y^2 \left(1 - \frac{1}{1+y}\right)^{y+1} \to +\infty \text{ as } y \to +\infty$$

since

$$\partial_x f(x,y) = y^2 \left[(1-x)^y - yx(1-x)^{y-1}\right] = y^2(1-x)^{y-1}(1-x-xy)$$

$$0 = \partial_x f(\overline{x},y) \quad \Rightarrow \quad \overline{x}_1 = 1 \, , \quad \overline{x}_2 = \frac{1}{1+y}$$

$$f(\overline{x}_2,y) = y^2 \left(1 - \frac{1}{1+y}\right)^{y+1}$$

as $\lim_{y\to+\infty} \left(1 - \frac{1}{1+y}\right)^{y+1} = e^{-1}$, it follows that $\lim_{y\to+\infty} f(\overline{x}_2,y) = +\infty$. Thus $\lim_{y\to+\infty} \sup_{x\in[0,1]} |f(x,y) - 0| = +\infty$, and $f(x,y)$ doesn't converge uniformly to 0.

Riemann integrable function

$$\int_{x=a}^{b} f(x)\ dx = \lim_{\Delta x \to +\infty} \sum_{n=1}^{N} f(\xi_n)(x_n - x_{n-1}) \ ,$$

with $\xi_n \in [x_n, x_{n+1}]$ with $x_0 = a$, $x_N = b$.

Riemann criterion for integrability

8.2.2 Lebesgue integral

…

Counter-example. $f(x,y) = y \cdot 1_{\left(0,\frac{1}{y}\right)}(x)$ for $x \in [0,1]$, for $y \to +\infty$.

8.3 Interchange limit and derivative

- I **open** interval
- $f(\cdot, y)$ differentaible on I for $\forall y \in V_{y_0}$, punctured

Suppose:

1. $f(x_1, y)$ converges as $y \to y_0$ for at least one point $x_1 \in I$
2. $f_x(x,y) \to h(x)$ uniformly as $y \to y_0$ for x in every compact subinterval $J \subseteq I$ *(This is required for differentiation as I is open)*

Then:

- $f(x,y)$ converges locally uniformly for $\forall x \in I$ to some $g(x)$, as $y \to y_0$
- $g(x)$ is differentiable on I and

$$\frac{d}{dx}\left(\lim_{y \to y_0} f(x,y)\right) = g'(x) = h(x) = \lim_{y \to y_0} \frac{\partial f}{\partial x}(x,y) \ .$$

Proof

Let (and fix) $J \subset I$ compact, with $x_1 \in J$, and length $|J|$.

Step 1. 1. and 2. implies that $f(\cdot, y)$ converges uniformly to some $g(x)$ for $x \in J$. By 2., for all $\varepsilon > 0$, $\exists V_{y_0,\delta}$ so that (Cauchy criterion)

$$\sup_{\xi \in J} |f_x(\xi, y) - f_x(\xi, y')| < \frac{\varepsilon}{2(|J|+1)} \ , \tag{8.1}$$

for all $y,\ y' \in V_{y_0}$. Applying Mean Value Theorem to $\Phi(x) := f(x,y) - f(x,y')$, differentiable for $x \in I$, there's a $\xi \in [x, x_1]$ so that

$$\Phi(x) - \Phi(x_1) = \Phi'(\xi)\,(x - x_1) = [f'_x(\xi, y) - f'_x(x,y)]\,(x - x_1) \ ,$$

and

$$|f(x,y) - f(x,y')| \le |f(x_1,y) - f(x_1,y')| + |[f'_x(\xi,y) - f'_x(x,y)]\,(x - x_1)|$$

By 1., for $\forall \varepsilon > 0$, there's a V_{y_0,δ_1} ($\subseteq V_{y_0,\delta}$, so that the uniform convergence of $f_x(\cdot, y)$ still holds) so that $|f(x_1, y) - f(x_1, y')| < \frac{\varepsilon}{2}$. Using (8.1), the second term is bounded by $\frac{\varepsilon}{2(|J|+1)}|J| < \frac{\varepsilon}{2}$, and thus for all $x \in J$

$$|f(x, y) - f(x, y')| < \varepsilon \, ,$$

i.e. $f(x, y)$ converges uniformly for $x \in J$ (to some $g(x)$) as $y \to y_0$. **todo** *Discuss how $g(x)$ can be defined "interval-wise" through uniform convergence on all the $J \subset I$.*

Step 2. Differentiability of g, through Moore-Osgood. As the derivative can be defined as the limit of the incremental ratio, the limit of a derivative and the derivative of a limit involve two limits on different variables. Thus, the possibility of interchanging limit and derivative operators can be proved trhough Moore-Osgood theorem.

Fix $x \in I$. For $t \neq 0$ so that $[x, x+t] \in I$ (or $[x+t, x] \in I$), define the incremental ratios

$$Q(t, y) := \frac{f(x+t, y) - f(x, y)}{t} \quad , \quad \varphi(t) := \frac{g(x+t) - g(x)}{t} \, .$$

Let's check the assumptions of Moore-Osgood theorem for the limits $\lim_{t\to 0}$, $\lim_{y\to y_0}$ applied to Q

- **Assumption 1. of M-O theorem** For every $y \in V_{y_0}$, $\lim_{t\to 0} Q(t, y)$ exists for all $x \in J \subset I$, and it's equal (by definition) to $\lim_{t\to 0} Q(t, y) = f_x(x, y)$
- **Assumption 2. of M-O theorem** ($Q(t, y) \to \varphi(t)$ uniformly as $y \to y_0$ for $t \in T_0$ punctured). For every fixed t, $Q(t, y) \to \varphi(y)$ as $f(\cdot, y) \to g(\cdot)$ as $y \to y_0$ (and this convergence holds for x, $x + t$). In order to proof that the convergence is uniform in t, let's apply MVT to $f(x + t, y) - f(x + t, y')$

  $$|Q(t, y) - Q(t, y')| = \left| \frac{f(x+t, y) - f(x, y) - f(x+t, y') + f(x, y')}{t} \right| = |f_x(\xi, y) - f_x(\xi, y')| \, ,$$

  for some $\xi \in [x, x+t]$. By 2., for $\forall \eta > 0$ $\exists V_{y_0}$ (and Cauchy criterion) so that $|f_x(\xi, y) - f_x(\xi, y')| < \eta$ for $y, y' \in V_{y_0}$, independently of t. Thus $Q(t, \cdot)$ converges uniformly in t as $y \to y_0$. Since $Q(t, y) \to \varphi(t)$ pointwise for $y \to y_0$, and $Q(t, y)$ converges uniformly, thus $Q(t, y) \to \varphi(t)$ uniformly in t for $y \to y_0$.

As the two assumptions of Moore-Osgood theorem holds for $Q(t, y)$, the theorem implies that

- $\lim_{y\to y_0} f_x(x, y)$ exists
- $\lim_{t\to 0} \varphi(t)$ exists
- and

$$h(x) := \lim_{y\to y_0} f_x(x, y) = \lim_{t\to 0} \varphi(t) = \frac{d}{dx} g(x) = \frac{d}{dx} \left(\lim_{y\to y_0} f(x, y) \right) \, . \quad \blacksquare$$

Counter-example. $f(x, y) = \frac{\sin(yx)}{\sqrt{y}}$, with $x \in \mathbb{R}$ and $y \to +\infty$.

The function $f(x, y) \to 0$ uniformly as $y \to +\infty$ (it converges pointwise to 0 and $\sup |f(x, y) - 0| = \frac{1}{\sqrt{y}} \to 0$ as $y \to +\infty$).

The derivative $\partial_x f(x, y) = \sqrt{y} \cos(yx)$ doesn't converge to any function as $y \to +\infty$, and thus it doesn't converge uniformly neither. It meaningless even to try evaluating its derivative.

8.4 Extra

Uniqueness of the theorem. If $\lim_{x \to x_0} f(x) = L_1$, and $\lim_{x \to x_0} f(x) = L_2$, then $L_1 = L_2$.

Part III

Differential Geometry

CHAPTER

NINE

INTRODUCTION TO DIFFERENTIAL GEOMETRY

Geometric entities embedded in 3-dimensional Euclidean space

- *Lines*
- *Surfaces*

Definition 9.1 (Euclidean space, E^3.)

A Euclidean space is a space whose points can be globally represented with a set of **Cartesian coordinates** $\{x^i\}_{i=1:3}$, i.e. $\mathbf{r}(x^i)$.

Some details about Cartesian coordinates

As shown in the introduction to *tensor calculus*, the *natural basis* induced by a set of coordinates $\{q^a\}_a$ is

$$\mathbf{b}_a := \frac{\partial \mathbf{r}}{\partial q^a}(q^b) \, .$$

For a set of Cartesian coordinates, the induced natural basis - whose vectors are often denoted with the letters $\hat{\mathbf{e}}_i$, or $\hat{\mathbf{x}}_i$ - is **orthonormal**, and uniform (constant, independent from the coordinates) in space, i.e.

$$\mathbf{b}_a = \hat{\mathbf{x}}_a = \frac{\partial \mathbf{r}}{\partial x^a} \, .$$

Thus,

- the **position vector r** can be written as

$$\mathbf{r} = x^a \hat{\mathbf{x}}_a \, ,$$

- since the vectors of the natural basis are not functions of the coordinates, then their space derivatives, i.e. the **second-order derivatives** of the position w.r.t. the coordinates - and the **Christoffell symbols** - are identically zero,

$$0 = \frac{\partial \hat{\mathbf{x}}_a}{\partial x^b} = \frac{\partial^2 \mathbf{r}}{\partial x^a \partial x^b} \quad , \quad \Gamma^c_{ab} \equiv 0 \, ,$$

- as the basis is orthonormal, $\hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j = \delta_{ij}$, then the components of the **metric tensor** are

$$[g_{ab}] = [g^{ab}] = \mathbf{I}_3 \, ,$$

- from the definition of the reciprocal basis, $\mathbf{b}^a = g^{ab}\mathbf{b}_b$, it follows that the **reciprocal basis** of a Cartesian basis coincides the basis itself.

9.1 Lines

A line γ is a geometric entity that has a 1-parameter parametric representation. Its points in 3-dimensional (Euclidean) space E^3 can be represented with the function

$$\gamma : \mathbf{r}(q) , \quad q \in [q_0, q_1] .$$

A parametrization is regular if the parameter q is monotonic along the curve.

9.1.1 Differential

The differential $d\mathbf{r}$ represents the elementary vector on the line. From the very definition of the differential of a function, it follows

$$d\mathbf{r}(q) = \mathbf{r}'(q)\, dq .$$

Definition 9.1.1 (Arc-length parameter)

The arc-length parameter s is defined by the condition $ds = |d\mathbf{r}(s)|$, or by the equivalent conditions

$$|d\mathbf{r}(s)| = |\mathbf{r}'(s)|\,|ds| \qquad \rightarrow \qquad |\mathbf{r}'(s)| = 1 \qquad \rightarrow \qquad \mathbf{r}'(s) = \hat{\mathbf{t}}(s) .$$

9.1.2 Frenet basis

Using **arc-length parameter** *Definition 9.1.1*, Frenet basis is naturally defined as the set $\{\hat{\mathbf{t}}, \hat{\mathbf{n}}, \hat{\mathbf{b}}\}$:

- tangent unit vector, $\hat{\mathbf{t}}(s) = \mathbf{r}'(s)$,
- normal unit vector, $\hat{\mathbf{n}}$, so that $\hat{\mathbf{r}}''(s) = \hat{\mathbf{t}}'(s) =: \kappa(s)\,\hat{\mathbf{n}}(s)$, with $\kappa(s)$ local curvature
- binormal unit vector, $\hat{\mathbf{b}}(s) = \hat{\mathbf{t}}(s) \times \hat{\mathbf{n}}(s)$

Using a general parameter, t, with some abuse of notation $\mathbf{r}(t) = \mathbf{r}(s(t))$ and indicating $\dot{(_)} = \frac{d}{dt}$,

- $\dot{\mathbf{r}} = \frac{ds}{dt}\frac{d\mathbf{r}}{ds} = \dot{s}\hat{\mathbf{t}}$
- $\ddot{\mathbf{r}} = \dfrac{d}{dt}\dot{\mathbf{r}} = \dfrac{d}{dt}\left(\dot{s}\hat{\mathbf{t}}\right) = \ddot{s}\hat{\mathbf{t}} + \dfrac{ds}{dt}\dfrac{d}{ds}\hat{\mathbf{t}} = \ddot{s}\hat{\mathbf{t}} + \dot{s}^2\kappa\,\hat{\mathbf{n}}$

Definition 9.1.2 (Osculator circle)

The osculator circle relative to a point P of a curve γ, its the circle that:

- is tangent to the curve γ in P
- has the same curvature of the curve γ in P

Osculator circle - Details

Circle with $R(s) = \frac{1}{\kappa(s)}$, in plane orthogonal to $\hat{\mathbf{b}}(s)$, passing through $\mathbf{r}(s)$, and thus center in $\mathbf{r}_C(s) = \mathbf{r}(s) + \hat{\mathbf{n}}R(s)$. Its parametric representation using its arc-length parameter p, with $\mathbf{r}(p=0) = \mathbf{r}(s)$ reads

$$\mathbf{r}(p) = \mathbf{r}_C(s) + R(s)\left[-\cos\left(\frac{p}{R(s)}\right)\hat{\mathbf{n}}(s) + \sin\left(\frac{p}{R(s)}\right)\hat{\mathbf{t}}(s)\right] .$$

Its first and second order derivatives w.r.t. the arc-length p evaluated in $p = 0$, i.e. $\mathbf{r} = \mathbf{r}(s)$ read:

- first derivative in $p = 0$,

$$\hat{\mathbf{t}}(p)\big|_{p=0} = \mathbf{r}'(p)\big|_{p=0} = \left[\sin\left(\frac{p}{R(s)}\right)\hat{\mathbf{n}}(s) + \cos\left(\frac{p}{R(s)}\right)\hat{\mathbf{t}}(s)\right]\Bigg|_{p=0} = \hat{\mathbf{t}}(s) ,$$

  i.e. the osculator circle has the same tangent as the curve in the point.

- second derivative in $p = 0$,

$$\kappa(p)\hat{\mathbf{n}}(p)\big|_{p=0} = \mathbf{r}''(p)\big|_{p=0} = \frac{1}{R(s)}\left[\cos\left(\frac{p}{R(s)}\right)\hat{\mathbf{n}}(s) - \sin\left(\frac{p}{R(s)}\right)\hat{\mathbf{t}}(s)\right]\Bigg|_{p=0} = \frac{1}{R(s)}\hat{\mathbf{n}}(s) = \kappa(s)\hat{\mathbf{n}}(s) ,$$

  i.e. the osculator circle has the same normal vector and curvature as the curve in the point.

9.2 Surfaces

A surface S is a geometric entity that has a 2-parameter parametric representation. Its points in 3-dimensional (Euclidean) space E^3 can be represented with the function

$$S : \mathbf{r}(q^1, q^2) .$$

9.2.1 Differential

The differential $d\mathbf{r}$ represents an elementary vector on - tangent to - the surface. From the very definition of the differential of a function, it follows

$$d\mathbf{r} = \frac{\partial \mathbf{r}}{\partial q^1}\, dq^1 + \frac{\partial \mathbf{r}}{\partial q^2}\, dq^2 = \mathbf{b}_1\, dq^1 + \mathbf{b}_2\, dq^2$$

9.2.2 Normal vector

The vectors $\mathbf{b}_1, \mathbf{b}_2$ are tangent to the surface S. A **regular parametrization** has $\mathbf{b}_1 \times \mathbf{b}_2 \neq \mathbf{0}$ everywhere. A third vector $\mathbf{b}_3 := \hat{\mathbf{n}}$ can be defined so that $|\hat{\mathbf{n}}| = 1$ and $\hat{\mathbf{n}} \cdot \mathbf{b}_i = 0$, $i = 1 : 2$. This is a **unit normal vector** to the surface. If the parametrization is regular, then the set of vectors $\{\mathbf{b}_1, \mathbf{b}_2, \hat{\mathbf{n}}\}$ forms a local(natural) basis of 3-dimensional vectors space.

The components of the metric tensor are

$$[g_{ab}] = \begin{bmatrix} g_{11} & g_{12} & \cdot \\ g_{21} & g_{22} & \cdot \\ \cdot & \cdot & 1 \end{bmatrix} ,$$

so that the vectors of the reciprocal basis, $\mathbf{b}^a = g^{ab}\mathbf{b}_b$, read

$$\begin{cases} \mathbf{b}^1 = g^{11}\mathbf{b}_1 + g^{12}\mathbf{b}_2 \\ \mathbf{b}^2 = g^{21}\mathbf{b}_1 + g^{22}\mathbf{b}_2 \\ \mathbf{b}^3 = \hat{\mathbf{n}} . \end{cases}$$

9.2.3 Derivatives of the natural basis

For $i = 1:2$, $j = 1:2$

$$\frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^j} = \frac{\partial \mathbf{b}_i}{\partial q^j} = \Gamma^k_{ij}\mathbf{b}_k = \Gamma^1_{ij}\mathbf{b}_1 + \Gamma^2_{ij}\mathbf{b}_2 + \Gamma^3_{ij}\mathbf{b}_3 \,,$$

so that

$$\Gamma^k_{ij} = \mathbf{b}^k \cdot \frac{\partial \mathbf{b}_i}{\partial q^j} \,.$$

Remark. Comparing two different choices of the orthogonal coordinate q^3, (1) unit normal vector $\hat{\mathbf{b}}_3 = \hat{\mathbf{n}}$, (2) non-unitary normal vector $\hat{\mathbf{b}}_3 = \mathbf{n}$, $\hat{\mathbf{b}}^3 = \frac{\mathbf{n}}{|\mathbf{n}|^2}$, it's easy to prove that the Christoffel coefficients with upper index $k = 3$, Γ^3_{ij}, are independent from this choice.

9.2.4 Geometric entities

Normal vector.

$$\mathbf{n}(q^1, q^2) = \frac{\partial \mathbf{r}}{\partial q^1}(q^1, q^2) \times \frac{\partial \mathbf{r}}{\partial q^2}(q^1, q^2) = \mathbf{b}_1(q^1, q^2) \times \mathbf{b}_2(q^1, q^2)$$

First-order approximation and the tangent plane. The equation of the points of the tangent plane π in point $\mathbf{r}\,(q^1, q^2)$ follows the orthogonality condition

$$\pi : \; (\mathbf{r} - \mathbf{r}(q^1, q^2)) \cdot \mathbf{n}(q^1, q^2) = 0 \,.$$

Length of elementary segment on the plane.

$$\begin{aligned} |d\mathbf{r}|^2 = d\mathbf{r} \cdot d\mathbf{r} = \\ = (\mathbf{b}_1\, dq^1 + \mathbf{b}_2\, dq^2) \cdot (\mathbf{b}_1\, dq^1 + \mathbf{b}_2\, dq^2) = \\ = g_{11}\, dq^1\, dq^1 + g_{12}\, dq^1\, dq^2 + g_{21}\, dq^2\, dq^1 + g_{22}\, dq^2\, dq^2 = g_{ij}\, dq^i\, dq^j \end{aligned}$$

Second order approximation and the curvature tensor.

$$\begin{aligned} \mathbf{r}(q^1 + dq^1, q^2 + dq^2) = \mathbf{r}(q_1, q_2) + \frac{\partial \mathbf{r}}{\partial q^i}\, dq^i + \frac{1}{2}\frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^j}\, dq^i dq^j + o\left(|dq^i|^2\right) = \\ = \mathbf{r}(q_1, q_2) + \mathbf{b}_i\, dq^i + \frac{1}{2}\mathbf{b}_k \Gamma^k_{ij}\, dq^i\, dq^j + \frac{1}{2}\hat{\mathbf{n}}\, \Gamma^3_{ij}\, dq^i dq^j + o\left(|dq^i|^2\right) \,, \end{aligned}$$

so that

$$\begin{aligned} [\mathbf{r}(q^1 + dq^1, q^2 + dq^2) - \mathbf{r}(q^1, q^2)] \cdot \hat{\mathbf{n}} = \frac{1}{2}\Gamma^3_{ij}\, dq^i\, dq^j = \\ = \frac{1}{2}\hat{\mathbf{n}} \cdot \frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^j}\, dq^i\, dq^j = \\ = \frac{1}{2}\hat{\mathbf{n}} \cdot \frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^j}\, \mathbf{b}^i \cdot \mathbf{b}_k dq^k\, \mathbf{b}^j \cdot \mathbf{b}_l dq^l = \\ = \frac{1}{2}\, \underbrace{dq^k \mathbf{b}_k}_{d\mathbf{r}} \cdot \left[\left(\hat{\mathbf{n}} \cdot \frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^j} \right) \mathbf{b}^i \otimes \mathbf{b}^j \right] \cdot \underbrace{dq^l \mathbf{b}_l}_{d\mathbf{r}} = \\ = \frac{1}{2} d\mathbf{r} \cdot \mathbb{b} \cdot d\mathbf{r} \,, \end{aligned}$$

having defined the **curvature tensor** $\mathbb{b}$,

$$\mathbb{b} = b_{ij}\mathbf{b}^i \otimes \mathbf{b}^j = \left(\hat{\mathbf{n}} \cdot \frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^j} \right) \mathbf{b}^i \otimes \mathbf{b}^j \,.$$

Curvature tensor

From the second-order Taylor expansion around a point $\mathbf{r}(q^1, q^2)$, the projection of the displacement onto the unit normal vector $\hat{\mathbf{n}}$ measures the distance from the surface to its tangent plane π:

$$[\mathbf{r}(q^1 + dq^1, q^2 + dq^2) - \mathbf{r}(q^1, q^2)] \cdot \hat{\mathbf{n}} = \frac{1}{2}\Gamma^3_{ij} dq^i dq^j = \frac{1}{2} b_{ij} dq^i dq^j.$$

The symmetric covariant tensor of rank 2, defined by its components

$$b_{ij} := \Gamma^3_{ij} = \hat{\mathbf{n}} \cdot \frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^j} = -\frac{\partial \hat{\mathbf{n}}}{\partial q^i} \cdot \mathbf{b}_j,$$

is the **second fundamental form** (or curvature tensor) of the surface embedded in E^3.

The quadratic form $b_{ij} dq^i dq^j$ characterizes the local deviation of the surface from the tangent plane π. Let $w_n(q^1, q^2) = d\mathbf{r} \cdot \mathbb{b} \cdot d\mathbf{r}$ the normal deviation of the surface w.r.t. the tangent plane, then

$$w_n(q^1, q^2) = b_{ij} dq^i dq^j \ .$$

The curvature tensor $\mathbb{b}$ is a **symmetric** order-2 tensor, locally built on a 2-dimensional vector space. Its **spectral decomposition**

$$\mathbb{b} = \kappa_1 \mathbf{v}_1 \otimes \mathbf{v}_1 + \kappa_2 \mathbf{v}_2 \otimes \mathbf{v}_2 \ ,$$

being $\kappa_1, \kappa_2 \in \mathbb{R}$ the real egivenvalues, and $\mathbf{v}_1$, $\mathbf{v}_2$ the unit orthogonal basis defined by the eigenvectors of the eigenvalue problem

$$\mathbb{b} \cdot \mathbf{v}_i = \kappa_i \mathbf{v}_i \ , \qquad i \in \{1, 2\} \ .$$

The eigenvectors $\mathbf{v}_{1,2}$ defines the **principal directions**, while the eigenvalues $\kappa_{1,2}$ are defined as the **principal curvatures**.

Eigenvalue problem

$$\begin{aligned}
\mathbb{b} \cdot \mathbf{v}_{(\alpha)} &= \kappa_{(\alpha)} \mathbf{v}_{(\alpha)} \\
\left(b_{ij} \mathbf{b}^i \otimes \mathbf{b}^j\right) \cdot \left(v^l_{(\alpha)} \mathbf{b}_l\right) &= \kappa_{(\alpha)} v^m_{(\alpha)} \mathbf{b}_m \\
b_{ij} v^j_{(\alpha)} \mathbf{b}^i &= \kappa_{(\alpha)} v^m_{(\alpha)} \mathbf{b}_m \\
g^{im} b_{ij} v^j_{(\alpha)} \mathbf{b}_m &= \kappa_{(\alpha)} v^m_{(\alpha)} \mathbf{b}_m \\
b^m{}_j v^j_{(\alpha)} \mathbf{b}_m &= \kappa_{(\alpha)} v^m_{(\alpha)} \mathbf{b}_m
\end{aligned}$$

i.e. in components $b^m{}_j v^j_{(\alpha)} = \kappa_{(\alpha)} v^m_{(\alpha)}$, or

$$\left(b^m{}_j - \kappa_{(\alpha)} \delta^m{}_j\right) v^j_{(\alpha)} = 0 \ .$$

Even more explicitly, the condition $\det\left(b^m{}_j - \kappa_{(\alpha)} \delta^m{}_j\right) = 0$ becomes

$$0 = \left| \begin{bmatrix} b^1{}_1 - \kappa_{(\alpha)} & b^1{}_2 \\ b^2{}_1 & b^2{}_2 - \kappa_{(\alpha)} \end{bmatrix} \right| = \kappa^2_{(\alpha)} - \left(b^1{}_1 + b^2{}_2\right) \kappa_{(\alpha)} + b^1{}_1 b^2{}_2 - b^1{}_2 b^2{}_1 = \kappa^2_{(\alpha)} - \mathrm{tr}\{\mathbf{B}\} \kappa_{(\alpha)} + \det\{\mathbf{B}\} \ .$$

The eigenvalues are

$$\kappa_{(\alpha)} = \frac{\mathrm{tr}\,\{\mathbf{B}\}}{2} \mp \sqrt{\left(\frac{\mathrm{tr}\,\{\mathbf{B}\}}{2}\right)^2 - \det\{\mathbf{B}\}} \ .$$

with $\mathbf{B}$ the matrix collecting the mixed components of the curvature tensor, $[\mathbf{B}]_{ij} = \left[b^i{}_j\right]$. It's easy to prove that

$$\text{tr}\{\mathbf{B}\} = \kappa_1 + \kappa_2 \ ,$$

and

$$\det\{\mathbf{B}\} = \kappa_1 \cdot \kappa_2 \ .$$

Remark. Eigenvalues are real, as

… **todo** *check and uncomment*

Mean curvature

The mean curvature is defined as the arithmetic average of the principal curvatures, $H := \frac{1}{2}(\kappa_1 + \kappa_2)$.

$$\begin{aligned} H &= \frac{1}{2}(\kappa_1 + \kappa_2) = \\ &= \frac{\text{tr}\{[b^i{}_j]\}}{2} = \frac{1}{2} b^i{}_i = \\ &= \frac{1}{2} g^{ij} b_{ij} = \frac{1}{2}\left(g^{11}b_{11} + g^{12}b_{12} + g^{21}b_{21} + g^{22}b_{22}\right) = \\ &= \frac{1}{2}[g_{ij}]^{-1} b_{ij} = \frac{1}{2\left(g_{11}g_{22} - g_{12}^2\right)}\left(g_{22}b_{11} - g_{12}b_{12} - g_{21}b_{21} + g_{11}b_{22}\right) \end{aligned}$$

Gaussian curvature

Gaussian curvature is defined as the determinant of $b^i{}_j$,

$$K = \det\{b^i{}_j\} = \frac{\text{tr}\{b_{ij}\}}{\text{tr}\{g_{ij}\}} \ ,$$

as $b_{ij} = g_{il} b^l{}_j$.

Principal Curvatures and Directions

The extreme values of the normal curvature $\kappa_n = \frac{b_{ij} dq^i dq^j}{g_{ij} dq^i dq^j}$ are the **principal curvatures** κ_1 and κ_2, which are the eigenvalues of the tensor b^i_j:

$$\det(b^i_j - \kappa \delta^i_j) = 0 \quad \Longrightarrow \quad \kappa^2 - 2H\kappa + K = 0.$$

Thus, the principal curvatures satisfy:

$$\kappa_{1,2} = H \pm \sqrt{H^2 - K}.$$

- The Gaussian curvature is the product of the principal curvatures: $K = \kappa_1 \kappa_2$.
- The Mean curvature is the average of the principal curvatures: $H = \frac{1}{2}(\kappa_1 + \kappa_2)$.

…

9.3 Curvature

This section is devoted to curvature, and its different descriptions:

- ***Extrinsic curvature***, i.e. the description of the curvature of a n-manifold embedded in a $n+1$-space, as it's usually done in the description of *2-dimensional surfaces in the 3-dimensional Euclidean space*.
- ***Intrinsic curvature***, i.e. the description of a curvature of a n-manifold without the need for an embedding in a higher-dimensional space. This approach exploits the concepts of parallel transport, that - when applied to a vector field on a closed loop, see *Parallel transport around a closed loop and the Riemann tensor* - provides the definition of Riemann tensor, *Definition 9.3.1*. This is the common approach to Einstein General Relativity: spacetime is a 4-dimensional manifold, and it's not embedded in a 5-dimensional space in the original formulation of GR theory.

9.3.1 Extrinsic curvature

Extrinsic curvature of a n-dimensional manifold needs the manifold to be embedded in a $n+1$-dimensional space.

From the second order approximation of the n-dimensional manifold $\mathbf{r}\left(q^i\right)$ in a $n+1$-dimensional space

$$\begin{aligned}\mathbf{r}\left(q^i+dq^i\right)-\mathbf{r}\left(q^i\right) &= \frac{\partial \mathbf{r}}{\partial q^i}dq^i+\frac{1}{2}\frac{\partial^2\mathbf{r}}{\partial q^i\partial q^k}dq^idq^k+o(|d\mathbf{q}|^2)= \\ &= \mathbf{b}_i\,dq^i+\frac{1}{2}\Gamma^\alpha_{ik}\mathbf{b}_\alpha\,dq^idq^k= \\ &= \mathbf{b}_i\,dq^i+\frac{1}{2}\left(\Gamma^l_{ik}\mathbf{b}_l+\Gamma^{n+1}_{ik}\mathbf{n}\right)\,dq^idq^k\,,\end{aligned}$$

with the Greek indices for sum over the $n+1$ dimensions, and latin indices for the indices in $\{1,\dots,n\}$. Here $\mathbf{n}:=\mathbf{b}_{n+1}$ s.t. $\mathbf{n}\cdot\mathbf{b}_i=0$, for $n\in\{1,\dots,n\}$.

The extrinsic curvature tensor, already used for *differential geometry of surfaces*, is defined as $\mathbb{b}:=\Gamma^{n+1}_{ik}\mathbf{b}^i\otimes\mathbf{b}^k$.

Derivatives of the normal vector - Weingarten formula

The spatial derivatives of the unit normal vector $\hat{\mathbf{n}}(q^i)$ reads

$$\frac{\partial\hat{\mathbf{n}}}{\partial q^i}=-b_i{}^l\mathbf{b}_l\,. \tag{9.1}$$

Proof

Consider an n-dimensional hypersurface embedded in an $(n+1)$-dimensional space with normal vector $\hat{\mathbf{n}}$ satisfying $\hat{\mathbf{n}}\cdot\mathbf{b}_i=0$. The derivative of basis vectors decomposes into tangential and normal components:

$$\frac{\partial\mathbf{b}_i}{\partial q^k}=\Gamma^\alpha_{ik}\mathbf{b}_\alpha=\Gamma^l_{ik}\mathbf{b}_l+\Gamma^{n+1}_{ik}\hat{\mathbf{n}}$$

Differentiating the normality condition $\hat{\mathbf{n}}\cdot\mathbf{b}_k=0$ with respect to q^i,

$$\begin{aligned}0 &= \frac{\partial\hat{\mathbf{n}}}{\partial q^i}\cdot\mathbf{b}_k+\hat{\mathbf{n}}\cdot\frac{\partial\mathbf{b}_k}{\partial q^i}= \\ &= \frac{\partial\hat{\mathbf{n}}}{\partial q^i}\cdot\mathbf{b}_k+\hat{\mathbf{n}}\cdot\left(\Gamma^l_{ik}\mathbf{b}_l+\Gamma^{n+1}_{ik}\hat{\mathbf{n}}\right)\,,\end{aligned}$$

or

$$\frac{\partial \hat{\mathbf{n}}}{\partial q^i} \cdot \mathbf{b}_k = -\hat{\mathbf{n}} \cdot \left(\Gamma^l_{ik} \mathbf{b}_l + \Gamma^{n+1}_{ik} \hat{\mathbf{n}} \right) = -\Gamma^{n+1}_{ik}$$

From the latter expression, it immediately follows

$$\frac{\partial \hat{\mathbf{n}}}{\partial q^i} = -\Gamma^{n+1}_{ik} \mathbf{b}^k = -g^{kl} \Gamma^{n+1}_{ik} \mathbf{b}_l = -b_i{}^l \mathbf{b}_l \ ,$$

where $b_{ik} = \Gamma^{n+1}_{ik}$ is the **second fundamental form**.

9.3.2 Intrinsic curvature

Intrinsic curvature of a n-dimensional manifold doesn't need any embedding in a higher dimensional space, but it's built from the metric $g_{\mu\nu}$ and its derivatives. It measures…

Riemann curvature tensor

Vector fields as differential operators, and directional or covariant derivatives

A vector field $X(f)$ acting on a scalar function f defines a directional derivative:

$$X(f) = X^\mu \partial_\mu f \ . \tag{9.2}$$

The definition of the vector field X as a differential operator can be recast as the scalar product of the vector field $\mathbf{X}$ and the gradient of the function f,

$$\nabla_X f = \mathbf{X} \cdot \nabla f \ .$$

The **directional derivative** of the function f along a curve γ with element $d\mathbf{r}$ is $d\mathbf{r} \cdot \nabla f$. So, for a scalar field f,

$$d\mathbf{r} \cdot \nabla f = (dq^\nu \mathbf{b}_\nu) \cdot \mathbf{b}^\mu \partial_\mu f = dq^\mu \partial_\mu f \ .$$

For a vector field $\mathbf{v}$,

$$\begin{aligned} d\mathbf{r} \cdot \nabla \mathbf{v} &= (dq^\nu \mathbf{b}_\nu) \cdot \mathbf{b}^\mu \partial_\mu \mathbf{v} = \\ &= dq^\mu \partial_\mu \mathbf{v} = \\ &= dq^\mu \left(\partial_\mu v^\nu + \Gamma^\nu_{\xi\mu} v^\xi \right) \mathbf{b}_\nu \ . \end{aligned} \tag{9.3}$$

Parallel transport

A vector field $\mathbf{v}$ is parallel transported along a curve γ with element $d\mathbf{r}$ if $d\mathbf{r} \cdot \nabla f = \mathbf{0}$. Thus, using the relation (9.3), parallel transport implies

$$\partial_\mu v^\nu = -\Gamma^\nu_{\xi\mu} v^\xi \ . \tag{9.4}$$

To first order, the transported vector v^μ from q^η to $q^\eta + dq^\eta$ satisfies:

$$v^\nu(q^\eta + dq^\eta) \sim v^\nu(q^\eta) + dq^\mu \partial_\mu v^\nu(q^\eta) = v^\nu(q^\eta) - dq^\mu \Gamma^\nu_{\mu\chi}(q^\eta) v^\chi(q^\eta) \ .$$

Lie bracket

The Lie bracket of two vector fields X and Y measures their non-commutativity, i.e.

$$[X,Y]f = X(Y(f)) - Y(X(f)) \,.$$

In component notation, applying the definition (9.2) of a vector field as differential operator twice,

$$\begin{aligned}[X,Y]\, f &= X^\mu \partial_\mu (Y^\nu \partial_\nu f) - Y^\nu \partial_\nu (X^\mu \partial_\mu f) = \\ &= X^\mu \partial_\mu Y^\nu \partial_\nu f + X^\mu Y^\nu \partial_{\mu\nu} f - Y^\nu \partial_\nu (X^\mu \partial_\mu f) - X^\mu Y^\nu \partial_{\mu\nu} f = \\ &= (X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu) \partial_\mu f \,,\end{aligned}$$

so that the components of the vector field $[X,Y]$ in the natural basis are

$$[X,Y]^\mu = X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu \,.$$

Parallel transport around a closed loop and the Riemann tensor

Elementary loop

The elementary loop is build with 2 vector fields X, Y. In general, the elementary loops need 5 sides if the vector fields X, Y don't commute. Let (**todo** *Add some details, and pay attention to notation, as it can be non-trivial for curved spaces*)

- point **A** be in $\mathbf{r}_A$
- point **B** be in $\mathbf{r}_B = \mathbf{r}_A + \mathbf{X}(\mathbf{r}_A)$
- point **C** be in $\mathbf{r}_C = \mathbf{r}_B + \mathbf{Y}(\mathbf{r}_B) = \mathbf{r}_A + \mathbf{X}(\mathbf{r}_A) + \mathbf{Y}(\mathbf{r}_A + \mathbf{X}(\mathbf{r}_A))$
- point **D** be in $\mathbf{r}_D = \mathbf{r}_A + \mathbf{Y}(\mathbf{r}_A)$
- point **E** be in $\mathbf{r}_E = \mathbf{r}_D + \mathbf{X}(\mathbf{r}_D) = \mathbf{r}_A + \mathbf{Y}(\mathbf{r}_A) + \mathbf{X}(\mathbf{r}_A + \mathbf{Y}(\mathbf{r}_A))$

The - maybe unexpected fifth side - is required when $\mathbf{r}_E \neq \mathbf{r}_C$, i.e. when

$$\begin{aligned}\mathbf{0} \neq \mathbf{r}_E - \mathbf{r}_C &= \{\mathbf{r}_A + \mathbf{Y}(\mathbf{r}_A) + \mathbf{X}(\mathbf{r}_A) + \mathbf{Y}(\mathbf{r}_A) \cdot \nabla \mathbf{X}(\mathbf{r}_A)\} + \\ &\quad - \{\mathbf{r}_A + \mathbf{X}(\mathbf{r}_A) + \mathbf{Y}(\mathbf{r}_A) + \mathbf{X}(\mathbf{r}_A) \cdot \nabla \mathbf{Y}(\mathbf{r}_A)\} + o(|\mathbf{X}||\mathbf{Y}|) = \\ &= \mathbf{Y}(\mathbf{r}_A) \cdot \nabla \mathbf{X}(\mathbf{r}_A) - \mathbf{X}(\mathbf{r}_A) \cdot \nabla \mathbf{Y}(\mathbf{r}_A) \,.\end{aligned}$$

or, in components

$$\begin{aligned}\mathbf{0} \neq \mathbf{r}_E - \mathbf{r}_C &= Y^\nu \partial_\nu \left(X^\mu \mathbf{b}_\mu\right) - X^\nu \partial_\nu \left(Y^\mu \mathbf{b}_\mu\right) = \\ &= \left[Y^\nu \left(\partial_\nu X^\mu + \Gamma^\mu_{\nu\xi} X^\xi\right) - X^\nu \left(\partial_\nu Y^\mu + \Gamma^\mu_{\nu\xi} Y^\xi\right)\right] \mathbf{b}_\mu = \\ &= -(X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu) \mathbf{b}_\mu = \\ &= -[X,Y]^\mu \mathbf{b}_\mu = -[X,Y] \,,\end{aligned}$$

i.e. when X, Y don't commute, as stated above.

Example 9.3.1 (Coordinates vector fields commute, and thus only 4 sides are required to build the loop)

…

$$[X,Y] = 0$$

…

The discrepancy (holonomy) after completing the loop gives rise to curvature.

Parallel transport along a closed path

Path A to C: first along $\mathbf{X}(\mathbf{r})$ then along to $\mathbf{Y}(\mathbf{r}+\mathbf{X}(\mathbf{r}))$.

$$\begin{aligned}\left(\mathbf{v}_{||\mathbf{X}}\right)_{||\mathbf{Y}} &= \mathbf{b}_\nu\Big[v^\nu - X^\mu\Gamma^\nu_{\mu\xi}v^\xi - Y^\mu\Gamma^\nu_{\mu\xi}v^\xi - X^\alpha\left(\partial_\alpha Y^\mu + \Gamma^\mu_{\alpha\phi}Y^\phi\right)\Gamma^\nu_{\mu\xi}v^\xi + \\ &\quad - Y^\mu X^\alpha\partial_\alpha\Gamma^\nu_{\mu\xi}v^\xi + Y^\mu X^\alpha\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta + X^\alpha X^\mu(\partial_\alpha Y^\mu + \Gamma^\mu_{\alpha\phi}Y^\phi)\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta\Big] = \\ &\sim \mathbf{b}_\nu\Big[v^\nu - X^\mu\Gamma^\nu_{\mu\xi}v^\xi - Y^\mu\Gamma^\nu_{\mu\xi}v^\xi - X^\alpha\left(\partial_\alpha Y^\mu + \Gamma^\mu_{\alpha\phi}Y^\phi\right)\Gamma^\nu_{\mu\xi}v^\xi + \\ &\quad - Y^\mu X^\alpha\partial_\alpha\Gamma^\nu_{\mu\xi}v^\xi + Y^\mu X^\alpha\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta\Big]\end{aligned}$$

Details

First transport along $\mathbf{X}$, using (9.4),

$$\mathbf{v}_{||\mathbf{X}} = \mathbf{b}_\nu\left(v^\nu + X^\mu\left\{\partial_\mu v^\nu\right\}_\parallel\right) = \mathbf{b}_\nu\left(v^\nu - X^\mu\Gamma^\nu_{\mu\xi}v^\xi\right)\ .$$

and then along $\mathbf{Y}(\mathbf{r}+\mathbf{X})$. Expanding $\left(\mathbf{v}_{||\mathbf{X}}\right)_{||\mathbf{Y}}$ to second order in components:

$$\left(\mathbf{v}_{||\mathbf{X}}\right)_{||\mathbf{Y}} = \mathbf{b}_\nu\left[v^\nu - X^\mu\Gamma^\nu_{\mu\xi}v^\xi - \left(Y^\mu + X^\alpha(\partial_\alpha Y^\mu + \Gamma^\mu_{\alpha\phi}Y^\phi)\right)\left(\Gamma^\nu_{\mu\xi} + X^\alpha\partial_\alpha\Gamma^\nu_{\mu\xi}\right)\left(v^\xi - X^\mu\Gamma^\xi_{\mu\eta}v^\eta\right)\right]$$

Expanding terms explicitly:

$$\begin{aligned}\left(\mathbf{v}_{||\mathbf{X}}\right)_{||\mathbf{Y}} &= \mathbf{b}_\nu\Big[v^\nu - X^\mu\Gamma^\nu_{\mu\xi}v^\xi - Y^\mu\Gamma^\nu_{\mu\xi}v^\xi - X^\alpha\left(\partial_\alpha Y^\mu + \Gamma^\mu_{\alpha\phi}Y^\phi\right)\Gamma^\nu_{\mu\xi}v^\xi + \\ &\quad - Y^\mu X^\alpha\partial_\alpha\Gamma^\nu_{\mu\xi}v^\xi + Y^\mu X^\alpha\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta + X^\alpha X^\mu(\partial_\alpha Y^\mu + \Gamma^\mu_{\alpha\phi}Y^\phi)\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta\Big]\end{aligned}$$

A to C to E, to second order.

$$\begin{aligned}\left(\left(\mathbf{v}_{||\mathbf{X}}\right)_{||\mathbf{Y}}\right)_{||[Y,X]} &= \mathbf{b}_\nu\left[\left\{\left(\mathbf{v}_{||\mathbf{X}}\right)_{||\mathbf{Y}}\right\}^\nu + [X,Y]^\mu\Gamma^\nu_{\mu\xi}\left\{\left(\mathbf{v}_{||\mathbf{X}}\right)_{||\mathbf{Y}}\right\}^\xi\right] = \\ &\sim \mathbf{b}_\nu\Big[v^\nu - X^\mu\Gamma^\nu_{\mu\xi}v^\xi - Y^\mu\Gamma^\nu_{\mu\xi}v^\xi - X^\alpha\left(\partial_\alpha Y^\mu + \Gamma^\mu_{\alpha\eta}Y^\eta\right)\Gamma^\nu_{\mu\xi}v^\xi + \\ &\quad - Y^\mu X^\alpha\partial_\alpha\Gamma^\nu_{\mu\xi}v^\xi + Y^\mu X^\alpha\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta + \left(X^\alpha\partial_\alpha Y^\mu - Y^\alpha\partial_\alpha X^\mu\right)\Gamma^\nu_{\mu\xi}v^\xi\Big]\end{aligned}$$

Path A to E: first along $\mathbf{Y}(\mathbf{r})$ then along to $\mathbf{X}(\mathbf{r}+\mathbf{Y}(\mathbf{r}))$. Just swapping the X, Y

$$\begin{aligned}\left(\mathbf{v}_{||\mathbf{Y}}\right)_{||\mathbf{X}} &= \mathbf{b}_\nu\Big[v^\nu - Y^\mu\Gamma^\nu_{\mu\xi}v^\xi - X^\mu\Gamma^\nu_{\mu\xi}v^\xi - Y^\alpha\left(\partial_\alpha X^\mu + \Gamma^\mu_{\alpha\eta}X^\eta\right)\Gamma^\nu_{\mu\xi}v^\xi + \\ &\quad - X^\mu Y^\alpha\partial_\alpha\Gamma^\nu_{\mu\xi}v^\xi + X^\mu Y^\alpha\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta + Y^\alpha Y^\mu(\partial_\alpha X^\mu + \Gamma^\mu_{\alpha\eta}X^\eta)\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta\Big] = \\ &\sim \mathbf{b}_\nu\Big[v^\nu - Y^\mu\Gamma^\nu_{\mu\xi}v^\xi - X^\mu\Gamma^\nu_{\mu\xi}v^\xi - Y^\alpha\left(\partial_\alpha X^\mu + \Gamma^\mu_{\alpha\eta}X^\eta\right)\Gamma^\nu_{\mu\xi}v^\xi + \\ &\quad - X^\mu Y^\alpha\partial_\alpha\Gamma^\nu_{\mu\xi}v^\xi + X^\mu Y^\alpha\Gamma^\nu_{\mu\xi}\Gamma^\xi_{\alpha\eta}v^\eta\Big]\end{aligned}$$

Difference path A $\to$ B $\to$ C $\to$ E - path A $\to$ D $\to$ E to second order.

Subtracting Path B from Path A yields, as the travelling direction during the closed path is the opposite,:

$$
\begin{aligned}
(A-B)^{\nu} = & \underbrace{-X^{\alpha}\partial_{\alpha}Y^{\mu}\Gamma^{\nu}_{\mu\xi}v^{\xi}}_{i_{[X,Y],2}} - \underbrace{X^{\alpha}Y^{\beta}\Gamma^{\mu}_{\alpha\beta}\Gamma^{\nu}_{\mu\xi}v^{\xi}}_{1} + \underbrace{Y^{\alpha}\partial_{\alpha}X^{\mu}\Gamma^{\nu}_{\mu\xi}v^{\xi}}_{i_{[X,Y],1}} + \underbrace{X^{\alpha}Y^{\beta}\Gamma^{\mu}_{\beta\alpha}\Gamma^{\nu}_{\mu\xi}v^{\xi}}_{1} + \\
& + \underbrace{\left(X^{\alpha}\partial_{\alpha}Y^{\mu} - Y^{\alpha}\partial_{\alpha}X^{\mu}\right)\Gamma^{\nu}_{\mu\xi}v^{\xi}}_{i_{[X,Y]}} + \\
& + X^{\alpha}Y^{\beta}\Gamma^{\nu}_{\beta\eta}\Gamma^{\eta}_{\alpha\xi}v^{\xi} - X^{\alpha}Y^{\beta}\Gamma^{\nu}_{\alpha\eta}\Gamma^{\eta}_{\beta\xi}v^{\xi} - X^{\alpha}Y^{\beta}\partial_{\alpha}\Gamma^{\nu}_{\beta\xi}v^{\xi} + X^{\alpha}Y^{\beta}\partial_{\beta}\Gamma^{\nu}_{\alpha\xi}v^{\xi} + \text{higher order terms} = \\
\sim & -\left[\partial_{\alpha}\Gamma^{\nu}_{\beta\xi} - \partial_{\beta}\Gamma^{\nu}_{\alpha\xi} + \Gamma^{\nu}_{\alpha\eta}\Gamma^{\eta}_{\beta\xi} - \Gamma^{\nu}_{\beta\eta}\Gamma^{\eta}_{\alpha\xi}\right]X^{\alpha}Y^{\beta}v^{\xi} = \\
= & -R^{\nu}{}_{\alpha\beta\xi}X^{\alpha}Y^{\beta}v^{\xi} \, ,
\end{aligned}
$$

having introduced the definition of the **Riemann tensor**, $R^{\nu}{}_{\alpha\beta\xi}$.

Definition 9.3.1 (Riemann tensor)

$$
R^{\nu}{}_{\alpha\beta\xi} := \partial_{\alpha}\Gamma^{\nu}_{\beta\xi} - \partial_{\beta}\Gamma^{\nu}_{\alpha\xi} + \Gamma^{\nu}_{\alpha\eta}\Gamma^{\eta}_{\beta\xi} - \Gamma^{\nu}_{\beta\eta}\Gamma^{\eta}_{\alpha\xi} \, .
$$

Remark. The contribution of the Lie bracket, required to close the loop, annihilates with other two terms.

Remark. Exploiting the anti-symmetry of the Riemann tensor, $R^{\nu}{}_{\alpha\beta\xi} = -R^{\nu}{}_{\beta\alpha\xi}$, it's possible to recast the expression of the parallel transport of the vector field **v** along the elementary closed path $\gamma_{X,Y}$ built with the vector fields X, Y as

$$
\begin{aligned}
H^{\nu} = \Delta v^{nu} := & \oint_{\gamma_{X,Y}} dq^{\mu} \left\{\partial_{\mu}v^{\nu}\right\}_{||} = \\
:= & -\oint_{\gamma_{X,Y}} dq^{\mu}\Gamma^{\nu}_{\mu\xi}v^{\xi} = \\
= & -\frac{1}{2}R^{\nu}{}_{\alpha\beta\xi}\left(X^{\alpha}Y^{\beta} - X^{\beta}Y^{\alpha}\right)v^{\xi} + \text{higher order terms} = \\
= & -\frac{1}{2}R^{\nu}{}_{\alpha\beta\xi}\, A^{\alpha\beta}\, v^{\xi} + \text{higher order terms} \, ,
\end{aligned}
$$

having introduced the **signed areas** $A^{\alpha\beta}$ of the projection of the area enclosed in the elementary loop onto the plane defined by the coordinates q^{α} and q^{β}.

todo

- discuss the projected areas
- discuss the higher-order correction to the area from Lie-bracket cut at the corner
- check the common standard for the order of the indices of the Riemann tensor

...

todo *Link the following sections to what's done so far.*

Riemann Curvature Tensor Definition

For arbitrary vector fields X, Y, Z:

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

If X, Y are coordinate vector fields ($[X,Y]=0$), this simplifies to:

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z$$

Component Expression

Computing $\nabla_i \nabla_j Z - \nabla_j \nabla_i Z$ for $Z = Z^m \mathbf{b}_m$:

$$R^k_{mij} = \partial_i \Gamma^k_{jm} - \partial_j \Gamma^k_{im} + \Gamma^l_{jm}\Gamma^k_{il} - \Gamma^l_{im}\Gamma^k_{jl}$$

Relation between Riemann curvature tensor and extrinsic curvature

If the parametrization of the space is regular enough, second-order mixed derivatives have the same value, i.e. $\frac{\partial^2 \mathbf{r}}{\partial q^i \partial q^k} = \frac{\partial^2 \mathbf{r}}{\partial q^k \partial q^i}$ that implies $\Gamma^\alpha_{ik} = \Gamma^\alpha_{ki}$. Similar relation holds for the mixed derivatives of vector $\mathbf{b}_i$ of the natural basis, $\frac{\partial^2 \mathbf{b}_i}{\partial q^j \partial q^k} = \frac{\partial^2 \mathbf{b}_i}{\partial q^k \partial q^j}$. As

$$\begin{aligned}
\frac{\partial^2 \mathbf{b}_i}{\partial q^j \partial q^k} &= \frac{\partial}{\partial q^j}\frac{\partial \mathbf{b}_i}{\partial q^k} = \\
&= \frac{\partial}{\partial q^j}\left(\Gamma^\alpha_{ik}\mathbf{b}_\alpha\right) = \\
&= \frac{\partial}{\partial q^j}\left(\Gamma^l_{ik}\mathbf{b}_l + \Gamma^{n+1}_{ik}\mathbf{n}\right) = \\
&= \partial_j \Gamma^l_{ik}\mathbf{b}_l + \Gamma^l_{ik}\left(\Gamma^m_{lj}\mathbf{b}_m + \Gamma^{n+1}_{lj}\mathbf{n}\right) + \partial_j \Gamma^{n+1}_{ik}\mathbf{n} + \Gamma^{n+1}_{ik}\partial_j \mathbf{n} = \\
&= \left(\partial_j \Gamma^l_{ik} + \Gamma^m_{ik}\Gamma^l_{mj} - \Gamma^{n+1}_{ik} b^l_j\right)\mathbf{b}_l + \left(\partial_j \Gamma^{n+1}_{ik} + \Gamma^l_{ik}\Gamma^{n+1}_{lj}\right)\mathbf{n}\,,
\end{aligned}$$

having used the **Weingarten equation** (9.1), $\partial_j \mathbf{n} = -b_j{}^l \mathbf{b}_l$. Equating every tangential components of $\partial_{jk}\mathbf{b}_i$ and $\partial_{kj}\mathbf{b}_i$, it follows for the component l,

$$\partial_j \Gamma^l_{ik} + \Gamma^m_{ik}\Gamma^l_{mj} - \Gamma^{n+1}_{ik} b^l_j = \partial_k \Gamma^l_{ij} + \Gamma^m_{ij}\Gamma^l_{mk} - \Gamma^{n+1}_{ij} b^l_k\,,$$

or, rearranging

$$\underbrace{\partial_j \Gamma^l_{ik} - \partial_k \Gamma^l_{ij} + \Gamma^m_{ik}\Gamma^l_{mj} - \Gamma^m_{ij}\Gamma^l_{mk}}_{=R^l{}_{ijk}} = \Gamma^{n+1}_{ik} b^l_j - \Gamma^{n+1}_{ij} b^l_k\,,$$

having introduced the definition of **Riemann curvature tensor** $R^l{}_{ijk}$. Writing $\Gamma^{n+1}_{ij} = b_{ij}$, it follows

$$R^l{}_{ijk} = b_{ij} b^l_j - b_{ij} b^l_k\,.$$

See also General Relativity: Differential Geometry: Curvature Tensor

Examples

Example 9.3.2 (2-d Euclidean space is flat, independently from the set of coordinates used for the description)

While it's trivial to prove that the 2-dimensional Euclidean space is flat using Cartesian coordinates, in this result is also proved using polar coordinates. This should be an expected result, as an intrinsic/tensor relation holds independently from an arbitrary choice of coordinates or vector basis.

Cartesian coordinates. Cartesian coordinates have uniform (constant in space) vectors of the natural basis, and thus all the Christoffel symbols equal to zero - being the components of their space derivatives - and the partial derivatives of the Christoffell symbols equal to zero as well. Thus,

$$R^k_{\ mij} = 0 \, .$$

Polar coordinates

**Polar Coordinates (r, θ). Let $q^1 = r, q^2 = \theta$. Thus, the natural basis and its reciprocal - using the common unit-length orhogonal polar basis - are

$$\begin{cases} \mathbf{b}_1 = \partial_1 \mathbf{r} = \hat{r} \\ \mathbf{b}_2 = \partial_2 \mathbf{r} = r\hat{\theta} \end{cases} \quad , \quad \begin{cases} \mathbf{b}^1 = \hat{r} \\ \mathbf{b}^2 = \dfrac{1}{r}\hat{\theta} \end{cases} \, .$$

Metric tensor, $g_{ij} = \mathbf{b}_i \cdot \mathbf{b}_j$.

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix}$$

Non-vanishing Christoffel Symbols. Using $\Gamma^l_{ik} = \frac{1}{2} g^{lj} \left(\partial_i g_{jk} + \partial_k g_{ji} - \partial_j g_{ik} \right)$:

1. $\Gamma^1_{22} = \frac{1}{2} g^{11} (-\partial_1 g_{22}) = \frac{1}{2}(1)(-2r) = -r$
2. $\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{2} g^{22} (\partial_1 g_{22}) = \frac{1}{2} \left(\frac{1}{r^2} \right) (2r) = \frac{1}{r}$

All other $\Gamma^k_{ij} = 0$.

Non-vanishing derivatives of Christoffel Symbols.

- $\partial_1 \Gamma^2_{12} = \partial_r \left(\frac{1}{r} \right) = -\frac{1}{r^2}$
- $\partial_1 \Gamma^1_{22} = \partial_r (-r) = -1$

Calculation of Riemann Tensor Components. Riemann tensor of a 2-dimensional manifold has 16 components. Not all the components are independent Exploiting symmetries of the Riemann tensor, $R^k_{\ mij} = \partial_i \Gamma^k_{jm} - \partial_j \Gamma^k_{im} + \Gamma^l_{jm} \Gamma^k_{il} - \Gamma^l_{im} \Gamma^k_{jl}$, it's easy to realized that

$$R^k_{\ mij} = -R^k_{\ mji} \, .$$

Thus, all the components with $i = j$ are identically zero. So, only 4 components can be independent, setting as an example $i = 1, j = 2$, and evaluating the 4 terms with $(k, m) \in \{1, 2\}^2$.

- **Component R^1_{212}:**

$$\begin{aligned} R^1_{212} &= \partial_1 \Gamma^1_{22} - \partial_2 \Gamma^1_{12} + \Gamma^l_{22} \Gamma^1_{1l} - \Gamma^l_{12} \Gamma^1_{2l} = \\ &= (-1) - 0 + (\Gamma^1_{22} \Gamma^1_{11} + \Gamma^2_{22} \Gamma^1_{12}) - (\Gamma^1_{12} \Gamma^1_{21} + \Gamma^2_{12} \Gamma^1_{22}) = \\ &= -1 - (0) - \left(\frac{1}{r} \cdot (-r) \right) = -1 - (-1) = 0 \end{aligned}$$

- **Component R^2_{112}:**

$$R^2_{112} = \partial_1\Gamma^2_{21} - \partial_2\Gamma^2_{11} + \Gamma^l_{21}\Gamma^2_{1l} - \Gamma^l_{11}\Gamma^2_{2l}$$

$$R^2_{112} = \left(-\frac{1}{r^2}\right) - 0 + (\Gamma^2_{21}\Gamma^2_{12}) - 0 = -\frac{1}{r^2} + \left(\frac{1}{r}\cdot\frac{1}{r}\right) = 0$$

- **Component R^1_{112}:**

...

- **Component R^2_{212}:**

...

Derivatives of the components of the metric tensor

Let $g_{ij} = \mathbf{b}_i \cdot \mathbf{b}_j$, $\mathbf{b}_i \cdot \mathbf{b}^j = \delta^j_i$ and $\partial_k \mathbf{b}_i = \Gamma^l_{ik}\mathbf{b}_l$, then

$$\partial_k g_{ij} = \partial_k \mathbf{b}_i \cdot \mathbf{b}_j + \mathbf{b}_i \cdot \partial_k \mathbf{b}_j = \Gamma^l_{ik} g_{lj} + \Gamma^l_{ik} g_{il} \, .$$

Thus, swapping indices

$$\begin{aligned}
\partial_k g_{ij} &= \Gamma^l_{ik} g_{lj} + \Gamma^l_{jk} g_{il} \\
\partial_i g_{jk} &= \Gamma^l_{ji} g_{lk} + \Gamma^l_{ki} g_{jl} \\
\partial_j g_{ki} &= \Gamma^l_{kj} g_{li} + \Gamma^l_{ij} g_{kl}
\end{aligned}$$

From the latter three relations, it's possible to retrieve the expression of Christoffel coefficients as a function of the components of the metric tensor and their derivatives. As an example, coefficient Γ^l_{ik} is found summing the first and the second relation, and subtracting the third one,

$$\Gamma^l_{ik} g_{jl} = \frac{1}{2}\left(\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ki}\right) \, ,$$

or

$$\Gamma^l_{ik} = \frac{1}{2} g^{lj}\left(\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ki}\right) \, .$$

Example 9.3.3 (Parallel transport of a vector on a sphere)

Coordinates and metric tensor

Using spherical coordinates $(q^1, q^2) = (\theta, \phi)$,[1] the **metric tensor** reads

$$\left[g_{ij}\right] = \begin{bmatrix} R^2 \sin^2 q^2 & \cdot \\ \cdot & R^2 \end{bmatrix} \quad , \quad \left[g^{ij}\right] = \begin{bmatrix} \frac{1}{R^2 \sin^2 q^2} & \cdot \\ \cdot & \frac{1}{R^2} \end{bmatrix} \, ,$$

with constant R, i.e. the radius of the sphere.

[1] Extrinsic approach. Recalling that $\mathbf{r} = r\hat{\mathbf{r}} = \hat{\mathbf{x}}\,(r\sin\phi\cos\theta) + \hat{\mathbf{y}}\,(r\sin\phi\sin\theta) + \hat{\mathbf{z}}\,r\cos\phi$, thus $\mathbf{b}_1 = \partial_\theta \mathbf{r} = -r\sin\phi\sin\theta\,\hat{\mathbf{x}} + r\sin\phi\cos\theta\,\hat{\mathbf{y}}$, and $\mathbf{b}_2 = \partial_\phi \mathbf{r} = r\cos\phi\cos\theta\,\hat{\mathbf{x}} + r\cos\phi\sin\theta\,\hat{\mathbf{y}} - r\sin\theta\,\hat{\mathbf{z}}$. The covariant components of the metric tensor are $g_{11} = r^2\sin^2\phi$, $g_{12} = 0$, $g_{22} = r^2$.

Christoffel symbols

The only non-zero derivative of the components of the metric tensor is

$$\partial_2 g_{11} = 2R^2 \sin q^2 \cos q^2 \ .$$

Non-zero Christoffel symbols are

- with upper index = 1:

$$\begin{aligned}
\Gamma^1_{ik} &= \frac{1}{2} g^{1j} \left(\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ki} \right) = \quad \left(j = 1,\ g^{11} = \frac{1}{R^2 \sin^2 q^2} \right) \\
&= \frac{1}{2} g^{11} \left(\partial_k g_{i1} + \partial_i g_{1k} - \partial_1 g_{ki} \right) = \quad (\partial_1 g_{ik} = 0) \\
&= \frac{1}{2} g^{11} \left(\partial_k g_{i1} + \partial_i g_{1k} \right) \ ,
\end{aligned}$$

  then

$$\Gamma^1_{12} = \Gamma^1_{21} = \frac{1}{2} \frac{1}{R^2 \sin^2 q^2} 2R^2 \sin q^2 \cos q^2 = \frac{\cos q^2}{\sin q^2} \ .$$

- with upper index = 2:

$$\begin{aligned}
\Gamma^2_{ik} &= \frac{1}{2} g^{2j} \left(\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ki} \right) = \quad \left(j = 2,\ g^{22} = \frac{1}{R^2} \right) \\
&= \frac{1}{2} g^{22} \left(\partial_k g_{i2} + \partial_i g_{2k} - \partial_2 g_{ki} \right) = \quad (\partial g_{2\cdot} = 0,\ \partial_2 g_{11} = 2R^2 \sin q^2 \cos q^2) \\
&= -\frac{1}{2} g^{22} \partial_2 g_{ik} \ ,
\end{aligned}$$

  then

$$\Gamma^2_{11} = -\sin q^2 \cos q^2 \ .$$

Non-zero derivatives of the Christoffel symbols

$$\partial_2 \Gamma^1_{12} = \partial_2 \Gamma^1_{21} = -\frac{1}{\sin^2 q^2} \ ,$$

$$\partial_2 \Gamma^2_{11} = \sin^2 q^2 - \cos^2 q^2 \ .$$

Parallel transport along a meridian, $\theta = $ const., i.e. $q^1 = $ const.

A meridian can be parametrized with $q^2 = \phi$ as a parameter, as the coordinate $q^1 = \theta$ is constant, and thus $dq^1 = 0$. The parallel transport of the vector field thus becomes a system of 2 equations in 2 unknowns $v^1(q^2)$, $v^2(q^2)$, namely

$$\begin{cases}
d_2 v^1 = -\Gamma^1_{12} v^1 - \Gamma^1_{22} v^2 = -\cot\left(q^2\right) v^1(q^2) \\
d_2 v^2 = -\Gamma^2_{12} v^1 - \Gamma^2_{22} v^2 = 0
\end{cases}$$

Thus the solution reads

$$\begin{cases}
v^1(\phi) = v^1_0 \dfrac{\sin \phi_0}{\sin \phi} \\
v^2(\phi) = v^2_0
\end{cases}$$

Details. Thus $v_2\left(q^2\right)=\overline{v^2}$, and v^1 can be evaluated with **separation of variables**,

$$\begin{aligned}\int_{v_1}^{v_2}\frac{dv}{v}&=-\int_{x_1}^{x_2}\cot(x)\,dx\\ \ln\left(v\right)\big|_{v_1}^{v_2}&=-\ln\left(\sin\phi\right)\big|_{\phi_1}^{\phi_2}\\ \frac{v_2}{v_1}&=\frac{\sin\phi_1}{\sin\phi_2}\\ v_2\sin\phi_2&=v_1\sin\phi_1\ .\end{aligned}$$

being

$$\int\cot(x)dx=\int\frac{\cos x}{\sin x}dx=\int\frac{d\sin x}{\sin x}=\ln\left(\sin x\right)\ .$$

Remark. The first equation of the system can be written as $v'(x)=-\frac{\cos(x)}{\sin(x)}v(x)$, or

$$0=v'(x)\sin(x)+v(x)\cos(x)=\tfrac{d}{dx}\left(v(x)\sin(x)\right)\ .$$

Parallel transport along a parallel, $\phi=$ const., i.e. $q^2=$ const.

A parallel can be parametrized with $q^1=\theta$ as a parameter, as the coordinate $q^2=\overline{\phi}$ is constant, and thus $dq^2=0$. The parallel transport of the vector field thus becomes a system of 2 equations in 2 unknowns $v^1(q^1)$, $v^2(q^1)$, namely

$$\begin{cases}d_1v^1(q^1)=-\Gamma^1_{11}v^1-\Gamma^1_{21}v^2=-\cot\left(q^2\right)v^2(q^1)\\ d_1v^2(q^1)=-\Gamma^2_{11}v^1-\Gamma^2_{21}v^2=\sin\left(q^2\right)\cos\left(q^2\right)\ v^1(q^1)\end{cases}$$

As $q^2=\overline{\phi}$ is constant,

$$\begin{cases}v^1(\theta)=v_1\dfrac{\sin\phi_1}{\sin\phi}\\ v^2(\theta)=0\end{cases}$$

Details.

$$\begin{cases}\dfrac{dv^1}{d\theta}(\theta)=-\dfrac{\cos(\phi)}{\sin(\phi)}\,v^2(\theta)\\ \dfrac{dv^2}{d\theta}(\theta)=\sin(\phi)\cos(\phi)\,v^1(\theta)\end{cases}$$

Taking the derivative $\frac{d}{d\theta}$ of the first equation, and replacing $\frac{dv^2}{d\theta}$ with the second equation,

$$\begin{aligned}0&=\left(v^1\right)''+\frac{\cos(\phi)}{\sin(\phi)}\sin(\phi)\cos(\phi)\,v^1(\theta)=\\ &=\left(v^1\right)''(\theta)+\cos^2(\phi)\,v^1(\theta)\ ,\end{aligned}$$

and thus

- if $\cos(\phi)\neq0$

$$\begin{cases}v^1(\theta)=A\cos\left(\cos(\phi)\theta\right)+B\sin\left(\cos(\phi)\theta\right)\\ v^2(\theta)=-\dfrac{\sin(\phi)}{\cos(\phi)}\left[-A\cos(\phi)\sin\left(\cos(\phi)\theta\right)+B\cos(\phi)\cos\left(\cos(\phi)\theta\right)\right]\\ \qquad=A\sin(\phi)\sin\left(\cos(\phi)\theta\right)-B\sin(\phi)\cos\left(\cos(\phi)\theta\right)\end{cases}$$

Choosing the reference of θ, to be $\theta = 0$ at the beginning of the path, then the initial conditions simplify to

$$\begin{cases} v_0^1 = A \\ v_0^2 = -\sin(\phi) B \,, \end{cases} \quad \rightarrow \quad \begin{cases} A = v_0^1 \\ B = -\dfrac{v_0^2}{\sin\phi} \end{cases}$$

- if $\cos(\phi) = 0$, then $v^1(\theta) = v_0^1$, $v^2(\theta) = v_0^2$ constant.

Parallel transport from the equator to the pole and back

Let a vector $\mathbf{v}$ on the equator and parallel to the local meridian, i.e. $\mathbf{v} = v^\phi \mathbf{b}_2$ in $(\theta, \phi) = (0, \frac{\pi}{2})$.

1. Parallel transport along the equator, $\phi = \frac{\pi}{2}$, θ from 0 to $\overline{\theta}$. The equator is described by the coordinate $\phi = \frac{\pi}{2}$, and thus $\cos\phi = 0$. The solution of the parallel transport equation on the equator gives constant $v^1(\theta)$ and $v^2(\theta)$. Thus,

$$v^1\left(\overline{\theta}\right) = 0 \quad , \quad v^2\left(\overline{\theta}\right) = v^\phi \,.$$

2. Parallel transport along the meridian, $\theta = \overline{\theta}$, ϕ from $\frac{\pi}{2}$ to $\delta\phi$.

$$v^1\left(0\right) = 0 \quad , \quad v^2\left(\overline{\theta}\right) = v^\phi \,.$$

3. *Parallel transport along a small circle around the pole to avoid singularity.*

$$\begin{cases} v^1(\theta) = v_0^1 \cos\left(\cos(\phi)\theta\right) - \dfrac{v_0^2}{\sin\phi} \sin\left(\cos(\phi)\theta\right) = \\ \qquad = -\dfrac{v^\phi}{\sin\delta\phi} \sin\left(\cos(\delta\phi)\theta\right) \\ v^2(\theta) = v_0^1 \sin(\phi) \sin\left(\cos(\phi)\theta\right) + v_0^2 \cos\left(\cos(\phi)\theta\right) = \\ \qquad = v^\phi \cos\left(\cos(\delta\phi)\theta\right) \end{cases}$$

The final values are $v^1 = \frac{v^\phi}{\sin\delta\phi} \sin(\cos(\delta\phi)\theta)$, $v^2 = v^\phi \cos(\cos(\delta\phi)\theta)$. *Exchanging the reference value for* θ *should be enough.* **todo** *check!*

4. Parallel transport along the meridian, $\theta = 0$, ϕ from $\delta\phi$ to $\frac{\pi}{2}$.

$$\begin{aligned} v^1\left(0\right) &= \frac{\sin(\delta\phi)}{\sin(\pi/2)} \frac{v^\phi}{\sin\delta\phi} \sin(\cos(\delta\phi)\theta) = \\ &= v^\phi \sin(\cos(\delta\phi)\theta) \\ &\rightarrow v^\phi \sin(\theta) \quad \text{for } \delta\phi \rightarrow 0 \\ v^2\left(0\right) &= v^\phi \cos(\cos(\delta\phi)\theta) \\ &\rightarrow v^\phi \cos(\theta) \quad \text{for } \delta\phi \rightarrow 0 \,. \end{aligned}$$

Result. The final vector after the parallel transport over the closed path is the initial vector rotated by the angle θ.

Example 9.3.4 (Riemann tensor on the 2-dimensional sphere)

With $(q^1, q^2) = (\theta, \phi)$,

$$\Gamma^1_{12} = \Gamma^1_{21} = -\sin\phi\cos\phi \quad , \quad \Gamma^2_{11} = -\sin\phi\cos\phi$$

$$\partial_2\Gamma^1_{12} = \partial_2\Gamma^1_{21} = -\frac{1}{\sin^2\phi} \quad , \quad \partial_2\Gamma^2_{11} = \sin^2\phi - \cos^2\phi$$

Part IV

Vector and Tensor Algebra and Calculus

CHAPTER

TEN

TENSOR ALGEBRA

This section introduces tensor algebra

Warning: This introduction is meant for **spaces with inner product** that allows to introduce quite naturally the useful concept of the *reciprocal basis* of a given basis of $\mathcal{V}$, belonging to the same space $\mathcal{V}$, somehow avoiding the complications coming from the introduction of dual space $\mathcal{V}^*$ and basis, and everything is required for that.

10.1 Vector space $\mathcal{V}$

A vector space is an algebraic structure with:

- a **set** $\mathcal{V}$, whose elements are called **vectors**, here indicated with bold symbols $\mathbf{v} \in \mathcal{V}$
- a **field** K (usually $\mathbb{R}$ or $\mathbb{C}$), whose elements are calles **scalars**,
- 2 **operations**, closed w.r.t. the set $\mathcal{V}$:
  - **vector sum**, $\mathbf{u} + \mathbf{v} \in \mathcal{V}$ for $\forall \mathbf{u},\ \mathbf{v} \in \mathcal{V}$
  - **multiplication of a scalar and a vector**, $a\mathbf{v} \in \mathcal{V}$ for $\forall a \in K,\ \mathbf{u} \in \mathcal{V}$

  with properties discussed below **todo** …

10.1.1 Operations (I)

Sum

…

Multiplication by a scalar

…

Definition 10.1.1 (Linear combination)

A linear combination of n vectors $\{\mathbf{v}_i\}_{i=1:n}$, $\mathbf{v}_i \in \mathcal{V}$, is the weighted sum

$$a^1\mathbf{v}_1 + a^2\mathbf{v}_2 + \cdots + a^n\mathbf{v}_n = a^i\mathbf{v}_i \in \mathcal{V}\ ,$$

having used Einstein's summation convention over repeated indices. Here the position of the indices has no particular meaning, but it'll have soon in the following sections.

Inner product

The existence of an inner product is not a requirement of a vector space

…

An inner product in a vectors space $\mathcal{V}$ over field K is an operation $\cdot : \mathcal{V} \times \mathcal{V} \to K$,

$$\mathbf{u} \cdot \mathbf{v} \,,$$

with the following properties: **todo**

10.1.2 Basis of a vector space $\mathcal{V}$

Definition 10.1.2 (Basis)

A basis is a minimal set of vectors of $\mathcal{V}$ that can represent all the elements of $\mathcal{V}$.

Definition 10.1.3 (Dimension of a vector space)

The dimension of a vector space $\mathcal{V}$ is the number of the elements of a basis of the space.

Definition 10.1.4 (Reciprocal basis)

In a inner product space, the reciprocal basis of a given basis $\{\mathbf{b}_a\}_{a=1:d}$ is the set of vectors $\{\mathbf{b}^b\}_{b=1:d}$, s.t.

$$\mathbf{b}^b \cdot \mathbf{b}_a = \delta^b_a \,.$$

Definition 10.1.5 ("Metric tensor")

$$g_{ab} := \mathbf{b}_a \cdot \mathbf{b}_b$$
$$g^{ab} := \mathbf{b}^a \cdot \mathbf{b}^b$$

The following holds

$$\begin{aligned} \mathbf{b}_a &= g_{ab}\mathbf{b}^b \quad (1) \\ \mathbf{b}^a &= g^{ab}\mathbf{b}_b \quad (2) \end{aligned} \tag{10.1}$$

"Proof"

Taking the dot product with $\mathbf{b}_c$ of the relation (10.1)(1),

$$\mathbf{b}_c \cdot \mathbf{b}_a = g_{ab} \underbrace{\mathbf{b}_c \cdot \mathbf{b}^b}_{=\delta_c^b} = g_{ac} \ .$$

10.1.3 Change of basis

Let T_a^b the elements of the matrix of change of basis, representing the vectors of the basis $\{\tilde{b}_a\}$ as linear combinations of the vectors of the basis $\{\mathbf{b}_b\}$,

$$\tilde{\mathbf{b}}_b = T_b^a \mathbf{b}_a \ .$$

Inverse transformation. Let $\widetilde{T}$ be the elements of the inverse transformation,

$$\mathbf{b}_c = \widetilde{T}_c^b \tilde{\mathbf{b}}_b = \widetilde{T}_c^b T_b^a \mathbf{b}_a \ ,$$

and thus

$$\widetilde{T}_c^b T_b^a = \delta_c^a \ .$$

Transformation of the reciprocal basis. todo

Transformation of components

A vector $\mathbf{v}$ can be represented in different basis, as different linear combinations of the elements of those bases,

$$\mathbf{v} = v^a \mathbf{b}_a = \tilde{v}^b \tilde{\mathbf{b}}_b \ .$$

Given the rules of change of basis, the rule of transformation of components immediately follwos

$$\begin{aligned} \tilde{\mathbf{b}}_b &= T_b^a \mathbf{b}_a \quad & \tilde{v}^b &= \widetilde{T}_a^b v^a \\ \mathbf{b}_b &= \widetilde{T}_b^a \tilde{\mathbf{b}}_a \quad & v^b &= T_a^b \tilde{v}^a \end{aligned}$$

Proof

$$\mathbf{v} = v^a \mathbf{b}_a = \underbrace{v^a \widetilde{T}_a^b}_{\tilde{v}^b} \tilde{\mathbf{b}}_b = \tilde{v}^b \tilde{\mathbf{b}}_b$$

or

$$\mathbf{v} = \tilde{v}^b \tilde{\mathbf{b}}_b = \underbrace{\tilde{v}^b \widetilde{T}_c^b}_{v^c} \mathbf{b}_c = v^c \mathbf{b}_c$$

It's clear that the vectors of the bases and the components follow inverse transformations to preserve the **invariance of vector w.r.t. a change of basis**: the vector $\mathbf{v}$ doesn't change if we change our description of it, by changing a basis.

Matrix of change of basis as a tensor (todo maybe later? Tensor not introduced yet here)

The rule of transformation between different basis can be interpreted using dot product between tensors and vectors

$$\tilde{\mathbf{b}}_b = T_b^a \mathbf{b}_a = T_c^a \mathbf{b}_a \delta_b^c = T_c^a \mathbf{b}_a \otimes \mathbf{b}^c \cdot \mathbf{b}_b = (T_c^a \mathbf{b}_a \otimes \mathbf{b}^c) \cdot \mathbf{b}_b = \mathbb{T} \cdot \mathbf{b}_b$$

Inverse and transpose

Interpreting the indices of the transformation matrix as the indices of rows and columns of a 2x2 matrix, transformation of components involves the transpose of the inverse matrix, as the indices a, b are swapped.

Example 10.1.1

Example 10.1.2 (Rotation)

As the inverse of a rotation is its transpose, $\widetilde{T}_a^b = T_b^a$. So the rule of transformation of components follows the same rule of change of basis. As an example, let the transformation between two Cartesian bases be the rotation

$$\begin{aligned} \tilde{\mathbf{x}} &= \cos\theta\,\mathbf{x} + \sin\theta\,\mathbf{y} \\ \tilde{\mathbf{y}} &= -\sin\theta\,\mathbf{x} + \cos\theta\,\mathbf{y} \end{aligned} \quad , \quad \begin{aligned} \mathbf{x} &= \cos\theta\,\tilde{\mathbf{x}} - \sin\theta\,\tilde{\mathbf{y}} \\ \mathbf{y} &= \sin\theta\,\tilde{\mathbf{x}} + \cos\theta\,\tilde{\mathbf{y}} \end{aligned}$$

Let a vector

$$\begin{aligned} \mathbf{v} &= v_x\mathbf{x} + v_y\mathbf{y} = \\ &= v_x\left(\cos\theta\,\tilde{\mathbf{x}} - \sin\theta\,\tilde{\mathbf{y}}\right) + v_y\left(\sin\theta\,\tilde{\mathbf{x}} + \cos\theta\,\tilde{\mathbf{y}}\right) = \\ &= \tilde{v}_x\tilde{\mathbf{x}} + \tilde{v}_y\tilde{\mathbf{y}} = \\ &= \tilde{v}_x\left(\cos\theta\,\mathbf{x} + \sin\theta\,\mathbf{y}\right) + \tilde{v}_y\left(-\sin\theta\,\mathbf{x} + \cos\theta\,\mathbf{y}\right) \end{aligned}$$

and thus

$$\begin{aligned} \tilde{v}_x &= \cos\theta\,v_x + \sin\theta\,v_y \\ \tilde{v}_y &= -\sin\theta\,v_x + \sin\theta\,v_y \end{aligned} \quad , \quad \begin{aligned} v_x &= \cos\theta\,\tilde{v}_x - \sin\theta\,\tilde{v}_y \\ v_y &= \sin\theta\,\tilde{v}_x + \sin\theta\,\tilde{v}_y \end{aligned}$$

10.1.4 Operations

Tensor product of vectors

$$\mathbf{v}_{(1)} \otimes \mathbf{v}_{(2)} \otimes \cdots \otimes \mathbf{v}_{(r)}$$

or writing each vector as a linear combination of the elements of a basis $\mathbf{b}_a$,

$$\mathbf{v}_{(i)} = v_{(i)}^{a_i}\mathbf{b}_{a_i}\ , \ldots$$

$$\begin{aligned} \mathbf{v}_{(1)} \otimes \mathbf{v}_{(2)} \otimes \cdots \otimes \mathbf{v}_{(3)} &= \left(v_{(1)}^{a_1}\mathbf{b}_{a_1}\right) \otimes \left(v_{(2)}^{a_2}\mathbf{b}_{a_2}\right) \otimes \cdots \otimes \left(v_{(r)}^{a_r}\mathbf{b}_{a_r}\right) = \\ &= v_{(1)}^{a_1}\, v_{(2)}^{a_2}\, \ldots\, v_{(r)}^{a_r}\mathbf{b}_{a_1} \otimes \mathbf{b}_{a_2} \otimes \cdots \otimes \mathbf{b}_{a_r} \end{aligned}$$

The result of the tensor product of r vectors is a rank-r tensor, $\mathbb{T}$, as it will be clear below, with components

$$T^{a_1 a_2 \ldots a_r} = v_{(1)}^{a_1}\, v_{(2)}^{a_2}\, \ldots\, v_{(r)}^{a_r}\ .$$

10.2 Space of tensors

Definition 10.2.1 (Tensor as a multilinear map)

A rank-r tensor $\mathbf{A}$ can be defined as a multilinear map, acting on r vectors[1] $\mathbf{v}^{(i)} \in \mathcal{V}$, $i = 1 : r$

$$\mathbf{A}\left(\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(r)}\right) : \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{r \text{ times}} \to K.$$

todo

- Tensor definition by components, and action on the vectors
- Reference to a similar definition of a vector: a vector $\mathbf{v} \in \mathcal{V}$, vector space with inner product, is uniquely determined by the value of its scalar product $\mathbf{u} \cdot \mathbf{v}$ with every vector $\mathbf{u} \in \mathcal{V}$ or, analogously, with every vector in a set of n linearly independent vectors $\left\{\mathbf{u}^{(i)}\right\}$, $i = 1 : n$, with $n = \dim(\mathcal{V})$.
- Definition of components w.r.t. a basis

10.2.1 Operations (I)

Sum

Sum of tensors with the same rank, $\mathbf{A}$, $\mathbf{B} \in \mathcal{V}$,

$$\mathbf{A} + \mathbf{B} \in \mathcal{V}^r$$

todo Components

Multiplication by a scalar

Multiplication by a scalar $a \in K$, of a r-rank tensor $\mathbf{A}$,

$$a\mathbf{A} \in \mathcal{V}^r$$

todo Components

Vector space of tensors

The set of tensors of a given rank with the operations of sum and multiplication by a scalar defined above forms a vector space.

[1] This introduction has no ambition iof being the most general — and precise? — introduction to tensors: as already stated somewhere else, existence of inner product is assumed even it's not necessary for the most general treatment of tensors; moreover, no distinction is made between dual space $\mathcal{V}^*$ of covectors — space of linear forms on V — and V itself;…

10.2.2 Basis

Change of basis and rule of transformation of components - classical definition of a tensor

$$
\begin{aligned}
\tilde{\mathbf{b}}_{i_a} &= T_{i_b}^{i_a} \mathbf{b}_{i_b} \\
\mathbf{b}_{i_a} &= \widetilde{T}_{i_a}^{i_b} \tilde{\mathbf{b}}_{i_b} \\
\mathbf{A} = A^{i_1 \dots i_p} \mathbf{b}_{i_1} \dots \mathbf{b}_{i_p} &= \\
= A^{i_1 \dots i_p} \left(T_{i_1}^{j_1} \tilde{\mathbf{b}}_{j_1} \right) \dots \left(T_{i_p}^{j_p} \tilde{\mathbf{b}}_{j_p} \right) &= \\
= A^{i_1 \dots i_p} T_{i_1}^{j_1} \dots T_{i_p}^{j_p} \tilde{\mathbf{b}}_{j_1} \dots \tilde{\mathbf{b}}_{j_p} &= \\
= \widetilde{A}^{j_1 \dots j_p} \tilde{\mathbf{b}}_{j_1} \dots \tilde{\mathbf{b}}_{j_p} &,
\end{aligned}
$$

with

$$
\widetilde{A}^{j_1 \dots j_p} = A^{i_1 \dots i_p} T_{i_1}^{j_1} \dots T_{i_p}^{j_p} \, . \tag{10.2}
$$

Definition 10.2.2 (Classical definition of a tensor)

Relation (10.2) is the "historical" definition of a tensor, through the law of transformation of its components following a change of basis.

10.2.3 Operations (II)

Tensor product

The tensor product of a p-rank tensor $\mathbf{A}$ and a q-rank tensor $\mathbf{B}$ is the $(p+q)$-rank tensor $\mathbf{A} \otimes \mathbf{B} = \mathbf{AB}$, that can be defined using component representation in a given basis,

$$
\begin{aligned}
\mathbf{A} \otimes \mathbf{B} &= \left(A^{a_1 \dots a_p} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_p} \right) \otimes \left(B^{b_1 \dots b_q} \mathbf{b}_{b_1} \dots \mathbf{b}_{b_q} \right) = \\
&= A^{a_1 \dots a_{p-1} a_p} \, B^{b_1 \, b_2 \dots b_q} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_p} \mathbf{b}_{b_1} \dots \mathbf{b}_{b_q}
\end{aligned}
$$

Dot product

The dot product of a p-rank tensor $\mathbf{A}$ and a q-rank tensor $\mathbf{B}$ is the $(p+q-2)$-rank tensor $\mathbf{A} \cdot \mathbf{B}$, that can be defined using component representation in a given basis,

$$
\begin{aligned}
\mathbf{A} \cdot \mathbf{B} &= \left(A^{a_1 \dots a_p} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_p} \right) \cdot \left(B_{b_1 \dots b_q} \mathbf{b}^{b_1} \dots \mathbf{b}^{b_q} \right) = \\
&= A^{a_1 \dots a_p} \, B_{b_1 \dots b_q} \mathbf{b}_{a_1} \dots \underbrace{\mathbf{b}_{a_p} \cdot \mathbf{b}^{b_1}}_{= \delta_{a_p}^{b_1}} \dots \mathbf{b}^{b_q} = \\
&= A^{a_1 \dots a_{p-1} k} \, B_{k \, b_2 \dots b_q} \mathbf{b}_{a_1} \dots \mathbf{b}_{a_{p-1}} \mathbf{b}^{b_2} \dots \mathbf{b}^{b_q}
\end{aligned}
$$

Contraction

Contraction of a pair of index of a p-rank tensor $\mathbf{A}$ returns a $p-2$-rank tensor defined as

$$C_a^b(\mathbf{A}) = ...$$

Exterior product

todo *see exterior algebra*

10.2.4 Invariants

10.3 Exterior algebra

$$\wedge$$

10.3.1 Exterior product

Generalization of the vector product

CHAPTER

ELEVEN

TENSOR CALCULUS IN EUCLIDEAN SPACES

This section deals with tensor calculus in Euclidean space or on manifolds embedded in Euclidean spaces, focusing on d-dimensional spaces with $d \leq 3$, with *inner product*.

This section may rely on results of *differential geometry*.

11.1 Coordinates

A set of parameters $\{q^a\}_{a=1:d}$ to represent vector (or point) in space,

$$\vec{r}(q^a)$$

if $\vec{r} \in E^d$, $a = 1 : d$.

In E^3,

- **Coordinate lines**, 2-parameter family of lines, keeping 2 coordinates constant. As an example, coordinate lines with constant q^2, q^3

$$\vec{r}_1(q^1) = \vec{r}(q^1, \bar{q}^2, \bar{q}^3) \, .$$

- **Coordinate surfaces,** 1-parameter family of surfaces, keeping 1 coordinate constant. As an example, coordinate surfaces with constant q^1,

$$\vec{r}_{23}(q^2, q^3) = \vec{r}(\bar{q}^1, q^2, q^3) \, .$$

Definition 11.1.1 (Regular parametrization)

If $\frac{\partial \vec{r}}{\partial q^a} \neq 0$.

11.1.1 Natural basis

Definition 11.1.2 (Natural basis)

The vectors of the natural basis induced by the set of coordinates $\{q^a\}$ are defined as the first derivative of the position $\vec{r}$ w.r.t. the coordinates

$$\vec{b}_a := \frac{\partial \vec{r}}{\partial q^a} \, .$$

Definition 11.1.3 (Reciprocal basis (todo move to Tensor Algebra))

Given a basis $\{\vec{b}_a\}_a$, its reciprocal basis the set of vector $\{\vec{b}^b\}_b$ defined as

$$\vec{b}^b \cdot \vec{b}_a = \delta_a^b \, ,$$

being δ_a^b Kronecker delta.

Definition 11.1.4 (Christoffel symbols)

Christoffel symbols (of the 2nd kind) are defined as the components of the partial derivatives of the vectors of a natural basis w.r.t. the coordinates referred to the natural basis itself,

$$\frac{\partial \vec{b}_a}{\partial q^b} = \Gamma_{ab}^c \, \vec{b}_c \tag{11.1}$$

Properties of Christoffel symbols

Exploiting the definition of reciprocal basis, Christoffel symbols can be written as

$$\Gamma_{ab}^c = \vec{b}^c \cdot \frac{\partial \vec{b}_a}{\partial q^b} \, .$$

Symmetry. Symmetry os the lower indices

$$\Gamma_{ab}^c = \Gamma_{ba}^c \, ,$$

readily follows Schwartz theorem about partial derivatives

$$\frac{\partial \vec{b}_a}{\partial q^b} = \frac{\partial}{\partial q^c}\frac{\partial \vec{r}}{\partial q^a} = \frac{\partial}{\partial q^a}\frac{\partial \vec{r}}{\partial q^b} = \frac{\partial \vec{b}_b}{\partial q^a}$$

11.2 Fields

Function of the points in space $F : E^d \to V^r$, being V^r a space of tensors of order r.

11.3 Differential operators

11.3.1 Directional derivative

$$F(\vec{r}) = F\left(\vec{r}\left(q^a\right)\right) = f(q^a)$$

$$f(q^a + \beta \Delta q^a) = F(\vec{r}(q^a + \beta \Delta q^a))$$

$$\vec{r}(q^a) + \alpha \vec{v} = \vec{r}(q^a + \beta \Delta q^a) \sim \vec{r}(q^a) + \frac{\partial \vec{r}}{\partial q^b} \beta \Delta q^b$$

$$\alpha\vec{v} \sim \beta\frac{\partial\vec{r}}{\partial q^b}(q^a)\,\Delta q^b = \beta\vec{b}_b(q^a)\Delta q^b \qquad \rightarrow \qquad \Delta q^b = \frac{\alpha}{\beta}\vec{b}^b(q^a)\cdot\vec{v}$$

The directional derivative for an arbitrary vector $\vec{v} \in V$

$$\frac{d}{d\alpha}F(\vec{r}+\alpha\vec{v})\bigg|_{\alpha=0}$$

is evaluated as the limit for $\alpha \to 0$ of the incremental ratio

$$\begin{aligned}\frac{F(\vec{r}+\alpha\vec{v})-F(\vec{r})}{\alpha} &\sim \frac{f(q^a+\beta\Delta q^a)-f(q^a)}{\alpha} = \\ &\sim \frac{1}{\alpha}\frac{\partial f}{\partial q^b}(q^a)\beta\Delta q^b = \\ &\sim \vec{v}\cdot\vec{b}^b(q^a)\frac{\partial f}{\partial q^b}(q^a) = \\ &= \vec{v}\cdot\nabla F(\vec{r})\end{aligned}$$

11.3.2 Gradient

The gradient is the differential operator is the first-order differential operator appearing in the definition of the directional derivative, $\nabla F(\vec{r})$. It takes a tensor field $F(\vec{r})$ of order r and gives a tensor field $\nabla F(\vec{r})$ of order $r+1$. Given a set of coordinates $\{q^a\}_{a=1:d}$, the gradient can be written using the reciprocal basis of the natural basis as

$$\nabla F(\vec{r}) = \vec{b}^b(\vec{r})\frac{\partial F}{\partial q^b}(\vec{r}) \tag{11.2}$$

Examples. …

Example 11.3.1 (Gradient of a scalar field - with general coordinates q^a)

Applying the definition (11.2) of gradient operator, it readily follows

$$\nabla F = \vec{b}^a\frac{\partial F}{\partial q^a}$$

Example 11.3.2 (Gradient of a vector field - with general coordinates q^a)

Applying the definition (11.2) of gradient operator, rule for the derivative of a product and the definition (11.1) of Christoffel symbols to write derivatives of base vectors,

$$\begin{aligned}\nabla F &= \vec{b}^a\frac{\partial}{\partial q^a}\left(F^b\vec{b}_b\right) = \\ &= \vec{b}^a\left[\frac{\partial F^b}{\partial q^a}\vec{b}_b + F^b\frac{\partial\vec{b}_b}{\partial q^a}\right] = \\ &= \vec{b}^a\left[\frac{\partial F^b}{\partial q^a}\vec{b}_b + F^b\,\Gamma^c_{ab}\,\vec{b}_c\right] = \\ &= \vec{b}^a\otimes\vec{b}_b\left[\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac}\,F^c\right].\end{aligned}$$

Example 11.3.3 (Gradient of a 2nd-order tensor field - with general coordinates q^a)

Applying the definition (11.2) of gradient operator, rule for the derivative of a product and the definition (11.1) of Christoffel symbols to write derivatives of base vectors,

$$\begin{aligned}\nabla F &= \vec{b}^{\,a}\frac{\partial}{\partial q^a}\left(F^{bc}\vec{b}_b\otimes\vec{b}_c\right) = \\ &= \vec{b}^{\,a}\left[\frac{\partial F^{bc}}{\partial q^a}\vec{b}_b\,\vec{b}_c + F^{bc}\frac{\partial\vec{b}_b}{\partial q^a}\,\vec{b}_c + F^{bc}\vec{b}_b\frac{\partial\vec{b}_c}{\partial q^a}\right] = \\ &= \vec{b}^{\,a}\left[\frac{\partial F^{bc}}{\partial q^a}\vec{b}_b\,\vec{b}_c + F^{bc}\Gamma^d_{ab}\,\vec{b}_d\,\vec{b}_c + F^{bc}\,\Gamma^{ac}_d\,\vec{b}_b\,\vec{b}_d\right] = \\ &= \vec{b}^{\,a}\otimes\vec{b}_b\otimes\vec{b}_c\left[\frac{\partial F^{bc}}{\partial q^a} + \Gamma^b_{ad}\,F^{dc} + \Gamma^c_{ad}\,F^{bd}\right] .\end{aligned}$$

11.3.3 Divergence

Divergence opearator is a first-order differential operator that can be defined as the contraction of the first two indices of the gradient,

$$\nabla\cdot F = C_1^2\left(\nabla F\right) .$$

It takes a tensor field $F(\vec{r})$ of order $r\geq 1$ and gives a tensor field $\nabla\cdot F(\vec{r})$ of order $r-1\geq 0$.

Example 11.3.4 (Divergence of a vector field - with general coordiantes q^a)

Applying contraction to the gradient of a vector field, it readily follows,

$$\begin{aligned}\nabla\cdot\left(F^b\vec{b}_b\right) &= C_1^2\left(\nabla F\right) = \\ &= C_1^2\left(\vec{b}^{\,a}\otimes\vec{b}_b\left[\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac}\,F^c\right]\right) = \\ &= \frac{\partial F^a}{\partial q^a} + \Gamma^a_{ac}\,F^c\end{aligned}$$

Example 11.3.5 (Divergence of a 2nd-order tensor field - with general coordiantes q^a)

Applying contraction to the gradient of a vector field, it readily follows,

$$\begin{aligned}\nabla\cdot\left(F^{bc}\vec{b}_b\otimes\vec{b}_c\right) &= C_1^2\left(\nabla F\right) = \\ &= C_1^2\left(\vec{b}^{\,a}\otimes\vec{b}_b\otimes\vec{b}_c\left[\frac{\partial F^{bc}}{\partial q^a} + \Gamma^b_{ad}\,F^{dc} + \Gamma^c_{ad}\,F^{bd}\right]\right) = \\ &= \vec{b}_c\left[\frac{\partial F^{ac}}{\partial q^a} + \Gamma^a_{ad}\,F^{dc} + \Gamma^c_{ad}\,F^{ad}\right]\end{aligned}$$

11.3.4 Laplacian

Laplacian operator is second-order differential operator that can be defined as the divergence of the gradient,

$$\Delta F = \nabla^2 F = \nabla \cdot \nabla F \, .$$

Example 11.3.6 (Laplacian of a scalar field - with general coordinates q^a)

$$\begin{aligned}
\nabla \cdot \nabla F &= C_1^2 \left[\nabla \left(\nabla F \right) \right] = \\
&= C_1^2 \left[\nabla \left(\vec{b}^a \frac{\partial F}{\partial q^a} \right) \right] = \\
&= C_1^2 \left[\nabla \left(\vec{b}_b \, g^{ab} \, \frac{\partial F}{\partial q^a} \right) \right] = \\
&= C_1^2 \left[\vec{b}^c \frac{\partial}{\partial q^c} \left(\vec{b}_b \, g^{ab} \, \frac{\partial F}{\partial q^a} \right) \right] = \\
&= C_1^2 \left\{ \vec{b}^c \left[\vec{b}_b \, \frac{\partial}{\partial q^c} \left(g^{ab} \, \frac{\partial F}{\partial q^a} \right) + g^{ab} \frac{\partial F}{\partial q^a} \frac{\partial \vec{b}_b}{\partial q^c} \right] \right\} = \\
&= C_1^2 \left\{ \vec{b}^c \left[\vec{b}_b \, \frac{\partial}{\partial q^c} \left(g^{ab} \, \frac{\partial F}{\partial q^a} \right) + g^{ab} \frac{\partial F}{\partial q^a} \, \Gamma^d_{bc} \, \vec{b}_d \right] \right\} = \\
&= C_1^2 \left\{ \vec{b}^c \, \vec{b}_b \left[\frac{\partial}{\partial q^c} \left(g^{ab} \, \frac{\partial F}{\partial q^a} \right) + g^{ad} \, \Gamma^b_{cd} \, \frac{\partial F}{\partial q^a} \right] \right\} = \\
&= \frac{\partial}{\partial q^b} \left(g^{ab} \, \frac{\partial F}{\partial q^a} \right) + g^{ad} \, \Gamma^b_{bd} \, \frac{\partial F}{\partial q^a} \, .
\end{aligned}$$

Example 11.3.7 (Laplacian of a vector field - with general coordinates q^a)

$$\begin{aligned}
\nabla \cdot \nabla F &= C_1^2 \left[\nabla \left(\nabla F \right) \right] = \\
&= C_1^2 \left\{ \nabla \left[\vec{b}^a \, \vec{b}_b \left(\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac} F^c \right) \right] \right\} = \\
&= C_1^2 \left\{ \nabla \left[\vec{b}_c \, \vec{b}_b \, g^{ac} \left(\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac} F^c \right) \right] \right\} = \\
&= C_1^2 \left\{ \nabla \cdot \left((\nabla F)^{cb} \, \vec{b}_c \, \vec{b}_b \right) \right\} = \\
&= C_1^2 \left\{ \vec{b}^a \, \vec{b}_c \, \vec{b}_b \left[\frac{\partial (\nabla F)^{cb}}{\partial q^a} + \Gamma^c_{ad} (\nabla F)^{db} + \Gamma^b_{ad} (\nabla F)^{cd} \right] \right\} = \\
&= \vec{b}_b \left[\frac{\partial (\nabla F)^{ab}}{\partial q^a} + \Gamma^a_{ad} (\nabla F)^{db} + \Gamma^b_{ad} (\nabla F)^{ad} \right] = \, .
\end{aligned}$$

11.3.5 Curl

11.4 Integrals in E^d, $d \leq 3$

11.4.1 Line integrals

Density

Integrals

$$\int_{\vec{r}\in\gamma} F(\vec{r})$$

represent the summation of contributions $F(\vec{r})$ over elementary segments of path γ, whose dimension is $|d\vec{r}|$, i.e. implicitly means

$$\int_{\vec{r}\in\gamma} F(\vec{r}) = \int_{\vec{r}\in\gamma} F(\vec{r})\,|d\vec{r}|\ .$$

Given a regular parametrization of the curve $\vec{r}(q^1)$ (with increasing q^1 so that $|dq^1| = dq^1$), and the differential $d\vec{r} = \vec{r}\,'(q^1)\,dq^1$, the integral can be written as an integral in the parameter q^1

$$\int_{q=q_a^1}^{q_b^1} F(\vec{r}(q^1))\,|\vec{r}\,'(q^1)|\,dq^1\ ,$$

with $\vec{r}(q_a^1)$, $\vec{r}(q_b^1)$ the extreme points of path γ.

Work

Integrals

$$\int_{\vec{r}\in\gamma} F(\vec{r})\cdot\hat{t}(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in\gamma} F(\vec{r})\cdot\hat{t}(\vec{r}) = \int_{\vec{r}\in\gamma} F(\vec{r})\cdot\hat{t}(\vec{r})|d\vec{r}| = \int_{\vec{r}\in\gamma} F(\vec{r})\cdot d\vec{r}\ ,$$

as $\hat{t} = \frac{d\vec{r}}{|d\vec{r}|}$. Given a regular parametrization of the curve $\vec{r}(q^1)$ (with increasing q^1 so that $|dq^1| = dq^1$), and the differential $d\vec{r} = \vec{r}\,'(q^1)\,dq^1$, the integral can be written as an integral in the parameter q^1

$$\int_{q^1=q_a^1}^{q_b^1} F(\vec{r}(q^1))\cdot\vec{r}\,'(q^1)\,dq^1$$

11.4.2 Surface integrals

Given two coordinates $q^1,\ q^2$ describing a surface, $\vec{r}(q^1,q^2)$ the elementary surface with unit normal reads

$$\hat{n}\,dS = d\vec{r}_1\times d\vec{r}_2 = \frac{\partial\vec{r}}{\partial q^1}\times\frac{\partial\vec{r}}{\partial q^2}\,dq^1\,dq^2\ ,$$

and the elementary surface thus reads

$$|dS| = |\hat{n}dS| = \left|\frac{\partial\vec{r}}{\partial q^1}\times\frac{\partial\vec{r}}{\partial q^2}\,dq^1\,dq^2\right|$$

Density

Integrals

$$\int_{\vec{r}\in S} F(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in S} F(\vec{r}) = \int_{\vec{r}\in S} F(\vec{r})|dS| \,.$$

Given regular parametrization of the surface, $\vec{r}(q^1,\, q^2),\ (q^1, q^2) \,\in\, Q^{12}$, the integral can be written as the multi-dimensional integral in coordinates $q^1,\ q^2$,

$$\int_{\vec{r}\in S} F(\vec{r}) = \int_{(q^1,q^2)\in Q^{12}} F(\vec{r}(q^1,q^2)) \left|\frac{\partial\vec{r}}{\partial q^1} \times \frac{\partial\vec{r}}{\partial q^2}\, dq^1\, dq^2\right|$$

Flux

Integrals

$$\int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r}) = \int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r})|dS|$$

Given regular parametrization of the surface, $\vec{r}(q^1,\, q^2),\ (q^1, q^2) \,\in\, Q^{12}$, the integral can be written as the multi-dimensional integral in coordinates $q^1,\ q^2$,

$$\int_{\vec{r}\in S} \hat{n}(\vec{r}) \cdot F(\vec{r}) = \int_{(q^1,q^2)\in Q^{12}} \frac{\partial\vec{r}}{\partial q^1} \times \frac{\partial\vec{r}}{\partial q^2} \cdot F(\vec{r}(q^1,q^2))\, dq^1\, dq^2$$

11.4.3 Volume

$$dV = \frac{\partial\vec{r}}{\partial q^1} \cdot \frac{\partial\vec{r}}{\partial q^2} \times \frac{\partial\vec{r}}{\partial q^3}\, dq^1\, dq^2\, dq^3 \,.$$

Density

Integrals

$$\int_{\vec{r}\in V} F(\vec{r})$$

implicitly mean

$$\int_{\vec{r}\in V} F(\vec{r}) = \int_{\vec{r}\in V} F(\vec{r})\,|dV| \,.$$

Given regular parametrization of the volume, $\vec{r}(q^1,\, q^2,\, q^3),\ (q^1, q^2, q^3) \in Q$, the integral can be written as the multi-dimensional integral in coordinates $q^1,\ q^2,\ q^3$,

$$\int_{\vec{r}\in V} F(\vec{r})|dV| = \int_{(q^1,q^2,q^3)\in Q} F(\vec{r}(q^1,q^2,q^3)) \left|\frac{\partial\vec{r}}{\partial q^1} \cdot \frac{\partial\vec{r}}{\partial q^2} \times \frac{\partial\vec{r}}{\partial q^3}\, dq^1\, dq^2\, dq^3\right| \,.$$

11.4.4 Theorems

Two useful lemmas

The next lemma forms the foundation of the well-known *divergence theorem* and *gradient theorem*:
the proof of these two theorems is based on a straightforward repeated use of this lemma.
Given how simple this lemma is and how frequently it is applied in writing balances and, more generally, in integration by parts, it is very convenient to remember this simple result.

Theorem 11.4.1 (Lemma 1.)

Under the assumptions of *Green's lemma in the plane*,

$$\int_V \frac{\partial A}{\partial x_i} = \oint_S A n_i$$

Proof

The reasoning closely follows the one used for the proof of *Green's lemma in the plane*.
For $\partial A/\partial z$:

$$\begin{aligned}\int_V \frac{\partial A}{\partial z} &= \int_R \int_{z=f_1(x,y)}^{z=f_2(x,y)} \frac{\partial A}{\partial z} dz dx dy = \\ &= \int_R [A(x,y,f_2(x,y)) - A(x,y,f_1(x,y))] dx dy\end{aligned}$$

The most complex step is transitioning from the integral over $(x,y) \in R$ to the surface integral over S, the boundary of volume V: the infinitesimal area element dR in the xy-plane is equal to $dR = dxdy$; the drawing and the proof refer to a *simple* volume (as in the case of Green's lemma in the plane, the results can be generalized to domains of arbitrary shape).
It is possible to divide the surface S into two "halves" $S^+ : z = f_2(x,y)$ and $S^- : z = f_1(x,y)$ such that
$S^+ \cup S^- = S$, and the outward normal has positive and negative z-components respectively ($S^+ : \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} > 0$, $S^- : \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} < 0$).
The surface element dR is also the projection of the surface element dS onto the xy-plane: in general, dS will not be parallel to the xy-plane and thus will be larger than dR. It's not difficult to show that:

$$dxdy = dR = \begin{cases} dS \hat{\mathbf{z}} \cdot \hat{\mathbf{n}} & \text{on } S^+ \\ -dS \hat{\mathbf{z}} \cdot \hat{\mathbf{n}} & \text{on } S^- \end{cases}$$

We can now continue the proof:

$$\begin{aligned}&\int_R [A(x,y,f_2(x,y)) - A(x,y,f_1(x,y))] dx dy = \\ &= \int_{S^+} A \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} dS + \int_{S^-} A \hat{\mathbf{n}} \cdot \hat{\mathbf{z}} dS = \\ &= \oint_S A \hat{\mathbf{z}} \cdot \hat{\mathbf{n}} dS = \\ &= \oint_S A n_z dS\end{aligned}$$

Just as the previous lemma forms the basis for the proof of the *gradient theorem* and the *divergence theorem*,
the following lemma forms the basis for the proof of the *curl theorem*.

Theorem 11.4.2 (Lemma 2.)

Under the assumptions of *Green's lemma in the plane*,

$$\int_S [\nabla \times (A\hat{\mathbf{e}}_\mathbf{i})] \cdot \hat{\mathbf{n}} = \oint_\gamma A dx_i$$

Proof

For $A\hat{\mathbf{e}}_\mathbf{x}$, we have $\nabla \times (A\hat{\mathbf{e}}_\mathbf{x}) = \partial A/\partial z \hat{\mathbf{e}}_\mathbf{y} - \partial A/\partial y \hat{\mathbf{e}}_\mathbf{z}$. The surface S is written in parametric form as: $\mathbf{r} = x\hat{\mathbf{e}}_\mathbf{x} + y\hat{\mathbf{e}}_\mathbf{y} + z(x,y)\hat{\mathbf{e}}_\mathbf{z}$. The vector $\partial \mathbf{r}/\partial y = \hat{\mathbf{e}}_\mathbf{y} + \partial z/\partial y \hat{\mathbf{e}}_\mathbf{z}$ is tangent to the surface S and hence perpendicular to the normal $\hat{\mathbf{n}}$:

$$0 = \hat{\mathbf{n}} \cdot \left(\hat{\mathbf{e}}_\mathbf{y} + \frac{\partial z}{\partial y} \hat{\mathbf{e}}_\mathbf{z} \right)$$

Now writing $[\nabla \times (A\hat{\mathbf{e}}_\mathbf{x})] \cdot \hat{\mathbf{n}}$:

$$[\nabla \times (A\hat{\mathbf{e}}_\mathbf{x})] \cdot \hat{\mathbf{n}} = \frac{\partial A}{\partial z} \hat{\mathbf{e}}_\mathbf{y} \cdot \hat{\mathbf{n}} - \frac{\partial A}{\partial y} \hat{\mathbf{e}}_\mathbf{z} \cdot \hat{\mathbf{n}} = - \left[\frac{\partial A}{\partial z} \frac{\partial z}{\partial y} + \frac{\partial A}{\partial y} \right] \hat{\mathbf{e}}_\mathbf{z} \cdot \hat{\mathbf{n}}$$

Recognizing that $\frac{\partial A(x,y,z(x,y))}{\partial y} = \frac{\partial A}{\partial z} \frac{\partial z}{\partial y} + \frac{\partial A}{\partial y}$, we can write:

$$\int_S [\nabla \times (A\hat{\mathbf{e}}_\mathbf{x})] \cdot \hat{\mathbf{n}} = - \int_S \frac{\partial A}{\partial y} \underbrace{\hat{\mathbf{e}}_\mathbf{z} \cdot \hat{\mathbf{n}} dS}_{dR = dx dy} = - \int_R \frac{\partial A}{\partial y} dx dy = \int_\gamma A dx$$

Gradient theorem

…**todo** *assumptions*…

$$\int_V \nabla f = \oint_{\partial V} f \hat{n}$$

Proof for simple domains V

This result immediately follows from Lemma 1 *Theorem 11.4.1*

$$\oint_{\partial V} f \hat{n} = \hat{x}_i \oint_{\partial V} f n_i = \hat{x}_i \int_V \partial_i f = \int_V \hat{x}_i \partial_i f = \int_V \nabla \vec{f} \,,$$

having (1) exploited the freedom to put unit vectors of the Cartesian basis inside the integrals, as they're unifrom - constant in space -, and (2) recognized the expression of the *gradient of a scalar field expressed using Cartesian coordinates*, as shown in *Example 11.5.1*.

Divergence theorem

…**todo** *assumptions*…

$$\int_V \nabla \cdot \vec{f} = \oint_{\partial V} \vec{f} \cdot \hat{n}$$

Proof

This result immediately follows from Lemma 1 *Theorem 11.4.1*

$$\oint_{\partial V} \vec{f} \cdot \hat{n} = \oint_{\partial V} f_i n_i = \int_V \partial_i f_i = \int_V \nabla \cdot \vec{f} \,,$$

having (1) exploited the freedom to put unit vectors of the Cartesian basis inside the integrals, as they're unifrom - constant in space -, and (2) recognized the expression of the *divergence of a vector field expressed using Cartesian coordinates*, as shown in *Example 11.5.4*.

Curl theorem

…**todo** *assumptions*…

$$\int_S \left[\nabla \times \vec{f}\right] \cdot \hat{n} = \oint_{\partial S} \vec{f} \cdot \hat{t}$$

Proof

This proof seamlessly follows from Lemma *Theorem 11.4.2*, applied to all the Cartesian contributions of the vector field

$$\vec{f} = f_x \hat{x} + f_y \hat{y} + f_z \hat{z} \,,$$

as

$$\int_S \left(\nabla \times \vec{f}\right) \cdot \hat{n} = \int_S (\nabla \times f_i \hat{x}_i) \cdot \hat{n} = \oint_{\partial S} f_i t_i = \oint_{\partial S} \vec{f} \cdot \hat{t} \,.$$

11.5 Tensor Calculus in Euclidean Spaces - Cartesian coordinates in E^3

Using Cartesian coordinates $(q^1, q^2, q^3) = (r, \theta, z)$ and Cartesian base vectors (uniform in space, so that their derivatives are zero), a point in Euclidean vector space E^3 can be represented as

$$\vec{r} = x\,\hat{x} + y\,\hat{y} + z\,\hat{z} \,.$$

11.5.1 Natural basis, reciprocal basis, metric tensor, and Christoffel symbols

Cartesian coordinates in Euclidean spaces are a very special coordinate system, with reciprocal basis everywhere coinciding with natural basis, with uniform basis in space (zero second-order derivative of space w.r.t. coordinates, and thus zero first order derivative of base vectors, and thus identically zero Christoffel symbols), and components of the metric tensor equal to the identity matrix

$$\begin{cases} \vec{b}_1 = \vec{b}^1 = \hat{x} \\ \vec{b}_2 = \vec{b}^2 = \hat{y} \\ \vec{b}_3 = \vec{b}^3 = \hat{z} \end{cases}$$

$$[g_{ab}] = [g^{ab}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Gamma^c_{ab} = 0 \qquad , \qquad \forall a, b, c = 1 : 3 \,.$$

11.5.2 Differential operators

Gradient

Example 11.5.1 (Gradient of a scalar field)

$$\nabla F = \hat{x}\,\partial_x F + \hat{y}\,\partial_y F + \hat{z}\,\partial_z F_z$$

Example 11.5.2 (Gradient of a vector field)

$$\begin{aligned}\nabla F &= \nabla(F_x\hat{x} + F_y\hat{y} + F_z\hat{z}) = \\ &= \hat{x}\otimes\hat{x}\,\partial_x F_x + \hat{x}\otimes\hat{y}\,\partial_x F_y + \hat{x}\otimes\hat{z}\,\partial_x F_z + \\ &+ \hat{y}\otimes\hat{x}\,\partial_y F_x + \hat{y}\otimes\hat{y}\,\partial_y F_y + \hat{y}\otimes\hat{z}\,\partial_y F_z + \\ &+ \hat{z}\otimes\hat{x}\,\partial_z F_x + \hat{z}\otimes\hat{y}\,\partial_z F_y + \hat{z}\otimes\hat{z}\,\partial_z F_z +\end{aligned}$$

Example 11.5.3 (Gradient of a 2nd-order tensor field)

Directional derivative

Divergence

Example 11.5.4 (Divergence of a vector field)

$$\begin{aligned}\nabla\cdot F = \nabla\cdot\left(F_x\,\hat{x} + F_y\,\hat{y} + F_z\,\hat{z}\right) = \\ = \partial_x F_x + \partial_y F_y + \partial_z F_z\,.\end{aligned}$$

Example 11.5.5 (Divergence of a 2nd-order tensor field)

$$\begin{aligned}\nabla\cdot F &= \nabla\cdot(F_{ab}\vec{e}_a\otimes\vec{e}_b) = \\ &= \vec{e}_c\frac{\partial F_{ab}}{\partial q^a} = \\ &= \hat{x}\left[\partial_x F_{xx} + \partial_y F_{yx} + \partial_z F_{zx}\right] + \\ &+ \hat{y}\left[\partial_x F_{xy} + \partial_y F_{yy} + \partial_z F_{zy}\right] + \\ &+ \hat{z}\left[\partial_x F_{xz} + \partial_y F_{yz} + \partial_z F_{zz}\right]\,.\end{aligned}$$

Laplacian

Example 11.5.6 (Laplacian of a scalar field)

$$\nabla^2 F = \partial_{xx} F + \partial_{yy} F + \partial_{zz} F$$

Example 11.5.7 (Laplacian of a vector field)

11.6 Tensor Calculus in Euclidean Spaces - cylindrical coordinates in E^3

11.6.1 Cylindrical coordiantes and cylindrical coordinates

Using cylindrical coordinates $(q^1, q^2, q^3) = (r, \theta, z)$ and cylindrical base vectors (uniform in space, so that their derivatives are zero), a point in Euclidean vector space E^3 can be represented as

$$\vec{r} = r\cos\theta\,\hat{x} + r\sin\theta\,\hat{y} + z\,\hat{z}\,.$$

11.6.2 Natural basis, reciprocal basis, metric tensor, and Christoffel symbols

Natural basis

Natural basis reads

$$\begin{cases} \vec{b}_1 = \dfrac{\partial\vec{r}}{\partial q^1} = \dfrac{\partial\vec{r}}{\partial r} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \\ \vec{b}_2 = \dfrac{\partial\vec{r}}{\partial q^2} = \dfrac{\partial\vec{r}}{\partial \theta} = -r\sin\theta\,\hat{x} + r\cos\theta\,\hat{y} \\ \vec{b}_3 = \dfrac{\partial\vec{r}}{\partial q^3} = \dfrac{\partial\vec{r}}{\partial z} = \hat{z} \end{cases}$$

Metric tensor

Covariant components of metric tensors,

$$g_{ab} = \vec{b}_a \cdot \vec{b}_b\,,$$

can be collected in the diagonal matrix

$$[g_{ab}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\,,$$

while its contra-variant components can be collected in the inverse matrix (easy to compute, since $[g_{ab}]$ is diagonal),

$$[g^{ab}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}\,.$$

Reciprocal basis

Reciprocal basis is readily evaluated using $\vec{b}^a = g^{ab}\,\vec{b}_b$,

$$\begin{cases} \vec{b}^1 = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \\ \vec{b}^2 = -\dfrac{1}{r}\sin\theta\,\hat{x} + \dfrac{1}{r}\cos\theta\,\hat{y} \\ \vec{b}^3 = \hat{z} \end{cases}$$

Physical basis

Since metric tensor is diagonal, the cylindrical coordinate system is orthogonal, and its natural and reciprocal basis are orthogonal. A unit orthogonal basis, usually named **physical basis** with unit vector with no physical dimension, is evalated by normalization process,

$$\begin{cases} \hat{r} = \hat{b}_1 = \dfrac{\vec{b}_1}{g_{11}} = \dfrac{\vec{b}^1}{g^{11}} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \\ \hat{\theta} = \hat{b}_2 = \dfrac{\vec{b}_2}{g_{22}} = \dfrac{\vec{b}^2}{g^{22}} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y} \\ \hat{z} = \hat{b}_3 = \dfrac{\vec{b}_3}{g_{33}} = \dfrac{\vec{b}^3}{g^{33}} = \hat{z}\,. \end{cases}$$

Derivatives of natural basis and Christoffel symbols

Derivatives of the natural basis read

$$\begin{aligned} \frac{\partial\vec{b}_1}{\partial q^1} &= \vec{0} \\ \frac{\partial\vec{b}_2}{\partial q^2} &= -r\cos\theta\,\hat{x} - r\sin\theta\,\hat{y} = -q^1\,\vec{b}_1 \\ \frac{\partial\vec{b}_3}{\partial q^3} &= \vec{0} \\ \frac{\partial\vec{b}_2}{\partial q^1} = \frac{\partial\vec{b}_1}{\partial q^2} &= -\sin\theta\,\hat{x} + \cos\theta\hat{y} = \frac{1}{q^1}\,\vec{b}_2 \\ \frac{\partial\vec{b}_3}{\partial q^1} = \frac{\partial\vec{b}_1}{\partial q^3} &= \vec{0} \\ \frac{\partial\vec{b}_3}{\partial q^2} = \frac{\partial\vec{b}_2}{\partial q^3} &= \vec{0} \end{aligned}$$

so that non-zero Christoffel symbols of a cylindrical coordinate system are

$$\begin{aligned} \Gamma^2_{12} = \Gamma^2_{21} &= \frac{1}{q^1} \\ \Gamma^1_{22} &= -q^1\,. \end{aligned}$$

11.6.3 Differential operators

Gradient

Example 11.6.1 (Gradient of a scalar field)

$$\begin{aligned}\nabla F = \vec{b}^a \frac{\partial F}{\partial q^a} = \\ = \hat{b}_a \, g^{aa} \frac{\partial F}{\partial q^a} = \\ = \hat{r} \, \frac{\partial F}{\partial r} + \hat{\theta} \, \frac{1}{r} \, \frac{\partial F}{\partial \theta} + \hat{z} \frac{\partial F}{\partial z} \, .\end{aligned}$$

Example 11.6.2 (Gradient of a vector field)

$$\begin{aligned}\nabla F = \vec{b}^a \otimes \vec{b}_b \left[\frac{\partial F^b}{\partial q^a} + \Gamma^b_{ac} \, F^c \right] = \\ = \cdots = \\ = \vec{b}^1 \otimes \vec{b}_1 \, \partial_1 F^1 \quad + \vec{b}^1 \otimes \vec{b}_2 \, [\partial_1 F^2 + \Gamma^2_{12} F^2] \quad + \vec{b}^1 \otimes \vec{b}_3 \, \partial_1 F^3 \\ + \vec{b}^2 \otimes \vec{b}_1 \, [\partial_2 F^1 + \Gamma^1_{22} F^2] \quad + \vec{b}^2 \otimes \vec{b}_2 \, [\partial_2 F^2 + \Gamma^2_{21} F^1] \quad + \vec{b}^2 \otimes \vec{b}_3 \, \partial_2 F^3 \\ + \vec{b}^3 \otimes \vec{b}_1 \, \partial_3 F^1 \quad + \vec{b}^3 \otimes \vec{b}_2 \, \partial_3 F^2 \quad + \vec{b}^3 \otimes \vec{b}_3 \, \partial_3 F^3 \\ = \hat{r} \otimes \hat{r} \, \partial_r F_r \quad + \hat{r} \otimes \hat{\theta} \, \frac{1}{r} \, [\partial_r (r F_\theta) + F_\theta] \quad + \hat{r} \otimes \hat{z} \, \partial_r F_z \\ + \hat{\theta} \otimes \hat{r} \, \frac{1}{r} \, [\partial_\theta F_r - r F_\theta] \quad + \hat{\theta} \otimes \hat{\theta} \, \left[\partial_\theta \left(\frac{F_\theta}{r} \right) + \frac{F_r}{r} \right] \quad + \hat{\theta} \otimes \hat{z} \, \frac{1}{r} \partial_\theta F_z \\ + \hat{z} \otimes \hat{r} \, \partial_z F_x \quad + \hat{z} \otimes \hat{\theta} \, \frac{1}{r} \partial_\theta F_y \quad + \hat{z} \otimes \hat{z} \, \partial_z F_z \, .\end{aligned}$$

Example 11.6.3 (Gradient of a 2nd-order tensor field)

Directional derivative

Divergence

Example 11.6.4 (Divergence of a vector field)

$$\begin{aligned}\nabla \cdot \vec{F} = \frac{\partial F^a}{\partial q^a} + \Gamma^a_{ac} \, F^c = \\ = \frac{\partial F_r}{\partial r} + \frac{\partial}{\partial \theta} \left(\frac{F_\theta}{r} \right) + \frac{F_\theta}{r} + \frac{\partial F_z}{\partial z} \, .\end{aligned}$$

Divergence of a 2nd-order tensor field

Using the general formula of the divergence of a 2nd-order tensor field (see *Divergence*)

$$\begin{aligned}\nabla \cdot \left(F^{bc}\vec{b}_b \otimes \vec{b}_c\right) &= C_1^2\left(\nabla F\right) = \\ &= \vec{b}_c\left[\frac{\partial F^{ac}}{\partial q^a} + \Gamma^a_{ad}\,F^{dc} + \Gamma^c_{ad}\,F^{ad}\right]\end{aligned}$$

the contravariant components in the natural basis induced by cylindrical coordinates of the divergence of a second order tensor reads

$$\begin{aligned}\nabla \cdot \mathbb{F} = \\ &= \vec{b}_1\left[\frac{\partial F^{11}}{\partial q^1} + \frac{\partial F^{21}}{\partial q^2} + \frac{\partial F^{31}}{\partial q^3} + \Gamma^2_{21}F^{11} + \Gamma^1_{22}F^{22}\right] + \\ &+ \vec{b}_2\left[\frac{\partial F^{12}}{\partial q^1} + \frac{\partial F^{22}}{\partial q^2} + \frac{\partial F^{32}}{\partial q^3} + \Gamma^2_{21}F^{12} + \Gamma^2_{21}F^{12} + \Gamma^2_{21}F^{21}\right] + \\ &+ \vec{b}_3\left[\frac{\partial F^{13}}{\partial q^1} + \frac{\partial F^{23}}{\partial q^2} + \frac{\partial F^{33}}{\partial q^3} + \Gamma^2_{21}F^{13}\right] = \\ &= \vec{b}_1\left[\frac{\partial F^{11}}{\partial q^1} + \frac{\partial F^{21}}{\partial q^2} + \frac{\partial F^{31}}{\partial q^3} + \frac{1}{q^1}F^{11} - q^1F^{22}\right] + \\ &+ \vec{b}_2\left[\frac{\partial F^{12}}{\partial q^1} + \frac{\partial F^{22}}{\partial q^2} + \frac{\partial F^{32}}{\partial q^3} + \frac{2}{q^1}F^{12} + \frac{1}{q^1}F^{21}\right] + \\ &+ \vec{b}_3\left[\frac{\partial F^{13}}{\partial q^1} + \frac{\partial F^{23}}{\partial q^2} + \frac{\partial F^{33}}{\partial q^3} + \frac{1}{q^1}F^{13}\right].\end{aligned}$$

Next, it's easy to exploit the definition of the coordinates $(q^1, q^2, q^3) = (r, \theta, z)$ and the relation between natural and physical basis and components to get

$$\begin{aligned}\nabla \cdot \mathbb{F} = \\ &= \hat{r}\left[\frac{\partial\left(F^{rr}\right)}{\partial r} + \frac{\partial\left(\frac{1}{r}F^{\theta r}\right)}{\partial\theta} + \frac{\partial F^{zr}}{\partial z} + \frac{1}{r}F^{rr} - r\frac{1}{r^2}F^{\theta\theta}\right] + \\ &+ r\hat{\theta}\left[\frac{\partial\left(\frac{1}{r}F^{r\theta}\right)}{\partial r} + \frac{1}{r}\frac{\partial\left(\frac{1}{r}F^{\theta\theta}\right)}{\partial\theta} + \frac{\partial F^{z\theta}}{\partial z} + \frac{2}{r}\frac{1}{r}F^{r\theta} + \frac{1}{r}\frac{1}{r}F^{\theta r}\right] + \\ &+ \hat{z}\left[\frac{\partial F^{rz}}{\partial r} + \frac{\partial\left(\frac{1}{r}F^{\theta z}\right)}{\partial\theta} + \frac{\partial F^{zz}}{\partial z} + \frac{1}{r}F^{rz}\right].\end{aligned}$$

or, after few algebraic manipulations,

$$\begin{aligned}\nabla \cdot \mathbb{F} = \\ &= \hat{r}\left[\frac{\partial F^{rr}}{\partial r} + \frac{1}{r}\frac{\partial F^{\theta r}}{\partial\theta} + \frac{\partial F^{zr}}{\partial z} + \frac{1}{r}F^{rr} - \frac{1}{r}F^{\theta\theta}\right] + \\ &+ \hat{\theta}\left[\frac{\partial F^{r\theta}}{\partial r} + \frac{1}{r}\frac{\partial F^{\theta\theta}}{\partial\theta} + \frac{\partial F^{z\theta}}{\partial z} + \frac{1}{r}F^{r\theta} + \frac{1}{r}F^{\theta r}\right] + \\ &+ \hat{z}\left[\frac{\partial F^{rz}}{\partial r} + \frac{1}{r}\frac{\partial F^{\theta z}}{\partial\theta} + \frac{\partial F^{zz}}{\partial z} + \frac{1}{r}F^{rz}\right].\end{aligned}$$

Laplacian

Example 11.6.5 (Laplacian of a scalar field)

Example 11.6.6 (Laplacian of a vector field)

11.7 Tensor Calculus in Euclidean Spaces - Spehrical coordinates in E^3

Using spherical coordinates $(q^1, q^2, q^3) = (r, \phi, \theta)$ and spherical base vectors (uniform in space, so that their derivatives are zero), a point in Euclidean vector space E^3 can be represented as

$$\vec{r} = r\cos\theta\sin\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\phi\,\hat{z}\,.$$

11.7.1 Natural basis, reciprocal basis, metric tensor, and Christoffel symbols

11.7.2 Differential operators

Gradient

Example 11.7.1 (Gradient of a scalar field)

Example 11.7.2 (Gradient of a vector field)

Example 11.7.3 (Gradient of a 2nd-order tensor field)

Directional derivative

Divergence

Example 11.7.4 (Divergence of a vector field)

Example 11.7.5 (Divergence of a 2nd-order tensor field)

Laplacian

Example 11.7.6 (Laplacian of a scalar field)

Example 11.7.7 (Laplacian of a vector field)

CHAPTER

TWELVE

TENSOR INVARIANTS

12.1 Rank-2 tensors

- Characteristic polynomial (*this definition needs that determinant exists*)

$$0 = \det\left(\mathbf{A} - s\mathbf{I}\right) \ .$$

Using a unit normal basis (**todo** generalization required?)

$$0 = \begin{vmatrix} A_{11} - s & A_{12} & A_{13} \\ A_{21} & A_{22} - s & A_{23} \\ A_{31} & A_{32} & A_{33} - s \end{vmatrix} = -s^3 + s^2 I_1 - s I_2 + I_3 \ ,$$

with

$$I_1 = \mathrm{tr}(\mathbf{A}) = A_{11} + A_{22} + A_{33}$$
$$I_2 = \frac{1}{2}\left(\mathrm{tr}(\mathbf{A})^2 - \mathrm{tr}(\mathbf{A}^2)\right) = \cdots = A_{11}A_{22} + A_{11}A_{33} + A_{22}A_{33} - A_{31}A_{13} - A_{12}A_{21} - A_{32}A_{23}$$
$$I_3 = \det(\mathbf{A}) = ...$$

The coefficients of the characteristic polynomial are invariant under transformation of coordinates (**todo** here only referring to Cartesian basis, and thus orthogonal transformations?)

- **Trace**…
- Determinant…

CHAPTER

THIRTEEN

UNITARY AND ROTATION TENSORS

Some notes in the introduction to classical mechanics: Rotations: Tensor formalism for rotations.

todo Move the mathematical treatment here?

todo Discuss vector operations that are invariant under 3-dimensional rotation and other unitary tensor applications (i.e. reflections). These properties are useful to prove the general expression of *isotropic tensors*.

13.1 Invariant operations under rotations and unitary transformations

Proof that/if:

- the only independent invariant operations producing a scalar with **two** vectors **u**, **v** are

$$\mathbf{u}\cdot\mathbf{u} = |\mathbf{u}|^2$$
$$\mathbf{v}\cdot\mathbf{v} = |\mathbf{v}|^2$$
$$\mathbf{u}\cdot\mathbf{v}$$

- the only independent invariant operations producing a scalar with **three** vectors **u**, **v**, **w** are

$$\mathbf{u}\times\mathbf{v}\cdot\mathbf{w}$$

- the only independent invariant operations producing a scalar with **four** vectors $\{\mathbf{u}^{(i)}\}_{i=1:4}$ are

$$\mathbf{u}^{(k_1)}\cdot\mathbf{u}^{(k_2)}\ \mathbf{u}^{(k_3)}\cdot\mathbf{u}^{(k_4)}\ ,$$

  with every index k_j independently ranging from 1 to 4.

CHAPTER

FOURTEEN

ISOTROPIC TENSORS

This section provides the definition of isotropic tensors, the general forms of low-rank tensors (from rank-2 to rank-4 tensors) that may be useful in some fields of science (e.g. in continuum mechanics, in the constitutive law relating stress tensor with strain tensor in isotropic linear elastic media or stress tensor with strain velocity tensor in Newtonian fluids).

Definition 14.1 (Isotropic tensor)

An isotropic tensor is …

The general form of the low-rank isotropic tensors in three-dimensional spaces, $\dim(\mathcal{V}) = 3$ is proved using the definition of a rank-r tensor as a multilinear map acting on r vectors, and exploiting the invariance of some vector operations under rotations.

14.1 Rank-2 isotropic tensors

The most general rank-2 isotropic tensor is a scalar multiple of the identity tensor, $\mathbf{A} = a\mathbf{I}$.

Proof

The action of a rank-2 isotropic tensor in $\mathcal{V}$ on every pair of vectors $\mathbf{u}$, $\mathbf{v}$ can be a function of $\mathbf{u} \cdot \mathbf{v}$ only, i.e. the only vector operation:

- producing a scalar
- invariant under rotations
- involving the two vectors $\mathbf{u}$, $\mathbf{v}$ (i.e. the tensor is a multilinear map, $\mathbf{A}(\mathbf{u}, \mathbf{v})$)

Now, using coordinates, the general expression of the isotropic tensor is retrieved comparing the expression built under the isotropic assumption,

$$\begin{aligned}\mathbf{A}(\mathbf{u}, \mathbf{v}) = a\mathbf{u} \cdot \mathbf{v} = \\ = a\left(u^i \mathbf{b}_i\right) \cdot \left(v^j \mathbf{b}_j\right) = \\ = a u^i v^j g_{ij}\end{aligned}$$

with the general expression of the action of the stress tensor on the vectors $\mathbf{u}$, $\mathbf{v}$

$$\mathbf{A}(\mathbf{u}, \mathbf{v}) = A_{ij}\mathbf{b}^i\mathbf{b}^j\left(u^k\mathbf{b}_k, v^l\mathbf{b}_l\right) = A_{ij}\delta^i_k\delta^j_l u^k v^l = A_{ij}u^i v^j \ ,$$

so that the *covariant components* read

$$A_{ij} = a g_{ij}$$

and the tensor can be written as

$$\begin{aligned}\mathbf{A} = A_{ij}\mathbf{b}^i\mathbf{b}^j = ag_{ij}\mathbf{b}^i\mathbf{b}^j = \\ = a\mathbf{b}^i\mathbf{b}_i = a\delta_i^j\mathbf{b}^i\mathbf{b}_j = \\ = ag^{ij}\mathbf{b}_j\mathbf{b}_i = ag^{ij}\mathbf{b}_i\mathbf{b}_j\ .\end{aligned}$$

Components of the identity tensor

The identity tensor can be defined as a rank-2 tensor whose dot product with any arbitrary $\mathbf{v} \in \mathbf{V}$ gives

$$\mathbf{I} \cdot \mathbf{v} = \mathbf{v}\ .$$

The identity tensor can be written using any vector basis $\{\mathbf{b}_i\}_i$, and its reciprocal $\{\mathbf{b}^j\}$ as

$$\mathbf{I} = \mathbf{b}_i\mathbf{b}^i = \mathbf{b}^i\mathbf{b}_i = g_{ij}\mathbf{b}^i\mathbf{b}^j = g^{ij}\mathbf{b}_i\mathbf{b}_j\ ,$$

as it's immediate to prove by direct computation

$$\begin{aligned}\mathbf{b}_i\mathbf{b}^i \cdot \mathbf{v} = \mathbf{b}_i\mathbf{b}^i \cdot (v^k\mathbf{b}_k) = \mathbf{b}_i\delta_k^i v^k = v^i\mathbf{b}_i = \mathbf{v} \\ \mathbf{b}^i\mathbf{b}_i \cdot \mathbf{v} = \mathbf{b}^i\mathbf{b}_i \cdot (v_k\mathbf{b}^k) = \mathbf{b}^i\delta_i^k v_k = v_i\mathbf{b}^i = \mathbf{v} \\ g_{ij}\mathbf{b}^i\mathbf{b}^j \cdot \mathbf{v} = g_{ij}\mathbf{b}^i\mathbf{b}^j \cdot (v^k\mathbf{b}_k) = \mathbf{b}^i\delta_k^j v^k g_{ij} = \mathbf{b}^i \underbrace{v^k g_{ik}}_{=v_i} = \mathbf{v} \\ g^{ij}\mathbf{b}_i\mathbf{b}_j \cdot \mathbf{v} = g^{ij}\mathbf{b}_i\mathbf{b}_j \cdot (v_k\mathbf{b}^k) = \mathbf{b}_i\delta_j^k v_k g^{ij} = \mathbf{b}_i \underbrace{v_k g^{ik}}_{=v^i} = \mathbf{v}\end{aligned}$$

14.2 Rank-3 isotropic tensors

todo

- Invariant object: $\mathbf{u} \times \mathbf{v} \cdot \mathbf{w}$
- Retrieve components, to find Levi-Civita symbols
- Discuss invariance under rotation and symmetry (pseudo-tensor)

14.3 Rank-4 isotropic tensors

Proof

The action of a rank-4 isotropic tensor in $\mathcal{V}$ on every set of 4 vectors **u**, **v**, **w**, **x** can be a function of the products of scalar products

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v}\, \mathbf{w} \cdot \mathbf{x} \\ \mathbf{u} \cdot \mathbf{w}\, \mathbf{v} \cdot \mathbf{x} \\ \mathbf{u} \cdot \mathbf{x}\, \mathbf{v} \cdot \mathbf{w}\ ,\end{aligned}$$

i.e. any possible combinations through scalar products of different vectors

- producing a scalar
- invariant under rotations

- involving all the 4 vectors **u**, **v**, **w**, **x** (i.e. the tensor is a multilinear map, $\mathbf{A}(\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x})$)

Now, using coordinates, the general expression of the isotropic tensor is retrieved comparing the expression built under the isotropic assumption,

$$\begin{aligned}\mathbf{A}(\mathbf{u}, \mathbf{v}) &= a\mathbf{u}\cdot\mathbf{v}\,\mathbf{w}\cdot\mathbf{x} + b\mathbf{u}\cdot\mathbf{w}\,\mathbf{v}\cdot\mathbf{x} + c\mathbf{u}\cdot\mathbf{x}\,\mathbf{v}\cdot\mathbf{w} = \\ &= a\, g_{ij}g_{kl}u^i v^j w^k x^l + b\, g_{ij}g_{kl}u^i w^j v^k x^l + c\, g_{ij}g_{kl}x^i v^j v^k w^l = \\ &= \left(a\, g_{ij}g_{kl} + b\, g_{ik}g_{jl} + c\, g_{il}g_{jk}\right) u^i v^j w^k x^l \ ,\end{aligned}$$

with the general expression of the action of the stress tensor on the vectors **u**, **v**

$$\begin{aligned}\mathbf{A}(\mathbf{u}, \mathbf{v}) &= A_{ijkl}\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l\left(u^m\mathbf{b}_m, v^n\mathbf{b}_n, u^p\mathbf{b}_p, v^q\mathbf{b}_q\right) = \\ &= A_{ijkl}\delta^i_m\delta^j_n\delta^k_p\delta^l_q u^m v^n w^p x^q = A_{ijkl}u^i v^j w^k x^l \ ,\end{aligned}$$

so that the *covariant components* read

$$A_{ijkl} = a\, g_{ij}g_{kl} + b\, g_{ik}g_{jl} + c\, g_{il}g_{jk} \ ,$$

and the tensor can be written as

$$\mathbf{A} = A_{ijkl}\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l = \left(a\, g_{ij}g_{kl} + b\, g_{ik}g_{jl} + c\, g_{il}g_{jk}\right)\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l =$$

The most general liner isotropic relation between two rank-2 tensors reads **A**, **B**

$$\begin{aligned}\mathbf{A} &= a\,\mathrm{tr}\,(\mathbf{B})\,\mathbf{I} + b\,\mathbf{B} + c\,\mathbf{B}^T \\ &= a\,\mathrm{tr}\,(\mathbf{B})\,\mathbf{I} + \tilde{b}\,\mathbf{B}^s + \tilde{c}\,\mathbf{B}^a \ ,\end{aligned}$$

being $\mathbf{B}^s = \frac{1}{2}\left(\mathbf{B} + \mathbf{B}^T\right)$, and $\mathbf{B}^a = \frac{1}{2}\left(\mathbf{B} - \mathbf{B}^T\right)$ the symmetric and the anti-symmetric parts of the tensor **B** respectively.

Linear isotropic relation between two rank-2 tensors

$$\begin{aligned}\mathbf{A} &= \mathbf{Q} : \mathbf{B} \\ \mathbf{Q} : \mathbf{B} &= \left(Q_{ijkl}\mathbf{b}^i\mathbf{b}^j\mathbf{b}^k\mathbf{b}^l\right) : \left(B^{mn}\mathbf{b}_m\mathbf{b}_n\right) = \\ &= Q_{ijkl}B^{kl}\mathbf{b}^i\mathbf{b}^j = \\ &= \left(a\, g_{ij}g_{kl} + b\, g_{ik}g_{jl} + c\, g_{il}g_{jk}\right)B^{kl}\mathbf{b}^i\mathbf{b}^j = \\ &= \left(a\, g_{ij}B_l{}^l + b\, B_{ij} + c\, B_{ji}\right)\mathbf{b}^i\mathbf{b}^j = \\ &= \left(a\, g_{ij}B_l{}^l + b\, B_{ij} + c\, B_{ji}\right)\mathbf{b}^i\mathbf{b}^j =\end{aligned}$$

depends on the three constant a, b, c of the isotropic tensor.

The most general liner isotropic relation between two symmetric rank-2 tensors reads **A**, **B**

$$\mathbf{A} = a\,\mathrm{tr}\,(\mathbf{B})\,\mathbf{I} + \tilde{b}\,\mathbf{B} \ ,$$

as the anti-symmetric part of a symmetric tensor is identically zero.

Linear isotropic relation between two symmetric rank-2 tensors

If $B_{ij} = B_{ji}$, the double-dot product becomes

$$\mathbf{A} = \mathbf{Q} : \mathbf{B} = \left(a\, g_{ij}B_l{}^l + d\, B_{ij}\right)\mathbf{b}^i\mathbf{b}^j \ ,$$

depends only on two constants a, d, having defined $d = b + c$. This is a common condition in continuum mechanics, where some constitutive law links two symmetric rank-2 tensors, like the stress and strain tensors for linear elastic media, or the stress and strain velocity tensors for Newtonian fluids.

CHAPTER

FIFTEEN

TIME DERIVATIVE OF INTEGRALS OVER MOVING DOMAINS

Some results about time derivatives of integrals over moving domains are collected here. These results are useful for writing balance equations of physical quantities in integral form over arbitrary domains, like:

- integral form of balance equations in continuum mechanics, see Continuum Mechanics:Governing Equations:Integral balance equations for arbitrary domains
- integral form of Maxwell's equations governing classical electromagnetism, see Electromagnetism:Principles of Classical Electromagnetism:derivation of balance equations for arbitary volume, starting from equations for a control volume

Link to hand-written notes.

15.1 Volume density

Reynolds transport theorem. Given a volume $V(t)$ with boundary $\partial V(t)$, whose points $\vec{r} \in \partial V(t)$ have velocity $\vec{v}_b$,

$$\frac{d}{dt}\int_{V(t)} f = \int_{V(t)} \frac{\partial f}{\partial t} + \oint_{\partial V(t)} f \vec{v}_b \cdot \hat{n} \, .$$

"Proof"

$$\begin{aligned}
\frac{d}{dt}\int_{v(t)} f(\vec{r},t)\,dt &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{v(t+\Delta t)} f(\vec{r},t+\Delta t) - \int_{v(t)} f(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{v(t)} \{f(\vec{r},t+\Delta t) - f(\vec{r},t)\} + \int_{\Delta v(t;\Delta t)} f(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{v(t)} \left\{\Delta t\,\frac{\partial f}{\partial t}(\vec{r},t) + o(\Delta t)\right\} + \oint_{\partial v(t)} \{\Delta t\,\vec{v}_b\cdot\hat{n}\,f(\vec{r},t) + o(\Delta t)\}\right] = \\
&= \int_{v(t)} \frac{\partial f}{\partial t} + \oint_{\partial v(t)} f\vec{v}_b\cdot\hat{n}\, .
\end{aligned}$$

15.2 Flux across a surface

$$\frac{d}{dt}\int_{S(t)} \vec{f}\cdot\hat{n} = \int_{S(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{n} + \int_{S(t)} \nabla\cdot\vec{f}\,\vec{v}_b\cdot\hat{n} - \int_{\partial S(t)} \vec{v}_b\times\vec{f}\cdot\hat{t}$$

"Proof"

$$\begin{aligned}
\frac{d}{dt}\int_{s(t)} \vec{f}(\vec{r},t)\cdot\hat{n}\,dt &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{s(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t+\Delta t) - \int_{s(t)} f(\vec{r},t)\cdot\hat{n}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{s(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t+\Delta t) - \int_{s(t)} f(\vec{r},t)\cdot\hat{n}(\vec{r},t) + \right. \\
&\qquad \left. - \int_{s(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t) + \int_{s(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{n}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{s(t)} \Delta t\frac{\partial \vec{f}}{\partial t}\cdot\hat{n} + o(\Delta t) + \oint_{\partial\Delta v(t;\Delta t)} \vec{f}\cdot\hat{n} - \oint_{\partial s(t)} \Delta t\vec{f}\cdot\hat{t}\times\vec{v}_b\right] = \\
&= \int_{s(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{n} + \int_s \nabla\cdot\vec{f}\,\vec{v}_b\cdot\hat{n} - \oint_{\partial s(t)} \vec{v}_b\times\vec{f}\cdot\hat{t}\,.
\end{aligned}$$

having used

$$\oint_{\partial\Delta v(t;\Delta t)} \vec{f}\cdot\hat{n} = \int_{\Delta v} \nabla\cdot\vec{f} = \Delta t\int_s \nabla\cdot\vec{f}\,\hat{n}\cdot\vec{v}_b + o(\Delta t)$$

15.3 Work line integral along a line

$$\frac{d}{dt}\int_{\ell(t)} \vec{f}\cdot\hat{t} = \int_{\ell(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{t} + \int_{\ell(t)} \nabla\times\vec{f}\cdot\vec{v}_b\times\hat{t} + \vec{f}_B\cdot\vec{v}_B - \vec{f}_A\cdot\vec{v}_A$$

"Proof"

$$\begin{aligned}
\frac{d}{dt}\int_{\ell(t)} \vec{f}(\vec{r},t)\cdot\hat{t}(\vec{r},t)\,dt &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{\ell(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t+\Delta t) - \int_{\ell(t)} \vec{f}(\vec{r},t)\cdot\hat{t}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{\ell(t+\Delta t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t+\Delta t) - \int_{\ell(t)} f(\vec{r},t)\cdot\hat{t}(\vec{r},t) + \right. \\
&\qquad \left. - \int_{\ell(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t) + \int_{\ell(t)} \vec{f}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t)\right] = \\
&= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\left[\int_{\ell(t+\Delta t)} \Delta t\,\frac{\partial \vec{f}}{\partial t}(\vec{r},t+\Delta t)\cdot\hat{t}(\vec{r},t+\Delta t) + \right. \\
&\qquad \left. + \oint_{\partial\Delta s(t)} f(\vec{r},t)\cdot\hat{t}(\vec{r},t) - \Delta t\vec{f}_A\cdot\vec{v}_A + \Delta t\vec{f}_B\cdot\vec{v}_B + o(\Delta t)\right] = \\
&= \int_{\ell(t)} \frac{\partial \vec{f}}{\partial t}\cdot\hat{t} + \int_{\ell(t)} \nabla\times\vec{f}\cdot\vec{v}_b\times\hat{t} + \vec{f}_B\cdot\vec{v}_B - \vec{f}_A\cdot\vec{v}_A\,.
\end{aligned}$$

having used

$$\oint_{\partial \Delta s(t;\Delta t)} \vec{f} \cdot \hat{t} = \int_{\Delta s} \hat{n} \cdot \nabla \times \vec{f} = \Delta t \int_{\ell} \nabla \times \vec{f} \cdot \hat{v}_b \times \vec{t} + o(\Delta t)$$

CHAPTER

SIXTEEN

CALCULUS IDENTITIES

Here some calculus identities are listed and proved, often using Cartesian coordinates and index notation. While these identities are independent on the choice of the coordinates - as calculus is - a generic set of coordinates can be used to prove them, and Cartesian coordinates are the most convenint choice.

$$\Delta\vec{v} = \nabla(\nabla\cdot\vec{v}) - \nabla\times\nabla\times\vec{v}\,. \tag{16.1}$$

Proof.

Uusing the identity

$$\varepsilon_{ijk}\varepsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}$$

and index notation for Cartesian coordinates,

$$\begin{aligned}
\left(\nabla\times\nabla\times\vec{v}\right)_i &= \hat{e}_i\cdot\left(\nabla\times\nabla\times\vec{v}\right) = \\
&= \varepsilon_{ijk}\,\partial_j(\varepsilon_{klm}\,\partial_l v_m) = \\
&= \varepsilon_{kij}\,\varepsilon_{klm}\partial_{jl}v_m = \\
&= \left(\delta_{il}\,\delta_{jm} - \delta_{im}\,\delta_{jl}\right)\partial_{jl}v_m = \\
&= \partial_{ij}\,v_j - \partial_{jj}\,v_i = \\
&= \hat{e}_i\cdot\left(\nabla(\nabla\cdot\vec{v}) - \Delta\vec{v}\right) = \qquad = \left(\nabla(\nabla\cdot\vec{v}) - \Delta\vec{v}\right)_i\,.
\end{aligned}$$

Part V

Functional Analysis

CHAPTER

SEVENTEEN

INTRODUCTION TO FUNCTIONAL ANALYSIS

- Lebesgue integral
- L^p, H^p function spaces
- Banach and Hilbert spaces

CHAPTER

EIGHTEEN

DIRAC'S DELTA

Dirac's delta $\delta(x)$ is a distribution, or generalized function, with the following properties

1.
$$\int_D \delta(x - x_0)\, dx = 1 \quad \text{if } x_0 \in D \tag{18.1}$$

2.
$$\int_D f(x)\delta(x - x_0)\, dx \quad \text{if } x_0 \in D \tag{18.2}$$

for $\forall f(x)$ "regular" **todo** *what does regular mean?*

18.1 Dirac's delta in terms of regular functions

18.1.1 Piece-wise constant

For $\varepsilon \to 0$,

$$\delta(x) \sim r_\varepsilon(x) = \begin{cases} \frac{1}{\varepsilon} & x \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}] \\ 0 & \text{otherwise} \end{cases}$$

Properties - proof.

1. Unitariety
$$\int_{x=-\infty}^{\infty} r_\varepsilon(x - x_0)\, dx = \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{1}{\varepsilon}\, dx = 1\,,$$
for $\forall \varepsilon$;

2. Shift property, using mean-value theorem of continuous functions
$$\int_{x=-\infty}^{\infty} r_\varepsilon(x - x_0) f(x)\, dx = \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{1}{\varepsilon} f(x)\, dx = \frac{1}{\varepsilon}\varepsilon f(\xi)\,,$$
with $\xi \in [x_0 - \frac{\varepsilon}{2}, x_0 + \frac{\varepsilon}{2}]$, for the mean value theorem. As $\varepsilon \to 0$, $\xi \to x_0$, and thus
$$\int_{x=-\infty}^{\infty} r_\varepsilon(x - x_0) f(x)\, dx \to f(x_0)$$

18.1.2 Piecewise-linear

For $\varepsilon \to 0$,

$$\delta(x) \sim t_\varepsilon(x) = \begin{cases} \frac{2}{\varepsilon}\left(1 - \frac{2|x|}{\varepsilon}\right) & x \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}] \\ 0 & \text{otherwise} \end{cases}$$

Properties - proof

1. Unitariety

$$\int_{x=-\infty}^{\infty} t_\varepsilon(x - x_0)\, dx = \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{2}{\varepsilon}\left(1 - \frac{2|x|}{\varepsilon}\right) dx = \frac{1}{2}\varepsilon\frac{2}{\varepsilon} = 1\,,$$

   for $\forall\varepsilon$;

2. Shift property, using mean-value integration scheme in $x \in [x_0 - \frac{\varepsilon}{2}, x_0]$, $x \in [x_0, x_0 + \frac{\varepsilon}{2}]$ (**todo** *why?*)

$$\begin{aligned}
\int_{x=-\infty}^{\infty} t_\varepsilon(x - x_0) f(x)\, dx &= \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0+\frac{\varepsilon}{2}} \frac{2}{\varepsilon}\left(1 - \frac{2|x - x_0|}{\varepsilon}\right) f(x)\, dx = \\
&= \int_{x=x_0-\frac{\varepsilon}{2}}^{x_0} \frac{2}{\varepsilon}\left(1 - \frac{2|x - x_0|}{\varepsilon}\right) f(x)\, dx + \int_{x=x_0}^{x_0+\frac{\varepsilon}{2}} \frac{2}{\varepsilon}\left(1 - \frac{2|x - x_0|}{\varepsilon}\right) f(x)\, dx = \\
&= \frac{\varepsilon}{2}\frac{2}{\varepsilon}\left(1 - \frac{2}{\varepsilon}\frac{\varepsilon}{4}\right) f\left(x_0 - \frac{\varepsilon}{4}\right) dx + \frac{\varepsilon}{2}\frac{2}{\varepsilon}\left(1 - \frac{2}{\varepsilon}\frac{\varepsilon}{4}\right) f\left(x_0 + \frac{\varepsilon}{4}\right) dx = \\
&= \frac{1}{2} f\left(x_0 - \frac{\varepsilon}{4}\right) + \frac{1}{2} f\left(x_0 + \frac{\varepsilon}{4}\right)
\end{aligned}$$

   As $\varepsilon \to 0$

$$\int_{x=-\infty}^{\infty} t_\varepsilon(x - x_0) f(x)\, dx \to f(x_0)$$

18.1.3 Gaussian approximation

For $\alpha \to +\infty$,

$$\varphi_\alpha(x) = \sqrt{\frac{\alpha}{\pi}} e^{-\alpha x^2} \sim \delta(x)$$

Properties - proof

Fourier transform of $\varphi_\alpha(x)$ reads

$$\begin{aligned}
\mathcal{F}\{\varphi_\alpha(x)\}(k) &= \int_{x=-\infty}^{+\infty} \varphi_\alpha(x) e^{-ikx}\, dx = \\
&= \int_{x=-\infty}^{+\infty} \sqrt{\frac{\alpha}{\pi}} e^{-\alpha x^2} e^{-ikx}\, dx = \\
&= \sqrt{\frac{\alpha}{\pi}} \int_{x=-\infty}^{+\infty} e^{-\alpha\left(x + i\frac{k}{2\alpha}\right)^2} dx\, e^{-\frac{k^2}{4\alpha}} = \\
&= \sqrt{\frac{\alpha}{\pi}} \sqrt{\frac{\pi}{\alpha}}\, e^{-\frac{k^2}{4\alpha}} = e^{-\frac{k^2}{4\alpha}}\,,
\end{aligned}$$

for $\alpha \to +\infty$,

$$\mathcal{F}\{\varphi_\alpha(x)\}(k) \to 1$$

Fourier transform of Dirac's delta is 1, as shown in (22.3), thus $\varphi_\alpha(x) \to \delta(x)$ for $\alpha \to +\infty$.

18.1.4 Fourier anti-transform

For $a \to +\infty$,

$$\delta(x) \sim \int_{y=-a}^{+a} e^{i2\pi yx}\, dy = \frac{1}{2\pi} \int_{k=-2\pi a}^{2\pi a} e^{ikx}\, dk\,, \tag{18.3}$$

or

$$\delta \sim 2 \int_{y=0}^{a} \cos(2\pi yx)\, dy\,.$$

Proof of the equilvanece

$$\delta(x) \sim \frac{1}{2\pi} \int_{k=-2\pi a}^{2\pi a} e^{ikx}\, dk = \frac{1}{2\pi} \left(\int_{k=-2\pi a}^{0} e^{ikx}\, dk + \int_{0}^{k=2\pi a} e^{ikx}\, dk \right) = \frac{1}{2\pi} \int_{k=0}^{2\pi a} \left(e^{ikx} + e^{ikx}\right)\, dx = \frac{1}{\pi} \int_{x=0}^{2\pi a} \cos(kx)\, dk$$
$$= \int_{y=-a}^{+a} e^{i2\pi yx}\, dy = \cdots = \int_{y=0}^{a} (e^{i2\pi yx} + e^{i2\pi yx})\, dy = 2 \int_{y=0}^{a} \cos(2\pi yx)\, dy\,.$$

18.1.5 sinc(x) approximation

For $a \to +\infty$

$$\delta(x) \sim \frac{\sin(2\pi xa)}{\pi x}$$

Proof

Directly follows from integral of the approximation (18.3)

$$\int_{y=-a}^{+a} e^{i2\pi yx}\, dy = \frac{1}{i2\pi x}\, e^{i2\pi yx}\Big|_{y=-a}^{+a} = \frac{1}{\pi x} \frac{e^{i2\pi ax} - e^{-i2\pi ax}}{2i} = \frac{\sin(2\pi xa)}{\pi x}$$

18.1.6 $\{$sinc$(c)\}^2$ approximation

For $t \to +\infty$,

$$\left\{ \frac{\sin(\omega t/2)}{\omega} \right\}^2 \sim \frac{\pi}{2} t\, \delta(\omega)$$

This approximation is used in **Fermi's golden rule in QM**, see the Modern Physics notes.

Proof

todo *Check and uncomment*

18.1.7 Fourier series

For $x \in [-\pi, \pi]$, and $N \to +\infty$, *Fourier series* of Dirac's delta (train with period 2π) reads

$$\delta(x) \sim \frac{1}{2\pi} \sum_{n=-N}^{N} e^{inx} = \frac{1}{2\pi} \frac{\sin\left(\left(N+\frac{1}{2}\right)x\right)}{\sin\left(\frac{x}{2}\right)}$$

or the T-periodic Dirac's delta train,

$$\delta(x) \sim \frac{1}{T} \sum_{n=-N}^{N} e^{in\frac{2\pi}{T}x} .$$

todo *Write the proof of the last expression, using the relation between complex exponentials and cosine and sine*

Proof

Coefficients of the Fourier series of Dirac's delta (train with period $T = 2\pi$) are evaluated using the expression (22.2)

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(t) e^{-in\frac{2\pi}{2\pi}t} = \frac{1}{2\pi} ,$$

and thus the complex Fourier series (22.1) of Dirac's delta reads

$$\delta(x) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}x} = \frac{1}{2\pi} \sum_{n=-\infty}^{+\infty} e^{inx}$$

Obs. here, integration interval $[-\pi, \pi]$ to "avoid troubles" with Dirac's delta on the extreme points of the interval (it would give $1/2$ and $1/2$ contributions on both extremes…)

It's possible to write the T-periodic Dirac's delta train as

$$\delta(x) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}x} = \frac{1}{T} \sum_{n=-\infty}^{+\infty} e^{in\frac{2\pi}{T}x}$$

Integral $I = \int_{-\infty}^{+\infty} e^{-\alpha x^2}\, dx$

$$\begin{aligned}
I^2 &= \int_{x=-\infty}^{+\infty} e^{-\alpha x^2}\, dx \int_{y=-\infty}^{+\infty} e^{-\alpha y^2}\, dy = \\
&= \int_{x=-\infty}^{+\infty} \int_{y=-\infty}^{+\infty} e^{-\alpha(x^2+y^2)}\, dx\, dy = \\
&= \int_{\theta=0}^{2\pi} \int_{r=0}^{+\infty} e^{-\alpha r^2}\, r\, dr\, d\theta = \\
&= 2\pi \frac{1}{2\alpha} \int_{r=0}^{+\infty} e^{-\alpha r^2}\, d\left(\alpha r^2\right) = \\
&= \frac{\pi}{\alpha} \left[-e^{\alpha r^2}\right] \Big|_{r=0}^{+\infty} = \frac{\pi}{\alpha} .
\end{aligned}$$

Part VI

Complex Calculus

CHAPTER

NINETEEN

COMPLEX ALGEBRA

19.1 Fundamental Theorem of Algebra

See Teorema fondamentale dell'algebra

CHAPTER

TWENTY

COMPLEX ANALYSIS

20.1 Complex functions, $f : \mathbb{C} \to \mathbb{C}$

A complex function f of complex variable $z = x + iy$, $f : \mathbb{C} \to \mathbb{C}$, can be written as

$$f(z) = \tilde{u}(z) + i\tilde{v}(z) = u(x, y) + iv(x, y) \ ,$$

as the sum of its real part $u(z)$ and i times its imaginary part $v(x, y)$. Here $x, y \in \mathbb{R}$, while $\tilde{u}(z), \tilde{v}(z) : \mathbb{C} \to \mathbb{R}$ and $u(x, y), v(x, y) : \mathbb{R}^2 \to \mathbb{R}$. With some abuse of notation, tilde won't be always explicitly written when arguments of real and imaginary parts of f functions won't be written.

20.1.1 Limit

$$\lim_{z \to z_0} f(z) = f(z_0) \qquad , \qquad \forall \varepsilon > 0 \ \exists \delta > 0 \ \text{ s.t. } \ |f(z) - f(z_0)| < \delta \ \forall z \text{ s.t. } |z - z_0| < \varepsilon, \ z \neq z_0 \ .$$

20.1.2 Derivative

Using the definition of *limit of complex functions*, the derivative of a function $f : \mathbb{C} \to \mathbb{C}$, if it exists, is the limit of incremental ratio,

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \ .$$

20.1.3 Line Integrals

Given a line $\gamma \in \mathbb{C}$, whose parametric form is $z(s)$, with regular parametrization with parameter $s \in [s_0, s_1]$,

$$\int_\gamma f(z) \, dz = \int_{s=s_0}^{s_1} f(z(s)) \, z'(s) \, ds \ .$$

20.2 Holomorphic Functions - Analytic Functions

Definition 20.2.1

A holomorphic function is a function whose *derivative* exists.

Examples of analytic functions. todo...

20.2.1 Cauchy-Riemann conditions

For a holomorphic function $f(z) = u(x,y) + iv(x,y)$, Cauchy-Riemann conditions

$$\begin{cases} u_{/x} = v_{/y} \\ u_{/y} = -v_{/x} \end{cases}$$

hold. The evaluation of the derivative once with $\Delta z = \Delta x$ and once with $\Delta z = i\Delta y$

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} =$$
$$= \begin{cases} \lim\limits_{\Delta x \to 0} \dfrac{f(x+\Delta x, y) - f(x,y)}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{u(x+\Delta x, y) + iv(x+\Delta x, y) - u(x,y) - iv(x,y)}{\Delta x} = u_{/x} + iv_{/x} \\ \lim\limits_{\Delta y \to 0} \dfrac{f(x, y+\Delta y) - f(x,y)}{i\Delta y} = \lim\limits_{\Delta y \to 0} \dfrac{u(x, y+\Delta y) + iv(x, y+\Delta y) - u(x,y) - iv(x,y)}{i\Delta y} = -iu_{/y} + v_{/y} \end{cases}$$

provides the proof.

20.2.2 Cauchy Theorem

For a holomorphic function f, $f : \Omega \subseteq \mathbb{C} \to \mathbb{C}$

$$\oint_\gamma f(z)\, dz = 0\,,$$

for $\forall \gamma \subset \Omega$. Proof follows from *Green's lemma*, and *Cauchy-Riemann conditions*

$$\begin{aligned} \oint_\gamma f(z)dz &= \oint_\gamma (u(x,y) + iv(x,y))\,(dx + idy) = \\ &= \oint_\gamma (udx - vdy) + i \oint_\gamma (udy + vdx) = \\ &= -\int_S \left(\underbrace{u_{/y} + v_{/x}}_{=0} \right) dx\, dy + i \int_S \left(\underbrace{u_{/x} - v_{/y}}_{=0} \right) dx\, dy = 0\,. \end{aligned}$$

20.3 Useful integrals

20.3.1 Independence of line integral for holomorphic functions

For a function $f(z)$ analytic in D, the line integral on paths $\ell_{ab,i}$ with the same extreme points a, b contained in D is independent on the path, but only depends on the extreme points a, b,

$$\int_{\ell_{ab,1}} f(z)\,dz = \int_{\ell_{ab,2}} f(z)\,dz$$

The proof readily follows, using *Cauchy theorem* applied to a function $f(z) : D \subseteq \mathbb{C} \to \mathbb{C}$, analytic in D, and splitting the closed path γ into two paths ℓ_1, ℓ_2 with the same extreme points, $\gamma = \ell_1 \cup (-\ell_2)$

$$0 = \oint_\gamma f(z)\,dz = \int_{\ell_1} f(z)\,dz + \int_{-\ell_2} f(z)\,dz = \int_{\ell_1} f(z)\,dz - \int_{\ell_2} f(z)\,dz\,.$$

20.3.2 Sum and difference of line integrals

20.3.3 Integral of z^n

Given a path γ embracing $z = 0$ only once in counter-clockwise direction, and $n \in \mathbb{Z}$

$$\oint_\gamma z^n\,dz = \begin{cases} 2\pi i & \text{if } n = -1 \\ 0 & \text{otherwise} \end{cases}$$

Since z^n is analytic everywhere (**todo** *prove it! Add a section with proofs for common functions*) except for $z = 0$, it's possible to evaluate the integral on a circle with center $z = 0$ and radius R. Using polar expression of the complex numbers on the circle, $z = Re^{i\theta}$, $\theta \in [0, 2\pi]$, R const, the differential becomes $dz = iRe^{i\theta}d\theta$ and the integral

$$\begin{aligned}
\oint_\gamma z^n\,dz &= \int_{\theta=0}^{2\pi} \left(Re^{i\theta}\right)^n iRe^{i\theta}d\theta = \\
&= i\int_{\theta=0}^{2\pi} R^{n+1}e^{i(n+1)\theta}d\theta = \\
&= \begin{cases} \text{if } n = -1 \quad : \quad i2\pi \\ \text{otherwise} \quad : \quad iR^{n+1}\dfrac{1}{i(n+1)}\,e^{i(n+1)\theta}\Big|_{\theta=0}^{2\pi} = \dfrac{R^{n+1}}{n+1}(1-1) = 0 \end{cases}
\end{aligned}$$

20.4 Meromorphic functions

Definition 20.4.1

A meromorphic function in a domain is a function holomorphic everywhere except for a (finite?) number of poles. **check**

20.4.1 Singularities

Definition 20.4.2 (Pole)

A pole of order n of a function $f(z)$ is a complex number a so that

$$f(z) = \frac{\phi(z)}{(z-a)^n} ,$$

with $\phi(z)$ holomorphic in $\phi(a) \neq 0$

Examples. …

Definition 20.4.3 (Branch)

Examples. $f(z) = z^{\frac{1}{2}}$

Definition 20.4.4 (Removable singularities)

Example. $f(z) = \frac{\sin z}{z}$

Other irregularities.

20.4.2 Laurent Series

Given a function $f(z)$, in a disk $D_{a,\varepsilon} : 0 < |z-a| < \varepsilon$, its Laurent series centered in a is the convergent (to $f(z)$, **todo** *which type of convergnence?*) series

$$f(z) \sim \sum_{n=-\infty}^{+\infty} a_n (z-a)^n , \tag{20.1}$$

with

$$a_n = \frac{1}{2\pi i} \int_\gamma f(z)\,(z-a)^{-(n+1)}\,dz \tag{20.2}$$

and γ embracing $z = a$ once counter-clockwise. Proof follows immediately inserting the expressions of the coefficients a_n and using the *integral of* z^n. Evaluating the integral (20.2) of the coefficients of the Laurent series, using (20.1) to replace $f(z)$ with its series

$$\begin{aligned} a_n &= \frac{1}{2\pi i} \oint_\gamma \sum_{m=-\infty}^{+\infty} a_m (z-a)^m (z-a)^{-(n+1)} = \\ &= \frac{1}{2\pi i} \oint_\gamma \sum_{m=-\infty}^{+\infty} a_m (z-a)^{m-n-1}\,dz = \\ &= \frac{1}{2\pi i} \oint_\gamma a_n\, z^{-1}\,dz = \\ &= a_n . \end{aligned}$$

todo *Some freestyle with function and its convergent series…add some detail, and the meaning of convergence*

20.4.3 Cauchy formula

For an analytic function $f(z)$,

$$f(a) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z-a}\, dz$$

Proof readily follows using the *integral of z^n* on the Taylor series of $\frac{f(z)}{z-a}$ whose 0th order term reads $f(a)$,

$$\frac{1}{2\pi i} \oint_\gamma \frac{f(a) + \sum_{m=1}^{+\infty} f'(a)(z-a)^m}{z-a}\, dz = \frac{1}{2\pi i} \oint_\gamma \frac{f(a)}{z-a}\, dz = f(a)\frac{2\pi i}{2\pi i} = f(a)\ .$$

20.4.4 Residues

Definition 20.4.5 (Residue)

The residue of function f in a, $\text{Res}(f,a)$ is a complex number R so that $f(z) - \frac{R}{(z-a)}$ has analytic antiderivative in a disk $D_{a,\varepsilon} : 0 < |z-a| < \varepsilon$.

todo Explain this definition. Couldn't be possible to use $\text{Res}(f,a) = \frac{1}{2\pi i} \oint_\gamma f(z)\, dz = a_{-1}$ instead?

Properties.

- If $f(z)$ is analytic in $D_{a,\varepsilon}$ and has a pole of order n in $z=a$, its Laurent series has $a_m = 0$ for $m < n$ and reads

$$f(z) = \sum_{m=-n}^{+\infty} a_m (z-a)^m\ , \tag{20.3}$$

  with $a_{-n} \neq 0$. Since $f(z)$ has a pole of order n in $z=a$, it can be written as

$$f(z) = \frac{\phi(z)}{(z-a)^n}\ ,$$

  with $\phi(z)$ analytic in $D_{a,\varepsilon}$ and $\phi(a) \neq 0$. Since $\phi(z)$ is analytic, it has a Taylor series (or a Laurent series with non-negative powers),

$$\phi(z) \sim \sum_{m=0}^{+\infty} b_m (z-a)^m\ ,$$

  (**todo** *prove it! Extension of the real case. Add a link to the proof*) and thus

$$f(z) \sim \sum_{m=0}^{+\infty} b_m (z-a)^{m-n} = \sum_{m=-n}^{+\infty} b_{m+n} (z-a)^m = \sum_{m=-n}^{+\infty} a_m (z-a)^m\ ,$$

  with $a_m = b_{m+n}$.

- For simple closed path γ (embracing a only once counter-clokwise) in $D_{a,\varepsilon}$,

$$\oint_\gamma f(z)\, dz = 2\pi i a_{-1} = 2\pi i \text{Res}(f,a) \tag{20.4}$$

  The proof readily follows, using the *integral of z^n* and Laurent series (20.1) of $f(z)$,

$$\oint_\gamma f(z)\, dz = \oint_\gamma \sum_{m=-\infty}^{+\infty} a_m (z-a)^m\, dz = 2\pi i a_{-1}\ .$$

- For a pole a of order n, the following holds

$$a_{-1} = \frac{1}{(n+1)!} \lim_{z \to a} \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n f(z) \right]$$

The proof follows using Laurent series {eq}`eq:laurent:pole-n} for a function with pole of order n, and evaluating the $(n-1)^{th}$ order derivative

$$\begin{aligned}
\frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n f(z) \right] &= \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n \sum_{m=-n}^{+\infty} a_n (z-a)^m \right] = \\
&= \frac{d^{n-1}}{dz^{n-1}} \left[\sum_{m=-n}^{+\infty} a_n (z-a)^{m+n} \right] = \\
&= \frac{d^{n-1}}{dz^{n-1}} \left[\sum_{m=0}^{+\infty} a_{m-n} (z-a)^{m} \right] = \\
&= \frac{d^{n-2}}{dz^{n-2}} \left[\sum_{m=0}^{+\infty} m a_{m-n} (z-a)^{m-1} \right] = \\
&= \frac{d^{n-3}}{dz^{n-3}} \left[\sum_{m=0}^{+\infty} m(m-1) a_{m-n} (z-a)^{m-2} \right] = \\
&= \cdots = \\
&= \left[\sum_{m=0}^{+\infty} m! \, a_{m-n} (z-a)^{m-n+1} \right]
\end{aligned}$$

and then letting $z \to a$, so that only the term with $m - n + 1 = 0$ survives

$$\lim_{z \to a} \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n \sum_{m=-n}^{+\infty} a_n (z-a)^m \right] = (n-1)! \, a_{-1} \, .$$

20.4.5 Residue Theorem

Theorem 20.4.1 (Residue Theorem)

Given $f(z)$ with a finite number of poles $p_n \in D$, then

$$\int_\gamma f(z) \, dz = 2\pi i \, \sum_n I(\gamma, p_n) \text{Res}(f, p_n) \, ,$$

being γ a path in D, and $I(\gamma, p_n)$ the winding index of the path γ around pole p_n (+1 for each counter-clockwise loop, -1 for each clockwise loop).

The proof readily follows extending the result for a single pole (20.4) to general number of poles and general paths γ embracing (with sign) each pole p_n $I(\gamma, p_n)$ times, with the same techinques shown in section *Sum and difference of line integrals*.

20.4.6 Cauchy argument principle

20.4.7 Evaluation of integrals

20.4.8 Inverse Laplace Transform

Given Laplace transform

$$F(s) := \mathcal{L}\{f(t)\}(s) := \int_{t=0^-}^{+\infty} f(t)e^{-st}\,dt\,,$$

the inverse transform can be evaluated as

$$f(t) = \mathcal{L}^{-1}\{F(s)\}(t) := \lim_{T\to+\infty} \frac{1}{2\pi i}\int_{s=a-iT}^{a+iT} e^{st}F(s)\,ds\,,$$

with $a > \text{Re}\{p_n\}$ (**todo** *why?*) for each pole of the function $F(s)$, evaluated on the vertical line $s = a+iy$, $y \in [-T, T]$, $ds = idy$,

$$\begin{aligned}
\lim_{T\to+\infty} \frac{1}{2\pi i}\int_{s=a-iT}^{a+iT} e^{st}F(s)\,ds &= \lim_{T\to+\infty} \frac{1}{2\pi i}\int_{s=a-iT}^{a+iT} e^{st}\int_{\tau=0^-}^{+\infty} f(\tau)e^{-s\tau}\,d\tau\,ds = \\
&= \lim_{T\to+\infty} \frac{1}{2\pi i}\int_{y=-T}^{T} e^{(a+iy)t}\int_{\tau=0^-}^{+\infty} f(\tau)e^{-(a+iy)\tau}\,d\tau\,idy = \\
&= \lim_{T\to+\infty} \frac{1}{2\pi}\int_{y=-T}^{T}\int_{\tau=0^-}^{+\infty} e^{iy(t-\tau)}e^{a(t-\tau)}f(\tau)\,d\tau\,dy = \\
&= \ldots \\
&= \int_{\tau=0^-}^{+\infty} \delta(t-\tau)e^{a(t-\tau)}f(\tau)d\tau = f(t)\,.
\end{aligned}$$

having used the transform of *Dirac's delta* $\delta(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{+\infty} e^{-j\omega t}\,d\omega$.

todo *Ohter approach: if $a > Re\{p_n\}$, the contour built with the vertical line with real part a and the arc of circumference on its…*

20.5 Sequences and Series in the Complex Field

The extension of sequences and series from the real line $\mathbb{R}$ to the complex plane $\mathbb{C} \cong \mathbb{R}^2$ preserves many fundamental operational properties while introducing deeper geometric and analytic structures. Most notably, the equivalence between complex differentiability (**holomorphy**) and power series convergence (**analyticity**) forms the cornerstone of complex analysis.

20.5.1 Complex Numerical Sequences and Series

Complex Numerical Sequences

A complex sequence $\{z_n\}_{n\in\mathbb{N}} \subset \mathbb{C}$ can be decomposed into its real and imaginary components:

$$z_n = x_n + iy_n, \quad \text{where } x_n, y_n \in \mathbb{R}$$

Convergence

Definition: Convergence in $\mathbb{C}$

A sequence $\{z_n\}$ **converges** to $L = u + iv \in \mathbb{C}$ (written $\lim_{n\to\infty} z_n = L$) if:

$$\lim_{n\to\infty} |z_n - L| = 0 \, .$$

Or, with the -definition: given $\varepsilon > 0$, $\exists N$ s.t. $|z_n - L| < \varepsilon$ for $\forall n > N$.

Equivalently, convergence in $\mathbb{C}$ under the standard Euclidean metric can be written component-wise, with $z_n = x_n + iy_n$,

$$\lim_{n\to\infty} z_n = L \iff \left(\lim_{n\to\infty} x_n = u \quad \text{and} \quad \lim_{n\to\infty} y_n = v\right)$$

Complex Numerical Series

A complex infinite series is defined as the formal sum:

$$\sum_{n=0}^{+\infty} z_n, \quad z_n \in \mathbb{C}$$

It converges to $S \in \mathbb{C}$ if the sequence of partial sums $S_N = \sum_{n=0}^{N} z_n$ converges to S.

Convergence

Convergence. Using the *convergence of a sequence*, a series is convergent if the sequence of the partial sums $S_N = \sum_{n=0}^{N} z_n$ converges.

Absolute Convergence: The series $\sum z_n$ converges absolutely if the real-valued series of moduli converges:

$$\sum_{n=0}^{+\infty} |z_n| < +\infty$$

Property: Absolute convergence in $\mathbb{C}$ implies standard convergence.

Conditional Convergence: The series converges, but $\sum |z_n| = +\infty$. As an example:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ...$$

Tests for absolute convergence

Test	Statement
Ratio Test	If $\lim_{n\to\infty} \left\lvert \frac{z_{n+1}}{z_n} \right\rvert = L$, then: • $L < 1 \implies$ Absolutely Convergent • $L > 1 \implies$ Divergent
Root Test	If $\limsup_{n\to\infty} \sqrt[n]{\lvert z_n \rvert} = L$, then: • $L < 1 \implies$ Absolutely Convergent • $L > 1 \implies$ Divergent

---

20.5.2 Complex Sequences and Series of Functions

Let $\Omega \subseteq \mathbb{C}$ be an open set and let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of complex-valued functions $f_n : \Omega \to \mathbb{C}$.

Convergence types of Sequences

1. **Pointwise Convergence:** $f_n \to f$ pointwise on Ω if for every $z \in \Omega$:

$$\lim_{n\to\infty} |f_n(z) - f(z)| = 0$$

   (or for $\forall z \in \Omega$, $\forall \varepsilon > 0$, $\exists N$ so that $|f_n(z) - f(z)| < \varepsilon$ for $\forall n > N$.)

2. **Uniform Convergence:** $f_n \rightrightarrows f$ uniformly on $E \subseteq \Omega$ if:

$$\lim_{n\to\infty} \sup_{z\in E} |f_n(z) - f(z)| = 0$$

   (or for $\forall \varepsilon > 0$, $\exists N$ so that $|f_n(z) - f(z)| < \varepsilon$ for $\forall n > N$, $\forall z \in \Omega$.)

3. **Locally Uniform (Compact) Convergence:** $f_n \to f$ uniformly on every compact subset $K \subset \Omega$.

Weierstrass theorem

todo *Uncomment?*

---

todo

CHAPTER

TWENTYONE

LAPLACE TRANSFORM

Definition of Laplace transform. Definition of Laplace transform, inverse transform, properties and theorems.

Applications of Laplace transform. Solution of ODEs,…

21.1 Definition and Properties

21.1.1 Definition of Laplace transform

Two-sided Laplace transform.

$$\mathcal{L}\left\{f(t)\right\}(s) := \int_{t=-\infty}^{+\infty} e^{-st} f(t)\, dt = F(s)\ .$$

One-sided Laplace transform.

$$\mathcal{L}\left\{f(t)\right\}(s) := \int_{t=0^-}^{+\infty} e^{-st} f(t)\, dt = F(s)\ .$$

One-sided transform can be thought as the two-sided transform of **causal functions**, i.e. functions that are equal to zero before a starting time $t = 0$.

21.1.2 Inverse transform

$$f(t) = \mathcal{L}^{-1}\left\{F(s)\right\} = ...$$

see *here*

21.1.3 Elementary one-sided transforms

Linearity.

$$\mathcal{L}\{af(t) + bg(t)\}(s) = aF(s) + bG(s)$$

***Dirac delta*.**

$$\mathcal{L}\left\{\delta(t)\right\} = \int_{t=0^-}^{+\infty} \delta(t)\, e^{st}\, dt = 1$$

Time delay. If $f(t) = 0$ for $t < 0$ ("causality"), for $\tau > 0$,

$$\mathcal{L}\{f(t-\tau)\}(s) = e^{-s\tau}F(s)$$

Proof readily follows direct computation with change of variable $z = t - \tau$, $dt = dz$

$$\mathcal{L}\{f(t-\tau)\}(s) = \int_{t=0^-}^{+\infty} f(t-\tau)e^{-st}\,dt = \int_{z=-\tau}^{+\infty} f(z)e^{-sz}\,dz\,e^{-s\tau} = \int_{z=0}^{+\infty} f(z)e^{-sz}\,dz\,e^{-s\tau} = e^{-s\tau}F(s)\,.$$

"Frequency shift"

$$\mathcal{L}\{f(t)e^{at}\}(s) = F(s-a)$$

Direct computation gives

$$\mathcal{L}\{f(t)e^{at}\}(s) = \int_{t=0^-}^{+\infty} f(t)e^{at}e^{-st}\,dt = \int_{t=0^-}^{+\infty} f(t)e^{-(s-a)t}\,dt = F(s-a)$$

Derivative.

$$\mathcal{L}\{f'(t)\}(s) = sF(s) - f(0^-)\,.$$

Proof readily follows direct computation, with integration by parts

$$\mathcal{L}\{f'(t)\}(s) = \int_{t=0^-}^{+\infty} f'(t)e^{-st}\,dt = [f(t)e^{-st}]\,|_{t=0^-}^{+\infty} + s\int_{t=0^-}^{+\infty} f(t)e^{-st}\,dt = sF(s) - f(0^-)\,,$$

provided that $\lim_{s\to+\infty} f(t)e^{-st} = 0$.

Integral.

$$\mathcal{L}\left\{\int_{\tau=0}^{t} f(\tau)\,d\tau\right\}(s) = \frac{1}{s}F(s)\,.$$

Proof readily follows direct computation, with integration by parts

$$\mathcal{L}\left\{\int_{\tau=0^-}^{t} f(\tau)\,d\tau\right\}(s) = \int_{t=0^-}^{+\infty}\int_{\tau=0^-}^{t} f(\tau)\,d\tau e^{-st}\,dt = \left[-\frac{e^{-st}}{s}\int_{\tau=0^-}^{t} f(\tau)\,d\tau\right]_{t=0}^{+\infty} + \frac{1}{s}\int_{t=0}^{+\infty} f(t)e^{-st}\,dt = \frac{1}{s}F(s)\,,$$

provided that $\int_{\tau=0^-}^{0} f(\tau)d\tau = 0$ and $\lim_{t\to+\infty} \frac{e^{-st}}{s}\int_{\tau=0^-}^{+\infty} f(\tau)\,d\tau = 0$.

Convolution.

$$\begin{aligned}
\mathcal{L}\left\{f(t) * g(t)\right\} &= \int_{t=0^-}^{+\infty}\int_{\tau=-\infty}^{+\infty} f(t-\tau)g(\tau)\,d\tau\,e^{-st}\,dt = \qquad (1)\\
&= \int_{\tau=-\infty}^{+\infty}\int_{z=-\tau^-}^{+\infty} f(z)g(\tau)\,e^{-s(z+\tau)}\,d\tau\,dz = (2)\\
&= \int_{z=0^-}^{+\infty} f(z)\,e^{-sz}\,dz\int_{\tau=0^-}^{+\infty} g(\tau)e^{-s\tau} =\\
&= \mathcal{L}\{f(t)\}(s)\,\mathcal{L}\{g(t)\}(s)\,.
\end{aligned} \tag{21.1}$$

having performed the change of coordinates $z = t - \tau$, $\tau = \tau$, with unitary Jacobian,

$$\frac{\partial(t,\tau)}{\partial(z,\tau)} = \partial_z t\partial_\tau\tau - \partial_z\tau\partial_z t = 1\cdot 1 - 1\cdot 0 = 1,$$

given the proper description of the domain of integration summarised in the extremes of integration in (1), and causality - i.e. all the functions $f(t)$ are identically zero for $t < 0$ - in (2).

Initial value. If …

$$f(0^+) = \lim_{s \to +\infty} sF(s)$$

From direct computation,

$$\begin{aligned}
\lim_{s \to +\infty} sF(s) &= \lim_{s \to +\infty} s \int_{t=0^-}^{+\infty} f(t)\, e^{-st}\, dt = \\
&= \lim_{s \to +\infty} \left\{ \left[s \left(-\frac{e^{-st}}{s} \right) f(t) \right] \Big|_{t=0}^{+\infty} + \int_{t=0}^{+\infty} e^{-st} f'(t)\, dt \right\} = \\
&= \lim_{s \to +\infty} \left\{ \left[-e^{-st} f(t) \right] \Big|_{t=0}^{+\infty} + \int_{t=0}^{+\infty} e^{-st} f'(t)\, dt \right\} = \\
&= f(0) \,,
\end{aligned}$$

provided that $\lim_{s \to +\infty} \lim_{t \to +\infty} e^{-st} f(t) = 0$ and $\lim_{s \to +\infty} \int_{t=0}^{+\infty} e^{-st} f'(t)\, dt = 0$.

Final value. If …

$$f(+\infty) = \lim_{s \to 0} sF(s)$$

From direct computation (**todo** *check and/or explain proof*),

$$\begin{aligned}
\lim_{s \to 0} sF(s) &= \lim_{s \to 0} s \int_{t=0^-}^{+\infty} f(t)\, e^{-st}\, dt = \\
&= \lim_{s \to 0} \left\{ \left[s \left(-\frac{e^{-st}}{s} \right) f(t) \right] \Big|_{t=0}^{+\infty} + \int_{t=0}^{+\infty} e^{-st} f'(t)\, dt \right\} = \\
&= \lim_{s \to 0} \left\{ \left[-e^{-st} f(t) \right] \Big|_{t=0}^{+\infty} + \int_{t=0}^{+\infty} e^{-st} f'(t)\, dt \right\} = \\
&= f(0) + f(+\infty) - f(0) = f(+\infty) \,,
\end{aligned}$$

provided that $\lim_{s \to 0} \lim_{t \to +\infty} e^{-st} f(t) = 0$.

21.2 Applications of Laplace Transform

21.2.1 Ordinary differential equations

Exploiting the properties of transforming derivation into product by the complex variable s, Laplace transform can be a useful tool in solving *Ordinary differential equations*.

Linear ODEs are treated in details in the section about *Linear Time-Invariant Systems*.

First-order linear differential equation with constant coefficients

A Cauchy problem governed by a first-order linear differential equation with constant coefficients reads

$$\begin{cases} \dot{u} + au = f(t) \quad , \quad \text{for } t > 0 \\ u(0^-) = u_0 \, . \end{cases}$$

Comments. 1) For linear differential equations with constant coefficients, a time shift is always possible to set initial conditions in $t_0 = 0$; 2) dealing with **impulsive forces** (more on this in *LTI systems: impulsive forces*), some attention needs to be paid to the time of initial conditions: if impulsive forces at $t = 0$ may exist, initial conditions need to be set at $t = 0^-$; the effect of the impulsive force is equivalent to a jump in initial conditions from $t = 0^-$ and $t = 0^+$.

General solution in time domain

Multiplying the ODE by e^{at},

$$e^{at} f(t) = e^{at} \left(\dot{u} + au \right) = \frac{d}{dt} \left(e^{at} u(t) \right)$$

and integrating from 0^- to a generic time t — after changing the dummy integration variable from τ to t —

$$e^{at} u(t) - u(0^-) = \int_{\tau=0^-}^{t} e^{a\tau} f(\tau) d\tau \, ,$$

and thus

$$u(t) = e^{-at} u(0^-) + \int_{\tau=0^-}^{t} e^{-a(t-\tau)} f(\tau) d\tau \, .$$

The solution of the problem can be also computed using Laplace transform:

1. transforming the problem from time to Laplace domain: the differential problem with unknown $u(t)$ in time is transformed into an algebraic problem $\hat{u}(s)$. Using Laplace **transform of the time derivative**

$$s\hat{u}(s) + u_0 + a\hat{u}(s) = \hat{f}(s) \, ,$$

2. solving the algebraic problem in Laplace domain for $\hat{u}(s)$

$$\hat{u}(s) = \frac{1}{s+a} u_0 + \frac{1}{s+a} f(s) \, .$$

3. transforming back the solution in time domain, $u(t) = \mathcal{L}^{-1}\{\hat{u}(s)\}$,

$$\begin{aligned} u(t) = \mathcal{L}^{-1}\{\hat{u}(s)\}(t) = \\ = \mathcal{L}^{-1}\left\{ \frac{u_0}{s+a} \right\} + \mathcal{L}^{-1}\left\{ \frac{1}{s+a} \hat{f}(s) \right\} = \\ = e^{-at} u_0 + \int_{0^-}^{t} e^{-a(t-\tau)} f(\tau) d\tau \, , \end{aligned}$$

having used the inverse transform of the exponential $\mathcal{L}\{e^{ct}\} = \int_{t=0^-}^{+\infty} e^{ct} e^{-st} \, dt = \frac{1}{s-c}$, and the inverse transform of the convolution

$$\hat{f}(s)\hat{g}(s) = \mathcal{L}\{f(t) * g(t)\}(s) = \int_{t=0^-}^{+\infty} \left\{ \int_{\tau=0^-}^{t} f(\tau) g(t-\tau) d\tau \right\} e^{-st} dt \, .$$

CHAPTER

TWENTYTWO

FOURIER TRANSFORMS

Fourier transforms are linear transformations of functions usually relating a physical domain of time and/or space, with a domain of frequency and/or wave-vectors.

Contents.

Fourier series. Fourier series is defined for finite-domain or periodic, time-continuous functions, or - more generally - continuous functions in the physical domain.

Fourier transform. Fourier transform is defined for infinite-domain non-periodic, time-continuous functions, or - more generally - continuous functions in the physical domain.

Relations between Fourier transforms and sampling. Fourier series, Fourier transform, discrete time Fourier transform and discrete Fourier transforms are presented, and their relations discussed. Fundamental results about **evenly-spaced sampling** seamlessly follows, as **Shannon-Nyquist theorem**, *Theorem 22.3.1*, shows.

Different Fourier transforms exist, depending if the original function is:

- time discrete/time continuous
- periodic/non-periodic

Transform	Time domain	Frequency domain
Fourier transform	Continuous function in infinite domain	Continuous function in infinite domain
Fourier series	Continuous function in finite domain (or periodic function in infinite domain)	Discrete function in compact range
Discrete-time Fourier transform	Discrete function (or continuous function sampled with Dirac's comb) in infinite domain	Continous periodic spectrum in infinite domain
Discrete Fourier transform	Discrete (or continuous, sampled with Dirac's comb) function in finite domain (or periodic in infinite domain)	Discrete periodic spectrum in infinite domain

Fourier transforms can be useful in:

- highlighting the frequency content of functions
- solving problems: sometimes, it can be easier to transform a problem in frequency domain, solve it in frequency domain, and transform the solution back to the physical domain

22.1 Fourier Series

For a T-periodic function,

$$g(t) \sim \frac{a_0}{2} + \sum_{n=1}^{+\infty} \left[a_n \cos\left(n\frac{2\pi}{T}t\right) + b_n \sin\left(n\frac{2\pi}{T}t\right) \right] ,$$

todo Prove it with properties of integrals of sin and cos over $t \in [0, T]$; prove convergence to average value at jumps

The exponential form reads

$$g(t) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}t} , \tag{22.1}$$

where

$$c_n = \frac{1}{T}\int_{t=0}^{T} f(t)\, e^{-in\frac{2\pi}{T}t} . \tag{22.2}$$

Proof

Exploiting the properties of integrals of complex exponentials with $k \in \mathbb{Z}$

$$\int_{t=0}^{T} e^{ik\frac{2\pi}{T}t}\, dt = \begin{cases} \frac{1}{ik\frac{2\pi}{T}} \left[e^{ik\frac{2\pi}{T}t}\right]\Big|_{t=0}^{T} = 0 & \text{if } k \neq 0 \\ T & \text{if } k = 0 \end{cases}$$

$$\int_{t=0}^{T} f(t) e^{-im\frac{2\pi}{T}t}\, dt \sim \int_{t=0}^{T} \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}t} e^{-im\frac{2\pi}{T}t} \sim \sum_{n=-\infty}^{+\infty} c_n \sim \int_{t=0}^{T} e^{i(n-m)\frac{2\pi}{T}t} \sim T\, c_m .$$

22.2 Fourier Transform

Contents: *definition*; *properties*; *inverse transform*; *Plancherel's theorem*; *uncertainty relation*

22.2.1 Definition

…

$$\mathcal{F}\left\{g(t)\right\}(f) := \int_{t=-\infty}^{+\infty} g(t)\, e^{-i2\pi ft}\, dt.$$

22.2.2 Properties

Linearity

***Dirac delta*.**

$$\mathcal{L}\left\{\delta(t)\right\} = \int_{t=-\infty}^{+\infty} \delta(t)\, e^{-i2\pi ft}\, dt = 1 \tag{22.3}$$

Time delay.

Derivative.

Integral.

Initial value.

Final value.

Property 22.2.1 (Transform of convolution)

$$\mathcal{F}\left\{a * b(t)\right\}(f) = \mathcal{F}\left\{a(t)\right\}(f)\,\mathcal{F}\left\{b(t)\right\}(f)$$

Proof.

$$\begin{aligned}\mathcal{F}\left\{a * b(t)\right\}(f) &= \int_{t=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} a(\tau)b(t-\tau)\,d\tau\,e^{-i2\pi ft}\,dt = \qquad (1)\\ &= \int_{\tau=-\infty}^{+\infty} a(\tau)\,e^{-i2\pi f\tau}\,d\tau\int_{z=-\infty}^{+\infty} b(z)\,e^{-i2\pi fz} = \\ &= \mathcal{F}\{a(t)\}(f)\,\mathcal{F}\{b(t)\}(f)\,,\end{aligned}$$

having used (1) transformation of coordinates $(z,\tau) = (t-\tau,\tau)$ with unit Jacobian

$$\frac{\partial(z,\tau)}{\partial(t,\tau)} = \begin{vmatrix}1 & -1\\ 0 & 1\end{vmatrix} = 1\,.$$

22.2.3 Inverse Fourier Transform

Under the assumptions …**todo**, the inverse Fourier transform reads

$$\mathcal{F}^{-1}\left\{G(f)\right\}(t) := \int_{f=-\infty}^{+\infty} G(f)\,e^{i2\pi ft}\,df.$$

Proof using Dirac's delta expression.

$$\begin{aligned}\mathcal{F}^{-1}\left\{G(f)\right\}(t) := \int_{f=-\infty}^{+\infty} G(f)\,e^{i2\pi ft}\,df &= \int_{f=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} g(\tau)e^{-i2\pi f\tau}\,e^{i2\pi ft}\,df = \\ &= \int_{f=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} g(\tau)e^{-i2\pi f\tau}\,e^{i2\pi ft}\,df = \\ &= \int_{f=-\infty}^{+\infty}\int_{\tau=-\infty}^{+\infty} g(\tau)e^{i2\pi f(t-\tau)}\,df = \\ &= \int_{\tau=-\infty}^{+\infty} g(\tau)\delta(t-\tau)\,d\tau = g(t)\,.\end{aligned}$$

Proof using dominated convergence theorem and Fubini's lemma.

Proof. By the *dominated convergence theorem*, it follows that

$$\begin{aligned}\int_{\mathbb{R}} e^{i2\pi x\xi} F(\xi)\, d\xi &= \lim_{\varepsilon\to 0} \int_{\mathbb{R}} \underbrace{e^{-\pi\varepsilon^2\xi^2 + i2\pi x\xi}}_{G(\xi;x,\varepsilon)} F(\xi)\, d\xi = \\ &= \lim_{\varepsilon\to 0} \int_{\mathbb{R}} g(y;x,\varepsilon) f(y)\, dy = \\ &= \lim_{\varepsilon\to 0} \int_{\mathbb{R}} \varphi_\varepsilon(x-y)\, f(y)\, dy = \\ &= \int_{\mathbb{R}} \delta(x-y)\, f(y)\, dy = f(x)\end{aligned}$$

Lemma 1. The Fourier transform of function $\varphi(t) := e^{-\pi|t|^2}$ reads

$$\begin{aligned}\mathcal{F}\{\varphi(t)\}(\omega) &= \int_{t=-\infty}^{+\infty} \varphi(t) e^{-i\omega t}\, dt = \\ &= \int_{t=-\infty}^{+\infty} e^{-\pi|t|^2} e^{-i\omega t}\, dt = \\ &= \int_{t=-\infty}^{+\infty} e^{-\pi\left(t^2 + i\frac{\omega}{\pi}t - \frac{\omega^2}{4\pi^2}\right)}\, dt\, e^{-\frac{\omega^2}{4\pi^2}} = \\ &= \int_{t=-\infty}^{+\infty} e^{-\pi\left(t + i\frac{\omega}{2\pi}\right)^2}\, dt\, e^{-\frac{\omega^2}{4\pi}} = \\ &= e^{-\frac{\omega^2}{4\pi}},\end{aligned}$$

having evaluated *the integral* $\int_{-\infty}^{+\infty} e^{-\alpha x^2}$ with $\alpha = \pi$. **todo** *justify the result for complex exponential. Use Bromwich contour integrals*

Lemma 2. Fourier transform of $f(\alpha t)$, $\alpha > 0$

$$\mathcal{F}\{f(\alpha t)\}(\omega) = \int_{\mathbb{R}} f(\alpha t) e^{-j\omega t}\, dt = \int_{\tau\in\mathbb{R}} f(\tau) e^{-j\frac{\omega}{\alpha}\tau}\, d\tau \frac{1}{\alpha} = \frac{1}{\alpha} F\left(\frac{\omega}{\alpha}\right)$$

Lemma 3. $\frac{1}{\varepsilon}\varphi\left(\frac{t}{\varepsilon}\right) \to \delta(x)$ for $\varepsilon \to 0$

$$\mathcal{F}\left\{\frac{1}{\varepsilon}\varphi\left(\frac{t}{\varepsilon}\right)\right\}(\omega) = \frac{1}{\varepsilon}\varepsilon e^{-\frac{\omega^2}{4\pi\varepsilon^2}} = e^{-\frac{\omega^2}{4\pi\varepsilon^2}}$$

0. Fourier transform

$$G(f) = \int_{t=-\infty}^{\infty} e^{-i\omega t} g(t)\, dt$$

1.

$$g(t) = e^{i\alpha t}\psi(t)$$

$$\mathcal{F}\{g(t)\}(\omega) = \int_{t=-\infty}^{+\infty} g(t) e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} \psi(t) e^{i\alpha t} e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} \psi(t) e^{-i(\omega-\alpha)t}\, dt = \mathcal{F}\{\psi(t)\}(\omega-\alpha)\,.$$

2.

$$\psi(t) = \phi(\alpha t)$$

$$\mathcal{F}\{\psi(t)\} = \int_{t=-\infty}^{+\infty} \psi(t) e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} \phi(\alpha t) e^{-i\omega t}\, dt = \int_{\tau=-\infty}^{+\infty} \phi(\tau) e^{-i\frac{\omega}{\alpha}\tau}\, \frac{d\tau}{\alpha} = \frac{1}{\alpha}\mathcal{F}\{\phi(t)\}\left(\frac{\omega}{\alpha}\right)\,.$$

3. Fubini's theorem
4. $$\varphi(t) := e^{-\pi t^2}$$
$$\mathcal{F}\{\varphi(t)\} = \int_{t=-\infty}^{+\infty} \varphi(t) e^{-i\omega t}\, dt = \int_{t=-\infty}^{+\infty} e^{-\pi t^2} e^{-i\omega t}\, dt$$
$$0 = \oint_{\gamma} e^{-\alpha |z|^2}\, dz = \int_{\dots} \dots$$
$$z = Re^{i\theta}, \quad dz = iRe^{i\theta} d\theta$$
$$\int_{C/4} e^{-\alpha |z|^2}\, dz = \int_{\theta=0}^{\frac{\pi}{2}} e^{-\alpha R^2} iRe^{i\theta} d\theta = iRe^{-\alpha R^2} \frac{e^{-i\theta}}{i} |_{\theta=0}^{\frac{\pi}{2}}$$
$$\int_{t=0}^{+\infty} e^{-\pi t^2} e^{-i\omega t}\, dt = \int_{t=0}^{+\infty} e^{-\left(\pi t^2 + i\omega t - \frac{\omega^2}{4\pi}\right)}\, dt\, e^{-\frac{\omega^2}{4\pi}} =$$
$$= \int_{t=0}^{+\infty} e^{-\pi\left(t + i\frac{\omega}{2\pi}\right)^2}\, dt\, e^{-\frac{\omega^2}{4\pi}}$$
5. $\varphi_\varepsilon(t) = \frac{1}{\varepsilon^n} \varphi\left(\frac{t}{\varepsilon}\right)$, $t \in \mathbb{R}^n$, is an approximation of Dirac's delta for $\varepsilon \to 0$, so that
$$\lim_{\varepsilon \to 0} \int_{t=-\infty}^{+\infty} \varphi_\varepsilon(t-\tau) f(t)\, dt = f(\tau)$$
$$\lim_{\varepsilon \to 0} \int_{t=-\infty}^{+\infty} \varphi_\varepsilon(t)\, dt = 1$$

   As the Fourier transform $\mathcal{F}\left\{\varphi_\varepsilon(t)\right\}(\omega) \to 1$ for $\varepsilon \to 0$, then $\varphi_\varepsilon(t) \to \delta(t)$.

22.2.4 Plancherel's theorem

…assumptions…**todo**

$$\int_{f=-\infty}^{+\infty} |G(f)|^2\, df = \int_{t=-\infty}^{+\infty} |g(t)|^2\, dt \tag{22.4}$$

and

$$\int_{f=-\infty}^{+\infty} A^*(f)\, G(f)\, df = \int_{t=-\infty}^{+\infty} a^*(t)\, g(t)\, dt\,. \tag{22.5}$$

Proof of Plancherel's thm for the magnitude

$$\begin{aligned}
\int_{f=-\infty}^{+\infty} |G(f)|^2\, df &= \int_{f=-\infty}^{+\infty} G(f)^* G(f)\, df = \\
&= \int_{f=-\infty}^{+\infty} \left(\int_{t_1=-\infty}^{+\infty} g(t_1) e^{-i2\pi f t_1} dt_1 \right)^* \left(\int_{t_2=-\infty}^{+\infty} g(t_2) e^{-i2\pi f t_2} dt_2 \right) df = \\
&= \int_{t_1,t_2=-\infty}^{+\infty} g^*(t_1)\, g(t_2) \int_{f=-\infty}^{+\infty} e^{i2\pi f(t_1-t_2)}\, df\, dt_1\, dt_2 = \qquad (1) \\
&= \int_{t_1,t_2=-\infty}^{+\infty} g^*(t_1)\, g(t_2)\, \delta(t_1-t_2)\, dt_1\, dt_2 = \qquad (2) \\
&= \int_{t_1=-\infty}^{+\infty} g^*(t_1)\, g(t_1)\, dt_1 = \qquad (3) \\
&= \int_{t_1=-\infty}^{+\infty} |g(t_1)|^2\, dt_1 \,.
\end{aligned}$$

having used (1) the approximation (18.3) of Dirac's delta, and (2) property (18.2) of Dirac's delta, and (3) the expression of the absolute value of complex functions $g^*(t_1)g(t_1) = |g(t_1)|^2$.

Proof of Plancherel's thm for the product of functions

$$\begin{aligned}
\int_{f=-\infty}^{+\infty} A^*(f)\, G(f)\, df &= \int_{f=-\infty}^{+\infty} G(f)^* G(f)\, df = \\
&= \int_{f=-\infty}^{+\infty} \left(\int_{t_1=-\infty}^{+\infty} a(t_1) e^{-i2\pi f t_1} dt_1 \right)^* \left(\int_{t_2=-\infty}^{+\infty} g(t_2) e^{-i2\pi f t_2} dt_2 \right) df = \\
&= \int_{t_1,t_2=-\infty}^{+\infty} a^*(t_1)\, g(t_2) \int_{f=-\infty}^{+\infty} e^{i2\pi f(t_1-t_2)}\, df\, dt_1\, dt_2 = \qquad (1) \\
&= \int_{t_1,t_2=-\infty}^{+\infty} a^*(t_1)\, g(t_2)\, \delta(t_1-t_2)\, dt_1\, dt_2 = \qquad (2) \\
&= \int_{t_1=-\infty}^{+\infty} a^*(t_1)\, g(t_1)\, dt_1 \,.
\end{aligned}$$

having used (1) the approximation (18.3) of Dirac's delta, and (2) property (18.2) of Dirac's delta.

22.2.5 Uncertainty relation

An uncertainty relation holds linking standard deviations of a probability density function in time domain and a probability density function built with its Fourier transform. From this very same relation, Heisenberg uncertainty relation between position and momentum in Quantum Mechanics seamlessly follows.

Given a function $g(t)$ whose square of the absolute value is normalized to one, and thus it can be used as a probability density function in time domain,

$$\int_{t=-\infty}^{+\infty} |g(t)|^2\, dt = \int_{t=-\infty}^{+\infty} g^*(t)\, g(t)\, dt = 1 \,.$$

for *Plancherel's theorem*, the square of the magnitude of Fourier transform $G(f)$ is unitary as well,

$$\int_{f=-\infty}^{+\infty} |G(f)|^2\, df = \int_{f=-\infty}^{+\infty} G^*(f)\, G(f)\, df = 1 \,,$$

and thus it can be interpreted as a probability density function in frequency domain. The following uncertainty relation holds

$$\sigma_{t,g}^2 \sigma_{f,G}^2 \geq \left(\frac{1}{2\pi}\frac{1}{2}\right)^2 ,$$

or in terms of pulsation $\omega = 2\pi\, f$,

$$\sigma_{t,g}^2 \sigma_{\omega,G}^2 \geq \left(\frac{1}{2}\right)^2 ,$$

Heisenberg uncertainty relation in quantum mechanics

Space and momentum representation of the state function Ψ are related by the transformation,

$$\langle x|\Psi\rangle := \psi(x,t) = \int_{p=-\infty}^{+\infty} \psi_p(p,t)\, e^{i\frac{p}{\hbar}x}\, dp\ ,$$

as it's shown in the section Quantum Mechanics:From position to momentum representation. The wave number reads $k = \frac{p}{\hbar}$. Starting from the uncertainty relation between the space coordinate x and the wave number k,

$$\sigma_x \sigma_k \geq \frac{1}{2}\ ,$$

Heisenberg uncertainty principle for position and momentum (for the same Cartesian coordinates) reads

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}\ .$$

Proof of the uncertainty relation

Assuming zero average $\overline{t} = 0$, $\overline{f} = 0$ (see below for proof without this assumption)

$$\begin{aligned}
\sigma_{t,g}^2 \sigma_{f,G}^2 &= \int_{t=-\infty}^{+\infty} |t|^2\ g^*(t)\, g(t)\, dt \int_{f=-\infty}^{+\infty} |f|^2\ G^*(f)\, G(f)\, df = \\
&= \int_{t=-\infty}^{+\infty} |t\, g(t)|^2\ dt \int_{f=-\infty}^{+\infty} |f\, G(f)|^2\ df = \qquad (1)\\
&= \int_{t=-\infty}^{+\infty} |t\, g(t)|^2\ dt \int_{t=-\infty}^{+\infty} \left|-\frac{i}{2\pi}\, \dot{g}(t)\right|^2\ dt \geq \qquad (2)\\
&= \left|\int_{t=-\infty}^{+\infty} -t\, g^*(t) \frac{i}{2\pi}\, \dot{g}(t)\, dt\right|^2 \geq \qquad (3)\\
&= \left(\frac{1}{2\pi}\right)^2 \left(\frac{1}{2}\right)^2
\end{aligned}$$

having used in

(1)

$$\mathcal{F}\{\dot{g}(t)\}(f) = \int_{t=-\infty}^{+\infty} \dot{g}(t) e^{-i2\pi ft}\, dt = \cdots = i2\pi\, f\, G(f)\ ,$$

$$f\, G(f) = -i\, \frac{\mathcal{F}\{\dot{g}(t)\}}{2\pi}$$

and thus *Plancherel's theorem*

$$\int_{f=-\infty}^{+\infty} |f\, G(f)|^2\, df = \int_{f=-\infty}^{+\infty} \left| -\frac{i}{2\pi}\, \mathcal{F}\{\dot{g}(t)\} \right|^2 df =$$

$$= \int_{t=-\infty}^{+\infty} \left| -\frac{i}{2\pi}\, \dot{g}(t) \right|^2 dt$$

in (2) Cauchy-Schwartz inequality,

(3)

$$a := \int_{t=-\infty}^{+\infty} t g^*(t) \dot{g}(t)\, dt =$$

$$= \underbrace{[t\, g^*(t)\, g(t)]\, |_{-\infty}^{+\infty}}_{=0} - \int_{t=-\infty}^{+\infty} \frac{d}{dt}\, (t g^*(t))\, g(t)\, dt =$$

$$= -\int_{t=-\infty}^{+\infty} g^*(t)\, g(t)\, dt - \int_{t=-\infty}^{+\infty} t \dot{g}^*(t) g(t)\, dt =$$

$$= -1 - a^*\ .$$

$$-1 = a + a^* = 2\,\mathrm{re}\{a\}\ ,$$

and thus

$$|a|^2 \geq \mathrm{re}\{a\}^2 = \frac{1}{4}\ .$$

If $\overline{t} \neq 0$, or $\overline{f} \neq 0$,

$$\sigma_{t,g}^2 \sigma_{f,G}^2 = \int_{t=-\infty}^{+\infty} \left| t - \overline{t} \right|^2 g^*(t)\, g(t)\, dt \int_{f=-\infty}^{+\infty} \left| f - \overline{f} \right|^2 G^*(f)\, G(f)\, df =$$

$$= \int_{t=-\infty}^{+\infty} \left| (t - \overline{t})\, g(t) \right|^2 dt \int_{f=-\infty}^{+\infty} \left| (f - \overline{f})\, G(f) \right|^2 df = \qquad (1)$$

$$= \ldots$$

22.3 Relations between Fourier transforms

Warning: Some freestyle in changing order of summations and integrals, and use of generalized functions here…check it!

Different Fourier transforms

Different Fourier transforms exist, depending if the original function is:

- time discrete/time continuous
- periodic/non-periodic

namely,

- FS, Fourier series: time continuous, periodic function (or finite domain, with a periodic extension)
- FT, Fourier transform: time continuous, non-periodic function
- DTFT, discrete-time Fourier transform: time didscrete, infinite-length sequence
- DFT, discrete Fourier transform: time discrete, finite-length sequence (and then with a periodic extension)

Different forms of Fourier transforms exist, depending on the features of the original function and its domain. The bridge between continous and discrete domains is the ***Dirac' comb***, $\text{III}_{\Delta x}(x) := \sum_{m=-\infty}^{+\infty} \delta(x - m\Delta x)$, that can be used to **model sampling** of continous functions into sequences of discrete values, both in time and frequency domain.

Here

- the standard **Fourier transform** of integrable functions is recalled
- the Fourier transform of an infinite sum of T-shifted integrable function is evaluated, resulting in the sampled (and scaled) Fourier transform of the original function, with sampling frequency $\Delta\nu = \frac{1}{T}$. Here T plays the role of the sampling period. …**Fourier series**
- the Fourier transform of a Δt-sampled (and scaled) integrable function is evaluated, resulting in the infinite sum of $\overline{\nu}$-shifted Fourier transform of the original function, with $\nu_s = \frac{1}{\Delta t}$. Here Δt plays the role of the discrete time-step, inverse of the sampling frequency…**Discrete-time Fourier transform**
- putting the last two points together, the Fourier transform of a infinite sum of T-shifted Δt-sampled function is evaluated resulting in…**DFT**

22.3.1 Fourier transform of integrable functions

$$F(\nu) := \mathcal{F}\left\{f(t)\right\}(\nu) := \int_{t=-\infty}^{+\infty} f(t)e^{-i2\pi\nu t}\,dt\,,$$

…

this is the starndard Fourier transform.

22.3.2 Fourier series

Fourier transform of the sum of shifted integrable functions

The infinite sum of a shifted integrable function is defined as

$$\tilde{f}_T(t) = \sum_{n=-\infty}^{+\infty} f(t - nT)\,.$$

Its Fourier transform reads

$$\begin{aligned}\mathcal{F}\left\{\tilde{f}_T(t)\right\}(\nu) &= \int_{t=-\infty}^{+\infty} \tilde{f}_T(t) e^{-i2\pi\nu t}\, dt = \\ &= \sum_{n=-\infty}^{+\infty} \int_{t=-\infty}^{+\infty} f(t-nT) e^{-i2\pi\nu t}\, dt = \quad (1) \\ &= \sum_{n=-\infty}^{+\infty} F(\nu) e^{-i2\pi\nu nT} = \quad (2) \\ &= F(\nu) \sum_{n=-\infty}^{+\infty} e^{-i2\pi\nu nT} = \quad (3) \\ &= \Delta\nu\, F(\nu)\, \mathrm{III}_{\Delta\nu}(\nu)\ ,\end{aligned}$$

having used properties of Fourier transform of shifted function in (1), and the properties of Dirac's comb in (3), having defined the frequency resolution

$$\Delta\nu := \frac{1}{T}\ .$$

This Fourier transform is proportional to the Fourier transform of the original function, sampled in frequency with elementary frequency $\Delta\nu$.

Infinite shifted sum and Fourier series

$$\tilde{f}_T(t) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{T}t}\ ,$$

with $c_n = \frac{1}{T}\int_{t=0}^{T} \tilde{f}(t) e^{-in\frac{2\pi}{T}t}$.

22.3.3 Discrete-time Fourier transform

Fourier transform of an integrable function sampled with a Dirac comb

Fourier transform of the original function sampled with $\Delta t\, \mathrm{III}_{\Delta t}(t)$ reads

$$\begin{aligned}\mathcal{F}\left\{\Delta t\, f(t)\, \mathrm{III}_{\Delta t}(t)\right\} &= \Delta t \int_{t=-\infty}^{+\infty} f(t)\, \mathrm{III}_{\Delta t}(t) e^{-i2\pi\nu t}\, dt = \\ &\sim \Delta t \frac{1}{\Delta t} \int_{t=-\infty}^{+\infty} f(t) \sum_{n=-\infty}^{+\infty} e^{in\frac{2\pi}{\Delta t}t} e^{-i2\pi\nu t}\, dt = \\ &= \sum_{n=-\infty}^{+\infty} \int_{t=-\infty}^{+\infty} f(t)\, e^{-i2\pi(\nu - n\nu_s)t}\, dt = \quad (22.6) \\ &= \sum_{n=-\infty}^{+\infty} F\left(\nu - n\nu_s\right) = \mathrm{DTFT}(f(t); \Delta t)\ ,\end{aligned}$$

i.e. equals the periodic sum of the Fourier of the original function, with period

$$\nu_s := \frac{1}{\Delta t}\ .$$

From this last sentence and from the *symmetry properties of Fourier transform*, **Nyquist-Shannon sampling theorem** follows seamlessly.

Theorem 22.3.1 (Nyquist-Shannon sampling theorem)

In order to **avoid aliasing** the sampling frequency must be twice the maxiumum[1] frequency in the signal,

$$\nu_s \geq 2\nu_{max} \ .$$

Alternative form of DTFT

todo check alternative expressions if using the definition of train of impulses instead of the Fourier series of Dirac's comb.

$$\begin{aligned} &= \Delta t \int_{t=-\infty}^{+\infty} f(t) \sum_{k=-\infty}^{+\infty} \delta(t - k\Delta t)\, e^{-i2\pi\nu t}\, dt = \\ &= \Delta t \sum_{k=-\infty}^{+\infty} f(k\Delta t) e^{-i2\pi\nu k\Delta t} = \text{DTFT}\left(f(t); \Delta t\right)(\nu) \ . \end{aligned} \tag{22.7}$$

22.3.4 Discrete Fourier transform

The discrete Fourier transform is a Fourier transform of a finite sequence of N elements $\{f_n\}_{n=0:N-1}$, defined as

$$F_k := \sum_{n=0}^{N-1} f_n\, e^{-i2\pi\frac{k}{N}n} \ , \quad \text{for } k = 0 : N-1 \ ,$$

whose inverse transform reads

$$f_n := \frac{1}{N} \sum_{k=0}^{N-1} F_k\, e^{i2\pi\frac{k}{N}n} \ , \quad \text{for } n = 0 : N-1 \ ,$$

Proof of inverse DFT

$$\begin{aligned} \sum_{k=0}^{N-1} F_k e^{i2\pi\frac{kn}{N}} &= \sum_{k=0}^{N-1}\sum_{m=0}^{N-1} f_m e^{i2\pi\frac{k(n-m)}{N}} = \\ &= \sum_{m=0}^{N-1} f_m \sum_{k=0}^{N-1} e^{i2\pi\frac{k(n-m)}{N}} = \\ &= \sum_{m=0}^{N-1} f_m \begin{Bmatrix} 0 & , & m \neq n \\ N & , & m = n \end{Bmatrix} = \\ &= \sum_{m=0}^{N-1} f_m\, N\, \delta_{mn} = N\, f_n \ . \end{aligned}$$

[1] Usually there's no such a frequency above which the signal is exactly zero, but usually there's a frequency above which the spectrum of the signal is approximately zero, i.e. below a threshold where it can be treated as zero, and introduce no aliasing.

Fourier transform of the sum of a shifted integral function sampled with a Dirac comb

Fourier transform of the sum of a shifted integral function sampled with a Dirac comb

$$\Delta t\, \tilde{f}(t)\, \mathrm{III}_{\Delta t}(t) = \Delta t \sum_{n=-\infty}^{+\infty} f(t - nT)\, \mathrm{III}_{\Delta t}(t)$$

reads

$$\mathcal{F}\left\{\Delta t\, \tilde{f}(t)\, \mathrm{III}_{\Delta t}(t)\right\}(\nu) = \Delta t \int_{t=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} f(t - nT) \sum_{k=-\infty}^{+\infty} \delta(t - k\Delta t)\, e^{-i2\pi\nu t}\, dt =$$
$$= \Delta t \sum_{n=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} f(k\Delta t - nT)\, e^{-i2\pi\nu k\Delta t} =$$

and defining $k\Delta\tau_n := k\Delta t - nT$,

$$= \Delta t \sum_{n=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} f(k\Delta\tau_n) e^{-i2\pi\nu k\Delta\tau_n} e^{-i2\pi\nu nT} =$$
$$= \underbrace{\Delta t \sum_{k=-\infty}^{+\infty} f(k\Delta\tau_n) e^{-i2\pi\nu k\Delta\tau_n}}_{=\mathrm{DTFT}(f(t),\Delta t)(\nu)} \underbrace{\sum_{n=-\infty}^{+\infty} e^{-i2\pi\nu nT}}_{=\Delta\nu\, \mathrm{III}_{\Delta\nu}(\nu)} =$$
$$= \mathrm{DTFT}(f(t), \Delta t)(\nu)\, \Delta\nu\, \mathrm{III}_{\Delta\nu}(\nu)\ .$$

Using the relation between sampling time Δt and sampling period $T = N\Delta t$, $\Delta\nu = \frac{1}{T}$, and thus

$$\Delta t\, \Delta\nu = \Delta t\, \frac{1}{T} = \frac{1}{N}\ ,$$

it follows

$$\mathcal{F}\left\{\Delta t\, \tilde{f}(t)\, \mathrm{III}_{\Delta t}(t)\right\}(\nu) = \mathrm{DTFT}(f(t), \Delta t)(\nu)\, \Delta\nu\, \mathrm{III}_{\Delta\nu}(\nu) =$$
$$= \Delta t\Delta\nu \sum_{k=-\infty}^{+\infty} f(k\Delta\tau_n) e^{-i2\pi\nu k\Delta\tau_n} \mathrm{III}_{\Delta\nu}(\nu) =$$
$$= \frac{1}{N} \sum_{k=-\infty}^{+\infty} f(k\Delta\tau_n) e^{-i2\pi\nu k\Delta\tau_n}\, \mathrm{III}_{\Delta\nu}(\nu)\ .$$

The function $\tilde{f}(t) = \sum_{n=-\infty}^{+\infty} f(t - nT)$ is T-periodic by definition. Thus, it can be written as its Fourier series

Relation between DFT and sampling

22.3.5 Useful properties

Dirac's comb $\mathrm{III}_{\Delta t}(t)$

Dirac comb $\mathrm{III}_{\Delta t}(t)$ is defined as a train of Dirac's delta

$$\mathrm{III}_{\Delta t}(t) = \sum_{m=-\infty}^{+\infty} \delta(t - m\Delta t)\ .$$

Dirac comb $\text{III}_{\Delta t}(t) isa \Delta t$-periodic (generalized) function. Thus (…), it can be represented as its Fourier series, here in the exponential form (22.1), as

$$\text{III}_{\Delta t}(t) \sim \sum_{n=-\infty}^{+\infty} c_n e^{in\frac{2\pi}{\Delta t}t} \, .$$

with coefficients given by (22.2)

$$\begin{aligned} c_n &= \frac{1}{\Delta t} \int_{t=-\frac{\Delta t}{2}}^{\frac{\Delta t}{2}} \text{III}_{\Delta t}(t)\, e^{-in\frac{2\pi}{\Delta t}t} = \\ &= \frac{1}{\Delta t} \int_{t=-\frac{\Delta t}{2}}^{\frac{\Delta t}{2}} \sum_{m=-\infty}^{+\infty} \delta(t - m\Delta t)\, e^{-in\frac{2\pi}{\Delta t}t} = \\ &= \frac{1}{\Delta t} \int_{t=-\frac{\Delta t}{2}}^{\frac{\Delta t}{2}} \delta(t)\, e^{-in\frac{2\pi}{\Delta t}t} = \frac{1}{\Delta t} \, . \end{aligned}$$

The Fourier series of Dirac comb $\text{III}_{\Delta t}(t)$ reads

$$\text{III}_{\Delta t}(t) = \sum_{m=-\infty}^{+\infty} \delta(t - m\Delta t) \sim \frac{1}{\Delta t} \sum_{n=-\infty}^{+\infty} e^{in\frac{2\pi}{\Delta t}t} \, ,$$

i.e. as an infinite sum of uniform-amplitude harmonics with discrete frequencies ν_n, that are integer multiple of a fundamental freqeuncy $\nu_s = \frac{1}{\Delta t}$, being Δt the sampling time (time-step of the discrete sequence), inverse of sampling frequency.

Symmetry of Fourier transform

22.4 Practice on Fourier analysis on finite-time discrete-time signal

In this section, some application of Fourier analysis of discrete-time signals with finite-time window observation. Outside this observation window, the signal can (or is, by some trasnformation) considered as periodic. Discrete-time signals are common after sampling.

Contents.

- Fourier transforms, and DFT
- Spectral Leackage
- Spectral Aliasing
- Realizations of signals with given PSD (or average ESD? for finite domains)

22.4.1 Fourier transforms

Discrete Fourier Transform (DFT)

The function `numpy.fft.fft()` computes the one-dimensional, n-point discrete Fourier Transform, with the **FFT algorithm**.

Discrete Fourier transform of a sequence of N complex numbers $\mathbf{x}_N = \{x_0, x_1, \dots, x_{n-1}\}$ is defined as

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi \frac{k}{N} n} ,$$

whose inverse transform reads

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{i2\pi \frac{n}{N} k} .$$

The DFT can be thought as … **todo**: link to Fourier transform of Dirac's comb sampled infinite sum of shifted function.

22.4.2 Leakage

Spectral leakage occurs in discrete-time signals when

- the period of the harmonic contents is not an integer multiple of the sampling time t_s
- the sampling time is a multiple of the period of the harmonics,

so that the sampled signal is not periodic on the finite-time interval.

Finite sampling time is equivalent to the multiplication of the signal with a box window. Fourier transform of this signal is equal to the convolution of the Fourier transform of the original signal and the Fourier transform of the box window.

```
%reset -f

import numpy as np
import matplotlib.pyplot as plt


#> Reference signal as a sum of harmonics
f1, f2 = 2., 0.5
phi1, phi2 = .0, .0
A1, A2 = 1., 2.

T1, T2 = 1/f1, 1/f2

x_fun = lambda t: A1*np.sin(2*np.pi*f1*t + phi1) + A2*np.sin(2*np.pi*f2*t + phi2)

print(f"Frequencies: {f1}, {f2}")
print(f"Periods    : {T1}, {T2}")
```

```
Frequencies: 2.0, 0.5
Periods    : 0.5, 2.0
```

```
samplings = [
    { 'fs': 9.2739138, 'tend': 5.  },
    { 'fs': 10., 'tend': 10. },
]

#> Add label to sampling dict, for plots
for s in samplings:
  s['label'] = f"fs={s['fs']}, tend={s['tend']}"
```

```
#> Check different samplings
xx, ff, tt, oo = [], [], [], []

for sampling in samplings:

  fs = sampling['fs']
  ts = 1 / fs
  tend = sampling['tend']

  t = np.arange(0, tend, ts)
  nt = len(t)

  #> Sampled signal
  x = x_fun(t)

  #> Fourier transform of the signal
  f = np.fft.fft(x) / nt * 2         # * ts for scaling, to get the actual amplitudes
  dom = 1. / tend
  om = np.arange(nt) * dom

  xx += [ x ]
  ff += [ f ]
  tt += [ t ]
  oo += [ om ]

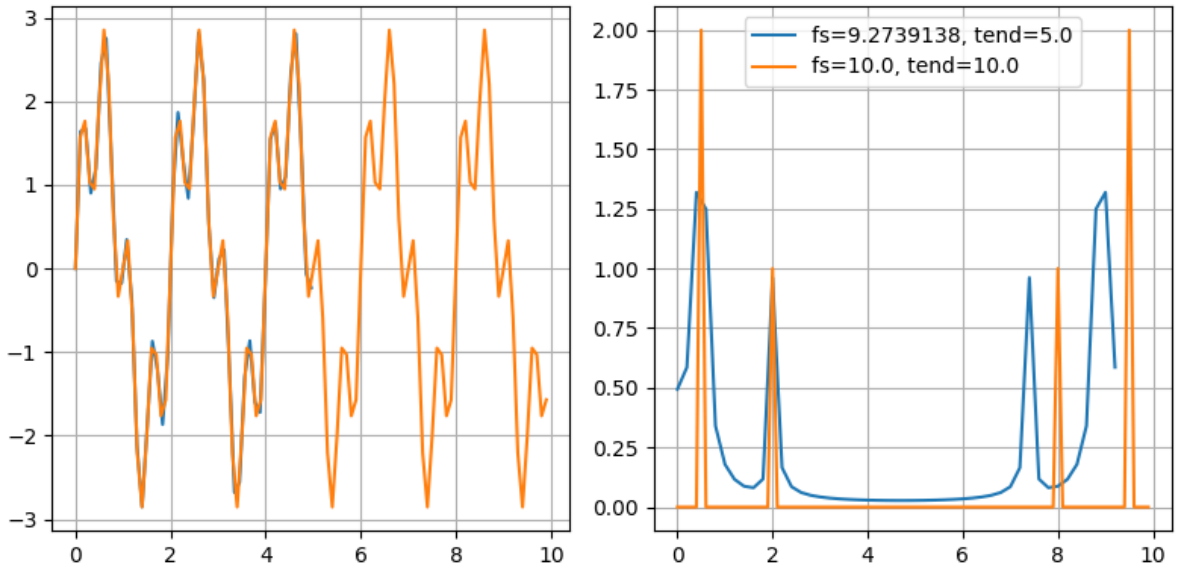
# print(om)
```

```
#> Raw Fourier transform
fig, ax = plt.subplots(1,2, figsize=(8,4))

for i, x in enumerate(xx):
  ax[0].plot(tt[i], xx[i])
  ax[1].plot(oo[i], np.abs(ff[i]), label=samplings[i]['label'])

ax[0].grid()
ax[1].grid()# ;#   ax[1].set_xlabel(f"$\omega$")
ax[1].legend()

fig.tight_layout()
```



22.4.3 Aliasing

```
%reset -f

import numpy as np
import matplotlib.pyplot as plt


#> Reference signal as a sum of harmonics
f1, f2 = 2.5, 0.5
phi1, phi2 = np.random.rand(), np.random.rand()
A1, A2 = 1., 2.

T1, T2 = 1/f1, 1/f2

x_fun = lambda t: A1*np.sin(2*np.pi*f1*t + phi1) + A2*np.sin(2*np.pi*f2*t + phi2)

print(f"Frequencies: {f1}, {f2}")
print(f"Periods    : {T1}, {T2}")
```

```
Frequencies: 2.5, 0.5
Periods    : 0.4, 2.0
```

```
samplings = [
    { 'fs': 10. , 'tend': 10. },
    { 'fs':  7. , 'tend': 10. },
    { 'fs':  4. , 'tend': 10. },
    { 'fs':  1.5, 'tend': 10. },
]

#> Add label to sampling dict, for plots
for s in samplings:
  s['label'] = f"fs={s['fs']}, tend={s['tend']}"
```

```
#> Check different samplings
xx, ff, tt, oo = [], [], [], []

max_om = 0. # for plot

for sampling in samplings:

  fs = sampling['fs']
  ts = 1 / fs
  tend = sampling['tend']

  t = np.arange(0, tend, ts)
  nt = len(t)

  #> Sampled signal
  x = x_fun(t)

  #> Fourier transform of the signal
  f = np.fft.fft(x) / nt * 2         # * ts for scaling, to get the actual amplitudes
  dom = 1. / tend
  om = np.arange(nt) * dom

  xx += [ x ]
  ff += [ f ]
  tt += [ t ]
  oo += [ om ]

  max_om = np.max([ max_om, nt * dom ])
```

```
#> Raw Fourier transform
n_samplings = len(samplings)
fig, ax = plt.subplots(n_samplings,2, figsize=(8,2*n_samplings))

# max_om = np.max(np.array(oo))
dt_high_res = .001

for i, x in enumerate(xx):

  t_high_res = np.arange(0, tt[i][-1], dt_high_res)
  x_high_res = x_fun(t_high_res)

  ax[i,0].plot(tt[i], xx[i], ':o')
  ax[i,0].plot(t_high_res, x_high_res, '--', color='black')
  ax[i,1].plot(oo[i], np.abs(ff[i]), label=samplings[i]['label'], marker='o')

  ax[i,0].grid()
  ax[i,1].grid()# ;#   ax[1].set_xlabel(f"$\omega$")
  ax[i,1].set_xlim([0, max_om])
  ax[i,0].set_ylabel(r"$f(t)$")
  ax[i,1].set_ylabel(r"$|F(\nu)|$")

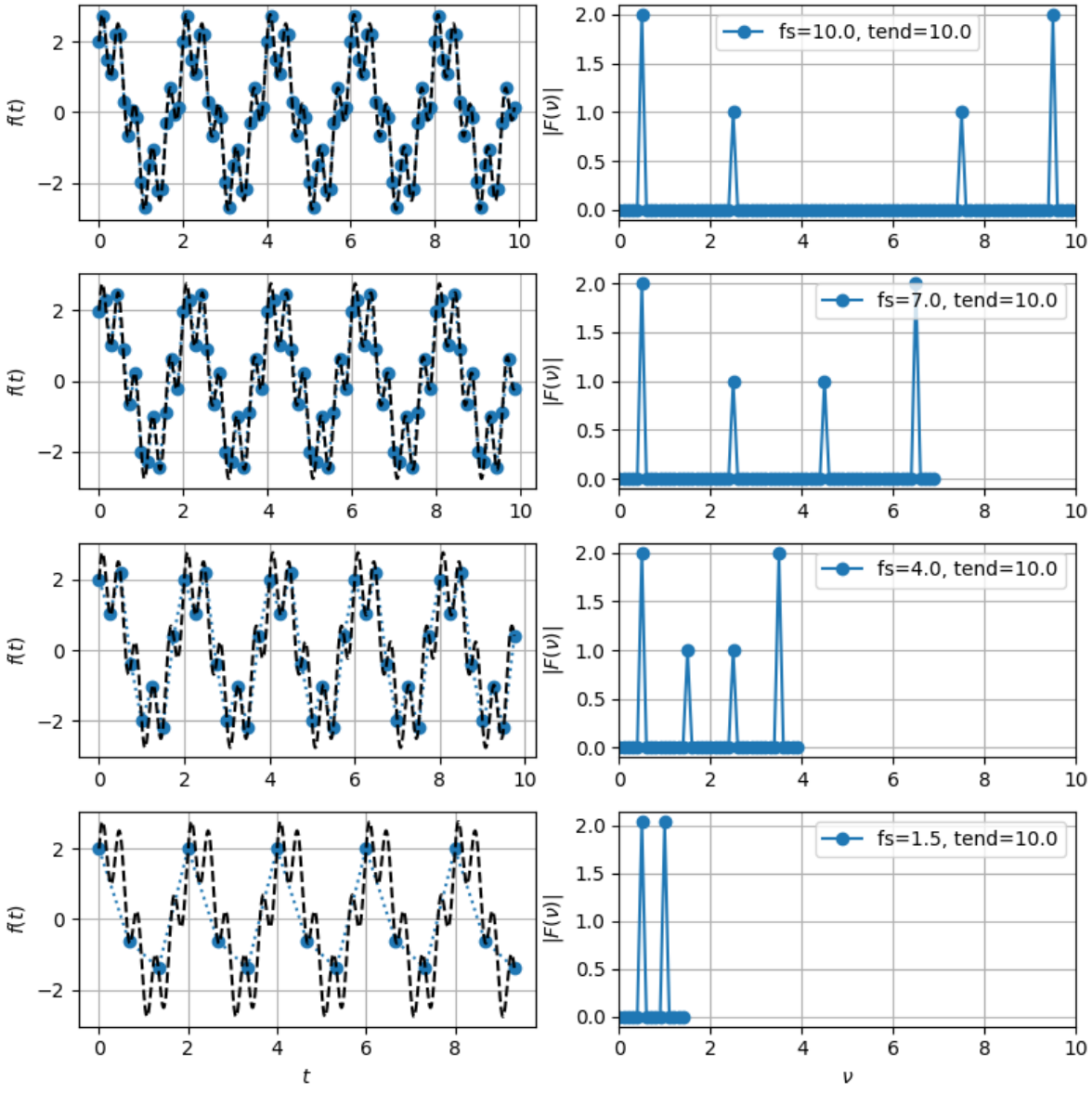
  ax[i,1].legend()


ax[n_samplings-1,0].set_xlabel(r"$t$")
ax[n_samplings-1,1].set_xlabel(r"$\nu$")
```

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```
fig.tight_layout()
```



22.4.4 Realizations of PSD

Stationary stochastic signals. For stationary stochastic signals, $PSD(\nu)$ is defined as the Fourier transform of it auto-correlation, i.e.

$$\Phi_{xx}(\nu) := \mathcal{F}\{R_{xx}(\tau)\}(\nu) = \int_{t=-\infty}^{+\infty} R_{xx}(\tau)\, e^{-i\nu\tau}\, d\tau \ ,$$

with

$$R_{xx}(\tau) := \mathbb{E}\left[x(t)x(t-\tau)\right] \ .$$

$$\begin{aligned}
\Phi_{xx}(\nu) &= \int_{\tau=-\infty}^{+\infty} R_{xx}(\tau)e^{-i\nu\tau}d\tau = \\
&= \int_{\tau=-\infty}^{+\infty} \mathbb{E}[x(t)x(t+\tau)]e^{-i\nu\tau}d\tau = \\
&= \int_{\tau=-\infty}^{+\infty} \lim_{T\to+\infty} \int_{t=-T}^{T} x(t)x(t+\tau)\, dt\, e^{-i\nu\tau}d\tau = \\
&= \int_{\tau=-\infty}^{+\infty} \lim_{T\to+\infty} \int_{t=-T}^{T} x(t)x(z)e^{-i\nu z}e^{i\nu t}dtdz = \\
&= \left(\int_{\tau=-\infty}^{+\infty} x(z)e^{-i\nu z}dz\right)\left(\lim_{T\to+\infty}\int_{t=-T}^{T} x(t)e^{i\nu t}dt\right) = \\
&= X(\nu)X^*(\nu) = \\
&= \left|X(\nu)\right|^2 \ .
\end{aligned}$$

todo Is it possible to switch the expected value and integral operator?

Periodic signals. Let $f(t)$ a T-periodic signal. Its spectrum is discrete with **freqeuncy resolution** $\Delta\nu = \frac{1}{T}$. If $f(t)$ is discrete with sampling time Δt, its spectrum is ν_s-periodic with **sampling frequency** $\nu_s = \frac{1}{\Delta t}$.

…

Here, given $PSD_x(\nu_k) := |X(\nu_k)|^2$, for all the avaliable frequencies, i.e.

$$\nu_k = k\nu_1 \ , \quad k = 0 : N-1$$

with $\nu_1 = \frac{1}{T}$ frequency resolution, find a realization of the process in time domain.

Remark. In order to have a valid $PSD_x(\nu)$, this function needs to satisfy some conditions, as the elements of the DFT $X(\nu_k)$ needs to satisfy symmetry conditions to be the **transform of a real signal**.

- $X(0)$ is real
- symmetry for $k \neq 0$
  - If N is even,

$$\begin{aligned} X[k] &= X^*[N-k-1] \\ X[N/2] &\in \mathbb{R} \end{aligned}$$

  - If N is odd,

$$X[k] = X^*[N-k]$$

Exact realization

Frequency discretization of spectrum implies time discretization, as spectral resolution is the inverse of the sampling period, and maximum frequency (of the mirrored spectrum) is the inverse of the time-step.

```
%reset -f

import numpy as np
import matplotlib.pyplot as plt
```

```
#> Time vector
dt, nt = .01, 1000

tv = np.arange(nt) * dt
tend = dt * nt

#> Frequency vector
df = 1. / tend
fv = np.arange(nt) * df
```

```
def psd_to_time_realization(psd_fun, fv, **params):
    """
    psd_fun(fv): function returning psd
    fv: vector of frequencies
    """
    N = len(fv)

    #> magnitude u(f;T) = |A(f;T)|^2 = A*(f;T) A(f;T)
    magnitude = np.sqrt(psd_fun(fv, **params))
    #> add random phase
    phase = np.exp(1j * np.random.uniform(0, 2*np.pi, N))
    # phase = np.zeros(N)
    spectrum = magnitude[1:] * phase[1:]
    #> Built a vector of transformed signal with the required symmetries to get a␣
↪real ifft()
    xf = np.concatenate([ [magnitude[0]], spectrum, [magnitude[-1]], np.
↪conj(spectrum[::-1]) ])
    xt = np.fft.ifft(xf)

    fmax = 2 * N * ( fv[1] - fv[0] )
    dt = 1. / fmax
    tv = np.arange(2*N)*dt

    return tv, xt
```

```
#> Fourier transform of the signal
"""
#> White noise approximation. Discretization sets the frequency resolution and the
# maximum frequency of the signal. With constant amplidute, this is a fmax-windowed
# and fs-sampled spectrum of a white noise.
tf_fun = lambda f: np.ones(np.shape(f))
"""
#> Low pass filter
tf_fun = lambda f: 1. / ( 1. + 1j * f )
```

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```
"""
#> ~ Resonant system
# It has a prevailing harmonic content close to the "resonance", producing an
# "approximately harmonic" signal,
f_n, xi = 3., 0.01
tf_fun = lambda f: 1. / ( 1. + 2 * xi / f_n * 1j * f + ( (1j * f)/f_n )**2 )
"""
"""
#> ~ Resonant system + derivator
# It has a prevailing harmonic content close to the "resonance", producing an
# "approximately harmonic" signal,
f_n, xi = 3., 0.01
tf_fun = lambda f: 1j * f / ( 1. + 2 * xi / f_n * 1j * f + ( (1j * f)/f_n )**2 )
"""

#> PSD of the signal
psd_fun = lambda f: np.abs(tf_fun(f))**2

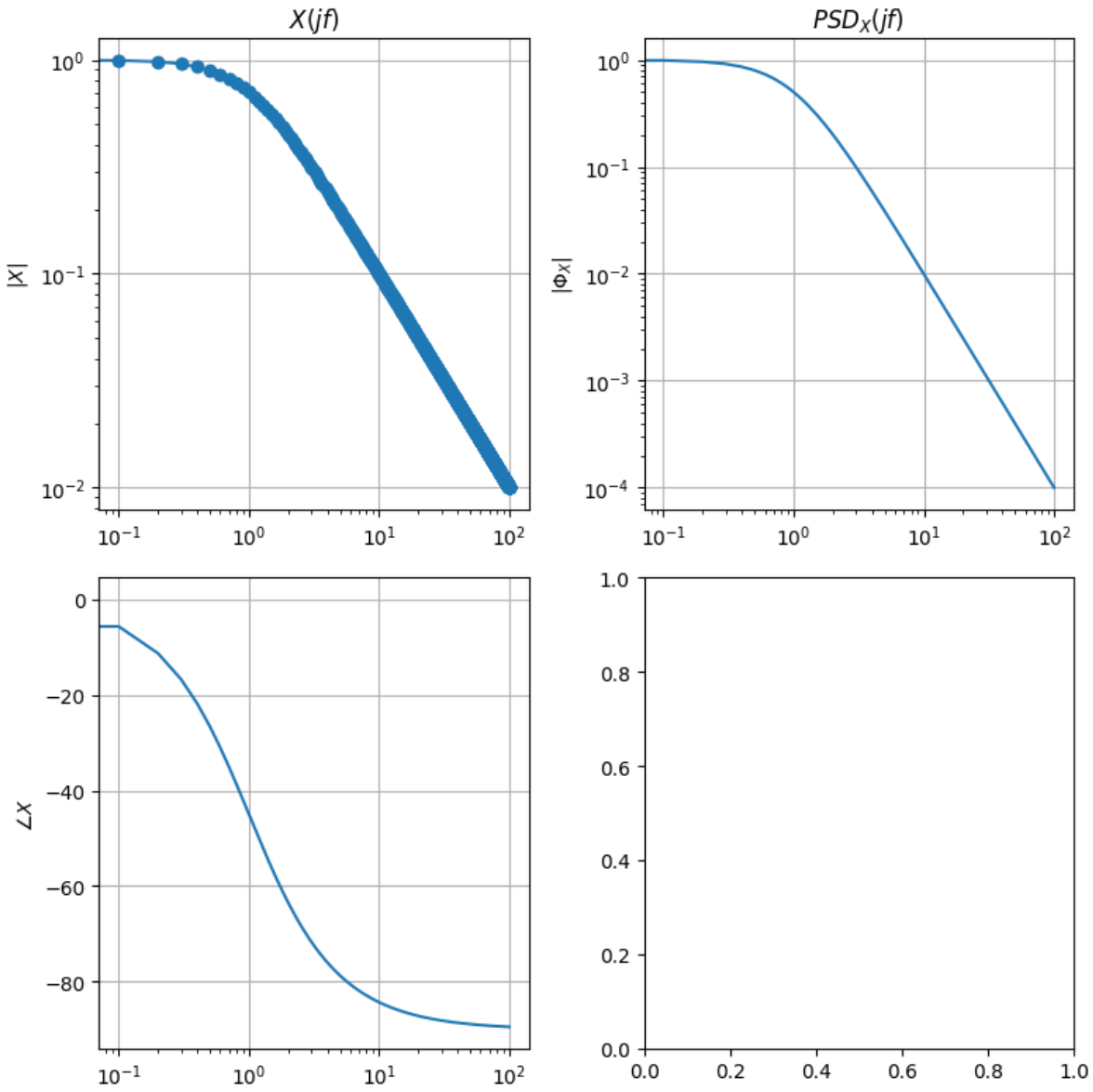
#> Plots of Fourier transform of the signal, and its PSD
tf_v = tf_fun(fv)
psd_v = tf_fun(fv) * np.conj(tf_fun(fv))

fig, ax = plt.subplots(2, 2, figsize=(8, 8))

ax[0,0].plot(fv, np.abs(tf_v), '-o')
ax[0,0].set_title(r"$X(j f)$")
ax[0,0].set_ylabel(r"$|X|$")
ax[0,0].set_xscale("log");  ax[0,0].set_yscale("log");  ax[0,0].grid()
ax[1,0].plot(fv, np.angle(tf_v) * 180/np.pi)
ax[1,0].set_ylabel(r"$\angle X$")
ax[1,0].set_xscale("log");  ax[1,0].grid()

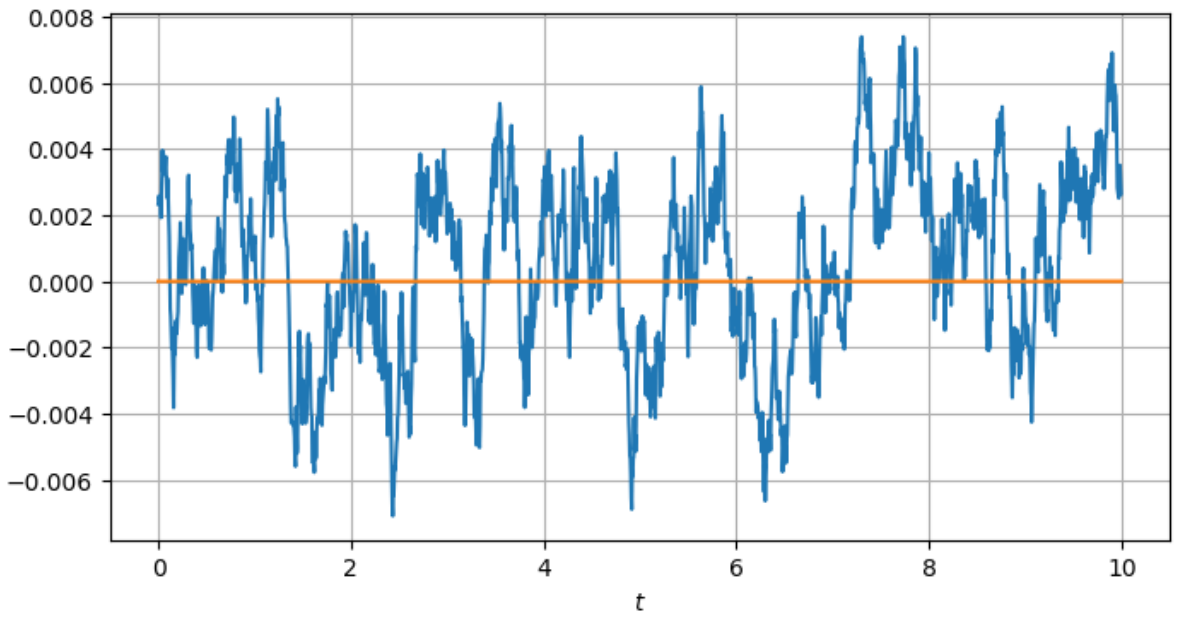
ax[0,1].plot(fv, np.abs(psd_v))
ax[0,1].set_title(r"$PSD_X(j f)$")
ax[0,1].set_ylabel(r"$|\Phi_X|$")
ax[0,1].set_xscale("log");  ax[0,1].set_yscale("log");  ax[0,1].grid()

fig.tight_layout()
```




```
tv_real, xt_real = psd_to_time_realization(psd_fun, fv)
```

```
fig, ax = plt.subplots(1, 1, figsize=(8,4))
ax.plot(tv_real, np.real(xt_real))
ax.plot(tv_real, np.imag(xt_real))
ax.set_xlabel(r"$t$")
ax.grid()
```




```
#> Check fft of the realized signal
xf_check = np.fft.fft(xt_real)

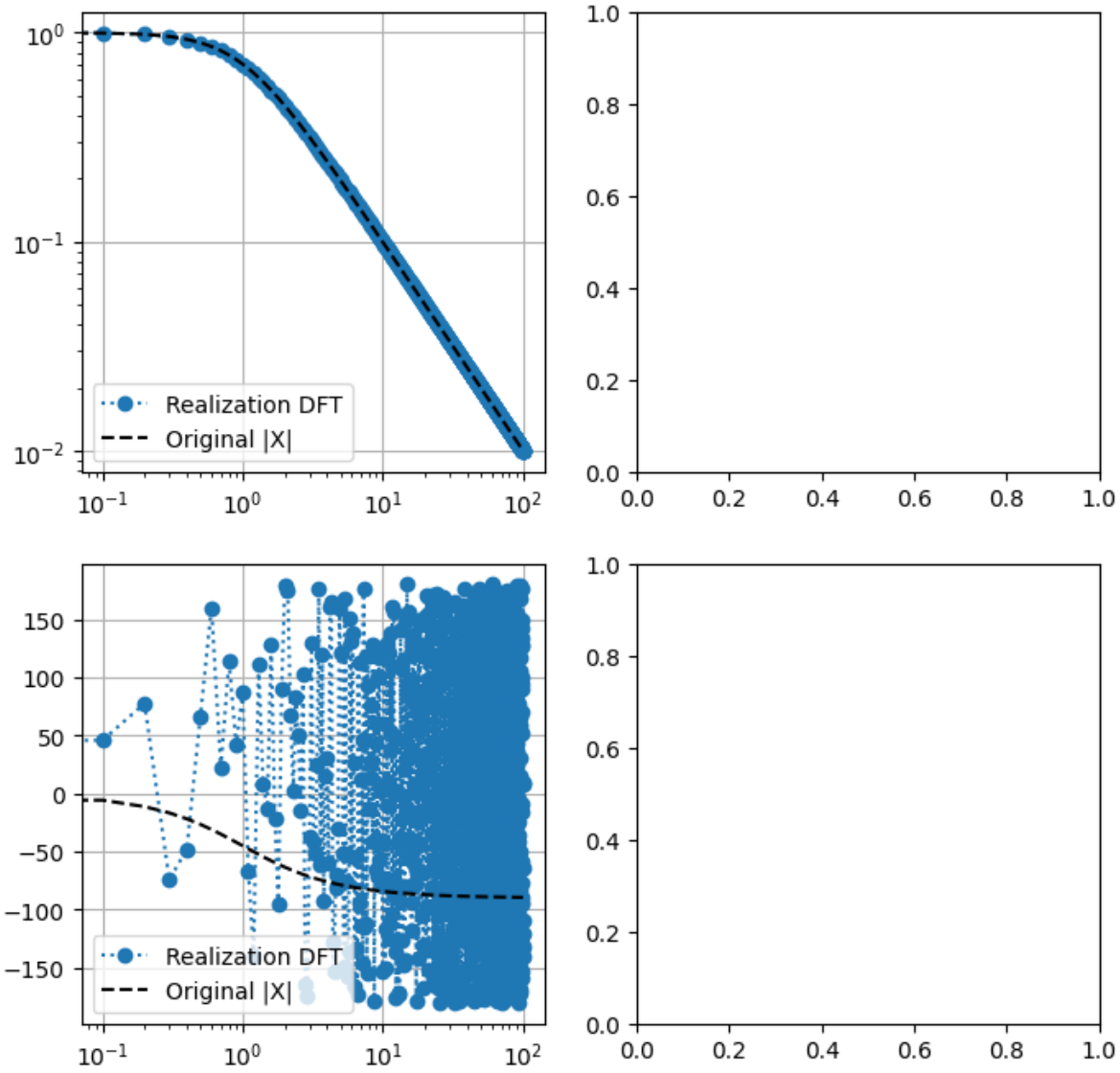
dt_real = tv_real[1] - tv_real[0]
tend_real = dt_real * len(tv_real)
fmax_real = 1 / dt_real
df_real = 1 / tend_real
fv_check = np.arange(len(tv_real)) * df_real

n_plot = len(tv_real) // 2

fig, ax = plt.subplots(2, 2, figsize=(8,8))
ax[0,0].plot(fv_check[:n_plot], np.abs(xf_check[:n_plot]), ':o', label="Realization␣
↪DFT")
ax[0,0].plot(fv, np.abs(tf_v), '--', color="black", label="Original |X|")
ax[0,0].set_xscale("log");  ax[0,0].set_yscale("log");  ax[0,0].grid(), ax[0,0].
↪legend()


ax[1,0].plot(fv_check[:n_plot], np.angle(xf_check[:n_plot]) * 180/np.pi, ':o', label=
↪"Realization DFT")
ax[1,0].plot(fv, np.angle(tf_v) * 180/np.pi, '--', color="black", label="Original |X|
↪")
ax[1,0].set_xscale("log");  ax[1,0].grid(), ax[1,0].legend()
```

```
(None, <matplotlib.legend.Legend at 0x789bf5127a40>)
```



While magnitude is exactly represented in the realization of the PSD, it contains no information about the phase of the Fourier transform of the original signal, as $\Phi_X(f) = |X(f)|^2$.

22.4.5 Shape filter

Time-continuous domain

A shape filter usually takes a white noise w as an input, to produce a

$$u(s) = G(s)w(s)$$

Time-discrete domain

22.4.6 Statistics of realizations

Given a PSD defined for $\omega \in [0, \omega_{max}]$, a realization has amplitude (equal for all the realizations)

$$|X_k(\omega)| = PSD_X^{1/2}(\omega)$$

and random phases ϕ_{kl} sampled for each $\omega_l \in [0, \omega_{max}]$ from a uniform probability distribution in $[-\pi, \pi]$, with probability density function

$$p(\phi) = \frac{1}{2\pi} , \quad \text{for } \phi \in [-\pi, \pi]$$

The kth realization at frequency ω_l has (discrete) spectrum

$$X_k(\omega_l) = \sqrt{PSD(\omega_l)} e^{i\phi_{kl}} ,$$

being ϕ_{kl} the arbitrary phase (no effect on the PSD) of the lth harmonics of the kth realization.

Single realization. Spectrum and PSD of a realization

$$\begin{aligned} X_k(\omega) &= \text{Re}\{X_k(\omega)\} + i\text{Im}\{X_k(\omega)\} = \\ &= |X_k(\omega)| \, e^{i\angle X_k(\omega)} = |X_k(\omega)| \cos\phi_{X_k}(\omega) + i|X_k(\omega)| \sin\phi_{X_k}(\omega) \\ PSD_{X_k}(\omega) &= |X_k(\omega)|^2 = X_k^*(\omega) X_k(\omega) = \left(\text{Re}\{X_k(\omega)\}\right)^2 + \left(\text{Im}\{X_k(\omega)\}\right)^2 = PSD_X(\omega) , \end{aligned}$$

i.e. while every individual realization may have different values of real and imaginary parts (due to arbitrary phase), they all have the same PSD.

Statistics of realizations.

- magnitude, deterministic not random

$$|X(\omega)| = \sqrt{PSD(\omega)}$$

- phase, uniform distribution for $\phi \in [-\pi, \pi]$
  - average value, variance (**does it have any meaning?**):

$$\begin{aligned} \mathbb{E}[\phi] &= \int_{-\pi}^{\pi} \phi p(\phi) \, d\phi = 0 \\ \mathbb{E}[\phi^2] &= 2 \int_0^{\pi} p(\phi) \phi^2 \, d\phi = \frac{\pi^2}{3} \end{aligned}$$

- real part and imaginary part

$$\begin{aligned} \mathbb{E}[\text{Re}\{X_k(\omega)\}] &= |X(\omega)| \int_{\phi=-\pi}^{\pi} p(\phi) \, \cos\phi \, d\phi = 0 \\ \mathbb{E}\left[\left(\text{Re}\{X_k(\omega)\}\right)^2\right] &= |X(\omega)|^2 \int_{-\pi}^{\pi} p(\phi) \cos^2\phi \, d\phi = \frac{1}{2}|X(\omega)|^2 \\ \mathbb{E}[\text{Im}\{X_k(\omega)\}] &= 0 \\ \mathbb{E}\left[\left(\text{Im}\{X_k(\omega)\}\right)^2\right] &= \frac{1}{2}|X(\omega)|^2 \end{aligned}$$

s.t.

\begin{aligned} \mathbb{E}[X_k(\omega)] & = 0 \ \mathbb{E}[|X_k(\omega)|^2] & = \mathbb{E}\left[\left(\text{Re}{ X_k(\omega) } \right)^2 + \left(\text{Im}{ X_k(\omega) } \right)^2 \right] = |X(\omega)|^2 = PSD(\omega) \ , \end{aligned}

as expected, as all the realizations come from the same PSD.

Sample average of N i.i.d. realizations,

$$\overline{X}(\omega) = \frac{1}{N} \sum_{n=1}^{N} X_n(\omega)$$

- average

$$\mathbb{E}\left[\overline{X}(\omega)\right] = 0$$

- PSD, "variance"

$$\begin{aligned} \mathbb{E}\left[\overline{X}^*(\omega)\overline{X}(\omega)\right] = \mathbb{E}\left[\frac{1}{N}\sum_{n=1}^{N} {X_n}^*(\omega)\,\frac{1}{N}\sum_{m=1}^{N} X_m(\omega)\right] = \quad \text{(i.i.d.)} \\ = \mathbb{E}\left[\frac{1}{N^2}\sum_{n=1}^{N} |X_n|^2(\omega)\right] = \\ = \frac{1}{N}\mathbb{E}\left[|X_k(\omega)|^2\right] \end{aligned}$$

…

- Sum of (zero-average) independent spectra, $X_1(\omega)$, $X_2(\omega)$,…(it's not needed that they're identically distributed),

$$S(\omega) := \sum_{n=1}^{N} X_n(\omega)\ .$$

The average of the PSD of the sum of the spectra equals the sum of the average of the PSD of the spectra,

$$\begin{aligned} \mathbb{E}\left[\text{PSD}_S(\omega)\right] := \mathbb{E}\left[\left(\sum_{n=1}^{N} X_n(\omega)\right)^* \left(\sum_{m=1}^{N} X_m(\omega)\right)\right] = \quad \text{(independent)} \\ = \sum_{n=1}^{N} \mathbb{E}\left[|X_n(\omega)|^2\right] = \\ = \sum_{n=1}^{N} \mathbb{E}\left[PSD_{X_n}(\omega)\right]\ . \end{aligned}$$

CHAPTER

TWENTYTHREE

Z-TRANSFORM

Contents

- Definition
- Elementary transforms
- Region of convergence
- Applications
  - Solution of difference equations
  - Analysis of stability of numerical integration methods
  - Discrete-time design of filters and discrete-time systems. Relation between continuous-time and discrete-time systems due to sampling. Relation with Laplace transform.

23.1 Definition

Z-transform can be considered as a discrete-time counterpart of the *Laplace transform*, as section Laplace transform of a sampled function tries to show. Z-transform converts a discrete-time signal $\{x_n\}$ into a complex-valued function $X(z)$. Similarly to the *definition of Laplace transform*, Z-transform can be either two-sided

$$X(z) = \mathcal{Z}\{x_n\}(z) = \sum_{n=-\infty}^{+\infty} x_n z^{-n} \,,$$

or one-sided

$$X(z) = \mathcal{Z}\{x_n\}(z) = \sum_{n=0}^{+\infty} x_n z^{-n} \,,$$

that can be also thought as the two-sided transform of **causal time-series**, i.e. $x_k = 0, \forall k < 0$.

23.1.1 Z-transform and discrete-time Fourier transform (DTFT)

The *discrete-time Fourier transform* of a sequence, that can be written as (22.6),

$$\mathrm{DTFT}(x(t); \Delta t)(\nu) = \Delta t \sum_{n=-\infty}^{+\infty} x(n\Delta t) e^{-i2\pi\nu n\Delta t} \,,$$

can be thought as a scaled Z-transform of the signal, evaluated in $z = i\omega\Delta t = i2\pi\nu\Delta t$,

$$\mathrm{DTFT}(x(t); \Delta t)(\nu) = \Delta t \left[\sum_{n=-\infty}^{+\infty} x_n z^{-n} \right] \Bigg|_{z=e^{i\omega\Delta t}} = \Delta t X(z = e^{i\omega\Delta t}) \,,$$

23.1.2 Laplace transform of a sampled function

Delta sampling. Let $\tilde{x}(t) = x(t)\text{III}_{\delta t}(t) = x(t)\sum_{m=-\infty}^{+\infty}\delta(t-m\Delta t)$ the sampling of the continuous-time function $x(t)$ with Dirac's comb. Its one-sided Laplace transform reads

$$\begin{aligned}
\mathcal{L}\{\tilde{x}(t)\}(s) &= \int_{t=0^-}^{+\infty}\tilde{x}(t)e^{-st}dt = \\
&= \sum_{m=-\infty}^{+\infty}\int_{t=0^-}^{+\infty}x(t)\delta(t-m\Delta t)e^{-st}dt = \\
&= \sum_{m=0}^{+\infty}x(m\Delta t)e^{-sm\Delta t} = \\
&= \sum_{m=0}^{+\infty}x(t_m)e^{-st_m} = \\
&= \sum_{m=0}^{+\infty}x_m z^{-m}\,,
\end{aligned} \tag{23.1}$$

with $x_m = x(t_m)$, $t_m = m\Delta t$, and $z = e^{s\Delta t}$. Here the summation runs over $m = 0 : +\infty$, as the integral of $\delta(t-m\Delta t)$ produces a non-zero result only for $m \geq 0$. The relation (23.1) shows that the Laplace transform of the Dirac delta sampled function is equivalent to the Z-transform of its samples, if the relations between t_m and m and z and $s\Delta t$ hold.

Sample and Hold.

$$\begin{aligned}
\mathcal{L}\{\tilde{x}(t)\}(s) &= \int_{t=0^-}^{+\infty}\tilde{x}(t)e^{-st}dt = \\
&= \sum_{m=-\infty}^{+\infty}\int_{t=t_m}^{t_{m+1}}x(t_m)e^{-st}dt = \\
&= \sum_{m=-\infty}^{+\infty}x(t_m)\frac{1}{s}\left[e^{-st_m}-e^{-st_{m+1}}\right] = \\
&= \frac{1-e^{-s\Delta t}}{s}\sum_{m=-\infty}^{+\infty}x_m e^{-st_m} = \\
&= \ldots
\end{aligned} \tag{23.2}$$

with $z = e^{s\Delta t}$, $s = \frac{1}{\Delta t}\ln z$.

23.2 Inverse transform

…

23.3 Elementary one-sided transforms

In this section, some elementary Z-transforms are collected. These can be interpreted as the discrete-time counterpart of the *elementary Laplace transforms*

Linearity.

Kronecker delta. $x_n = \delta_{0n} = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$

$$\mathcal{Z}\{\delta_{0n}\}(z) = \sum_{n=0}^{+\infty} \delta_{n0} z^{-n} = 1 \, .$$

"Time" delay. Z-transform of a causal sequence x_n (with $x_n = 0$ for $n < 0$), x_{n-k} with $k \geq 0$

$$\mathcal{Z}\{x_{n-k}\}(z) = z^{-k} X(z)$$

Proof

$$\begin{aligned} \mathcal{Z}\{x_{n-k}\}(z) &= \sum_{n=0}^{+\infty} x_{n-k} z^{-n} = \\ &= \sum_{n=k}^{+\infty} x_{n-k} z^{-(n-k)} z^{-k} = \\ &= \sum_{n=0}^{+\infty} x_n z^{-n} z^{-k} = \\ &= z^{-k} X(z) \, . \end{aligned}$$

First difference backward. Directly follows from the Z-transform of time delay

$$\mathcal{Z}\left\{x_n - x_{n-1}\right\} = \left(1 - z^{-1}\right) X(z)$$

First difference forward.

Accumulation.

Convolution.

Initial value. If x_n is causal, i.e. $x_n = 0$ for $n < 0$, then

$$x_0 = \lim_{z \to +\infty} X(z) \, .$$

Proof

$$X(z) = x_0 + \sum_{n=1}^{+\infty} x_n z^{-n}$$

As

$$\left| \sum_{n=1}^{+\infty} x_n z^{-n} \right| \leq \sum_{n=1}^{+\infty} |x_n z^{-n}| \leq M \sum_{n=1}^{+\infty} |z|^{-n} = M \frac{1}{|z|} \frac{1}{1 - \frac{1}{|z|}} ,$$

for every z, with $|z| > 1$ s.t. the geometric series is convergent. As $|z| \to +\infty$, the series goes to zero, and thus

$$\lim_{z \to +\infty} X(z) = x_0 .$$

Final value. If *the poles of* $(z-1)X(z)$ *are inside the unit circle (condition for as.stability)*, then

$$x_{+\infty} = \lim_{z \to 1} (z-1) X(z) .$$

Proof

…

23.4 Applications

23.4.1 Solution of difference equations

First-order reduction.

Reduction to first order k-dimensional system of a k-order ODE

From k-order difference equation to system of 1-order difference equations, as it can be done with ODEs. For ODEs

$$x^{(k)} + a_{k-1} x^{(k-1)} + \cdots + a_1 \dot{x} + a_0 x = f(t)$$

may be written as a first-order k-dimensional system

$$\dot{\mathbf{z}} = \mathbf{f}(\mathbf{z}, t)$$

$$\begin{bmatrix} \dot{z}_0 \\ \dot{z}_1 \\ \cdots \\ \dot{z}_{k-1} \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ \cdots \\ -a_{k-1} z_{k-1} - \cdots - a_1 z_1 - a_0 z_0 + f(t) \end{bmatrix}$$

Reduction to first order k-dimensional system of a k-order difference equation

For a difference equation

$$x_n + a_1 x_{n-1} + \cdots + a_{k-1} x_{n-k+1} = f_n$$

may be written as

$$\mathbf{z}_{n+1} = \mathbf{f}(\mathbf{z}_n, t_n)$$

$$\begin{bmatrix} z_{0,n} \\ z_{1,n} \\ \cdots \\ z_{k-1,n} \end{bmatrix} = \begin{bmatrix} z_{1,n-1} \\ z_{2,n-1} \\ \cdots \\ f_n - a_1 z_{k-1,n-1} \cdots - a_{k-1} z_{0,n-1} \end{bmatrix}$$

with $z_{j,n} = x_{n-k+1+j}$, for $j = 0 : k-1$.

Ansatz. The solution of a homogeneous linear ODE can be solved as a linear combination of elementary solutions with expression $x(t) = Xe^{st}$ - except for the case of *multiple roots in the characteristic polynomial*

$$x^{(k)} + a_{k-1}x^{(k-1)} + \cdots + a_1\dot{x} + a_0 x = 0 \,,$$

with values of s as the roots of the polynomial

$$s^k + a_{k-1}s^{k-1} + \cdots + a_1 s + a_0 = 0 \,.$$

The solution is then

$$x(t) = \sum_{a=1}^{k} X_a e^{s_a t} \,,$$

with the integration constants to be determined with the initial conditions provided along with the differential equation. *Asymptotical stability conditions* depend on the real part of the eigenvalues, and on the multiplicity of the imaginary eigenvalues.

A linear difference equation can be solved as a linear combination of elementary solutions with expression $x_n = Xz^n$

$$x_n + a_{k-1}x_{n-1} + \cdots + a_1 x_{n-k+2} + a_0 x_{n-k+1} = 0 \,,$$

with values of z as the roots of the polynomial

$$z^k + a_{k-1}z^{k-1} + \cdots + a_1 z + a_0 = 0 \,.$$

The solution is then

$$x_n = \sum_{a=1}^{k} X_a {z_a}^n \,,$$

with the integration constants to be determined with the initial conditions provided along with the difference equation.

Stability conditions. …

- asymptotically stable if all the eigenvalues are $|z_a| < 1$
- unstable if…
- marginally stable if…

todo *treat multiple roots*

23.4.2 Analysis of stability of numerical integration methods for ODEs

Part VII

Approximation of Functions

CHAPTER

TWENTYFOUR

INTRODUCTION TO FUNCTION APPROXIMATION

Lagrange interpolation

Useful, because derivatives and integrals of polynomial functions are easy to evaluate exactly. Used in:

- numerical methods for ODEs, e.g. *multi-step methods*
- numerical methods for PDEs, e.g. base functions of *FEM*.
  - Gaussian quadrature provides exact integration of integrals.
  - FEM uses **piece-wise** Lagrangian interpolation of the fields
- …

CHAPTER

TWENTYFIVE

LAGRANGE INTERPOLATION

Let $p_n(t)$ a polynomial interpolation of the points $\{(t_k, y_k)\}_{k=0:n}$, with $y_k = y(t_k)$. Lagrange interpolation of $n+1$ points requires a polynomial of degree n.

25.1 Lagrangian basis, and polynomial interpolation

Lagrangian basis. The Lagrangian basis is a set of $n+1$ functions $\ell_k(t)$, so that

$$\ell_k(t_j) = \delta_{jk} \, .$$

In order to satisfy these n+1 conditions, the kth function of the Lagrange basis reads

$$\ell_j(t) = \prod_{\substack{0 \leq m \leq n \\ m \neq j}} \frac{t - t_m}{t_j - t_m} \, .$$

Properties.

- $\ell_k(t_j) = \delta_{jk}$
- $\sum_k \ell_k(t) = 1$ **todo** *Prove it!*

Polynomial approximation. The Lagrange interpolating polynomial immediately follows as the linear combination

$$L_n \left\{ y(t) \right\} (t) = \sum_{j=0}^{n} y_j \ell_j(t) \, .$$

25.2 Low-degree approximations with constant steps

Constant-step sampling means $t_k = t_0 + kh$, being h the step.

- One-node approximation, (t_0, y_0)

  $$\ell_0(t) = 1$$

  so that the constant approximation reads $L_0(t) = y_0 \ell_0(t) = y_0$.

- Two-node approximation, (t_0, y_0), (t_1, y_1)

  $$\ell_0(t) = \frac{t - t_1}{t_0 - t_1} = -\frac{t - t_1}{h}$$
  $$\ell_1(t) = \frac{t - t_0}{t_1 - t_0} = \frac{t - t_0}{h}$$

so that the 1-degree approximation reads $L_1(t) = y_0\ell_0(t) + y_1\ell_1(t)$.

- Three-node approximation, (t_0, y_0), (t_1, y_1), (t_2, y_2)

$$\ell_0(t) = \frac{(t-t_1)(t-t_2)}{(t_0-t_1)(t_0-t_2)} = \frac{(t-t_1)(t-t_2)}{2h^2}$$
$$\ell_1(t) = \frac{(t-t_0)(t-t_2)}{(t_1-t_0)(t_1-t_2)} = -\frac{(t-t_0)(t-t_2)}{h^2}$$
$$\ell_2(t) = \frac{(t-t_0)(t-t_1)}{(t_2-t_0)(t_2-t_0)} = \frac{(t-t_0)(t-t_1)}{2h^2}$$

so that the 2-degree approximation reads $L_2(t) = y_0\ell_0(t) + y_1\ell_1(t) + y_2\ell_2(t)$.

- Four-node approximation, (t_0, y_0), (t_1, y_1), (t_2, y_2), (t_3, y_3)

$$\ell_0(t) = -\frac{(t-t_1)(t-t_2)(t-t_3)}{6h^3}$$
$$\ell_1(t) = \frac{(t-t_0)(t-t_2)(t-t_3)}{2h^3}$$
$$\ell_2(t) = -\frac{(t-t_0)(t-t_1)(t-t_3)}{2h^3}$$
$$\ell_3(t) = \frac{(t-t_0)(t-t_1)(t-t_2)}{6h^3}$$

so that the 3-degree approximation reads $L_3(t) = \sum_{k=0}^{3} y_k\ell_k(t)$.

25.3 Applications

25.3.1 Lagrange interpolation and multi-step methods for IVPs governed by ODEs

Adams-Bashforth

AB1. Using $n = 0$ approximation with points (t_0, f_0) to integrate from t_0 to t_1

$$y_1 - y_0 = \int_{t=t_0}^{t_1} f_0 dt = \\ = hf_0$$

and thus, AB1 is the **explicit Euler** method

$$y_{n+1} = y_n + hf(t_n, y_n)\ .$$

AB2. Using $n = 1$ approximation with points (t_0, f_0), (t_1, f_1) to integrate from t_1 to t_2

$$y_2 - y_1 = \int_{t=t_1}^{t_2} \left\{-f_0\frac{t-t_1}{h} + f_1\frac{t-t_0}{h}\right\} dt = \\ = \frac{1}{h}\left\{-f_0\frac{(t-t_1)^2}{2} + f_1\frac{(t-t_0)^2}{2}\right\}\Bigg|_{t_1}^{t_2} = \\ = \frac{1}{h}\left\{-f_0\frac{h^2}{2} + f_1\frac{3h^2}{2}\right\}\ ,$$

and thus

$$y_{n+1} = y_n + h\left(\frac{3}{2}f(t_n, y_n) - \frac{1}{2}f(t_{n-1}, y_{n-1})\right)$$

Adams-Moulton

AM1. Using $n = 0$ approximation with points (t_1, f_1) to integrate from t_0 to t_1

$$\begin{aligned} y_1 - y_0 = \int_{t=t_0}^{t_1} f_1 dt = \\ = h f_0 \end{aligned}$$

and thus, AB1 is the **implicit Euler** method

$$y_{n+1} = y_n + h f(t_{n+1}, y_{n+1}) \, .$$

AM2. Using $n = 1$ approximation with points (t_1, f_1), (t_2, f_2) to integrate from t_1 to t_2

$$\begin{aligned} y_2 - y_1 &= \int_{t=t_1}^{t_2} \left\{ -f_1 \frac{t-t_2}{h} + f_2 \frac{t-t_1}{h} \right\} dt = \\ &= \frac{1}{h} \left\{ -f_1 \frac{(t-t_2)^2}{2} + f_2 \frac{(t-t_1)^2}{2} \right\} \Bigg|_{t_1}^{t_2} = \\ &= \frac{1}{h} \left\{ f_1 \frac{h^2}{2} + f_2 \frac{h^2}{2} \right\} \, , \end{aligned}$$

and thus

$$y_{n+1} = y_n + h \left(\frac{1}{2} f(t_{n+1}, y_{n+1}) + \frac{1}{2} f(t_n, y_n) \right)$$

Backward Differentiation Formulas

BDF1 Using $n = 1$ approximation

$$\begin{aligned} f_1 = \frac{d}{dt} y(t) \Big|_{t=t_1} \simeq \\ = \frac{d}{dt} \left[-y_0 \frac{t-t_1}{h} + y_1 \frac{t-t_0}{h} \right] \Big|_{t=t_1} = \\ = \frac{1}{h} \left(-y_0 + y_1 \right) \, , \end{aligned}$$

and thus, BDF1 is the **implicit Euler** method,

$$y_{n+1} = y_n + h f(t_{n+1}, y_{n+1}) \, .$$

BDF2 Using $n = 2$ approximation

$$\begin{aligned} f_2 = \frac{d}{dt} y(t) \Big|_{t=t_2} \simeq \\ = \frac{d}{dt} \left[y_0 \frac{(t-t_1)(t-t_2)}{2h^2} - y_1 \frac{(t-t_0)(t-t_2)}{h^2} + y_2 \frac{(t-t_0)(t-t_1)}{2h^2} \right] \Bigg|_{t=t_2} = \\ = \frac{1}{h^2} \left[\frac{y_0}{2} (2t - t_1 - t_2) - y_1 (2t - t_0 - t_2) + \frac{y_2}{2} (2t - t_0 - t_1) \right] \Big|_{t=t_2} = \\ = \frac{1}{h^2} \left[\frac{y_0}{2} h - y_1 \cdot 2h + \frac{y_2}{2} \cdot 3h \right] \, , \end{aligned}$$

and thus

$$\frac{3}{2}y_{n+1} - 2y_n + \frac{1}{2}y_{n-1} = hf(t_{n+1}, y_{n+1})$$

i.e.

$$y_{n+1} - \frac{4}{3}y_n + \frac{1}{3}y_{n-1} = \frac{2}{3}hf(t_{n+1}, y_{n+1})\,.$$

Part VIII

Calculus of Variations

CHAPTER

TWENTYSIX

INTRODUCTION TO CALCULUS OF VARIATIONS

Contents

- ***Basics***. Definition of a functional, definition of a variation, stationariety principle: Lagrange equations, Euler-Beltrami equation
- ***Examples***. Shortest path between two points; Fermat principle; Lagrangian formulation of classical mechanics, and special relativity
- ***Extra***.
  - Variational principle must have an **invariant form**, i.e. it must not depent on arbitrary choices of coordinates or parameters that have no geometrical or physical invariant nature
  - Parametrization of the integral with a parameters whose value depends on the integral itself. E.g.: arc-length parametrization of the length of a curve; arc-length parametrization of time in Fermat principle;…

26.1 Basics of calculus of variations

Calculus of variation deals with variations - i.e. “small changes” - of functions and functionals.

The meaning of the term functional may vary on the subfield of interest. In the field of calculus of variation, a **functional** can be defined as a function of function, i.e. a function whose argument is another function.

Fields and applications

Fields and applications related to calculus of variations (give some examples below):

- gradient-based techniques like some methods in:
  - optimization, either free or constrained (via Lagrange multiplier methods)
  - sensitivity
- classical mechanics and physics in general:
  - analytical mechanics: Lagrangian formulation and Hamiltonian formulation of classical mechanics
- …

Examples

- Lagrange equations for general problem
- examples:
  - brachistochrone for minimum time,…
  - catenary, i.e. static solution of wire and cables with neglibile bending stiffness
  - isoperimetric inequality, i.e. circle is the plane closed curve with given perimeter enclosing the largest area
- sensitivity of results to parameters. Some interesting sensitivity, both in time and trasnformed domains
  - characteristics of a system:
    - equilibria
    - eigenvalues
    - …
- optimal control methods

26.1.1 Lagrange equations

Given the functional S, with arguments a function $q(t)$ and the independent variable t,

$$S[q(t), t] = \int_{t=t_0}^{t_1} L(\dot{q}(t),\, q(t),\, t)\, dt$$

its variation w.r.t. the function $q(t)$ reads

$$\delta S[q(t), t] = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \left(S[q(t) + \varepsilon w(t),\, t] - S[q(t),\, t] \right) \ ,$$

where the function $w(t)$ is arbitrary, among those satisfying the constraint of the problems: as an example here, if the function $q(t)$ has prescribed values q^* for some values of the independent variable, t^*, the variation $w(t)$ of the function $q(t)$ is zero there, $w(t^*)$ so that the variated function $q(t) + \varepsilon w(t)$ satisfies the constraint as well, i.e. $q(t^*) + \varepsilon w(t^*) = q^*$.

Variation involves only small changes of function arguments, since these ones are the elements that can be effectively changed, while the independent variable is not.

Direct computation of the variation gives

$$\begin{aligned}
\delta S[q(t),t] &= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\left(S[q(t)+\varepsilon w(t),\,t]-S[q(t),\,t]\right)= \\
&= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\left(\int_{t=t_0}^{t_1} L(\dot q(t)+\varepsilon w(t),\,q(t)+\varepsilon w(t),\,t)-\int_{t=t_0}^{t_1} L(\dot q(t),\,q(t),\,t)\,dt\right)= \\
&= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\int_{t=t_0}^{t_1}\left(L(\dot q(t)+\varepsilon w(t),\,q(t)+\varepsilon w(t),\,t)-L(\dot q(t),\,q(t),\,t)\right)\,dt= \\
&= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\int_{t=t_0}^{t_1}\left\{L(\dot q(t),\,q(t),\,t)+\varepsilon\left[\frac{\partial L}{\partial \dot q}\dot w(t)+\frac{\partial L}{\partial q}w(t)\right]+o(\varepsilon)-L(\dot q(t),\,q(t),\,t)\right\}\,dt= \\
&= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\int_{t=t_0}^{t_1}\left\{\varepsilon\left[\frac{\partial L}{\partial \dot q}\dot w(t)+\frac{\partial L}{\partial q}w(t)\right]+o(\varepsilon)\right\}\,dt= \\
&= \int_{t=t_0}^{t_1}\left\{\frac{\partial L}{\partial \dot q}\dot w(t)+\frac{\partial L}{\partial q}w(t)\right\}\,dt= \\
&= \left[w(t)\frac{\partial L}{\partial \dot q}\right]\Big|_{t=t_0}^{t_1}+\int_{t=t_0}^{t_1}\left\{-\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q}\right)+\frac{\partial L}{\partial q}\right\}w(t)\,dt\,.
\end{aligned}$$

The solution depends on the boundary conditions at the extreme points t_0, t_1. **If** the value of the function $q(t)$ is prescribed in t_0 and t_1, $q(t_0)=q_0$, $q(t_1)=q_1$, then its variation is zero, $w(t_0)=w(t_1)=0$, for the reason that has been discussed above. The variation of the functional with prescribed boundary values of the argument function thus reads

$$\delta S[q(t),t]=\int_{t=t_0}^{t_1}\left\{-\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q}\right)+\frac{\partial L}{\partial q}\right\}\delta q(t)\,dt\,,$$

having called $w(t)=:\delta q(t)$ to stress that is the variation of function $q(t)$. This notation - it's just notation, it has no special properties - could be useful if the functional depends on several arguments.

Stationary conditions, $\delta S = 0$. Stationary condition of the functional S implies that $\delta S = 0$ for all the possible variations of the argument function, $\forall \delta q(t)$. This condition implies that the integrand is identically zero, i.e. **Lagrange equations**,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot q}\right)-\frac{\partial L}{\partial q}=0\,,$$

Higher-order derivatives

Method 1. If the Lagrangian function L depends on higher order derivatives,

$$L\left(q^{(n)}(t),\,q^{(n-1)}(t),\,\dots,\,q'(t),\,q(t),\,t\right)$$

it's possible to recast the problem defining the n-dimensional function, $\mathbf{q}(t)$,

$$\mathbf{q}(t)=\left(q^0(t),q^1(t),\dots,q^{n-1}(t)\right):=\left(q(t),q'(t),\dots,q^{(n-1)}(t)\right)\,.$$

With some abuse of notation in L, the functional S can be recasted as

$$\begin{aligned}
S[q(t),t] &= \int_{t=t_0}^{t_1} L(q^{(n)}(t),\,\dots,\,q(t),\,t)\,dt= \\
&= \int_{t=t_0}^{t_1} L(\dot{\mathbf{q}}(t),\,\mathbf{q}(t),\,t)\,dt\,.
\end{aligned}$$

todo *Add constraints on components of* $\mathbf{q}(t)$*?*

Repeating the computation, the variation of the functional reads

$$\delta S[\mathbf{q}(t),t] = \left[\delta\mathbf{q}^T(t)\frac{\partial L}{\partial\dot{\mathbf{q}}}\right]\Big|_{t=t_0}^{t_1} + \int_{t=t_0}^{t_1}\delta\mathbf{q}^T(t)\left\{-\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\mathbf{q}}}\right)+\frac{\partial L}{\partial\mathbf{q}}\right\}dt\,.$$

Method 2. …

Euler-Beltrami equation

If the Lagrangian function L is not an explicit function of the independent variable, $L(q'(x),q(x))$, Euler-Berltrami equation follows from the derivative of the Lagrangian,

$$\begin{aligned}\frac{dL}{dx} &= \frac{\partial L}{\partial q'}q''(x)+\frac{\partial L}{\partial q}q' = \\ &= \frac{\partial L}{\partial q'}q''+\frac{d}{dx}\left(\frac{\partial L}{\partial q'}\right)q' = \\ &= \frac{d}{dx}\left(\frac{\partial L}{\partial q'}q'\right)\,,\end{aligned}$$

and thus

$$\frac{d}{dx}\left[L-q'\frac{\partial L}{\partial q'}\right]=0 \qquad\rightarrow\qquad L-q'\frac{\partial L}{\partial q'}=C \quad \text{const.}$$

Note 1. While Lagrange equations are a set of N equations if the functional depends on N argument functions $q_k(t)$, $k=1:N$, Euler-Beltrami equation is an equation only. Indeed for multiple argument functions

$$\begin{aligned}\frac{dL}{dx} &= \frac{\partial L}{\partial q_k'}q_k''(x)+\frac{\partial L}{\partial q_k}q_k' = \\ &= \frac{\partial L}{\partial q_k'}q_k''+\frac{d}{dx}\left(\frac{\partial L}{\partial q_k'}\right)q_k' = \frac{d}{dx}\left(\frac{\partial L}{\partial q_k'}q_k'\right)\,,\end{aligned}$$

where Einstein's summation notation of repeated index is used. Euler-Beltrami thus reads

$$L(q_l'(x),q_l(x))-q_k'(x)\frac{\partial L}{\partial q_k'}(q_l'(x),q_l(x))=C\,.$$

Note 2. If the Lagrangian function is an explicit function of the independent variable x, $L(q'(x),q(x),x)$, it's not hard to realize that the derivative of the Lagrangian function, along with the use of the Lagrange equation, gives

$$\frac{d}{dx}\left[L-q'\frac{\partial L}{\partial q'}\right]=\frac{\partial L}{\partial x}\,.$$

Example 26.1.1 (Euler-Beltrami with $L(q'(x),q(x),x)$, Hamiltonian, energy and E.Noether)

Euler-Beltrami equation shows that if $L(q'(x),q(x))$, thus $L-q'\partial_{q'}L$ is constant, (or an integral of motion in dynamics). In analytical mechanics (Lagrange mechanics, Hamiltonian mechanics), Lagrangian and Hamiltonian functions of a system read

$$\begin{aligned}L(\dot{q}_k(t),q_k(t),t) &= T(\dot{q},q,t)+U(q,t)\\ H(p,q,t) &:= p_k\dot{q}_k-L=\dot{q}_k\frac{\partial L}{\partial\dot{q}^k}-L\,,\end{aligned}$$

having used the common definition of the generalized momenta $p_k := \frac{\partial L}{\partial \dot{q}^k}$. It should be immediate to realize that the Hamiltonian is just the quantity appearing in Euler-Beltrami equation (or in its "modified version" if $\partial_t L \neq 0$), and thus

$$\frac{dH}{dt} = \frac{\partial L}{\partial t} \, .$$

In mechanics, if $\partial_t L = 0$, the Hamiltonian is a constant of motion. In this case, it can be prove that the Hamiltonian is equal to the eneergy of the system.

Classical examples.

Example 26.1.2 (Brachistochrone)

Find the trajectory…

- Elementary length: $ds = v\,dt$
- Energy: $E(y) = \frac{1}{2}mv^2 - mgy + C$. Setting $E = 0$ at starting point, from rest, at $y_0 = 0$, it implies $C = 0$; thus $v = \sqrt{2gy}$
- $x(s)$, $y(s)$,
- $ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + y'^2(x)}\,dx$

$$T = \int_{t_0}^{t_1} dt = \int_{s_0}^{s_1} \frac{ds}{v} = \int_{x_0}^{x_1} \frac{\sqrt{1 + y'^2(x)}}{\sqrt{2gy(x)}}\,dx$$

The Lagrangian doesn't explicitly depend on x, thus Euler-Beltrami equation can be used. Partial derivative of the Lagrangian function w.r.t. q' reads

$$\frac{\partial L}{\partial y'} = ... = \frac{1}{\sqrt{2gy}\sqrt{1 + y'^2}} y' \, ,$$

and thus Euler-Beltrami equation reads

$$C = L - q' \frac{\partial L}{\partial y'} = \frac{1 + y'^2 - y'^2}{\sqrt{2gy}\sqrt{1 + y'^2}} = \frac{1}{\sqrt{2gy}\sqrt{1 + y'^2}}$$

Squaring $2gC^2 = \frac{1}{y(1+y'^2)}$, it's possible to write

$$y(x) = \frac{1}{2gC^2(1 + y'^2(x))} \, ,$$

Making the substitution $y(x)' = ...$

Example 26.1.3 (Catenary)

Example 26.1.4 (Isoperimetric problem)

26.2 Examples

Contents

Geometry. *Shortest path*

Physics.

- *Fermat principle* in optics;
- *Lagrange equations in classical mechanics*;
- *Lagrange equations in special relativity*;
- *Lagrange equations in quantum mechanics*
- Lagrange approach to classical electromagnetism [**external reference, basics: classical electromagnetism**]

26.2.1 Shortest path

Shortest path between two given points A, B. The length of a curve γ_{AB} connecting the two points reads

$$L_{\gamma_{AB}} = \int_{\gamma_{AB}} ds \,,$$

with s the arc-length parameter. The value of s at the extremes of integration **is not independent** on the curve γ_{AB}, and thus on the result. In order to write the integrals w.r.t. a parameter with given values at the extreme points, a change of parameter is required. Let's define a parameter ℓ, so that the curve in space can be represented as $\gamma : \mathbf{r}(t), t \in [t_A, t_B]$, with t_A, t_B given, and $\mathbf{r}(t_A) = \mathbf{r}_A$, $\mathbf{r}(t_B) = \mathbf{r}_B$ given. The elementary length ds becomes

$$|d\mathbf{r}| = \underbrace{|\mathbf{r}'(s)|}_{=1,\text{ by def of } s} ds = |\mathbf{r}'(\ell)|\, d\ell \,.$$

The integral thus becomes

$$L_{\gamma_{AB}} = L[\mathbf{r}(\ell)] = \int_{\ell_A}^{\ell_B} |\mathbf{r}'(\ell)|\, d\ell \,,$$

where the dependence on the curve $\mathbf{r}(\ell)$ is made explicit, and the values of the parameter ℓ at the extreme is given. Variation of the functional reads

$$\begin{aligned}
\delta L &= \delta \int_{\ell_A}^{\ell_B} |\mathbf{r}'(\ell)|\, d\ell = \\
&= \int_{\ell_A}^{\ell_B} \delta |\mathbf{r}'(\ell)|\, d\ell = \\
&= \int_{\ell_A}^{\ell_B} \delta\mathbf{r}' \cdot \frac{\mathbf{r}'}{|\mathbf{r}'|} d\ell = \\
&= \underbrace{\delta\mathbf{r} \cdot \hat{\mathbf{t}}\big|_{\ell_A}^{\ell_B}}_{=0} - \int_{\ell_A}^{\ell_B} \delta\mathbf{r} \cdot \frac{d\hat{\mathbf{t}}}{d\ell} d\ell \,,
\end{aligned}$$

being $\hat{\mathbf{t}}(\ell) = \frac{\mathbf{r}'(\ell)}{|\mathbf{r}'(\ell)|}$, the unit-length tangent vector to the curve. Stationariety of L, for any possible variation $\delta\mathbf{r}(\ell)$ implies

$$\frac{d\hat{\mathbf{t}}}{d\ell} = \mathbf{0} \,,$$

Thus, the solution reads $\mathbf{r}(\ell) = \alpha\hat{\mathbf{t}}\,\ell + \mathbf{r}_0$, and prescribing the boundary conditions

$$\mathbf{r}(\ell) = \mathbf{r}_A + \frac{\ell - \ell_A}{\ell_B - \ell_A}\left(\mathbf{r}_B - \mathbf{r}_A\right) \ .$$

Details about the integration

26.2.2 Fermat principle

In geometrical optics, Fermat principles asserts that a light ray between two points represent the path with the shortest travelling time connecting them. Let γ a curve in space. The travelling time of a light ray with speed c reads

$$T = \int_{\gamma_{AB}} dt = \int_{\gamma_{AB}} \frac{1}{c} ds \ ,$$

begin $ds = cdt$. Let ℓ be a parameter, so that $\gamma : \mathbf{r}(\ell)$ is a parametrization of the curve, with the extreme points independent from the result. The speed of light may depend on the position in space, $c(\mathbf{r})$, and it can be written as a function of the speed of light in vacuum and the refractive index $n(\mathbf{r})$ as $c(\mathbf{r}) = \frac{c_0}{n(\mathbf{r})}$. Making the dependence on $\mathbf{r}(\ell)$, $\mathbf{r}'(\ell)$ explicit in the functional,

$$T[\mathbf{r}(\ell)] = \int_{\ell_A}^{\ell_B} n(\mathbf{r}(\ell))\,|\mathbf{r}'(\ell)|\,d\ell \ .$$

Fermat principle reads

$$\begin{aligned}
0 = \delta T = \\
&= \delta \int_{\ell_A}^{\ell_B} n(\mathbf{r}(\ell))\,|\mathbf{r}'(\ell)|\,d\ell = \\
&= \int_{\ell_A}^{\ell_B} \left\{ \delta\mathbf{r}\cdot\nabla_{\mathbf{r}} n|\mathbf{r}|' + \delta\mathbf{r}'\cdot\frac{\mathbf{r}'}{|\mathbf{r}'|} n(\mathbf{r}) \right\} d\ell = \\
&= \int_{\ell_A}^{\ell_B} \delta\mathbf{r}\cdot\left\{ \nabla_{\mathbf{r}} n\,|\mathbf{r}|' - \frac{d}{d\ell}\left(\frac{\mathbf{r}'}{|\mathbf{r}'|} n(\mathbf{r})\right) \right\} d\ell + \underbrace{\delta\mathbf{r}\cdot\frac{\mathbf{r}'}{|\mathbf{r}'|} n(\mathbf{r})\Big|_{\ell_A}^{\ell_B}}_{=0} ,
\end{aligned}$$

From the arbitrarieness of $\delta\mathbf{r}$,

$$\begin{aligned}
0 = \frac{d}{d\ell}\left(n(\mathbf{r}(\ell))\frac{\mathbf{r}'(\ell)}{|\mathbf{r}'(\ell)|}\right) - |\mathbf{r}'(\ell)|\,\nabla n(\mathbf{r}(\ell)) = \\
&= \frac{\mathbf{r}'}{|\mathbf{r}|}\mathbf{r}'\cdot\nabla n - n\frac{d}{d\ell}\left(\frac{\mathbf{r}'}{|\mathbf{r}'|}\right) - |\mathbf{r}'|\nabla n = \\
&= |\mathbf{r}'|\left[\mathbb{I} - \frac{\mathbf{r}'\cdot\mathbf{r}'}{|\mathbf{r}'|^2}\right]\cdot\nabla n - n\frac{d}{d\ell}\left(\frac{\mathbf{r}'}{|\mathbf{r}'|}\right) \ .
\end{aligned}$$

and thus

$$\mathbb{P}_{\perp\hat{\mathbf{t}}}\cdot\nabla n = \frac{n}{|\mathbf{r}'|}\frac{d\hat{\mathbf{t}}}{d\ell} \ ,$$

or, re-introducting the "physical" variable s (the arc-length is the parameter with physical, geometrical, non arbitrary meaning; the equations should be invariant from the parametrization, so we should be happy of the following result, written in invariant form),

$$\mathbb{P}_{\perp\hat{\mathbf{t}}}\cdot\nabla n = n\frac{d\hat{\mathbf{t}}}{ds} \ . \tag{26.1}$$

being $\mathbb{P}_{\perp\hat{\mathbf{t}}}$ the orthogonal projector in the direction perpendicular to the unit tangent vector $\hat{\mathbf{t}}$. Using the results of *geometry of curves*, the derivative $\hat{\mathbf{t}}'(s) = \kappa(s)\hat{\mathbf{n}}(s)$, being $\hat{\mathbf{n}}$ the unit normal vector pointing towards the local center of curvature (center of the osculator circle, tangent with the same second order derivative), and $\kappa(s) = \frac{1}{R(s)}$ is the local curvature, and $R(s)$ the radius of curvature (the radius of the osculator circle). Thus

$$\kappa\hat{\mathbf{n}} = \left[\mathbb{I} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}\right] \cdot \frac{\nabla n}{n}$$

Projecting this equation on the local Frenet basis $\{\hat{\mathbf{t}}, \hat{\mathbf{n}}, \hat{\mathbf{b}}\}$,

$$\begin{cases} t : 0 = \dfrac{1}{n}\hat{\mathbf{t}} \cdot \nabla n \\ n : \kappa = \dfrac{1}{n}\hat{\mathbf{n}} \cdot \nabla n \\ b : 0 = \dfrac{1}{n}\hat{\mathbf{b}} \cdot \nabla n \end{cases}$$

26.2.3 Lagrange equations in classical mechanics

For a detailed treatment of system of point masses and rigid bodies in classical mechanics, see [Basics: Classical Mechanics], in particular Classical Mechanics: Analytical Mechanics: Lagrange Equations of the Second Kind, and its subsection. Here, only the Lagrangian approach to the a point mass system is shown.

Let $\mathbf{r}(q^k(t), t)$ the position of the mass point, as a function of a set of generalized coordinates $q^k(t)$ and time t. The velocity of the mass can be written as

$$\mathbf{v} = \dot{\mathbf{r}}(t) = \dot{q}^k \partial_{q^k}\mathbf{r} + \partial_t \mathbf{r} \,.$$

The velocity field can be interpreted as a function $\mathbf{v}\left(q^k, \dot{q}^k, t\right)$. From the latter relation, as $\partial_{q^k}\mathbf{r}$ is independent from the $\dot{q}^k$, it follows

$$\partial_{q^k}\mathbf{r} = \partial_{\dot{q}^k}\dot{\mathbf{r}} \,.$$

Following Newton's mechanics, the equation of motion of a point mass with mass m in a force field with potential $V(\mathbf{r})$ reads

$$m\ddot{\mathbf{r}} = -\nabla_{\mathbf{r}} V(\mathbf{r}) \,.$$

Multiplyig the equation of motion by $\partial_{q^k}\mathbf{r}$,

$$\begin{aligned} 0 &= \partial_{q^k}\mathbf{r} \cdot \{m\ddot{\mathbf{r}} + \nabla_{\mathbf{r}} V\} = \\ &= \frac{d}{dt}\left(\partial_{q^k}\mathbf{r} \cdot m\dot{\mathbf{r}}\right) - \frac{d}{dt}\partial_{q^k}\mathbf{r} \cdot m\dot{\mathbf{r}} + \partial_{q^k} V = \qquad (1) \\ &= \frac{d}{dt}\left(\partial_{\dot{q}^k}\dot{\mathbf{r}} \cdot m\dot{\mathbf{r}}\right) - \partial_{q^k}\dot{\mathbf{r}} \cdot m\dot{\mathbf{r}} + \partial_{q^k} V = \\ &= \frac{d}{dt}\frac{\partial}{\partial\dot{q}^k}\left(\frac{1}{2}m\dot{\mathbf{r}} \cdot \dot{\mathbf{r}}\right) - \frac{\partial}{\partial q^k}\left(\frac{1}{2}m\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} - V\right) = \\ &= \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}^k} - \frac{\partial\mathcal{L}}{\partial q^k} \,, \end{aligned}$$

with the Lagrangian function $\mathcal{L}(\mathbf{r}; \dot{\mathbf{r}}, t) = \frac{1}{2}m|\dot{\mathbf{r}}|^2 - V(\mathbf{r})$. Here, in (1) the details about mixed derivatives are used to write $\frac{d}{dt}\partial_{q^k}\mathbf{r} = \partial_{q^k}\mathbf{v}$.

Details about mixed derivatives

$$\begin{aligned}\frac{d}{dt}\partial_{q^k}\mathbf{r}\left(q^l\left(\mathbf{r},t\right),t\right) &= \dot{q}^l\partial_{q^l}\partial_{q^k}\mathbf{r}+\partial_t\partial_{q^k}\mathbf{r} = \\ &= \partial_{q^k}\left\{\dot{q}^l\partial_{q^l}\mathbf{r}+\partial_t\mathbf{r}\right\} = \\ &= \partial_{q^k}\mathbf{v}\,.\end{aligned}$$

26.2.4 Lagrange equations in special relativity

Using tensor formalism

Equations of motion and physical principles **must** be written using physical properties, and thus be independent from an arbitrary parametrization.

Strong formulation. Let the equation of motion of a particle be

$$m\frac{d\mathbf{U}}{d\tau}=\mathbf{K}\,,$$

with m the rest mass, τ the proper time, $\mathbf{U}=\frac{d\mathbf{X}}{d\tau}$ the 4-velocity, and $\mathbf{K}$ the 4-force.

Weak formulation. Multiplying by an arbitrary test 4-vector $\mathbf{W}(\tau)$ and integrating over an arbitrary interval $\tau\in[\tau_0,\tau_1]$,

$$0=\int_{\tau_0}^{\tau_1}\mathbf{W}\cdot\left\{m\frac{d\mathbf{U}}{d\tau}-\mathbf{K}\right\}d\tau\,.$$

The parametrization of the trajectory is changed from τ to an arbitrary parameter λ, so that $\mathbf{X}_{0,1}=\mathbf{X}(\tau_{0,1})=\mathbf{X}(\tau(\lambda_{0,1}))$ are prescribed for given values $\lambda_{0,1}$. As the invariant $d\tau$ is defined through

$$c^2d\tau^2=ds^2=d\mathbf{X}\cdot d\mathbf{X}=\mathbf{X}'(\lambda)\cdot\mathbf{X}'(\lambda)\,d\lambda^2\,,$$

the relation between the differentials and the rule of derivation of composite functions read

$$d\tau=\frac{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}{c}\,d\lambda\qquad,\qquad\frac{d}{d\tau}=\frac{d\lambda}{d\tau}\frac{d}{d\lambda}=\frac{c}{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}\frac{d}{d\lambda}\,.$$

The velocity vector becomes

$$\mathbf{U}=\frac{d\mathbf{X}}{d\tau}=\frac{d\lambda}{d\tau}\frac{d\mathbf{X}}{d\lambda}=\frac{c}{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}\mathbf{X}'\,.$$

The integral thus becomes

$$\begin{aligned}0&=\int_{\tau_0}^{\tau_1}\mathbf{W}\cdot\left\{m\frac{d\mathbf{U}}{d\tau}-\mathbf{K}\right\}d\tau= \\ &=\int_{\lambda_0}^{\lambda_1}\mathbf{W}\cdot\left\{m\frac{c}{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}\left(\frac{c}{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}\mathbf{X}'\right)'-\mathbf{K}\right\}\frac{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}{c}d\lambda= \\ &=\int_{\lambda_0}^{\lambda_1}\mathbf{W}\cdot\left(\frac{mc}{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}\mathbf{X}'\right)'d\lambda-\int_{\lambda_0}^{\lambda_1}\mathbf{W}\cdot\mathbf{K}\frac{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}{c}d\lambda=\end{aligned}$$

Lagrange mechanics immediately follows choosing the test function $\mathbf{W}=\delta\mathbf{X}$.

- Free particle. For a free particle, $\mathbf{K} = \mathbf{0}$.

$$
\begin{aligned}
0 = & \int_{\lambda_0}^{\lambda_1} \delta\mathbf{X} \cdot \left(\frac{mc}{\sqrt{\mathbf{X}' \cdot \mathbf{X}'}} \mathbf{X}' \right)' \, d\lambda = \\
= & \underbrace{\left[\delta\mathbf{X} \cdot \frac{mc}{\sqrt{\mathbf{X}' \cdot \mathbf{X}'}} \mathbf{X}' \right) \Big|_{\lambda_0}^{\lambda_1}}_{=0} - \int_{\lambda_0}^{\lambda_1} \delta\mathbf{X}' \cdot \frac{mc}{\sqrt{\mathbf{X}' \cdot \mathbf{X}'}} \mathbf{X}' \, d\lambda = \\
= & -\delta \int_{\lambda_0}^{\lambda_1} mc \sqrt{\mathbf{X}' \cdot \mathbf{X}'} \, d\lambda = \\
= & -\delta \int_{\tau_0}^{\tau_1} mc^2 \, d\tau = \\
= & -\delta \int_{s_0}^{s_1} mc \, ds = \\
= & \, \delta S \, ,
\end{aligned}
$$

  where the change of the independent parameters is made after the variation is put outside the integral $\int_{\lambda_0}^{\lambda_1}$, with given extreme values.

- Particle subjecd to Lorentz force,

$$
\mathbf{K} = q\mathbf{F}(\mathbf{X}) \cdot \mathbf{U} = q\mathbf{F}(\mathbf{X}) \frac{c}{\sqrt{\mathbf{X}' \cdot \mathbf{X}'}} \mathbf{X}' \, .
$$

  The second integral becomes

$$
\begin{aligned}
-\int_{\lambda_0}^{\lambda_1} \delta\mathbf{X} \cdot \mathbf{K} \frac{\sqrt{\mathbf{X}' \cdot \mathbf{X}'}}{c} d\lambda = & -q \int_{\lambda_0}^{\lambda_1} \delta\mathbf{X} \cdot \mathbf{F} \cdot \mathbf{U} \frac{\sqrt{\mathbf{X}' \cdot \mathbf{X}'}}{c} d\lambda = \\
& = -q \int_{\lambda_0}^{\lambda_1} \delta\mathbf{X} \cdot \mathbf{F} \cdot \mathbf{X}' \, d\lambda = \\
& = -q \int_{\lambda_0}^{\lambda_1} \delta\mathbf{X} \cdot \left[\nabla\mathbf{A} - \nabla^T \mathbf{A} \right] \cdot \mathbf{X}' \, d\lambda = \quad \text{(see details, below)} \\
& = -\delta \int_{\lambda_0}^{\lambda_1} q\mathbf{A}(\mathbf{X}) \cdot \mathbf{X}' \, d\lambda = \\
& = -\delta \int_{\tau_0}^{\tau_1} q\mathbf{A}(\mathbf{X}) \cdot \mathbf{U} \, d\tau
\end{aligned}
$$

  The variational principle thus reads

$$
\begin{aligned}
0 = \delta S = & \\
= & \, \delta \int_{\tau_0}^{\tau_1} \left\{ -mc^2 - q\mathbf{A}(\mathbf{X}(\tau)) \cdot \mathbf{U}(\tau) \right\} \, d\tau = \\
= & \, \delta \int_{\lambda_0}^{\lambda_1} \left\{ -mc \sqrt{\mathbf{X}'(\lambda) \cdot \mathbf{X}'(\lambda)} - q\mathbf{A}\left(\mathbf{X}(\lambda)\right) \cdot \mathbf{X}'(\lambda) \right\} \, d\lambda \, .
\end{aligned}
$$

EM field force - details

$$\begin{aligned}\delta\int_{\lambda_0}^{\lambda_1}\mathbf{A}(\mathbf{X})\cdot\mathbf{X}'\,d\lambda &= \int_{\lambda_0}^{\lambda_1}\delta\mathbf{X}\cdot\nabla\mathbf{A}(\mathbf{X})\cdot\mathbf{X}'\,d\lambda+\int_{\lambda_0}^{\lambda_1}\mathbf{A}(\mathbf{X})\cdot\delta\mathbf{X}'\,d\lambda\\ &= \int_{\lambda_0}^{\lambda_1}\delta\mathbf{X}\cdot\nabla\mathbf{A}(\mathbf{X})\cdot\mathbf{X}'\,d\lambda+[\mathbf{A}(\mathbf{X})\cdot\delta\mathbf{X}]_{\lambda_0}^{\lambda_1}-\int_{\lambda_0}^{\lambda_1}\frac{d}{d\lambda}\mathbf{A}(\mathbf{X}(\lambda))\cdot\delta\mathbf{X}=\\ &= \int_{\lambda_0}^{\lambda_1}\delta\mathbf{X}\cdot\nabla\mathbf{A}(\mathbf{X})\cdot\mathbf{X}'\,d\lambda-\int_{\lambda_0}^{\lambda_1}\mathbf{X}'\cdot\mathbf{A}(\mathbf{X}(\lambda))\cdot\delta\mathbf{X}=\\ &= \int_{\lambda_0}^{\lambda_1}\delta\mathbf{X}\cdot\left[\nabla\mathbf{A}(\mathbf{X})-\nabla^T\mathbf{A}(\mathbf{X})\right]\cdot\mathbf{X}'\,d\lambda\end{aligned}$$

From the variational principle to the equations of motion

$$\begin{aligned}0 &= \delta\int_{\lambda_0}^{\lambda_1}\mathcal{L}\left(\mathbf{X}(\lambda),\mathbf{X}'(\lambda),\lambda\right)\,d\lambda=\\ &= \int_{\lambda_0}^{\lambda_1}\left\{\delta\mathbf{X}'(\lambda)\cdot\nabla_{\mathbf{X}'}\mathcal{L}+\delta\mathbf{X}(\lambda)\cdot\nabla_{\mathbf{X}}\mathcal{L}\right\}\,d\lambda=\\ &= \underbrace{\left[\delta\mathbf{X}\cdot\nabla_{\mathbf{X}'}\mathcal{L}\right]|_{\lambda_0}^{\lambda_1}}_{=0}-\int_{\lambda_0}^{\lambda_1}\delta\mathbf{X}\cdot\left\{\frac{d}{d\lambda}\left(\nabla_{\mathbf{X}'}\mathcal{L}\right)-\nabla_{\mathbf{X}}\mathcal{L}\right\}\,d\lambda\,,\end{aligned}$$

and, since $\delta\mathbf{X}$ must be arbitary, Lagrange equations follow

$$\frac{d}{d\lambda}\left(\nabla_{\mathbf{X}'}\mathcal{L}\right)-\nabla_{\mathbf{X}}\mathcal{L}=\mathbf{0}\,.$$

If the Lagrangian funcion is

$$\mathcal{L}(\mathbf{X},\mathbf{X}',\lambda)=-mc\sqrt{\mathbf{X}'(\lambda)\cdot\mathbf{X}'(\lambda)}-q\mathbf{A}\left(\mathbf{X}(\lambda)\right)\cdot\mathbf{X}'(\lambda)\,,$$

its derivatives are

$$\nabla_{\mathbf{X}'}\mathcal{L}=-\frac{mc}{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}\mathbf{X}'-q\mathbf{A}(\mathbf{X})=-m\mathbf{U}-q\mathbf{A}$$

$$\nabla_{\mathbf{X}}\mathcal{L}=-q\nabla\mathbf{A}\cdot\mathbf{X}'=-q\nabla\mathbf{A}\cdot\mathbf{U}\frac{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}{c}$$

and

$$\begin{aligned}\frac{d}{d\lambda}\nabla_{\mathbf{X}'}\mathcal{L} &= \frac{d\tau}{d\lambda}\frac{d}{d\tau}\left(-m\mathbf{U}-q\mathbf{A}\right)=\\ &= \frac{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}{c}\frac{d}{d\tau}\left(-m\mathbf{U}-q\mathbf{A}(\mathbf{X})\right)=\\ &= \frac{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}{c}\left(-m\frac{d}{d\tau}\mathbf{U}-q\mathbf{U}\cdot\nabla\mathbf{A}(\mathbf{X})\right)\,.\end{aligned}$$

Putting together all the pieces of the Lagrange equations, and dividing by $\frac{\sqrt{\mathbf{X}'\cdot\mathbf{X}'}}{c}$ - different from zero, if the 3-velocity of the particle is $|\mathbf{v}|<c$ -,

$$0=-m\frac{d\mathbf{U}}{d\tau}-q\mathbf{U}\cdot\nabla\mathbf{A}+q\nabla\mathbf{A}\cdot\mathbf{U}\,,$$

and thus

$$m\frac{d\mathbf{U}}{d\tau}=q\left[\nabla\mathbf{A}-\nabla^T\mathbf{A}\right]\cdot\mathbf{U}\,.$$

Using coordinates

With $\mathbf{X}\left(q^{\mu}(\lambda)\right)$,...

26.2.5 Quantum Mechanics

Schrodinger equation

$$i\hbar\frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi\rangle \ ,$$

Schrodinger equation for a particle with mass m in a conservative force field with potential energy $V(\mathbf{r})$,in position representation, becomes

$$i\hbar\partial_t\Psi = -\frac{\hbar^2}{2m}\nabla^2\Psi + V(\mathbf{r})\Psi \ ,$$

with $\Psi(\mathbf{r},t) : \Omega \times T \to \mathbb{C}$.

Derivative w.r.t. a complex number and its complex conjugate

See *Holomorphic functions*.

Derivative w.r.t. the conjugate conjugate variable

Let a function $f(z) = u(z) + iv(z) = U(x,y) + iV(x,y)$, with x, y independent variables. With a change of variables, and treating z and $w := z^*$ as independent variables[1]

$$\begin{cases} z = x + iy \\ z^* = x - iy \end{cases} \quad , \quad \begin{cases} x = \dfrac{1}{2}\left(z + z^*\right) \\ y = \dfrac{1}{2i}\left(z - z^*\right) \end{cases}$$

a function of z, and thus x, y is written as a function of z, z^*

$$\begin{aligned} u(z,z^*) &= U(x(z,z^*), y(z,z^*)) \\ v(z,z^*) &= V(x(z,z^*), y(z,z^*)) \end{aligned}$$

The partial derivatives of a function $f = u + iv = F(x,y) = f(z,z^*)$ read

$$\begin{aligned} \partial_z f|_{z^*} &= \partial_z x|_{z^*}\partial_x f|_y + \partial_z y|_{z^*}\partial_y f|_x = \\ &= \frac{1}{2}\partial_x f|_y - i\frac{1}{2}\partial_y f|_x = \\ &= \frac{1}{2}\left(\partial_x u + \partial_y v\right) + i\frac{1}{2}\left(\partial_x v - \partial_y u\right) \\ \partial_{z^*} f|_z &= \partial_{z^*} x|_z\partial_x f|_y + \partial_{z^*} y|_z\partial_y f|_x = \\ &= \frac{1}{2}\partial_x f|_y + i\frac{1}{2}\partial_y f|_x = \\ &= \frac{1}{2}\left(\partial_x u - \partial_y v\right) + i\frac{1}{2}\left(\partial_x v + \partial_y u\right) = \\ &= \frac{1}{2}\left(\partial_x u - \partial_y v\right) - i\frac{1}{2}\left(-\partial_x v - \partial_y u\right) \ . \end{aligned}$$

[1] This is the process leading to Wirtinger's derivative, see Wikipedia: Wirtinger derivative and Stack Exchange: What is the intuition behind the Wirtinger derivatives?. Does this assumption have any consequence? $x, y \in \mathbb{R}$, while $z, z^* \in \mathbb{C}$, and $z, w := z^*$ treated as independent.

Remark 1. If $f^{holo}(z)$ is **holomorphic**, then *Cauchy-Riemann conditions* hold, $u_{/x} = v_{/y}$, $u_{/y} = -v_{/x}$ and thus

$$\partial_z f^{holo}|_{z^*} = \partial_x u + i\partial_x v = \partial_y v - i\partial_y u \qquad , \qquad \partial_{z^*} f^{holo}|_z = 0 \, .$$

Remark 2. Comparing the expressions of $\partial_z f|_{z^*}$, and $\partial_{z^*} f|_z$, then

$$\left(\frac{\partial f}{\partial z^*}\right)\Big|_z = \left(\frac{\partial f^*}{\partial z}\right)^*\Big|_z \, .$$

Thus, if f^{real} is real-valued, then $f^{real} = f^{real\,*}$,

$$\left(\frac{\partial f^{real}}{\partial z^*}\right)\Big|_z = \left(\frac{\partial f^{real}}{\partial z}\right)^*\Big|_z \, .$$

Free-style approach

Multiplying Schrodinger equation by the c.c. of a test function $f(z)$, and integrating in both time and space domain,

$$\begin{aligned} 0 &= \int_T \int_\Omega f^* \left\{ -i\hbar\partial_t \Psi - \frac{\hbar^2}{2m}\partial_{kk}\Psi + V\Psi \right\} d\mathbf{r}\, dt = \\ &= \int_T \int_\Omega \left\{ -i\hbar\, f^* \partial_t \Psi + \frac{\hbar^2}{2m}\partial_k f^*\, \partial_k \Psi + f^* V \Psi \right\} d\mathbf{r}\, dt - \int_T \oint_{\partial\Omega} \frac{\hbar^2}{2m} f^* n_k \partial_k \Psi \, d\mathbf{r}\, dt \, . \end{aligned}$$

Now, let's

- add the complex conjugate
- define $f = \partial\Psi$
- set the test function equal to zero on $\partial\Omega$

$$\begin{aligned} 0 = &\int_T \int_\Omega \left\{ -i\hbar\, \delta\Psi^* \partial_t \Psi + \frac{\hbar^2}{2m}\partial_k \delta\Psi^*\, \partial_k \Psi + \delta\Psi^* V \Psi \right\} d\mathbf{r}\, dt + \\ &+ \int_T \int_\Omega \left\{ i\hbar\, \delta\Psi \partial_t \Psi^* + \frac{\hbar^2}{2m}\partial_k \delta\Psi\, \partial_k \Psi^* + \delta\Psi V \Psi^* \right\} d\mathbf{r}\, dt = \end{aligned}$$

Last two pairs of terms can be written as… while the first pair

$$\begin{aligned} \delta\left(\Psi^*\partial_t\Psi - \Psi\partial_t\Psi^*\right) &= \delta\Psi^*\partial_t\Psi + \Psi^*\delta\partial_t\Psi - \delta\Psi\partial_t\Psi^* - \Psi\partial_t\delta\Psi^* = \\ &= \delta\Psi^*\partial_t\Psi + \partial_t\left(\Psi^*\delta\Psi\right) - \delta\Psi^*\partial_t\Psi - \delta\Psi\partial_t\Psi^* - \partial_t\left(\Psi\delta\Psi^*\right) + \partial_t\Psi\delta\Psi^* = \\ &= 2\left(\delta\Psi^*\partial_t\Psi - \delta\Psi\partial_t\Psi^*\right) + \partial_t\left(\dots\right) \, . \end{aligned}$$

having used integration by parts and switched the variation and the partial derivative w.r.t. time (**but**, is that legal? See below (26.6)). Thus the variational principle - with prescribed $\delta\Psi$ at boundary of space and time domain - reads

$$0 = -\delta \int_T \int_\Omega \left\{ i\frac{\hbar}{2}\left(\Psi^*\partial_t\Psi - \Psi\partial_t\Psi^*\right) - \frac{\hbar^2}{2m}\nabla\Psi^* \cdot \nabla\Psi - \Psi^* V(\mathbf{r})\Psi \right\} d\mathbf{r}\, dt \, .$$

Lagrange equations

The function inside the integral is defined as the **Lagrangian function**

$$\mathcal{L}\left(\Psi, \partial_t\Psi, \partial_k\Psi; \Psi^*, \partial_t\Psi^*, \partial_k\Psi^*\right) = i\frac{\hbar}{2}\left(\Psi^*\partial_t\Psi - \Psi\partial_t\Psi^*\right) - \frac{\hbar^2}{2m}\nabla\Psi^* \cdot \nabla\Psi - \Psi^* V(\mathbf{r})\Psi \, , \tag{26.2}$$

treating the wave function, its derivatives and the complex conjugate functions as **independent variables**, as done for Wirtinger derivatives. Schrodinger equation can be retrieved from Lagrange equation

$$0 = \partial_t \left(\frac{\partial \mathcal{L}}{\partial \left(\partial_t \Psi^* \right)} \right) + \partial_k \left(\frac{\partial \mathcal{L}}{\partial \left(\partial_k \Psi^* \right)} \right) - \frac{\partial \mathcal{L}}{\partial \Psi^*} \, ,$$

taking the derivatives of the Lagrangian function $\mathcal{L}\left(\Psi, \partial_t \Psi, \partial_k \Psi\right)$ in formula (26.2).

Proof.

The partial derivatives w.r.t. the complex conjugate functions,

$$\begin{aligned}
\partial_t \left(\frac{\partial \mathcal{L}}{\partial \left(\partial_t \Psi^* \right)} \right) &= \partial_t \left(-i \frac{\hbar}{2} \Psi \right) = -i \frac{\hbar}{2} \partial_t \Psi \\
\partial_t \left(\frac{\partial \mathcal{L}}{\partial \left(\partial_k \Psi^* \right)} \right) &= \partial_t \left(-\frac{\hbar^2}{2m} \Psi_k \right) = -\frac{\hbar^2}{2m} \partial_{kk} \Psi \\
\frac{\partial \mathcal{L}}{\partial \Psi^*} &= i \frac{\hbar}{2} \partial_t \Psi - V(\mathbf{r}) \Psi \, ,
\end{aligned}$$

it immediately follows

$$\begin{aligned}
0 = \partial_t \left(\frac{\partial \mathcal{L}}{\partial \left(\partial_t \Psi^* \right)} \right) + \partial_k \left(\frac{\partial \mathcal{L}}{\partial \left(\partial_k \Psi^* \right)} \right) - \frac{\partial \mathcal{L}}{\partial \Psi^*} = \\
= -i \frac{\hbar}{2} \partial_t \Psi - \frac{\hbar^2}{2m} \partial_{kk} \Psi - \left(i \frac{\hbar}{2} \partial_t \Psi - V(\mathbf{r}) \Psi \right) = \\
= -i \hbar \partial_t \Psi - \frac{\hbar^2}{2m} \partial_{kk} \Psi + V(\mathbf{r}) \Psi \, .
\end{aligned}$$

A more rigorous approach

Let $\psi(\mathbf{r}, t) = \Psi(q^j(\mathbf{r}, t), \mathbf{r}, t)$ be a function of generalized coordinates $q^j(\mathbf{r}, t)$, function of the independent variables $\mathbf{r}$, t. Its partial time and space derivatives read

$$\begin{aligned}
\partial_t \psi|_{\mathbf{r}} &= \partial_t \Psi|_{\mathbf{r}} = \partial_t q^j \, \partial_{q^j} \Psi|_{\mathbf{r},t} + \partial_t \Psi|_{\mathbf{q},\mathbf{r}} \\
\partial_k \psi|_{t} &= \partial_k \Psi|_{t} = \partial_k q^j \, \partial_{q^j} \Psi|_{\mathbf{r},t} + \partial_k \Psi|_{\mathbf{q},\mathbf{r}}
\end{aligned}$$

Thus, the derivatives of the wave function $\partial_t \Psi$, and $\partial_k \Psi$ are functions of the independent variables $\mathbf{r}$, t, the generalized functions q^j and their derivatives $\partial_t q^j$, $\partial_k q^j$ respectively. The following relations follow

$$\begin{aligned}
\partial_{q^j} \Psi &= \frac{\partial \left(\partial_t \psi \right)}{\partial \left(\partial_t q^j \right)} \\
\partial_{q^j} \Psi &= \frac{\partial \left(\partial_k \psi \right)}{\partial \left(\partial_k q^j \right)} \, .
\end{aligned}$$

Multiplying Schrodinger equation

$$
\begin{aligned}
0 &= \partial_{q^j}\Psi^* \left\{ -i\hbar\partial_t\psi - \frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r})\psi \right\} + \partial_{q^j}\Psi \left\{ i\hbar\partial_t\psi^* - \frac{\hbar^2}{2m}\nabla^2\psi^* + V(\mathbf{r})\psi^* \right\} = \\
&= -i\hbar\partial_{q^j}\Psi^*\partial_t\psi - \frac{\hbar^2}{2m}\partial_{q^j}\Psi^*\partial_{kk}\psi + \partial_{q^j}\Psi^* V(\mathbf{r})\psi + i\hbar\partial_{q^j}\Psi\partial_t\psi^* - \frac{\hbar^2}{2m}\partial_{q^j}\Psi\partial_{kk}\psi^* + \partial_{q^j}\Psi V(\mathbf{r})\psi^* = \\
&= -i\hbar\left(\partial_{q^j}\Psi^*\partial_t\psi - \partial_{q^j}\Psi\partial_t\psi^*\right) - \frac{\hbar^2}{2m}\left(\partial_{q^j}\Psi^*\partial_{kk}\psi + \partial_{q^j}\Psi\partial_{kk}\psi^*\right) + \partial_{q^j}\Psi^* V\Psi + \Psi^* V\partial_{q^j}\Psi = \\
&= \partial_{q^j}\left\{ -\frac{i\hbar}{2}\left(\Psi^*\partial_t\Psi - \Psi\partial_t\Psi^*\right) + \frac{\hbar^2}{2m}\left(\nabla\Psi\cdot\nabla\Psi^*\right) + \Psi^* V(\mathbf{r})\Psi \right\} + \\
&\quad - \partial_t\left\{ \frac{\partial}{\partial(\partial_t q^i)}\left\{ -\frac{i\hbar}{2}\left(\Psi^*\partial_t\Psi - \Psi\partial_t\Psi^*\right) \right\} \right\} + \\
&\quad - \partial_k\left\{ \frac{\partial}{\partial(\partial_k q^i)}\left\{ \frac{\hbar^2}{2m}\nabla\Psi\cdot\nabla\Psi^* \right\} \right\} = \\
&= -\partial_{q^j}\mathcal{L} + \partial_t\frac{\partial\mathcal{L}}{\partial(\partial_t q^i)} + \partial_k\frac{\partial\mathcal{L}}{\partial(\partial_k q^i)} \,,
\end{aligned}
$$

where the Lagrangian function $\mathcal{L}$ defined as

$$
\mathcal{L}(q^j(\mathbf{r},t), \partial_t q^j(\mathbf{r},t), \partial_k q^j(\mathbf{r},t), \mathbf{r}, t) = i\frac{\hbar}{2}\left(\Psi^*\partial_t\Psi - \Psi\partial_t\Psi^*\right) - \frac{\hbar^2}{2m}\nabla\Psi^*\cdot\nabla\Psi - \Psi^* V(\mathbf{r})\Psi \,.
$$

Details are given in the following box. The second and the third term in the Lagrangian function don't depend on $\partial_t q^i$, the first and the third term don't depend on $\partial_k q^i$.

Details

- First pair of terms

$$
\begin{aligned}
\partial_{q^j}\Psi^*\partial_t\psi - \partial_{q^j}\Psi\partial_t\psi^* &= \frac{1}{2}2\left(\partial_{q^j}\Psi^*\partial_t\psi - \partial_{q^j}\Psi\partial_t\psi^*\right) = \\
&= \frac{1}{2}\left(\partial_{q^j}\left(\Psi^*\partial_t\psi\right) - \Psi^*\partial_{q^j}\partial_t\Psi - \partial_t\left(\Psi^*\partial_{q^j}\Psi\right) + \partial_t\partial_{q^j}\Psi^*\psi - \text{c.c.}\right) = \\
&= \frac{1}{2}\left(\partial_{q^j}\left(\Psi^*\partial_t\psi\right) - \Psi^*\partial_{q^j}\partial_t\Psi - \partial_t\left(\Psi^*\partial_{q^j}\Psi\right) + \partial_t\partial_{q^j}\Psi^*\psi - \text{c.c.}\right) = \\
&= \frac{1}{2}\left(\partial_{q^j}\left(\Psi^*\partial_t\psi - \Psi\partial_t\psi^*\right) - \partial_t\left(\Psi^*\partial_{q^j}\Psi - \Psi\partial_{q^j}\Psi^*\right)\right) = \\
&= \frac{1}{2}\left(\partial_{q^j}\left(\Psi^*\partial_t\psi - \Psi\partial_t\psi^*\right) - \partial_t\left(\Psi^*\frac{\partial(\partial_t\Psi)}{\partial(\partial_t q^j)} - \Psi\frac{\partial(\partial_t\Psi^*)}{\partial(\partial_t q^j)}\right)\right) = \\
&= \frac{1}{2}\partial_{q^j}\left(\Psi^*\partial_t\psi - \Psi\partial_t\psi^*\right) - \frac{1}{2}\partial_t\frac{\partial}{\partial(\partial_t q^j)}\left(\Psi^*\partial_t\Psi - \Psi\partial_t\Psi^*\right) =
\end{aligned}
$$

- Second pair of terms

$$
\begin{aligned}
-\partial_{q^j}\Psi^*\partial_{kk}\psi - \partial_{q^j}\Psi\partial_{kk}\psi^* &= -\partial_k\left(\partial_{q^j}\Psi^*\partial_k\psi\right) + \partial_k\partial_{q^j}\Psi^*\partial_k\psi - \partial_k\left(\partial_{q^j}\Psi\partial_k\psi^*\right) + \partial_k\partial_{q^j}\Psi\partial_k\psi^* = \\
&= -\partial_k\left(\partial_{q^j}\Psi^*\partial_k\psi + \partial_{q^j}\Psi\partial_k\psi^*\right) + \partial_k\partial_{q^j}\Psi^*\partial_k\psi + \partial_k\partial_{q^j}\Psi\partial_k\psi^* = \\
&= -\partial_k\left(\frac{\partial(\partial_k\psi^*)}{\partial(\partial_k q^j)}\partial_k\psi + \frac{\partial(\partial_k\psi)}{\partial(\partial_k q^j)}\partial_k\psi^*\right) + \partial_k\partial_{q^j}\Psi^*\partial_k\psi + \partial_k\partial_{q^j}\Psi\partial_k\psi^* = \\
&= -\partial_k\frac{\partial}{\partial(\partial_k q^j)}\left(\partial_k\psi^*\partial_k\psi\right) + \partial_{q^j}\partial_k\psi^*\partial_k\psi + \partial_{q^j}\partial_k\psi\partial_k\psi^* = \\
&= -\partial_k\frac{\partial}{\partial(\partial_k q^j)}\left(\partial_k\psi^*\partial_k\psi\right) + \partial_{q^j}\left(\partial_k\psi^*\partial_k\psi\right) \,.
\end{aligned}
$$

- Third pair of terms

$$\partial_{q^j}\Psi^* V\psi + \partial_{q^j}\Psi V\psi^* = \partial_{q^j}\Psi^* V\Psi + \partial_{q^j}\Psi V\Psi^* = \\ = \partial_{q^j}\left\{\Psi^* V\Psi\right\} \ .$$

The variation of function Ψ

$$\delta\Psi = \delta q^j \partial_{q^j}\Psi \ .$$

Mixed derivatives

$$\partial_k\left(\partial_{q^j}\Psi\big|_{\mathbf{r},t}\right)\Big|_t = \partial_k q^m \partial_{q^m}\partial_{q^j}\Psi + \partial_k\partial_{q^j}\Psi = \\ = \partial_{q^j}\left(\partial_k q^m \partial_{q^m}\Psi + \partial_k\Psi\right) = \\ = \partial_{q^j}\left(\partial_k\psi\big|_t\right)\Big|_{\mathbf{r},t} \ .$$

Variation of a function

Function of 1-variable t**.** Let f be a function of t, $q(t)$ and its first n derivatives $q^k(t)$, i.e.

$$f\left(q^{(n)}(t), \dots, q(t), t\right) \ .$$

The variation of this function w.r.t. $q(t)$ is defined through an arbitrary variation $\delta q(t)$ as

$$\delta f = \sum_k \delta q^{(k)}\, \partial_{q^{(k)}} f \ . \tag{26.3}$$

…**todo** *Add the definitions for multi-dimensional functions, and independent variables,…*

Exchanging derivatives and variation of the function $f\left(q^i(x^k), x^k\right)$

Let $F\left(x^k\right) = f(q^i(x^k), x^k)$ be a function of the x^k, through q^k - or not, if there's no explicit dependence on x^k,

$$\partial_{x^k} f = \partial_{x^k} q^j \partial_{q^j} f + \partial_{x^k} f \\ \delta f = \delta q^i \partial_{q^i} f$$

The derivative $\partial_{x^k} f$ is a function of x^k, $q^i(x^k)$, $\partial_{x^m} q^i(x^k)$. Thus, using the definition (26.3), it follows

$$\delta\partial_{x^k} f = \delta\left(\partial_{x^k} q^j \partial_{q^j} f + \partial_{x^k} f\right) = \\ = \delta q^i \partial_{q^i}\left(\dots\right) + \partial_{x^m}\delta q^i\, \partial_{\partial_{x^m} q^i}\left(\dots\right) = \\ = \delta q^i\left(\partial_x q^j \partial_{q^i q^j} f + \partial_{q^i x^k} f\right) + \partial_{x^m}\delta q^i \delta^m_k \delta^j_i \partial_{q^j} f = \\ = \delta q^i\left(\partial_x q^j \partial_{q^i q^j} f + \partial_{q^i x^k} f\right) + \partial_{x^k}\delta q^i \partial_{q^i} f \ . \tag{26.4}$$

$$\partial_{x^k}\delta f = \partial_{x^k}\left(\delta q^i \partial_{q^i} f\right) = \\ = \partial_{x^k}\delta q^i \partial_{q^i} f + \delta q^i\left(\partial_{q^j q^i} f \partial_{x^k} q^j + \partial_{x^k q^i} f\right) \ . \tag{26.5}$$

Comparing the expressions (26.4) and (26.5), it follows that

$$\delta\partial_{x^k} f = \partial_{x^k}\delta f \ .$$

If the variation is performed only on the function $q^i(x^k)$, a similar property holds for any pair of indices i, k

$$\delta_{q^i}\partial_{x^k} f = \partial_{x^k}\delta_{q^i} f \,, \tag{26.6}$$

with the definition $\delta q^i f = \delta q^i \partial_{q^i} f$, wtihout summing on the pair of repeated.

26.3 Extra

26.3.1 Parametrization of the integral with the arc-length

Parametrization of the curve $\gamma_{AB} : \mathbf{r}(s)$, with a physical parameter $s \in [s_A, s_B]$ whose values at extreme points may depend on the solution itself: i.e. even though $\mathbf{r}_A$, $\mathbf{r}_B$ are prescribed, the values s_A, s_B, s.t. $\mathbf{r}_A = \mathbf{r}(s_A)$, $\mathbf{r}_B = \mathbf{r}(s_B)$, are not. The functional reads

$$S_{\gamma_{AB}} = S[\mathbf{r}(s)] = \int_{\mathbf{r}\in\gamma_{AB}} \mathcal{L}(\mathbf{r}, s)\, ds = \int_{s_A}^{s_B} \mathcal{L}(\mathbf{r}(s), s)\, ds$$

Method 1.

$$\begin{aligned}\Delta S_{\gamma_{AB}} &= S[\mathbf{r}(s_\delta) + \varepsilon\delta\mathbf{r}(s_\delta)] - S[\mathbf{r}(s) + \varepsilon\delta\mathbf{r}(s)] = \\ &= \int_{\mathbf{r}\in\mathbf{r}(s_\delta)+\varepsilon\delta\mathbf{r}(s_\delta)} \mathcal{L}(\mathbf{r}, s_\delta)\, ds_\delta - \dots = \end{aligned}$$

Here ds_δ is the arc-length of the elementary vector of the curve γ_δ, $d\mathbf{r}_\delta = d\mathbf{r} + \varepsilon d\delta\mathbf{r}$, i.e.

$$\begin{aligned} ds_\delta^2 &= (d\mathbf{r} + \varepsilon d\delta\mathbf{r}) \cdot (d\mathbf{r} + \varepsilon d\delta\mathbf{r}) = \\ &= |d\mathbf{r}|^2 + 2\varepsilon d\mathbf{r}\cdot d\delta\mathbf{r} + o(\varepsilon) = \\ &\simeq |d\mathbf{r}|^2 \left(1 + 2\varepsilon \frac{d\delta\mathbf{r}\cdot d\mathbf{r}}{|d\mathbf{r}|^2}\right) + o(\varepsilon)\,, \end{aligned}$$

so that first-order approximation reads

$$\begin{aligned} ds_\delta &= |d\mathbf{r}| \left(1 + \varepsilon \frac{d\delta\mathbf{r}\cdot d\mathbf{r}}{|d\mathbf{r}|^2}\right) = \\ &= ds + \varepsilon\hat{\mathbf{t}}\cdot d\delta\mathbf{r}\,. \end{aligned}$$

The first integral in the difference thus becomes

$$\begin{aligned} &\int_{s_\delta=s_{\delta,A}}^{s_{\delta,B}} \mathcal{L}(\mathbf{r}_\delta(s_\delta), s_\delta)\left(ds + \varepsilon\hat{\mathbf{t}}\cdot d\delta\mathbf{r}\right) \\ &\quad= \int_{s_\delta=s_{\delta,A}}^{s_{\delta,B}} \left[\mathcal{L}(\mathbf{r}(s_\delta), s_\delta) + \varepsilon\delta\mathbf{r}\cdot\nabla\mathcal{L}(\mathbf{r}(s_\delta), s_\delta) + o(\varepsilon)\right]\left(\frac{ds}{ds_\delta} + \varepsilon\hat{\mathbf{t}}\cdot\frac{d\delta\mathbf{r}}{ds_\delta}\right) ds_\delta = \\ &\quad= \int_{s=s_A}^{s_B} \mathcal{L}(\mathbf{r}(s), s)\, ds + \varepsilon\int_{s=s_A}^{s_B} \delta\mathbf{r}\cdot\nabla\mathcal{L}(\mathbf{r}(s), s)\, ds + \\ &\quad+ \varepsilon\left[\delta\mathbf{r}\cdot\hat{\mathbf{t}}\mathcal{L}\right]\Big|_A^B - \varepsilon\int_{s_\delta=s_{\delta,A}}^{s_{\delta,B}} \delta\mathbf{r}\cdot\frac{d}{ds}\left(\hat{\mathbf{t}}\mathcal{L}\right) ds_\delta + o(\varepsilon) = \\ &\quad= \int_{s=s_A}^{s_B} \mathcal{L}(\mathbf{r}(s), s)\, ds + \varepsilon\int_{s=s_A}^{s_B} \delta\mathbf{r}\cdot\left\{\nabla\mathcal{L}(\mathbf{r}(s), s) - \frac{d}{ds_\delta}\left(\hat{\mathbf{t}}\mathcal{L}\right)\right\} ds + \\ &\quad+ \varepsilon\left[\delta\mathbf{r}\cdot\hat{\mathbf{t}}\mathcal{L}\right]\Big|_A^B + o(\varepsilon)\,. \end{aligned}$$

Thus, the variation becomes

$$\delta_{\mathbf{r}} S = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \Delta_{\mathbf{r}} S = \\ = \int_{s=s_A}^{s_B} \delta\mathbf{r} \cdot \left\{ \nabla \mathcal{L}(\mathbf{r}(s), s) - \frac{d}{ds} \left(\hat{\mathbf{t}} \mathcal{L} \right) \right\} ds + \left[\delta\mathbf{r} \cdot \hat{\mathbf{t}} \mathcal{L} \right] \Big|_A^B$$

having assumed a function $s(s_\delta)$ associating every point on curve γ_{AB} to a value of the parameter s_δ, and having used the rules for changing independent variables in the integrals, $\frac{ds}{ds_\delta} ds_\delta = d_s$, and for the derivative of compsite functions $\frac{d}{ds_\delta} = \frac{ds}{ds_\delta} \frac{d}{ds}$.

Fermat principle

Following *Fermat principle*, the differential equations of the trajectory of a light ray come from the condition

$$\delta \int_\gamma \frac{1}{c} \, ds = \frac{1}{c_0} \delta \int_\gamma n \, ds \, .$$

Here the integrand is $\mathcal{L}(\mathbf{r}(s), s) = n(\mathbf{r}(s), s)$, and following the general formulation of the variational problem with prescribed ends $\delta\mathbf{r}|_{A,B} = \mathbf{0}$ gives

$$\mathbf{0} = \nabla \mathcal{L} - \frac{d}{ds} \left(\hat{\mathbf{t}} \mathcal{L} \right) = \\ = \nabla \mathcal{L} - \kappa \hat{\mathbf{n}} \mathcal{L} - \hat{\mathbf{t}} \hat{\mathbf{t}} \cdot \mathcal{L} = \\ = \mathbb{P}_{\perp \hat{\mathbf{t}}} \cdot \nabla \mathcal{L} - \kappa \hat{\mathbf{n}} \mathcal{L} \, ,$$

i.e. the very same equations as (26.1), having recalled that the derivative of the tangent unit vector to a line w.r.t. the arc-length is proportional to the unit normal vector, through the curvature, $\frac{d\hat{\mathbf{t}}}{ds} = \kappa \hat{\mathbf{n}}$.

Method 2.

26.3.2 Integral with non prescirbed ends

Part IX

Ordinary Differential Equations

CHAPTER

TWENTYSEVEN

INTRODUCTION TO ORDINARY DIFFERENTIAL EQUATIONS

todo *Introduction.*

Initial value problems.

- *Linear ODEs with constant coefficients*
- *Linear ODEs in state-space representation*
- *Numerical solution of initial value problems*

Boundary value problems. *todo*

- link to numerical methods for PDEs*: steady-state version of PDEs often leads to a boundary value problem governed by a ODE, FDM, FEM, FVM*

Links

- *System theory*
- *Control theory*

CHAPTER

TWENTYEIGHT

LINEAR EQUATIONS

28.1 Linear equations with constant coefficients

28.1.1 Definition

28.1.2 General solution

$$y(x) = y_o(x) + y_p(x)$$

Solution of the homogeneous equation

Independent solutions for non-multiple root

$$\begin{aligned} 0 &= a_n y^{(n)}(x) + \dots a_1 y'(x) + a_0 y(x) = \\ &= a_n \left(-s_1 + \frac{d}{dx}\right) \dots \left(-s_n + \frac{d}{dx}\right) y(x) \,, \end{aligned}$$

If all the roots are non-multiple, then n independent solutions are found setting the result of every individual operator $\left(-s_k + \frac{d}{dx}\right)$ on $y(x)$ equal to zero,

$$0 = \left(-s_k + \frac{d}{dx}\right) y(x) = -s_k y(x) + y'(x) \,,$$

whose solution is $y_k(x) = A_k e^{s_k x}$.

Independent solutions for roots with miltiplicity $p > 1$

Independent solutions for root with multiplicity $p = 2$

$$0 = \left(-s_k + \frac{d}{dx}\right)^2 y(x) = \left(-s_k + \frac{d}{dx}\right)\left(-s_k + \frac{d}{dx}\right) y(x) \,.$$

In order to satisfy this equation, one of the the following two conditions must hold

$$\begin{aligned} -sy(x) + y'(x) &= 0 \\ -sy(x) + y'(x) &= A\, e^{sx} \end{aligned}$$

The solution of the first condition is

$$-sy(x) + y'(x) = 0 \qquad \rightarrow \qquad y(t) = Ce^{sx}$$

The solution of the second condition is

$$\begin{aligned} -sy(x) + y'(x) &= A\,e^{sx} \\ e^{-sx}\left(-sy(x) + y'(x)\right) &= A \\ \frac{d}{dx}\left(e^{-sx}\,y(x)\right) &= A \\ \int \frac{d}{dx}\left(e^{-sx}y(x)\right) &= A\,x + B \\ e^{-sx}y(x) = A\,x + B \qquad &\rightarrow \qquad y(x) = A\,x\,e^{sx} + B\,e^{sx}\,, \end{aligned}$$

Independent solutions for root with multiplicity $p = 3$

$$0 = \left(-s_k + \frac{d}{dx}\right)^3 y(x) = \left(-s_k + \frac{d}{dx}\right)\left(-s_k + \frac{d}{dx}\right)\left(-s_k + \frac{d}{dx}\right) y(x)\,.$$

In order to satisfy this equation, one of the the following two conditions must hold

$$\begin{aligned} -sy(x) + y'(x) &= 0 \\ -sy(x) + y'(x) &= A\,e^{sx} \\ -sy(x) + y'(x) &= A\,e^{sx} + B\,x\,e^{sx} \end{aligned}$$

The solution of the first condition is…

The solution of the second condition is…

The solution of the third condition is (and contains the solution of the first and the second condition)

$$\begin{aligned} -sy(x) + y'(x) &= A\,e^{sx} + B\,x\,e^{sx} \\ e^{-sx}\left(-sy(x) + y'(x)\right) &= A + Bx \\ \frac{d}{dx}\left(e^{-sx}\,y(x)\right) &= A + Bx \\ \int \frac{d}{dx}\left(e^{-sx}y(x)\right) &= Ax + \widetilde{B}x^2 + C \\ e^{-sx}y(x) = A\,x + B \qquad &\rightarrow \qquad y(x) = C_0 e^{sx} + C_1\,x\,e^{sx} + C_2\,x^2\,e^{sx}\,. \end{aligned}$$

Independent solutions for root with multiplicity p

$$y(x) = C_0 e^{sx} + C_1\,x\,e^{sx} + \ldots C_{p-1}\,x^{p-1}\,e^{sx}$$

Asymptotic stability of the homogeneous equation

The asymptotic stability of the homogeneous equation with non-zero initial conditions is equivalent to the asymptotic stability of the equation to an impulsive force, see *LTI: impulsive force*.

todo *how many equilibrium points? If only one equilibrium exists, it's possible to talk about the stability of the whole system, instead of the stability of the individual equilibrium point (without the need of the representation of the phase space)*

todo *General definition of asymptotic stability of an equilibrium.*

- stable equilibrium: $y(x) \to \overline{y}$, as $x \to +\infty$
- unstable equilibrium: $|y(x) - \overline{y}| \to +\infty$, as $x \to +\infty$
- marginally stable equilibrium: if not stable or unstable, i.e. $|y(x) - \overline{y}| < M$, without being asymptotically stable

The conditions for asymptotic stability immediately follows from the general solution of the homogeneous equation:

- asymptotically stable: all the eigenvalues have negative real part, so that

$$|e^{s_k x}| = |e^{\text{re}\{s\}x} e^{j\text{im}\{s\}}| = e^{\text{re}\{s\}x} \to 0 \quad , \quad \text{as } x \to +\infty \text{ if re}\{s\} < 0$$

- unstable: there is either
  - an eigenvalue with positive real part

$$|e^{s_k x}| = |e^{\text{re}\{s\}x} e^{j\text{im}\{s\}}| = e^{\text{re}\{s\}x} \to +\infty \quad , \quad \text{as } x \to +\infty \text{ if re}\{s\} < 0$$

  - or a multiple eigenvalue with zero real part. As an example a root $s = j\omega$ with multiplicity $p = 2$

$$\left|e^{j\omega x}(C_0 + C_1 x)\right| = |C_1 x| \to +\infty \quad , \quad \text{as } x \to +\infty$$

    if $C_1 \neq 0$ (but this condition is purely ideal in practical applications, as a perfect zero hardly exists in nature). The same relation holds, when evaluating the pair oc conjugate complex elementary solution, $e^{j\omega}(C_0 + C_1 x) + e^{-j\omega}(C_0^* + C_1^* x)$

- marginally-stable, if all the eigenvalues have negative real part, except for a set of imaginary eigenvalues with multiplicity $p = 1$. The asymptotic solution becomes

$$y(x) \to \sum_{k \in \text{marginal}} C_k e^{j\omega_k x}$$

Particular solution

CHAPTER

TWENTYNINE

LINEAR SYSTEMS

29.1 General solution

The general solution of the initial value problem

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{x}(t_0) &= \mathbf{x}_0 \end{aligned} \tag{29.1}$$

with no impulsive forcing in t_0 reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x_0} + \int_{\tau=t_0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \tag{29.2}$$

with

$$\begin{cases} \frac{\partial}{\partial t}\Phi(t, t_0) = \mathbf{A}(t)\Phi(t, t_0) \\ \Phi(t, t_0) = \mathbf{I}\,, \end{cases}$$

or alternatively, $\partial_{t_0}\Phi(t, t_0) = \Phi(t, t_0)\mathbf{A}(t_0)$.

General solution - details

Pre-multiplication of equation (29.1) by $\phi(t)$, with $\dot{\phi}(t) = -\phi(t)\mathbf{A}(t)$ gives

$$\begin{aligned} \phi(t)\dot{\mathbf{x}} - \phi(t)\mathbf{A}\mathbf{x} &= \phi(t)\mathbf{B}\mathbf{u} \\ \phi(t)\dot{\mathbf{x}} + \dot{\phi}(t)\mathbf{x} &= \phi(t)\mathbf{B}\mathbf{u} \\ \frac{d}{dt}\left(\phi(t)\mathbf{x}\right) &= \phi(t)\mathbf{B}\mathbf{u} \end{aligned}$$

and integration - without impulsive forces in $t = t_0$, that would cause jump in initial conditions, see *section about impulsive force* -

$$\phi(t)\mathbf{x} - \phi(t_0)\mathbf{x_0} = \int_{\tau=t_0}^{t} \phi(\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau$$

and thus

$$\mathbf{x}(t) = \phi^{-1}(t)\phi(t_0)\mathbf{x_0} + \int_{\tau=t_0}^{t} \phi^{-1}(t)\phi(\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau$$

or, defining $\Phi(a,b) := \phi^{-1}(a)\phi(b)$,

$$\mathbf{x}(t) = \Phi(t,t_0)\mathbf{x_0} + \int_{\tau=t_0}^{t} \Phi(t,\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau$$

with $\Phi(t,t_0)$ satisfying the initial value problem

$$\begin{cases} \frac{\partial}{\partial t}\Phi(t,t_0) = \mathbf{A}(t)\Phi(t,t_0) \\ \Phi(t,t_0) = \mathbf{I}\,. \end{cases}$$

The initial conditions follows from the solution, as the integral identically vanishes as $t \to t_0$ if there's no impulsive force in t_0. The matrix differential equation follows from the definition of the function $\Phi(t,\tau)$ in terms of $\phi(t)$ and from the time derivative of the inverse of a matrix

$$\frac{d}{dt}\phi^{-1} = -\phi^{-1}\dot{\phi}\phi^{-1}\,,$$

as $\mathbf{0} = \frac{d}{dt}\mathbf{I} = \frac{d}{dt}\left(\phi^{-1}\phi\right)$.

Time derivative of $\Phi(t,\tau)$ w.r.t. the first argument reads

$$\begin{aligned} \partial_t\Phi(t,\tau) &= \partial_t\left(\phi^{-1}(t)\phi(\tau)\right) = \\ &= \frac{d}{dt}\phi^{-1}(t)\,\phi(\tau) = \\ &= -\phi^{-1}(t)\dot{\phi}(t)\phi^{-1}(t)\,\phi(\tau) = \\ &= \underbrace{\phi^{-1}(t)\phi(t)}_{=\mathbf{I}}\mathbf{A}(t)\,\underbrace{\phi^{-1}(t)\,\phi(\tau)}_{\Phi(t,\tau)} = \\ &= \mathbf{A}(t)\Phi(t,\tau)\,. \end{aligned}$$

Derivative w.r.t. the second argument simply reads

$$\begin{aligned} \partial_\tau\Phi(t,\tau) &= \phi^{-1}(t)\frac{d}{d\tau}\phi(\tau) = \\ &= \phi^{-1}(t)\phi(\tau)\mathbf{A}(\tau) = \\ &= \Phi(t,\tau)\mathbf{A}(\tau)\,. \end{aligned}$$

29.2 Stochastic response

29.2.1 Correlation matrix

Using the general expression of the solution (29.2) of the linear system (29.1), the following relation holds for the correlation $\mathbf{R_{xx}}(t,\tau) := \mathbb{E}[\mathbf{x}(t)\mathbf{x}^*(\tau)]$,

$$\mathbf{R_{xx}}(t,\tau) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R_{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\,d\chi\,,$$

or for $\mathbf{P_{xx}}(t) := \mathbf{R_{xx}}(t,t)$

$$\mathbf{P_{xx}}(t) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{t} \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R_{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(t,\chi)d\lambda\,d\chi\,.$$

Details

$$
\begin{aligned}
\mathbf{R_{xx}}(t,\tau) &= \mathbb{E}\left[\mathbf{x}(t)\mathbf{x}^*(\tau)\right] = \\
&= \mathbb{E}\left[\Phi(t,t_0)\mathbf{x_0}\mathbf{x_0}^*\Phi^*(t,t_0)\right] + \\
&+ \mathbb{E}\left[\int_{\lambda=t_0}^{t}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{u}(\lambda)d\lambda\,\mathbf{x_0}^*\Phi^*(t,t_0)\right] + \\
&+ \mathbb{E}\left[\Phi(t,t_0)\mathbf{x_0}\int_{\lambda=t_0}^{t}\mathbf{u}^*(\lambda)\mathbf{B}^*(\lambda)\Phi^*(t,\lambda)d\lambda\right] + \\
&+ \mathbb{E}\left[\int_{\lambda=t_0}^{t}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{u}(\lambda)d\lambda\left(\int_{\chi=t_0}^{\tau}\Phi(\tau,\chi)\mathbf{B}(\chi)\mathbf{u}(\chi)d\chi\right)^*\right] ,
\end{aligned}
$$

and applying the expectation opearator only on stochastic functions,

$$
\begin{aligned}
\mathbf{R_{xx}}(t,\tau) &= \Phi(t,t_0)\mathbb{E}\left[\mathbf{x_0}\mathbf{x_0}^*\right]\Phi^*(t,t_0) + \\
&+ \int_{\lambda=t_0}^{t}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbb{E}\left[\mathbf{u}(\lambda)\mathbf{x}_0^*\right]d\lambda\Phi^*(t,t_0) + \\
&+ \Phi(t,t_0)\int_{\lambda=t_0}^{t}\mathbb{E}\left[\mathbf{x_0}\mathbf{u}^*(\lambda)\right]\mathbf{B}^*(\lambda)\Phi^*(t,\lambda)d\lambda + \\
&+ \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbb{E}\left[\mathbf{u}(\lambda)\mathbf{u}^*(\chi)\right]\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\,d\chi .
\end{aligned}
$$

Assuming uncorrelated input noise and uncertainty on the initial conditions, $\mathbb{E}[\mathbf{u}(t)\mathbf{x}_0^*] = \mathbf{0}$, and thus

$$
\mathbf{R_{xx}}(t,\tau) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R_{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\,d\chi ,
$$

with the definition

$$
\mathbf{P_{xx}}(t) := \mathbf{R_{xx}}(t,t) .
$$

If the input is a white noise, $\mathbf{R_{uu}}(t,\tau) = \mathbf{U}(t)\delta(t-\tau)$,

$$
\mathbf{P_{xx}}(t) = \Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda)d\lambda
$$

Details

$$
\begin{aligned}
&\int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{R_{uu}}(\lambda,\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\,d\chi = \\
&= \int_{\lambda=t_0}^{t}\int_{\chi=t_0}^{\tau}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\delta(\lambda-\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\lambda\,d\chi = \\
&= \int_{\lambda=t_0}^{\min\{t,\tau\}}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda)d\lambda ,
\end{aligned}
$$

as

$$
\int_{\chi=t_0}^{\tau}\delta(\lambda-\chi)\mathbf{B}^*(\chi)\Phi^*(\tau,\chi)d\chi = \begin{cases}\mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda) & , \quad \text{if } \lambda\in[t_0,\tau] \\ 0 & , \quad \text{otherwise}\end{cases}
$$

Lyapunov equation of the correlation matrix

Matrix $\mathbf{P}(t)$ satisfies the following Lyapunov equation,

$$\dot{\mathbf{P}} = \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^* + \mathbf{B}\mathbf{U}\mathbf{B}^* \,. \tag{29.3}$$

Details

$$\begin{aligned}
\frac{d\mathbf{P}}{dt} &= \partial_t\Phi\mathbf{P}_0\Phi + \Phi\mathbf{P}_0\partial_t\Phi + \underbrace{\Phi(t,t)}_{\mathbf{I}}\mathbf{B}(t)\mathbf{U}(t)\mathbf{B}^*(t)\,\underbrace{\Phi(t,t)}_{=\mathbf{I}} + \\
&+ \int_{\lambda=t_0}^{t} \left\{\partial_t\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi^*(\tau,\lambda) + \Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\partial_t\Phi^*(\tau,\lambda)\right\} d\lambda = \\
&= \mathbf{B}(t)\mathbf{U}(t)\mathbf{B}^*(t) + \mathbf{A}(t)\underbrace{\left[\Phi(t,t_0)\mathbf{P}_0\Phi^*(t,t_0) + \int_{\lambda=t_0}^{t}\Phi(t,\lambda)\mathbf{B}(\lambda)\mathbf{U}(\lambda)\mathbf{B}^*(\lambda)\Phi(t,\lambda)\right]}_{=\mathbf{P}(t)} + [\ldots]^*\,\mathbf{A}^*(t) = \\
&= \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^* + \mathbf{B}\mathbf{U}\mathbf{B}^* \,.
\end{aligned}$$

CHAPTER

THIRTY

INTEGRATION SCHEMES FOR CAUCHY PROBLEMS

OLD-Contents

Schemes.

- Euler methods, CN
- RK
- Multi-step: AB, AM, BDF,…
- Other methods:
  - Newmark-beta, Verlet (Leapfrog),…

Concepts.

- Convergence:
- Consistence:
- Stability:
  - zero-stability
  - A-stability
  - L-stability
- Stiffness

Theoretical results

- Dahlquist barriers
- Lax-Richtmyer

References

- Advantage of L-stability compared to A-stability, Mathematics Stack Exchange.

Many models of real-life systems are governed by Ordinary Differential Equations (ODEs). While simple linear equations yield elegant analytical solutions, as shown in sections about *Linear ODEs with constant coefficients* and *Linear ODEs in state-space representation*, real-world engineering problems demand numerical approximations, as no practical closed-form solution exists.

The choice of a time-integration scheme is far more than a software implementation detail; it represents a fundamental trade-off between **accuracy**, **computational cost**, and **fidelity**: a naive choice of solver can damp out physically vital high-frequency oscillations, generate artificial energy out of nothing, or completely explode due to numerical instability.

This chapter tries to provide some rigorous mathematical foundation needed to analyze, select, (and build? Do we need any new algorithm? Do not re-invent the wheel!) robust time-integration schemes.

Contents.

Foundations of numerical integration of initial value problems

First, the canonical problem[1]

$$\begin{cases} \dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y}(t)) \\ \mathbf{y}(0) = \mathbf{y}_0 \end{cases}$$

is discussed, along with the definition of

- **local truncation error** and **consistency**
- **global truncation error** and **convergence**

Common integration schemes

For first-order equation.

- One-step methods(I):
  - Explicit: Explicit Euler, Heun (special case of RK)
  - Implicit: Implicit Euler, Crank-Nicolson
- One-step methods(II): Runge-Kutta (RK) methods
- Linear multi-step methods:
  - Adams-Bashforth (AB)
  - Adams-Moulton (AM)
  - Backward Differentiation Formulas (BDF)

Stability analysis

Linear stability on test equations, $\dot{y} = \lambda y$ for the first-order system, $\ddot{y} = ...$ for the second-order system. Through one-step transfer function $G(z)$, $z = \lambda\, dt$, $y_{n+1} = G(z) y_n$,

- Stability region of numerical methods
- 0-ztability
- A-stability
- L-stability

Theorems and limits

- Dahlquist barriers
- Lax theorem: consistency + stability = convergence

[1] A second-order canonical problem $\mathbf{M}\ddot{\mathbf{y}} + \mathbf{C}\dot{\mathbf{y}} + \mathbf{K}\mathbf{y} = \mathbf{f}(t, \mathbf{y}(t))$, with proper initial conditions, is also discussed. This problem often arises with some interesting properties - symmetry of matrices - usually arises from a Lagrangian approach to classical mechanics systems. Some numerical methods targeting this form of equations exist, like [Leapfrog], [Verlet], or [Newmark] methods.

Integration and conservation laws

Some physical systems satisfy some conservation laws. As an example, **Hamiltonian systems** satisfy **energy conservation** or - equivalently - volume conservation in phase space.

Simplectic integrators are numerical methods that preserves these conservation laws of the original system. …

Examples

- *Example: torsional dynamics of an elastic beam*: finite element method of the continuous model is built; spectral decomposition of the discrete problem is discussed and compared with the eigenvalues of the continuous system; diagonal structural damping is introduced; previous results are used for an aware numerical integration in time of the dynamics of the beam subject to a harmonic torsional moment, using fast mode truncation and their static recovery.

30.1 Foundations

…

30.1.1 Local truncation error and Consistency

Let

$$y_{n+1} = \Psi(t_{n+1}; y_n, ..., y_{n-k+1}; h, f(\cdot)) = y_n + hA(t_{n+1}; y_n, ..., y_{n-k+1}; h, f(\cdot)) \ ,$$

the numerical method for the solution of the initial value problem

$$\begin{aligned} \dot{y}(t) &= f(y(t), t) \\ y(0) &= y_0 \ . \end{aligned}$$

The **local truncation error** is the error produced by **one step of the method**, i.e.

$$\delta^h_{n+k} := y_{n+k} - y(t_{n+k}) \ ,$$

being y_{n+k} the result of one step of the numerical method, and $y(t_{n+k})$ the exact solution using the same initial conditions (assuming, there's no error in previous steps). A numerical method is defined **consistent** if

$$\lim_{h \to 0} \frac{\delta^h_{n+k}}{h} = 0 \ .$$

The method has **order** p if

$$\delta^h_{n+k} = O(h^{p+1}) \quad , \quad \text{for } h \to 0 \ .$$

30.1.2 Global truncation error and Convergence

The **global truncation error** e_n is the accumulation of the error up to time t_n,

$$e_n^h := y_n - y(t_n) \, .$$

A numerical method is defined **convergent** if

$$\lim_{h \to 0} \max_n |e_n| = 0 \, .$$

Recurrence relation for One-step methods

A one-step method reads

$$y_{n+1} = \Psi(t_{n+1}; y_n; h, f(\cdot)) = y_n + hA(t_{n+1}; y_n; h, f(\cdot)) \, ,$$

A recurrence relation exists between the global and the local truncation error

…

this implies some bound of the global error, see wiki

30.2 Families of integration methods

Methods for the first-order IVP

$$\begin{cases} \dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y}) \\ \mathbf{y}(0) = \mathbf{y}_0 \, . \end{cases} \tag{30.1}$$

30.2.1 One-step methods (I): Euler, Crank-Nicolson

30.2.2 Multi-step methods

Adams-Bashforth and Adams-Moulton. The integration of the first-order equation (30.1) between t_n and t_{n+1} reads

$$\mathbf{y}(t_{n+1}) - \mathbf{y}(t_n) = \int_{\tau=t_n}^{t_{n+1}} \mathbf{f}(\tau, \mathbf{y}(\tau)) \, d\tau \, .$$

AB and AM methods replace the function $\mathbf{f}(t, \mathbf{y}(t))$ with a polynomial Lagrangian approximation $\mathbf{p}_s(t) \simeq \mathbf{f}(t)$ built with values of $\mathbf{f}(t_{n-k}, \mathbf{y}(t_{n-k}))$, with $k_{AB} = 0 : s-1$ (AB are explicit methods) and $k_{AM} = -1 : s-1$ (AM are implicit methods).

Once the original function is replaced by its polynomial approximation, the integration can be performed exactly.

Backward Differentiation Formulas. BDF methods exploits the derivative of the polynomial approximation $\mathbf{q}_s(t) \simeq \mathbf{y}(t)$. The value of the derivative $\dot{\mathbf{q}}_s(t)$ evaluated in $(t_{n+1}, \mathbf{y}_{n+1})$ can be written as a linear combinations of the $\{\mathbf{y}_{n-k}\}_{k=-1:s-1}$ values. BDF immediately follows from the comparison

$$\mathbf{f}(t_{n+1}, \mathbf{y}_{n+1}) = \dot{\mathbf{y}}(t_{n+1}) \simeq \dot{\mathbf{q}}_s(t) = \sum_{k=-1}^{s-1} a_k \mathbf{y}_{n-k}$$

Adams-Bashforth (AB)

For the derivation of the low-order AB methods, see *Lagrange interpolation: Applications: Adams-Bashforth*

Adams-Moulton (AM)

For the derivation of the low-order AB methods, see *Lagrange interpolation: Applications: Adams-Moulton*

Backward Differentiation Formulas (BDF)

For the derivation of the low-order AB methods, see *Lagrange interpolation: Applications: BDF*

30.2.3 One-step methods (II): Runge-Kutta (RK)

Method.

$$y_{n+1} = y_n + h \sum_{m=1}^{s} b_m k_m \ ,$$

with

$$\begin{aligned} k_1 &= f(t_n, y_n) \\ k_2 &= f(t_n + hc_2, y_n + ha_{21}k_1) \\ &\dots \\ k_s &= f\left(t_n + hc_s, y_n + h\sum_{j=1}^{s-1} a_{sj}k_j\right) \ . \end{aligned}$$

Rationale

Starting from a n-order Taylor series

$$y(t+h) = \sum_{p=0}^{n} \frac{h^p}{p!} y^{(p)}(t) \ ,$$

and matching it with a linear combination of evaluations of the first-order derivative,

$$y_{n+1} = y_n + h \sum_{m=1}^{s} b_m k_m \ ,$$

with $t_{n+1} = t_n + h$.

Butcher tableau

The coefficients of a RK method can be organized in Butcher tableau

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$

$$\begin{array}{c|ccccc} 0 & & & & & \\ c_2 & a_{21} & & & & \\ c_3 & a_{31} & a_{32} & & & \\ \vdots & \vdots & \vdots & \ddots & & \\ c_s & a_{s1} & a_{s2} & \dots & a_{s,s-1} & \\ \hline & b_1 & b_2 & \dots & b_{s-1} & b_s \end{array}$$

Constraints

$$\sum_{j=1}^{s} b_j = 1 \, .$$

Example: RK2

RK2 methods include mid-point, Heun and Ralston methods.

Derivation.

Starting from a 2-order Taylor series, and matching it with a linear combination of evaluations of the first-order derivative.

$$\begin{aligned} y(t+h) &= y(t) + hy'(t) + \frac{h^2}{2} y''(t) + o(h^2) = \\ &= y(t) + hf(t, y(t)) + \frac{h^2}{2} \left[f_t(t, y(t)) + y'(t) f_y(t, y(t)) \right] + o(h^2) = \\ &= y(t) + hf(t, y(t)) + \frac{h^2}{2} \left[f_t(t, y(t)) + f(t, y(t)) f_y(t, y(t)) \right] + o(h^2) \, , \end{aligned}$$

with $y'(t) = f(t, y(t))$, so that $y''(t) = f_t(t, y(t)) + y'(t) f_y(t, y(t))$.

RK approximation.

$$y_{n+1} = y_n + h(b_1 k_1 + b_2 k_2) \, ,$$

with

$$\begin{aligned} k_1 &= f(t_n, y_n) \\ k_2 &= f(t_n + hc_2, y_n + ha_{21} k_1) \, . \end{aligned}$$

Expanding in Taylor series k_2,

$$\begin{aligned} k_2 &= f(t_n + hc_2, y_n + ha_{21} k_1) = \\ &= f(t_n, y_n) + hc_2 f_t(t_n, y_n) + ha_{21} k_1 f_y(t_n, y_n) + O(h^2) \, , \end{aligned}$$

inserting in the RK2 scheme

$$\begin{aligned} y_{n+1} &= y_n + hb_1 f(t_n, y_n) + hb_2 \left[f(t_n, y_n) + hc_2 f_t(t_n, y_n) + ha_{21} k_1 f_y(t_n, y_n) \right] + o(h^2) = \\ &= y_n + h \left[b_1 + b_2 \right] f(t_n, y_n) + h^2 b_2 c_2 f_t(t_n, y_n) + h^2 b_2 a_{21} f(t_n, y_n) f_y(t_n, y_n) + o(h^2) \, , \end{aligned}$$

and matching coefficients of Taylor expansions, the coefficients of the RK method are computed as the solution of a **underdetermined system** of 3 equations with 4 unknowns

$$\begin{cases} b_1 + b_2 = 1 \\ b_2 c_2 = \frac{1}{2} \\ b_2 a_{21} = \frac{1}{2} \end{cases} \quad , \quad \begin{array}{c|cc} 0 & & \\ c_2 & a_{21} & \\ \hline & b_1 & b_2 \end{array}$$

or leaving $\alpha = c_2 = a_{21}$ as a free parameter,

$$\begin{cases} b_1 = 1 - \frac{1}{2\alpha} \\ b_2 = \frac{1}{2\alpha} \\ c_2 = \alpha \\ a_{21} = \alpha \end{cases} \quad , \quad \begin{array}{c|cc} 0 & & \\ \alpha & \alpha & \\ \hline & 1 - \frac{1}{2\alpha} & \frac{1}{2\alpha} \end{array}$$

This system has infinite solutions. Different methods correspond to different choices among these solutions:

- mid-point method, $\alpha = \frac{1}{2}$
- Heun method, $\alpha = 1$
- Ralston's method (minimum truncation error **todo** *prove it!*), $\alpha = \frac{2}{3}$

30.3 Stability of integration schemes

30.3.1 Stability of a first-order linear system

Linear homogeneous equation (or system?)

$$\begin{aligned} \dot{y} &= \lambda y \\ y(0) &= y_0 \ , \end{aligned}$$

is a first test for the stability of a numerical method. The problem has the exact solution

$$y(t) = y_0 e^{\lambda t} \ ,$$

and is **asymptotically stable** if $\mathrm{re}\{\lambda\} < 0$, as

$$\lim_{t \to +\infty} y(t) = \lim_{t \to +\infty} y_0 e^{\lambda t} = \lim_{t \to +\infty} y_0 e^{\mathrm{re}\{\lambda\} t} e^{j\,\mathrm{im}\{\lambda\} t} \ .$$

Forced linear equation (or system?)

$$\dot{y} = \lambda y + u \ .$$

The exact solution converges to the forced solution if the system is asymptotically stable, For periodic forcing, the behavior can be evaluated in frequency (Laplace or Fourier) domain

$$y(s) = \frac{1}{s - \lambda} u(s) \ .$$

A-stability

A-stability ensures the numerical method doesn't explode in the simulation of a asymptotically stable system.

A numerical method is defined A-stable if its stability region includes the whole left half plane.

L-stability

Let $z := h\lambda$. For **one-step methods**, it's possible to define a **scalar transfer function** $G(z)$ between y_n and y_{n+1},

$$y_{n+1} = G(z)y_n \ .$$

For **multi-step methods**… proerties of characteristic polynomial and its roots

For **systems**… amplification matrix…

L-stability ensures that the method damps extremely fast - **stiff** - modes, that are usually physically irrelevant but that can be numerically troublesome. A method is defined L-stable if

$$\lim_{z\to-\infty} G(z) = 0 \ .$$

0-stability

0-stability (of a numerical method for a certain differential equation on a given interval) if a perturbation in the starting value of size ε causes the numerical solution over the time interval to change less than $K\varepsilon$ for some K that doesn't depend on the time-step h.

It's enough to check the conditions for the differential equation $y' = 0$ **todo** *why? check it!*

A linear multi-step method is zero-stable **iif** the root condition (analogous to at least marginally stable system, i.e. roots with absolute value $|z| < 1$, or $|z| = 1$ with multiplicity $= 1$) is satisfied.

30.3.2 Some schemes

Homogeneous equation

$$\dot{y} = f(t, y) = \lambda y \ .$$

Explicit Euler.

$$y_{n+1} = y_n + hf(y_n, t_n) = y_n + h\lambda y_n = (1 + h\lambda)y_n \ .$$

The solution of this difference equation with initial condition y_0 reads

$$y_n = (1 + h\lambda)^n y_0 \ .$$

Implicit Euler.

$$y_{n+1} = y_n + hf(y_{n+1}, t_{n+1}) = y_n + h\lambda y_{n+1} \ .$$

The solution of this difference equation with initial condition y_0 reads

$$y_n = \frac{1}{(1 - h\lambda)^n} y_0 \ .$$

Crank-Nicolson.

$$y_{n+1} = y_n + \frac{h}{2}\left[f(y_n, t_n) + f(y_{n+1}, t_{n+1})\right] = y_n + \frac{h}{2}\left[\lambda y_n + \lambda y_{n+1}\right] \ .$$

$$y_n = \left[\frac{1 + \lambda h/2}{1 - \lambda h/2}\right]^n y_0 \ .$$

BDF 2. BDF 2 is a multi-step approach, so either Z-transform of the conversion to a one-step 2-dimensional system is required to discuss its stability properties, and transfer function.

$$y_{n+1} - \frac{4}{3}y_n + \frac{1}{3}y_{n-1} = \frac{2}{3}hf(t_{n+1}, y_{n+1})$$

$$y_{n+1} - \frac{4}{3}y_n + \frac{1}{3}y_{n-1} = \frac{2}{3}h\lambda y_{n+1}$$

Z-transform. Using the definition $\zeta := \lambda h$, and applying the Z-transform,

$$0 = \left[\left(1 - \frac{2}{3}z\right) - \frac{4}{3}\zeta^{-1} + \frac{1}{3}\zeta^{-2}\right] Y(\zeta)$$

the characteristic polynomial reads

$$0 = \pi(\zeta; z) = (3 - 2z)\,\zeta^2 - 4\zeta + 1\ .$$

The roots of the polynomial depend on $z \in \mathbb{C}$, and the system is stable if all the roots have $|\zeta_k| \leq 1$. This condition can be summarized with the condition of **spectral radius** of the matrix $\mathbf{A}$ of the one-step system, $\rho(\mathbf{A}) = \max|\lambda_k| < 1$.

Reduction of difference equation to a one-step system. The 2-step difference equation above

$$y_{n+1} - \frac{4}{3-2z}y_n + \frac{1}{3-2z}y_n = 0$$

can be recast as a system, $y_n = z_{1,n}$, $y_{n-1} = z_{0,n}$

$$\begin{aligned} z_{0,n+1} &= z_{1,n} \\ z_{1,n+1} &= \frac{4}{3-2z}z_{1,n} - \frac{1}{3-2z}z_{0,n} \end{aligned}$$

or using matrix formalism

$$\mathbf{z}_{n+1} = \mathbf{A}\mathbf{z}_n\ ,$$

with $\mathbf{A} = \begin{bmatrix} \cdot & 1 \\ -\frac{1}{3-2z} & \frac{4}{3-2z} \end{bmatrix}$. The eigenvalues of the system follows from the condition

$$0 = \zeta\left(\zeta - \frac{4}{3-2z}\right) + \frac{1}{3-2z}\ ,$$

i.e. the same condition as $\pi(\zeta; z) = 0$.

L-stability. As $\to +\infty$, the polynomials goes to $\pi(z \to +\infty; \zeta) \sim -\frac{2}{3}z\zeta^2$ and thus the roots goes to $z \to +\infty$.

Methods for second-order equations.

$$\ddot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})\ ,$$

or as a first-order system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{f}(\mathbf{x}, t) \end{bmatrix}\ .$$

A test second-order homogeneous equation may be the dynamical equation of an undapmed mass-spring system,

$$\ddot{x} + \Omega^2 x = 0\ ,$$

being $(d_t + i\Omega)(d_t - i\Omega)x = 0$. This analysis can be performed with a system with conjugate complex eigenvalues $\lambda = \sigma + j\omega$ and $\lambda^* = \sigma - j\omega$, sot that the test equation becomes

$$\begin{aligned} 0 &= (d_t - \sigma - i\omega)(d_t - \sigma + i\omega)x = \\ &= d_{tt}x - 2\sigma d_t x + (\sigma^2 + \omega^2)\ . \end{aligned}$$

Leapfrog method.

$$\begin{cases} x_{i+1} = x_i + hv_i + \dfrac{h^2}{2} f(x_i) \\ v_{i+1} = v_i + \dfrac{h}{2}\left(f(x_i) + f(x_{i+1})\right) \ . \end{cases}$$

Verlet method. Position Verlet is algebrically equivalent to the Leapfrom method, so it shares the same stability properties.

Newmark method. …

Test equation has $f(x) = -\Omega^2 x$, so that Leapfrog method applied to the test problem reads

$$\begin{cases} x_{i+1} = x_i + hv_i - \dfrac{h^2\Omega^2}{2} x_i \\ v_{i+1} = v_i - \Omega^2 \dfrac{h}{2}\left(x_i + x_{i+1}\right) \ . \end{cases}$$

so that the $\mathbf{z}_{i+1}$ state can be written as a function of the $\mathbf{z}_i$ state

$$\begin{bmatrix} 1 & \cdot \\ \frac{h\Omega^2}{2} & 1 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}_{i+1} = \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2} & h \\ -\frac{h\Omega^2}{2} & 1 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}_i$$

or

$$\begin{aligned} \begin{bmatrix} x \\ hv \end{bmatrix}_{i+1} &= \begin{bmatrix} 1 & \cdot \\ -\frac{h\Omega^2}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2} & 1 \\ -\frac{h^2\Omega^2}{2} & 1 \end{bmatrix} \begin{bmatrix} x \\ hv \end{bmatrix}_i = \\ &= \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2} & 1 \\ -h^2\Omega^2\left(1 - \frac{h^2\Omega^2}{4}\right) & 1 - \frac{h^2\Omega^2}{2} \end{bmatrix} \begin{bmatrix} x \\ hv \end{bmatrix}_i \ . \end{aligned}$$

Generalized test equation - with general pair of complex eigenvalues, not only imaginary eigenvalues - has $f(x, v) = 2\sigma v - (\sigma^2 + \omega^2)x = -2\xi\Omega v - \Omega^2 x$, with

$$\begin{aligned} \Omega^2 &= \sigma^2 + \omega^2 \\ \sigma &= -\xi\Omega \qquad \rightarrow \qquad \xi = -\frac{\sigma}{\Omega} = -\frac{\sigma}{\sqrt{\sigma^2 + \omega^2}} \end{aligned}$$

Leapfrog method applied to the test problem reads

$$\begin{cases} x_{i+1} = x_i + hv_i + \dfrac{h^2}{2}\left(-\Omega^2 x_i - 2\xi\Omega v_i\right) \\ v_{i+1} = v_i + \dfrac{h}{2}\left[-\Omega^2\left(x_i + x_{i+1}\right) - 2\xi\Omega\left(v_i + v_{i+1}\right)\right] \ . \end{cases}$$

todo *check!*

$$\begin{bmatrix} 1 & \cdot \\ \frac{h^2\Omega^2}{2} & 1 + \xi h\Omega \end{bmatrix} \begin{bmatrix} x \\ hv \end{bmatrix}_{i+1} = \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2} & 1 - \xi h\Omega \\ -\frac{h^2\Omega^2}{2} & 1 - \xi h\Omega \end{bmatrix} \begin{bmatrix} x \\ hv \end{bmatrix}_i$$

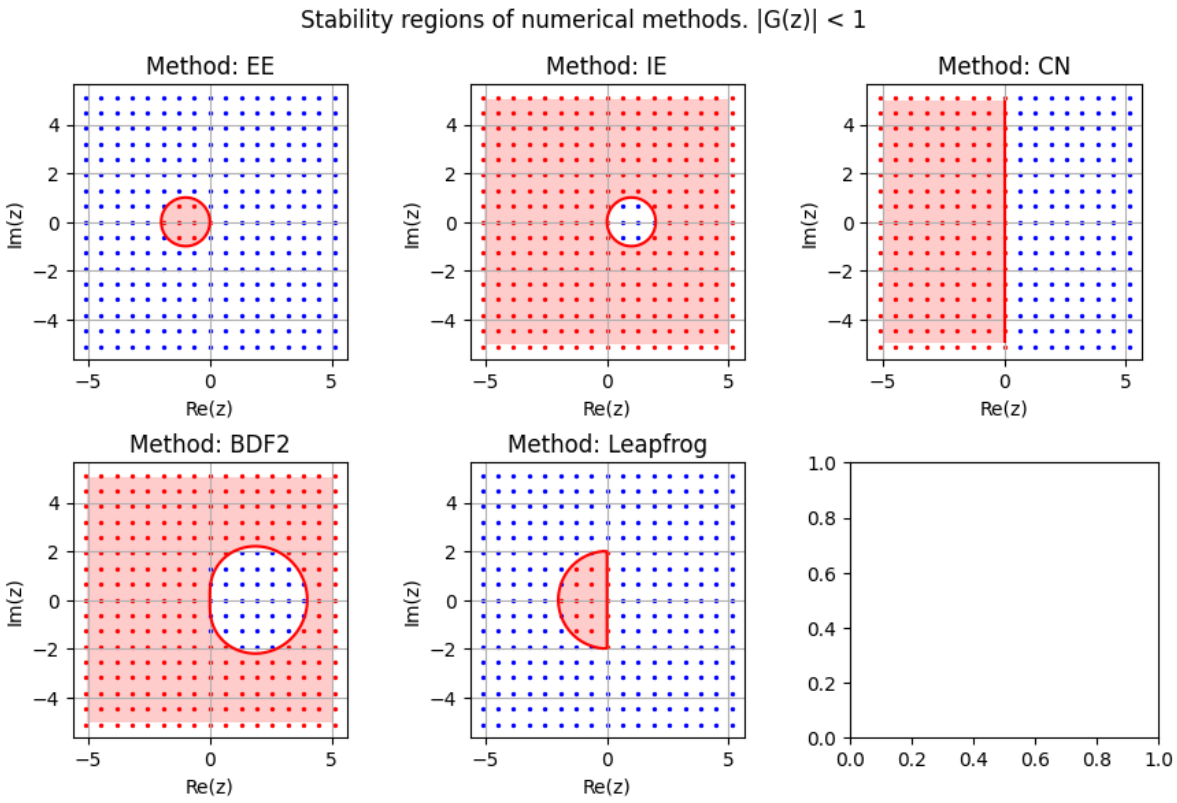
$$\begin{aligned} \begin{bmatrix} x \\ hv \end{bmatrix}_{i+1} &= \begin{bmatrix} 1 & \cdot \\ -\frac{h^2\Omega^2}{2(1+\xi h\Omega)} & \frac{1}{1+\xi h\Omega} \end{bmatrix} \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2} & 1 - \xi h\Omega \\ -\frac{h^2\Omega^2}{2} & 1 - \xi h\Omega \end{bmatrix} \begin{bmatrix} x \\ hv \end{bmatrix}_i = \\ &= \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2} & 1 - \xi h\Omega \\ -\frac{h^2\Omega^2}{1+\xi h\Omega}\left(1 - \frac{h^2\Omega^2}{4}\right) & \frac{1-\xi h\Omega}{1+\xi h\Omega}\left(1 - \frac{h^2\Omega^2}{2}\right) \end{bmatrix} \begin{bmatrix} x \\ hv \end{bmatrix}_i \ . \end{aligned}$$

- if $\xi^2 \geq 1$, then the eigenvalues are real
- if $\xi^2 \leq 1$, then the eigenvalues are conjugate complex, $s_{1,2} = -\xi\Omega \mp j\Omega\sqrt{1 - \xi^2}$

Forced equation

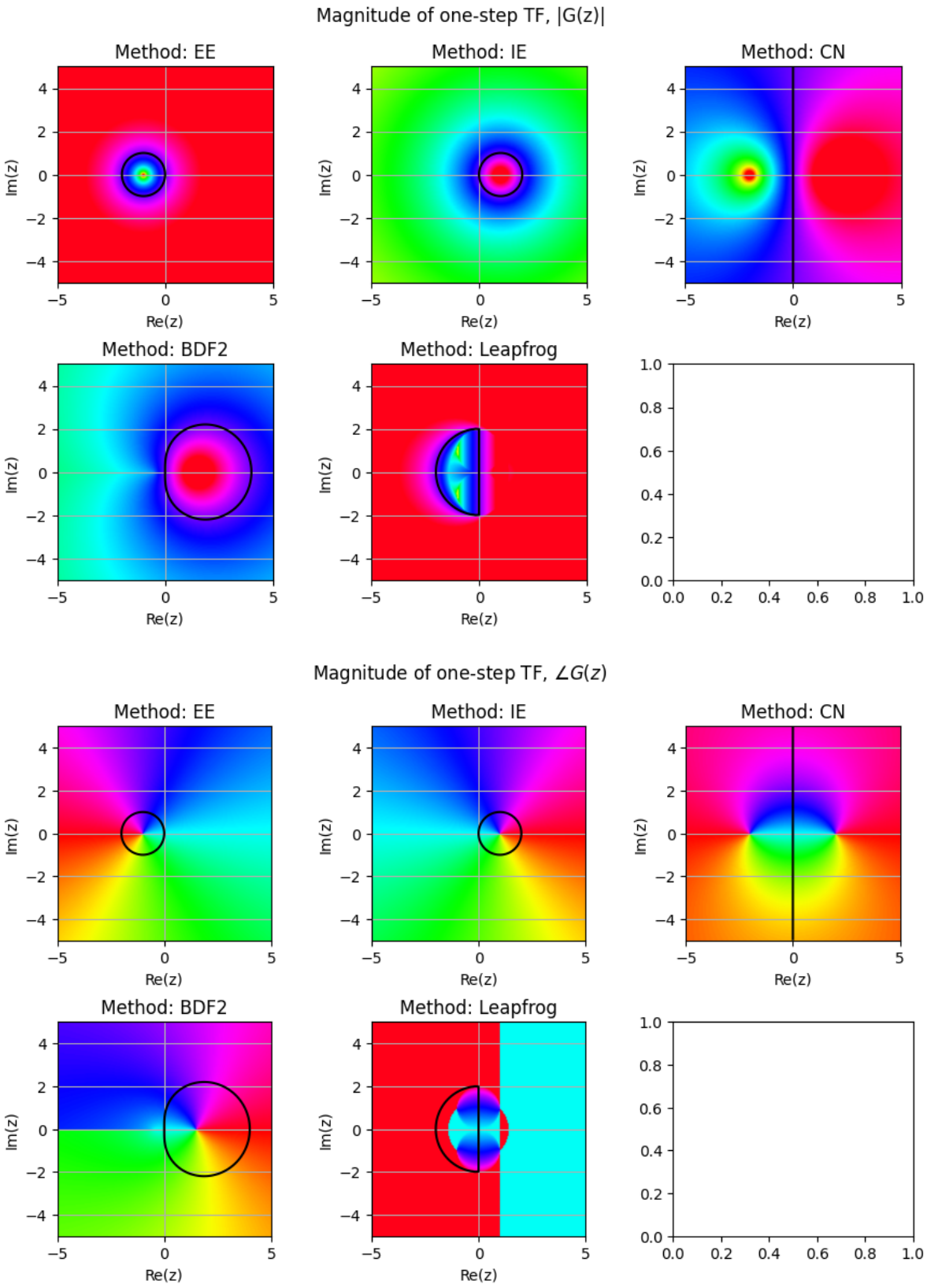
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A-stability

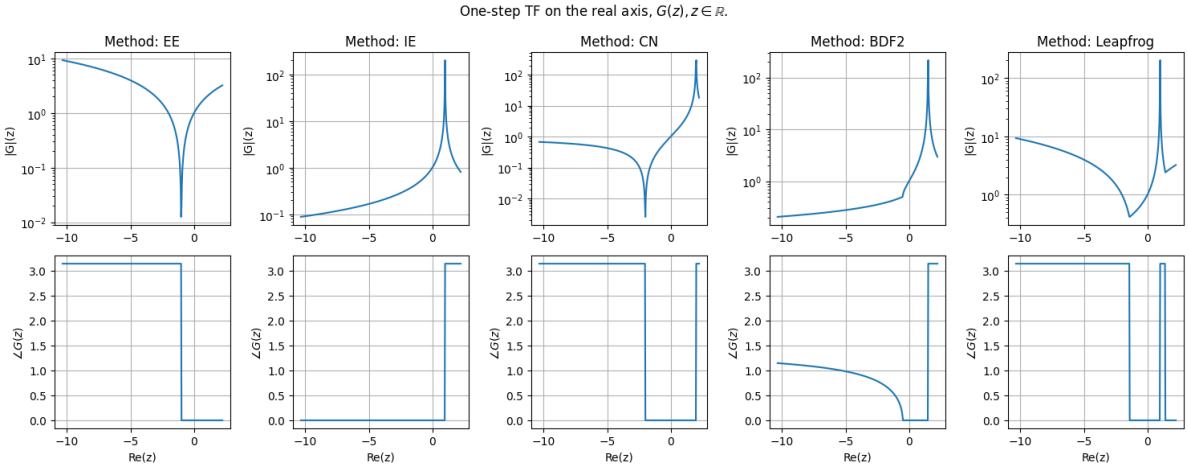


A-stability. Implicit Euler and Crank-Nicolson methods are A-stable, while explicit Euler is not.

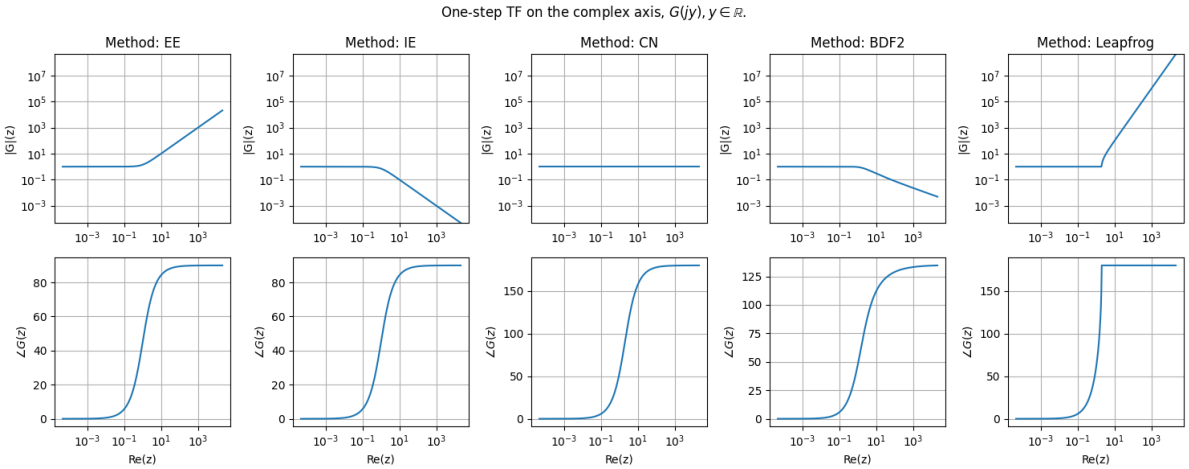
G(z): absolute value and phase



G(z) on the real axis



G(z) on the imaginary axis



Remark. For multiple-steps methods, like BDF2 and Leapfrog, the magnitude and phase of the eigenvalue of the one-step matrix $\mathbf{G}(z)$ is shown here.

L-stable methods. Among the methods shown here, only IE and BDF2 are L-stable, i.e. with $|G(z \to +\infty)| \to 0$

Dahlquist barriers - for linear multi-step methods

First Dahlquist barrier.

- A zero-stable and linear q-step multis-tep method has maximum order of convergence $q + 1$ if q is odd, $q + 2$ if q is even. If the method is explicit, maximum order of convergence is q.

Second Dahlquist barrier.

- No explicit linear multi-step method is A-stable
- Maximum order of an (implicit) A-stable linear multi-step method is 2
- Among the A-stable multi-step methods of order 2, the trapezoidal rule has the smallest error constant

A **linear multi-step method** for the equation $y' = f(t, y)$ with initial conditions $y(t_0) = y_0$, reads

$$\sum_{j=0}^{s} a_j y_{n+j} = h \sum_{j=0}^{s} b_j f(t_{n+j}, y_{n+j})$$

If $b_0 = 0$, the method is explicit, otherwise it's implicit as the unknown y_{n+s} appears in both the LHS and the RHS.

A-stability. For the equation $y' = \lambda y$, i.e. $f(t, y) = \lambda y$, the Z-transformed integration step becomes

$$0 = \sum_{j=0}^{s} (a_j - h\lambda b_j)\zeta^{j-s} = \rho(\zeta) - z\sigma(\zeta) = \pi(\zeta, z) \ ,$$

with $z := h\lambda$.

For explicit methods, $b_0 = 0$, the degree of $\sigma(\zeta)$ is lower than the degree of $\rho(\zeta)$

…

30.4 Simplectic integrators

For Hamiltonian systems (see Classical Mechanics: Hamiltonian Mechanics), simplectic integrators preserve the volume in phase space.

$$\dot{\mathbf{p}} = -\partial_{\mathbf{q}} H$$
$$\dot{\mathbf{q}} = \partial_{\mathbf{p}} H$$

Preservation of volume in phase space for Hamiltonian systems

Let V_t a material domain in phase space. The derivative of its volume reads

$$\begin{aligned}
\frac{d}{dt}\int_{V_t} 1 = \int_{V_t}\int_{V_t} \partial_t 1 + \oint_{\partial V_t} \mathbf{u}\cdot\hat{\mathbf{n}} = \\
= \oint_{\partial V_t} (\partial_{\mathbf{p}} H, -\partial_{\mathbf{q}} H)\cdot\hat{\mathbf{n}} = \\
= \int_{V_t} \nabla_{\mathbf{q}}\cdot\nabla_{\mathbf{p}} H - \nabla_{\mathbf{p}}\cdot\nabla_{\mathbf{q}} H = \\
= 0 \ .
\end{aligned}$$

having applied Reynolds' transport theroem in the phase space, with $\mathbf{u} = \frac{d\mathbf{r}}{dt} = (\partial_{\mathbf{p}} H, -\partial_{\mathbf{q}} H)$ the "velocity" in the phase space, with the state $\mathbf{r} = (\mathbf{q}, \mathbf{p})$, and $\hat{\mathbf{n}}$ the unit normal vector pointing outwards. In the last steps, Hamilton equations are used to write velocity components as a function of partial derivatives of the Hamiltonian function, and divergence theorem is used in the phase space to transform the surface integral to a volume integral. The last step reads in components reads

$$\nabla_{\mathbf{q}}\cdot\nabla_{\mathbf{p}} H - \nabla_{\mathbf{p}}\cdot\nabla_{\mathbf{q}} H = \frac{\partial^2 H}{\partial q^i \partial p_i} - \frac{\partial^2 H}{\partial p^i \partial q_i} = 0 \ ,$$

under the assumption of sufficient regularity required by the Schwartz theorem about mixed partial derivatives.

Algorithms - examples:

- Verlet: simplectic, time reversible
- …

Applications - examples:

- Integration of spatial dynamics: motion of planets,…
- …

30.5 Energy conservation of integration schemes

Energy conservation can be discussed with:

- the Jacobian of the transformation induced by one-step of the method, $\mathbf{z}_n \mapsto \mathbf{z}_{n+1}$
- the change of energy $\Delta E = E_{n+1} - E_n$

Energy conservation is discussed here for Hamiltonian systems, see *simplectic integrators*

$$\begin{cases} \dot{\mathbf{q}} = \partial_{\mathbf{p}} H \\ \dot{\mathbf{p}} = -\partial_{\mathbf{q}} H \end{cases}$$

or for second-order mechanical systems

$$\mathbf{M}\ddot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$$

with $\mathbf{f} = -\partial_{\mathbf{x}} V(\mathbf{x})$.

The Hamiltonian function $H(\mathbf{q}, \mathbf{p})$ can be thought as the energy as a function of the generalized coordinates $\mathbf{q}$ and the generalized momenta $\mathbf{p}$. For Hamiltoninan system, where the Hamiltonian function is independent from time, the Hamiltonian (the energy) is conserved along the trajectories of the system,

$$\begin{aligned} \dot{H} &= \dot{\mathbf{q}} \cdot \partial_{\mathbf{q}} H + \dot{\mathbf{p}} \cdot \partial_{\mathbf{p}} H = \\ &= \partial_{\mathbf{p}} H \cdot \partial_{\mathbf{q}} H - \partial_{\mathbf{q}} H \cdot \partial_{\mathbf{p}} H = 0 \,. \end{aligned}$$

The second-order mechanical system is conservative if with $\mathbf{f} = -\partial_{\mathbf{x}} V(\mathbf{x})$, as the total mechanical energy

$$E = \frac{1}{2} \dot{\mathbf{x}}^* \mathbf{M} \dot{\mathbf{x}} + V(\mathbf{x})$$

is conserved along the evolution of the system. This second-order system can be written - if $\mathbf{M}$ is invertible - as a first-order system as

$$\frac{d}{dt} \begin{bmatrix} \mathbf{x} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{M}^{-1} \mathbf{f}(\mathbf{x}, t) \end{bmatrix} \,,$$

or as a first-order system with Hamiltonian form, with $H(\mathbf{p}, \mathbf{q}) = \frac{1}{2} \mathbf{p}^* \mathbf{M}^{-1} \mathbf{p} + V(\mathbf{q})$, having defined $\mathbf{p} := \mathbf{M}\dot{\mathbf{x}}$, $\mathbf{q} := \mathbf{x}$,

$$\frac{d}{dt} \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix} = \begin{bmatrix} \mathbf{M}^{-1} \mathbf{p} \\ -\partial_{\mathbf{q}} V(\mathbf{q}) \end{bmatrix} \,.$$

.

30.5.1 Examples

Verlet

For the mechanical system governed by the differential equation $\ddot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$, the numerical scheme reads

$$\begin{aligned} \tilde{\mathbf{x}}_{n+1} &= \mathbf{x}_n + \frac{h}{2} \mathbf{v}_n \\ \mathbf{v}_{n+1} &= \mathbf{v}_n + h \mathbf{f}(\tilde{\mathbf{x}}_{n+1}) \\ \mathbf{x}_{n+1} &= \tilde{\mathbf{x}}_{n+1} + \frac{h}{2} \mathbf{v}_{n+1} \end{aligned}$$

so that the transformation $\mathbf{z}_n \mapsto \mathbf{z}_{n+1}$ reads

$$\begin{aligned}\mathbf{v}_{n+1} &= \mathbf{v}_n + h\mathbf{f}(\tilde{\mathbf{x}}_{n+1}(\mathbf{x}_n, \mathbf{v}_n)) \\ \mathbf{x}_{n+1} &= \tilde{\mathbf{x}}_{n+1}(\mathbf{x}_n, \mathbf{v}_n) + \frac{h}{2}\mathbf{v}_{n+1} = \tilde{\mathbf{x}}_{n+1}(\mathbf{x}_n, \mathbf{v}_n) + \frac{h}{2}\left[\mathbf{v}_n + h\mathbf{f}(\tilde{\mathbf{x}}_{n+1}(\mathbf{x}_n, \mathbf{v}_n))\right]\end{aligned}$$

or

$$\begin{aligned}\mathbf{v}_{n+1} &= \mathbf{v}_n + h\mathbf{f}\left(\mathbf{x}_n + \frac{h}{2}\mathbf{v}_n\right) \\ \mathbf{x}_{n+1} &= \mathbf{x}_n + h\mathbf{v}_n + \frac{h^2}{2}\mathbf{f}\left(\mathbf{x}_n + \frac{h}{2}\mathbf{v}_n\right)\end{aligned}$$

Jacobian of the one-step transformation

Thus, the Jacobian reads

$$\mathbf{J} = \partial_{\mathbf{z}_n}\mathbf{z}_{n+1} = \begin{bmatrix}\mathbf{I} + \frac{h^2}{2}\partial_{\tilde{\mathbf{x}}}\mathbf{f} & h + \frac{h^3}{4}\partial_{\tilde{\mathbf{x}}}\mathbf{f} \\ h\partial_{\tilde{\mathbf{x}}}\mathbf{f} & \mathbf{I} + \frac{h^2}{2}\partial_{\tilde{\mathbf{x}}}\mathbf{f}\end{bmatrix}$$

and the value of its determinant can be evaluated - for h "small enough" the upper-left block is approximately the identity and thus invertible; all the blocks commute, as they're linear combinations of zero and first power of $\partial_{\tilde{\mathbf{x}}}\mathbf{f}$ - as

$$|\mathbf{J}| = \left|\left(\mathbf{I} + \frac{h^2}{2}\partial_{\tilde{\mathbf{x}}}\mathbf{f}\right)^2 - h^2\partial_{\tilde{\mathbf{x}}}\mathbf{f}\left(\mathbf{I} + \frac{h^2}{4}\partial_{\tilde{\mathbf{x}}}\mathbf{f}\right)\right| = |\mathbf{I}| \,.$$

Determinant of block matrices - with Schur complement

The determinant of the block matrices - with **Schur complement**, if it exists -

$$\begin{bmatrix}A & B \\ C & D\end{bmatrix} = \begin{bmatrix}I & 0 \\ CA^{-1} & I\end{bmatrix}\begin{bmatrix}A & 0 \\ 0 & D - CA^{-1}B\end{bmatrix}\begin{bmatrix}I & A^{-1}B \\ 0 & I\end{bmatrix}$$

reads - just take the determinants and exploit the property of determinant of the product is equal to the product of the determinants, and the determinant of a block-diagonal matrix -

$$\left|\begin{bmatrix}A & B \\ C & D\end{bmatrix}\right| = |A||D - CA^{-1}B| \,.$$

If A, B, C, D have proper dimensions and $AC = CA$, i.e. commute, it follows

$$\begin{aligned}\left|\begin{bmatrix}A & B \\ C & D\end{bmatrix}\right| &= |A||D - CA^{-1}B| = \\ &= |AD - ACA^{-1}B| = \\ &= |AD - CAA^{-1}B| = \\ &= |AD - CB| \,.\end{aligned}$$

Proof of inverse of block matrix by direct computation

$$\begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ C & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix} = \\ = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

One-step energy difference

Explicit Euler

For the mechanical system governed by the differential equation $\ddot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$, or by the first-order system

$$\frac{d}{dt} \begin{bmatrix} \mathbf{x} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{f}(\mathbf{x}) \end{bmatrix}$$

the numerical scheme, and the transformation $\mathbf{z}_n \mapsto \mathbf{z}_{n+1}$, reads

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h\mathbf{v}_n \\ \mathbf{v}_{n+1} = \mathbf{v}_n + h\mathbf{f}(\mathbf{x}_n)$$

Jacobian of the one-step transformation

Thus, the Jacobian reads

$$\mathbf{J} = \partial_{\mathbf{z}_n} \mathbf{z}_{n+1} = \begin{bmatrix} \mathbf{I} & h\mathbf{I} \\ h\partial_{\tilde{\mathbf{x}}}\mathbf{f} & \mathbf{I} \end{bmatrix}$$

and the value of its determinant can be evaluated - for h "small enough" the upper-left block is approximately the identity and thus invertible; all the blocks commute, as they're linear combinations of zero and first power of $\partial_{\tilde{\mathbf{x}}}\mathbf{f}$ - as

$$|\mathbf{J}| = \left|\mathbf{I} - h^2 \partial_{\tilde{\mathbf{x}}}\mathbf{f}\right| \ .$$

First-derivative of the force w.r.t. $\mathbf{x}$ can be related to the stiffness matrix around equilibria,

$$\mathbf{f}(\overline{\mathbf{x}} + \delta\mathbf{x}) \simeq \mathbf{f}(\overline{\mathbf{x}}) + \delta\mathbf{x} \cdot \partial_{\mathbf{x}}\mathbf{f}(\overline{\mathbf{x}}) = \mathbf{K}\delta\mathbf{x} \ .$$

For marginally-stable systems, the stiffness matrix has imaginary eigenvalues, $\lambda_k(\mathbf{K}) = \mp i\Omega_k$. For the numerical integration of system around an equilibrium, it follows

$$|\mathbf{J}| = \left|\mathbf{I} - h^2\mathbf{K}\right| \ ,$$

and thus - recalling that the determinant is equal to the product of the eigenvalues -

$$|\mathbf{J}| = \prod_k \lambda_k(\mathbf{I} - h^2\mathbf{K}) = \prod_k \left(1 - h^2\lambda_k(\mathbf{K})\right) = \prod_k (1 + h^2\Omega_k^2) \geq 1 \ .$$

Eigenproblem of $\mathbf{A} + b\mathbf{I}$

If $\mathbf{A}\mathbf{v}_k = \alpha_k \mathbf{v}_k$, then the matrix $\mathbf{A} + b\mathbf{I}$ has the same eigenvectors $\mathbf{v}_k$ with eigenvalues $\lambda_k = \alpha_k + b$

One-step energy difference

Implicit Euler

For the mechanical system governed by the differential equation $\ddot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$, or by the first-order system

$$\frac{d}{dt}\begin{bmatrix}\mathbf{x}\\ \mathbf{v}\end{bmatrix} = \begin{bmatrix}\mathbf{v}\\ \mathbf{f}(\mathbf{x})\end{bmatrix}$$

the numerical scheme, and the transformation $\mathbf{z}_n \mapsto \mathbf{z}_{n+1}$, reads

$$\begin{aligned}\mathbf{x}_{n+1} &= \mathbf{x}_n + h\mathbf{v}_{n+1}\\ \mathbf{v}_{n+1} &= \mathbf{v}_n + h\mathbf{f}(\mathbf{x}_{n+1})\end{aligned}$$

Jacobian of the one-step transformation

The Jacobian of the inverse transformation $\mathbf{z}_{n+1} \mapsto \mathbf{z}_n$ reads

$$\mathbf{J}^{-1} = \partial_{\mathbf{z}_{n+1}}\mathbf{z}_n = \begin{bmatrix}\mathbf{I} & -h\mathbf{I}\\ -h\partial_{\tilde{\mathbf{x}}}\mathbf{f} & \mathbf{I}\end{bmatrix}$$

and the value of its determinant can be evaluated - for h "small enough" the upper-left block is approximately the identity and thus invertible; all the blocks commute, as they're linear combinations of zero and first power of $\partial_{\tilde{\mathbf{x}}}\mathbf{f}$ - as

$$|\mathbf{J}^{-1}| = \left|\mathbf{I} - h^2\partial_{\tilde{\mathbf{x}}}\mathbf{f}\right| \ .$$

First-derivative of the force w.r.t. $\mathbf{x}$ can be related to the stiffness matrix around equilibria,

$$\mathbf{f}(\overline{\mathbf{x}} + \delta\mathbf{x}) \simeq \mathbf{f}(\overline{\mathbf{x}}) + \delta\mathbf{x}\cdot\partial_{\mathbf{x}}\mathbf{f}(\overline{\mathbf{x}}) = \mathbf{K}\delta\mathbf{x}\ .$$

For marginally-stable systems, the stiffness matrix has imaginary eigenvalues, $\lambda_k(\mathbf{K}) = \mp i\Omega_k$. For the numerical integration of system around an equilibrium, it follows

$$|\mathbf{J}^{-1}| = \left|\mathbf{I} - h^2\mathbf{K}\right| \ ,$$

and thus - recalling that the determinant is equal to the product of the eigenvalues -

$$|\mathbf{J}^{-1}| = \prod_k \lambda_k(\mathbf{I} - h^2\mathbf{K}) = \prod_k \left(1 - h^2\lambda_k(\mathbf{K})\right) = \prod_k (1 + h^2\Omega_k^2) \geq 1\ ,$$

and thus

$$|\mathbf{J}| = \left|\mathbf{J}^{-1}\right|^{-1} = \left[\prod_k (1 + h^2\Omega_k^2)\right]^{-1} \leq 1\ .$$

One-step energy difference

Crank-Nicolson

30.6 Time-integration of a mechanical system

Reference to:

- Modal and numerical methods for mechanical systems - continuum mechanics
- Natural modes in structures - blog

30.6.1 Example equation: torsion of elastic beam

Here, the PDE governing the torsional dynamics of a linear homogeneous elastic beam is used as a toy problem

$$I\partial_{tt}\theta - \partial_x \left(GJ\partial_x\theta\right) = m$$

supplied with proper boundary and initial conditions. Here the clamped-free beam is studied, with prescribed torsional moment $M(t)$ ath teh "free" end so that the boundary conditions read

$$\begin{aligned} \theta(x=0,t) &= 0 \\ GJ\partial_x\theta(x=\ell,t) &= M(t)\ . \end{aligned}$$

See

- Continuum mechanics: Beams
- Continuum mechanics: Timoshenko beam
- Finite element methods for beam structures

Exact solution

Finite element model

Semi-discretized model in space

$$\begin{aligned} \int_{z=0}^{\ell} w(z)m(z,t)\,dz &= \int_{z=0}^{\ell} w(z)\left\{I\partial_{tt}\theta - \partial_x\left(GJ\partial_x\theta\right)\right\}\,dz = \\ &= \int_{z=0}^{\ell} w(z)I\partial_{tt}\theta(z,t)\,dz + \int_{z=0}^{\ell} \partial_x w(z)GJ\partial_x\theta(z,t)\,dz - \left[w(x)GJ\partial_x\theta(x,t)\right]\Big|_{z=0}^{\ell}\ . \end{aligned}$$

With the boundary conditions of the model, the FEM approximation of the problem becomes

$$\int_{z=0}^{\ell} w(z)m(z,t)\,dz + w(\ell)M(t) = \int_{z=0}^{\ell} w(z)I\partial_{tt}\theta(z,t)\,dz + \int_{z=0}^{\ell} \partial_x w(z)GJ\partial_x\theta(z,t)\,dz\ .$$

30.6.2 Boundary conditions

Matrix partition or augmented system.

Matrix partition

$$\mathbf{f} = \mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x}$$

$$\begin{bmatrix}\mathbf{f}_f \\ \mathbf{f}_d\end{bmatrix} = \begin{bmatrix}\mathbf{M}_{ff} & \mathbf{M}_{fd} \\ \mathbf{M}_{df} & \mathbf{M}_{dd}\end{bmatrix}\begin{bmatrix}\ddot{\mathbf{x}}_f \\ \ddot{\mathbf{x}}_d\end{bmatrix} + \begin{bmatrix}\mathbf{K}_{ff} & \mathbf{K}_{fd} \\ \mathbf{K}_{df} & \mathbf{K}_{dd}\end{bmatrix}\begin{bmatrix}\mathbf{x}_f \\ \mathbf{x}_d\end{bmatrix}$$

Here the value of the kinematic variables $\mathbf{x}_d$, and the actions $\mathbf{f}_f$ is known. The unknowns are $\mathbf{x}_f$ (the displacement of the free d.o.f.s of the system), and $\mathbf{f}_d$ (the reactions at the constraints). Separating the unknowns from the known quantities,

$$\mathbf{M}_{ff}\ddot{\mathbf{x}}_f + \mathbf{K}_{ff}\mathbf{x}_f = \mathbf{f}_f - \mathbf{M}_{fd}\ddot{\mathbf{x}}_d - \mathbf{K}_{fd}\mathbf{x}_d$$

$$\mathbf{f}_d = -\begin{bmatrix}\mathbf{M}_{df} & \mathbf{M}_{dd}\end{bmatrix}\begin{bmatrix}\ddot{\mathbf{x}}_f \\ \ddot{\mathbf{x}}_d\end{bmatrix} - \begin{bmatrix}\mathbf{K}_{df} & \mathbf{K}_{dd}\end{bmatrix}\begin{bmatrix}\mathbf{x}_f \\ \mathbf{x}_d\end{bmatrix}$$

This system is decoupled, as the reactions at the constraints are not involved in the dynamical systems for $\mathbf{x}_f$, and thus they can be computed a posteriori.

Augmented system

See the discussion about Poisson equation - i.e. the static version of the dynamical equation used here, with no inertia contribution - in Math: Numerical methods for PDEs

- FEM: Poisson equation. Augmenting the linear system
- FVM: Poisson equation. Augmenting the linear system

Spectral decomposition of the undamped system

Eigenvalue problem focuses on the free response of the system, so all the forcing terms - both prescribed actions in $\mathbf{f}_f$ and prescribed kinetic variables in $\mathbf{x}_d$ - are set equal to zero. Thus, it's possible to focus on the f-part of the partitioned system only, $\mathbf{M}_{ff}\ddot{\mathbf{x}}_f + \mathbf{K}_{ff}\mathbf{x}_f = ...$ In the following lines, pedices are dropped.

Either generalized eigenvalue problem of the undamped system as a second-order n-dimensional system

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{f}\,,$$

$$s^2\mathbf{M}\mathbf{x} + \mathbf{K}\mathbf{x} = \mathbf{0}$$

or the generalized eigenvalue problem as a first-order $2n$-dimensional system

$$\begin{bmatrix}\mathbf{I} & \cdot \\ \cdot & \mathbf{M}\end{bmatrix}\begin{bmatrix}\dot{\mathbf{x}} \\ \dot{\mathbf{v}}\end{bmatrix} + \begin{bmatrix}\cdot & -\mathbf{I} \\ \mathbf{K} & \cdot\end{bmatrix}\begin{bmatrix}\mathbf{x} \\ \mathbf{v}\end{bmatrix} = \begin{bmatrix}\mathbf{0} \\ \mathbf{f}\end{bmatrix}\,.$$

$$s\mathbf{A}\mathbf{z} + \mathbf{B}\mathbf{z} = \mathbf{0}$$

Matrix symmetries and routines.

- In the second-orded system, matrices of a mechanical system are usually symmetric. It's possible to use the routine for Hermitian matrices `.eigsh()`
- In the first-orded system, the augmented matrices are not symmetric. It's not possible to use the routine for Hermitian matrices. Use the routine for general matrix `.eigs()` instead

Solution of the eigenvalue problem, and diagonalization - modal form. Routines returns mass normalized eigenvectors, so that

$$\mathbf{V}^*\mathbf{M}\mathbf{V} = \mathbf{I}\,.$$

As the matrix form of the generalized eigenvalue problem reads $\mathbf{M}\mathbf{V}\mathbf{S}^2 + \mathbf{K}\mathbf{V} = \mathbf{0}$.

Changing basis from "nodal" to "modal" variables, $\mathbf{x} = \mathbf{V}\mathbf{q}$, and projecting on the eigenvectors, the dynamical equation becomes

$$\begin{aligned}\mathbf{V}^*\mathbf{f} &= \mathbf{V}^*\left(\mathbf{M}\mathbf{V}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{V}\mathbf{u}\right)\\ \mathbf{V}^*\mathbf{f} &= \ddot{\mathbf{u}} + \mathrm{diag}\left\{-s_i^2\right\}\mathbf{u} =\\ &= \ddot{\mathbf{u}} + \mathrm{diag}\left\{\Omega_i^2\right\}\mathbf{u}\,.\end{aligned}$$

For a mechanical system with definite positive $\mathbf{M}$ and $\mathbf{K}$, the eigenvalues of the system are imaginary so that $s_i^2 = -\Omega_i^2$.

```
Mass-normalization of the eigenvectors within the tol
```

Modal structural damping

See

- Continuum mechanics: structural damping
- Sensitivity of eigenvalues and eigenvectors in a second-order system
- Sensitivity of spectral decomposition

Modal damping for the simultaneous diagonalization of mass, damping and stiffness matrices.

$$\mathbf{V}^*\mathbf{f} = \ddot{\mathbf{u}} + \mathrm{diag}\{2\xi_i\Omega_i\}\dot{\mathbf{u}} + \mathrm{diag}\{\Omega_i^2\}\mathbf{u}$$

i.e. with

$$\mathbf{V}^*\mathbf{C}\mathbf{V} = \mathrm{diag}\{2\xi_i\Omega_i\}$$

so that the damping matrix in "nodal" variables reads

$$\mathbf{C} = \mathbf{V}\mathrm{diag}\{2\xi_i\Omega_i\}\mathbf{V}^*$$

The eigenvalues of the damped system, for $\xi_i < 1$, become

$$s_i = -\xi_i\Omega_i \mp j\Omega_i\sqrt{1-\xi_i^2}\,.$$

Exact solution.

$$\begin{aligned}&I\ddot{\theta} - GJ\theta'' = 0\\ &\theta(0,t) = 0\\ &GJ\theta'(\ell,t) = 0\,.\end{aligned}$$

$$\theta(z,t) = f(x)g(t)$$

$$\frac{I}{GJ}\frac{\ddot{g}}{g} = \frac{f''}{f} = -k^2$$

$$f(z) = A\cos(kz) + B\sin(kz)$$

with boundary conditions

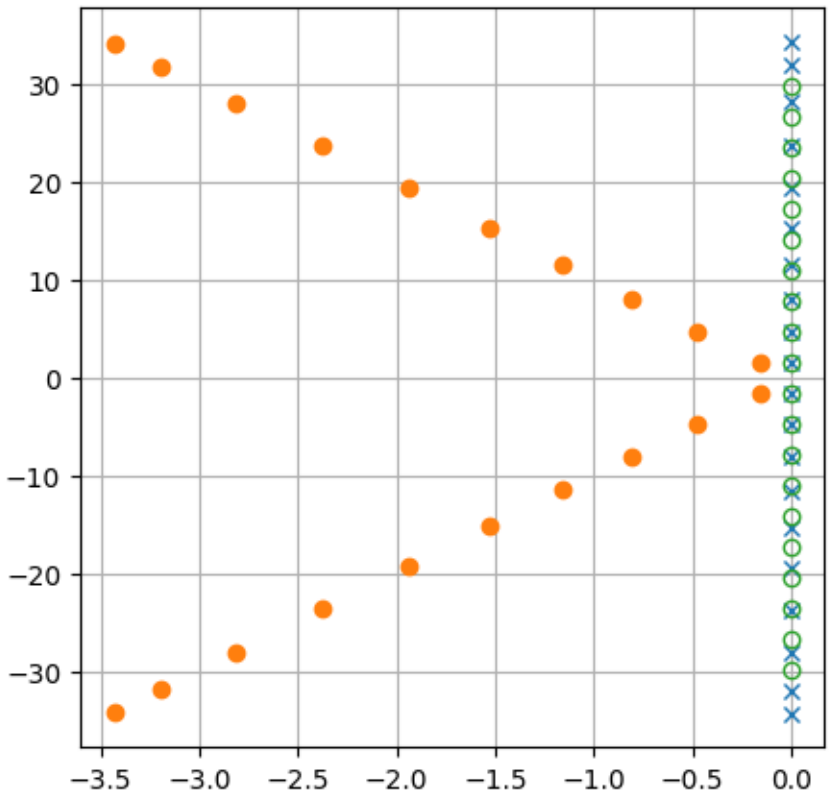
$$0 = f(0) = A$$
$$0 = f'(\ell) = Bk\cos(k\ell)$$

so that $\cos(k\ell) = 0$, and thus $k_n\ell = \frac{\pi}{2} + n\pi$, $n \in \mathbb{N}$, i.e. the wave numbers of the modes are

$$k_n = \frac{\pi}{\ell}\left(n + \frac{1}{2}\right) .$$

Pulsations of the modes are

$$\omega_n = k_n\sqrt{\frac{GJ}{I}} = \frac{\pi}{\ell}\sqrt{\frac{GJ}{I}}\left(n + \frac{1}{2}\right) .$$




```
 Numerical evals     Exact evals
-------------------------------------
 0.0000 -1.5724 i   0.0000 -1.5708 i
 0.0000 -4.7561 i   0.0000 -4.7124 i
 0.0000 -8.0571 i   0.0000 -7.8540 i
 0.0000-11.5542 i   0.0000-10.9956 i
 0.0000-15.3203 i   0.0000-14.1372 i
 0.0000-19.4002 i   0.0000-17.2788 i
 0.0000-23.7547 i   0.0000-20.4204 i
 0.0000-28.1465 i   0.0000-23.5619 i
 0.0000-31.9858 i   0.0000-26.7035 i
 0.0000-34.3236 i   0.0000-29.8451 i
```

Eigenvalues of the first modes show a good match with the exact eigenvalues. For larger wave-number modes, the modes of the discrete system have higher frequencies than the modes of the continuous system: numerical approximation introduces "numerical constraints" (as it's not capable of representing the exact solution, but only a system with a finite-number dofs), and thus makes the system "stiffer": higher frequencies of the numerical eigenvalues agree with this intuition.

30.6.3 Response to a torsional moment at the "free" end

With force at the free end, the forcing term of the discrete system reads

$$\mathbf{f}_f = \begin{bmatrix} 0 \\ \dots \\ 0 \\ M(t) \end{bmatrix} = \begin{bmatrix} 0 \\ \dots \\ 0 \\ 1 \end{bmatrix} M(t) \, ,$$

being the last node the free node. In modal variables

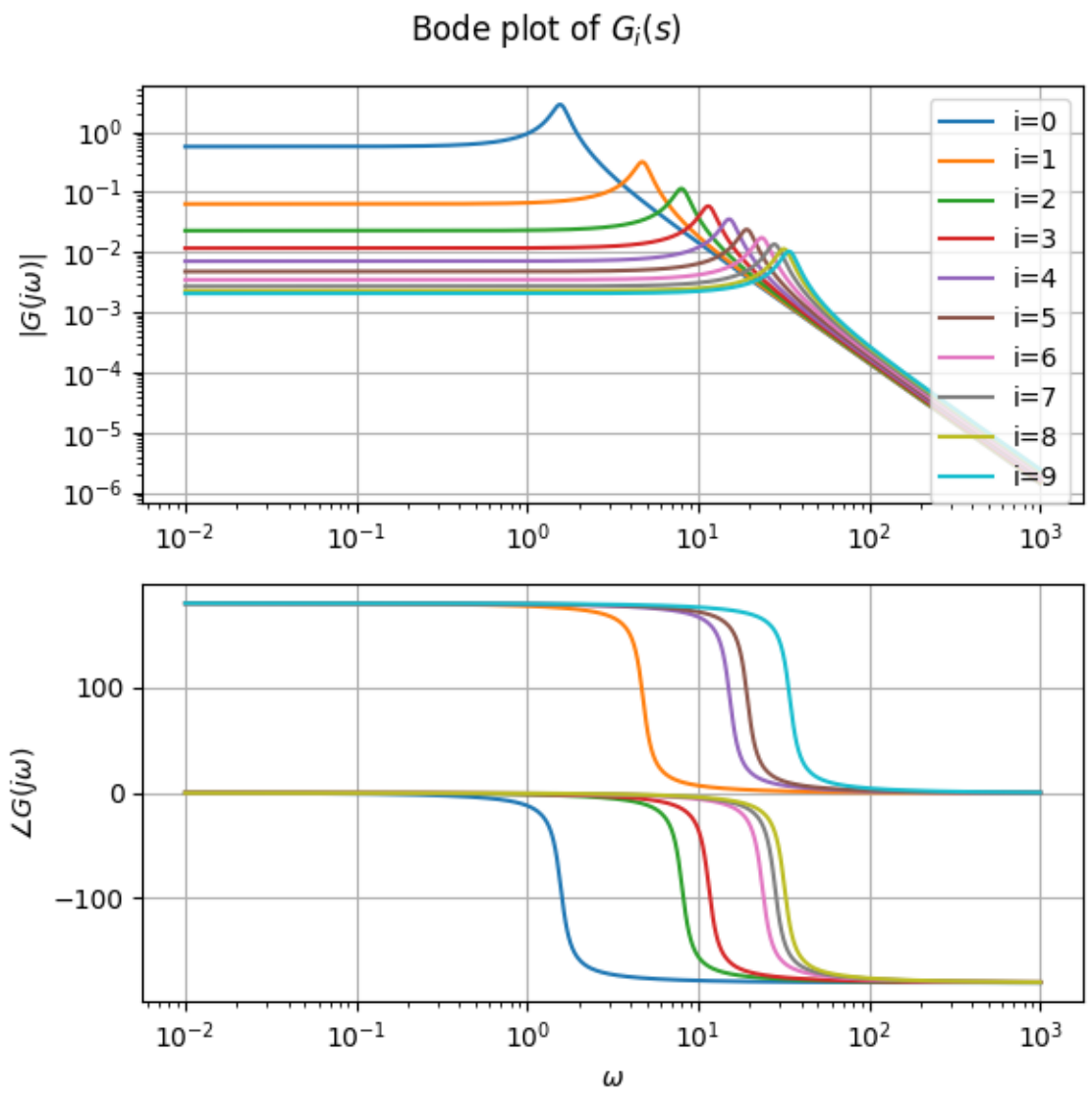
$$\ddot{u}_i + 2\xi_i \Omega_i \dot{u}_i + \Omega_i^2 u_i = \mathbf{v}_i^* \mathbf{f}_f \, .$$

The input-output relation from the input $M(t)$ and the $N-1$-dimensional output u_i reads

$$u_i(s) = \frac{\mathbf{v}_i^* \mathbf{F}}{s^2 + 2\xi_i \Omega_i s + \Omega_i^2} M(t) \, ,$$

i.e. the transfer function between the torsional moment at the free end $M(t)$ and the amplitude of the ith mode is

$$G_i(s) = \frac{\mathbf{v}_i^* \mathbf{F}}{s^2 + 2\xi_i \Omega_i s + \Omega_i^2} \, .$$

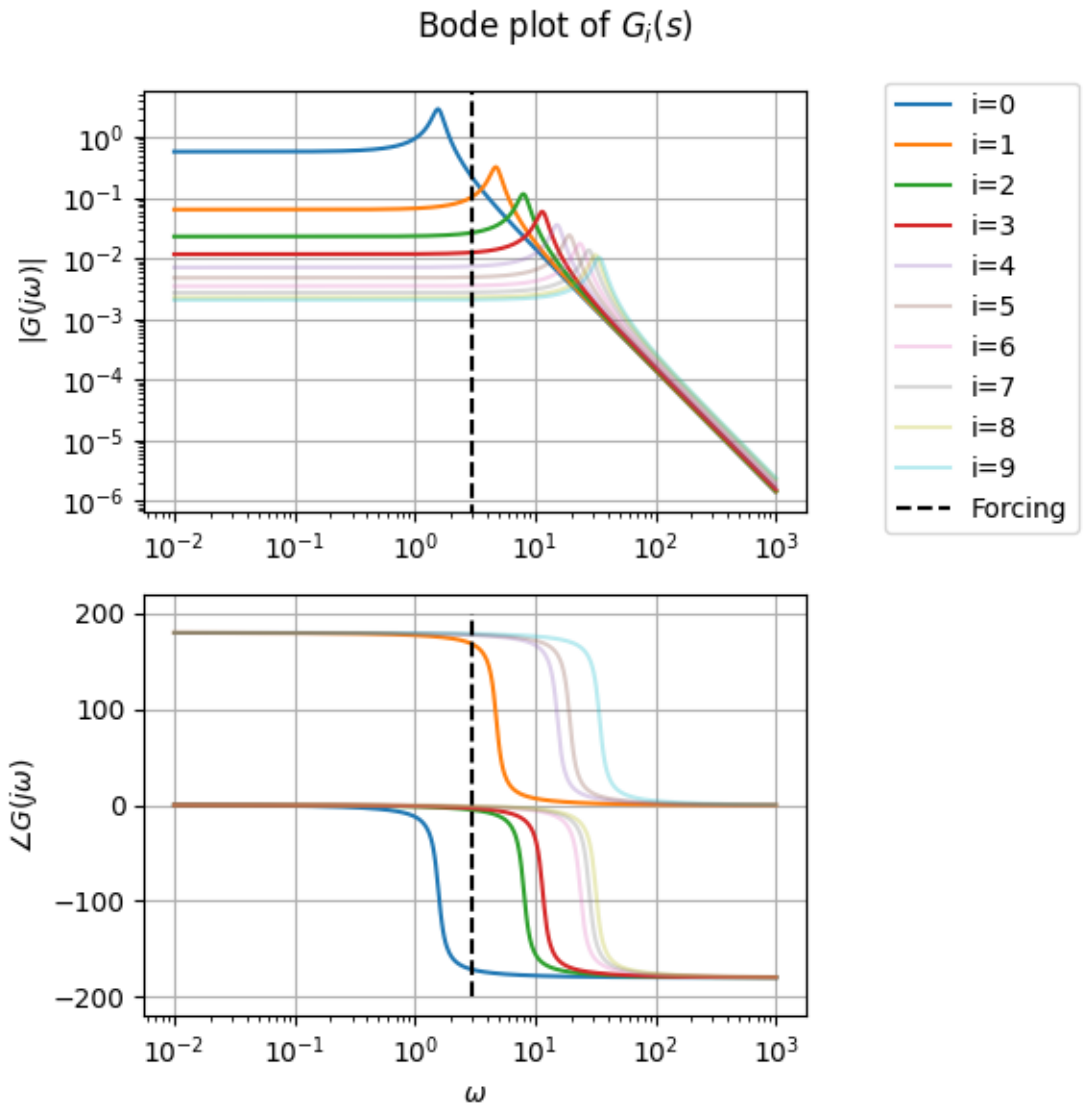


30.6.4 Time simulations

Here different approaches are evaluated:

- Truncation and direct recovery of loads
- Mode acceleration and static recovery of fast modes

```
Maximum frequency in forcing spectrum : 3.0
Threshold for slow/fast mode splitting: 12.0
Number of slow modes                  : 4
```



Mode acceleration and static recovery of fast modes.

$$\begin{bmatrix} \mathbf{u} \\ \mathbf{Ku} \\ \dot{\mathbf{q}}_s \\ \ddot{\mathbf{q}}_s \end{bmatrix} = \begin{bmatrix} \mathbf{U}_s & \mathbf{0} \\ \mathbf{KU}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \\ -\text{diag}\{\omega_s^2\} & -\text{diag}\{2\xi_s\omega_s\} \end{bmatrix} \begin{bmatrix} \mathbf{q}_s \\ \dot{\mathbf{q}}_s \end{bmatrix} + \begin{bmatrix} \mathbf{K}^{-1} - \mathbf{U}_s\text{diag}\{(m_s\omega_s^2)^{-1}\}\mathbf{U}_s^* \\ \mathbf{I} - \mathbf{MU}_s\text{diag}\{m_s^{-1}\}\mathbf{U}_s^* \\ \mathbf{0} \\ \text{diag}\{m_s^{-1}\}\mathbf{U}_s^T \end{bmatrix} \mathbf{f} \,.$$

Time integration of the slow part

First order system

$$\frac{d}{dt}\begin{bmatrix} \mathbf{q}_s \\ \dot{\mathbf{q}}_s \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\text{diag}\{\omega_s^2\} & -\text{diag}\{2\xi_s\omega_s\} \end{bmatrix} \begin{bmatrix} \mathbf{q}_s \\ \dot{\mathbf{q}}_s \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \text{diag}\{m_s^{-1}\}\mathbf{U}_s^T \end{bmatrix} \mathbf{f} \,.$$

$$\dot{\mathbf{z}} = \mathbf{Az} + \mathbf{B}M(t) = \mathbf{f}(t, \mathbf{z})$$

Second order system

$$\ddot{\mathbf{q}}_s = -\text{diag}\{\omega_s^2\}\mathbf{q}_s - \text{diag}\{2\xi\omega_s\}\dot{\mathbf{q}}_s + \text{diag}\{m_s^{-1}\}\mathbf{U}_s^*\mathbf{f}$$

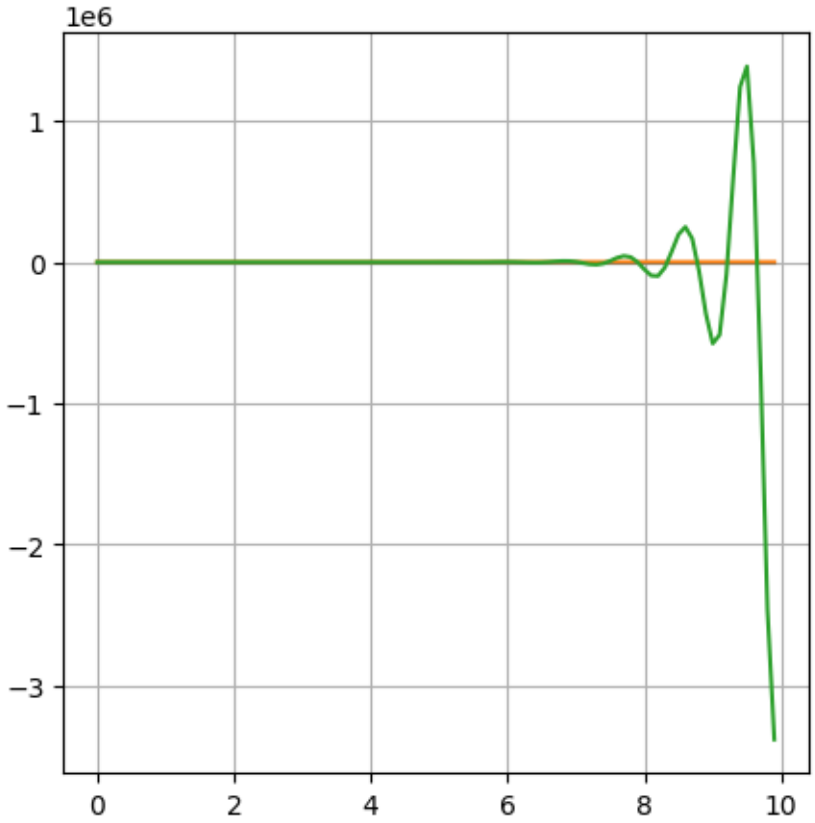
Explicit Euler

Explicit Euler.

$$\mathbf{z}_{n+1} = \mathbf{z}_n + h\mathbf{f}(t_n, \mathbf{z}_n)$$

Explicit Euler numerical integration is unstable

…

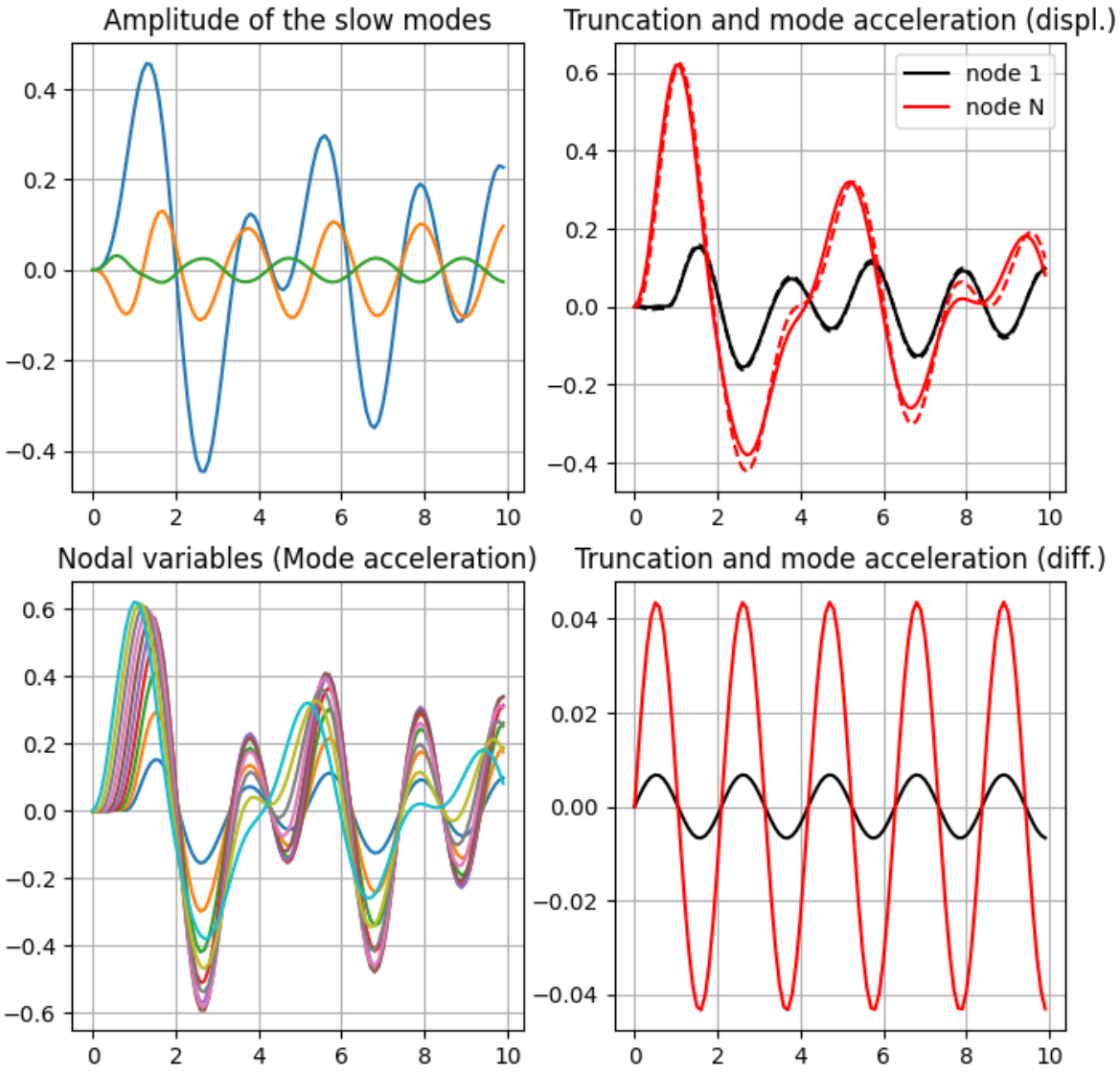


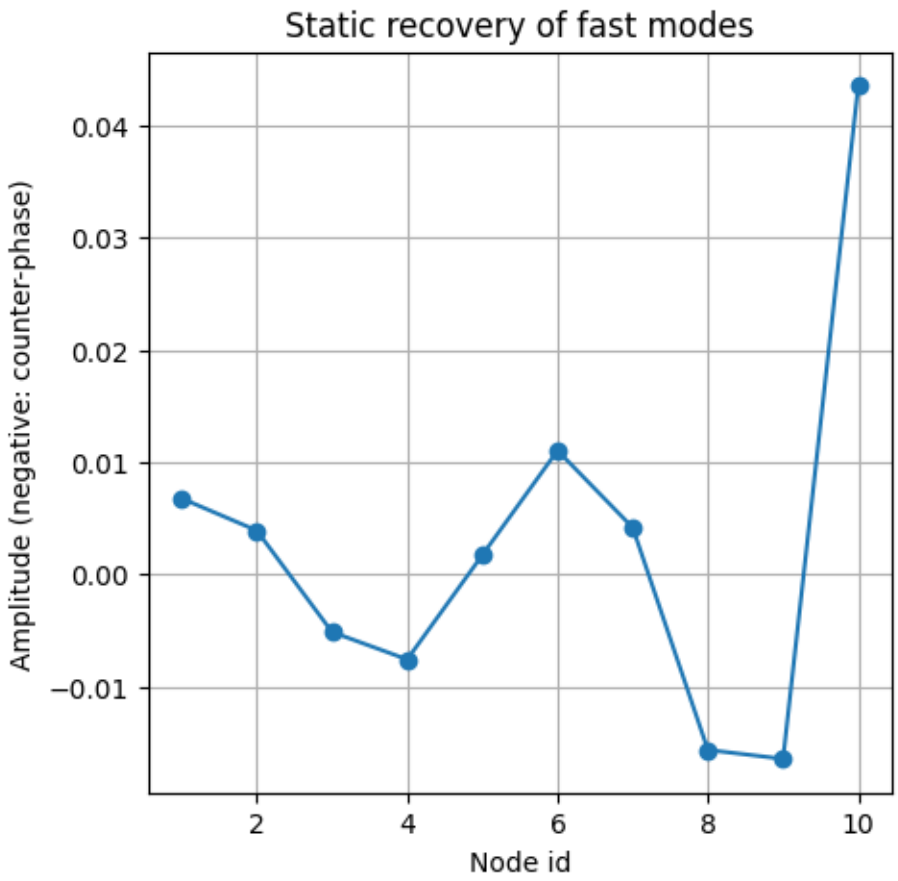
RK4(5)

Runge-Kutta. Exploiting **scipy.integrate.solve_ivp**. RK4(5) is a variable time-step method.

RK4(5) and high-frequencies. This method is not suited for systems with high-frequency contents, as the method uses smaller and smaller time-step to keep local truncation error small (and the method stable). Extreme small values of dt makes the number of iterations explodes and the performance of the method so poor that it becomes useless.

RK4(5) with fast mode truncation and mode acceleration recovery. Using the fast mode truncation and their static recovery, the dynamical part of the system involves only slow modes. Thus, this approach makes RK4(5) method suitable.


Amplitude of the slow modes
Truncation and mode acceleration (displ.)
node 1
node N
Nodal variables (Mode acceleration)
Truncation and mode acceleration (diff.)


Static recovery of fast modes
Amplitude (negative: counter-phase)
0.04
0.03
0.02
0.01
0.00
−0.01
2
4
6
8
10
Node id

Part X

System Theory

CHAPTER

THIRTYONE

INTRODUCTION TO SYSTEM THEORY

System theory shifts the focus from solving individual equations to analyzing the interaction between inputs, states, and outputs. …

Links

- *Ordinary differential equations* …
- *Control theory* aims at getting the desired behavior from a given model

- **State-space representation.**
- **Linear systems.**
- **Stability**
- **Controllability and Observability.**
- **Time, Laplace and Fourier domains.** Realizations,…

CHAPTER

THIRTYTWO

LINEAR TIME-INVARIANT SYSTEMS

A linear time invariant system is governed by a linear ODE with constant coefficients. These equations can be recast as a first order system of ODEs,

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \\ \mathbf{x}(0^-) = \mathbf{x}_0 \end{cases}$$

Exploiting the *properties of matrix exponential* the general expression of the state can be written as the sum of the free response to initial condition and the forced response.

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau$$
$$\mathbf{y}(t) = \mathbf{C}e^{\mathbf{A}t}\mathbf{x}_0 + \mathbf{C}\int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau + \mathbf{D}\mathbf{u}(t)$$

Proof in time domain

Multipying by $e^{-\mathbf{A}t}$,

$$e^{-\mathbf{A}t}(\dot{x}(t) - \mathbf{A}\mathbf{x}(t)) = e^{-\mathbf{A}t}\mathbf{B}\mathbf{u}(t)$$
$$\frac{d}{dt}\left(\mathbf{x}e^{-\mathbf{A}t}\right) = e^{-\mathbf{A}t}\mathbf{B}\mathbf{u}(t)$$

and integrating from 0^- to a generic time value t,

$$e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}_0 = \int_{\tau=0^-}^{t} e^{-\mathbf{A}\tau}\mathbf{B}\mathbf{u}(\tau)\, d\tau$$

The state $\mathbf{x}(t)$ can be written as the sum of the free response and a force response. The general expression of the state and the output as a function reads

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau$$
$$\mathbf{y}(t) = \mathbf{C}e^{\mathbf{A}t}\mathbf{x}_0 + \mathbf{C}\int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau + \mathbf{D}\mathbf{u}(t)$$

Laplace domain. The *Laplace transform* of the problem reads

$$\begin{cases} s\hat{\mathbf{x}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\hat{\mathbf{u}} + \mathbf{x}_0 \\ \hat{\mathbf{y}} = \mathbf{C}\hat{\mathbf{x}} + \mathbf{D}\hat{\mathbf{u}} \end{cases}$$

$$(s\mathbf{I}-\mathbf{A})\hat{\mathbf{x}} = \mathbf{B}\hat{\mathbf{u}} + \mathbf{x}_0$$
$$\hat{\mathbf{x}}(s) = (s\mathbf{I}-\mathbf{A})^{-1}\mathbf{x}_0 + (s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}\hat{\mathbf{u}}(s)$$
$$\hat{\mathbf{y}}(s) = \mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{x}_0 + \left[\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+\mathbf{D}\right]\hat{\mathbf{u}}(s)$$

Performing inverse Laplace transform allows to go back to time domain (just use Laplace inverse transform of a matrix exponential, and the formula (21.1) for Laplace transform of convolution).

32.1 Impulsive force

The effect of an impulsive force $\mathbf{u}_\delta\delta(t)$ at time $t=0$ is equivalent to an instantaneous change in the initial state, $\Delta\mathbf{x}_0 = \mathbf{B}\mathbf{u}_\delta$, from time 0^- before the impulse to time 0^+ after the impulse. Splitting the input $\mathbf{u}(t)$ as the sum of impulsive input and regular input,

$$\mathbf{u}(t) = \mathbf{u}_r(t) + \mathbf{u}_\delta\delta(t)$$
$$\hat{\mathbf{u}}(s) = \hat{\mathbf{u}}_r(s) + \mathbf{u}_\delta$$

the solution in **time domain** reads

$$\begin{aligned}\mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\left(\mathbf{u}_r(\tau)+\mathbf{u}_\delta\delta(\tau)\right)\,d\tau = \\ &= e^{\mathbf{A}t}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right) + \int_{\tau=0^-}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}_r(\tau)\,d\tau \\ \mathbf{y}(t) &= \mathbf{C}e^{\mathbf{A}t}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right) + \int_{\tau=0^-}^{t} \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}_r(\tau)\,d\tau + \mathbf{D}\mathbf{u}_r(t) + \mathbf{D}\mathbf{u}_\delta(t)\end{aligned}$$

while in **Laplace domain** reads

$$\hat{\mathbf{x}}(s) = (s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}\hat{\mathbf{u}}_r(s) + (s\mathbf{I}-\mathbf{A})^{-1}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right)$$
$$\hat{\mathbf{y}}(s) = \left[\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+\mathbf{D}\right]\hat{\mathbf{u}}_r(s) + \mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\left(\mathbf{x}_0+\mathbf{B}\mathbf{u}_\delta\right) + \mathbf{D}\mathbf{u}_\delta$$

32.2 Properties

Matrix exponential.

$$e^{\mathbf{A}t} = \sum_{k=0}^{+\infty}\frac{\mathbf{A}^k t^k}{k!}\,.$$

Assuming it's possible swap derivative operator and summation (when?), it's possible to write

$$\frac{d}{dt}e^{\mathbf{A}t} = \frac{d}{dt}\sum_{k=0}^{+\infty}\frac{\mathbf{A}^k t^k}{k!} = \sum_{k=1}^{+\infty} k t^{k-1}\frac{\mathbf{A}^k t^{k-1}}{k!} = \mathbf{A}e^{\mathbf{A}t}\,.$$

Laplace transform of exponential matrix.

$$\begin{aligned}\mathcal{L}\left\{e^{\mathbf{A}t}\right\}(s) &:= \int_{t=0^-}^{+\infty} e^{\mathbf{A}t}e^{-st}\,dt = \\ &= \int_{t=0^-}^{+\infty} e^{(-s\mathbf{I}+\mathbf{A})t}\,dt = \\ &= (-s\mathbf{I}+\mathbf{A})^{-1}\,e^{(-s\mathbf{I}+\mathbf{A})t}\Big|_{t=0^-}^{+\infty} = \\ &= (s\mathbf{I}-\mathbf{A})^{-1}\,,\end{aligned}$$

for all the values of s for which $-s\mathbf{I} + \mathbf{A}$ is asymptotically stable, i.e. has all the eignevalues (thus, assuming that the matrix $\mathbf{A}$ can be diagonalizable. What happens if not? Exploit other matrix decompositions to draw conclusions) with negative real parts, and thus for all the values of $s > \max \mathrm{re}\{s_k(\mathbf{A})\}$, as it's shown in *Example 32.2.1*

Example 32.2.1 (Asymptotic stability of a matrix A)

An $N \times N$ diagnonalizable matrix $\mathbf{A}$,

$$\mathbf{A}\mathbf{v}_k = \mathbf{v}_k \, s_k \tag{32.1}$$

has all the eigenvalues with negative real part, $\mathrm{re}\,\{s_k\} < 0$, $\forall k = 1 : N$.

The eigenvalues of a matrix $a\mathbf{I} + \mathbf{A}$ are $a + s_k$, while the eigenvectors are the same as those of the matrix $\mathbf{A}$. This can be easily proved adding $a\mathbf{I}\mathbf{v}_k$ to both sides of equation (32.1),

$$(a\mathbf{I} + \mathbf{A})\, \mathbf{v}_k = \mathbf{v}_k(a + s_k) \; .$$

Transform of the convolution.

CHAPTER

THIRTYTHREE

COORDINATE TRANSFORMATION

33.1 General transformation

In this section, a coordinate transformation of the state of a linear system

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}$$

is discussed. Let the coordinate transformation be $\mathbf{x} = \mathbf{T}\hat{\mathbf{x}}$ be invertible, then the linear system using the new set of coordinates becomes

$$\begin{cases} \dot{\hat{\mathbf{x}}} = \hat{\mathbf{A}}\hat{\mathbf{x}} + \hat{\mathbf{B}}\mathbf{u} \\ \mathbf{y} = \hat{\mathbf{C}}\hat{\mathbf{x}} + \hat{\mathbf{D}}\mathbf{u} \end{cases}$$

with

$$\begin{aligned} \hat{\mathbf{A}} &= \mathbf{T}^{-1}\mathbf{A}\mathbf{T} \quad & \hat{\mathbf{B}} &= \mathbf{T}^{-1}\mathbf{B} \\ \hat{\mathbf{C}} &= \mathbf{C}\mathbf{T} & \hat{\mathbf{D}} &= \mathbf{D}\,. \end{aligned}$$

33.1.1 Characteristic polynomial

The characteristic polynomial of the system doesn't change, as

$$\left|s\mathbf{I} - \hat{\mathbf{A}}\right| = \left|s\mathbf{I} - \mathbf{T}^{-1}\mathbf{A}\mathbf{T}\right| = \left|\mathbf{T}^{-1}\left(s\mathbf{I} - \mathbf{A}\right)\mathbf{T}\right| = \underbrace{\left|\mathbf{T}^{-1}\right|}_{=|\mathbf{T}|^{-1}} \left|s\mathbf{I} - \mathbf{A}\right| \left|\mathbf{T}\right| = \left|s\mathbf{I} - \mathbf{A}\right| \ .$$

33.1.2 Gramians

The observability Gramian (*Definition 36.1.1*) and controllability Gramian (*Definition 37.1.1*) of the transformed system read

$$\begin{aligned} \hat{\mathbf{W}}_o &= \mathbf{T}^T\mathbf{W}_c\mathbf{T} \\ \hat{\mathbf{W}}_c &= \mathbf{T}^{-1}\mathbf{W}_c\mathbf{T}^{-T} \end{aligned}$$

Proof

$$\hat{\mathbf{W}}_o := \int_{\tau=0}^{t} e^{\hat{\mathbf{A}}^T(t-\tau)}\hat{\mathbf{C}}^T(\tau)\hat{\mathbf{C}}(\tau)e^{\hat{\mathbf{A}}(t-\tau)}d\tau$$
$$\hat{\mathbf{W}}_c := \int_{\tau=0}^{t} e^{\hat{\mathbf{A}}(t-\tau)}\hat{\mathbf{B}}(\tau)\hat{\mathbf{B}}^T(\tau)e^{\hat{\mathbf{A}}^T(t-\tau)}d\tau$$

33.2 Example of transformations

33.2.1 Diagonalization

Spectral decomposition. If the spectral decomposition of matrix $\mathbf{A} \in \mathbb{C}^{n,n}$ exists, there are vector basis $\{\mathbf{v}_i\}_{i=1:n}$, $\{\mathbf{w}_i\}_{i=1:n}$ of $\mathbb{C}^n$ so that

$$\mathbf{A}\mathbf{v}_i = s_i\mathbf{v}_i$$
$$\mathbf{w}_i^*\mathbf{A} = s_i\mathbf{w}_i^* ,$$

being s_i the eigenvalues of the matrix, $\mathbf{v}_i$ the right-eigenvectors and $\mathbf{w}_i$ the left eigenvectors. Using matrix formalism

$$\mathbf{A}\mathbf{V} = \mathbf{V}\Lambda$$
$$\mathbf{W}\mathbf{A} = \Lambda\mathbf{W} .$$

Left- and right-eigenvectors associated with different eigenvalues are orthogonal.

Left- and right-eigenvectors orthogonality.

$$\mathbf{w}_k^*\mathbf{A}\mathbf{v}_i = s_i\mathbf{w}_k^*\mathbf{v}_i$$
$$\mathbf{w}_k^*\mathbf{A}\mathbf{v}_i = s_k\mathbf{w}_k^*\mathbf{v}_i$$

implies (subtraction) $(s_i - s_k)\mathbf{w}_k^*\mathbf{v}_i = 0$. If $s_i \neq s_k$, then $\mathbf{w}_k^*\mathbf{v}_i = 0$.

- If $s_i, s_k \neq 0$, this also implies $\mathbf{w}_k^*\mathbf{A}\mathbf{v}_i = 0$.
- This relation when the indices are equal can be used for normalization condition
- A unit orthogonal basis can be computed for eigenvalues with algebraic multiplicity > 1

If normalization condition reads $\mathbf{w}_i^*\mathbf{v}_i = 1$ (no sum), $\mathbf{W}^*\mathbf{V} = \mathbf{I}$, then

$$\mathbf{W}^*\mathbf{A}\mathbf{V} = \Lambda .$$

Diagonalization. Let $\mathbf{x} = \mathbf{V}\hat{\mathbf{x}}$ the representation of the state with the vector basis $\mathbf{V}$, and the projection of the state dynamical equation onto $\mathbf{W}$. The state-space representation of the system reads

$$\begin{cases} \dot{\hat{\mathbf{x}}} = \Lambda\hat{\mathbf{x}} + \mathbf{W}^*\mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{V}\hat{\mathbf{x}} + \mathbf{D}\mathbf{u} , \end{cases}$$

with Λ diagonal. I.e. in components, with $\hat{\mathbf{B}} := \mathbf{W}^*\mathbf{B}$,

$$\dot{\hat{x}}_i = \lambda_i\hat{x}_i + \sum_{j=1}^{m}\hat{B}_{ij}u_j = \lambda_i\hat{x}_i + \hat{\mathbf{b}}_i^*\mathbf{u} ,$$

or in Laplace domain, the ith component in the diagonal representation reads

$$\hat{x}_i = \frac{1}{s-\lambda_i}\hat{\mathbf{b}}_i^* \mathbf{u} \, .$$

the state in the original coordinates reads

$$\mathbf{x} = \mathbf{V}\hat{\mathbf{x}} = \sum_{i=1}^{n} \mathbf{v}_i \hat{x}_i = \sum_{i=1}^{n} \mathbf{v}_i \frac{1}{s-\lambda_i}\hat{\mathbf{b}}_i^* \mathbf{u} \, .$$

and the output reads

$$\begin{aligned}
\mathbf{y} &= \mathbf{C}\mathbf{V}\hat{\mathbf{x}} + \mathbf{D}\mathbf{u} = \\
&= \hat{\mathbf{C}}\hat{\mathbf{x}} + \mathbf{D}\mathbf{u} = \\
&= \sum_{i=1}^{n} \hat{\mathbf{c}}_i \hat{x}_i + \mathbf{D}\mathbf{u} = \\
&= \left[\sum_{i=1}^{n} \frac{1}{s-\lambda_i} \hat{\mathbf{c}}_i \hat{\mathbf{b}}_i^* + \mathbf{D} \right] \mathbf{u} = \\
&= \mathbf{H}(s)\mathbf{u} \, .
\end{aligned}$$

Proof

…

Transfer function. Thus, the transfer function between the input and the output can be written as

$$\hat{\mathbf{H}}(s) = \sum_{i=1}^{n} \frac{1}{s-\lambda_i} \hat{\mathbf{c}}_i \hat{\mathbf{b}}_i^* + \mathbf{D} \, .$$

This is a linear combination of factors $\frac{1}{s-\lambda_i}$. Thus, the denominators of these fractions appear in the denominatore of the transfer function if both $\mathbf{b}_i \neq \mathbf{0}$ and $\mathbf{c}_i \neq \mathbf{0}$. **todo** *Further cancellations may occur. It's better to use* observability *or* controllability *vector basis or* Kalman decomposition *to link cancellations and observability/controllability*

33.2.2 Kalman decomposition

$$\mathbf{x} = \mathbf{T}\mathbf{x} = \begin{bmatrix} \mathbf{U}_1 & \mathbf{U}_2 & \mathbf{U}_3 & \mathbf{U}_4 \end{bmatrix} \mathbf{z} = \begin{bmatrix} \mathbf{U}_1 & \mathbf{U}_2 & \mathbf{U}_3 & \mathbf{U}_4 \end{bmatrix} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} \, ,$$

Kalman decomposition of a linear system reads

$$\begin{cases} \dfrac{d}{dt} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} & \mathbf{A}_{14} \\ \mathbf{0} & \mathbf{A}_{22} & \mathbf{0} & \mathbf{A}_{24} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{33} & \mathbf{A}_{34} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{44} \end{bmatrix} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \mathbf{u} \\ \mathbf{y} = \begin{bmatrix} \mathbf{0} & \mathbf{C}_2 & \mathbf{0} & \mathbf{C}_4 \end{bmatrix} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} + \mathbf{D}\mathbf{u} \end{cases}$$

Transfer function

Asymptotical stable systems have the free response decaying to zero after a certain time range, so that only the forced response survives. Forced response in Laplace domain reads

$$
\begin{aligned}
(s\mathbf{I} - \mathbf{A}_{44})\mathbf{z}_4 &= \mathbf{0} \\
\mathbf{z}_3 &= (s\mathbf{I} - \mathbf{A}_{33})^{-1}\mathbf{A}_{34}\mathbf{z}_4 \\
\mathbf{z}_2 &= (s\mathbf{I} - \mathbf{A}_{22})^{-1}\left(\mathbf{A}_{24}\mathbf{z}_4 + \mathbf{B}_2\mathbf{u}\right) \\
\mathbf{z}_1 &= (s\mathbf{I} - \mathbf{A}_{11})^{-1}\left(\mathbf{A}_{12}\mathbf{z}_2 + \mathbf{A}_{13}\mathbf{z}_3 + \mathbf{A}_{14}\mathbf{z}_4 + \mathbf{B}_1\mathbf{u}\right)
\end{aligned}
$$

and thus

$$
\begin{aligned}
\mathbf{z}_4 &= \mathbf{0} \\
\mathbf{z}_3 &= \mathbf{0} \\
\mathbf{z}_2 &= (s\mathbf{I} - \mathbf{A}_{22})^{-1}\mathbf{B}_2\mathbf{u} \\
\mathbf{z}_1 &= (s\mathbf{I} - \mathbf{A}_{11})^{-1}\left[\mathbf{A}_{12}(s\mathbf{I} - \mathbf{A}_{22})^{-1}\mathbf{B}_2 + \mathbf{B}_1\right]\mathbf{u}
\end{aligned}
$$

and the output reads

$$
\mathbf{y} = \mathbf{C}_2(s\mathbf{I} - \mathbf{A}_{22})^{-1}\mathbf{B}_2\mathbf{u}\ .
$$

From the input-output relation, it's clear that only the reachable and observable part of the system contributes to the transfer function, i.e. to the input-output relation for asymptotically stable systems after the free response due to non-zero initial conditions has decayed.

CHAPTER

THIRTYFOUR

LTI SYSTEM RESPONSE

Usually the response of LTI to 3 different input are studied

- integrable input, being response to non-zero initial condition equivalent to an impulsive force at time $t = 0$, see *equivalence impulsive force - istantanteous change of state*
- periodic input
- stochastic input

34.1 LTI

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}$$

with initial conditions, $\varkappa(0) = \mathbf{x}_0$.

34.2 Response of LTI systems

The response of a LTI can be written as the sum of a free and forced response.

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau \\ \mathbf{y}(t) &= \mathbf{C}\, e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} \mathbf{C}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\, d\tau + \mathbf{D}\mathbf{u}(t) \end{aligned} \tag{34.1}$$

Proof

todo

Impulse response, $\mathbf{H}(t)$

Functions

$$\begin{aligned} \mathbf{H}_x(t) &:= e^{\mathbf{A}t}\mathbf{B} \\ \mathbf{H}(t) &:= \mathbf{C}e^{\mathbf{A}t}\mathbf{B} + \mathbf{D}\delta(t) \end{aligned}$$

are defined **impulse response functions** for the state $\mathbf{x}$ and the output $\mathbf{y}$, as they determine with the evolution of the state and the output of a system with zero initial condition $\mathbf{x}_0 = \mathbf{0}$ as a consequence of a impulsive force $\mathbf{u}_\delta(t) = \mathbf{u}\delta(t)$. Using Lagrange formula (34.1)

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{B}\mathbf{u} &&= \mathbf{H}_x(t)\mathbf{u} \\ \mathbf{y}(t) &= \left[\mathbf{C}e^{\mathbf{A}t}\mathbf{B} + \mathbf{D}\delta(t)\right]\mathbf{u} &&= \mathbf{H}(t)\mathbf{u} \end{aligned} \tag{34.2}$$

Convolution

Integral in Lagrange formula (34.1) is the convolution of the impulse response function and the input. Changing the integration variable $z = t - \tau$, $dz = -d\tau$, $\tau = 0 : z = t$, $\tau = t$, $z = 0$ (and going back from z to τ as the symbol for the dummy integration variable), it's possible to swap the arguments of the impulsive force and the forcing,

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} \mathbf{H}_x(\tau)\mathbf{u}(t-\tau)\,d\tau \\ \mathbf{y}(t) &= \mathbf{C}\,e^{\mathbf{A}t}\mathbf{x}_0 + \int_{\tau=0}^{t} \mathbf{H}(\tau)\mathbf{u}(t-\tau)\,d\tau \end{aligned} \tag{34.3}$$

Linear but not time-invariant

todo *The response of linear system, but not time-invariant can be written with the same expression used by Lagrange fomrula (34.1), with $\mathbf{H}(t)$ not equal to …, but solution of the ODE…*

$$\begin{aligned} \Phi\left(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}\right) &= \Phi\mathbf{B}\mathbf{u} \\ \frac{d}{dt}\left(\Phi\mathbf{x}\right) &= \Phi\mathbf{B}\mathbf{u}\ , \end{aligned}$$

with $\dot{\Phi} = -\Phi\mathbf{A}$, and proper initial conditions $\Phi(t,0) = \mathbf{I}$.

…

$$\mathbf{x}(t) = \Phi(t,0)\mathbf{x}_0 + \int_{\tau=0}^{t} \Phi(t,\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)\,d\tau\ .$$

34.3 Equilibria

Equilibrium solution of a ODE is a stationary solution, independent from time, $\dot{\bar{\mathbf{x}}} = \mathbf{0}$. With a steady forcing $\bar{\mathbf{u}}$,

$$\begin{cases} \mathbf{0} = \mathbf{A}\bar{\mathbf{x}} + \mathbf{B}\bar{\mathbf{u}} \\ \mathbf{y} = \mathbf{C}\bar{\mathbf{x}} + \mathbf{D}\bar{\mathbf{u}}\ . \end{cases}$$

If A is not singular, for every steady forcing $\bar{\mathbf{u}}$ only one equilibrium exists for the system, $\bar{\mathbf{x}} = \mathbf{A}^{-1}\mathbf{B}\bar{\mathbf{u}}$.

If A is singular…**todo** *rely on* Linear algebra

34.4 Stability

Asymptotic stability to initial perturbations. A system is stable under perturbation of initial conditions if the free response goes to zero as $t \to +\infty$,

$$\lim_{t \to +\infty} \mathbf{x}_{free}(t) = \mathbf{0} \, .$$

todo *for all perturbations*(?).

If matrix $\mathbf{A}$ is diagonalizable, a system is asymptotically stable if all the eigenvalues of the matrix $\mathbf{A}$ have negative real part, $\text{re}\{\lambda_i(\mathbf{A})\} < 0$.

todo *rely on* Linear algebra

34.5 Response to integrable input - and Laplace transform

$$\begin{aligned} \mathbf{x}(s) &= (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}_0 + \mathbf{H}_x(s)\mathbf{u}(s) \\ \mathbf{y}(s) &= \mathbf{C} \, (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{x}_0 + \mathbf{H}(s)\mathbf{u}(s) \end{aligned} \tag{34.4}$$

34.6 Response to periodic input of stable systems - and Fourier transform

Forced response to periodic input of an stable system can be represented in Fourier domain replacing s with $j\omega$ (and dropping j once it's clear we're dealing with Fourier analysis)

$$\begin{aligned} \mathbf{x}(\omega) &= \mathbf{H}_x(\omega) \, \mathbf{u}(\omega) \\ \mathbf{y}(\omega) &= \mathbf{H}(\omega) \, \mathbf{u}(\omega) \end{aligned} \tag{34.5}$$

34.7 Response to stochastic input of stable systems

34.7.1 Assumptions

…

34.7.2 Expected value

$$\mu_{\mathbf{x}}(t) := \mathbb{E}[\mathbf{x}(t)]$$

Taking expected value of LTI equations

$$\begin{cases} \mathbb{E}[\dot{\mathbf{x}}] = \mathbf{A}\mathbb{E}[\mathbf{x}] + \mathbf{B}\mathbb{E}[\mathbf{u}] \\ \mathbb{E}[\mathbf{y}] = \mathbf{C}\mathbb{E}[\mathbf{x}] + \mathbf{D}\mathbb{E}[\mathbf{u}] \end{cases}$$

todo

- *is it possible to swap time derivative and expected value operators?*

- *does time derivative of a stochastic process always exist? What's the right way to interpret time derivative in stochastic equations? As an increment?*

$$d\mathbf{x} = \mathbf{A}\mathbf{x}\,dt + \mathbf{B}\mathbf{u}\,dt$$

so that if $\mathbf{u}(t) \sim \xi(t)$, then $\mathbf{u}\,dt \sim d\mathbf{W}$

34.8 Analysis of response of LTI in Fourier domain

Energy or power. For

- time integrable signals: don't average in time; energy
- periodic or stochastic non-integrable signals: average in time; power

Stationary stochastic process. Strong-sense (joint probability independent from time) or wide-sense (moments up to order m are independent on time)

Ergodic process. Average over random process can be replaced with average over time.

Example of stationary non-ergodic process

As an example of stationary, but not ergodic process is a random process whose realizations are

$$X_t = \theta\,, \qquad \forall t \in [0, T]$$

with θ drawn from a random variable Θ, with average μ. Average in time over a realization gives θ, while average over realizations gives μ for all t,

$$\begin{aligned} \text{Avg. in time over a realization:} \quad & \frac{1}{T}\int_{t=0}^{T} X_t\,dt = \frac{1}{T}\int_{t=0}^{T} \theta\,dt = \theta \\ \text{Avg. in probability over realizations:} \quad & \mathbb{E}[X_t] = \mathbb{E}[\Theta] = \mu\,. \end{aligned}$$

34.8.1 Stationary Ergodic processes

Auto-correlation

$$\mathbf{R}_{XX}(t_1, t_2) = \mathbb{E}\left[\mathbf{X}_{t_1}\,\mathbf{X}_{t_2}^*\right]$$

Auto-covariance

$$\mathbf{K}_{XX}(t_1, t_2) = \mathbb{E}\left[(\mathbf{X}_{t_1} - \mu_{t_1})(\mathbf{X}_{t_2} - \mu_{t_2})^*\right]$$

For stationary processes, both auto-correlation and auto-covariance don't depend on two time instant t_1, t_2 but only on the time-shift between them, $t_2 = t_1 + \tau$,

$$\mathbf{R}_{XX}(\tau) = \mathbb{E}\left[\mathbf{X}_t\,\mathbf{X}_{t+\tau}^*\right]\,, \qquad \forall t\,.$$

Autocorrelation and ESD of time integrable processes

With causal, $\mathbf{x}(t \leq 0) = 0$, and integrable processes, autocorrelation is defined as

$$\tilde{\mathbf{R}}_{xx}(\tau) = \int_{t=0}^{+\infty} \mathbf{x}(t)\mathbf{x}^*(t+\tau)\, dt \,,$$

Energy Spectral Density (ESD) is defined as its Fourier transform

$$\tilde{\Phi}_{xx}(f) = \mathcal{F}\left\{\mathbf{R}_{xx}(\tau)\right\}(f) \,,$$

Autocorrelation and PSD of infinite-time processes

Using average over time as an approximation of the probability average

$$\mathbb{E}[f(t)] \sim \lim_{T\to+\infty} \frac{1}{T} \int_{t=0}^{T} f(\tau)\, d\tau \,,$$

Power Spectral Density (PSD), defined as the Fourier transform of the auto-correlation, reads

$$\begin{aligned}\Phi_{YY}(f) &= \mathcal{F}[\mathbf{R}_{YY}(\tau)](f) = \\ &= \int_{\tau=-\infty}^{+\infty} \mathbb{E}\left[\mathbf{y}(t)\mathbf{y}^*(t+\tau)\right] e^{-i2\pi f\tau}\, d\tau \,.\end{aligned}$$

For asymptotically stable systems subject to periodic forcing, response to non-homogenoeus initial conditions vanishes while the forced response can be written either in Fourier or in time domain, exploiting the property of the transform of convolution *Property 22.2.1*, as

$$\mathbf{y}(f) = \mathbf{H}(f)\,\mathbf{u}(f) \qquad , \qquad \mathbf{y}(t) = \mathbf{H} * \mathbf{u}(t) = \int_{\tau=-\infty}^{+\infty} \mathbf{H}(\tau)\mathbf{u}(t-\tau)\, d\tau \,.$$

Inserting harmonic response into the expression of PSD, it becomes

$$\begin{aligned}\Phi_{YY}(f) &= \int_{\tau=-\infty}^{+\infty} \mathbb{E}\left[\left(\int_{\xi=-\infty}^{+\infty} \mathbf{H}(\xi)\,\mathbf{u}(t-\xi)\, d\xi\right)\left(\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\chi)\,\mathbf{u}(t+\tau-\chi)\, d\chi\right)^*\right] e^{-i2\pi f\tau}\, d\tau = \quad (1)\\ &= \int_{\tau=-\infty}^{+\infty}\int_{\xi=-\infty}^{+\infty}\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\xi)\mathbb{E}\left[\mathbf{u}(t-\xi)\mathbf{u}^*(t+\tau-\chi)\right]\mathbf{H}^*(\chi)\, e^{-i2\pi f\tau}\, d\tau\, d\chi\, d\xi = \quad (2)\\ &= \int_{\tau=-\infty}^{+\infty}\int_{\xi=-\infty}^{+\infty}\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\xi)\mathbb{E}\left[\mathbf{u}(t)\mathbf{u}^*(t+\tau-\chi+\xi)\right]\mathbf{H}^*(\chi)\, e^{-i2\pi f\tau}\, d\tau\, d\chi\, d\xi = \quad (3)\\ &= \int_{z=-\infty}^{+\infty}\int_{\xi=-\infty}^{+\infty}\int_{\chi=-\infty}^{+\infty} \mathbf{H}(\xi)\mathbb{E}\left[\mathbf{u}(t)\mathbf{u}^*(t+z)\right]\mathbf{H}^*(\chi)\, e^{-i2\pi f(z-\xi+\chi)}\, dz\, d\chi\, d\xi = \\ &= \int_{\xi=-\infty}^{+\infty} \mathbf{H}(\xi)\, e^{i2\pi f\xi}\, d\xi \int_{z=-\infty}^{+\infty} \mathbf{R}_{uu}(z)\, e^{-i2\pi fz}\, dz \int_{\chi=-\infty}^{+\infty} \mathbf{H}^*(\chi)\, e^{-i2\pi f\chi}\, d\chi = (4)\\ &= \overline{\mathbf{H}}(f)\,\Phi_{uu}(f)\,\mathbf{H}^T(f) \,.\end{aligned}$$

having used (1) the fact that the impulse response function is deterministic, (2) time shift in the autocorrelation for a **stationary** random process $\mathbf{u}(t)$ as an input, and (3) the change of coordinates $(z, \xi, \chi) = (\tau - \chi + \xi, \xi, \chi)$, and (4) a slight notation abuse ndicating Fourier transform with the same symbols of the functions in time domain.

CHAPTER

THIRTYFIVE

LTI: STABILITY AND FEEDBACK

Contents

- Stability of a SISO LTI, w.r.t. non-zero i.c. or impulsive input
-
-

35.1 Stability of a LTI - SISO

35.1.1 Transfer function

Let the SISO input-output relation of a LTI system be represented in Laplace domain by the transfer function $G(s)$,

$$y(s) = G(s)u(s)$$

For rational transfer functions

$$G(s) = k\frac{\prod_{n=1}^{N}(z_n - s)}{\prod_{d=1}^{D}(p_d - s)} = \frac{\sum_{n=1}^{N} a_n s^n}{\sum_{d=0}^{D} b_d s^d} \ ,$$

with z_n the zeros, p_d the poles, and $k = \frac{a_0}{b_0} = \lim_{s\to 0} G(s)$ the static gain. If the system is strictly proper, $N < D$, and the TF can be written as a **sum of partial functions**. As an example, for a transfer function with simple poles

$$G(s) = \sum_{d=1}^{N} \frac{A_d}{(p_d - s)} \ ,$$

while for a pole p_d with multiplicity $m_d > 1$ all the terms $\propto \frac{1}{(p_d - s)^{e_d}}$, with $e_d = 1 : m_d$ must be included, see example below.

$$G(s) = \sum_{d=1}^{N} \frac{A_d}{(p_d - s)^{e_d}} \ ,$$

Example 35.1.1 (Sum of partial fractions of a TF with poles with multiplicity > 1)

Let's write the rational function $G(s) = \frac{s+1}{(s+2)^3}$ as a sum of partial functions,

$$\begin{aligned} G(s) = \frac{s+1}{(s+2)^3} = \\ = \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{(s+2)^3} = \\ = \frac{A(s+2)^2 + B(s+2) + C}{(s+2)^3} = \\ = \frac{s^2(A) + s(4A+B) + 4A + 2B + C}{(s+2)^3} , \end{aligned}$$

The value of the coefficients A, B, C is computed comparing the first and the last expression of the numerator of $G(s)$

$$\begin{cases} s^2 : & A = 0 \\ s & : & 4A + B = 1 \\ 1 & : & 4A + 2B + C = 1 \end{cases} \qquad \rightarrow \qquad \begin{cases} A = 0 \\ B = 1 \\ C = -1 \end{cases}$$

and thus

$$G(s) = \frac{s+1}{(s+2)^3} = \frac{1}{(s+2)^2} - \frac{1}{(s+2)^3} .$$

35.1.2 Stability w.r.t. non-zero initial conditions - or w.r.t. implusive input - in time domain

Transfer function $G(s)$ represents the free the function w.r.t. impulsive input. Response in time domain can be evaluated as the inverse Laplace transform of the transfer function,

$$y(t) = \mathcal{L}^{-1}\left\{y(s)\right\} = \mathcal{L}^{-1}\left\{\sum_{n_d=1}^{D} \frac{R_d}{(s-p_d)}\right\} = \sum_{d=1}^{D} R_d\, e^{p_d t} .$$

If $\text{re}\{p_d\} < 0$ for $\forall d$, then the response is asymptotically stble for $t \rightarrow +\infty$, as $|e^{p_d t}| = e^{\text{re}\{p_d\} t} \rightarrow 0$, for $t \rightarrow +\infty$.

CHAPTER

THIRTYSIX

OBSERVABILITY AND DETECTABILITY

36.1 Time-continuous linear systems

The solution of the linear system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

with initial conditions $\mathbf{x}(0) = \mathbf{x}_0$ reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 + \int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \ ,$$

and the output of the system reads

$$\begin{aligned} \mathbf{y}(t) &= \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t) = \\ &= \mathbf{C}(t)\Phi(t, t_0)\mathbf{x}_0 + \mathbf{C}(t)\int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau + \mathbf{D}(t)\mathbf{u}(t) \ . \end{aligned} \tag{36.1}$$

36.1.1 Observability

A system is observable if the initial state $\mathbf{x}_0$ can be uniquely determined, knowing the output $\mathbf{y}(t)$ and the input $\mathbf{u}(t)$ over a finite time interval.

A relation between initial conditions as a function of the output and the input of the system can be retrieved from the expression (36.1),

$$\mathbf{C}(t)\Phi(t, t_0)\mathbf{x}_0 = \mathbf{y}(t) - \left[\mathbf{C}(t)\int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau + \mathbf{D}(t)\mathbf{u}(t)\right] =: \tilde{\mathbf{y}}(t) \ ,$$

with the definition of the known combination of input and output on the RHS as $\tilde{\mathbf{y}}(t)$. This system of equations is usually not square, and only involves two time instants t_0, t. If the latter equation is multiplied by $\Phi^T(\tau, t_0)\mathbf{C}^T(\tau)$ (least square solution; L_2 norm;...) and integrated from t_0 to t in order to use all the information from t_0 to t

$$\int_{\tau=t_0}^{t} \Phi^T(\tau, t_0)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\Phi(\tau, t_0)d\tau \, \mathbf{x}_0 = \int_{\tau=t_0}^{t} \Phi^T(\tau, t_0)\mathbf{C}^T(\tau)\tilde{\mathbf{y}}(\tau)d\tau \ .$$

$$\mathbf{W}_o(t) \, \mathbf{x}_0 = \int_{\tau=t_0}^{t} \Phi^T(\tau, t_0)\mathbf{C}^T(\tau)\tilde{\mathbf{y}}(\tau)d\tau \ .$$

todo

- Discuss the **pullback** of the output to the state space (before $\sim$ least-square inversion)

Definition 36.1.1 (Observability Gramian)

$$\mathbf{W}_o(t) := \int_{\tau=t_0}^{t} \Phi^T(\tau, t_0)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\Phi(\tau, t_0)d\tau \tag{36.2}$$

36.1.2 Detectability

A system is observable if all the unobservable modes are naturally stable, i.e. they decay to zero.

todo *Check definition, and requirement for finite or infinite time range*

Optimal input and controllability Gramian

Let's solve the optimal output defined as the one with the maximum objective function $\frac{1}{2}\int_{\tau=0}^{t} |\mathbf{y}(\tau)|^2$, subject to the equations of motion as constraints. **todo** Here the final time t is assumed to be known, but in general it's not: include t in the optimization, or run several optimizations to find optimal t^*. The solution of the free system - or the forced system, with the $\tilde{\mathbf{y}}(t)$ defined as the combination of output and input above - reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 \ ,$$

so that the output is $\mathbf{y}(t) = \mathbf{C}\Phi(t, t_0)\mathbf{x}_0$. The objective function read

$$\begin{aligned} J(\mathbf{x}_0) &= \frac{1}{2}\int_{\tau=0}^{t} \mathbf{y}^T(\tau)\mathbf{y}(\tau)d\tau = \\ &= \frac{1}{2}\mathbf{x}_0^T \int_{\tau=0}^{t} \Phi^T(t, t_0)\mathbf{C}^T\mathbf{C}\Phi(t, t_0)\mathbf{x}_0 = \\ &= \frac{1}{2}\mathbf{x}_0^T\mathbf{W}_o(t)\mathbf{x}_0 \ . \end{aligned}$$

- If $\mathbf{x}_0 \in K(\mathbf{W}_o(t))$, i.e. belongs to the kernel of the **observatility Gramian**, thus $J(\mathbf{x}_0) = 0$, i.e. the initial conditions has no influence on the output.
- If $\mathbf{x}_0 \notin K(\mathbf{W}_o(t))$, i.e. does not belong to the kernel of the observatility Gramian, thus $J(\mathbf{x}_0) > 0$.
- using singular value decomposition, it's possible to find the initial condition producing the largest output. Let $\mathbf{W}_o = \mathbf{U}\Sigma\mathbf{U}^*$, the optimal initial condition with given L_2 norm, e.g. $\|\mathbf{x}_0\|_2 = 1$, is the singular vector $\mathbf{u}_1$ associated with the largest singular value σ_1, $\mathbf{x}_0 = \mathbf{u}_1$, and the optimal output is

$$\max_{\|\mathbf{x}_0\|_2=1} J(\mathbf{x}_0) = J(\mathbf{x}_0 = \mathbf{u}_1) = \frac{1}{2}\sigma_1 \ .$$

Observability Gramian satisfies the *Lyapunov equation*

$$\dot{\mathbf{W}}_o(t) = \mathbf{C}^T(t)\mathbf{C}(t) + \mathbf{A}^T(t)\mathbf{W}_o(t) + \mathbf{W}_o(t)\mathbf{A}(t) \ ,$$

with initial conditions $\mathbf{W}_o(0) = \mathbf{0}$, as the integral vanishes with the same values of the lower and upper extremes of integration.

Lyapunov equation for the controllability Graminan

$$\frac{d}{dt}\mathbf{W}_o(t) = \frac{d}{dt}\int_{\tau=0}^{t} \Phi^T(t,\tau)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\Phi(t,\tau)d\tau =$$
$$= \mathbf{C}(t)\mathbf{C}^T(t) + \int_{\tau=0}^{t} \partial_t\Phi^T(t,\tau)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\Phi(t,\tau)d\tau + \int_{\tau=0}^{t} \Phi^T(t,\tau)\mathbf{C}^T(\tau)\mathbf{C}(\tau)\partial_t\Phi(t,\tau)d\tau =$$
$$= \mathbf{C}(t)\mathbf{C}^T(t) + \mathbf{A}(t)\mathbf{W}_c(t) + \mathbf{W}_c(t)\mathbf{A}^T(t) \ ,$$

as $\Phi(t,t) = \mathbf{I}$, and $\partial_t\Phi(t,\tau) = \mathbf{A}(t)\Phi(t,\tau)$.

Observability for LTI systems

For LTI systems $\Phi(t,t_0) = e^{\mathbf{A}(t-t_0)}$. The controllability Gramian becomes

$$\mathbf{W}_o(t) = \int_{\tau=0}^{t} e^{\mathbf{A}^T(t-\tau)}\mathbf{C}^T\mathbf{C}e^{\mathbf{A}(t-\tau)}d\tau$$

A vector $\mathbf{v}$ belongs to the null space of $\left(e^{\mathbf{A}^T t}\mathbf{C}^T\right)^T$ if

$$0 = \mathbf{C}e^{\mathbf{A}t}\mathbf{v} \ .$$

Using the definition of the exponential matrix $e^{\mathbf{A}t} := \sum_{n=0}^{+\infty} \frac{\mathbf{A}^n t^n}{n!}$, it follows

$$0 = \mathbf{C}\sum_{k=0}^{+\infty} \frac{\mathbf{A}^k t^k}{k!}\mathbf{v}$$

Thus…, a vector $\mathbf{v}$ is in the null-space of $\mathbf{W}_o$ (symmetric! Be precise and use *orthogonality between range and kernel of the adjoint* if it's orthogonal to the rows of the matrix

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \dots \\ \mathbf{CA}^{n-1} \end{bmatrix} \ ,$$

i.e. $\mathcal{O}\mathbf{v} = \mathbf{0}$. If the row space of matrix $\mathcal{O}$ spans $\mathbb{R}^n$, then the system is fully observable.

36.2 Time-discrete linear systems

CHAPTER

THIRTYSEVEN

REACHABILITY AND CONTROLLABILITY

37.1 Time-continuous linear systems

The solution of the linear system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

with initial conditions $\mathbf{x}(0) = \mathbf{x}_0$ reads

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 + \int_{\tau=0}^{t} \Phi(t, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \ ,$$

37.1.1 Reachability

A state $\mathbf{x}_1$ is reachable if there's a control input $\mathbf{u}(t)$ that can move the system from the initial state $\mathbf{x}_0 = \mathbf{0}$ to the final state $\mathbf{x}_1$ in finite time.

37.1.2 Controllability

A state $\mathbf{x}_0$ is reachable if there's a control input $\mathbf{u}(t)$ that can move the system from the initial state $\mathbf{x}_0$ to the final state $\mathbf{x}_1 = \mathbf{0}$ in finite time.

Optimal input and controllability Gramian

Let's solve the optimal input defined as the one with the minimal cost $\frac{1}{2}\int_{\tau=0}^{t_1} |\mathbf{u}(\tau)|^2$, subject to the equations of motion as constraints. **todo** Here the final time t_1 is assumed to be known, but in general it's not: include t_1 in the optimization, or run several optimizations to find optimal t_1^*.

The cost function of the constrained problem reads

$$\widetilde{J}(\mathbf{u}; \lambda) = \frac{1}{2}\int_{\tau=0}^{t_1} \mathbf{u}^T(\tau)\mathbf{u}(\tau)d\tau + \lambda^T \left(\mathbf{x}_1 - \int_{\tau=0}^{t_1} \Phi(t_1, \tau)\mathbf{B}(\tau)\mathbf{u}(\tau)d\tau \right) \ .$$

Variation w.r.t. λ gives the solution of the equation, i.e. the constraints. Variation w.r.t. the control $\mathbf{u}(\tau)$ gives the optimal control as a function of λ

$$\mathbf{u}(\tau) = \mathbf{B}^T(\tau)\Phi^T(t_1, \tau)\lambda \ .$$

Replacing this expression in the solution of the dynamical system, an expression of λ in terms of the final state $\mathbf{x}_1$ follows,

$$\mathbf{x}_1 = \int_{\tau=0}^{t_1} \Phi(t_1,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t_1,\tau)d\tau\lambda = \mathbf{W}_c(t_1)\lambda \ ,$$

having introduced the definition of the **controllability Gramian**, $\mathbf{W}_c(t_1)$.

- If $\mathbf{x}_1 \in R(\mathbf{W}_c(t_1))$ is in the range of the controllability Gramian, at least a value of the Lagrange multiplier λ that satisfies the relation - and thus a control driving the system from $\mathbf{0}$ to $\mathbf{x}_1$ - exists.
- If the controllability Gramian is full-rank, it's invertible, the solution of the problem is unique and reads

  $$\lambda = \mathbf{W}_c^{-1}(t_1)\mathbf{x}_1 \ ,$$

  and the cost of the control is

  $$J = \frac{1}{2}\mathbf{x}_1^T\mathbf{W}_c^{-1}\int_{\tau=0}^{t_1} \Phi(t_1,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t_1,\tau)d\tau\mathbf{W}_c^{-1}\mathbf{x}_1 = \frac{1}{2}\mathbf{x}_1^T\mathbf{W}_c^{-1}\mathbf{x}_1 \ .$$

- In general, using singular value decomposition of the Gramian $\mathbf{W}_c = \mathbf{U}\Sigma\mathbf{U}^*$ (as the Gramian is symmetric, $\mathbf{V} = \mathbf{U}$)

  $$\mathbf{u}_{\mathbf{x}_1} = \Sigma\mathbf{u}_\lambda \ ,$$

  with $\mathbf{u}_\mathbf{v} \coloneqq \mathbf{U}^*\mathbf{v}$, the projection of the vector $\mathbf{v}$ onto the unit orthogonal base of $\mathbb{R}^n$. A solution $\mathbf{u}_\lambda$ exists if zero singular values $\sigma_i = 0$ correspond to null projection of the vector $\mathbf{x}_1$ onto the ith vector of the base $\mathbf{u}_i^*\mathbf{x}_1 = \left\{\mathbf{u}_{\mathbf{x}_1}\right\}_i = 0$. Otherwise a solution of the equation does not exist: in this situation, there's no control law driving the system into $\mathbf{x}_1$, an error in the final state exists, and the problem can be formulated as a optimization problem weighting both the control and the error (resulting in a least square solution of a linear system with no solution?).

Definition 37.1.1 (Controllability Gramian)

$$\mathbf{W}_c(t,t_0) \coloneqq \int_{\tau=t_0}^{t} \Phi(\tau,t_0)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(\tau,t_0)d\tau \tag{37.1}$$

Controllability Gramian satisfies the *Lyapunov equation*

$$\dot{\mathbf{W}}_c(t) = \mathbf{B}(t)\mathbf{B}^T(t) + \mathbf{A}(t)\mathbf{W}_c(t) + \mathbf{W}_c(t)\mathbf{A}^T(t) \ ,$$

with initial conditions $\mathbf{W}_c(0) = \mathbf{0}$, as the integral vanishes with the same values of the lower and upper extremes of integration.

Lyapunov equation for the controllability Graminan

$$\begin{aligned}\frac{d}{dt}\mathbf{W}_c(t) &= \frac{d}{dt}\int_{\tau=0}^{t} \Phi(t,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t,\tau)d\tau = \\ &= \mathbf{B}(t)\mathbf{B}^T(t) + \int_{\tau=0}^{t} \partial_t\Phi(t,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\Phi^T(t,\tau)d\tau + \int_{\tau=0}^{t} \Phi(t,\tau)\mathbf{B}(\tau)\mathbf{B}^T(\tau)\partial_t\Phi^T(t,\tau)d\tau = \\ &= \mathbf{B}(t)\mathbf{B}^T(t) + \mathbf{A}(t)\mathbf{W}_c(t) + \mathbf{W}_c(t)\mathbf{A}^T(t) \ ,\end{aligned}$$

as $\Phi(t,t) = \mathbf{I}$, and $\partial_t\Phi(t,\tau) = \mathbf{A}(t)\Phi(t,\tau)$.

Controllability for LTI systems

For LTI systems $\Phi(t, t_0) = e^{\mathbf{A}(t-t_0)}$. The controllability Gramian becomes

$$\mathbf{W}_c(t) = \int_{\tau=0}^{t} e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{B}^T e^{\mathbf{A}^T(t-\tau)} d\tau$$

A vector $\mathbf{v}$ belongs to the null space of $\left(e^{\mathbf{A}t}\mathbf{B}\right)^T$ if

$$0 = \mathbf{v}^T e^{\mathbf{A}t}\mathbf{B} \, .$$

Using the definition of the exponential matrix $e^{\mathbf{A}t} := \sum_{n=0}^{+\infty} \frac{\mathbf{A}^n t^n}{n!}$, it follows

$$0 = \mathbf{v}^T \sum_{n=0}^{+\infty} \frac{\mathbf{A}^n t^n}{n!}\mathbf{B}$$

Thus…, a vector $\mathbf{v}$ is in the null-space of $\mathbf{W}_c$ (symmetric! Be precise and use *orthogonality between range and kernel of the adjoint* if it's orthogonal to the columns of the matrix

$$\mathcal{C} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix} \, ,$$

i.e. $\mathbf{v}^T\mathcal{C} = \mathbf{0}$. If the column space of matrix $\mathcal{C}$ spans $\mathbb{R}^n$, then the system is fully controllable.

Obs. If $\mathbf{v}^T\mathbf{A}^n\mathbf{B}$ for $n = 0 : N-1$, thus it holds for all the integer exponent $n \geq N$, as a consequence of the *Cayley-Hamilton theorem*, i.e. $\mathbf{A}^n$ can be written as a linear combination of the powers $\mathbf{A}^k$, $k = 0 : n-1$.

37.2 Time-discrete linear systems

CHAPTER

THIRTYEIGHT

STATE-SPACE REALIZATIONS

State-Space RealizationsIn system theory, a realization is a state-space triplet $\{A, B, C, D\}$ that represents a given transfer function $H(s)$. For a strictly proper system, the relationship is defined by:

$$H(s) = C(sI - A)^{-1}B$$

Given a general n-th order transfer function:

$$H(s) = \frac{b_{n-1}s^{n-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

There are two primary canonical forms used for implementation and analysis.

38.1 Controller Canonical Form (CCF)

The Controller Canonical Form is structured to make the system's controllability obvious. It is often used in control design (e.g., pole placement) because the input $u(t)$ enters the "chain" of integrators at the final stage.

State-Space Representation. The matrices for an n-th order system in CCF are:

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} b_0 & b_1 & \dots & b_{n-1} \end{bmatrix}$$

Characteristics

- Structure: The states are typically chosen as the output and its derivatives ($x_1 = y, x_2 = \dot{y}, \dots$) in the absence of zeros.
- Controllability: A system in this form is always controllable, provided there are no pole-zero cancellations.

38.2 Observer Canonical Form (OCF)

The Observer Canonical Form is the dual of the CCF. It is structured such that the output $y(t)$ is directly influenced by all state variables, making the system's observability easily verifiable.

State-Space Representation.

The matrices for an n-th order system in OCF are:

$$A = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_{n-1} \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

Characteristics:

- Duality: The A matrix of the OCF is the transpose of the A matrix of the CCF ($A_{OCF} = A_{CCF}^T$), and the B and C matrices are swapped and transposed ($B_{OCF} = C_{CCF}^T$ and $C_{OCF} = B_{CCF}^T$).
- Observability: A system in this form is always observable.

38.3 Example: realization of a second-order system

From the specific case $H(s) = \frac{s}{s^2+a_1 s+a_0}$:

Controller Realization

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

Observer Realization

$$\dot{x} = \begin{bmatrix} 0 & -a_0 \\ 1 & -a_1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

38.4 Minimality and Similarity Transformations

A realization is minimal if and only if it is both controllable and observable. The dimension of a minimal realization equals the degree of the denominator of the transfer function (the McMillan degree).Any two realizations $\{A, B, C\}$ and $\{\tilde{A}, \tilde{B}, \tilde{C}\}$ of the same transfer function are related by a similarity transformation T such that:

$$\tilde{A} = TAT^{-1}, \quad \tilde{B} = TB, \quad \tilde{C} = CT^{-1}$$

CHAPTER

THIRTYNINE

KALMAN DECOMPOSITION

Controllability and *Non-observability* properties define two sub-spaces of the state space, $\mathbb{R}^n$. The complements of these subpsaces define subspaces as well.

Kalman decomposition applies a coordinate transformation in order to make explicit the existence of:

- X_1, non-observable and reachable sub-space, $X_1 := X_{\overline{o}} \cap X_r$
- X_2, the complement of X_1 w.r.t. the reachable sub-space, i.e. $X_1 \cup X_2 = X_r$
- X_3, the complement of X_1 w.r.t. the non-observable sub-space, i.e. $X_1 \cup X_3 = X_{\overline{o}}$
- X_4, the complement of X_1, X_2, X_3 w.r.t. the state-space $\mathbb{R}^n$.

A**-invariance** of Ran($\mathbf{W}_c$) and Ker($\mathbf{W}_o$) is exploited.

In this section, *singular value decomposition* of the Gramians of *observability* and *controllability* helps in the decomposition of a LTI system in its non-observable/observable and controllable/non-controllable parts.

As shown in the section about coordinate transformation, in the Kalman decomposition subsection, only the observable and controllable parts appear in the input-output transfer function of the system.

todo *SVD may help or not, but that's not the point here. We're supposed to know how to find an orthonormal basis, and if we can't remember how to do it, we're supposed to find a the method we like the most to do that. Orthonormal basis are not even strictly required here. Moreover, SVD of symmetric (semi) definite positive matrices coincides with spectral decomposition…*

todo *Infinite-horizon time, or steady-state conditions, are discussed here?*

39.1 Controllability

Controllability

Let $\mathbf{W}_c$ the controllability Gramian of a LTI system. Controllable states belong to the range of W_c. Controllable states define a sub-space of $\mathbb{R}^n$, being $\mathbf{x} \in \mathbb{R}^n$. Singular value decomposition (or spectral decomposition) of the semi-poisitive definite symmetric Gramian matrix,

$$\mathbf{W}_C = \mathbf{U}_C \Sigma_C \mathbf{U}_C^* \, ,$$

provides a decomposition of $\mathbb{R}^n$ into the controllable sub-space and its orthogonal complement, with

$$\mathbf{W}_C = [\, \mathbf{U}_c \,|\, \mathbf{U}_{\overline{c}} \,] \begin{bmatrix} \Sigma_c & \cdot \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \mathbf{U}_c^* \\ \mathbf{U}_{\overline{c}} \end{bmatrix} \, ,$$

with the columns of $\mathbf{U}_c$ as a base of the controllable sub-space. Introducing a change of variable of the state

$$\mathbf{x} = \mathbf{U}_C \mathbf{x}_1 \, ,$$

the LTI system becomes

$$\begin{aligned}\dot{\mathbf{x}}_1 &= \mathbf{U}_C^* \mathbf{A} \mathbf{U}_C \mathbf{x}_1 + \mathbf{U}_C^* \mathbf{B} \mathbf{u} \\ \mathbf{y} &= \mathbf{C} \mathbf{U}_C \mathbf{x}_1 + \mathbf{D} \mathbf{u} \; .\end{aligned}$$

and thus

$$\begin{aligned}\frac{d}{dt} \begin{bmatrix} \mathbf{x}_{1,c} \\ \mathbf{x}_{1,\overline{c}} \end{bmatrix} &= \begin{bmatrix} \mathbf{U}_c^* \mathbf{A} \mathbf{U}_c & \mathbf{U}_c^* \mathbf{A} \mathbf{U}_{\overline{c}} \\ \mathbf{0} & \mathbf{U}_{\overline{c}}^* \mathbf{A} \mathbf{U}_{\overline{c}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1,c} \\ \mathbf{x}_{1,\overline{c}} \end{bmatrix} + \begin{bmatrix} \mathbf{U}_c^* \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u} \\ \mathbf{y} &= \begin{bmatrix} \mathbf{C} \mathbf{U}_c & \mathbf{C} \mathbf{U}_{\overline{c}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1,c} \\ \mathbf{x}_{1,\overline{c}} \end{bmatrix} + \mathbf{D} \mathbf{u} \; .\end{aligned}$$

Properties

The columns of the matrix $\mathcal{C} = [\, \mathbf{B} \,|\, \mathbf{AB} \,|\, \ldots \,|\, \mathbf{A}^{n-1}\mathbf{B} \,]$ are linear combinations of the range of $\mathbf{W}_C$, i.e. of the columns of $\mathbf{U}_c$, and thus they're orthogonal w.r.t. the columns of $\mathbf{U}_{\overline{c}}$. It follows that

$$\begin{aligned}\mathbf{U}_{\overline{c}}^* \mathbf{B} &= \mathbf{0} \\ \mathbf{U}_{\overline{c}}^* \mathbf{A}^k \mathbf{B} &= \mathbf{0} \quad , \quad \forall k \in \mathbb{N}\end{aligned}$$

as $\mathbf{U}_{\overline{c}}^* \mathbf{A}^k \mathbf{B} = \mathbf{U}_{\overline{c}}^* \mathbf{U}_c \alpha$ and $\mathbf{U}_{\overline{c}}^* \mathbf{U}_c = \mathbf{0}$

39.2 Observability

Observability

Let $\mathbf{W}_o$ the observability Gramian of a LTI system. Non-observable states belong to the kernel of W_o. Non-observable states define a sub-space of $\mathbb{R}^n$, being $\mathbf{x} \in \mathbb{R}^n$. Singular value decomposition (or spectral decomposition) of the semi-poisitive definite symmetric Gramian matrix,

$$\mathbf{W}_O = \mathbf{U}_O \Sigma_O \mathbf{U}_O^* \; ,$$

provides a decomposition of $\mathbb{R}^n$ into the non-observable sub-space and its orthogonal complement, with

$$\mathbf{W}_O = [\, \mathbf{U}_o \,|\, \mathbf{U}_{\overline{o}} \,] \begin{bmatrix} \Sigma_o & \cdot \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \mathbf{U}_o^* \\ \mathbf{U}_{\overline{o}} \end{bmatrix} \; ,$$

with the columns of $\mathbf{U}_{\overline{o}}$ as a base of the non-observable sub-space. Introducing a change of variable of the state

$$\mathbf{x} = \mathbf{U}_o \mathbf{x}_1 \; ,$$

the LTI system becomes

$$\begin{aligned}\dot{\mathbf{x}}_1 &= \mathbf{U}_O^* \mathbf{A} \mathbf{U}_O \mathbf{x}_1 + \mathbf{U}_O^* \mathbf{B} \mathbf{u} \\ \mathbf{y} &= \mathbf{C} \mathbf{U}_O \mathbf{x}_1 + \mathbf{D} \mathbf{u} \; .\end{aligned}$$

and thus

$$\begin{aligned}\frac{d}{dt} \begin{bmatrix} \mathbf{x}_{1,o} \\ \mathbf{x}_{1,\overline{o}} \end{bmatrix} &= \begin{bmatrix} \mathbf{U}_o^* \mathbf{A} \mathbf{U}_o & \mathbf{0} \\ \mathbf{U}_{\overline{o}}^* \mathbf{A} \mathbf{U}_o & \mathbf{U}_{\overline{o}}^* \mathbf{A} \mathbf{U}_{\overline{o}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1,o} \\ \mathbf{x}_{1,\overline{o}} \end{bmatrix} + \begin{bmatrix} \mathbf{U}_o^* \mathbf{B} \\ \mathbf{U}_{\overline{o}}^* \mathbf{B} \end{bmatrix} \mathbf{u} \\ \mathbf{y} &= \begin{bmatrix} \mathbf{C} \mathbf{U}_o & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1,o} \\ \mathbf{x}_{1,\overline{o}} \end{bmatrix} + \mathbf{D} \mathbf{u} \; ,\end{aligned}$$

as $\mathbf{U}_o^* \mathbf{A} \mathbf{U}_{\overline{o}} = \mathbf{0}$ (see below).

Properties

The kernel of the observability Gramian is the sub-space of vectors $\mathbf{v}$ so that $\mathcal{O}\mathbf{v} = \mathbf{0}$, with

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \dots \\ \mathbf{CA}^{n-1} \end{bmatrix} .$$

It follows that

$$\begin{aligned} \mathbf{C}\mathbf{U}_{\overline{o}} &= \mathbf{0} \\ \mathbf{C}\mathbf{A}^k\mathbf{U}_{\overline{o}} &= \mathbf{0} \quad , \quad \forall k \in \mathbb{N} \end{aligned}$$

so that the rows of $\mathbf{C}$ and the rows of $\mathbf{CA}^k$ are linear combinations of the columns of $\mathbf{U}_o$, i.e. the columns of $\left(\mathbf{CA}^k\right)^*$ are linear combinations of the columns of $\mathbf{U}_o$,

$$\left(\mathbf{CA}^k\right)^* = \mathbf{U}_o\alpha \, ,$$

or

$$\mathbf{CA}^k = \alpha^*\mathbf{U}_o^* \, .$$

As the property holds for every $k \in \mathbb{N}$, the following holds as well

$$\mathbf{U}_o^*\mathbf{A}\mathbf{U}_{\overline{o}} \, .$$

39.3 Kalman decomposition

Let

- $\mathbf{U}_1$ spanning the **reachable** and **non-observable sub-psace**, $R \cap \overline{O}$
- $\mathbf{U}_2$ complementing $\mathbf{U}_1$ to get the reachable sub-sspace R
- $\mathbf{U}_3$ complementing $\mathbf{U}_3$ to get the non-observable sub-space O
- $\mathbf{U}_4$ complementing $\mathbf{U}_1$, $\mathbf{U}_2$, $\mathbf{U}_3$ to get $\mathbb{R}^n$

The columns of $\mathbf{U}_1$ and $\mathbf{U}_3$ form a vector basis of the non-observable sub-space. The columns of $\mathbf{U}_1$ and $\mathbf{U}_2$ form a vector basis of the reachable subspace.

A coordinate transfromation of the form

$$\mathbf{x} = \mathbf{T}\mathbf{x} = \begin{bmatrix} \mathbf{U}_1 & \mathbf{U}_2 & \mathbf{U}_3 & \mathbf{U}_4 \end{bmatrix} \mathbf{z} = \begin{bmatrix} \mathbf{U}_1 & \mathbf{U}_2 & \mathbf{U}_3 & \mathbf{U}_4 \end{bmatrix} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} \, ,$$

produce a transformed system with the following structure

$$\begin{cases} \dfrac{d}{dt} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} & \mathbf{A}_{14} \\ \mathbf{0} & \mathbf{A}_{22} & \mathbf{0} & \mathbf{A}_{24} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{33} & \mathbf{A}_{34} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{44} \end{bmatrix} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \mathbf{u} \\ \mathbf{y} = \begin{bmatrix} \mathbf{0} & \mathbf{C}_2 & \mathbf{0} & \mathbf{C}_4 \end{bmatrix} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \mathbf{z}_3 \\ \mathbf{z}_4 \end{bmatrix} + \mathbf{D}\mathbf{u} \end{cases}$$

Details

Columns of T. Here the columns of $\mathbf{U}_1$ are assumed to be a set of unit normal vectors, $\mathbf{U}_1^*\mathbf{U}_1 = \mathbf{I}$. Without this assumption, Gram-Schmidt orthogonalization process ensure a orthogonal basis Φ_1 exists and the basis in $\mathbf{U}_1$ can be written as a linear combination of the columns of Φ_1, i.e. $\mathbf{U}_1 = \Phi_1\alpha_1$, with non-singular α_1. Columns of $\mathbf{U}_2$ are linearly independent from the columns of $\mathbf{U}_1$ to form a basis of the reachable sub-space. Columns $\mathbf{U}_3$ are linearly independent from the columns of $\mathbf{U}_1$ to form a basis of the non-observable sub-space. In general, they're not unit orthogonal but a unit orthogonal basis can be found, to get

$$\begin{aligned}\mathbf{U}_2 &= \Phi_2\alpha_2 + \mathbf{U}_1\alpha_{12}\\ \mathbf{U}_3 &= \Phi_3\alpha_3 + \mathbf{U}_1\alpha_{13}\,.\end{aligned}$$

In general, the columns of Φ_2 and Φ_3 are not mutually orthogonal, as the reachable sub-space and the non-observable sub-space are not orthogonal, $\Phi_2^*\Phi_3 \neq \mathbf{0}$. The columns of the complement $\mathbf{U}_4$ are assumed to be unit normal vectors, orthogonal w.r.t. the columns of $\mathbf{U}_1$, $\mathbf{U}_2$, $\mathbf{U}_3$ (and of Φ_2, Φ_3).

Recast T.

$$\mathbf{T} = \begin{bmatrix}\mathbf{U}_1 & \Phi_2 & \Phi_3 & \mathbf{U}_4\end{bmatrix}\begin{bmatrix}\mathbf{I} & \alpha_{12} & \alpha_{13} & \cdot\\ \cdot & \alpha_2 & \cdot & \cdot\\ \cdot & \cdot & \alpha_3 & \cdot\\ \cdot & \cdot & \cdot & \mathbf{I}\end{bmatrix} = \Phi\alpha$$

Inverse $\mathbf{T}^{-1}$.

$$\mathbf{T}^{-1} = \alpha^{-1}\Phi^{-1}\,,$$

with

$$\alpha^{-1} = \begin{bmatrix}\mathbf{I} & -\alpha_{12}\alpha_2^{-1} & -\alpha_{13}\alpha_3^{-1} & \cdot\\ \cdot & \alpha_2^{-1} & \cdot & \cdot\\ \cdot & \cdot & \alpha_3^{-1} & \cdot\\ \cdot & \cdot & \cdot & \mathbf{I}\end{bmatrix}$$

$$\Phi^{-1} = \begin{bmatrix}\mathbf{U}_1^*\\ \left[\left(\mathbf{I}-\Phi_3\Phi_3^*\right)\Phi_2\left(\mathbf{I}-\Phi_2^*\Phi_3\Phi_3^*\Phi_2\right)^{-1}\right]^*\\ \left[\left(\mathbf{I}-\Phi_2\Phi_2^*\right)\Phi_3\left(\mathbf{I}-\Phi_3^*\Phi_2\Phi_2^*\Phi_3\right)^{-1}\right]^*\\ \mathbf{U}_4^*\end{bmatrix}$$

Details

Transformed matrices: matrix $\hat{\mathbf{C}}$.

$$\hat{\mathbf{C}} = \mathbf{C}\mathbf{T} = \mathbf{C}\Phi\alpha$$

$$\mathbf{C}\Phi = \mathbf{C}\begin{bmatrix}\mathbf{U}_1 & \Phi_2 & \Phi_3 & \mathbf{U}_4\end{bmatrix} = \begin{bmatrix}\mathbf{0} & \mathbf{C}\Phi_2 & \mathbf{0} & \mathbf{C}\mathbf{U}_4\end{bmatrix}\,,$$

as $\mathbf{C}\mathbf{U}_1 = \mathbf{0}$, $\mathbf{C}\Phi_3 = \mathbf{0}$. Looking at the structures of the matrices $\mathbf{C}\Phi$ and α,

$$\begin{bmatrix}\cdot & * & \cdot & *\end{bmatrix}\begin{bmatrix}* & * & * & \cdot\\ \cdot & * & \cdot & \cdot\\ \cdot & \cdot & * & \cdot\\ \cdot & \cdot & \cdot & *\end{bmatrix} = \begin{bmatrix}\cdot & * & \cdot & *\end{bmatrix}\,,$$

it's easy to show that this multiplication preserves the structure of the matrix $\mathbf{C}\Phi$. Explicitly evaluating the non-zero blocks, the transformed matrix reads

$$\hat{\mathbf{C}} = \begin{bmatrix} \mathbf{0} & \mathbf{C}\Phi_2\alpha_2 & \mathbf{0} & \mathbf{C}\mathbf{U}_4 \end{bmatrix}$$

Transformed matrices: matrix $\hat{\mathbf{B}}$.

$$\hat{\mathbf{B}} = \mathbf{T}^{-1}\mathbf{B} = \alpha^{-1}\Phi^{-1}\mathbf{B}$$

$$\Phi^{-1}\mathbf{B} = \begin{bmatrix} \mathbf{U}_1^* \\ \left[\left(\mathbf{I} - \Phi_3\Phi_3^*\right)\Phi_2\left(\mathbf{I} - \Phi_2^*\Phi_3\Phi_3^*\Phi_2\right)^{-1}\right]^* \\ \left[\left(\mathbf{I} - \Phi_2\Phi_2^*\right)\Phi_3\left(\mathbf{I} - \Phi_3^*\Phi_2\Phi_2^*\Phi_3\right)^{-1}\right]^* \\ \mathbf{U}_4^* \end{bmatrix}\mathbf{B}$$

as $\mathbf{U}_1^*\mathbf{B} = \mathbf{0}$, $\left[\left(\mathbf{I} - \Phi_2\Phi_2^*\right)\Phi_3\right]^*\mathbf{B} = \mathbf{0}$. The last relation holds, as $\left(\mathbf{I} - \Phi_2\Phi_2^*\right)\Phi_3$ is the orthogonal projection of Φ_3 perpendicular to the vectors Φ_2. Since this projection is perpendicular both to $\mathbf{U}_1$ and to Φ_2 (that generates the reachable sub-space), it follows that $\left[\left(\mathbf{I} - \Phi_2\Phi_2^*\right)\Phi_3\right]^*\mathbf{B} = \mathbf{0}$. The same argument gives $\mathbf{U}_4^*\mathbf{B} = \mathbf{0}$, and thus

$$\Phi^{-1}\mathbf{B} = \begin{bmatrix} * \\ * \\ \cdot \\ \cdot \end{bmatrix} .$$

Looking at the structure of the matrices α^{-1} and $\Phi^{-1}\mathbf{B}$, it's easy to show that their matrix multiplication preserves the structure

$$\begin{bmatrix} * & * & * & \cdot \\ \cdot & * & \cdot & \cdot \\ \cdot & \cdot & * & \cdot \\ \cdot & \cdot & \cdot & * \end{bmatrix}\begin{bmatrix} * \\ * \\ \cdot \\ \cdot \end{bmatrix} = \begin{bmatrix} * \\ * \\ \cdot \\ \cdot \end{bmatrix} .$$

Transformed matrices: matrix $\hat{\mathbf{A}}$.

The structure of the matrix arises from $\mathbf{A}$-invariance of reachability and controllability sub-spaces and from the properties

$$\vec{\mathbf{r}}^*\mathbf{A}\mathbf{r} = 0$$
$$\mathbf{o}^*\mathbf{A}\overline{\mathbf{o}} = 0 ,$$

and recalling that $\mathbf{U}_1 : r\overline{o}$, $(\mathbf{I} - \Phi_3\Phi_3)^*\Phi_2 : ro$, $(\mathbf{I} - \Phi_2\Phi_2)^*\Phi_3 : \overline{ro}$, $\mathbf{U}_4 : \overline{r}o$. Focusing on $\Phi^{-1}\mathbf{A}\Phi$ first

$$\Phi^{-1}\mathbf{A}\Phi = \begin{bmatrix} \mathbf{U}_1^* \\ \left[\left(\mathbf{I} - \Phi_3\Phi_3^*\right)\Phi_2\left(\mathbf{I} - \Phi_2^*\Phi_3\Phi_3^*\Phi_2\right)^{-1}\right]^* \\ \left[\left(\mathbf{I} - \Phi_2\Phi_2^*\right)\Phi_3\left(\mathbf{I} - \Phi_3^*\Phi_2\Phi_2^*\Phi_3\right)^{-1}\right]^* \\ \mathbf{U}_4^* \end{bmatrix}\mathbf{A}\begin{bmatrix} \mathbf{U}_1 & \Phi_2 & \Phi_3 & \mathbf{U}_4 \end{bmatrix} =$$
$$= \begin{bmatrix} * & * & * & * \\ \cdot & * & \cdot & * \\ \cdot & \cdot & * & * \\ \cdot & \cdot & \cdot & * \end{bmatrix}$$

Pre- and post-multiplication by α^{-1} and α preserves the structure of $\Phi^{-1}\mathbf{A}\Phi$ to have

$$\hat{\mathbf{A}} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} & \mathbf{A}_{13} & \mathbf{A}_{14} \\ \mathbf{0} & \mathbf{A}_{22} & \mathbf{0} & \mathbf{A}_{24} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{33} & \mathbf{A}_{34} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{44} \end{bmatrix} .$$

CHAPTER

FORTY

LYAPUNOV EQUATION

This page needs to be built, with some general results about Lyapunov equation, its solution,… some material is already here, but not structurally organized…

References.

- *Lyapunov equation of the correlation matrix*, and its general solution
- …

Examples.

- *Controllability* Gramian
- *Observability* Gramian
- In optimal control
- Variance of the state in linear system forced by white Gaussian noise
- …

CHAPTER

FORTYONE

ORDER REDUCTION OF DYNAMICAL SYSTEMS

Classes of order reduction methods:

- **Modal methods.** Modal methods are often useful for mechanical systems, whose governing equations are second order differential equations with symmetric mass and stiffness matrices.[1]

  The behavior of a mechanical system can be often represented as a superposition of independent mass-damper-spring systems, with their individual proper eigenvalue (for under-damped systems, frequency and damping, $s_k = \sigma_k + j\omega_k$), and their shape determined by the eigenvector $\mathbf{u}_k$ (or better, usually a pair of complex conjugate eigenvalues and eigenvectors).

  If the spectrum of the forcing is known, with $\omega_{f,max}$ as the maximum frequency of the forcing, it's possible to split the modes of the system in slow (or dynamical) modes - showing dynamical response to forcing, and whose natural frequency is $\omega < \omega_{f,max}$ -, and fast modes (or static) - showing static response to forcing, and thus whose natural frequency is $\omega \gg \omega_{f,max}$.

  Two methods of modal order reduction follows: 1) truncation of fast modes; 2) static recovery of fast modes. These methods are discussed in the notes about modal methods in structural mechanics.

- ***Balanced truncation*.** Balanced truncation relies on coordinate transformation that scales the problem to equally weight reachability and observability properties. Once the coordinates have been transformed, the reduced order model is obtained truncating every modes whose reachability/controllability is below a desired threshold.

- **Energy-based.** POD…

- *…Koopman,…*

These methods can be combined in sequences of order reductions.

41.1 Balanced truncation

While *Kalman decomposition* provides the theoretical results for the existence of 4 subspaces of a linear dynamical system, and the transformation of coordinates that partition the dynamical systems using reachability and (non-)observatilibty as criteria, this decomposition is likely to be almost useless in many applications: it relies on a sharp distinction between zero and non-zero observability/controllability while these properties can be represented with non-binary numbers but real numbers, and suffers from the limits of finite algebra (the result of a numerical algorithm involving real numbers is unlikely to be 0, and thus it's likely that we need to answer questions like "is 10^{-9} zero enough?" in some step of the algorithm). Moreover, the most interesting part of the system is likely to be the reachable and observable part[1]: balanced

[1] Modal methods can be used for non-symmetric systems as well, relying on spectral decomposition of the adjoint problem (or, equivalently, to left eigenvectors).

[1] The non-observable or non-controllable parts of a system should not contain any unstable dynamics, otherwise it can be measured by the sensors, and/or controlled by the actuators. Observability and controllability are strictly related to the choice of sensors and actuators. If any uncontrollable unstable dynamics exists, a different choice of actuators may solve this issue. If any unobservable unstable dynamics exists, a different choice of sensors may solve the issue.

truncation doesn't suffer from the issues of a full Kalman decomposition while ranking dynamics of the system using controllability and reachability as a criterion.

For finite-time horizon problems, controllability Gramian $\mathbf{W}_c$ and observability Gramian $\mathbf{W}_o$ can be computed as the solution of the Lyapunov equation

$$\begin{aligned}\mathbf{0} &= \mathbf{A}\mathbf{W}_c + \mathbf{W}_c\mathbf{A}^T + \mathbf{B}\mathbf{B}^T \\ \mathbf{0} &= \mathbf{A}^T\mathbf{W}_o + \mathbf{W}_o\mathbf{A} + \mathbf{C}^T\mathbf{C}\,.\end{aligned}$$

Balanced truncation exploits two pairs of combined scaling+rotation transformations, obtained with *singular value decomposition* of the Gramians of the system.

Method 1.

- Singular value decomposition of the controllability Gramian $\mathbf{W}_c = \mathbf{U}_c\Sigma_c^2\mathbf{U}_c^*$
- First coordinate transformation, $\mathbf{x} = \mathbf{T}_c\mathbf{x}_1$, with $\mathbf{T}_c = \mathbf{U}_c\Sigma_c$, so that $\mathbf{A}_1 = \mathbf{T}_c^{-1}\mathbf{A}\mathbf{T}_c$, and the observability Gramian of the transformed system can be expressed as $\mathbf{W}_{o,1} = \mathbf{T}_c^T\mathbf{W}_o\mathbf{T}_c$
- Singular value decomposition of $\mathbf{W}_{o,1} = \mathbf{U}_{o,1}\Sigma_{o,1}^2\mathbf{U}_{o,1}^*$
- Second transformation $\mathbf{x}_1 = \mathbf{T}_o\mathbf{x}_2$, with $\mathbf{T}_o = \mathbf{U}_{o,1}\Sigma_{o,1}^{-1/2}$
- Thus, the Gramians in the twice transformed system, with $\mathbf{T} = \mathbf{T}_c\mathbf{T}_o$, read

$$\begin{aligned}\mathbf{W}_{c,2} = \mathbf{T}^{-1}\mathbf{W}_c\mathbf{T}^{-T} &= \left(\Sigma_{o,1}^{1/2}\mathbf{U}_{o,1}^*\Sigma_c^{-1}\mathbf{U}_c^*\right)\left(\mathbf{U}_c\Sigma_c^2\mathbf{U}_c^*\right)\left(\mathbf{U}_c\Sigma_c^{-1}\mathbf{U}_{o,1}\Sigma_{o,1}^{1/2}\right) &&= \Sigma_{o,1} \\ \mathbf{W}_{o,2} = \mathbf{T}_o^T\mathbf{W}_{o,1}\mathbf{T}_o &= \left(\Sigma_{o,1}^{-1/2}\mathbf{U}_{o,1}^*\right)\left(\mathbf{U}_{o,1}\Sigma_{o,1}^2\mathbf{U}_{o,1}^*\right)\left(\mathbf{U}_{o,1}\Sigma_{o,1}^{-1/2}\right) &&= \Sigma_{o,1}\,,\end{aligned}$$

  i.e. they're equal. The pair of transformations makes Gramians equal and diagonal, and the elements of the diagonal - called **Hankel values** of the system - simultaneously measure the controllability and the observability of the system.

todo *here inverse of the diagonal singular value matrices are assumed to exists, i.e. there's no non-observable or non-controllable state.*

The transformed system is then partitioned as

$$\begin{cases}\dfrac{d}{dt}\begin{bmatrix}\mathbf{z}_1\\ \mathbf{z}_2\end{bmatrix} = \begin{bmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{bmatrix}\begin{bmatrix}\mathbf{z}_1\\ \mathbf{z}_2\end{bmatrix} + \begin{bmatrix}\mathbf{B}_1\\ \mathbf{B}_2\end{bmatrix}\mathbf{u} \\ \mathbf{y} = \begin{bmatrix}\mathbf{C}_1 & \mathbf{C}_2\end{bmatrix}\begin{bmatrix}\mathbf{z}_1\\ \mathbf{z}_2\end{bmatrix} + \mathbf{D}\mathbf{u}\end{cases}$$

with $\mathbf{x} = \mathbf{T}\mathbf{z}$, and thus

$$\mathbf{z} = \mathbf{T}^{-1}\mathbf{x} = \Sigma_{o,1}^{1/2}\mathbf{U}_{o,1}^*\Sigma_c^{-1}\mathbf{U}_c^*$$

$$\begin{aligned}\begin{bmatrix}\mathbf{z}_1\\ \mathbf{z}_2\end{bmatrix} &= \begin{bmatrix}\Sigma_{o,1;1}^{1/2} & \mathbf{0}\\ \mathbf{0} & \Sigma_{o,1;2}^{1/2}\end{bmatrix}\begin{bmatrix}\mathbf{U}_{o,1;11}^* & \mathbf{U}_{o,1;12}^*\\ \mathbf{U}_{o,1;21}^* & \mathbf{U}_{o,1;22}^*\end{bmatrix}\begin{bmatrix}\Sigma_{c;1}^{-1} & \mathbf{0}\\ \mathbf{0} & \Sigma_{c;2}^{-1}\end{bmatrix}\begin{bmatrix}\mathbf{U}_{c;11}^* & \mathbf{U}_{c;12}^*\\ \mathbf{U}_{c;21}^* & \mathbf{U}_{c;22}^*\end{bmatrix}\mathbf{x} = \\ &= \begin{bmatrix}\Sigma_{o,1;1}^{1/2}\mathbf{U}_{o,1;11}^* & \Sigma_{o,1;1}^{1/2}\mathbf{U}_{o,1;12}^*\\ \Sigma_{o,1;2}^{1/2}\mathbf{U}_{o,1;21}^* & \Sigma_{o,1;2}^{1/2}\mathbf{U}_{o,1;22}^*\end{bmatrix}\begin{bmatrix}\Sigma_{c;1}^{-1}\mathbf{U}_{c;11}^* & \Sigma_{c;1}^{-1}\mathbf{U}_{c;12}^*\\ \Sigma_{c;2}^{-1}\mathbf{U}_{c;21}^* & \Sigma_{c;2}^{-1}\mathbf{U}_{c;22}^*\end{bmatrix}\mathbf{x} =\end{aligned}$$

todo *Show if this algorithm allows to compute only the desired part of the svd, allowing the incremental computation, and not requiring the full svd computation*

Transformation of Gramians

$$\hat{\mathbf{W}}_c = \int e^{\hat{\mathbf{A}}t}\hat{\mathbf{B}}\hat{\mathbf{B}}^T e^{\hat{\mathbf{A}}^T t} = \mathbf{T}^{-1}\mathbf{W}_c\mathbf{T}^{-T}$$
$$\hat{\mathbf{W}}_o = \int e^{\hat{\mathbf{A}}^T t}\hat{\mathbf{C}}^T\hat{\mathbf{C}}e^{\hat{\mathbf{A}}t} = \mathbf{T}^T\mathbf{W}_o\mathbf{T}$$

as for $\mathbf{x} = \mathbf{T}\hat{\mathbf{x}}$, $\hat{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}$, $\hat{\mathbf{B}} = \mathbf{T}^{-1}\mathbf{B}$, $\hat{\mathbf{C}} = \mathbf{C}\mathbf{T}$, $\hat{\mathbf{D}} = \mathbf{D}$.

Method 2.

- Cholesky decomposition (for positive semi-definite matrices) of the controllability and observability Gramians $\mathbf{W}_c = \mathbf{L}_c\mathbf{L}_c^*$, $\mathbf{W}_o = \mathbf{L}_o\mathbf{L}_o^*$
- Compute the SVD of the product of the Cholesky factors, $\mathbf{L}_o^*\mathbf{L}_c = \mathbf{U}\Sigma\mathbf{V}^*$
- Partition

$$\Sigma = \begin{bmatrix}\Sigma_1 & \mathbf{0} \\ \mathbf{0} & \Sigma_2\end{bmatrix} \quad , \quad \mathbf{U} = \begin{bmatrix}\mathbf{U}_1 & \mathbf{U}_2\end{bmatrix} \quad , \quad \mathbf{V} = \begin{bmatrix}\mathbf{V}_1 & \mathbf{V}_2\end{bmatrix}$$

- Projection matrices $\mathbf{T} = \mathbf{L}_c\mathbf{V}_1\Sigma_1^{-1/2}$, $\tilde{\mathbf{T}} = \Sigma_1^{-1/2}\mathbf{U}_1^*\mathbf{L}_o^*$[2],

$$\mathbf{A}_r = \tilde{\mathbf{T}}\mathbf{A}\mathbf{T}$$
$$\mathbf{B}_r = \tilde{\mathbf{T}}\mathbf{B}$$
$$\mathbf{C}_r = \mathbf{C}\mathbf{T}$$
$$\mathbf{D}_r = \mathbf{D}$$

---

[2] This can be interpreted as a **Petrov-Galerkin** projection (**todo** *link to Finite Element Method?*), i.e. a model reduction with different base and test basis: here, base vectors are the columns of the matrix $\mathbf{T}$ while test vectors are the rows of the matrix $\tilde{\mathbf{T}}$.

CHAPTER

FORTYTWO

SHAPE FILTER

Response of a LTI to stochastic inputs. From the *response of LTI systems to stochastic input*, the auto-correlation of a zero-mean stationary process $x(t)$ (and thus its auto-covariance) reads

$$\mathbf{R}_{\mathbf{xx}}(\tau) = \mathbb{E}\left[\mathbf{x}_t \mathbf{x}_{t+\tau}^*\right] \ .$$

Fourier transform of this auto-correlation is defined as its **power spectral density, PSD**,

$$\Phi_{\mathbf{xx}}(f) := \mathcal{F}\{\mathbf{R}_{\mathbf{xx}}(\tau)\}(f) \ .$$

For a linear system whose input-output relation reads

$$\mathbf{y}(s) = \mathbf{H}(s)\mathbf{u}(s) \ ,$$

the PSD of the output can be written as a function of the PSD of the input and the Fourier transform of the transfer function as

$$\Phi_{\mathbf{yy}}(f) = \overline{\mathbf{H}}(f)\Phi_{\mathbf{uu}}(f)\mathbf{H}^T(f) \ . \tag{42.1}$$

SISO system. The transfer function is a scalar function for a SISO system, and thus

$$\Phi_{yy}(f) = \overline{H}(f)\Phi_{uu}(f)H(f) = |H(f)|^2 \Phi_{uu}(f) \ , \tag{42.2}$$

as the product of a complex number by its complex cojugate is equal to the square of its modulus.

White noise. A white noise $\xi(t)$ can be defined as a random process with

- zero expected value, $\mathbb{E}[\xi_t] = 0$
- Dirac's delta auto-correlation, $\mathbb{E}[\xi_t \xi_{t+\tau}] = \delta(\tau)$.

Fourier transform of a Dirac's delta is 1, and thus the PSD of a white noise is

$$\Phi_{\xi\xi}(f) = 1 \ .$$

SISO shape filter. A process $y(t)$ with the desired PSD $\Phi_{yy}(f)$ is the output of a shape filter - a dynamical system - with TF $H(f)$ satisfying the relation $|H(f)|^2 = \Phi_{yy}(f)$, fed by a white noise with unit autocovariance. This can be immediately proved inserting $\Phi_{uu}(f) = 1$ in the relation (42.2).

42.1 LTI system with non-white noise disturbance

Let the linear system

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r} \end{cases}$$

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{B}_n\mathbf{n} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} + \mathbf{D}_n\mathbf{n} \end{cases}$$

have non-white noise disturbances $\mathbf{n} = \begin{bmatrix} \mathbf{d} & \mathbf{r} \end{bmatrix}$, whose PSD is known and that can be modeled as the output of a shape filter fed by a white-noise $\mathbf{w}$,

$$\begin{cases} \dot{\mathbf{x}}_n = \mathbf{A}_n\mathbf{x}_n + \mathbf{B}_{nn}\mathbf{w}_n \\ \mathbf{n} = \mathbf{C}_{nn}\mathbf{x}_n \ , \end{cases}$$

then, the augmented system becomes

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_n\mathbf{C}_{nn}\mathbf{x}_n + \mathbf{B}\mathbf{u} \\ \dot{\mathbf{x}}_n = \mathbf{A}_n\mathbf{x}_n + \mathbf{B}_{nn}\mathbf{w}_n \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}_n\mathbf{C}_{nn}\mathbf{x}_n + \mathbf{D}\mathbf{u} \end{cases}$$

or

$$\begin{cases} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{x}}_n \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B}_n\mathbf{C}_{nn} \\ \cdot & \mathbf{A}_n \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_n \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \cdot \end{bmatrix} \mathbf{u} + \begin{bmatrix} \cdot \\ \mathbf{B}_{nn} \end{bmatrix} \mathbf{w}_n \\ \mathbf{y} = \begin{bmatrix} \mathbf{C} & \mathbf{D}_n\mathbf{C}_{nn} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_n \end{bmatrix} + \mathbf{D}\mathbf{u} \end{cases}$$

42.2 Example

A 1-dimensional non-white noise process d has PSD, $\Phi_{dd}(\omega) = |H(j\omega)|^2$ with

$$H(s) = \frac{s}{(s-p_1)(s-p_2)} = \frac{s}{s^2 + a_1 s + a_0} \ .$$

with $a_0 = p_1 p_2$, $a_1 = -(p_1 + p_2)$.

Using the *controller canonical realization*, a state-space representation of the shape filter follows

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \cdot & 1 \\ -a_0 & -a_1 \end{bmatrix} \begin{bmatrix} \cdot \\ 1 \end{bmatrix} w \\ y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \ . \end{cases}$$

Lyapunov equation for the coorelation matrix d. Correlation matrix of the state can be evaluated as the solution of the *Lyapunov equation for the correlation matrix*, for systems - like the shape filter - with a white noise input.

Non-strictly proper system. If a LTI fed by a white noise process as an input is not striclty proper, $\mathbf{D} \neq \mathbf{0}$, …

- the output has infinite variance (it can be split as a colored noise + white noise),
- PSD doesn't decay to zero as $\omega \to 0$
- Lyapunov equation for variance doesn't converge to a finite value solution

todo prove these statements

CHAPTER

FORTYTHREE

PADÉ APPROXIMATION OF TIME DELAY

43.1 Time delay

In time-domain

$$u_\tau(t) := u(t-\tau)\ , \quad \tau \geq 0\ ,$$

with causal function $u(t) = 0$, for $t < 0$.

In Laplace domain,

$$\begin{aligned}\mathcal{L}\left\{u(t-\tau)\right\} &:= \int_{t=0}^{+\infty} u(t-\tau)e^{-st}\,dt = \\ &= \int_{t'=0}^{+\infty} u(t')e^{-s(t'+\tau)}\,dt' = \\ &= e^{-s\tau}\int_{t'=0}^{+\infty} u(t')e^{-st'}\,dt' = \\ &= e^{-s\tau}U(s)\ .\end{aligned}$$

43.2 Padé approximation

Padè approximation is a rational approximation of the exponential transfer function of the delay, $e^{-s\tau}$.

- Padé approximation is a (reasonable) **low-frequency approximation** of time delay. This immediately follows as Padé approximation exploits Taylor expansion of the time-delay transfer function $e^{-s\tau}$ for $s = 0$.
- Padé is usually **required in time-domain methods**, in order to avoid delay differential equations

Examples.

- *First-order system w/ time delay - referece tracking*
- …

43.2.1 Examples

$(1,1)$**-Padé approximation.**

$$e^{-s\tau} \simeq P_{1,1}(s) = \frac{1 - \frac{\tau}{2}s}{1 + \frac{\tau}{2}s}$$

$(1,1)$-Padé approximation - details

As an example, a rational approximation with both numerator and denominator of first order

$$e^{-s\tau} \simeq \frac{1 + as}{1 + bs} \, .$$

Using Taylor approximation $e^{-s\tau}$

$$e^{-s\tau} \simeq 1 - \tau s + \frac{\tau^2 s^2}{2} + o(s^2) \, ,$$

and equating

$$\left(1 - \tau s + \frac{\tau^2 s^2}{2} + o(s^2)\right)(1 + bs) = 1 + as \, ,$$

the first terms

$$\begin{aligned} 1: &\quad 1 = 1 \\ s: &\quad -\tau + b = a \\ s^2: &\quad \frac{\tau^2}{2} - \tau b = 0 \, , \end{aligned}$$

it follows

$$\begin{aligned} b &= \frac{\tau}{2} \\ a &= -\frac{\tau}{2} \, , \end{aligned}$$

$(2,2)$**-Padé approximation.**

$$e^{-s\tau} \simeq P_{2,2}(s) = \frac{1 - \frac{\tau s}{2} + \frac{\tau^2 s^2}{12}}{1 + \frac{\tau s}{2} + \frac{\tau^2 s^2}{12}}$$

$(2,2)$-Padé approximation - details

As an example, a rational approximation with both numerator and denominator of first order

$$e^{-s\tau} \simeq \frac{1 + as + bs^2}{1 + cs + ds^2} \, .$$

Using Taylor approximation $e^{-s\tau}$

$$e^{-s\tau} \simeq 1 - \tau s + \frac{\tau^2 s^2}{2} - \frac{\tau^3 s^3}{3!} + \frac{\tau^4 s^4}{4!} + o(s^4) \, ,$$

and equating

$$\left(1 - \tau s + \frac{\tau^2 s^2}{2} - \frac{\tau^3 s^3}{3!} + \frac{\tau^4 s^4}{4!} + o(s^2)\right)(1 + cs + ds^2) = 1 + as + bs^2 \, ,$$

the first terms

$$\begin{aligned} 1: &\quad 1 = 1 \\ s: &\quad -\tau + c = a \\ s^2: &\quad \frac{\tau^2}{2} - \tau c + d = b \\ s^3: &\quad -\frac{\tau^3}{3!} + \frac{\tau^2}{2}c - \tau d = 0 \\ s^4: &\quad \frac{\tau^4}{4!} - \frac{\tau^3}{3!}c + \frac{\tau^2}{2}d = 0\,, \end{aligned}$$

it follows

$$\begin{aligned} c &= \frac{1}{2}\tau \\ d &= \frac{1}{12}\tau^2 \\ a &= -\frac{1}{2}\tau \\ b &= \frac{1}{12}\tau^2 \end{aligned}$$

as

$$\begin{aligned} \frac{1}{6}c - \frac{1}{2\cdot 3!}\tau = 0 \quad &\rightarrow \quad c = \frac{1}{2}\tau \\ d = \tau\frac{c}{2} - \frac{1}{6}\tau^2 \quad &\rightarrow \quad d = \frac{1}{12}\tau^2 \\ &\rightarrow \quad a = -\frac{1}{2}\tau \\ &\rightarrow \quad b = \frac{1}{12}\tau^2 \end{aligned}$$

43.2.2 Padé approximation in state-space representation

Let the LTI system

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t-\tau) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t-\tau)\,, \end{aligned}$$

A Padé approximation results in a LTI whose output is the approximate delayed input, and whose input is the input itself. As an example, the $(1,1)$-Padé approximation in Laplace domain reads

$$U_\tau(s) = e^{-s\tau}U(s) \simeq \frac{1-\frac{\tau s}{2}}{1+\frac{\tau s}{2}}U(s) = \left[\frac{-s+\frac{2}{\tau}}{s+\frac{2}{\tau}}\right]U(s) = \left[-1+\frac{\frac{4}{\tau}}{s+\frac{2}{\tau}}\right]U(s),$$

and it has a state-space realization using *controllable canonical form*,[1]

$$\begin{cases} \dot{\mathbf{x}}_\tau = \left[-\frac{2}{\tau}\right]\mathbf{x}_\tau + [1]\,u \\ u_\tau(t) = \left[\frac{4}{\tau}\right]\mathbf{x}_\tau + [-1]\,u \end{cases}.$$

[1] Usually, controllable canonical form for delay on input; observable canonical form for delay on output. Here, in this example with a $(1,1)$-Padé approximation, the state of the system is 1-dimensional, and the two canonical forms differ only by the "position" of the "$b_{n-1} = \frac{4}{\tau}$ element. The not-strictly proper TF is first written as a sum of a constant and a strictly proper TF, whose realization follows the rules of canonical realizations.

The **augmented system** reads

$$\begin{cases} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{x}}_\tau \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B}\mathbf{C}_\tau \\ \cdot & \mathbf{A}_\tau \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_\tau \end{bmatrix} + \begin{bmatrix} \mathbf{B}\mathbf{D}_\tau \\ \mathbf{B}_\tau \end{bmatrix} \mathbf{u} \\ \mathbf{y} = \begin{bmatrix} \mathbf{C} & \mathbf{D}\mathbf{C}_\tau \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_\tau \end{bmatrix} + \begin{bmatrix} \mathbf{D}\mathbf{D}_\tau \end{bmatrix} \mathbf{u} \end{cases}$$

.

Part XI

Control Theory

CHAPTER

FORTYFOUR

INTRODUCTION TO CONTROL METHODS

System behavior is treated in detailed in *System Theory* section. Control theory targets a desired behavior of the system, ensuring stability and performance (in terms of desired output, error, response velocity, input effort, robustness to noise and model uncertainty,…)

Links

- *Ordinary differential equations*
- *System theory*

Table of contents

- Frequency domain control of linear systems
- State estimation and observer
- Optimal control
- System identification

Features

Domain.

- Frequency domain for linear systems: root locus, Nyquist criterion and Bode criteria,…
- Time domain: *optimal control*, robust optimal control,…

Full-state, partial-state feedback.

- Full-state feedback: the complete state of the system can be observed and used for the feedback
- Partial-state feedback

Observers, e.g. Kalman filter, tries to estimate the state of a system from its output. Later, this estimatio can be used for feedback.

Model-based/model-free control.

CHAPTER

FORTYFIVE

CLOSED-LOOP CONTROL: REQUIREMENTS AND PERFORMANCE

Requirement/Performance	...	...
Stability	Nyquist criterion on the open-loop TF $L(s)$	
Robustness	Stability margins	
Reference tracking	Type of the system. Integrators in open-loop TF $L(s)$	
Measurement noise suppression		
Input load		
Input noise suppression		
Transient performance		

Let the transfer function of the system be

$$\mathbf{y} = \mathbf{G}\mathbf{u} + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n} \,,$$

linking the control input **u**, and the exogenous inputs (process disturbances **d**, and measurement noise **n**) to the output **y**. Let control disturbance $\Delta\mathbf{u}$ on the ideal control $\mathbf{u}_c$ be additive, $\mathbf{u} = \mathbf{u}_c + \Delta\mathbf{u}$. Let **F** be a feed-forward system from reference **r** to the reference output $\mathbf{y}_{\text{ref}}$ that is compared with the output of the system **y** to define the error $\mathbf{e} = \mathbf{y}_{\text{ref}} - \mathbf{y}$ feeding the regulator **R**.

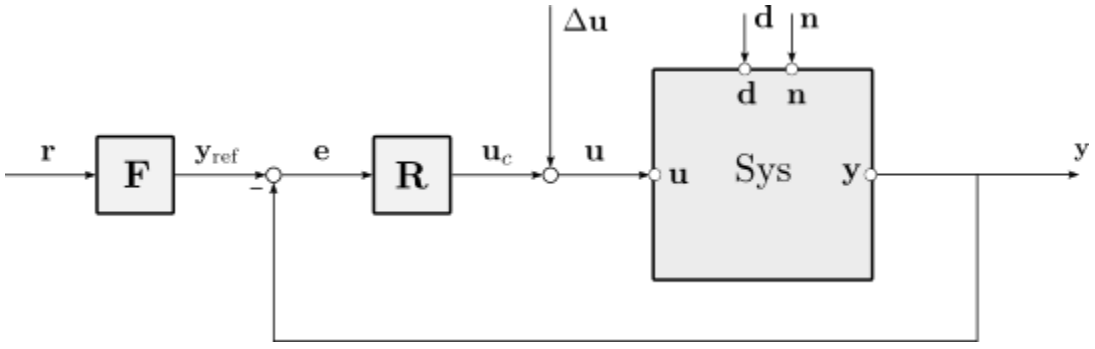


Fig. 45.1: Block diagram of the closed-loop system

Closed-loop performance usually deals with the relation between the output **y**, the tracking error **e**, and the ideal control input $\mathbf{u}_c$ (the output of he regulator; that's what we need to care about for controller response, performance, saturation,…) w.r.t. the reference signal **r** and the exogenous inputs $\Delta\mathbf{u}$, **d**, **r**, in terms of:

- tracking of the reference
- control performance
- disturbance suppression/noise rejection (on both the output and the input control)

For a MIMO

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{e} \\ \mathbf{u}_c \end{bmatrix} = \begin{bmatrix} [\mathbf{I}+\mathbf{GR}]^{-1} & \cdot & \cdot \\ \cdot & [\mathbf{I}+\mathbf{GR}]^{-1} & \cdot \\ \cdot & \cdot & [\mathbf{I}+\mathbf{RG}]^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{GRF} & \mathbf{G} & \mathbf{G}_d & \mathbf{G}_n \\ \mathbf{F} & -\mathbf{G} & -\mathbf{G}_d & -\mathbf{G}_n \\ \mathbf{RF} & -\mathbf{RG} & -\mathbf{RG}_d & -\mathbf{RG}_n \end{bmatrix} \begin{bmatrix} \mathbf{r} \\ \Delta\mathbf{u} \\ \mathbf{d} \\ \mathbf{n} \end{bmatrix} .$$

For a SISO

$$\begin{bmatrix} y \\ e \\ u_c \end{bmatrix} = \frac{1}{1+GR} \begin{bmatrix} GRF & G & G_d & G_n \\ F & -G & -G_d & -G_n \\ RF & -RG & -RG_d & -RG_n \end{bmatrix} \begin{bmatrix} r \\ \Delta u \\ d \\ n \end{bmatrix} .$$

Two-degrees of freedom problem. Typical design procedure:

- Design R to provide load/noise performance
- Design F to provide tracking performance

Simplified model. If

- there's no feed-forward, i.e. $\mathbf{F} = \mathbf{I}$ or $F = 1$ for a SISO,
- the measurement noise is additive on an ideal output, the TF is $\mathbf{y} = \mathbf{Gu} + \mathbf{G}_d\mathbf{d} + \mathbf{n}$, i.e.$\mathbf{G}_n = \mathbf{I}$ or $G_n(s) = 1$ for a SISO,

for a MIMO

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{e} \\ \mathbf{u}_c \end{bmatrix} = \begin{bmatrix} [\mathbf{I}+\mathbf{GR}]^{-1} & \cdot & \cdot \\ \cdot & [\mathbf{I}+\mathbf{GR}]^{-1} & \cdot \\ \cdot & \cdot & [\mathbf{I}+\mathbf{RG}]^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{GR} & \mathbf{G} & \mathbf{G}_d & \mathbf{I} \\ \mathbf{I} & -\mathbf{G} & -\mathbf{G}_d & -\mathbf{I} \\ \mathbf{R} & -\mathbf{RG} & -\mathbf{RG}_d & -\mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{r} \\ \Delta\mathbf{u} \\ \mathbf{d} \\ \mathbf{n} \end{bmatrix} ,$$

and for a SISO

$$\begin{bmatrix} y \\ e \\ u_c \end{bmatrix} = \frac{1}{1+GR} \begin{bmatrix} GR & G & G_d & 1 \\ 1 & -G & -G_d & -1 \\ R & -RG & -RG_d & -R \end{bmatrix} \begin{bmatrix} r \\ \Delta u \\ d \\ n \end{bmatrix} .$$

For a SISO, beside process noise **d** that may have its own dynamics, the performance of the closed loop system is defined by 4 transfer functions only, **"the gang of the 4"**,

- sensitivity function ($\mathbf{y}(\mathbf{n})$, $\mathbf{e}(\mathbf{r})$, $\mathbf{e}(\mathbf{n})$)

$$S = \frac{1}{1+GR} ,$$

- complementary sensitivity ($\mathbf{y}(\mathbf{r})$, $\mathbf{u}_c(\Delta\mathbf{u})$)

$$T = \frac{GR}{1+GR} ,$$

- load sensitivity ($\mathbf{y}(\Delta\mathbf{u})$, $\mathbf{e}(\Delta\mathbf{u})$)

$$PS = \frac{G}{1+GR} ,$$

- noise sensitivity ($\mathbf{u}_c(\mathbf{r})$, $\mathbf{u}_c(\mathbf{n})$)

$$CS = \frac{R}{1+GR} \, ,$$

Details

Output.

$$\begin{aligned}
\mathbf{y} &= \mathbf{G}\mathbf{u} + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n} = \\
&= \mathbf{G}\left(\mathbf{u}_c + \Delta\mathbf{u}\right) + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n} = \\
&= \mathbf{G}\mathbf{R}\mathbf{e} + \mathbf{G}\Delta\mathbf{u} + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n} = \\
&= \mathbf{G}\mathbf{R}\left(\mathbf{y}_{\text{ref}} - \mathbf{y}\right) + \mathbf{G}\Delta\mathbf{u} + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n} = \\
&= \mathbf{G}\mathbf{R}\mathbf{F}\mathbf{r} - \mathbf{G}\mathbf{R}\mathbf{y} + \mathbf{G}\Delta\mathbf{u} + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n} \, ,
\end{aligned}$$

and thus

$$\begin{aligned}
\mathbf{y} = &\left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{G}\mathbf{R}\mathbf{F}\,\mathbf{r} + \\
&+ \left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{G}\,\Delta\mathbf{u} + \\
&+ \left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{G}_d\,\mathbf{d} + \\
&+ \left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{G}_n\,\mathbf{n} \, .
\end{aligned}$$

Error.

$$\begin{aligned}
\mathbf{e} &= \mathbf{F}\mathbf{r} - \mathbf{y} = \\
&= \mathbf{F}\mathbf{r} - \left(\mathbf{G}\mathbf{u} + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n}\right) = \\
&= \mathbf{F}\mathbf{r} - \mathbf{G}\mathbf{u}_c - \mathbf{G}\Delta\mathbf{u} - \mathbf{G}_d\mathbf{d} - \mathbf{G}_n\mathbf{n} = \\
&= \mathbf{F}\mathbf{r} - \mathbf{G}\mathbf{R}\mathbf{e} - \mathbf{G}\Delta\mathbf{u} - \mathbf{G}_d\mathbf{d} - \mathbf{G}_n\mathbf{n} \, ,
\end{aligned}$$

and thus

$$\begin{aligned}
\mathbf{e} = &\left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{F}\,\mathbf{r} + \\
&- \left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{G}\,\Delta\mathbf{u} + \\
&- \left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{G}_d\,\mathbf{d} + \\
&- \left[\mathbf{I} + \mathbf{G}\mathbf{R}\right]^{-1}\mathbf{G}_r\,\mathbf{r} \, .
\end{aligned}$$

Input.

$$\begin{aligned}
\mathbf{u}_c &= \mathbf{R}\mathbf{F}\mathbf{r} - \mathbf{R}\mathbf{y} = \\
&= \mathbf{R}\mathbf{F}\mathbf{r} - \mathbf{R}\left(\mathbf{G}\mathbf{u} + \mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n}\right) = \\
&= \mathbf{R}\mathbf{F}\mathbf{r} - \mathbf{R}\mathbf{G}\mathbf{u}_c - \mathbf{R}\mathbf{G}\Delta\mathbf{u} - \mathbf{R}\left(\mathbf{G}_d\mathbf{d} + \mathbf{G}_n\mathbf{n}\right) \, ,
\end{aligned}$$

and thus

$$\begin{aligned}
\mathbf{u}_c = &\left[\mathbf{I} + \mathbf{R}\mathbf{G}\right]^{-1}\mathbf{R}\mathbf{F}\,\mathbf{r} + \\
&- \left[\mathbf{I} + \mathbf{R}\mathbf{G}\right]^{-1}\mathbf{R}\mathbf{G}\,\Delta\mathbf{u} + \\
&- \left[\mathbf{I} + \mathbf{R}\mathbf{G}\right]^{-1}\mathbf{R}\mathbf{G}_d\,\mathbf{d} + \\
&- \left[\mathbf{I} + \mathbf{R}\mathbf{G}\right]^{-1}\mathbf{R}\mathbf{G}_r\,\mathbf{r} \, .
\end{aligned}$$

Algebraic constraint on sensitivity and complementary sensitivity. It's not possible to make $S(s)$ and $T(s)$ small at the same time, as

$$S(s) + T(s) = 1$$

Usually,

- $|S(j\omega)|$ small at low frequency ω to get small error in the band of the reference and filtering low-frequency measurement noise on the inp
- $|T(j\omega)|$ small at high frequency ω to filter high-frequency noise from Δu to u (usually reference signal r has no high-frequency content, so there should be little issues in filtering out the effect of reference signal to the output).

45.1 Stability (SISO)

See *Nyquist* and *Bode* stability criterion in *Frequency domain control* design.

45.2 Reference tracking (SISO)

Setting $\mathbf{F} = \mathbf{I}$ in Fig. 45.1, and neglecting disturbances, the transfer functions between the output and the reference and between the tracking error and the reference signal are

$$E(s) = \frac{1}{1 + G(s)R(s)} Y_{\text{ref}}(s)$$
$$Y(s) = \frac{G(s)R(s)}{1 + G(s)R(s)} Y_{\text{ref}}(s)$$

Polynomial forcing.

Laplace transform of polynomial signals

Laplace transform of (casual) polynomial signals immediately follows from the transform of a Dirac's delta and the property of the Laplace transform of the integral of a function,

$$\mathcal{L}\left\{\int_{\tau=0}^{t} f(\tau)\, d\tau\right\} = \frac{F(s)}{s}\,,$$

being $F(s)$ the Laplace transform of function $f(t)$. Indeed, a ramp can be interpreted as the integral of a Dirac's delta, a casual parabolic signal is the integral of the ramp, and so on.

Dirac's delta.

$$F_{-1} := F_\delta = \mathcal{L}\{\delta(t)\} = \int_{t=0^-}^{+\infty} \delta(t) e^{-st} dt = 1\,.$$

Step. Step is the integral of the Dirac's delta.

$$F_0 := F_{\text{step}} = \mathcal{L}\{1\} = \frac{1}{s}\,.$$

Ramp. Step is the integral of the Dirac's delta.

$$F_1 := F_{\text{ramp}} = \mathcal{L}\{t\} = \frac{1}{s^2}\,.$$

Signal t^n, $n \in \mathbb{N}$.

$$\mathcal{L}\{t^n\} = \frac{1}{s^{n+1}}\,.$$

Limit of the tracking error of polynomial forcing, for $t \to +\infty$. Assuming the system is stable, the free-response to initial condition vanishes. Using **final value** property of Laplace transform (see *Laplace transform definition and properties*),

$$\lim_{t\to+\infty} e(t) = \lim_{s\to 0} sE(s) \ .$$

Let the open-loop transfer function $L(s) = G(s)R(s)$ be $L(s) = \frac{N(s)}{s^p D(s)}$, having explicitly written the integrators (the factors $\frac{1}{s}$ in the denominator of the TF), so that $D(0) \neq 0$, $N(s) \neq 0$. For casual systems $|N| \leq p+|D|$. The closed-loop function reads

$$\frac{1}{1+L(s)} = \frac{s^p D(s)}{s^p D(s) + N(s)} \ .$$

Let $Y_{\text{ref}}(s) = C_0 + \frac{C_1}{s} + \cdots + \frac{C_{m+1}}{s^{m+1}}$ the Laplace transform of a m-degree polynomial reference, the limit value of the tracking error reads

$$\begin{aligned}
\lim_{t\to+\infty} e(t) &= \lim_{s\to 0} sE(s) = \\
&= \lim_{s\to 0} \frac{s^{p+1} D(s)}{s^p D(s) + N(s)} \left(C_0 + \frac{C_1}{s} + \cdots + \frac{C_{m+1}}{s^{m+1}} \right) = \\
&= \lim_{s\to 0} \frac{s^{p-m} D(s)}{s^p D(s) + N(s)} \left(s^{m+1} C_0 + s^m C_1 + \cdots + C_{m+1} \right) = \\
&= \begin{cases} 0 & , \quad p > m \\ \dfrac{D(0)}{N(0)} & , \quad p = m \\ +\infty & , \quad p < m \end{cases}
\end{aligned}$$

Type of the system.

CHAPTER

FORTYSIX

FREQUENCY DOMAIN CONTROL

46.1 Stability of closed-loop SISO systems

Let $L(s) = G(s)R(s)$ the open-loop transfer function of the the system represented in Fig. 46.1: the system is explicitly represented as a combination of a plant with transfer function $G(s)$ and a regulator with transfer function $R(s)$. The output of the system y, and the error $e = y - y_{ref}$ are related with the reference signals as

$$\begin{aligned} y(s) &= \frac{L(s)}{1 + L(s)} y_{ref}(s) \\ e(s) &= \frac{1}{1 + L(s)} y_{ref}(s) \ . \end{aligned} \tag{46.1}$$

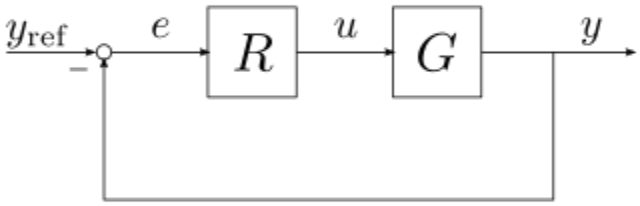


Fig. 46.1: Block diagram of the closed-loop system with retroaction on the output error $e = y_{ref} - y$. Here, $G(s)$ is the plant transfer function, $R(s)$ is the regulator transfer function (that can be designed to control a given plant), and $L(s) = G(s)R(s)$ is defined as the open-loop transfer function.

Details

In Laplace domain, neglecting here the initial conditions,

$$y = Gu = GRe = GR(y_{ref} - y) \ ,$$

and thus, with $L = GR$ and SISO system (so that TF are not matrix but scalar functions)

$$y(s) = \frac{L(s)}{1 + L(s)} y_{ref}(s) \ .$$

For the output error,

$$e(s) = y_{ref} - y = y_{ref} - GRe \ ,$$

and thus

$$e(s) = \frac{1}{1 + L(s)} y_{ref} \ .$$

46.1.1 Cauchy argument principle

For the *Cauchy argument principle*, the difference of argument of function $F(s)$ when s performs a counter-clockwise loop over th contour Γ enclosing poles p_n, and zeros z_d of the function $F(s)$ reads

$$\Delta \arg\{F(s)\} = \sum_n \arg(s - z_n) - \sum_d \arg(s - p_d) = Z2\pi - P2\pi = (Z - P)2\pi \ ,$$

and thus the the diagram of function $F(s)$ performs

$$N = Z - P \tag{46.2}$$

loops around the origin $0 + i0$ of the complex plane.

46.1.2 Nyquist stability criterion

Nyquist stability criterion provides the conditions for the stability of the closed loop transfer functions (46.1). As shown in section about *stability of LTI systems*, a linear system is asymptotically stable if all of its poles have negative real parts.

All the closed-loop transfer functions have the same poles (as they have the same denominator, except for *dangerous* cancellations not changing the stability but the observability or the controllability **todo** *Discuss*).

The poles of the closed loop systems are the zeros of the function $1 + L(s)$. If the closed-loop system is asymptotically stable, it must have no pole with positive real part. As the complex variable s performs a clockwise (and thus, the signs of the relation change w.r.t. Cauchy argument criterion) loop over **Nyquist path** (semicircle in the RHS half-plane; then **todo** discuss the case where poles and zeros are on the imaginary axis…deform the path…). No zero of $1 + G(s)$ with positive real part means $Z = 0$. Using the relation between poles, zeros and loops around the origin given by Cauchy argument principle (46.2), and recalling the opposite direction, it follows that

$$N = P \ ,$$

in order to have $Z = 0$, i.e. the diagram of $1 + G(s)$ must perform N counter-clockwise loops around $0 + i0$ equal to the number of its poles with positive real parts. As the poles of $1 + G(s)$ are the same as the poles of $G(s)$, it's possible to formulate Nyquist criterion looking at $G(s)$, as

Theorem 46.1.1 (Nyquist theorem)

In order for the closed-loop system to be asymptotically stable, the diagram of the open-loop transfer function $G(s)$ must perform a number N of counter-clocwise loops around the *critical point* $-1 + i\,0$ equal to the number of its zeros with positve real part.

46.1.3 Bode stability criterion for minimal phase systems

Definition 46.1.1 (Minimal phase systems)

…

…

Theorem 46.1.2 (Bode stability criterion)

…

46.1.4 Phase margin

46.1.5 Gain margin

CHAPTER

FORTYSEVEN

OPTIMAL CONTROL

- Deterministic/stochastic signals
- Full-state feedback, observer and full-state on observed state, output feedback
- Regulator (targeting zero state $\mathbf{x} = \mathbf{0}$), or *reference tracking* (targeting zero error of the output w.r.t. a reference input of the closed-loop system $\mathbf{y}_{\text{ref}}$)

Deterministic signals.

- *Full-state feedback*, $\mathbf{u} = -\mathbf{K}\mathbf{x}$
- Observer, for state estimation
- *Full-state feed back on the observed state*, $\mathbf{u} = -\mathbf{K}\mathbf{o}$
- *Output feedback*, $\mathbf{u} = -\mathbf{K}\mathbf{y}$

Stochastic signals.

- *Kalman filter*, i.e. optimal observer w.r.t. white noise system exogenous input and measurement noise

47.1 Full-state feedback

Optimal control can be recast as a **constrained optimization problem**, J, where an extreme - optimum - of an objective function must be found, subject to constraints that include the equations of motion. Some constraints may be included into an augmented objective function $\widetilde{J}$ with the methods of *Lagrange multipliers*.

Contents

- *Generic ODE without exogenous inputs*:
- *Linear system without exogenous inputs*:
- *Linear system without exogenous inputs - infinite time horizon*

Different models - governing equations

- Generic ODE

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$$

- Linear ODE for deterministic signals

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_u\mathbf{u} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}_u\mathbf{u} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r} \end{cases}$$

- Linear SDE for stochastic signals. Exogenous inputs like disturbances **d** and measuerement noise **r** can be treated as stochastic processes. Thus, the equations are stochastic equations. As an example, if these noise are **white noise** they can be interpreted as the time derivative of Wiener processes, and the SDE can be written as

$$d\mathbf{x} = \mathbf{A}\mathbf{x}\,dt + \mathbf{B}_u\mathbf{u}\,dt + \mathbf{B}_d d\mathbf{w}_d \ ,$$

  formally writing the differential of the Wiener process $d\mathbf{w}_d$ as a white noise **d** times dt, "$d\mathbf{w}_d = \mathbf{d}\,dt$". While this may look like a useless trick, it helps us recallling that $|d\mathbf{w}_d|^2 \sim dt$, when evaluating covariances.

Different approaches to the solution

Different approaches to the solution can be used, and help for a detailed comprehension of the topics.

- **Variational apporach** to constrained optimization,

$$J(\mathbf{x}, \mathbf{u}) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) - \int_{\tau=0}^{T} \lambda^T \left(\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}, \mathbf{u})\right) \ ,$$

  where the equations of motion are constraints inserted in the objective function with the method of Lagrange multipliers.

- **Dynamic programming approach**, leading to **Hamilton-Jacobi-Bellman (HJB) equation** for the **value function** $V(t)$

$$V(\mathbf{x}_t, t; \mathbf{u}) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) \ ,$$

  with the dynamics of the system subject to the governing equation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, and the initial condition for the value function - the tail cost function - $x(t) = x_t$. These constraints can be explicitly applied if an expression of the solution of the dynamical equation exists, or they can be added with the method of Lagrange multipliers.

- …

A common choice of the running cost $C(\mathbf{x}, \mathbf{u})$ and the final cost $D(\mathbf{x}(T))$ are

$$C(\mathbf{x}(\tau), \mathbf{u}(\tau)) = \frac{1}{2}\left\{\mathbf{x}^T(\tau)\mathbf{Q}\mathbf{x}(\tau) + \mathbf{u}^T(\tau)\mathbf{R}\mathbf{u}(\tau)\right\}$$
$$D(\mathbf{x}(T)) = \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) \ .$$

Finite time vs. Infinite time horizon.

… …

47.1.1 Generic ODE without exogenous inputs

Let a general general ODE, with $\mathbf{M}$ non singular (otherwise DAE…any issue with DAE?)

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \ ,$$

with initial condition $\mathbf{x}(0) = \mathbf{x}_0$.

Variational approach

The objective function combines (weights) the error on a desired performance and the control input, in order to get the desired behavior with feasible control (that can be provided by actuators, without saturation, avoiding unnecessary high power input and too sharp behavior,…)

As an example, if the goal of the control $\mathbf{u}$ is to keep the system around $\mathbf{x} = \mathbf{0}$, the cost function to be minimized can be designed as

$$J = \int_{t=0}^{T} \frac{1}{2} \begin{bmatrix} \mathbf{x}^T & \mathbf{u}^T \end{bmatrix} \begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} dt + \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T) \ ,$$

with $\mathbf{Q} \geq 0$, and $\mathbf{R} > 0$ and symmetric.

Constrained optimization

$$\widetilde{J}(\mathbf{x}, \mathbf{u}; \lambda) = \int_{\tau=0}^{T} C\left(\mathbf{x}(\tau), \mathbf{u}(\tau)\right) \, d\tau + D\left(\mathbf{x}(T)\right) - \int_{\tau=0}^{T} \lambda^T \left(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}(\tau), \mathbf{u}(\tau))\right) \, d\tau \ .$$

$$\begin{aligned} 0 = \delta\widetilde{J} = & \int_{t=0}^{T} \delta\mathbf{x}^T \left(\partial_{\mathbf{x}} C + \partial_{\mathbf{x}}\mathbf{f}^T\lambda + \mathbf{M}^T\dot{\lambda}\right) d\tau + \\ & + \int_{t=0}^{T} \delta\mathbf{u}^T \left(\partial_{\mathbf{u}} C + \partial_{\mathbf{u}}\mathbf{f}^T\lambda\right) d\tau + \\ & + \delta\mathbf{x}^T(T)\partial_{\mathbf{x}_T} D + \\ & + \int_{t=0}^{T} \delta\lambda^T(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}, \mathbf{u}))d\tau + \\ & - \delta\mathbf{x}\mathbf{M}^T\lambda\Big|_{t=0}^{T} \ . \end{aligned}$$

So that

$$\begin{aligned} \delta\lambda(t): \quad & \mathbf{M}\dot{\mathbf{x}} = \mathbf{f} \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ \delta\mathbf{x}(t): \quad & \mathbf{M}^T\dot{\lambda} = -\partial_{\mathbf{x}}\mathbf{f}^T\lambda - \partial_{\mathbf{x}} C \\ \delta\mathbf{x}(T): \quad & \mathbf{M}^T\lambda(T) = \partial_{\mathbf{x}_T} D \\ \delta\mathbf{u}(t): \quad & \mathbf{0} = \partial_{\mathbf{u}} C + \partial_{\mathbf{u}}\mathbf{f}^T\lambda \ . \end{aligned} \tag{47.1}$$

The equations can be recast after the definition of the Hamiltonian of the system $H(\mathbf{x}, \mathbf{u}, \lambda) = C(\mathbf{x}, \mathbf{u}) + \lambda^T\mathbf{f}(\mathbf{x}, \mathbf{u})$ as

$$\begin{aligned} \delta\lambda(t): \quad & \mathbf{0} = -\mathbf{M}\dot{\mathbf{x}} + \partial_{\lambda} H = -\mathbf{M}\dot{\mathbf{x}} + \mathbf{f} \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ \delta\mathbf{x}(t): \quad & \mathbf{M}^T\dot{\lambda} = -\partial_{\mathbf{x}} H \\ \delta\mathbf{x}(T): \quad & \mathbf{M}^T\lambda(T) = \partial_{\mathbf{x}_T} D \\ \delta\mathbf{u}(t): \quad & \mathbf{0} = \partial_{\mathbf{u}} H \ . \end{aligned} \tag{47.2}$$

Gradient descent

- Start from a control law $\mathbf{u}^{(0)}(t)$,
- the state equation is integrated forward in time to get $\mathbf{x}^{(0)}(t)$,
- Lagrange multiplier equation is integrated backward in time to get $\lambda^{(0)}(t)$,
- The control law is updated with an increment $\delta\mathbf{u}$ that's proportional to the gradient of the cost function (with "opposite direction" to get $\delta\widetilde{J} < 0$), i.e. $\mathbf{u}^{(1)}(t) = \mathbf{u}^{(0)}(t) + \delta\mathbf{u}^{(0)}(t)$ with $\delta\mathbf{u}^{(0)}(t) = -c\nabla_{\mathbf{u}(t)}J^{(0)}$, with a positive step $c > 0$ and $\nabla_{\mathbf{u}}J = \mathbf{S}^T\mathbf{x} + \mathbf{R}\mathbf{u} + \partial_{\mathbf{u}}\mathbf{f}^T\lambda$.
- Repeat previous steps with the updated control law, until convergence.

Dynamic programming approach

Dynamic programming approach for optimal control may be interpreted as the continuous-time counterpart fo the dynamic programming approach used in reinforcement learning (RL) on *Markov decision processes*. While RL usually produces a model-free policy - control law - and thus exploits *Bellman's equations* on the state-action value function $Q(s;a)$, optimal control is a model-based control system design that relies on a model of the system and uses Bellman's equation for the state value function $V(s)$.

HJB equation - Evaluation equation

Without dynamical equations as a constraint introduced with Lagrange multipliers

The value function $V(\mathbf{x}_t, t)$,

$$V(\mathbf{x}_t, t) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) \ ,$$

subject to the dynamical equations of motion in time range $\tau \in [t, T]$, is defined as a function of two arguments: the initial time t, the initial state $\mathbf{x}_t$ of the system at time t. If the initial state $\mathbf{x}_t$ is a function of t itself, the value function can be written as a function of time t only and its ordinary derivative w.r.t. time t reads

$$\frac{d}{dt}V(\mathbf{x}_t, t) = \partial_t V(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V\,\dot{\mathbf{x}} = \partial_t V(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V\,\mathbf{f} \ ,$$

or, using the definition and the rule of derivative for integrals

$$\frac{d}{dt}V(\mathbf{x}_t, t) = -C(\mathbf{x}(t), \mathbf{u}(t)) \ .$$

Comparing the two expressions of the time derivative, **Hamilton-Jacobi-Bellman equation** for evaluating a given control $\mathbf{u}$ follows

$$0 = \partial_t V(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V(\mathbf{x}_t, t)\,\mathbf{f}(\mathbf{x}_t, \mathbf{u}_t) + C(\mathbf{x}_t, \mathbf{u}_t) \ .$$

With dynamical equations as a constraint introduced with Lagrange multipliers

$$V(\mathbf{x}_t, t; \lambda) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) - \int_{\tau=t}^{T} \lambda^T(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f})d\tau \ ,$$

so that the variation (where's the variation w.r.t. t? Integration extreme is not prescribed, or is it?) reads

$$\begin{aligned}\delta V(\mathbf{x}_t, t; \mathbf{u}, \lambda) = &\int_{\tau=t}^{T} \left\{\delta\mathbf{x}^T\partial_{\mathbf{x}}C + \delta\mathbf{u}^T\partial_{\mathbf{u}}C\right\} d\tau + \delta\mathbf{x}_T\partial_{\mathbf{x}_T}D(\mathbf{x}(T)) + \\ &- \int_{\tau=t}^{T} \delta\lambda^T(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f})d\tau + \int_{\tau=t}^{T} \delta\mathbf{x}^T\mathbf{M}^T\dot{\lambda}\, d\tau + \\ &- \delta\mathbf{x}^T(T)\mathbf{M}^T\lambda(T) + \delta\mathbf{x}_t^T\mathbf{M}^T\lambda(t) + \\ &+ \int_{\tau=t}^{T} \left\{\delta\mathbf{x}^T\partial_{\mathbf{x}}\mathbf{f}^T\lambda + \delta\mathbf{u}^T\partial_{\mathbf{u}}\mathbf{f}^T\lambda\right\} d\tau\end{aligned}$$

All the variations but $\delta\mathbf{x}_t$ are identically zero if the equations (47.9) are satisfied, the variation w.r.t. $\delta\mathbf{x}_t$ shows that the sensitivity to the initial state in the value function $\mathbf{x}_t$ equals the value of the Lagrange multiplier $\lambda(t)$,

$$\nabla_{\mathbf{x}_t} V(\mathbf{x}_t, t; ...) = \mathbf{M}^T\lambda(t) \ , \tag{47.3}$$

with $\lambda(t)$ part of the solution of the equations (47.9). Thus the Lagrange multiplier $\lambda(t)$ provides an information (first order, sensitivity) about the change in the value function $V(\mathbf{x}_t, t)$ for small changes of the intial state $\mathbf{x}_t$, i.e.

$$\delta_{\mathbf{x}_t} V(\mathbf{x}_t, t) \sim \delta\mathbf{x}_t^T\nabla_{\mathbf{x}_t} V + o(|\delta\mathbf{x}_t|) = \delta\mathbf{x}_t^T\mathbf{M}^T\lambda(t) + o(|\delta\mathbf{x}_t|) \ .$$

todo

- useful to exploit adjoint info for sensitivity

HJB equation - Optimality equation

$$\begin{aligned}V^*(\mathbf{x}_t, t) = \min_{\mathbf{u}} V(\mathbf{x}_t, t) = = \min_{\mathbf{u}} &\left\{\int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T))\right\} = \\ &= \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}^*(\tau))d\tau + D(\mathbf{x}(T)) \ ,\end{aligned}$$

subject to the dynamical equations of motion. The value function is a function of two arguments, $\mathbf{x}_t$, and t, and the first argument is a function of t. Thus the ordinary derivative w.r.t. time t reads

$$\frac{d}{dt}V^*(\mathbf{x}_t, t) = \partial_t V^*(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V^*\,\dot{\mathbf{x}} = \partial_t V^*(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V^*\,\mathbf{f}(\mathbf{x}_t, \mathbf{u}_t^*) \ ,$$

or, using the definition and the rule of derivative for integrals

$$\frac{d}{dt}V^*(\mathbf{x}_t, t) = -C(\mathbf{x}(t), \mathbf{u}^*(t)) \ .$$

Comparing the two expressions of the time derivative, Hamilton-Bellman-Jacobi equation for evaluating a given control $\mathbf{u}$ follows

$$0 = \partial_t V^*(\mathbf{x}_t, t) + \partial_{\mathbf{x}} V^*(\mathbf{x}_t, t)\,\mathbf{f}(\mathbf{x}_t, \mathbf{u}_t^*) + C(\mathbf{x}_t, \mathbf{u}_t^*) \ ,$$

or

$$0 = \partial_t V^*(\mathbf{x}_t, t) + \min_{\mathbf{u}} \left\{\partial_{\mathbf{x}} V^*(\mathbf{x}_t, t)\,\mathbf{f}(\mathbf{x}_t, \mathbf{u}_t) + C(\mathbf{x}_t, \mathbf{u}_t)\right\} \ .$$

47.1.2 Linear system without exogenous inputs

Variational approach

Let the dynamical equation of a system be

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}_y\mathbf{x} + \mathbf{D}_y\mathbf{u} \\ \mathbf{z} = \mathbf{C}_z\mathbf{x} + \mathbf{D}_z\mathbf{u} \end{cases},$$

with measurment output $\mathbf{y}$ and performance output $\mathbf{z}$, and with running and final cost

$$\begin{aligned} C(\mathbf{x},\mathbf{u}) &= \frac{1}{2}\left(\mathbf{z}^T\mathbf{Q}\mathbf{z} + \mathbf{u}^T\mathbf{R}\mathbf{u}\right) = \\ &= \frac{1}{2}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}^T \begin{bmatrix}\mathbf{C}_z^T\mathbf{Q}\mathbf{C}_z & \mathbf{C}_z^T\mathbf{Q}\mathbf{D}_z \\ \mathbf{D}_z^T\mathbf{Q}\mathbf{C}_z & \mathbf{D}_z^T\mathbf{Q}\mathbf{D}_z + \mathbf{R}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} = \\ &= \frac{1}{2}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}^T \begin{bmatrix}\widetilde{\mathbf{Q}} & \mathbf{S} \\ \mathbf{S}^T & \widetilde{\mathbf{R}}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}, \end{aligned}$$

$$D(\mathbf{x}_T) = \frac{1}{2}\mathbf{x}^T(T)\mathbf{Q}_T\mathbf{x}(T)\,.$$

The equations (47.9) become

$$\begin{aligned} \delta\lambda(t): \quad & \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ \delta\mathbf{x}(t): \quad & \dot{\lambda} = -\mathbf{A}^T\lambda - \widetilde{\mathbf{Q}}\mathbf{x} - \mathbf{S}\mathbf{u} \\ \delta\mathbf{x}(T): \quad & \lambda(T) = \mathbf{Q}_T\mathbf{x}(T) \\ \delta\mathbf{u}(t): \quad & \mathbf{0} = \widetilde{\mathbf{R}}\mathbf{u} + \mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\,. \end{aligned} \tag{47.4}$$

The weights onf the control are definite positive, $\widetilde{\mathbf{R}} > 0$, and thus invertible. The control can be written as a function of the state and the co-state as

$$\mathbf{u} = -\widetilde{\mathbf{R}}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right)\,. \tag{47.5}$$

The expression (47.5) shows that the optimal control law is a control proportional to the state $\mathbf{x}$ and the co-state λ. As it's shown below, there's a proportionality relation between the state and the co-state $\lambda = \mathbf{P}\mathbf{x}$ as well, with $\mathbf{P}$ a symmetric, semi-positive definite matrix, satisfying some matrix equations. See:

- the relation $\lambda_t = \mathbf{P}_t\mathbf{x}_t$ arising by the relation (47.3) with $\mathbf{M} = \mathbf{I}$, $\lambda_t = \nabla_{\mathbf{x}_t}V$ in the *dynamic programming approach*
- the relation $\lambda_t = \mathbf{P}_t\mathbf{x}_t$ arising in the Hamiltonian form of the closed-loop system (47.6)

Hamiltonian form of the closed-loop system

Now the closed loop system reads

$$\begin{bmatrix}\dot{\mathbf{x}}\\ \dot{\lambda}\end{bmatrix} \begin{bmatrix}\mathbf{A} - \mathbf{B}\widetilde{\mathbf{R}}^{-1}\mathbf{S}^T & -\mathbf{B}\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T \\ -\widetilde{\mathbf{Q}} + \mathbf{S}\widetilde{\mathbf{R}}^{-1}\mathbf{S}^T & -\mathbf{A}^T + \mathbf{S}\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \lambda\end{bmatrix}, \tag{47.6}$$

with initial condition $\mathbf{x}(0) = \mathbf{x}_0$ and final condition $\lambda(T) = \mathbf{Q}_T\mathbf{x}(T)$. The solution of this linear system can be formally written using the propagator $\Psi(t,t_0)$ as

$$\begin{bmatrix}\mathbf{x}(t)\\ \lambda(t)\end{bmatrix} = \Psi(t,t_0)\begin{bmatrix}\mathbf{x}(t_0)\\ \lambda(t_0)\end{bmatrix} = \begin{bmatrix}\Psi_{xx}(t,t_0) & \Psi_{x\lambda}(t,t_0)\\ \Psi_{\lambda x}(t,t_0) & \Psi_{\lambda\lambda}(t,t_0)\end{bmatrix}\begin{bmatrix}\mathbf{x}(t_0)\\ \lambda(t_0)\end{bmatrix}$$

As the initial condition for λ is not known, it's not possible so far to integrate this equation forward in time.

Initial condition $\lambda(0)$. It's possible to find the initial condition for the co-state, using the propagator between 0 and T, and the final condition $\lambda(T) = \mathbf{Q}_T \mathbf{x}_T$. The state of the Hamiltonian system in T reads

$$\begin{bmatrix} \mathbf{x}(T) \\ \lambda(T) \end{bmatrix} = \begin{bmatrix} \Psi_{xx}(T, t_0) & \Psi_{x\lambda}(T, t_0) \\ \Psi_{\lambda x}(T, t_0) & \Psi_{\lambda\lambda}(T, t_0) \end{bmatrix} \begin{bmatrix} \mathbf{x}(t_0) \\ \lambda(t_0) \end{bmatrix}$$

so that (using the second block of the equations first, and then replacing $\mathbf{x}_T$ from the first block),

$$\begin{aligned} \mathbf{0} &= -\lambda_T + \Psi_{\lambda\lambda}(T, t_0)\lambda_0 + \Psi_{\lambda x}(T, t_0)\mathbf{x}_0 = \\ &= -\mathbf{Q}_T \mathbf{x}_T + \Psi_{\lambda\lambda}(T, t_0)\lambda_0 + \Psi_{\lambda x}(T, t_0)\mathbf{x}_0 = \\ &= -\mathbf{Q}_T \left[\Psi_{xx}(T, t_0)\mathbf{x}_0 + \Psi_{x\lambda}(T, t_0)\lambda_0\right] + \Psi_{\lambda\lambda}(T, t_0)\lambda_0 + \Psi_{\lambda x}(T, t_0)\mathbf{x}_0 \,, \end{aligned}$$

and thus

$$\lambda_0 = \left(\Psi_{\lambda\lambda}(T, t_0) - \mathbf{Q}_T \Psi_{x\lambda}(T, t_0)\right)^{-1} \left(\mathbf{Q}_T \Psi_{xx}(T, t_0) - \Psi_{\lambda x}(T, t_0)\right) \mathbf{x}_0 \,,$$

or

$$\lambda_0 = \mathbf{P}_0 \mathbf{x}_0 \,.$$

Relation between $\mathbf{x}(t)$ and $\lambda(t)$.

$$\begin{aligned} \mathbf{x}_t &= \Psi_{xx}(t, t_0)\mathbf{x}_0 + \Psi_{x\lambda}(t, t_0)\lambda_0 = \\ &= \left[\Psi_{xx}(t, t_0) + \Psi_{x\lambda}(t, t_0)\mathbf{P}_0\right] \mathbf{x}_0 \,, \\ \lambda_t &= \Psi_{\lambda x}(t, t_0)\mathbf{x}_0 + \Psi_{\lambda\lambda}(t, t_0)\lambda_0 = \\ &= \left[\Psi_{\lambda x}(t, t_0) + \Psi_{\lambda\lambda}(t, t_0)\mathbf{P}_0\right] \mathbf{x}_0 = \\ &= \left[\Psi_{\lambda x}(t, t_0) + \Psi_{\lambda\lambda}(t, t_0)\mathbf{P}_0\right] \left[\Psi_{xx}(t, t_0) + \Psi_{x\lambda}(t, t_0)\mathbf{P}_0\right]^{-1} \mathbf{x}_t \,. \end{aligned}$$

or

$$\lambda(t) = \mathbf{P}(t)\mathbf{x}(t) \,.$$

todo

- Does this inverse matrix exist? …

Dynamic programming approach

Value function and relation $\lambda = \mathbf{P}\mathbf{x}$

Let the value function be

$$V(\mathbf{x}_t, t; \mathbf{u}) = \frac{1}{2} \int_{\tau=t}^{T} \left\{ \mathbf{x}_\tau^T \widetilde{\mathbf{Q}}_\tau \mathbf{x}_\tau + \mathbf{x}_\tau^T \mathbf{S}_\tau \mathbf{u}_\tau + \mathbf{u}_\tau^T \mathbf{S}_\tau^T \mathbf{x}_\tau + \mathbf{u}_\tau^T \widetilde{\mathbf{R}}_\tau \mathbf{u}_\tau \right\} d\tau \, + \frac{1}{2} \mathbf{x}_T^T \mathbf{Q}_T \mathbf{x}_T \,,$$

subject to the equations of motion as constraints $\dot{\mathbf{x}}_\tau = \mathbf{A}_\tau \mathbf{x}_\tau + \mathbf{B}_\tau \mathbf{u}_\tau$, and the initial condition $\mathbf{x}(t) = \mathbf{x}_t$. This constraint can be introduced either 1) expressing the state $\mathbf{x}_\tau$ as a function of the initial state and the input

$$\mathbf{x}_\tau = \Phi(\tau, t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau, \xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \,,$$

or 2) with the methods of Lagrange multipliers. A **co-state** λ_t - corresponding to the Lagrange multiplier - can be evaluated as

$$\lambda_t = \nabla_{\mathbf{x}_t} V \,.$$

Method 1. Using the expression (29.2) of the *general solution of a linear system of ODEs* for $\tau > t$ with initial conditions $\mathbf{x}(t) = \mathbf{x}_t$ and input $\mathbf{u}(\tau)$,

$$
\begin{aligned}
V(\mathbf{x}_t, t; \mathbf{u}) = \frac{1}{2} \int_{\tau=t}^{T} \left\{ \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right)^T \widetilde{\mathbf{Q}}_\tau \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right) \right\} + \\
+ \frac{1}{2} \int_{\tau=t}^{T} \left\{ \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right)^T \mathbf{S}_\tau \mathbf{u}_\tau \right\} d\tau + \\
+ \frac{1}{2} \int_{\tau=t}^{T} \left\{ \mathbf{u}_\tau^T \mathbf{S}_\tau^T \left(\Phi(\tau,t)\mathbf{x}_t + \int_{\xi=t}^{\tau} \Phi(\tau,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right) \right\} d\tau + \\
+ \frac{1}{2} \int_{\tau=t}^{T} \mathbf{u}_\tau^T \widetilde{\mathbf{R}}_\tau \mathbf{u}_\tau \, d\tau + \\
+ \frac{1}{2} \left(\Phi(T,t)\mathbf{x}_t + \int_{\xi=t}^{T} \Phi(T,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right)^T \mathbf{Q}_T \left(\Phi(T,t)\mathbf{x}_t + \int_{\xi=t}^{T} \Phi(T,\xi)\mathbf{B}_\xi \mathbf{u}_\xi \, d\xi \right) =
\end{aligned}
$$

the relation $\lambda_t = \nabla_{\mathbf{x}_t} V$ gives

$$
\begin{aligned}
\lambda_t = \nabla_{\mathbf{x}_t} V = \\
= \int_{\tau=t}^{T} \left(\Phi^T(\tau,t)\widetilde{\mathbf{Q}}_\tau \mathbf{x}_\tau + \Phi^T(\tau,t)\mathbf{S}_\tau \mathbf{u}_\tau \right) d\tau + \Phi^T(T,t)\mathbf{Q}_T \mathbf{x}_T = \\
= \left\{ \int_{\tau=t}^{T} \Phi(\tau,t)^T \widetilde{\mathbf{Q}}_\tau \Phi(\tau,t) \, d\tau + \Phi^T(T,t)\mathbf{Q}_T \Phi(T,t) \right\} \mathbf{x}_t + \int_{\tau=t}^{T} \Phi^T(\tau,t)\mathbf{S}_\tau \mathbf{u}_\tau \, d\tau \, .
\end{aligned}
$$

From optimization, $\mathbf{u}_\tau = -\widetilde{\mathbf{R}}_\tau^{-1} \left(\mathbf{B}_\tau^T \lambda_\tau + \mathbf{S}_\tau^T \mathbf{x}_\tau \right)$,

Method 2.

Optimal control

HJB optimality equation

$$
0 = \partial_t V^*(\mathbf{x}_t, t) + \min_{\mathbf{u}} \left\{ \partial_{\mathbf{x}} V^*(\mathbf{x}_t, t) \, \mathbf{f}(\mathbf{x}_t, \mathbf{u}_t) + C(\mathbf{x}_t, \mathbf{u}_t) \right\} \, .
$$

$$
\begin{aligned}
\mathbf{0} = \delta_{\mathbf{u}} V = \\
= \int_{\tau=t}^{T} \left\{ \int_{\xi=t}^{\tau} \delta\mathbf{u}_\xi^T \mathbf{B}_\xi^T \Phi^T(\tau,\xi) \, d\xi \, \left(\widetilde{\mathbf{Q}}_\tau \mathbf{x}_\tau + \mathbf{S}_\tau \mathbf{u}_\tau \right) + \delta\mathbf{u}_\tau^T \mathbf{S}_\tau^T \mathbf{x}_\tau + \delta\mathbf{u}_\tau^T \widetilde{\mathbf{R}} \mathbf{u}_\tau + \mathbf{B}_\tau^T \Phi^T(T,\tau)\mathbf{Q}_T \mathbf{x}_T \right\} d\tau = \\
\\
= \text{TODO...details about variable swap in the integrals of the first term} = \\
\\
= \int_{\tau=t}^{T} \delta\mathbf{u}_\tau^T \left\{ \mathbf{B}_\tau^T \underbrace{\left(\int_{\xi=\tau}^{T} \Phi^T(\xi,\tau) \left(\widetilde{\mathbf{Q}}_\xi \mathbf{x}_\xi + \mathbf{S}_\xi \mathbf{u}_\xi \right) d\xi + \Phi^T(T,\tau)\mathbf{Q}\mathbf{x}_T \right)}_{\lambda_\tau} + \mathbf{S}_\tau^T \mathbf{x}_\tau + \widetilde{\mathbf{R}} \mathbf{u}_\tau \right\} d\tau \, ,
\end{aligned}
$$

and thus, from the arbitrariety of $\delta\mathbf{u}_\tau$, and since $\widetilde{\mathbf{R}}$ is required to be invertible

$$
\mathbf{u}_\tau = -\widetilde{\mathbf{R}}_\tau^{-1} \left(\mathbf{B}_\tau^T \lambda_\tau + \mathbf{S}_\tau^T \mathbf{x}_\tau \right) \, .
$$

...

If (**todo** why?) $\lambda = \mathbf{P}\mathbf{x}$,

Using HJB equation.

$$V(\mathbf{x}_t, t; \mathbf{u}) = \frac{1}{2}\int_{\tau=t}^{T} ... \, d\tau + \frac{1}{2}\mathbf{x}_T^T \mathbf{Q}_T \mathbf{x}_T + ...$$

Proportional control. This should not be an assumption, but a result of the problem *!!!*

If $\mathbf{u} = -\mathbf{G}\mathbf{x}$, the solution of the closed-loop system

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{G})\,\mathbf{x}$$

reads $\mathbf{x}(\tau) = \Psi_c(\tau, t)\mathbf{x}(t)$. The value function becomes

$$\begin{aligned} V(\mathbf{x}_t, t; \mathbf{G}) &= \frac{1}{2}\mathbf{x}_t^T \int_{\tau=t}^{T} \Phi_c^T(\tau, t)\left(\widetilde{\mathbf{Q}} - \mathbf{G}^T\mathbf{S}^T - \mathbf{S}\mathbf{G} + \mathbf{G}^T\widetilde{\mathbf{R}}\mathbf{G}\right)\Phi_c(\tau, t)\, d\tau\, \mathbf{x}_t + \frac{1}{2}\mathbf{x}_t^T \Phi_c^T(T, t)\mathbf{Q}_T\Phi_c(T, t)\mathbf{x}_t = \\ &= \frac{1}{2}\mathbf{x}_t^T \left\{ \int_{\tau=t}^{T} \Phi_c^T(\tau, t)\left(\widetilde{\mathbf{Q}} - \mathbf{G}^T\mathbf{S}^T - \mathbf{S}\mathbf{G} + \mathbf{G}^T\widetilde{\mathbf{R}}\mathbf{G}\right)\Phi_c(\tau, t)\, d\tau + \Phi_c^T(T, t)\mathbf{Q}_T\Phi_c(T, t) \right\} \mathbf{x}_t = \\ &= \frac{1}{2}\mathbf{x}_t^T \mathbf{P}(t)\mathbf{x}_t \, . \end{aligned}$$

Thus, the costate reads $\lambda = \nabla_{\mathbf{x}_t} V = \mathbf{P}\mathbf{x}$.

Matrix $\mathbf{P}$ satisfies a *Lyapunov equation*,

$$\dot{\mathbf{P}} = \mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} + \left(\widetilde{\mathbf{Q}} - \mathbf{G}^T\mathbf{S}^T - \mathbf{S}\mathbf{G} + \mathbf{G}^T\widetilde{\mathbf{R}}\mathbf{G}\right) \, ,$$

with final conditions

$$\mathbf{P}(T) = \mathbf{Q}_T \, ,$$

as it can be easily found by direct computation of the time derivative of $\mathbf{P}$.

Decoupled weights

Let $\mathbf{S} = \mathbf{0}$, then $\mathbf{u} = -\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T\lambda$, and thus the control is a linear combination of the co-state.

The transformation $\mathbf{v} := \mathbf{u} + \widetilde{\mathbf{R}}^{-1}\mathbf{S}^T\mathbf{x}$, make a coupled objective function for the original system a decoupeld objective function for a modified system.

todo

- Are Ψ matrices invertible?
- Properties of the Hamilton matrix…
- What's a conjugate point?
- …

Coupled state-input weights

$$J = \frac{1}{2}\int_{\tau=0}^{T} \begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix} \begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} d\tau + \frac{1}{2}\mathbf{x}_T^T \mathbf{Q}_T \mathbf{x}_T$$

Decoupled state-input weights

The problem can be decoupled with a transformation

$$\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} = \begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{T} & \mathbf{I}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}$$

so that

$$\begin{aligned}\begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix} \begin{bmatrix}\mathbf{x}\\ \mathbf{u}\end{bmatrix} &= \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{I} & \mathbf{T}^T\\ \mathbf{0} & \mathbf{I}\end{bmatrix} \begin{bmatrix}\mathbf{Q} & \mathbf{S}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix} \begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{T} & \mathbf{I}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix} = \\ &= \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q}+\mathbf{T}^T\mathbf{S}^T & \mathbf{S}+\mathbf{T}^T\mathbf{R}\\ \mathbf{S}^T & \mathbf{R}\end{bmatrix} \begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{T} & \mathbf{I}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix} = \\ &= \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q}+\mathbf{T}^T\mathbf{S}^T+\mathbf{S}\mathbf{T}+\mathbf{T}^T\mathbf{R}\mathbf{T} & \mathbf{S}+\mathbf{T}^T\mathbf{R}\\ \mathbf{S}^T+\mathbf{R}\mathbf{T} & \mathbf{R}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix},\end{aligned}$$

Extra-diagonal terms are identically zero if $\mathbf{T} = -\mathbf{R}^{-1}\mathbf{S}^T$. The coordinate transformation becomes

$$\begin{aligned}\mathbf{x} &= \tilde{\mathbf{x}}\\ \mathbf{u} &= -\mathbf{R}^{-1}\mathbf{S}^T\tilde{\mathbf{x}} + \tilde{\mathbf{u}}\end{aligned}$$

and the running cost reads

$$\begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\mathbf{Q}-\mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T & \mathbf{0}\\ \mathbf{0} & \mathbf{R}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}\\ \tilde{\mathbf{u}}\end{bmatrix}.$$

The linear system

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}\\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}\end{aligned}$$

becomes

$$\begin{aligned}\dot{\tilde{\mathbf{x}}} &= \left(\mathbf{A}-\mathbf{B}\mathbf{R}^{-1}\mathbf{S}^T\right)\tilde{\mathbf{x}} + \mathbf{B}\tilde{\mathbf{u}}\\ \mathbf{y} &= \left(\mathbf{C}-\mathbf{D}\mathbf{R}^{-1}\mathbf{S}^T\right)\tilde{\mathbf{x}} + \mathbf{D}\tilde{\mathbf{u}}\end{aligned}$$

Decoupled system

$$J = \frac{1}{2}\int_{\tau=0}^{T} \begin{bmatrix}\mathbf{x}\\ \tilde{\mathbf{u}}\end{bmatrix}^T \begin{bmatrix}\hat{\mathbf{Q}} & \mathbf{0}\\ \mathbf{0} & \mathbf{R}\end{bmatrix} \begin{bmatrix}\mathbf{x}\\ \tilde{\mathbf{u}}\end{bmatrix} d\tau + \frac{1}{2}\mathbf{x}_T^T\mathbf{Q}_T\mathbf{x}_T - \int_{\tau=0}^{t} \lambda^T\left(\dot{\mathbf{x}} - \hat{\mathbf{A}}\mathbf{x} + \mathbf{B}\tilde{\mathbf{u}}\right) d\tau.$$

Optimization gives

$$\begin{aligned}\delta\lambda(t):&\quad \dot{\mathbf{x}} = \hat{\mathbf{A}}\mathbf{x} + \mathbf{B}\tilde{\mathbf{u}}\\ &\quad \mathbf{x}(0) = \mathbf{x}_0\\ \delta\mathbf{x}(t):&\quad \dot{\lambda} = -\hat{\mathbf{A}}^T\lambda - \hat{\mathbf{Q}}\mathbf{x}\\ \delta\mathbf{x}(T):&\quad \lambda(T) = \mathbf{Q}_T\mathbf{x}(T)\\ \delta\mathbf{u}(t):&\quad \mathbf{0} = \mathbf{R}\tilde{\mathbf{u}} + \mathbf{B}^T\lambda\,.\end{aligned} \tag{47.7}$$

The modified optimal control reads

$$\tilde{\mathbf{u}} = -\mathbf{R}^{-1}\mathbf{B}^T\lambda$$

and thus

$$\mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right) .$$

$$\mathbf{x}(t) = \hat{\Phi}(t,t_0)\mathbf{x}_0 + \int_{\tau=t_0}^{t} \hat{\Phi}(t,\tau)\mathbf{B}_\tau\tilde{\mathbf{u}}(\tau)d\tau$$

$$\lambda_t = \partial_{\mathbf{x}_t} V = \left[\int_{\tau=t}^{T} \hat{\Phi}^T(\tau,t)\hat{\mathbf{Q}}_\tau\hat{\Phi}(\tau,t)\,d\tau + \hat{\Phi}^T(T,t)\mathbf{Q}_T\hat{\Phi}(T,t)\right]\mathbf{x}_t = \mathbf{P}(t,T)\mathbf{x}_t .$$

Lyapunov and Riccati equations

Control law in the uncoupled coordinates,

$$\begin{aligned}\tilde{\mathbf{u}} &= -\mathbf{R}^{-1}\mathbf{B}^T\lambda = \\ &= -\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}\mathbf{x} = \\ &= -\widetilde{\mathbf{K}}\mathbf{x} ,\end{aligned}$$

and

$$\begin{aligned}\mathbf{u} &= \tilde{\mathbf{u}} - \mathbf{R}^{-1}\mathbf{S}^T\mathbf{x} = \\ &= -\mathbf{R}^{-1}\left(\mathbf{B}^T\mathbf{P} + \mathbf{S}^T\right)\mathbf{x} = \\ &= -\mathbf{K}\mathbf{x} .\end{aligned}$$

so that $\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} = \mathbf{K} - \mathbf{R}^{-1}\mathbf{S}^T$.

Relation between the co-state an the state

$$\lambda = \mathbf{P}\mathbf{x} .$$

State and co-state dynamical equations

$$\begin{aligned}\dot{\mathbf{x}} &= \hat{\mathbf{A}}\mathbf{x} + \mathbf{B}\tilde{\mathbf{u}} \\ \dot{\lambda} &= -\hat{\mathbf{A}}^T\lambda - \hat{\mathbf{Q}}\mathbf{x}\end{aligned}$$

Riccati equation for λ.

$$-\hat{\mathbf{A}}^T\mathbf{P}\mathbf{x} - \hat{\mathbf{Q}}\mathbf{x} = \dot{\mathbf{P}}\mathbf{x} + \mathbf{P}\hat{\mathbf{A}}\mathbf{x} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}\mathbf{x} .$$

For arbitrary $\mathbf{x}$, Riccati equation follows

$$\begin{aligned}\mathbf{0} &= \dot{\mathbf{P}} + \mathbf{P}\hat{\mathbf{A}} + \hat{\mathbf{A}}^T\mathbf{P} + \hat{\mathbf{Q}} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} = \\ &= \dot{\mathbf{P}} + \mathbf{P}\left(\mathbf{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{S}^T\right) + \left(\mathbf{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{S}^T\right)^T\mathbf{P} + \left(\mathbf{Q} - \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T\right) - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} .\end{aligned}$$

Lyapunov equation for λ. Replacing $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}$,

$$\begin{aligned}\mathbf{0} &= \dot{\mathbf{P}} + \mathbf{P}\mathbf{A} - \mathbf{K}^T\mathbf{S}^T + \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T + \mathbf{A}^T\mathbf{P} - \mathbf{S}\mathbf{K} + \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T + \mathbf{Q} - \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T - \left(\mathbf{K} - \mathbf{R}^{-1}\mathbf{S}^T\right)^T\mathbf{R}\left(\mathbf{K} - \mathbf{R}^{-1}\mathbf{S}^T\right) = \\ &= \dot{\mathbf{P}} + \mathbf{P}\mathbf{A} + \mathbf{A}^T\mathbf{P} + \mathbf{Q} + \mathbf{S}\mathbf{R}^{-1}\mathbf{S} - \mathbf{S}\mathbf{K} - \mathbf{K}^T\mathbf{S}^T - \mathbf{K}^T\mathbf{R}\mathbf{K} + \mathbf{S}\mathbf{R}^{-1}\mathbf{R}\mathbf{K} + \mathbf{K}^T\mathbf{R}\mathbf{R}^{-1}\mathbf{S}^T - \mathbf{S}\mathbf{R}^{-1}\mathbf{R}\mathbf{R}^{-1}\mathbf{S}^T = \\ &= \dot{\mathbf{P}} + \mathbf{P}\mathbf{A} + \mathbf{A}^T\mathbf{P} + \mathbf{Q} - \mathbf{K}^T\mathbf{R}\mathbf{K} .\end{aligned}$$

...

$$\partial_t \hat{\Phi}(t,\tau) = \hat{\mathbf{A}}(t)\hat{\Phi}(t,\tau) \, .$$

47.1.3 Linear system without exogenous inputs - infinite time horizon

In the infinite-time horizon, under the assumption of stabilizability and observability (**todo** Prove it!), the optimal control law for a LTI system is a proportional control law,

$$\mathbf{u} = -\mathbf{K}\mathbf{x} = -\mathbf{R}^{-1}(\mathbf{B}^T\mathbf{P} + \mathbf{S}^T)\mathbf{x} \, ,$$

with the constant matrix $\mathbf{P}$ satisfying the **algebraic Riccation equation (ARE)**

$$\mathbf{0} = \mathbf{P}\hat{\mathbf{A}} + \hat{\mathbf{A}}^T\mathbf{P} + \hat{\mathbf{Q}} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} \, , \tag{47.8}$$

with $\hat{\mathbf{A}} = \mathbf{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{S}^T$ and $\hat{\mathbf{Q}} = \mathbf{Q} - \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T$.

Introducing the relation $\lambda(t) = \mathbf{P}\mathbf{x}(t)$ - with constant $\mathbf{P}$ - into the Hamiltonian system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\lambda} \end{bmatrix} \begin{bmatrix} \mathbf{A} - \mathbf{B}\widetilde{\mathbf{R}}^{-1}\mathbf{S}^T & -\mathbf{B}\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T \\ -\widetilde{\mathbf{Q}} + \mathbf{S}\widetilde{\mathbf{R}}^{-1}\mathbf{S}^T & -\mathbf{A}^T + \mathbf{S}\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \lambda \end{bmatrix} \, ,$$

it's possible to get 2 "decoupled" (the coupling lies in the relation $\lambda = \mathbf{P}\mathbf{x}$) equations for $\mathbf{x}$ and λ,

$$\begin{aligned} \dot{\mathbf{x}} &= \left(\hat{\mathbf{A}} - \mathbf{B}\widetilde{\mathbf{R}}^{-1}\mathbf{B}^T\mathbf{P}\right)\mathbf{x} \\ \dot{\lambda} &= -\left(\hat{\mathbf{A}}^T + \hat{\mathbf{Q}}\mathbf{P}^{-1}\right)\lambda \, . \end{aligned}$$

If the matrix $\mathbf{P}$ is chosen as the solution of the ARE (47.8), then the two equations have opposite eigenvalues, as

$$\begin{aligned} \hat{\mathbf{A}}^T + \hat{\mathbf{Q}}\mathbf{P}^{-1} &= \left(\hat{\mathbf{A}}^T\mathbf{P} + \hat{\mathbf{Q}}\right)\mathbf{P}^{-1} = \\ &= \left(-\mathbf{P}\hat{\mathbf{A}} + \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}\right)\mathbf{P}^{-1} = \\ &= -\mathbf{P}\left(\hat{\mathbf{A}} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}\right)\mathbf{P}^{-1} \, , \end{aligned}$$

the two matrices are related by a similarity transformation, through $\mathbf{P}$, with a minus sign.

Thus, the transition matrix of the state and the co-state in the closed-loop systems are dual, in the sense of

$$\Phi_{cl,x}(t,t_0) = \Phi^T_{cl,\lambda}(t_0,t) \, .$$

Lyapunov criterium for stability. Let $V(x) = \mathbf{x}^T\mathbf{P}\mathbf{x}$, its time derivative reads

$$\begin{aligned} \dot{V}(x(t)) &= \dot{\mathbf{x}}^T\mathbf{P}\mathbf{x} + \mathbf{x}^T\mathbf{P}\dot{\mathbf{x}} = \\ &= \mathbf{x}^T(\hat{\mathbf{A}} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P})^T\mathbf{P}\mathbf{x} + \mathbf{x}^T\mathbf{P}(\hat{\mathbf{A}} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P})\mathbf{x} = \\ &= \mathbf{x}^T(\hat{\mathbf{A}}^T\mathbf{P} + \mathbf{P}\hat{\mathbf{A}} - 2\mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P})\mathbf{x} = \\ &= -\mathbf{x}^T(\hat{\mathbf{Q}} + \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P})\mathbf{x} = \\ &\leq 0 \, , \end{aligned}$$

for all $\mathbf{x} \neq \mathbf{0}$, if the matrix $\hat{\mathbf{Q}} + \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}$ is a positive definite matrix.

Asymptotic stability and detectability of $(\hat{\mathbf{A}}, \hat{\mathbf{Q}}^{1/2})$

For asymptotic stability, we require the pair $(\hat{A}, \hat{Q}^{1/2})$ to be detectable (see *observability and detectability*. Here is the proof logic:

- define the null space: suppose there exists a trajectory $x(t)$ such that $\dot{V}(x(t)) = 0$ for all $t \geq 0$.
- analyze the terms: Since both $\hat{Q}$ and $PBR^{-1}B^TP$ are positive semi-definite, $\dot{V} = 0$ implies that $x^T\hat{Q}x = 0$ and $x^TPBR^{-1}B^TPx = 0$ simultaneously.
- vanishing control: If $x^TPBR^{-1}B^TPx = 0$, then the optimal control input $u = -R^{-1}B^TPx$ must be zero, as

$$0 = \mathbf{x}^T\mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{R}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}\mathbf{x} = \mathbf{u}^T\mathbf{R}\mathbf{u} \ ,$$

  forces $\mathbf{u} = \mathbf{0}$, as the weight matrix on the control must be positive definite, $\mathbf{R} > 0$ (update the condition for mixed weights, and/or weights on performance outputs with $\widetilde{\mathbf{R}}$).
- the residual dynamics: in this "unobservable" region where $\dot{V} = 0$, the dynamics simplify to $\dot{x} = \hat{A}x$.
- the constraint: for $\dot{V}$ to remain zero, we also require $\mathbf{x}^T\hat{\mathbf{Q}}\mathbf{x} = 0$, and with $\hat{\mathbf{Q}}^{1/2}$ as the *Choleski factor* of the semi-definite positive matrix, $\hat{\mathbf{Q}}^{1/2}\mathbf{x}(t) = 0$ for all t.

All the statements above can be then summarized as the condition of detectability of a linear system

$$\begin{cases} \dot{\mathbf{x}} = \hat{\mathbf{A}}\mathbf{x} \\ \mathbf{y} = \hat{\mathbf{Q}}^{1/2}\mathbf{x} \end{cases}$$

todo *check the following sentence*

If this system is detectable, then any state x that produces zero output ($y = 0$) must satisfy $\lim_{t\to\infty} x(t) = 0$. Since $y = 0$ is a requirement for $\dot{V} = 0$, we have proved that all trajectories must eventually vanish, achieving Global Asymptotic Stability.

Schur decoposition.

$$\mathbf{U}^T\mathbf{H}\mathbf{U} = \mathbf{T} = \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} \\ \mathbf{0} & \mathbf{T}_{22} \end{bmatrix}$$

47.2 Full-state feedback (OLD)

Optimal control can be recast as a **constrained optimization problem**, J, where an extreme - optimum - of an objective function must be found, subject to constraints that include the equations of motion. Some constraints may be included into an augmented objective function $\widetilde{J}$ with the methods of *Lagrange multipliers*.

Finite time vs. Infinite time horizon.

…

47.2.1 Generic ODE without exogenous inputs

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \ ,$$

with initial condition $\mathbf{x}(0) = \mathbf{x}_0$.

The objective function combines (weights) the error on a desired performance and the control input, in order to get the desired behavior with feasible control (that can be provided by actuators, without saturation, avoiding unnecessary high power input and too sharp behavior,…)

As an example, if the goal of the control $\mathbf{u}$ is to keep the system around $\mathbf{x} = \mathbf{0}$, the cost function to be minimized can be designed as

$$J = \int_{t=0}^{T} \frac{1}{2} \begin{bmatrix} \mathbf{x}^T & \mathbf{u}^T \end{bmatrix} \begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} dt + \frac{1}{2} \mathbf{x}^T(T) \mathbf{Q}_T \mathbf{x}(T) \ ,$$

with $\mathbf{Q} \geq 0$, and $\mathbf{R} > 0$ and symmetric.

Constrained optimization

$$\begin{aligned} \widetilde{J} = \frac{1}{2} \int_{t=0}^{T} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix}^T \begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} dt + \frac{1}{2} \mathbf{x}^T(T) \mathbf{Q}_T \mathbf{x}(T) - \int_{t=0}^{T} \lambda^T \left(\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}, \mathbf{u}) \right) \\ 0 = \delta \widetilde{J} = \\ = \int_{t=0}^{T} \delta \mathbf{x}^T \left(\mathbf{Q}\mathbf{x} + \mathbf{S}\mathbf{u} + \partial_{\mathbf{x}} \mathbf{f}^T \lambda + \mathbf{M}^T \dot{\lambda} \right) \\ + \int_{t=0}^{T} \delta \mathbf{u}^T \left(\mathbf{S}^T \mathbf{x} + \mathbf{R}\mathbf{u} + \partial_{\mathbf{u}} \mathbf{f}^T \lambda \right) \\ + \delta \mathbf{x}^T(T) \mathbf{Q}_T \mathbf{x}(T) \\ + \int_{t=0}^{T} \delta \lambda^T (\mathbf{M}\dot{\mathbf{x}} - \mathbf{f}(\mathbf{x}, \mathbf{u})) \\ - \delta \mathbf{x} \mathbf{M}^T \lambda \Big|_{t=0}^{T} \end{aligned}$$

So that

$$\begin{aligned} \delta\lambda(t): \quad & \mathbf{M}\dot{\mathbf{x}} = \mathbf{f} \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ \delta\mathbf{x}(t): \quad & \mathbf{M}^T \dot{\lambda} = -\partial_{\mathbf{x}} \mathbf{f}^T \lambda - \mathbf{S}\mathbf{u} - \mathbf{Q}\mathbf{x} \\ \delta\mathbf{x}(T): \quad & \mathbf{M}^T \lambda(T) = \mathbf{Q}_T \mathbf{x}(T) \\ \delta\mathbf{u}(t): \quad & \mathbf{u} = -\mathbf{R}^{-1} \left(\mathbf{S}^T \mathbf{x} + \partial_{\mathbf{u}} \mathbf{f}^T \lambda \right) \ . \end{aligned} \tag{47.9}$$

Solution of the problem

Replacing $\mathbf{u}$

$$\begin{bmatrix} \mathbf{M} & \\ & \mathbf{M}^T \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ -\partial_{\mathbf{x}} \mathbf{f}^T \lambda - \mathbf{S}\mathbf{u} - \mathbf{Q}\mathbf{x} \end{bmatrix} \ ,$$

with $\mathbf{u} = -\mathbf{R}^{-1} \left(\mathbf{S}^T \mathbf{x} + \partial_{\mathbf{u}} \mathbf{f}^T \lambda \right)$.

Gradient descent

- Start from a control law $\mathbf{u}^{(0)}(t)$,
- the state equation is integrated forward in time to get $\mathbf{x}^{(0)}(t)$,
- Lagrange multiplier equation is integrated backward in time to get $\lambda^{(0)}(t)$,
- The control law is updated with an increment $\delta\mathbf{u}$ that's proportional to the gradient of the cost function (with "opposite direction" to get $\delta\widetilde{J} < 0$), i.e. $\mathbf{u}^{(1)}(t) = \mathbf{u}^{(0)}(t) + \delta\mathbf{u}^{(0)}(t)$ with $\delta\mathbf{u}^{(0)}(t) = -c\nabla_{\mathbf{u}(t)} J^{(0)}$, with a positive step $c > 0$ and $\nabla_{\mathbf{u}} J = \mathbf{S}^T \mathbf{x} + \mathbf{R}\mathbf{u} + \partial_{\mathbf{u}} \mathbf{f}^T \lambda$.

- Repeat previous steps with the updated control law, until convergence.

47.2.2 LTI

$$J = \int_{t=0}^{T} \frac{1}{2} \begin{bmatrix} \mathbf{x}^T & \mathbf{u}^T \end{bmatrix} \begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}^T & \mathbf{R} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} dt + \frac{1}{2} \mathbf{x}^T(T) \mathbf{Q}_T \mathbf{x}(T)$$
$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases} \tag{47.10}$$

Constrained optimization

As $\mathbf{f}(\mathbf{x}, \mathbf{u}) = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$, here $\partial_{\mathbf{x}}\mathbf{f} = \mathbf{A}$ and $\partial_{\mathbf{u}}\mathbf{f} = \mathbf{B}$. For the linear system (47.10) $\mathbf{M} = \mathbf{I}$. Thus the equations (47.9) become

$$\begin{aligned} \delta\lambda(t): \quad & \mathbf{M}\dot{\mathbf{x}} = \mathbf{f} \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ \delta\mathbf{x}(t): \quad & \mathbf{M}^T\dot{\lambda} = -\mathbf{A}^T\lambda - \mathbf{S}\mathbf{u} - \mathbf{Q}\mathbf{x} \\ \delta\mathbf{x}(T): \quad & \mathbf{M}^T\lambda(T) = \mathbf{Q}_T\mathbf{x}(T) \\ \delta\mathbf{u}(t): \quad & \mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right) . \end{aligned} \tag{47.11}$$

Infinite-horizon full-state feedback

No need for an observer. The system is assumed to be stable. The augmented cost function reads

$$\widetilde{J}(\mathbf{x}, \mathbf{u}; \lambda) = \int_{t=0}^{+\infty} \left\{ \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} + \mathbf{x}^T\mathbf{S}\mathbf{u} + \frac{1}{2}\mathbf{u}^T\mathbf{R}\mathbf{u} - \lambda^T(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} - \mathbf{B}\mathbf{u}) \right\} dt ,$$

with given initial conditions $\mathbf{x}(0) = \mathbf{x}_0$, so that $\delta\mathbf{x}_0 = \mathbf{0}$. Using *calculus of variations*, the variations of the cost function w.r.t. $\mathbf{x}$, $\mathbf{u}$, λ read

$$\begin{aligned} \delta_{\mathbf{x}}: \quad & \mathbf{Q}\mathbf{x} + \mathbf{S}\mathbf{u} + \dot{\lambda} + \mathbf{A}^T\lambda = \mathbf{0} \\ \delta_{\mathbf{u}}: \quad & \mathbf{S}^T\mathbf{x} + \mathbf{R}\mathbf{u} + \mathbf{B}^T\lambda = \mathbf{0} \\ \delta_{\lambda}: \quad & \dot{\mathbf{x}} - \mathbf{A}\mathbf{x} - \mathbf{B}\mathbf{u} = \mathbf{0} \end{aligned}$$

From the variation w.r.t. $\mathbf{u}$, since $\mathbf{R} > 0$ and thus innvertible,

$$\mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T\mathbf{x} + \mathbf{B}^T\lambda\right)$$

Now, assuming the relation $\lambda = \mathbf{P}\mathbf{x}$, it follows

$$\begin{aligned} \dot{\lambda} &= -\mathbf{Q}\mathbf{x} - \mathbf{S}\mathbf{u} - \mathbf{A}^T\lambda = \\ &= \left\{-\mathbf{Q} + \mathbf{S}\mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right) - \mathbf{A}^T\mathbf{P}\right\}\mathbf{x} \\ \dot{\lambda} &= \dot{\mathbf{P}}\mathbf{x} + \mathbf{P}\dot{\mathbf{x}} = \\ &= \dot{\mathbf{P}}\mathbf{x} + \mathbf{P}\left(\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}\right) = \\ &= \left\{\dot{\mathbf{P}} + \mathbf{P}\left[\mathbf{A} - \mathbf{B}\mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right)\right]\right\}\mathbf{x} , \end{aligned}$$

and comparing the two different expressions of $\dot{\lambda}$, if the equality holds for any $\mathbf{x}$, the **dynamical Riccati equation** for $\mathbf{P}$ is derived as

$$\dot{\mathbf{P}} + \mathbf{P}\widetilde{\mathbf{A}} + \widetilde{\mathbf{A}}^T\mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} + \widetilde{\mathbf{Q}} = \mathbf{0} ,$$

where $\widetilde{\mathbf{A}} = \mathbf{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{S}^T$ and $\widetilde{\mathbf{Q}} = \mathbf{Q} - \mathbf{S}\mathbf{R}^{-1}\mathbf{S}^T$. Riccati equation is a non-linear dynamical matrix equation in $\mathbf{P}$. Algorithms for computing the solution of dynamical and algebraic equation exists, see *Example 47.2.1*.

Once $\mathbf{P}$ is evaluated, the control law reads

$$\mathbf{u} = -\mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right)\mathbf{x}\,.$$

For infinite-horizone, the algebraic equation (ARE) for the steady state is solved after setting $\dot{\mathbf{P}} = \mathbf{0}$, the solution for a LTI system is a constant matrix $\mathbf{P}$, and thus the control law is a proportional feedback on the full-state of the system,

$$\mathbf{u} = -\mathbf{G}\mathbf{x}\,,$$

with $\mathbf{G} = \mathbf{R}^{-1}\left(\mathbf{S}^T + \mathbf{B}^T\mathbf{P}\right)$.

Example 47.2.1 (Solution of Riccati equation)

…

Properties. todo

- $\mathbf{P}$ symmetric? definite positive? …
- …

47.3 Phase and gain margin of optimal control

Some results are available for phase and gain margin of optimal control, on the input side. **todo** *discuss* The relationship between the actual input to the plant u and an additive input error Δu reads

$$u = \Delta u + u_c = \Delta u + Ke = \Delta u + K(x_{ref} - x) = \Delta u + Kx_{ref} + KHu\,,$$

$$u = \frac{1}{1 + KH(s)}\Delta u(s) + \frac{K}{1 + KH(s)}x_{ref}(s)\,, \tag{47.12}$$

for a SISO or

$$\mathbf{u} = \left[\mathbf{I} + \mathbf{K}(s)\mathbf{H}(s)\right]^{-1}\Delta\mathbf{u} + \left[\mathbf{I} + \mathbf{K}(s)\mathbf{H}(s)\right]^{-1}\mathbf{K}(s)\mathbf{x}_{ref}\,, \tag{47.13}$$

for a MIMO system. The additive input error Δu can be thought as the difference between the ideal/design condition, due to un-modelled parts, failures, and any other form of discrepancy to the model altering the gain and the phase (adding delay) of the ideal control u_c.

Details

For a SISO system, the relation between u and Δu reads

$$u = \Delta u + u_c = \Delta u + Ke = \Delta u + K(x_{ref} - x) = \Delta u + Kx_{ref} + KHu\,,$$

and thus

$$u(s) = \frac{1}{1 + KH(s)}\Delta u(s) + \frac{K}{1 + KH(s)}x_{ref}(s)\,.$$

For a MIMO systems, TFs are matrix TFs and thus matrix inversion is required (as the division between matrix makes no sense at all),

$$\mathbf{u}(s) = \left[\mathbf{I} + \mathbf{KH}(s)\right]^{-1}\Delta\mathbf{u}(s) + \left[\mathbf{I} + \mathbf{KH}(s)\right]^{-1}\mathbf{K}\mathbf{x}_{ref}(s)\,.$$

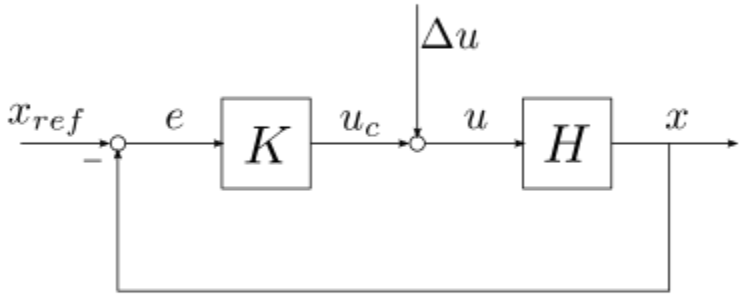


Fig. 47.1: Block diagram of optimal control closed-loop system with retroaction on the error $e = x_{ref} - x$ on the full state.

47.3.1 SISO

$$|1 + S(j\omega)|^2 \geq 1 \, .$$

Details

Starting from the algebraic Lyapunov equation for the matrix $\mathbf{P}$

$$\mathbf{0} = \mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{K}\mathbf{R}\mathbf{K}^T + \mathbf{Q} \, ,$$

with the optimal matrix of gains, $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}$. Adding and subtracting $s\mathbf{IP}$, and defining the matrix $\mathbf{G}(s) = (s\mathbf{I} - \mathbf{A})^{-1}$,

$$\begin{aligned}\mathbf{0} &= \mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{K}^T\mathbf{R}\mathbf{K} + \mathbf{Q} = \\ &= (\mathbf{A} + s\mathbf{I})^T\mathbf{P} + \mathbf{P}(\mathbf{A} - s\mathbf{I}) - \mathbf{K}^T\mathbf{R}\mathbf{K} + \mathbf{Q} = \\ &= \mathbf{G}^{-T}(-s)\mathbf{P} + \mathbf{P}\mathbf{G}^{-1}(s) - \mathbf{K}^T\mathbf{R}\mathbf{K} + \mathbf{Q} \, .\end{aligned}$$

Pre-multiplication by $\mathbf{B}^T\mathbf{G}^T(-s)$ and post-multiplication by $\mathbf{G}(s)\mathbf{B}$ gives

$$\begin{aligned}\mathbf{0} &= -\mathbf{B}^T\mathbf{P}\mathbf{G}(s)\mathbf{B} - \mathbf{B}^T\mathbf{G}^T(-s)\mathbf{P}\mathbf{B} - \mathbf{B}^T\mathbf{G}^T(-s)\mathbf{K}^T\mathbf{R}\mathbf{K}\mathbf{G}(s)\mathbf{B} + \mathbf{B}^T\mathbf{G}^T(-s)\mathbf{Q}\mathbf{G}(s)\mathbf{B} = \\ &= -\mathbf{R}\mathbf{K}\mathbf{G}(s)\mathbf{B} - \mathbf{B}^T\mathbf{G}^T(-s)\mathbf{K}^T\mathbf{R} - \mathbf{B}^T\mathbf{G}^T(-s)\mathbf{K}^T\mathbf{R}\mathbf{K}\mathbf{G}(s)\mathbf{B} + \mathbf{B}^T\mathbf{G}^T(-s)\mathbf{Q}\mathbf{G}(s)\mathbf{B} = \\ &= -\mathbf{R}\mathbf{K}\mathbf{H}(s) - \mathbf{H}^T(-s)\mathbf{K}^T\mathbf{R} - \mathbf{H}^T(-s)\mathbf{K}^T\mathbf{R}\mathbf{K}\mathbf{H}(s) + \mathbf{H}^T(-s)\mathbf{Q}\mathbf{H}(s) \, ,\end{aligned}$$

having used the expression of the optimal gain matrix to write $\mathbf{B}^T\mathbf{P} = \mathbf{R}\mathbf{K}$, and defined the open-loop *input* TF matrix $\mathbf{S}(s) := \mathbf{K}\mathbf{H}(s)$. Moving terms with or without $\mathbf{S}(s)$ on two different sides of the equality, and adding \mathbf{R}, and

collecting terms, it follows

$$\left[\mathbf{I}+\mathbf{S}^T(-s)\right]\mathbf{R}\left[\mathbf{I}+\mathbf{S}(s)\right]=\mathbf{R}+\mathbf{H}^T(-s)\mathbf{Q}\mathbf{H}(s)\ .$$

Moving to Fourier domain, $s=j\omega$, and recalling that $\mathbf{S}^T(-j\omega)=\mathbf{S}^*(j\omega)$,

$$\left[\mathbf{I}+\mathbf{S}^*(j\omega)\right]\mathbf{R}\left[\mathbf{I}+\mathbf{S}(j\omega)\right]=\mathbf{R}+\mathbf{H}^*(j\omega)\mathbf{Q}\mathbf{H}(j\omega)\ .$$

todo *Notation for semi-definite positive symmetric matrices*

$$\left[\mathbf{I}+\mathbf{S}^*(j\omega)\right]\mathbf{R}\left[\mathbf{I}+\mathbf{S}(j\omega)\right]\geq\mathbf{R}\ .$$

For a 1-dimensional state SISO system, $\mathbf{R}=r$, $\mathbf{S}(s)=S(s)$, and thus

$$\left|1+S(j\omega)\right|^2\geq 1\ .$$

47.3.2 MIMO

$$\left[\mathbf{I}+\mathbf{S}^*(j\omega)\right]\mathbf{R}\left[\mathbf{I}+\mathbf{S}(j\omega)\right]\geq\mathbf{R}\ .$$

with $\mathbf{H}(s)=(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}$, and $\mathbf{S}(s)=\mathbf{K}\mathbf{H}(s)$ the open-loop TF between $\Delta\mathbf{u}$ and the ideal control loop $\mathbf{u}_c$, and $\mathbf{I}+\mathbf{S}(s)$ the (inverse of the) transfer function between $\Delta\mathbf{u}$ and $\mathbf{u}$, as shown in (47.13).

47.4 Optimal observer for stochastic disturbance

In this section, all the machinery is developed for white-noise process and measurement disturbances. Non-white noise disturbances can be obtained as the output of shape filters with white noise inputs, as shown with the *augmented system approach*.

From the equations of the plant

$$\begin{cases}\dot{\mathbf{x}}=\mathbf{A}\mathbf{x}+\mathbf{B}\mathbf{u}+\mathbf{B}_d\mathbf{d}\\ \mathbf{y}=\mathbf{C}\mathbf{x}+\mathbf{D}\mathbf{u}+\mathbf{D}_d\mathbf{d}+\mathbf{D}_r\mathbf{r}\end{cases}$$

and the observer

$$\begin{cases}\dot{\mathbf{o}}=\mathbf{A}\mathbf{o}+\mathbf{B}\mathbf{u}+\mathbf{L}\left(\mathbf{y}-\mathbf{y}_o\right)\\ \mathbf{y}_o=\mathbf{C}\mathbf{o}+\mathbf{D}\mathbf{u}\end{cases}$$

the dynamical equation of the error, $\mathbf{e}=\mathbf{o}-\mathbf{x}$, follows

$$\begin{aligned}\dot{\mathbf{e}}&=\dot{\mathbf{o}}-\dot{\mathbf{x}}=\\&=\mathbf{A}\mathbf{o}+\mathbf{B}\mathbf{u}+\mathbf{L}\left[\mathbf{C}\mathbf{x}+\mathbf{D}\mathbf{u}+\mathbf{D}_d\mathbf{d}+\mathbf{D}_r\mathbf{r}-\mathbf{C}\mathbf{o}-\mathbf{D}\mathbf{u}\right]-\mathbf{A}\mathbf{x}-\mathbf{B}\mathbf{u}-\mathbf{B}_d\mathbf{d}=\\&=\left(\mathbf{A}-\mathbf{L}\mathbf{C}\right)\mathbf{e}+\left(\mathbf{L}\mathbf{D}_d-\mathbf{B}_d\right)\mathbf{d}+\mathbf{L}\mathbf{D}_r\mathbf{r}=\\&=\overline{\mathbf{A}}\mathbf{e}+\begin{bmatrix}\overline{\mathbf{B}}_d & \overline{\mathbf{B}}_r\end{bmatrix}\begin{bmatrix}\mathbf{d}\\ \mathbf{r}\end{bmatrix}=\\&=\overline{\mathbf{A}}\mathbf{e}+\overline{\mathbf{B}}\begin{bmatrix}\mathbf{d}\\ \mathbf{r}\end{bmatrix}\ .\end{aligned}$$

Let the distrubances be a white noise, then the *Lyapunov equation for the autocorrelation matrix of the state* reads

$$\dot{\mathbf{R}}_{\mathbf{ee}}=\overline{\mathbf{A}}(\mathbf{L})\mathbf{R}_{\mathbf{ee}}+\mathbf{R}_{\mathbf{ee}}\overline{\mathbf{A}}^T(\mathbf{L})+\overline{\mathbf{B}}(\mathbf{L})\mathbf{W}\overline{\mathbf{B}}^T(\mathbf{L})\ ,$$

with $\mathbf{W}$ the matrix of the cross-correlations of the white noise distrubances,

$$\mathbf{W} = \begin{bmatrix} \mathbf{W_{dd}} & \mathbf{W_{dr}} \\ \mathbf{W_{rd}} & \mathbf{W_{rr}} \end{bmatrix} .$$

If stationary conditions are reached, the autocorrelation matrix of the error must satisfy the algebraic version of the Lyapunov equation

$$\mathbf{0} = \overline{\mathbf{A}}(\mathbf{L})\mathbf{R_{ee}} + \mathbf{R_{ee}}\overline{\mathbf{A}}^T(\mathbf{L}) + \overline{\mathbf{B}}(\mathbf{L})\mathbf{W}\overline{\mathbf{B}}^T(\mathbf{L}) . \tag{47.14}$$

Optimal observer. The optimal observer is defined as the observer that makes the estimation error minimum. As an example, the error can be measured as the weighted trace of the auto-correlation of the error. The constrained optimization problem reads

$$\min_{\mathbf{L}} \operatorname{Tr}\left(\mathbf{W_e}\mathbf{R_{ee}}\right) ,$$

subject to Lyapunov equation as a constraint.

Details of the constrained optimization

The optimal solution is found, setting equal to zero the derivatives of the objective function

$$\widetilde{J} = \operatorname{Tr}\left(\mathbf{W_e}\mathbf{R_{ee}} + \Lambda\left\{(\mathbf{A}-\mathbf{LC})\mathbf{R} + \mathbf{R}(\mathbf{A}-\mathbf{LC})^T + \begin{bmatrix}\mathbf{LD}_d - \mathbf{B}_d & \mathbf{LD}_r\end{bmatrix}\mathbf{W}\begin{bmatrix}\mathbf{D}_d^T\mathbf{L}^T - \mathbf{B}_d^T \\ \mathbf{D}_r^T\mathbf{L}^T\end{bmatrix}\right\}\right) ,$$

w.r.t. the matrices $\mathbf{L}$, Λ, $\mathbf{R}$

$$\begin{aligned}
\mathbf{0} &= \frac{\partial\widetilde{J}}{\partial\mathbf{R}} = \mathbf{W_e}^T + (\mathbf{A}-\mathbf{LC})^T\Lambda^T + (\mathbf{A}-\mathbf{LC})\Lambda^T \\
\mathbf{0} &= \frac{\partial\widetilde{J}}{\partial\Lambda} = \text{algebraic Lyapunov equation for } \mathbf{R_{ee}} \\
\mathbf{0} &= \frac{\partial\widetilde{J}}{\partial\mathbf{L}} = -(\mathbf{CR}\Lambda)^T - \Lambda\mathbf{RC}^T + \\
&\quad + \Lambda^T\mathbf{L}\left(\mathbf{D}_d\mathbf{W}_{dd}\mathbf{D}_d^T\right)^T + \Lambda\mathbf{L}\left(\mathbf{D}_d\mathbf{W}_{dd}\mathbf{D}_d^T\right) + \\
&\quad + \Lambda^T\mathbf{L}\left(\mathbf{D}_r\mathbf{W}_{rr}\mathbf{D}_r^T\right)^T + \Lambda\mathbf{L}\left(\mathbf{D}_r\mathbf{W}_{rr}\mathbf{D}_r^T\right) + \\
&\quad + \Lambda^T\mathbf{L}\left(\mathbf{D}_r\mathbf{W}_{rd}\mathbf{D}_d^T\right)^T + \Lambda\mathbf{L}\left(\mathbf{D}_r\mathbf{W}_{rd}\mathbf{D}_d^T\right) + \\
&\quad + \Lambda^T\mathbf{L}\left(\mathbf{D}_d\mathbf{W}_{dr}\mathbf{D}_r^T\right)^T + \Lambda\mathbf{L}\left(\mathbf{D}_d\mathbf{W}_{dr}\mathbf{D}_r^T\right) + \\
&\quad - \Lambda\mathbf{B}_d\mathbf{W}_{dd}\mathbf{D}_d^T - \Lambda^T\mathbf{B}_d\mathbf{W}_{dd}\mathbf{D}_d^T + \\
&\quad - \Lambda\mathbf{B}_d\mathbf{W}_{dr}\mathbf{D}_r^T - \Lambda\mathbf{B}_d\mathbf{W}_{dr}\mathbf{D}_r^T
\end{aligned}$$

Collecing the symmetric (**todo** why this assumption?) matrix Λ (is it possible to collect this matrix and set the other factor to zero?)

$$\mathbf{L} = \left[\mathbf{R_{ee}}\mathbf{C}^T + \mathbf{B}_d\left(\mathbf{W}_{dd}\mathbf{D}_d^T + \mathbf{W}_{dr}\mathbf{D}_r^T\right)\right]\left[\mathbf{D}_d\mathbf{W}_{dd}\mathbf{D}_d^T + \mathbf{D}_d\mathbf{W}_{dr}\mathbf{D}_r^T + \mathbf{D}_r\mathbf{W}_{rd}\mathbf{D}_d^T + \mathbf{D}_r\mathbf{W}_{rr}\mathbf{D}_r^T\right]^{-1} , \tag{47.15}$$

Replacing the expression (47.15) of the $\mathbf{L}$ matrix into the algebraic Lyapunov equation (47.14) for the autocorrelation $\mathbf{R_{ee}}$, a Riccati equation in $\mathbf{R_{ee}}$ appears.

Derivatives w.r.t. a matrix

$$\frac{\partial \mathrm{Tr}(\mathbf{AB})}{\partial \mathbf{A}} = \frac{\partial \mathrm{Tr}(\mathbf{AB})}{\partial A_{ij}} = \frac{\partial (A_{ab}B_{ba})}{\partial A_{ij}} = \delta_{ai}\delta_{bj}B_{ba} = B_{ji} = \mathbf{B}^T$$

$$\frac{\partial \mathrm{Tr}(\mathbf{AB})}{\partial \mathbf{B}} = \frac{\partial \mathrm{Tr}(\mathbf{AB})}{\partial B_{ij}} = \frac{\partial (A_{ab}B_{ba})}{\partial B_{ij}} = A_{ab}\delta_{ib}\delta_{aj} = A_{ji} = \mathbf{A}^T$$

$$\frac{\partial \mathrm{Tr}(\mathbf{AB}^T)}{\partial \mathbf{B}} = \frac{\partial \mathrm{Tr}(\mathbf{AB})}{\partial B_{ij}} = \frac{\partial (A_{ab}B_{ab})}{\partial B_{ij}} = A_{ab}\delta_{ia}\delta_{bj} = A_{ij} = \mathbf{A}$$

$$\frac{\partial \mathrm{Tr}(\mathbf{ABA}^T)}{\partial \mathbf{A}} = \frac{\partial \mathrm{Tr}(\mathbf{ABA}^T)}{\partial A_{ij}} = \frac{\partial (A_{ab}B_{bc}A_{ac})}{\partial A_{ij}} = B_{jc}A_{ic} + A_{ib}B_{bj} = \mathbf{AB}^T + \mathbf{AB}\ .$$

$$\frac{\partial \mathrm{Tr}(\mathbf{ABC})}{\partial \mathbf{B}} = \frac{\partial \mathrm{Tr}(\mathbf{ABC})}{\partial B_{ij}} = \frac{\partial (A_{ab}B_{bc}C_{ca})}{\partial B_{ij}} = A_{ai}C_{ja} = \mathbf{A}^T\mathbf{C}^T\ .$$

$$\frac{\partial \mathrm{Tr}(\mathbf{CABA}^T)}{\partial \mathbf{A}} = \frac{\partial \mathrm{Tr}(\mathbf{CABA}^T)}{\partial A_{ij}} = \frac{\partial (C_{ab}A_{bc}B_{cd}A_{ad})}{\partial A_{ij}} = C_{ai}B_{jd}A_{ad} + C_{ib}A_{bc}B_{cj} = \mathbf{C}^T\mathbf{AB}^T + \mathbf{CAB}\ .$$

$$\mathrm{Tr}(\mathbf{AB}) = A_{ij}B_{ji} = \mathrm{Tr}(\mathbf{BA})$$

47.5 Optimal observer for deterministic disturbances

See next section about *optimal observer for stochastic disturbances*, or **Kalman filter**.

47.6 Sub-optimal control for output feedback

47.7 Combination of controller and observer

Full-state feed-back or the observed state builds the controller on the observed state. As an example, an optimal full-state controller is designed as the state was available, to find matrix $\mathbf{G}$; a state observer is designed to get the observed state $\mathbf{o}(t)$, through the design of a matrix $\mathbf{L}$ as in *Kalman filter*; the feed-back on the observed state produces the control $\mathbf{u}(t) = -\mathbf{G}\mathbf{o}(t)$.

Two different cases are discussed here:

1. the whole state is used to design the control law, as it's the case of small-systems with few states/degrees of freedom. In this situation, a **separation principle** holds, i.e. the state estimator design and the control design can be performed independently, as the eigenvalues of the closed-loop system are the eigenvalues of the closed-loop matrix $\mathbf{A} - \mathbf{BG}$ and the observer $\mathbf{A} - \mathbf{LC}$
2. a reduced state is used to design the control law, as it's the case of large-dimensional systems or continuous systems (as deformable bodies, whose nomber of degrees of freedom is infinite). For these systems, a subset of states is used for control design, in order to reduce the computational burden. In this situation, separation principle doesn't hold and **spill-over** may occur between states.

47.7.1 Separation principle

System

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_u\mathbf{u} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}_u\mathbf{u} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r} \end{cases}$$

Observer

$$\begin{cases} \dot{\mathbf{o}} = \mathbf{A}\mathbf{o} + \mathbf{B}_u\mathbf{u} + \mathbf{L}(\mathbf{y} - \mathbf{y}_o) \\ \mathbf{y}_o = \mathbf{C}\mathbf{o} + \mathbf{D}_u\mathbf{u} \,. \end{cases}$$

with the error dynamics, $\mathbf{e} := \mathbf{x} - \mathbf{o}$

$$\dot{\mathbf{e}} = (\mathbf{A} - \mathbf{L}\mathbf{C})\,\mathbf{e} + (\mathbf{B}_d - \mathbf{L}\mathbf{D}_d)\,\mathbf{d} - \mathbf{L}\mathbf{D}_r\mathbf{r}$$

and with the control law $\mathbf{u} = -\mathbf{G}\mathbf{o} = -\mathbf{G}\,(\mathbf{x} - \mathbf{e})$. The augmented system with state $\{\mathbf{x}, \mathbf{e}\}$ reads

$$\frac{d}{dt}\begin{bmatrix}\mathbf{x}\\ \mathbf{e}\end{bmatrix} = \begin{bmatrix}\mathbf{A} - \mathbf{B}_u\mathbf{G} & \mathbf{B}_u\mathbf{G} \\ \mathbf{0} & \mathbf{A} - \mathbf{L}\mathbf{C}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{e}\end{bmatrix} + \begin{bmatrix}\mathbf{B}_d & \mathbf{0} \\ \mathbf{B}_d - \mathbf{L}\mathbf{D}_d & -\mathbf{L}\mathbf{D}_r\end{bmatrix}\begin{bmatrix}\mathbf{d}\\ \mathbf{r}\end{bmatrix} \,.$$

The matrix of the system is a block triangular matrix, and thus its eigenvalues are the union of the eigenvalues of the diagonal blocks, i.e. the eigenvalues of the closed-loop matrix $\mathbf{A} - \mathbf{B}_u\mathbf{G}$, and the matrix of the observer $\mathbf{A} - \mathbf{L}\mathbf{C}$. Thus, for the modal stability of the system, the observer and the ccontroller can be designed independently.

47.7.2 Spillover

Design reduced system and full system

Let $\mathbf{x}_1$ the state of the reduced system used for the design, and $\mathbf{x}_2$ the "residual" state of the large-dimensional system used for verification, or as a detailed mathematical model of the system of interest. Let the dynamical system be

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{A}_1\mathbf{x}_1 + \mathbf{B}_{1u}\mathbf{u} \\ \dot{\mathbf{x}}_2 = \mathbf{A}_2\mathbf{x}_2 + \mathbf{B}_{2u}\mathbf{u} \\ \mathbf{y} = \mathbf{C}_1\mathbf{x}_1 + \mathbf{C}_2\mathbf{x}_2 + \mathbf{D}_u\mathbf{u} \end{cases}$$

and the optimal control designed on the reduced system, so that the input is $\mathbf{u} = -\mathbf{G}\mathbf{x}_1$. The equations governing the closed-loop dynamics of the detailed system reads

$$\frac{d}{dt}\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix} = \begin{bmatrix}\mathbf{A}_1 - \mathbf{B}_{1u}\mathbf{G} & \cdot \\ -\mathbf{B}_{2u}\mathbf{G} & \mathbf{A}_2\end{bmatrix}\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix}\mathbf{C}_1 - \mathbf{D}_u\mathbf{G} & \mathbf{C}_2\end{bmatrix}\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}$$

The matrix of the system is block triangular, so that the eigenvalues of the system are the union of the eigenvalues of the closed loop matrix designed on the reduced system $\mathbf{A}_1 - \mathbf{B}_{1u}\mathbf{G}$, and the matrix of the residual detailed system $\mathbf{A}_2$. In order to get a stable system, the reduced system must contain all the open-loop unstable dynamics, as the matrix $\mathbf{A}_2$ must be stable.

Design reduced system, observer and full system

Let $\mathbf{x}_1$ the state of the reduced system used for the design, and $\mathbf{x}_2$ the "residual" state of the large-dimensional system used for verification, or as a detailed mathematical model of the system of interest. Let the dynamical system be

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{A}_1\mathbf{x}_1 + \mathbf{B}_{1u}\mathbf{u} \\ \dot{\mathbf{x}}_2 = \mathbf{A}_2\mathbf{x}_2 + \mathbf{B}_{2u}\mathbf{u} \\ \mathbf{y} = \mathbf{C}_1\mathbf{x}_1 + \mathbf{C}_2\mathbf{x}_2 + \mathbf{D}_u\mathbf{u} \end{cases}$$

and the observer designed on the low-dimensional system

$$\begin{cases} \dot{\mathbf{o}} = \mathbf{A}_1\mathbf{o} + \mathbf{B}_{1u}\mathbf{u} + \mathbf{L}\left(\mathbf{y} - \mathbf{y}_o\right) \\ \mathbf{y}_o = \mathbf{C}_1\mathbf{o} + \mathbf{D}_{1u}\mathbf{u} \end{cases}$$

so that

$$\dot{\mathbf{o}} = \left(\mathbf{A}_1 - \mathbf{L}\mathbf{C}_1\right)\mathbf{o} + \mathbf{L}\mathbf{C}_1\mathbf{x}_1 + \mathbf{L}\mathbf{C}_2\mathbf{x}_2 + \left(\mathbf{B}_{1u} - \mathbf{L}\mathbf{D}_u\right)\mathbf{u} \,,$$

and the dynamical equation of the error $\mathbf{e} := \mathbf{x}_1 - \mathbf{o}$ is

$$\dot{\mathbf{e}} = \left(\mathbf{A}_1 - \mathbf{L}\mathbf{C}_1\right)\mathbf{e} - \mathbf{L}\mathbf{C}_2\mathbf{x}_2 \,.$$

The input is proportional to the observed state, $\mathbf{u} = -\mathbf{G}\mathbf{o}$. Thus, the augmented system reads

$$\frac{d}{dt}\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{e} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 - \mathbf{B}_{1u}\mathbf{G} & \cdot & \mathbf{B}_{1u}\mathbf{G} \\ -\mathbf{B}_{2u}\mathbf{G} & \mathbf{A}_2 & \mathbf{B}_{2u}\mathbf{G} \\ \cdot & -\mathbf{L}\mathbf{C}_2 & \mathbf{A}_1 - \mathbf{L}\mathbf{C}_1 \end{bmatrix}\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{e} \end{bmatrix} \,.$$

The matrix of the system is not block triangular, as the dynamics of the "residual" state of the large-dimensional system and the reconstruction error are coupled. The eigenvalue of the systems are non-trivial as they're not anymore the eigenvalue of the diagonal blocks.

47.8 Optimal control for reference tracking

In order to get perfect asymptotic tracking of polynomial reference signals with degree m with optimal control, the original system must be **augmented** with the dynamical equations of $m + 1$ **integral errors**. This process is shown for *tracking step reference* with the augmentation of the original system with a single integrator and for *tracking ramp reference* with the augmentation with a double integrator.

This procedure is somehow equivalent to the **addition of integrators in open-loop transfer function**, as discussed in the section about *reference tracking as a performance of a closed-loop system*.

An example about optimal control for reference tracking in a mass-damper-spring system is shown in the *next section*, as a **Jupyter notebook** that can be opened, run, and modified in **Colab**.

47.8.1 Augmented system for tracking step reference signal - integrator

Let $\mathbf{y}_{\text{ref}}$ a reference signal. An augmented system can be defined in order to use optimal control for tracking the desired reference signal.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

the state-space representation of the plant. Let $\mathbf{y}_{ref}$ a desired output and the integral error

$$\mathbf{e}(t) := \int_{\tau=0}^{t} \left\{ \mathbf{y}(\tau) - \mathbf{y}_{ref}(\tau) \right\} d\tau \ ,$$

as a new state with dynamical equation

$$\dot{\mathbf{e}} = \mathbf{y}(t) - \mathbf{y}_{ref}(t) = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{ref} \ .$$

The optimal control is applied to the augmented system

$$\underbrace{\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{e}} \end{bmatrix}}_{\dot{\mathbf{z}}} = \underbrace{\begin{bmatrix} \mathbf{A} & \cdot \\ \mathbf{C} & \cdot \end{bmatrix}}_{\hat{\mathbf{A}}} \underbrace{\begin{bmatrix} \mathbf{x} \\ \mathbf{e} \end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix} \mathbf{B} \\ \mathbf{D} \end{bmatrix}}_{\hat{\mathbf{B}}_u} \mathbf{u} + \underbrace{\begin{bmatrix} \cdot \\ -\mathbf{I} \end{bmatrix}}_{\hat{\mathbf{B}}_{ref}} \mathbf{y}_{ref}$$

Optimal control framework provides the opitmal gain matrix $\hat{\mathbf{K}}$, so that $\mathbf{u} = -\hat{\mathbf{K}}\mathbf{z}$ and the closed loop system becomes

$$\dot{\mathbf{z}} = \left(\hat{\mathbf{A}} - \hat{\mathbf{B}}_u \hat{\mathbf{K}} \right) \mathbf{z} + \hat{\mathbf{B}}_{ref} \mathbf{y}_{ref} \ .$$

$\lim_{t \to +\infty} e(t)$

Transfer function. From the reference signal to the tracking error,

$$\mathbf{e}(s) = \mathbf{y}_{\text{ref}} - \mathbf{G}_k(s) \frac{\mathbf{e}}{s}$$

$$\mathbf{e}(s) = \left[\mathbf{I} + \frac{1}{s} \mathbf{G}_k(s) \right]^{-1} \mathbf{y}_{\text{ref}}(s)$$

Final value theorem.

$$\lim_{t \to +\infty} \mathbf{e}(t) = \lim_{s \to 0} s\mathbf{e}(s)$$

Let $\mathbf{y}_{\text{ref}}$ a polynomial reference input, $\mathbf{y}_{\text{ref}} = \mathbf{C}_{-1}\delta(t) + \sum_{k=0}^{m} \mathbf{C}_k t^k$ then its Laplace transform reads $\mathbf{y}_{\text{ref}}(s) = \sum_{k=-1}^{m} \frac{\mathbf{C}_k}{s^{k+1}}$

$$\begin{aligned} \lim_{t \to +\infty} \mathbf{e}(t) &= \lim_{s \to 0} s\mathbf{e}(s) = \\ &= \lim_{s \to 0} s \left[\mathbf{I} + \frac{1}{s} \mathbf{G}_k(s) \right]^{-1} \left(\sum_{k=-1}^{m} \frac{\mathbf{C}_k}{s^{k+1}} \right) = \\ &= \lim_{s \to 0} s \cdot s\, \mathbf{G}_k^{-1}(s) \left(\sum_{k=-1}^{m} \frac{\mathbf{C}_k}{s^{k+1}} \right) \ , \end{aligned}$$

and thus, if

- $m = 0$ (i.e. linear combination of Dirac's delta and a step) the asymptotic value of the tracking error is zero;
- $m = 1$ (i.e. up to a ramp) the asymptotic value of the tracking error is constant, $\mathbf{G}_k^{-1}(0)\mathbf{C}_1$;
- $m > 1$ the tracking error goes to infinity.

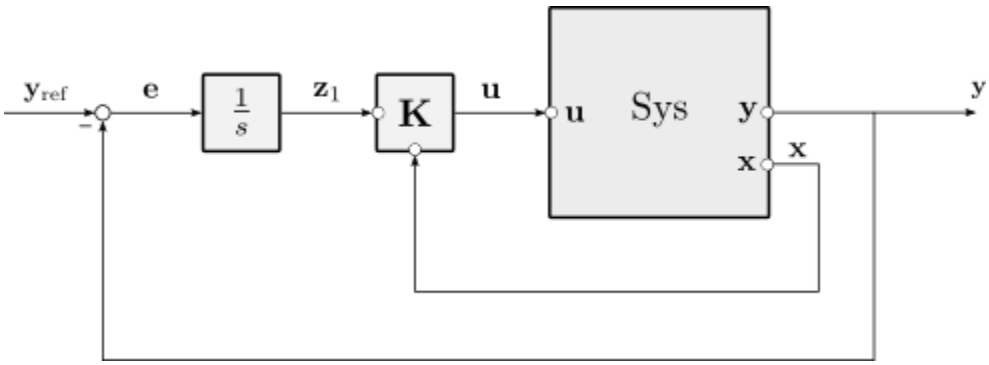


Fig. 47.2: Caption

47.8.2 Augmented system for tracking ramp reference signal - double integrator

Let $\mathbf{y}_{\text{ref}}$ a reference signal. An augmented system can be defined in order to used optimal control. Let

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

the state-space representation of the plant. Let $\mathbf{y}_{ref}$ a desired output and the integral error

$$\mathbf{e}_1(t) := \int_{\tau=0}^{t} \left\{\mathbf{y}(\tau) - \mathbf{y}_{ref}(\tau)\right\} d\tau \,,$$

as a new state with dynamical equation

$$\dot{\mathbf{e}}_1 = \mathbf{y}(t) - \mathbf{y}_{ref}(t) = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{ref} \,.$$

The second augmenting state is defined as

$$\mathbf{e}_2(t) := \int_{\tau=0}^{t} \mathbf{e}_1(\tau)\, d\tau \,,$$

so that the dynamical equation is

$$\dot{\mathbf{e}}_2 = \mathbf{e}_1 \,.$$

The optimal control is applied to the augmented system

$$\underbrace{\begin{bmatrix}\dot{\mathbf{x}}\\ \dot{\mathbf{e}}_1\\ \dot{\mathbf{e}}_2\end{bmatrix}}_{\dot{\mathbf{z}}} = \underbrace{\begin{bmatrix}\mathbf{A} & \cdot & \cdot\\ \mathbf{C} & \cdot & \cdot\\ \cdot & \mathbf{I} & \cdot\end{bmatrix}}_{\hat{\mathbf{A}}} \underbrace{\begin{bmatrix}\mathbf{x}\\ \mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix}\mathbf{B}\\ \mathbf{D}\\ \cdot\end{bmatrix}}_{\hat{\mathbf{B}}_u} \mathbf{u} + \underbrace{\begin{bmatrix}\cdot\\ -\mathbf{I}\\ \cdot\end{bmatrix}}_{\hat{\mathbf{B}}_{ref}} \mathbf{y}_{ref}$$

Optimal control framework provides the opitmal gain matrix $\hat{\mathbf{K}}$, so that $\mathbf{u} = -\hat{\mathbf{K}}\mathbf{z}$ and the closed loop system becomes

…

$\lim_{t\to+\infty} e(t)$

Transfer function. The closed-loop system reads

$$\begin{aligned} s\mathbf{x} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} = \\ &= (\mathbf{A} - \mathbf{B}\mathbf{K})\,\mathbf{x} - \mathbf{B}\mathbf{K}_1\mathbf{z}_1 - \mathbf{B}\mathbf{K}_2\mathbf{z}_2 \\ \mathbf{y} &= (\mathbf{C} - \mathbf{D}\mathbf{K}_u)\,\mathbf{x} - \mathbf{D}\mathbf{K}_1\mathbf{z}_1 - \mathbf{D}\mathbf{K}_2\mathbf{z}_2\ . \end{aligned}$$

so that

$$\begin{aligned} \mathbf{x} &= (s\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{K})^{-1}\left[-\mathbf{B}\mathbf{K}_1\mathbf{z}_1 - \mathbf{B}\mathbf{K}_2\mathbf{z}_2\right] = \\ &= \ldots \end{aligned}$$

and $\mathbf{y} = \mathbf{G}_1(s)\mathbf{K}_1\mathbf{z}_1 + \mathbf{G}_2(s)\mathbf{K}_2\mathbf{z}_2$.

From the reference signal to the tracking error,

$$\begin{aligned} \mathbf{e}(s) &= \mathbf{y}_{\text{ref}} - \mathbf{y} = \\ &= \mathbf{y}_{\text{ref}} - \mathbf{G}(s)\mathbf{K}_1\mathbf{z}_1 - \mathbf{G}(s)\mathbf{K}_2\mathbf{z}_2 = \\ &= \mathbf{y}_{\text{ref}} - \mathbf{G}_1(s)\frac{\mathbf{e}}{s} - \mathbf{G}_2(s)\frac{\mathbf{e}}{s^2}\ , \end{aligned}$$

$$\mathbf{e}(s) = \left[\mathbf{I} + \frac{1}{s}\mathbf{G}_1(s) + \frac{1}{s^2}\mathbf{G}_2(s)\right]^{-1}\mathbf{y}_{\text{ref}}(s)$$

Final value theorem.

$$\lim_{t\to+\infty}\mathbf{e}(t) = \lim_{s\to 0} s\mathbf{e}(s)$$

Let $\mathbf{y}_{\text{ref}}$ a polynomial reference input, $\mathbf{y}_{\text{ref}} = \mathbf{C}_{-1}\delta(t) + \sum_{k=0}^{m}\mathbf{C}_k t^k$ then its Laplace transform reads $\mathbf{y}_{\text{ref}}(s) = \sum_{k=-1}^{m}\frac{\mathbf{C}_k}{s^{k+1}}$

$$\begin{aligned} \lim_{t\to+\infty}\mathbf{e}(t) &= \lim_{s\to 0} s\mathbf{e}(s) = \\ &= \lim_{s\to 0} s\left[\mathbf{I} + \frac{1}{s}\mathbf{G}_1(s) + \frac{1}{s^2}\mathbf{G}_2(s)\right]^{-1}\left(\sum_{k=-1}^{m}\frac{\mathbf{C}_k}{s^{k+1}}\right) = \\ &= \lim_{s\to 0} s\cdot s^2\,\mathbf{G}_2^{-1}(s)\left(\sum_{k=-1}^{m}\frac{\mathbf{C}_k}{s^{k+1}}\right)\ , \end{aligned}$$

and thus, if

- $m = 1$ (i.e. linear combination of Dirac's delta, step, and ramp) the asymptotic value of the tracking error is zero;
- $m = 2$ (i.e. up to a parabolic input, $\cdots + \mathbf{C}_2 t^2$) the asymptotic value of the tracking error is constant, $\mathbf{G}_2^{-1}(0)\mathbf{C}_2$;
- $m > 2$ the tracking error goes to infinity.

47.8.3 Example of optimal control for tracking reference input

Mass-damper-spring system

$$\begin{bmatrix}\dot{x}\\ \dot{v}\end{bmatrix} = \begin{bmatrix}\cdot & 1\\ -\frac{k}{m} & -\frac{c}{m}\end{bmatrix}\begin{bmatrix}x\\ v\end{bmatrix} + \begin{bmatrix}\cdot\\ 1\end{bmatrix}u$$
$$y = \begin{bmatrix}1 & \cdot\end{bmatrix}\begin{bmatrix}x\\ v\end{bmatrix}$$

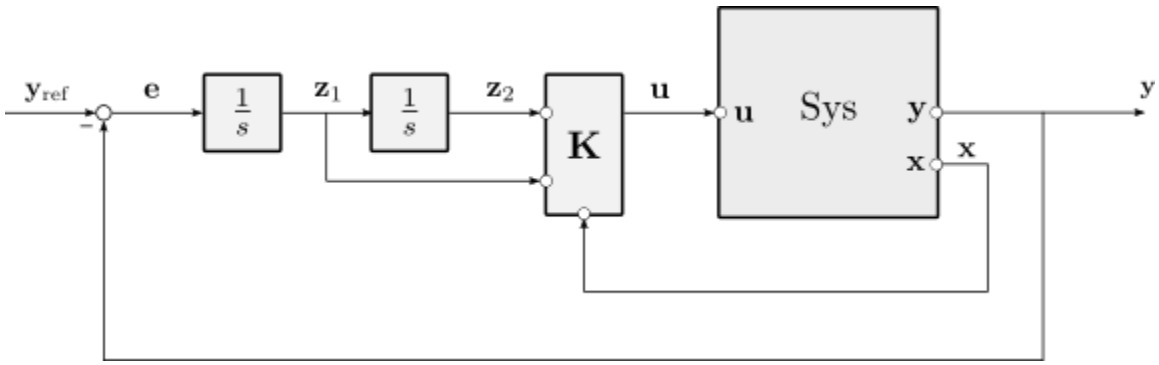


Fig. 47.3: Caption

i.e. the control u is a force per unit-mass, as the second component of the state equation is nothing but the second principle of Newton's mechanics,

$$m\ddot{x} + c\dot{x} + kx = F \ .$$

The first equation is the kinematic relation between position and velocity. The measurement output is the position x.

Infinite time horizon optimal problem is solved with the `control.lqr` routine, see https://python-control.readthedocs.io/en/latest/generated/control.lqr.html. The continuous-time LTI system is defined as a state space system with the function `control.ss`, see https://python-control.readthedocs.io/en/latest/generated/control.ss.html.

Augmented system for reference signal

Single integral for step tracking

Let x_{ref} a reference signal. An augmented system can be defined in order to used optimal control. Let

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}\\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}\end{aligned}$$

the state-space representation of the plant. Let $\mathbf{y}_{ref}$ a desired output and the integral error

$$\mathbf{e}(t) := \int_{\tau=0}^{t} \left\{\mathbf{y}(\tau) - \mathbf{y}_{ref}(\tau)\right\} d\tau \ ,$$

as a new state with dynamical equation

$$\dot{\mathbf{e}} = \mathbf{y}(t) - \mathbf{y}_{ref}(t) = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{ref} \ .$$

The optimal control is applied to the augmented system

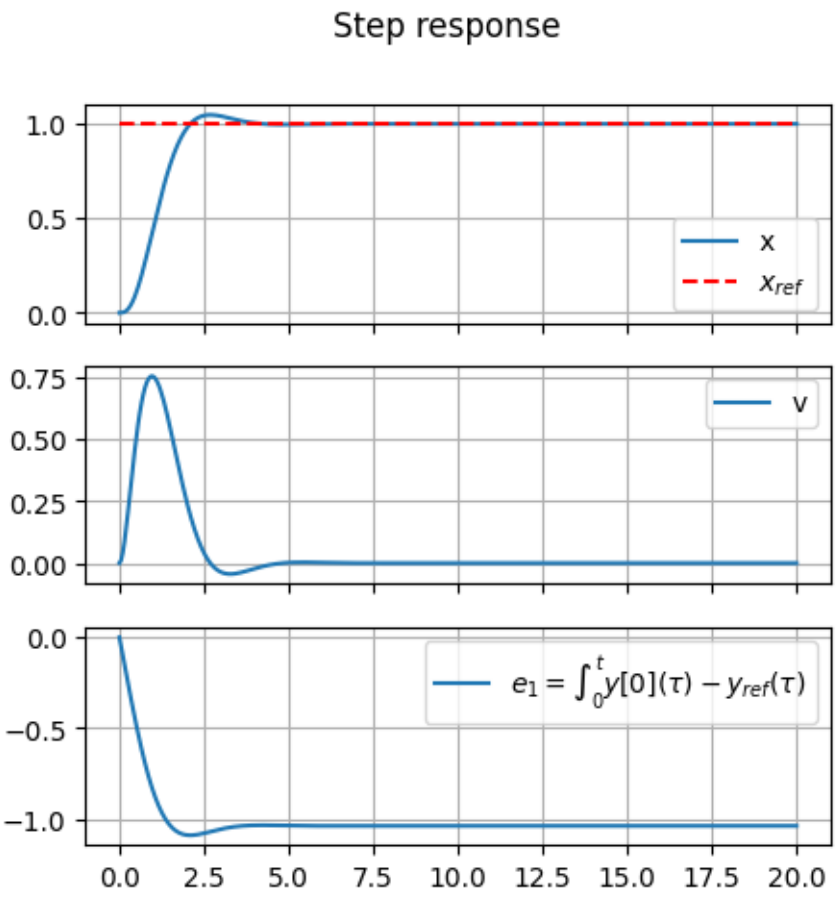
$$\underbrace{\begin{bmatrix}\dot{\mathbf{x}}\\ \dot{\mathbf{e}}\end{bmatrix}}_{\dot{\mathbf{z}}} = \underbrace{\begin{bmatrix}\mathbf{A} & \cdot\\ \mathbf{C} & \cdot\end{bmatrix}}_{\hat{\mathbf{A}}} \underbrace{\begin{bmatrix}\mathbf{x}\\ \mathbf{e}\end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix}\mathbf{B}\\ \mathbf{D}\end{bmatrix}}_{\hat{\mathbf{B}}_u} \mathbf{u} + \underbrace{\begin{bmatrix}\cdot\\ -\mathbf{I}\end{bmatrix}}_{\hat{\mathbf{B}}_{ref}} \mathbf{y}_{ref}$$

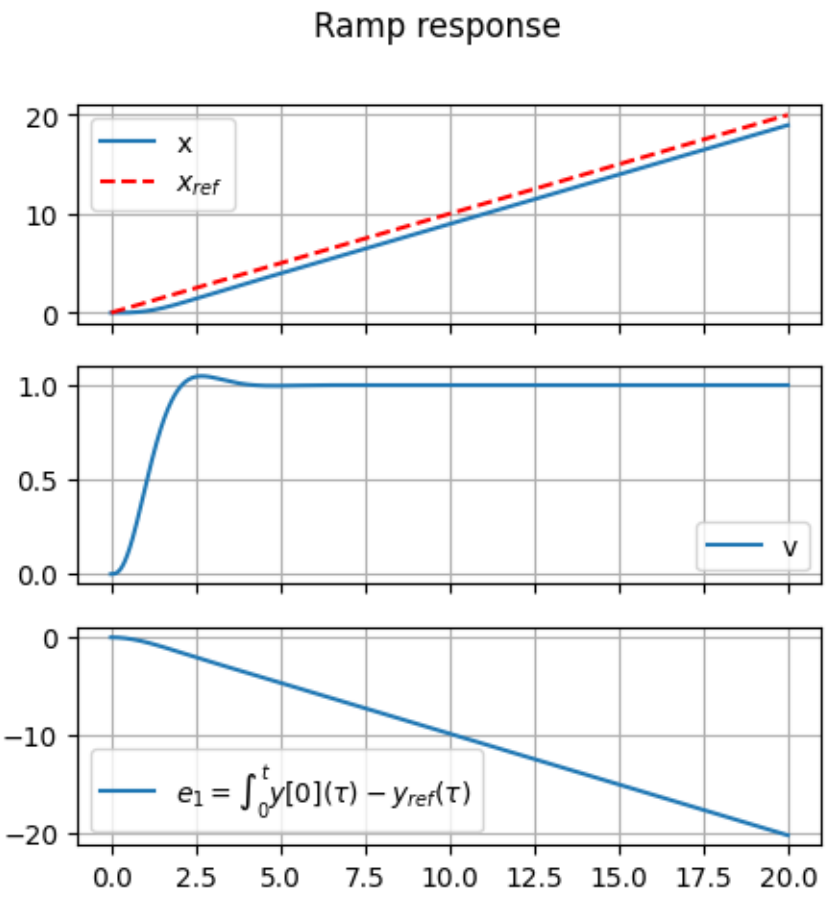
Optimal control framework provides the opitmal gain matrix $\hat{\mathbf{K}}$, so that $\mathbf{u} = -\hat{\mathbf{K}}\mathbf{z}$ and the closed loop system becomes

$$\dot{\mathbf{z}} = \left(\hat{\mathbf{A}} - \hat{\mathbf{B}}_u\hat{\mathbf{K}}\right)\mathbf{z} + \hat{\mathbf{B}}_{ref}\mathbf{y}_{ref} \ .$$

Optimal control design

Response





Double integral for ramp tracking

As the error never reaches a zero value in the ramp response of the augmented system with one integral, another integral is added to the system.

Let

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

the state-space representation of the plant. Let $\mathbf{y}_{ref}$ a desired output and the integral error

$$\mathbf{e}_1(t) := \int_{\tau=0}^{t} \left\{\mathbf{y}(\tau) - \mathbf{y}_{ref}(\tau)\right\} d\tau \ ,$$

as a new state with dynamical equation

$$\dot{\mathbf{e}}_1 = \mathbf{y}(t) - \mathbf{y}_{ref}(t) = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{ref} \ .$$

The second augmenting state is defined as

$$\mathbf{e}_2(t) := \int_{\tau=0}^{t} \mathbf{e}_1(\tau)\, d\tau \ ,$$

so that the dynamical equation is

$$\dot{\mathbf{e}}_2 = \mathbf{e}_1 \ .$$

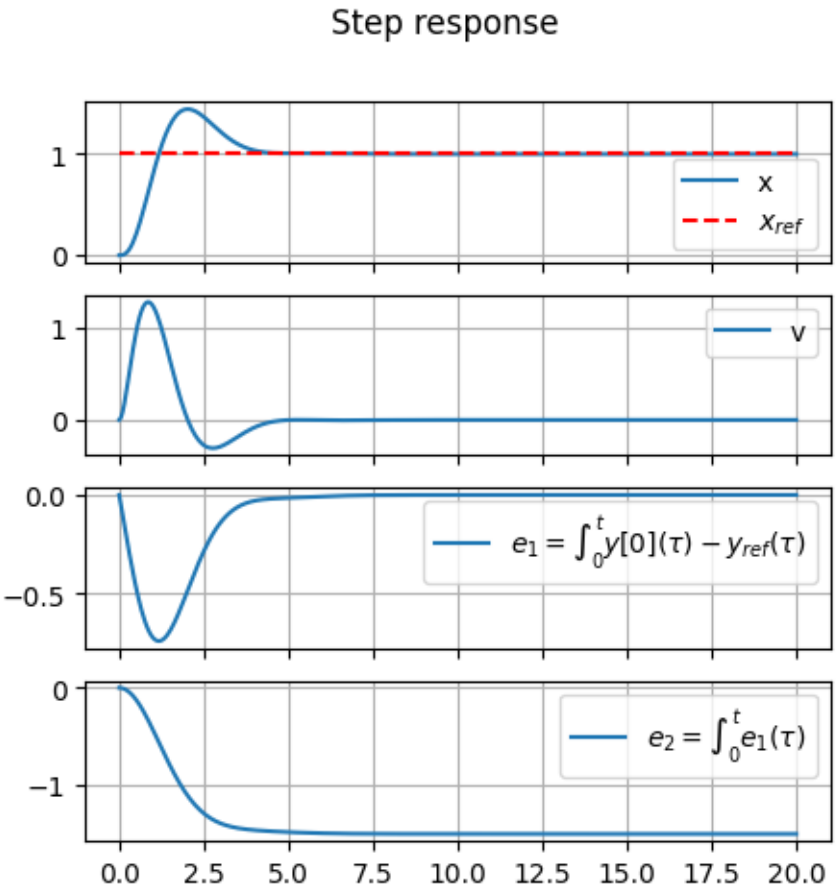
The optimal control is applied to the augmented system

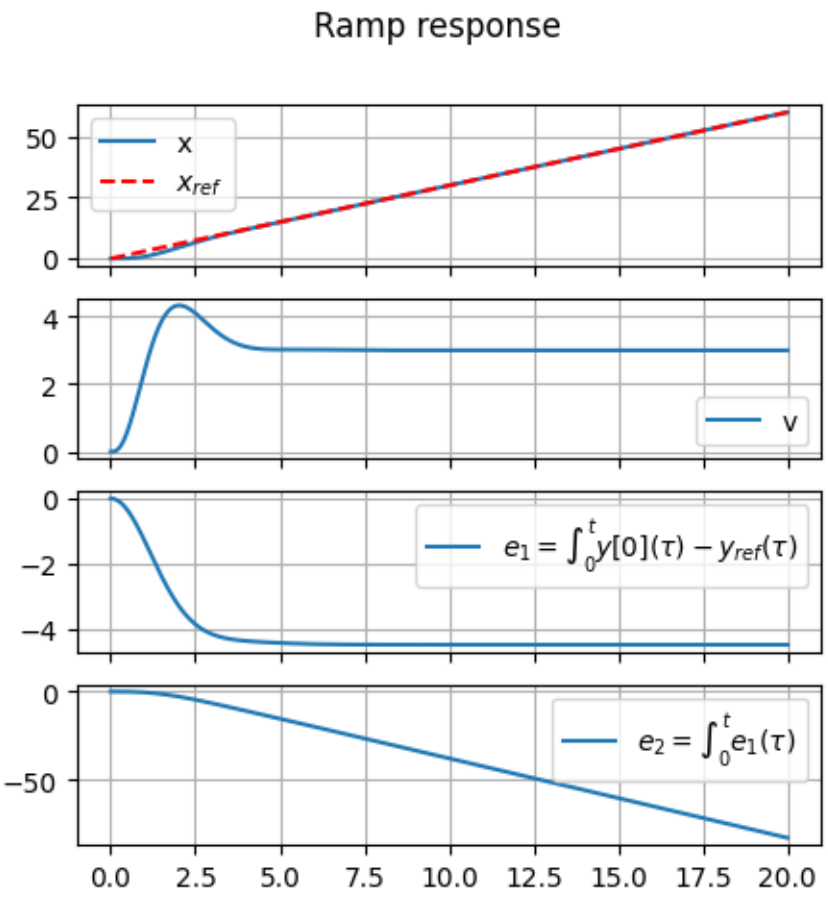
$$\underbrace{\begin{bmatrix}\dot{\mathbf{x}}\\ \dot{\mathbf{e}}_1\\ \dot{\mathbf{e}}_2\end{bmatrix}}_{\dot{\mathbf{z}}} = \underbrace{\begin{bmatrix}\mathbf{A} & \cdot & \cdot\\ \mathbf{C} & \cdot & \cdot\\ \cdot & \mathbf{I} & \cdot\end{bmatrix}}_{\hat{\mathbf{A}}} \underbrace{\begin{bmatrix}\mathbf{x}\\ \mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix}\mathbf{B}\\ \mathbf{D}\\ \cdot\end{bmatrix}}_{\hat{\mathbf{B}}_u} \mathbf{u} + \underbrace{\begin{bmatrix}\cdot\\ -\mathbf{I}\\ \cdot\end{bmatrix}}_{\hat{\mathbf{B}}_{ref}} \mathbf{y}_{ref}$$

Optimal control framework provides the opitmal gain matrix $\hat{\mathbf{K}}$, so that $\mathbf{u} = -\hat{\mathbf{K}}\mathbf{z}$.

Optimal control design

Response





47.9 Kalman filter

Kalman filter is an optimal state estimator, in terms of minimum-variance error reconstruction. It has the structure of a state observer.

- For continuous-time LTI systems with stochastic distrubances, see *optimal observer for continuous-time LTI with stochastic disturbances*
- …**todo** *discrete-time*, *deterministic distrubances*,…

47.10 Hamilton Bellman Jacobi equation

Let a dynamical system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \ ,$$

with initial conditions $\mathbf{x}(0) = \mathbf{x}_0$. Let a **value function** $V(\mathbf{x}_t, t)$ defined as

$$V(\mathbf{x}_t, t) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau)) d\tau + D(\mathbf{x}(T)) \ ,$$

quantifying a cost for $\tau \in [t, T]$, subject to the dynamical equations. Value function has two arguments, the "initial time" t and the state of the system at that time $\mathbf{x}_t$. The dynamical equations are a constraint that can be introduced into the value function with the method of Lagrange multipliers

$$\widetilde{V}(\mathbf{x}_t, t, \lambda) = \int_{\tau=t}^{T} C(\mathbf{x}(\tau), \mathbf{u}(\tau))d\tau + D(\mathbf{x}(T)) - \int_{\tau=t}^{T} \lambda^T(\tau) \left(\dot{\mathbf{x}}(\tau) - \mathbf{f}(\mathbf{x}(\tau), \mathbf{u}(\tau))\right) d\tau \, .$$

Method 1. Non-augmented value function

Method 2. Augmented value function

CHAPTER

FORTYEIGHT

POLE PLACEMENT

48.1 SISO

48.2 MIMO

CHAPTER

FORTYNINE

EXAMPLES

List of examples of scripts

Frequency domain.

Optimal control.

- *First-order stable system with optimal control for reference tracking*
- *First-order stable system with time delay optimal control for reference tracking*
- *Mass-damper-spring system. Second-order stable system with optimal control for reference tracking*
- *Inverted pendulum*
- *Inverted pendulum on cart, with force and torque actuation*
- *Inverted pendulum on cart, with only force actuation*

49.1 First order system - Reference tracking

The same as the example *first order system with time delays - reference tracking*, without time delays from the input to the output.

49.2 First order system w/ time delay - Reference tracking

The same as the example *first order system - reference tracking*, with a time delay from the input to the output.

todo

- Comparison between frequency domain methods (PI controller,…) and time-domain methods (optimal control,…)

Need for *Padé approximation* for time-domain method, in order to avoid dealing with delay differential equations.

49.2.1 Libraries

```
0.10.2
```

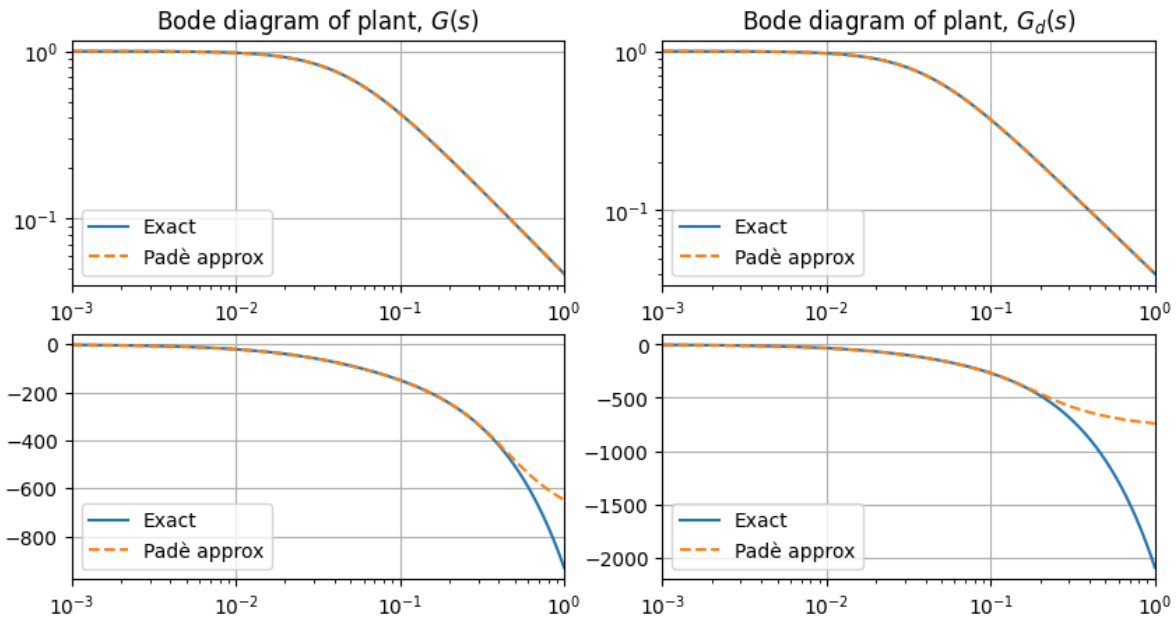
49.2.2 Plant model

Plant model from mathworks. https://www.mathworks.com/help/control/ug/temperature-control-in-a-heat-exchanger.html. First order + time delay. This example assumes **measurable disturbance** $d(t)$, so that they can feed a **feedforward** controller. This assumption is not made here. The plant is described by the following transfer functions between the output $Y(s) = T(s)$ (the measurement of temperature) and the control input $U(s) = V(s)$ (the voltage input to the heat exchanger) and the exogenous input D,

$$y(s) = \underbrace{\frac{e^{-\tau s}}{as+1}u(s)}_{\text{control input}} + \underbrace{\frac{e^{-\tau_d s}}{a_d s+1}d(s)}_{\text{exogenous input}} = G(s)u(s) + G_d(s)d(s) \ .$$

State-space realization, with $y(s) = x_1(s) + x_2(s)$, with x_1 from control input, and x_2 from disturbances,

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{a} & \cdot \\ \cdot & -\frac{1}{a_d} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{a} \\ \cdot \end{bmatrix} u(t-\tau) + \begin{bmatrix} \cdot \\ \frac{1}{a_d} \end{bmatrix} d(t-\tau_d) \\ \quad y = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{cases}$$



49.2.3 Control design

Here a low-order regulator with TF $R(s)$ is designed for tracking a desired step-input.

The regulator takes the tracking error $e(t)$ as an input and returns the plant control input $u(t)$ as its output. Using TF algebra in Laplace domain (for the forced response)

$$\begin{aligned} y &= Gu + G_d d = GRe + G_d d = GR(y_{\text{ref}} - y) + G_d d \\ e &= y_{\text{ref}} - y = y_{\text{ref}} - GRe - G_d d \end{aligned}$$

so that the closed-loop TFs are

$$\begin{aligned} y &= \frac{GR}{1+GR} y_{\text{ref}} + \frac{G_d}{1+GR} d =: T(s) y_{\text{ref}} + \frac{G_d}{1+GR} d \\ e &= \frac{1}{1+GR} y_{\text{ref}} - \frac{G_d}{1+GR} d =: S(s) y_{\text{ref}} - \frac{G_d}{1+GR} d \,. \end{aligned}$$

The control input can be written as a function of y_{ref} and d as well

$$u = Re = Ry_{\text{ref}} - Ry = Ry_{\text{ref}} - RGu - RG_d d \,,$$

so that

$$u = \frac{R}{1+RG} y_{\text{ref}} - \frac{RG_d}{1+RG} d \,.$$

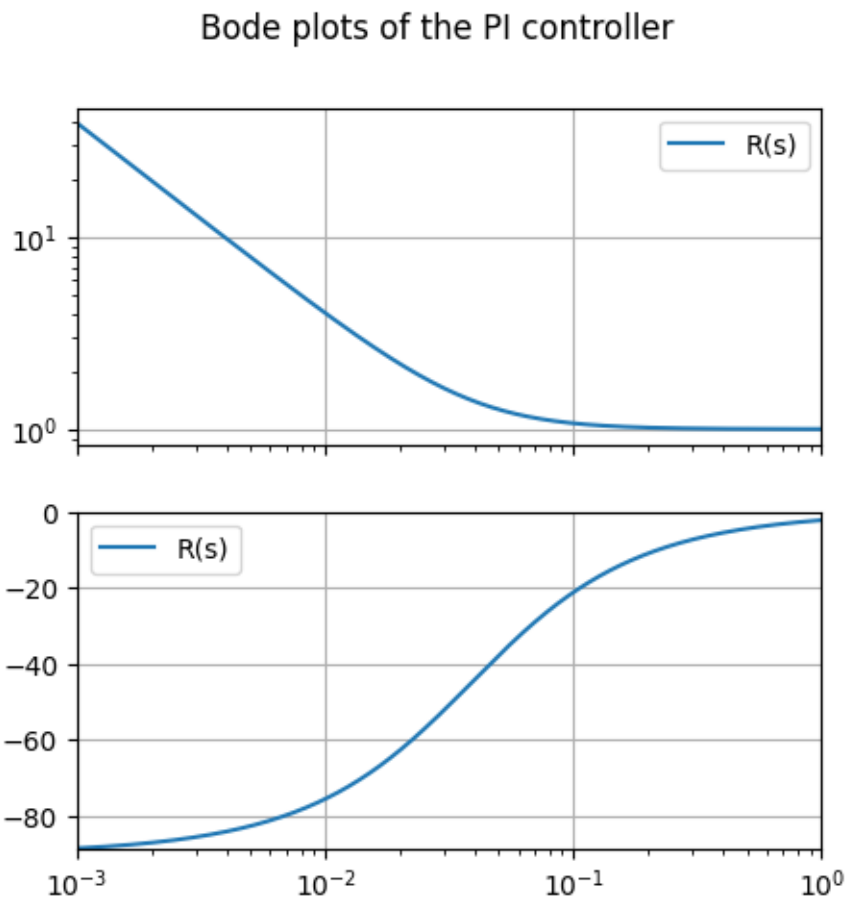
PI controller

With $e(t) := y_{\text{ref}}(t) - y(t)$

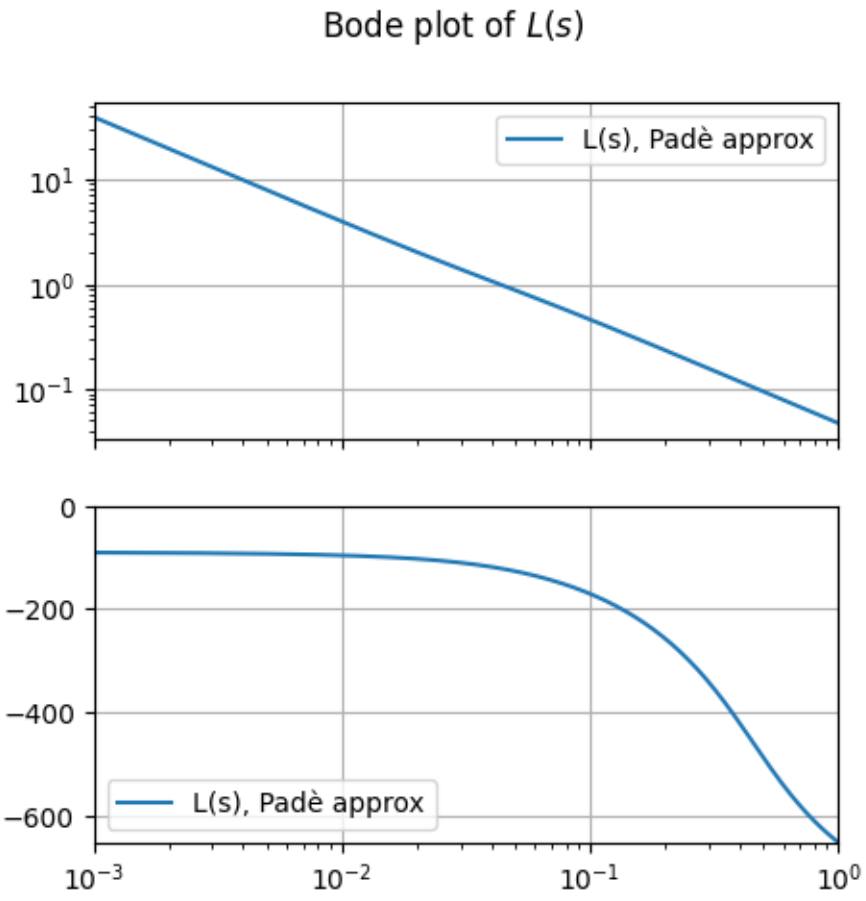
$$\begin{aligned} u(t) &= K_p e(t) + K_i \int_{\tau=0}^{t} e(\tau) d\tau \\ u(s) &= \left[K_p + \frac{K_i}{s} \right] e(s) = R(s) e(s) \end{aligned}$$

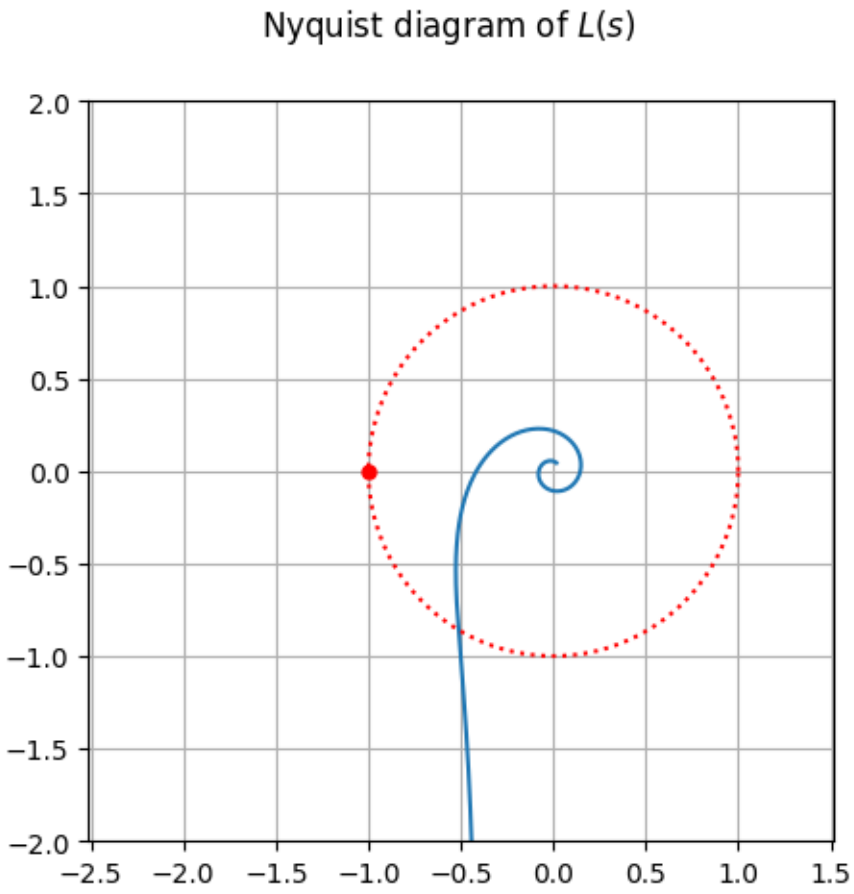
Open-loop TF from error to output $y(s) = L(s)e(s) = R(s)G(s)e(s)$ reads

$$L(s) = \frac{sK_p + K_i}{s} \frac{e^{-\tau s}}{as+1}$$




```
Ignoring fixed x limits to fulfill fixed data aspect with adjustable data limits.
```

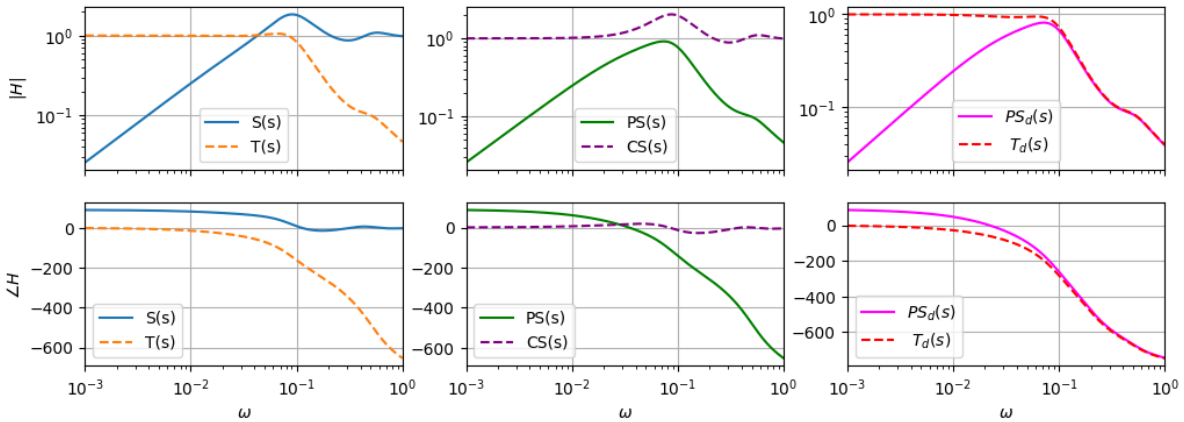

Bode plot of $L(s)$
L(s), Padè approx
10^1
10^0
10^{-1}
0
−200
−400
−600
L(s), Padè approx
10^{-3}
10^{-2}
10^{-1}
10^0



Optimal controller

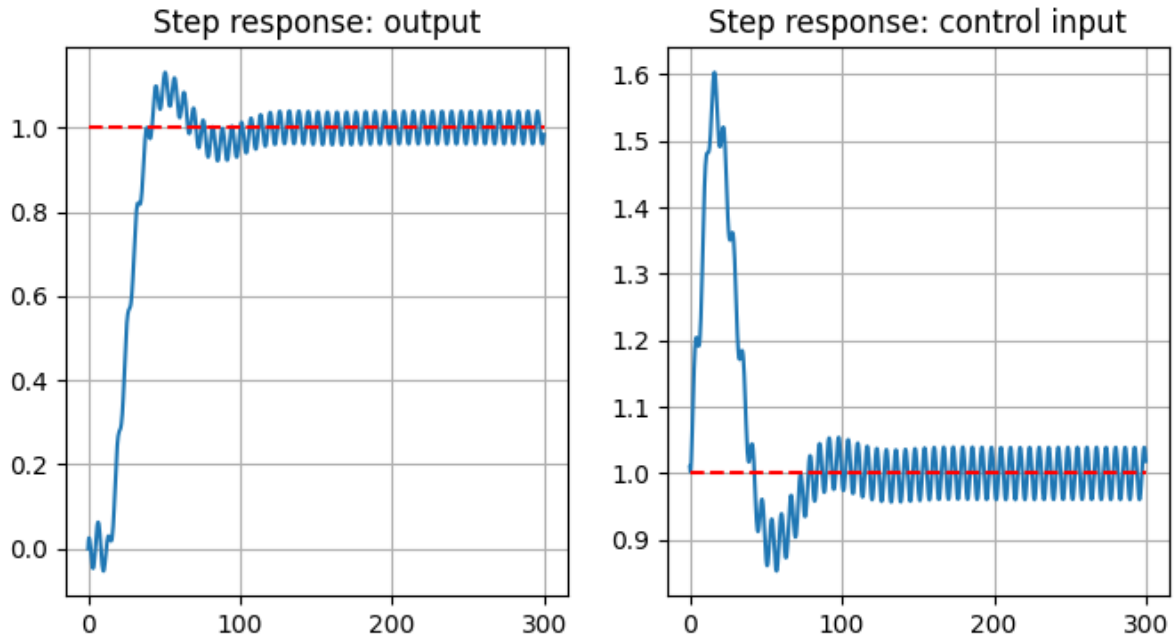
49.2.4 Closed-loop transfer functions - gang of the four

```
Text(0.5, 0, '$\\omega$')
```



Time-domain response

```
0.03996803834887157
```



Model error using Padé approximation in time simulation. Only Padé approximation for time delay is supported for state-space representation and time response simulation. Here, the response to a step of closed-loop system with Padé approximation of the time delay is non-zero in $t \in [0, \tau]$, but shows "small-amplitude" oscillations.

Effect of the disturbances. Here the harmonic disturbances have frequency ω_d much larger than the cutting frequency ω_c. Thus, d produces the output $y_d(\omega_d) = T_d(\omega_d)d(\omega_d)$, with gain $|G(\omega_d)| \simeq .04$ Thus, even with a disturbance d with the same amplitude as the reference step signal, the oscillations around the desired output have "small" amplitude, $y_d \simeq A_d \cos(\omega_d t + \varphi_d)$, with $A_d = |T_d(\omega_d)||d(\omega_d)| \simeq .04|d(\omega_d)|$.

49.3 Inverted pendulum

49.3.1 Equations of motion with Lagrangian mechanics

Non-linear equations

Second-order dynamical equation of the mechanical system reads

$$m\ell^2\ddot{\theta} + c\dot{\theta} = mg\ell \sin\theta + C \, .$$

Equilibrium conditions of unforced system ($C = 0$) are found for $\overline{\theta}_1 = 0$, $\overline{\theta}_2 = \pi$. The first equilibrium is unstable, the second equilibrium is stable. Around the second equilibrium, the linearized equation becomes

$$m\ell^2\ddot{\theta} + c\dot{\theta} + mg\ell\theta = C \, ,$$

so that, for under-damped systems, it's possible to define the natural frequency of the undamped system $\omega_n^2 = \frac{g}{\ell}$, and the damping coefficient ξ

as, by definition

$$2\xi\omega_n := \frac{c}{m\ell^2}$$
$$\xi = \frac{c}{2m\ell^2\omega_n} = \frac{c}{2m\ell^2\sqrt{\frac{g}{\ell}}}$$

Equilibria

Linearized equations around equilibria

Linearized equation around the stable equilibrium. First-order system equation around the stable equilibium becomes

$$\begin{bmatrix}\dot{\theta}\\ \dot{\Omega}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\omega_n^2 & -2\omega_n\xi\end{bmatrix}\begin{bmatrix}\theta\\ \Omega\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{m\ell^2}\end{bmatrix} C \,.$$

Linearized equation around the unstable equilibrium. First-order system equation around the **unstable equilibium** becomes

$$\begin{bmatrix}\dot{\theta}\\ \dot{\Omega}\end{bmatrix} = \begin{bmatrix}0 & 1\\ \omega_n^2 & -2\omega_n\xi\end{bmatrix}\begin{bmatrix}\theta\\ \Omega\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{m\ell^2}\end{bmatrix} C$$
$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}\theta\\ \Omega\end{bmatrix} \,.$$

This the equation we're interested in, when studying the inverted pendulum system.

Augmented sytem for tracking reference signal

Let $\mathbf{y}_{\text{ref}}$ a reference signal. An augmented system can be defined in order to used optimal control. Let

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

the state-space representation of the plant. Let $\mathbf{y}_{\text{ref}}$ a desired output and the integral error

$$\mathbf{e}_{\text{int}}(t) := \int_{\tau=0}^{t} \{\mathbf{y}(\tau) - \mathbf{y}_{\text{ref}}(\tau)\}\, d\tau \,,$$

as a new state with dynamical equation

$$\dot{\mathbf{e}}_{\text{int}} = \mathbf{y}(t) - \mathbf{y}_{\text{ref}}(t) = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{\text{ref}} \,.$$

The optimal control is applied to the augmented system

$$\underbrace{\begin{bmatrix}\dot{\mathbf{x}}\\ \dot{\mathbf{e}}_{\text{int}}\end{bmatrix}}_{\dot{\mathbf{z}}} = \underbrace{\begin{bmatrix}\mathbf{A} & \cdot\\ \mathbf{C} & \cdot\end{bmatrix}}_{\hat{\mathbf{A}}}\underbrace{\begin{bmatrix}\mathbf{x}\\ \mathbf{e}_{\text{int}}\end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix}\mathbf{B}\\ \mathbf{D}\end{bmatrix}}_{\hat{\mathbf{B}}_u}\mathbf{u} + \underbrace{\begin{bmatrix}\cdot\\ -\mathbf{I}\end{bmatrix}}_{\hat{\mathbf{B}}_{ref}}\mathbf{y}_{ref}$$

$$\mathbf{y} = \hat{\mathbf{C}}\mathbf{z} + \hat{\mathbf{D}}_u\mathbf{u} \,.$$

Optimal control framework provides the opitmal gain matrix $\hat{\mathbf{K}}$, so that $\mathbf{u} = -\hat{\mathbf{K}}\mathbf{z}$ and the closed loop system becomes

$$\dot{\mathbf{z}} = \left(\hat{\mathbf{A}} - \hat{\mathbf{B}}_u\hat{\mathbf{K}}\right)\mathbf{z} + \hat{\mathbf{B}}_{ref}\mathbf{y}_{ref}$$
$$\mathbf{y} = \left(\hat{\mathbf{C}} - \hat{\mathbf{D}}_u\mathbf{K}\right)\mathbf{z} \,.$$

If the output of the system is the angle $\theta(t)$, with reference signal $\mathbf{y}_{\text{ref}} = \theta_{\text{ref}}$, the dynamical system is a SISO system, whose state-space representation is

…

49.3.2 Libraries

```
0.10.2
```

49.3.3 Plant dynamical equations

```
[ 0.          +0.j -7.07395639+0.j  6.93388498+0.j]
```

49.3.4 Control

Optimal control for full-state feedback

State-space representation of the open-loop system from the tracking error $\mathbf{e}$ to the output $\mathbf{y}$ reads

$$\begin{cases} \dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_{\mathbf{x}})\,\mathbf{x} - \mathbf{B}\mathbf{K}_{\mathbf{e}}\mathbf{e}_{\text{int}} \\ \dot{\mathbf{e}}_{\text{int}} = \mathbf{e} \\ \mathbf{y} = (\mathbf{C} - \mathbf{D}\mathbf{K}_{\mathbf{x}})\,\mathbf{x} - \mathbf{D}\mathbf{K}_{\mathbf{e}}\mathbf{e}_{\text{int}} \end{cases},$$

or

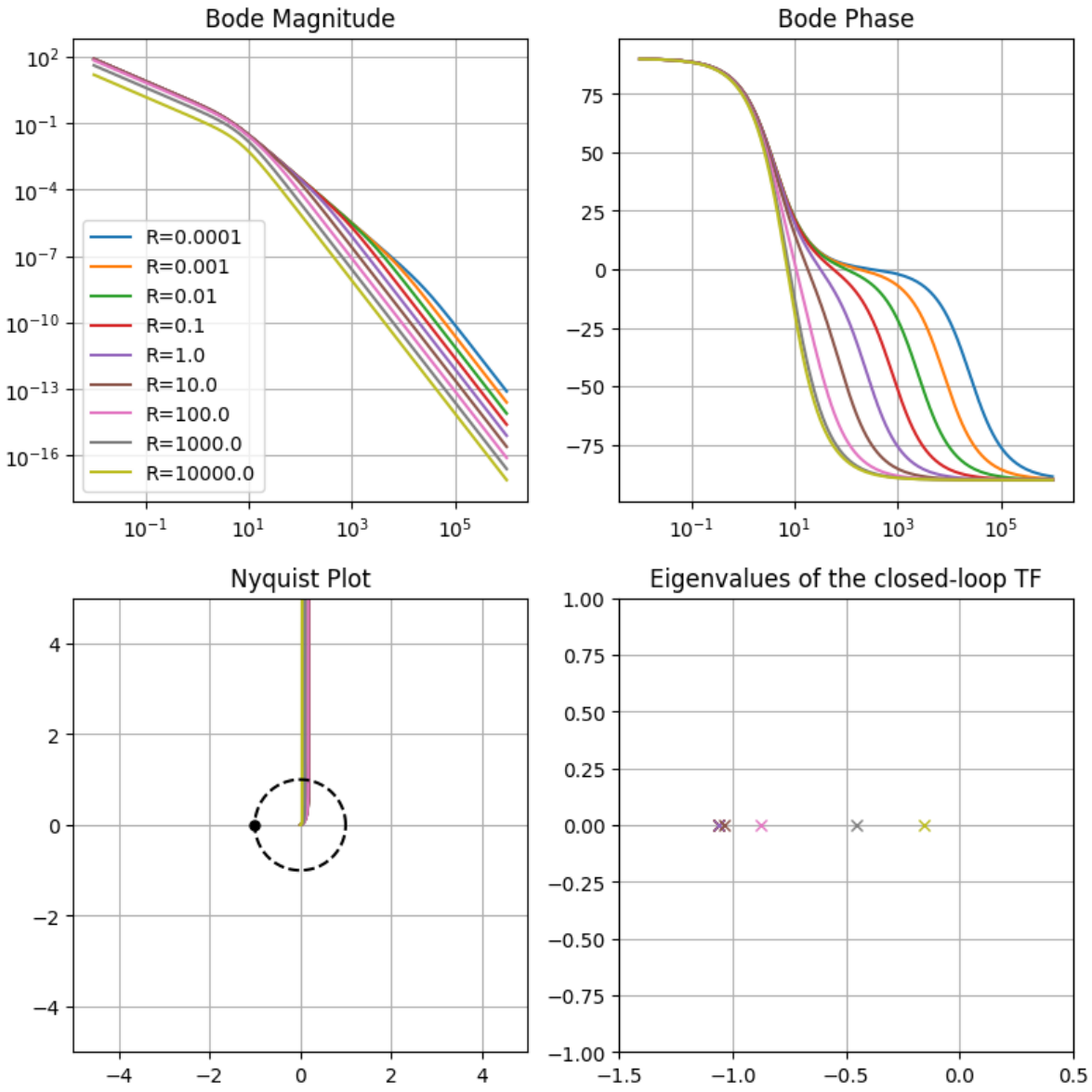
$$\begin{cases} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{e}}_{\text{int}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{K}_{\mathbf{x}} & -\mathbf{B}\mathbf{K}_{\mathbf{e}} \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{e}_{\text{int}} \end{bmatrix} + \begin{bmatrix} \cdot \\ \mathbf{I} \end{bmatrix} \mathbf{e} \\ \mathbf{y} = \begin{bmatrix} \mathbf{C} - \mathbf{D}\mathbf{K}_{\mathbf{x}} & -\mathbf{D}\mathbf{K}_{\mathbf{e}} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{e}_{\text{int}} \end{bmatrix} \end{cases}$$

Sensitivity to control weight

For a given weight matrix on the augmented state $\mathbf{Q}$, here the sensitivity to the value of the control weight $\mathbf{R} = [\, r \,]$ is investigated. Here, the output is the angle θ of the inverted pendulum w.r.t. the unstable equilibrium.

Bode and Nyquist diagram of the stabilized[1] open-loop TF are shown, along with the eigenvalues of the closed loop system, with the reference signal as input.

[1] After the optimal control gain matrix $\mathbf{K}$ has been computed, i.e. it's not surprising that this open-loop system is stable.



Optimal observer

System equations, without disturbances/noise

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases},$$

Observer (state estimator) equations

$$\begin{cases} \hat{\mathbf{y}} = \mathbf{C}\hat{\mathbf{x}} + \mathbf{D}\mathbf{u} \\ \dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}\left(\mathbf{y} - \hat{\mathbf{y}}\right) \\ \quad = \left(\mathbf{A} - \mathbf{L}\mathbf{C}\right)\hat{\mathbf{x}} + \mathbf{L}\mathbf{C}\mathbf{x} + \mathbf{B}\mathbf{u} \end{cases},$$

so that the dynamical equation of the error $\varepsilon := \hat{\mathbf{x}} - \mathbf{x}$ reads

$$\dot{\varepsilon} = \left(\mathbf{A} - \mathbf{L}\mathbf{C}\right)\varepsilon \,.$$

The dynamical equations of the augmented system, plant+observer, reads

$$\begin{bmatrix}\dot{\mathbf{x}}\\ \dot{\varepsilon}\end{bmatrix} = \begin{bmatrix}\mathbf{A} & \mathbf{0}\\ \mathbf{0} & \mathbf{A}-\mathbf{LC}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \varepsilon\end{bmatrix} + \begin{bmatrix}\mathbf{B}\\ \mathbf{0}\end{bmatrix}\mathbf{u}\,,$$

and the observer design here is the design of a matrix $\mathbf{L}$ that makes the dynamics of the error ε, and thus the matrix $\mathbf{A}-\mathbf{LC}$, asymptotically stable.

State estimator with process and measurement noise

$$\begin{cases}\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} + \mathbf{B}_d\mathbf{d}\\ \mathbf{y} = \mathbf{Cx} + \mathbf{Du} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r}\,,\end{cases}$$

The state estimator problem with $\mathbf{D}_d = \mathbf{0}$ and $\mathbf{D_r} = \mathbf{I}$ can be solved with the function `lqe` of the `control` library,

```
control.lqe(sys, Edd, Err, Edr)
control.lqe(A, Bd, C, Edd, Err, Edr)
```

```
[[13.87503766]
 [96.258335  ]]
```

```
"\n#> Method 2.\n# Use LQR duality: lqe(A, B, C, V, W) is dual to lqr(A.T, C.T, V,␣
↪W)\n# Note: we use C.T because it replaces B in the dual problem\nL_transposed,␣
↪P, E = ct.lqr(A.T, C.T, V, W)\n\n# The actual observer gain is the transpose of␣
↪the 'feedback' result\nL = L_transposed.T\n"
```

Combination of controller and observer - separation principle

Properties of control

…

49.3.5 Verifying control on non-linear system

Reference tracking on the non-linear system

Here the control system is tested on the non-linear system for tracking a reference square wave signal.

Non-linear equations of the plant

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\mathbf{u})$$
$$\mathbf{y} = \mathbf{Cx} + \mathbf{Du}$$

Observer

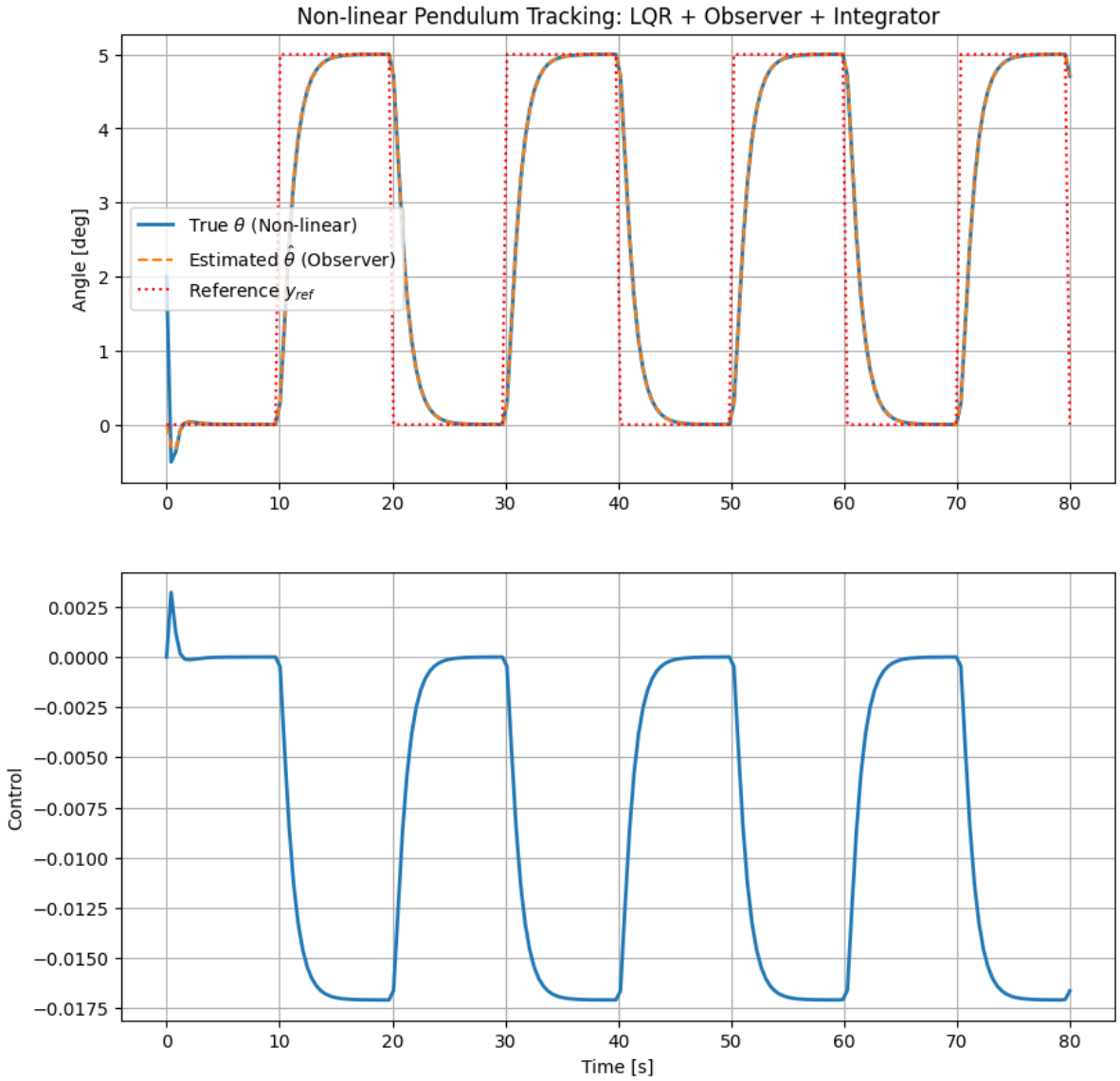
$$\dot{\varepsilon} = (\mathbf{A}-\mathbf{LC})\,\varepsilon$$

Integral error

$$\dot{\mathbf{e}}_{\text{int}} = \mathbf{y} - \mathbf{y}_{\text{ref}}$$

Proportional control

$$\mathbf{u} = -\mathbf{K}_\mathbf{x}\mathbf{x} - \mathbf{K}_\mathbf{x}\varepsilon - \mathbf{K}_\mathbf{e}\mathbf{e}_{\text{int}} \ .$$



49.4 Inverted pendulum on a cart

todo

- Update `control` library version, to have `ct.poles()`
- …

Contents.

- Equations of motion: non-linear equations, linearized equations around equilibria; augmented equations for reference tracking
- Control:
  - full-state feedback: this controller is used later in the observed-state feedback, exploiting *separation principle*
  - observed-state feedback: combination of observer and controller
- Verification:
  - test control laws on non-linear equations

Here, algebraic Riccati equations (ARE) are solved for infinite time-horizon problems of optimal full-state feedback and optimal observer for stochastic disturbances:

- *Optimal full-state feedback - infinite time-horizon*
- *Optimal observer - stochastic disturbances*

49.4.1 Equations of motion with Lagrangian mechanics

See Classical Mechanics: Lagrangian Mechanics

Non-linear equations

Second-order dynamical equation of the mechanical system reads

$$\begin{aligned}(M+m)\ddot{x} - m\ell\ddot{\theta}\cos\theta + m\ell\dot{\theta}^2\sin\theta &= F_x - c_x\dot{x}\\ -m\ell\ddot{x}\cos\theta + m\ell^2\ddot{\theta} - mg\ell\sin\theta &= C_\theta - c_\theta\dot{\theta}\,.\end{aligned}$$

or at the first order, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$,

$$\begin{cases}\dot{x} = v\\ \dot{\theta} = \omega\\ (M+m)\,\dot{v} - m\ell\cos\theta\,\dot{\omega} = -m\ell\omega^2\sin\theta + F_x - c_x v\\ -m\ell\cos\theta\,\dot{v} + m\ell^2\dot{\omega} = mg\ell\sin\theta + C_\theta - c_\theta\omega\,.\end{cases}$$

Linearized equations around equilibria

Linearized equation around the unstable equilibrium. First-order system equation around the **unstable equilibium** becomes

$$\begin{bmatrix}M+m & -m\ell\\ -m\ell & m\ell^2\end{bmatrix}\begin{bmatrix}\ddot{x}\\ \ddot{\theta}\end{bmatrix} + \begin{bmatrix}c_x & \cdot\\ \cdot & c_\theta\end{bmatrix}\begin{bmatrix}\dot{x}\\ \dot{\theta}\end{bmatrix} + \begin{bmatrix}\cdot & \cdot\\ \cdot & -mg\ell\end{bmatrix}\begin{bmatrix}\dot{x}\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}F_x\\ C_\theta\end{bmatrix}\,.$$

or

$$\mathbf{M}\ddot{\mathbf{z}} + \mathbf{C}\dot{\mathbf{z}} + \mathbf{K}\mathbf{z} = \mathbf{f}$$

State-space (first-order) representation. With state and input

$$\mathbf{x} = \begin{bmatrix}\mathbf{z}\\ \dot{\mathbf{z}}\end{bmatrix} \quad , \quad \mathbf{u} = \mathbf{f}\,,$$

the first-order state-space representation of the system reads

$$\begin{cases} \dot{\mathbf{x}} = \begin{bmatrix} \cdot & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \mathbf{x} + \begin{bmatrix} \cdot \\ \mathbf{M}^{-1} \end{bmatrix} \mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{x} \, . \end{cases}$$

This the equation we're interested in, when studying the inverted pendulum on a cart.

Augmented system for tracking reference signal

Let $\mathbf{y}_{\text{ref}}$ a reference signal. An augmented system can be defined in order to used optimal control. Let

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{aligned}$$

the state-space representation of the plant. Let $\mathbf{y}_{\text{ref}}$ a desired output and the integral error

$$\mathbf{e}_{\text{int}}(t) := \int_{\tau=0}^{t} \{\mathbf{y}(\tau) - \mathbf{y}_{\text{ref}}(\tau)\} \, d\tau \, ,$$

as a new state with dynamical equation

$$\dot{\mathbf{e}}_{\text{int}} = \mathbf{y}(t) - \mathbf{y}_{\text{ref}}(t) = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{\text{ref}} \, .$$

The optimal control is applied to the augmented system

$$\underbrace{\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{e}}_{\text{int}} \end{bmatrix}}_{\dot{\mathbf{z}}} = \underbrace{\begin{bmatrix} \mathbf{A} & \cdot \\ \mathbf{C} & \cdot \end{bmatrix}}_{\hat{\mathbf{A}}} \underbrace{\begin{bmatrix} \mathbf{x} \\ \mathbf{e}_{\text{int}} \end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix} \mathbf{B} \\ \mathbf{D} \end{bmatrix}}_{\hat{\mathbf{B}}_u} \mathbf{u} + \underbrace{\begin{bmatrix} \cdot \\ -\mathbf{I} \end{bmatrix}}_{\hat{\mathbf{B}}_{ref}} \mathbf{y}_{ref}$$

$$\mathbf{y} = \hat{\mathbf{C}}\mathbf{z} + \hat{\mathbf{D}}_u \mathbf{u} \, .$$

Optimal control framework provides the opitmal gain matrix $\hat{\mathbf{K}}$, so that $\mathbf{u} = -\hat{\mathbf{K}}\mathbf{z}$ and the closed loop system becomes

$$\begin{aligned} \dot{\mathbf{z}} &= \left(\hat{\mathbf{A}} - \hat{\mathbf{B}}_u \hat{\mathbf{K}}\right) \mathbf{z} + \hat{\mathbf{B}}_{\text{ref}} \mathbf{y}_{\text{ref}} \\ \mathbf{y} &= \left(\hat{\mathbf{C}} - \hat{\mathbf{D}}_u \mathbf{K}\right) \mathbf{z} \, . \end{aligned}$$

If the output of the system is the angle $\theta(t)$, with reference signal $\mathbf{y}_{\text{ref}} = \theta_{\text{ref}}$, the dynamical system is a SISO system, whose state-space representation is

…

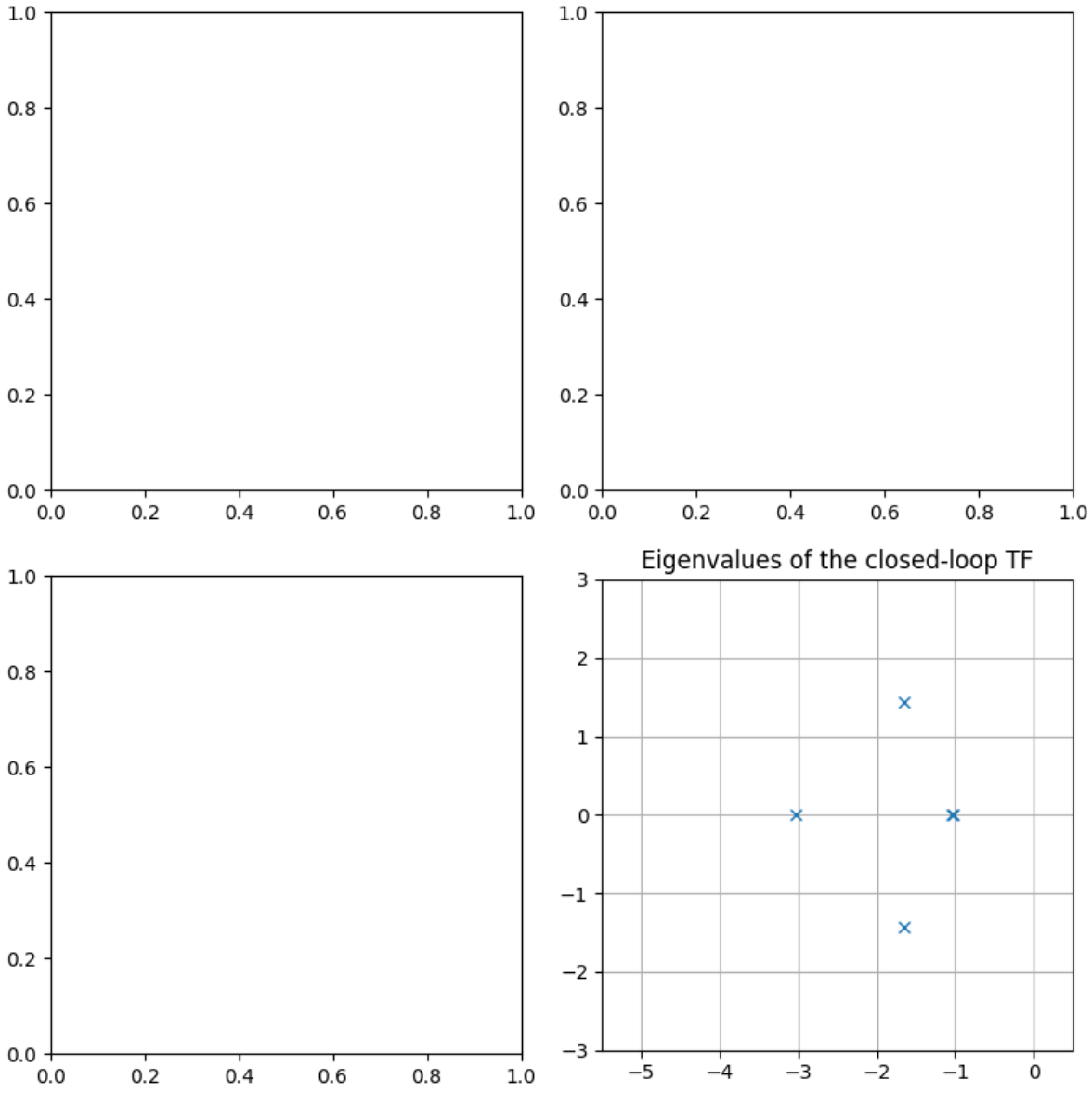
49.4.2 Libraries

```
[ 0.00000000e+00+0.j  0.00000000e+00+0.j  0.00000000e+00+0.j
 -1.01602637e+01+0.j  9.65526368e+00+0.j -4.99999873e-03+0.j]
```

49.4.3 Control

Full-state feedback

```
[-160.29249401+0.j            -3.03748266+0.j
   -1.65844928+1.43107063j    -1.65844928-1.43107063j
   -1.03177742+0.j            -1.03711996+0.j        ]
```



todo. Update `control` library

Feed-back on state estimate

Observer design

System equations, without disturbances/noise

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases},$$

Observer (state estimator) equations

$$\begin{cases} \hat{\mathbf{y}} = \mathbf{C}\hat{\mathbf{x}} + \mathbf{D}\mathbf{u} \\ \dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}\left(\mathbf{y} - \hat{\mathbf{y}}\right) \\ \quad = \left(\mathbf{A} - \mathbf{L}\mathbf{C}\right)\hat{\mathbf{x}} + \mathbf{L}\mathbf{C}\mathbf{x} + \mathbf{B}\mathbf{u} \end{cases},$$

so that the dynamical equation of the error $\varepsilon := \hat{\mathbf{x}} - \mathbf{x}$ reads

$$\dot{\varepsilon} = \left(\mathbf{A} - \mathbf{L}\mathbf{C}\right)\varepsilon \ .$$

The dynamical equations of the augmented system, plant+observer, reads

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\varepsilon} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{A} - \mathbf{L}\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \varepsilon \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u} \ ,$$

and the observer design here is the design of a matrix $\mathbf{L}$ that makes the dynamics of the error ε, and thus the matrix $\mathbf{A} - \mathbf{L}\mathbf{C}$, asymptotically stable.

Control design

Control design is performed as the state is known, and then applied to the estimated state $\hat{\mathbf{x}} = \mathbf{x} + \varepsilon$, as $\mathbf{u} = -\mathbf{K}\hat{\mathbf{x}}$.

Here the state is augmented with an integrator

$$\mathbf{e}_{\text{int}}(t) := \int_{\tau=0}^{t} \{\mathbf{y}(\tau) - \mathbf{y}_{\text{ref}}(\tau)\} d\tau \ ,$$

to get tracking of slow signals and then the state of the augmented system is $\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \mathbf{e}_{\text{int}} \end{bmatrix}$, and optimal control produces a feedback on the augmented state

$$\mathbf{u} = -\mathbf{K}\mathbf{z} = -\mathbf{K}_{\mathbf{x}}\mathbf{x} - \mathbf{K}_{\mathbf{e}}\mathbf{e}_{\text{int}} \ .$$

The integral error can be measured and needs no observer. Thus, the feedback on the observed state becomes

$$\mathbf{u} = -\mathbf{K}_{\mathbf{x}}\hat{\mathbf{x}} - \mathbf{K}_{\mathbf{e}}\mathbf{e}_{\text{int}} \ .$$

Combination observer + controller

State-space representation of the open-loop system from the tracking error $\mathbf{e}$ to the output $\mathbf{y}$ reads

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \\ \dot{\varepsilon} = \left(\mathbf{A} - \mathbf{L}\mathbf{C}\right)\varepsilon \\ \dot{\mathbf{e}}_{\text{int}} = \mathbf{e} = \mathbf{y} - \mathbf{y}_{\text{ref}} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{\text{ref}} \end{cases}$$

$$\hat{\mathbf{x}} = \mathbf{x} + \varepsilon$$

$$\begin{aligned}\mathbf{u} &= -\mathbf{K}_\mathbf{x}\hat{\mathbf{x}} - \mathbf{K}_\mathbf{e}\mathbf{e}_{\text{int}} \\ &= -\mathbf{K}_\mathbf{x}\mathbf{x} - \mathbf{K}_\mathbf{x}\varepsilon - \mathbf{K}_\mathbf{e}\mathbf{e}_{\text{int}}\end{aligned}$$

The **open-loop system** reads

$$\begin{bmatrix}\dot{\mathbf{x}} \\ \dot{\varepsilon} \\ \dot{\mathbf{e}}_{\text{int}}\end{bmatrix} = \begin{bmatrix}\mathbf{A} & \cdot & \cdot \\ \cdot & \mathbf{A}-\mathbf{LC} & \cdot \\ \mathbf{C} & \cdot & \cdot\end{bmatrix}\begin{bmatrix}\mathbf{x} \\ \varepsilon \\ \mathbf{e}_{\text{int}}\end{bmatrix} + \begin{bmatrix}\mathbf{B} \\ \cdot \\ \mathbf{D}\end{bmatrix}\mathbf{u} + \begin{bmatrix}\cdot \\ \cdot \\ -\mathbf{I}\end{bmatrix}\mathbf{y}_{\text{ref}}$$

$$\mathbf{y} = \begin{bmatrix}\mathbf{C} & \cdot & \cdot\end{bmatrix}\begin{bmatrix}\mathbf{x} \\ \varepsilon \\ \mathbf{e}_{\text{int}}\end{bmatrix} + \mathbf{Du}$$

while the **closed-loop system** becomes

$$\begin{bmatrix}\dot{\mathbf{x}} \\ \dot{\varepsilon} \\ \dot{\mathbf{e}}_{\text{int}}\end{bmatrix} = \begin{bmatrix}\mathbf{A}-\mathbf{BK}_\mathbf{x} & -\mathbf{BK}_\mathbf{x} & -\mathbf{BK}_\mathbf{e} \\ \cdot & \mathbf{A}-\mathbf{LC} & \cdot \\ \mathbf{C}-\mathbf{DK}_\mathbf{x} & -\mathbf{DK}_\mathbf{x} & -\mathbf{DK}_\mathbf{e}\end{bmatrix}\begin{bmatrix}\mathbf{x} \\ \varepsilon \\ \mathbf{e}_{\text{int}}\end{bmatrix} + \begin{bmatrix}\cdot \\ \cdot \\ -\mathbf{I}\end{bmatrix}\mathbf{y}_{\text{ref}}$$

$$\mathbf{y} = \begin{bmatrix}\mathbf{C}-\mathbf{DK}_\mathbf{x} & -\mathbf{DK}_\mathbf{x} & -\mathbf{DK}_\mathbf{e}\end{bmatrix}\begin{bmatrix}\mathbf{x} \\ \varepsilon \\ \mathbf{e}_{\text{int}}\end{bmatrix}.$$

If the system has no direct influence of input on measurement output, i.e. $\mathbf{D} = \mathbf{0}$, the system is triangular.

State estimator with process and measurement noise

$$\begin{cases}\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} + \mathbf{B}_d\mathbf{d} \\ \mathbf{y} = \mathbf{Cx} + \mathbf{Du} + \mathbf{D}_d\mathbf{d} + \mathbf{D}_r\mathbf{r}\end{cases},$$

The state estimator problem with $\mathbf{D}_d = \mathbf{0}$ and $\mathbf{D_r} = \mathbf{I}$ can be solved with the function `lqe` of the `control` library,

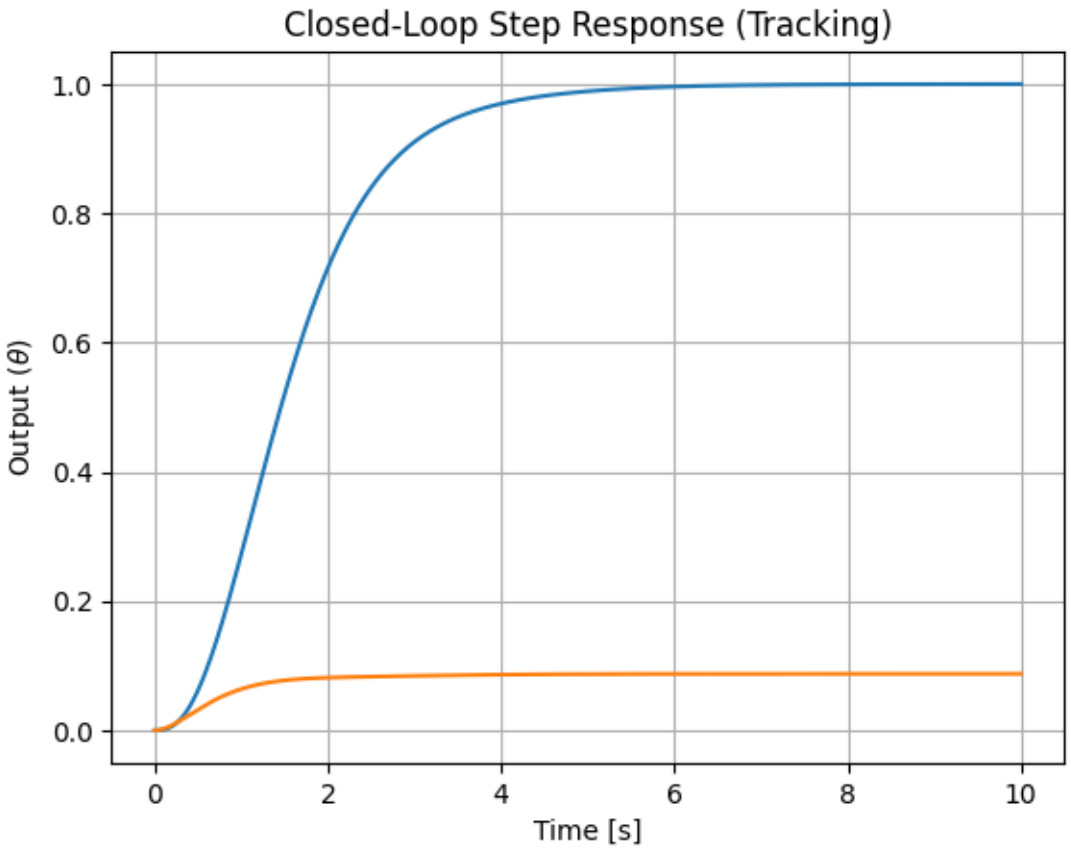
```
control.lqe(sys, Edd, Err, Edr)
control.lqe(A, Bd, C, Edd, Err, Edr)
```

```
Optimal observer matrix
[[  3.44960563   3.08045383]
 [  3.08045383  34.19623533]
 [ 10.69448739  58.84138659]
 [ 57.12488834 589.43585336]]
```

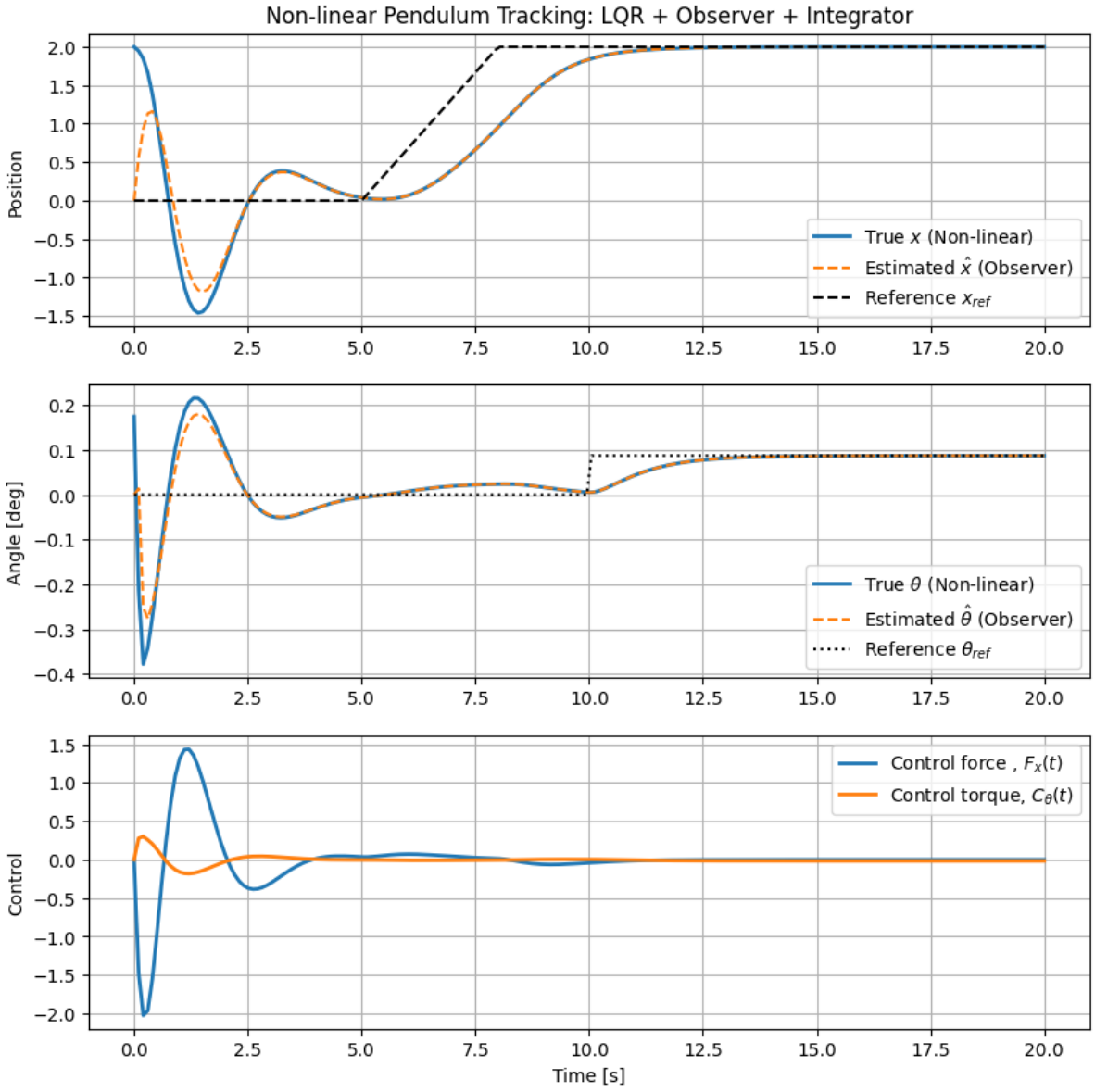
```
"\n#> Method 2.\n# Use LQR duality: lqe(A, B, C, V, W) is dual to lqr(A.T, C.T, V,␣
↪W)\n# Note: we use C.T because it replaces B in the dual problem\nL_transposed,␣
↪P, E = ct.lqr(A.T, C.T, V, W)\n\n# The actual observer gain is the transpose of␣
↪the 'feedback' result\nL = L_transposed.T\n"
```

49.4.4 Closed-loop testing

Linearized system



Non-linear system, around the unstable equilibrium



49.5 Inverted pendulum on a cart - only force actuation

todo

- Update `control` library version, to have `ct.poles()`
- …

This is the same system as the inverted pendulum on a cart, analyzed *here*, but without torque actuation.

As a result, this system is **under-actuated**. It can track a reference x_{ref} for $\theta_{\text{ref}} = 0$ (as it can achieve a stable dynamics of $\theta \to 0$), but it can't track an arbitrary combination of x_{ref}, θ_{ref} simultanouesly. In order to track a $\theta_{\text{ref}} \neq 0$ as an equilibrium of the system, there's the need to allow for x-drift: brute-force approach is likely to fail, while a reduced-

state model with state $\mathbf{x} = [v, \theta, \omega, e_{\theta,int}]$ should be able to track desired constant θ_{ref} with a deriving $x(t) = vt$ (**todo** *check it*).

Contents.

- Equations of motion: non-linear equations, linearized equations around equilibria; augmented equations for reference tracking
- Control:
  - full-state feedback: this controller is used later in the observed-state feedback, exploiting *separation principle*
  - observed-state feedback: combination of observer and controller
- Verification:
  - test control laws on non-linear equations

Here, algebraic Riccati equations (ARE) are solved for infinite time-horizon problems of optimal full-state feedback and optimal observer for stochastic disturbances:

- *Optimal full-state feedback - infinite time-horizon*
- *Optimal observer - stochastic disturbances*

49.5.1 Equations of motion with Lagrangian mechanics

See Classical Mechanics: Lagrangian Mechanics

…

Second-order dynamical equation of the mechanical system reads

$$\begin{aligned}(M+m)\ddot{x} - m\ell\ddot{\theta}\cos\theta + m\ell\dot{\theta}^2\sin\theta &= F_x - c_x\dot{x}\\ -m\ell\ddot{x}\cos\theta + m\ell^2\ddot{\theta} - mg\ell\sin\theta &= -c\dot{\theta}\,.\end{aligned}$$

or at the first order

$$\begin{cases}\dot{x} = v\\ \dot{\theta} = \omega\\ (M+m)\,\dot{v} - m\ell\cos\theta\,\dot{\omega} = -m\ell\omega^2\sin\theta + F_x - c_x v\\ -m\ell\cos\theta\,\dot{v} + m\ell^2\dot{\omega} = mg\ell\sin\theta - c\omega\,.\end{cases}$$

Linearized equation around the unstable equilibrium. First-order system equation around the **unstable equilibium** becomes

$$\begin{bmatrix}M+m & -m\ell\\ -m\ell & m\ell^2\end{bmatrix}\begin{bmatrix}\ddot{x}\\ \ddot{\theta}\end{bmatrix} + \begin{bmatrix}c_x & \cdot\\ \cdot & c_\theta\end{bmatrix}\begin{bmatrix}\dot{x}\\ \dot{\theta}\end{bmatrix} + \begin{bmatrix}\cdot & \cdot\\ \cdot & -mg\ell\end{bmatrix}\begin{bmatrix}\dot{x}\\ \dot{\theta}\end{bmatrix} = \begin{bmatrix}F_x\\ \cdot\end{bmatrix}\,.$$

or

$$\mathbf{M}\ddot{\mathbf{z}} + \mathbf{C}\dot{\mathbf{z}} + \mathbf{K}\mathbf{z} = \mathbf{f}$$

State-space (first-order) representation. With state and input

$$\mathbf{x} = \begin{bmatrix}\mathbf{z}\\ \dot{\mathbf{z}}\end{bmatrix} \quad , \quad \mathbf{u} = \mathbf{f}\,,$$

the first-order state-space representation of the system reads

$$\begin{cases}\dot{\mathbf{x}} = \begin{bmatrix}\cdot & \mathbf{I}\\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C}\end{bmatrix}\mathbf{x} + \begin{bmatrix}\cdot\\ \mathbf{M}^{-1}\end{bmatrix}\mathbf{u}\\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{x}\,.\end{cases}$$

This the equation we're interested in, when studying the inverted pendulum on a cart.

49.5.2 Augmented system for tracking reference signal

Let $\mathbf{y}_{\text{ref}}$ a reference signal. An augmented system can be defined in order to used optimal control. Let

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

the state-space representation of the plant. Let $\mathbf{y}_{\text{ref}}$ a desired output and the integral error

$$\mathbf{e}_{\text{int}}(t) := \int_{\tau=0}^{t} \{\mathbf{y}(\tau) - \mathbf{y}_{\text{ref}}(\tau)\}\, d\tau \,,$$

as a new state with dynamical equation

$$\dot{\mathbf{e}}_{\text{int}} = \mathbf{y}(t) - \mathbf{y}_{\text{ref}}(t) = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} - \mathbf{y}_{\text{ref}} \,.$$

The optimal control is applied to the augmented system

$$\underbrace{\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{e}}_{\text{int}} \end{bmatrix}}_{\dot{\mathbf{z}}} = \underbrace{\begin{bmatrix} \mathbf{A} & \cdot \\ \mathbf{C} & \cdot \end{bmatrix}}_{\hat{\mathbf{A}}} \underbrace{\begin{bmatrix} \mathbf{x} \\ \mathbf{e}_{\text{int}} \end{bmatrix}}_{\mathbf{z}} + \underbrace{\begin{bmatrix} \mathbf{B} \\ \mathbf{D} \end{bmatrix}}_{\hat{\mathbf{B}}_u} \mathbf{u} + \underbrace{\begin{bmatrix} \cdot \\ -\mathbf{I} \end{bmatrix}}_{\hat{\mathbf{B}}_{ref}} \mathbf{y}_{ref}$$

$$\mathbf{y} = \hat{\mathbf{C}}\mathbf{z} + \hat{\mathbf{D}}_u\mathbf{u} \,.$$

Optimal control framework provides the opitmal gain matrix $\hat{\mathbf{K}}$, so that $\mathbf{u} = -\hat{\mathbf{K}}\mathbf{z}$ and the closed loop system becomes

$$\dot{\mathbf{z}} = \left(\hat{\mathbf{A}} - \hat{\mathbf{B}}_u\hat{\mathbf{K}}\right)\mathbf{z} + \hat{\mathbf{B}}_{\text{ref}}\mathbf{y}_{\text{ref}}$$
$$\mathbf{y} = \left(\hat{\mathbf{C}} - \hat{\mathbf{D}}_u\mathbf{K}\right)\mathbf{z} \,.$$

If the output of the system is the angle $\theta(t)$, with reference signal $\mathbf{y}_{\text{ref}} = \theta_{\text{ref}}$, the dynamical system is a SISO system, whose state-space representation is

…

49.5.3 Libraries

```
# Install the control library if running in Colab
try:
    import control as ct
except ImportError:
    !python3.11 -m pip install control

import numpy as np
import scipy as sp
import control as ct
import matplotlib.pyplot as plt
```

```
print(ct.__version__)
```

```
0.10.2
```

```
#> Inverted pendulum plant (linearized) equations
#> Parameters

g = 9.81
m = .1
M = .1
l = .2
cx = 1e-3
ca = 1e-3

#> Matrices of the second-order mechanical system
MM = np.array([[M+m, -m*l], [-m*l, m*l**2]])
MM_inv = np.linalg.inv(MM)
CC = np.array([[cx, .0], [.0, ca]])
KK = np.array([[.0, .0], [.0, -m*g*l]])

#> Matrices of the state-space representation
A = np.block([[np.zeros((2,2)), np.eye(2)], [-MM_inv @ KK, -MM_inv @ CC]])
B = np.block([[np.zeros((2,1))], [MM_inv @ np.array([[1.], [.0]])]])
C = np.block([np.eye(2), np.zeros((2,2))])
D = np.zeros((2,1))

sys = ct.ss(A, B, C, D)
```

```
#> Augmented system for reference input
# dx     = A x + B u
# dz = e =   y[0] - yref = C[0,:] x + D[0,:] u - yref
#  y     =   x
# d [ x ] = [ A        . ][ x ] + [ B       ] u + [ . ] yref
#   [ z ]   [ C[:2,:]  . ][ z ]   [ D[:2,:] ]     [-I ]

Aa = np.block([
    [      A, np.zeros((4, 2))],
    [C[:2,:], np.zeros((2, 2))]
])
Ba = np.block([
    [B],
    [D[:2,:]]
])
Ca = np.block([
    [C, np.zeros((2,2))]
])
Da = np.zeros((2,1))

sys_aug = ct.ss(Aa, Ba, Ca, Da)

evals = ct.poles(sys_aug)
print(evals)
```

```
[ 0.00000000e+00+0.j  0.00000000e+00+0.j  0.00000000e+00+0.j
 -1.01602637e+01+0.j  9.65526368e+00+0.j -4.99999873e-03+0.j]
```

49.5.4 Full-state feedback

State-space representation of the open-loop system from the tracking error $\mathbf{e}$ to the output $\mathbf{y}$ reads

$$\begin{cases} \dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K}_\mathbf{x})\,\mathbf{x} - \mathbf{B}\mathbf{K}_\mathbf{e}\mathbf{e}_{\text{int}} \\ \dot{\mathbf{e}}_{\text{int}} = \mathbf{e} \\ \mathbf{y} = (\mathbf{C} - \mathbf{D}\mathbf{K}_\mathbf{x})\,\mathbf{x} - \mathbf{D}\mathbf{K}_\mathbf{e}\mathbf{e}_{\text{int}} \end{cases},$$

or

$$\begin{cases} \begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{e}}_{\text{int}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{K}_\mathbf{x} & -\mathbf{B}\mathbf{K}_\mathbf{e} \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{e}_{\text{int}} \end{bmatrix} + \begin{bmatrix} \cdot \\ \mathbf{I} \end{bmatrix} \mathbf{e} \\ \mathbf{y} = \begin{bmatrix} \mathbf{C} - \mathbf{D}\mathbf{K}_\mathbf{x} & -\mathbf{D}\mathbf{K}_\mathbf{e} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{e}_{\text{int}} \end{bmatrix} \end{cases}$$

```
# LQR Weights
Q = np.diag([10, 10, 1, 1, 10, 10]) #

R = np.diag([1])
# R_values = np.linspace(0.001, .01, 4)  # [0.001, 0.01, 0.1]
# R_values = np.logspace(-4, 4, 9)

omega_freq = np.logspace(-2, 2, 1000)

fig, ax = plt.subplots(2,2, figsize=(8, 8))

# for R in R_values:

# Compute LQR Gain
# K, S, E = ct.lqr(A, B, Q, R)
Ka, Sa, Ea = ct.lqr(sys_aug, Q, R)

# #> Loop transfer function L(s) = K * (sI - A)^-1 * B
# #> Open-loop TF, L(s) = G(s) R(s), with R(s) = K
# # from the error to the output
# A_ol = np.block([
#     [     A - B @ Ka[:,:2], - B @ Ka[:,2:]],
#     [      np.zeros((1,2)), np.zeros((1, 1))]
# ])
# B_ol = np.block([[.0], [.0], [1.]])
# C_ol = np.block([[ C - D @ Ka[:,:2], - D @ Ka[:,2]]])
# D_ol = np.zeros((1,1))
# sys_ol = ct.ss(A_ol, B_ol, C_ol, D_ol)

#> Closed-loop TF
# from the disturbance signal to the output
# Simulate closed-loop response with reference input
A_cl = Aa - Ba @ Ka
B_ref = np.block([[np.zeros((4,2))], [-np.eye(2)]])
# C_cl = np.block([[np.eye(2), np.zeros((2,4))]])
# D_cl = np.zeros((2, 2))
C_cl = np.eye(6)
D_cl = np.zeros((6,2))

sys_cl = ct.ss(A_cl, B_ref, C_cl, D_cl)

evals_cl = ct.poles(sys_cl)
```

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```
# #> Frequency response
# mag_L, phase_L, omega_L = ct.frequency_response(sys_ol, omega_freq)

#> Plots
# #> Bode plots (first line) of L(s)
# ax[0,0].loglog(omega_L, mag_L, label=f'R={R}')
# ax[0,1].semilogx(omega_L, np.degrees(phase_L), label=f'R={R}')
# #> Nyquist diagram of L(s)
# ax[1,0].plot(mag_L*np.cos(phase_L), mag_L*np.sin(phase_L), label=f'R={R}')
#> Eigenvalues of the closed-loop system
ax[1,1].plot(np.real(evals_cl), np.imag(evals_cl), 'x', label=f'R={R}')

#> Critical point and circle in Nyquist plot
theta_v = np.linspace(0, 2*np.pi, 100)
# ax[1,0].plot([-1], [0], 'o', ms=5, color="black")
# ax[1,0].plot(np.cos(theta_v), np.sin(theta_v), '--', color="black")

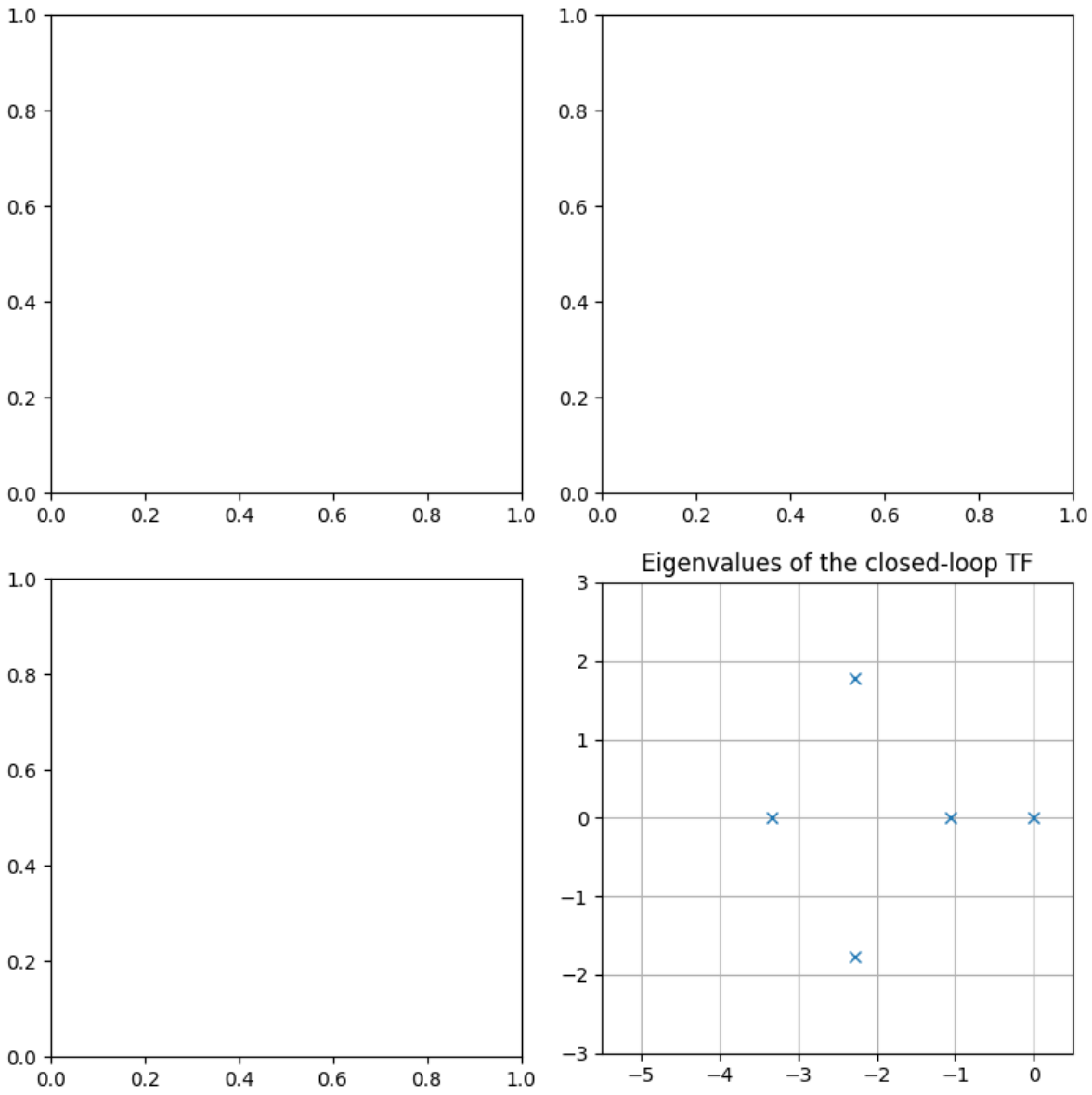
# #> Formatting plots
# ax[0,0].set_title('Bode Magnitude'); ax[0,0].grid(True); ax[0,0].legend()
# ax[0,1].set_title('Bode Phase'); ax[0,1].grid(True)
# ax[1,0].set_title('Nyquist Plot'); ax[1,0].grid(True); ax[1,0].set_aspect('equal');␣
 ↪ax[1,0].set_xlim([-5, 5]); ax[1,0].set_ylim([-5, 5])
ax[1,1].set_title('Eigenvalues of the closed-loop TF'); ax[1,1].grid(True); ax[1,1].
 ↪set_aspect('equal')
ax[1,1].set_xlim([-5.5, .5]); ax[1,1].set_ylim([-3., 3.])

fig.set_tight_layout(True)

print(evals_cl)

plt.show()
```

```
[-5.27272016e+01+0.j          -3.34300171e+00+0.j
 -2.27123174e+00+1.77694612j -2.27123174e+00-1.77694612j
 -1.05816059e+00+0.j           1.60427392e-21+0.j        ]
```




```
# Time vector
t = np.linspace(0, 10, 1000)

# Step response
# sys_cl is the system from your code (A_cl_obs, B_cl_obs, C_cl_obs, D_cl_obs)
# t_step, y_step = ct.step_response(sys_cl_obs, T=t)

theta_ref_deg = 10.
theta_ref = np.deg2rad(theta_ref_deg)
x_ref = 1.

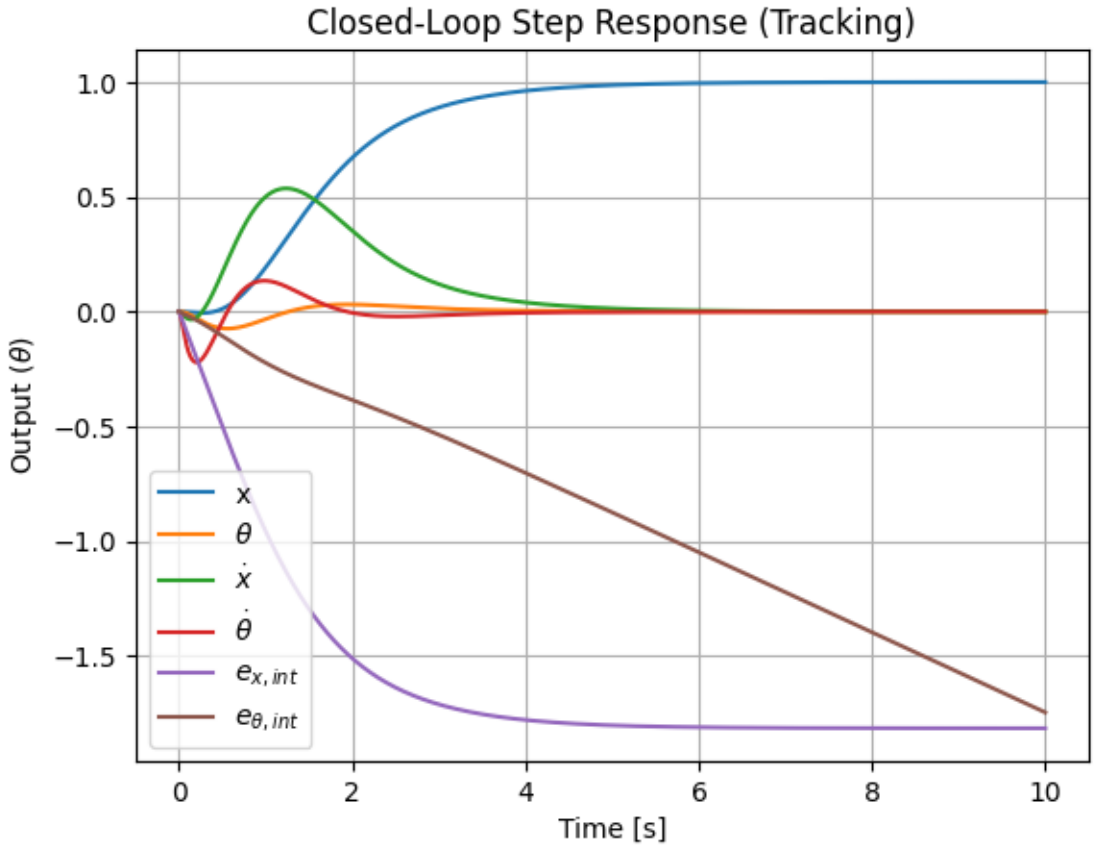
timepts = t.copy()
inputs = [ np.ones(len(t)) * x_ref, np.ones(len(t)) * theta_ref ]

t_step, y_step = ct.forced_response(sys_cl, timepts, inputs)
```

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```
plt.figure()
plt.plot(t_step, y_step.T, label=["x", r"$\theta$", r"$\dot{x}$", r"$\dot{\theta}$", r
↪"$e_{x,int}$", r"$e_{\theta,int}$"])
plt.title('Closed-Loop Step Response (Tracking)')
plt.xlabel('Time [s]')
plt.ylabel(r"Output ($\theta$)")
plt.legend()
plt.grid(True)
```




```
#> Optimal state estimator
Edd = .1 * np.eye(1)        # Process noise covariance
Err = .1 * np.eye(2)        # Measurement noise covariance

# If Edd = .01 * np.eye(2) ct.lqe() function returns an error,
# returning "non symmetric QN - i.e. Edd - matrix". There's no
# way to make it see as symmetric...

#> Method 1.
Bd = B.copy()
L, P, E = ct.lqe(A, Bd, C, Edd, Err)

print(L)

"""
#> Method 2.
```

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```
# Use LQR duality: lqe(A, B, C, V, W) is dual to lqr(A.T, C.T, V, W)
# Note: we use C.T because it replaces B in the dual problem
L_transposed, P, E = ct.lqr(A.T, C.T, V, W)

# The actual observer gain is the transpose of the 'feedback' result
L = L_transposed.T
"""
```

```
[[  3.21284488   1.75111149]
 [  1.75111149  19.74082023]
 [  6.69438184  21.04491708]
 [ 19.14950954 196.38318735]]
```

```
"\n#> Method 2.\n# Use LQR duality: lqe(A, B, C, V, W) is dual to lqr(A.T, C.T, V,␣
↪W)\n# Note: we use C.T because it replaces B in the dual problem\nL_transposed,␣
↪P, E = ct.lqr(A.T, C.T, V, W)\n\n# The actual observer gain is the transpose of␣
↪the 'feedback' result\nL = L_transposed.T\n"
```

```
#> Optimal controller
# R = 1e-4
# Ka, Sa, Ea = ct.lqr(sys_aug, Q, R)

A_cl_obs = np.block([
    [ A - B @ Ka[:,:4], - B @ Ka[:,:4],   - B @ Ka[:,4:]],
    [  np.zeros((4,4)),      A - L @ C, np.zeros((4, 2))],
    [ C - D @ Ka[:,:4], - D @ Ka[:,:4],   - D @ Ka[:,4:]]
])
B_cl_obs = np.block([[np.zeros((4,2))], [np.zeros((4,2))], [-np.eye(2)]])
C_cl_obs = np.block([ C - D @ Ka[:,:4], - D @ Ka[:,:4], - D @ Ka[:,4:]])
D_cl_obs = np.zeros((2, 2))

sys_cl_obs = ct.ss(A_cl_obs, B_cl_obs, C_cl_obs, D_cl_obs)
```

```
# Time vector
t = np.linspace(0, 10, 1000)

# Step response
# sys_cl is the system from your code (A_cl_obs, B_cl_obs, C_cl_obs, D_cl_obs)
# t_step, y_step = ct.step_response(sys_cl_obs, T=t)

theta_ref_deg = 5.
theta_ref = np.deg2rad(theta_ref_deg)
x_ref = 1.

timepts = t.copy()
inputs = [ np.ones(len(t)) * x_ref, np.ones(len(t)) * theta_ref ]

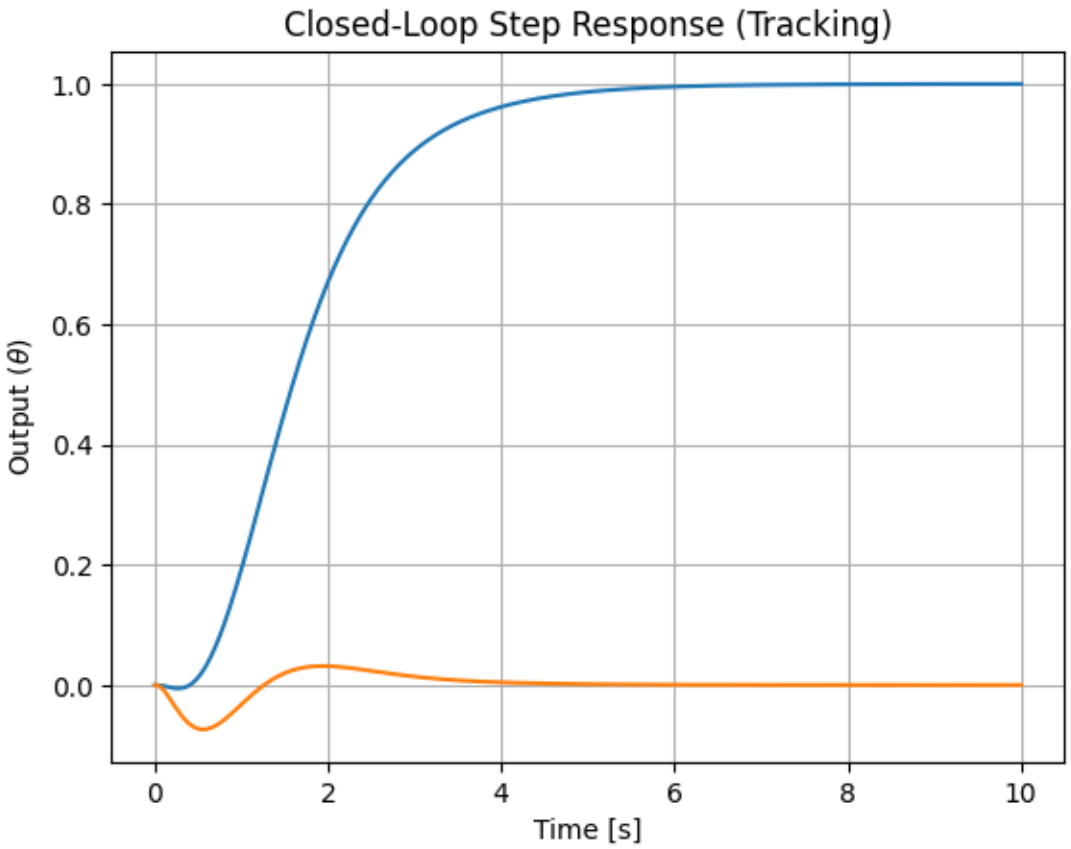
t_step, y_step = ct.forced_response(sys_cl_obs, timepts, inputs)

plt.figure()
plt.plot(t_step, y_step.T)
plt.title('Closed-Loop Step Response (Tracking)')
plt.xlabel('Time [s]')
```

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```
plt.ylabel(r"Output ($\theta$)")
plt.grid(True)
```



Non-linear problem

```
from scipy.integrate import solve_ivp
from scipy.interpolate import interp1d

# solve_ivp(fun, t_span, y0, method="RK45", ..., args=tuple)

# fun(t, z, args=tuple)
def f(t, z, args):
    """
    State, z = [ x, vareps, e_int ]
    """

    #> args is supped to be a 1-element tuple with a dict
    args_di = args

    #> System parameters
    g, l = args_di["g"], args_di["l"]
    m, M = args_di["m"],  args_di["M"]
    cx, ca = args_di["cx"], args_di["ca"]
```

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```
    #> Linearized system matrices
    A, B, C, D = args_di["A"], args_di["B"], args_di["C"], args_di["D"]

    #> Optimal control matrix
    Kx, Ke = args_di["Kx"], args_di["Ke"]
    #> State-estimator L matrix
    L = args_di["L"]

    #> Reference input: select the proper y_ref for the current time 't'
    y_ref_func = args_di["y_ref_func"]
    yref = y_ref_func(t) # Interpolates to get the value at time t
    # pos_ref, angle_ref = yref[0], yref[1]

    #> Motor physical limits
    motor_clipping = args_di["motor_clipping"]
    u_max = args_di["u_max"]

    #> State variables
    x, th, v, om, eps, e_int = z[0], z[1], z[2], z[3], z[4:8], z[8:]
    state = z[:4]

    #> Feed-back control
    u = - Kx @ state - Kx @ eps - Ke @ e_int

    if ( motor_clipping ):
        u = np.clip(u, -u_max, u_max)

        #> Need for anti-windup
        # if (u_ideal > u_max and (C @ x + D @ u - yref) > 0) or (u_ideal < -u_max␣
↪and (C @ x + D @ u - yref) < 0):
        #     de_int = 0 # Stop integrating...
        # else:
        #     de_int = C @ x + D @ u - yref

    M_vel = np.array([
         [            M+m, -m*l*np.cos(th)],
         [-m*l*np.cos(th),          m*l**2]
    ])

    dx     = v
    dth    = om
    d_vel  = np.linalg.inv(M_vel) @ \
             np.array([
                  [-m*l*om**2*np.sin(th) + u[0] - cx*v],
                  [ m*g*l*np.sin(th) - ca*om]
             ])
    deps   = ( A - L @ C ) @ eps
    de_int = C @ state + D @ u - yref

    return np.concatenate(([dx], [dth], d_vel.ravel(), deps, de_int))
```

```
#> Reference signal
t_span = [0, 20]
t_eval = np.linspace(t_span[0], t_span[1], 200)
```

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```
ref_theta_values = np.where(t_eval < 10, np.radians(0), np.radians(5))

# Transition settings
start_time = 5
duration = 3
end_time = start_time + duration
start_val = 0
end_val = 2
ref_x_values = np.piecewise(t_eval,
    [t_eval < start_time, (t_eval >= start_time) & (t_eval <= end_time), t_eval > end_
 ↪time],
    [start_val,
     lambda t: start_val + (end_val - start_val) * (t - start_time) / duration,
     end_val]
)

y_ref_values = np.column_stack((ref_x_values, ref_theta_values))

#> Create an interpolator to pass to the solve_ivb function
y_ref_func = interp1d(t_eval, y_ref_values, axis=0, kind='linear', fill_value=
 ↪"extrapolate")
```

```
# 1. Bundle all parameters into the dictionary
sim_config = {
    "g": 9.81, "l": l, "m": m, "cx": cx,       # Example physical params
    "M": M, "ca": ca,
    "A": A, "B": B, "C": C,  "D": D,           # Linearized matrices
    "Kx": Ka[:, :4], "Ke": Ka[:, 4:],          # Optimal gains (Kx is 1x2, Ke is 1x1)
    "L": L,                                    # Observer gain
    "y_ref_func": y_ref_func,                  # Ref input interpolator function
    "motor_clipping": False, "u_max": .05      # Motor clippint
}

# 2. Set Initial Conditions [x, theta, v, omega, eps_theta, eps_omega, e_int]
# Example: System starts at 2 degrees, observer thinks it's at 0
# Error is defined to be eps = x_hat - x
z0 = [2., np.radians(10), .0, .0, -2., -np.radians(10), .0, .0, .0, .0]

# 3. Execute the simulation
from scipy.integrate import solve_ivp
sol = solve_ivp(
    f,
    t_span,
    z0,
    args=(sim_config,),
    t_eval=t_eval,
    method='BDF'      # Default: "RK45" very slow for stiff problems(?)
)
```

```
import matplotlib.pyplot as plt

#> Output and control effort
fig, ax = plt.subplots(3, 1, figsize=(10, 10))

#> Estimation error, eps = x_hat - x, x_hat = x + eps
```

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```
hat_x     = sol.y[0] + sol.y[4]
hat_theta = sol.y[1] + sol.y[5]

# Plot True Pendulum Angle (convert back to degrees)
ax[0].plot(sol.t, sol.y[0]       , label='True $x$ (Non-linear)', linewidth=2)
 ↪              # output
ax[0].plot(sol.t, hat_x    , '--', label='Estimated $\\hat{x}$ (Observer)')
 ↪              # estimated output
ax[0].plot(sol.t, y_ref_func(sol.t)[:,0], '--', color="black", label=r'Reference $x_
 ↪{ref}$')        # reference

ax[1].plot(sol.t, sol.y[1]       , label='True $\\theta$ (Non-linear)', linewidth=2)
 ↪              # output
ax[1].plot(sol.t, hat_theta, '--', label='Estimated $\\hat{\\theta}$ (Observer)')
 ↪              # estimated output
ax[1].plot(sol.t, y_ref_func(sol.t)[:,1], ':' , color="black", label=r'Reference $\
 ↪theta_{ref}$')   # reference

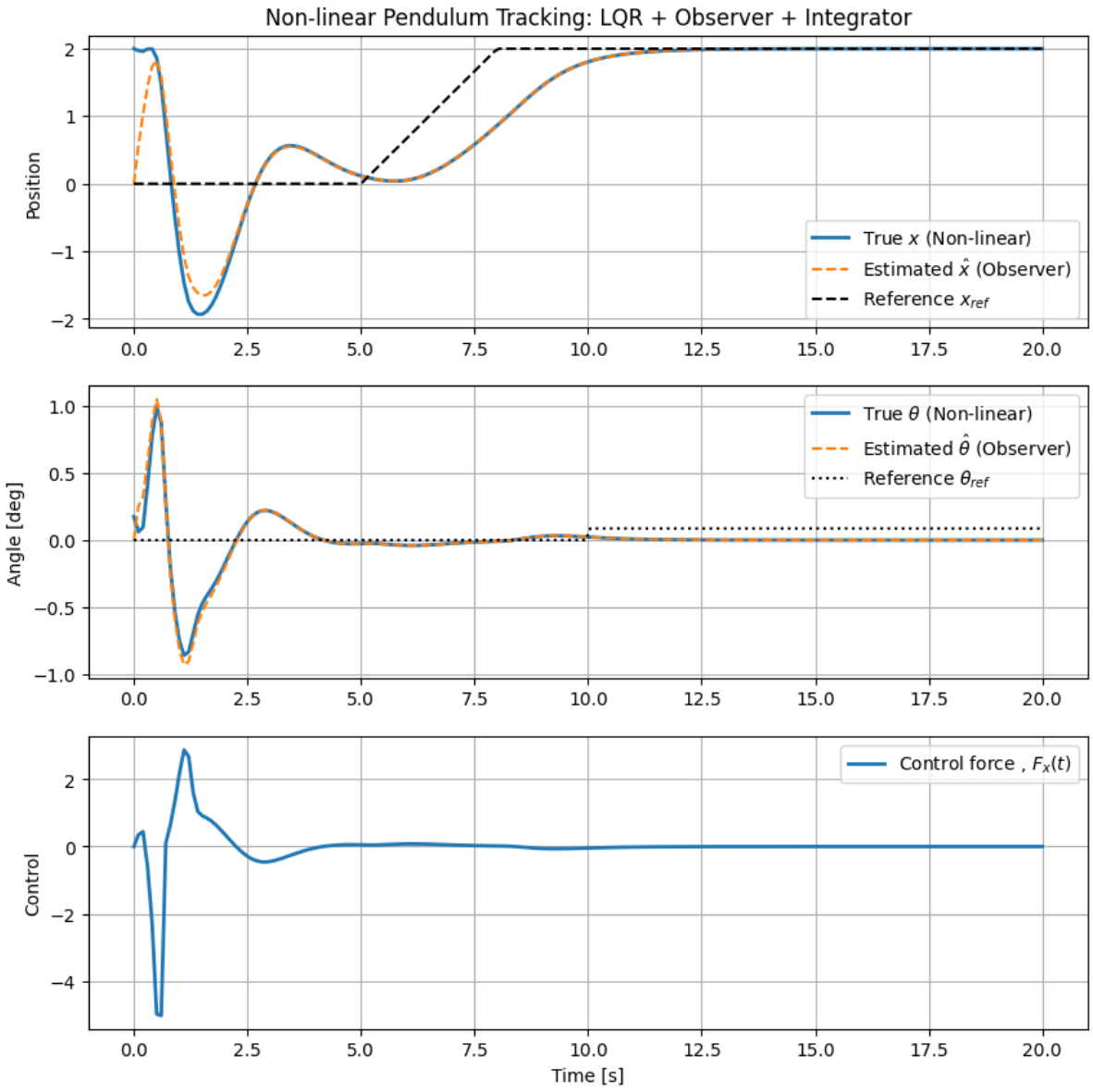
ax[0].set_title("Non-linear Pendulum Tracking: LQR + Observer + Integrator")
ax[0].set_ylabel("Position")
# ax[0].set_xlabel("Time [s]")
ax[1].set_ylabel("Angle [deg]")

u = -Ka[:,:4] @ sol.y[0:4] - Ka[:,:4] @ sol.y[4:8] - Ka[:,4:] @ sol.y[8:]
ax[2].plot(sol.t, u[0], label=r"Control force , $F_x(t)$", linewidth=2)      # output
ax[2].set_ylabel("Control")
ax[2].set_xlabel("Time [s]")

ax[0].legend();  ax[0].grid()
ax[1].legend();  ax[1].grid()
ax[2].legend();  ax[2].grid()

plt.show()

#>
```


Non-linear Pendulum Tracking: LQR + Observer + Integrator
Position
True x (Non-linear)
Estimated $\hat{x}$ (Observer)
Reference x_{ref}
Angle [deg]
True θ (Non-linear)
Estimated $\hat{\theta}$ (Observer)
Reference θ_{ref}
Control
Control force , $F_x(t)$
Time [s]

Part XII

Partial Differential Equations

CHAPTER

FIFTY

INTRODUCTION TO PARTIAL DIFFERENTIAL EQUATIONS

Partial differential equations usually comes from balance equations in **continuum mechanics**. Integral equations are the most general form of these equations, and an equivalent differential problem only exists if the fields involved in the equations are regular enough, for their derivatives to exist - and to apply theorems requiring some regularity of the functions.

Classical numerical methods:

- **FVM**: directly solves the **integral problem**, solving integral balance equations for cells in which the domain is divided
- **FDM**: given the problem in **differential form**, FDM directly approximates space derivatives of the **strong formulation** of the problem
- **FEM**: given the problem in **differential form**, FEM projects the **weak formulation** of the problem on a finite-dimensional space
- **BEM**: *integro-differential equation*, *singularities*,…
- **Spectral methods**,…
- **SEM**,…

50.1 Examples

In Physics:

- Advection equation

$$\partial_t u + \vec{a} \cdot \nabla u = f$$

- Diffusion equation

$$\partial_t u - \nu \nabla^2 u = f$$

- Hyperbolic equation/system of equations

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F}(\mathbf{u}) = \mathbf{f}$$

- Wave equation

$$\frac{1}{c^2} \partial_{tt} u - \nabla^2 u = f$$

50.2 Balance equations in physics

- Small-strain continuum mechanics

$$\rho\partial_{tt}\vec{s} = \rho_0\vec{g} + \nabla\cdot\boldsymbol{\sigma}(\boldsymbol{\varepsilon})$$

- Heat conduction
- Fluid dynamics
  - Navier-Stokes for compressible fluids (conservative or convective equations)

$$\{$$

  - Navier-Stokes for incompressible fluids (convective form,…)

$$\begin{cases}\rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\vec{u} - \mu\nabla^2\vec{u} + \nabla P = \rho\vec{g}\\ \nabla\cdot\vec{u} = 0\end{cases}$$

todo

- Different forms of equations may be more or less convenient for different solution approaches
- Most of the physical laws comes from integral balance equation of the form

$$\frac{d}{dt}\int_{V_t}\rho\mathbf{u} = \int_{V_t}\rho\mathbf{r} + \oint_{\partial V_t}\hat{n}\cdot\mathbf{T}(\mathbf{u})$$

  whose local - differential - form (in case of differentiable functions) readily follows from the application of Reynolds' transport theorem and divergence theorem to transform time derivative and boundary terms

$$\begin{aligned}\partial_t(\rho\mathbf{u}) + \nabla\cdot(\rho\mathbf{u}\vec{u}) &= \rho\mathbf{r} + \nabla\cdot\mathbf{T}(\mathbf{u})\\ \partial_t(\rho\mathbf{u}) + \nabla\cdot\mathbf{F}(\mathbf{u}) &= \rho\mathbf{r}\,,\end{aligned}$$

  and the physical meaning of each term is evident and readily expalinable as flux or volume or surface sources.

- Further manipulation/simplification may cover the clear meaning of the terms of the differential equation. As an example, the conservative form of Navier-Stokes equations for incompressible fluids with constant and uniform density read

$$\begin{cases}\partial_t(\rho\vec{u}) + \nabla\cdot(\rho\vec{u}\otimes\vec{u}) = \rho\vec{g} + \nabla\cdot\mathbb{T}\\ \nabla\cdot\vec{u} = 0\,,\end{cases}$$

  where the stress tensor for a Newtonian fluid reads

$$\begin{aligned}\mathbb{T} &= -p\mathbb{I} + 2\mu\mathbb{D} + \lambda(\nabla\cdot\vec{u})\,\mathbb{I}\\ &= -p\mathbb{I} + \mu(\nabla\vec{u} + \nabla^T\vec{u}) + \lambda(\nabla\cdot\vec{u})\,\mathbb{I}\end{aligned}$$

  Using the incompressibility constraint $\nabla\cdot\vec{u}$, and treating the density ρ as a constant and uniform parameter, the convective form of the Navier-Stokes equations reads

$$\begin{cases}\rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\,\vec{u} = \rho\vec{g} - \nabla P + 2\mu\nabla\cdot\mathbb{D}\\ \nabla\cdot\vec{u} = 0\,.\end{cases}$$

  The divergence of the viscous stress tensor becomes

$$2\mu\nabla\cdot\mathbb{D} = \mu\nabla\cdot(\nabla\vec{u} + \nabla^T\vec{u}) = \mu\left(\nabla^2\vec{u} + \nabla\underbrace{(\nabla\cdot\vec{u})}_{=0}\right) = \mu\nabla^2\vec{u}\,,$$

so that one of the most common form of incompressible Navier-Stokes equations follows

$$\begin{cases} \rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\vec{u} - \mu\nabla^2\vec{u} + \nabla P = \rho\vec{g} \\ \nabla\cdot\vec{u} = 0\,. \end{cases}$$

It should be evident that in the latter form of Navier-Stokes equations no divergence explicitly appears, so that the right expression of surface source terms can't be found immediately. In momentum equation, surface source terms come from surface stress acting on the boundary of the domain, whose expression reads

$$\begin{aligned} \vec{t}_n &= \hat{n}\cdot\mathbb{T} = \\ &= \hat{n}\cdot(-P\mathbb{I} + 2\mu\mathbb{D}) = \\ &= \hat{n}\cdot\left(-P\mathbb{I} + \mu\left(\nabla\vec{u} + \nabla^T\vec{u}\right)\right) = \\ &= -P\hat{n} + \hat{n}\cdot\left(\mu\left(\nabla\vec{u} + \nabla^T\vec{u}\right)\right) = \end{aligned}$$

As an example, in weak formulation of incompressible Navier-Stokes problem the **natural boundary condition** arising in the method depends on the expression of the strong formulation of the NS problem. If one needs to prescribe stress boundary conditions, it could be an idea to start from NS equations w/o extra simplifications.

CHAPTER

FIFTYONE

ELLIPTIC EQUATIONS

51.1 Poisson equation

Given the volume density source $f(\vec{r})$ and the diffusivity $\nu(\vec{r})$, Poisson equation for the scalar field $\phi(\vec{r})$ reads

$$-\nabla \cdot (\nu \nabla \phi) = f \qquad \vec{r} \in V$$

with proper boundary conditions on ∂V. As an example, tipical boundary conditions are:

$\phi(\vec{r}) = g(\vec{r})$	$\vec{r} \in S_D$	esserntial - Dirichlet b.c.
$\nu \hat{n} \cdot \nabla \phi(\vec{r}) = h(\vec{r})$	$\vec{r} \in S_N$	natural - Neumann b.c.
$a\phi(\vec{r}) + \nu \hat{n} \cdot \nabla \phi(\vec{r}) = b(\vec{r})$	$\vec{r} \in S_R$	Robin b.c.

Numerical methods

1-dimensional Poisson equation:

- *Finite Element Methods*
- *Finite Volume Methods*
- *Finite Difference Methods*
- *Boundary Element Methods*
- *Spectral Methods* and *Spectral Element Methods*

51.1.1 Weak formulation

For $\forall w \in$ ... (functional space, recall some results about existence and uniqueness of the solution, Lax-Milgram theorem,…)

$$\begin{aligned} 0 &= \int_V w \, \{\nabla \cdot (\nu \nabla \phi) + f\} = \\ &= \oint_{\partial V} w \hat{n} \cdot (\nu \nabla \phi) + \int_V \{-\nu \nabla \vec{w} \cdot \nabla \phi + wf\} = \end{aligned}$$

Splitting boundary contribution as the sum from single contributions from different regions, and applying boundary conditions, setting $w = 0$ for $\vec{r} \in S_D$ (see the ways to prescribe essential boundary conditions),

$$0 = \int_{S_D} \underbrace{w}_{=0} \hat{n} \cdot (\nu \nabla \phi) + \int_{S_N} w \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=h} + \int_{S_R} w \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=b-a\phi} + \int_V \{-\nu \nabla \vec{w} \cdot \nabla \phi + wf\} \ .$$

and rearranging the equation separating terms containing unknowns from known contributions,

$$\int_V \nu \nabla w \cdot \nabla \phi + \int_{S_R} wa\phi = \int_V wf + \int_{S_N} wh + \int_{S_R} wb \qquad \forall w \in \dots \ ,$$

and $\phi = g$, for $\vec{r} \in S_D$.

Different ways to prescribe essential boundary conditions

Strong formulation.

Using Lagrance multiplier - weak formulation of essential boundary conditions. Adding a the essential boundary condition as a constraint with Lagrange multipliers in the weak formulation of the problem,

$$\dots + \int_{S_D} w_D(\phi - g) \ ,$$

…

51.1.2 Existence and uniqueness

Assuming two solutions $u_1(\mathbf{r})$, $u_2(\mathbf{r})$ exist for the Poisson problem

$$\begin{cases} -\nabla \cdot (\nu \nabla u) = f & \mathbf{r} \in V \\ u|_{S_D} = g \\ \nu \hat{n} \cdot \nabla u|_{S_N} = h \end{cases}$$

Their difference $\delta u(\mathbf{r}) := u_2(\mathbf{r}) - u_1(\mathbf{r})$ then satisfies the homogeneous problem

$$\begin{cases} -\nabla \cdot (\nu \nabla \delta u) = 0 & \mathbf{r} \in V \\ \delta u|_{S_D} = 0 \\ \nu \hat{n} \cdot \nabla \delta u|_{S_N} = 0 \end{cases}$$

The norm of the gradient of the difference of the solution reads

$$\begin{aligned} \int_V \nu |\nabla \delta u|^2 &= \int_V \nu \nabla \delta u \cdot \nabla \delta u = \\ &= \int_V \nabla \cdot (\delta u \, \nu \nabla \delta u) - \int_V \delta u \, \underbrace{\nabla \cdot (\nu \nabla \delta u)}_{=0} = \\ &= \oint_{\partial V} \delta u \, \nu \hat{n} \cdot \nabla \delta u = \\ &= \int_{S_D} \underbrace{\delta u}_{\delta u|_{S_D}=0} \nu \hat{n} \cdot \nabla \delta u + \int_{S_N} \delta u \, \underbrace{\nu \hat{n} \cdot \nabla \delta u}_{\nu \hat{n} \cdot \nabla \delta u|_{S_N}=0} = \\ &= 0 \ . \end{aligned}$$

If $\nu(\mathbf{r}) > 0$ for $\forall \mathbf{r} \in V$[1], it follows that

$$\nabla \delta u = \nabla (u_2 - u_1) = 0 \ ,$$

[1] As an example, this condition appears in physical systems with non-zero and positive diffusion coefficients in diffusion problems, like thermal conduction or specie diffusion via Fick's law.

and thus the two solutions differs at most by an additive constant,

$$u_2(\mathbf{r}) - u_1(\mathbf{r}) = c \ .$$

If the Dirichlet boundary has non-null dimension, it forces the value of the functions to coincide on that boundary, $u_2(\mathbf{r}) = u_1(\mathbf{r})$ on S_D, and thus sets the value of the additive constant c to be zero, $c = 0$, and

$$u_2(\mathbf{r}) = u_1(\mathbf{r}) \ ,$$

thus proving the uniqueness of the solution of the Poisson problem, with non-negative diffusion coefficient and non-zero dimension of Dirichlet boundary.

CHAPTER

FIFTYTWO

PARABOLIC EQUATIONS

52.1 Heat equation

Heat equation for a scalar field $\phi(\vec{r},t)$ can be interpreted as the unsteady equation of a *Poisson equation*,

$$\partial_t \phi - \nabla \cdot (\nu \nabla \phi) = f \qquad (\vec{r},t) \in V \times [0,T] \,,$$

with proper boundary and initial conditions, $\phi(\vec{r},0) = \phi_0(\vec{r})$. Common boundary conditions are the same as the one discussed for Poisson problem.

52.1.1 Weak formulation

For $\forall w \in$ ... (functional space, recall some results about existence and uniqueness of the solution, Lax-Milgram theorem,…)

$$\begin{aligned} 0 &= \int_V w \, \{-\partial_t \phi + \nabla \cdot (\nu \nabla \phi) + f\} = \\ &= \oint_{\partial V} w \hat{n} \cdot (\nu \nabla \phi) + \int_V \{-\partial_t \phi - \nu \nabla \vec{w} \cdot \nabla \phi + wf\} = \end{aligned}$$

Splitting boundary contribution as the sum from single contributions from different regions, and applying boundary conditions, setting $w = 0$ for $\vec{r} \in S_D$ (see the ways to prescribe essential boundary conditions),

$$0 = \int_{S_D} \underbrace{w}_{=0} \hat{n} \cdot (\nu \nabla \phi) + \int_{S_N} w \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=h} + \int_{S_R} w \underbrace{\hat{n} \cdot (\nu \nabla \phi)}_{=k-\phi} + \int_V \{-\partial_t \phi - \nu \nabla \vec{w} \cdot \nabla \phi + wf\} \,.$$

and rearranging the equation separating terms containing unknowns from known contributions,

$$\int_V w \partial_t \phi + \int_V \nu \nabla w \cdot \nabla \phi + \int_{S_R} w \phi = \int_V wf + \int_{S_N} wh + \int_{S_R} wk \qquad \forall w \in \ldots \,,$$

and $\phi = g$, for $\vec{r} \in S_D$.

CHAPTER

FIFTYTHREE

HYPERBOLIC EQUATIONS

- Integral equations
- Jump conditions
- Differential equations
- Method of characteristics: domains of influence
- "Inviscid" limit, multiple solutions, entropy condition and unique physical solution

Some useful script on GDrive `basics-books/colabs/math/pde/fvm`. So far 1-dimensional problem only: linear equation, linear system (linearized p-sys), non linear system (p-sys).

Problems

1-dimensional domain

- *Scalar linear*
- *Vector linear*
- *Scalar non-linear*
- Vector non-linear

n-dimensional domain

- Scalar linear
- Vector linear
- Scalar non-linear
- Vector non-linear

todo

- hyperbolic problems as the **ideal limit of dissipative** (parabolic?) problems. In non-linear problems, multiple (infinite?) solutions of the hyperbolic problem exist, but there's only one (? Prove it! Or just provide the idea) solution that is the limit of the solution of the dissipative problem, in the *ideal* limit: when do jumps occur? When do rerefaction fans occur?
  - show multiple solutions
  - show limit of the dissipative solution

  on some simple problems, maybe starting from a scalar problem in 1-dimesnional domain, and then generalize to "more difficult" problems
- domains of **dependence** and **influence**. Show them in both unsteady and steady problems?

  - Example with compressible flows (or the p-sys) comparing unsteady and steady problems, and discussing the nature of the equations in the subsonic and supersonic regimes

53.1 Hyperbolic problems and conservation laws

todo

- In fluid dynamics: no viscosity, no heat conduction
- In this limit, the equation may produce non-physical equations, i.e. solutions that are not the limit of the solution of the full problem in the limit of $\mathbf{F}^d(\nabla \mathbf{u}) \to \mathbf{0}$
- How many equations in how many variables?
- Linear diffusion flux $\mathbf{F}^d = ...$
- Entropy condition, to select the physical solution from all the possible solutions

53.1.1 Integral equations

Many laws can be written as integral balance equations for a material volume V_t,

$$\frac{d}{dt}\int_{V_t} \mathbf{u} + \oint_{\partial V_t} \hat{\mathbf{n}} \cdot \mathbf{F}(\mathbf{u}, \nabla \mathbf{u}) = \int_{V_t} \mathbf{s}$$
$$\frac{d}{dt}\int_{V_t} u_i + \oint_{\partial V_t} n_j F_{ji} = \int_{V_t} s_i$$

or for generic volume v_t, using *Reynolds' transport theorem*,

$$\frac{d}{dt}\int_{v_t} \mathbf{u} + \oint_{\partial v_t} \mathbf{u}\,(\mathbf{v} - \mathbf{v}_b) \cdot \hat{\mathbf{n}} + \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{F}(\mathbf{u}, \nabla \mathbf{u}) = \int_{v_t} \mathbf{s}\,,$$

being $\mathbf{v}$ the velocity of the continuous medium and $\mathbf{v}_b$ the velocity of the boundary ∂v_t.

The flux can be often written as a sum of a *conservative part* and a *diffusion part* as

$$\mathbf{F}(\mathbf{u}, \nabla \mathbf{u}) = \mathbf{F}(\mathbf{u}) + \mathbf{F}^d(\nabla \mathbf{u})\,,$$

with slight abuse of notation in writing the conservative flux with the same letter as the full flux.

Under some conditions, the diffusion part goes to zero.

53.1.2 Jump relations

Collapsing the volume v_t on a surface, volume terms vanish faster than the surface terms. Jump conditions follow from the arbitrariety of this surface

$$\begin{aligned}\mathbf{0} &= \hat{\mathbf{n}}_1 \cdot [(\mathbf{v}_1 - \mathbf{v}_b)\mathbf{u} + \mathbf{F}_1] + \hat{\mathbf{n}}_2 \cdot [(\mathbf{v}_2 - \mathbf{v}_b)\mathbf{u} + \mathbf{F}_2] = \\ &= \hat{\mathbf{n}} \cdot [(\mathbf{v} - \mathbf{v}_b)\mathbf{u} + \mathbf{F}] = \\ &= \left[v_n^{rel}\mathbf{u} + \mathbf{F}_n\right]\,.\end{aligned}$$

53.1.3 Differential equations

If the functions are differentiable, with divergence theorem and exploiting the arbitrariety of the volume v_t

$$\partial_t u_i + \partial_j F_{ji} = s_i$$
$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F} = \mathbf{s} \, .$$

53.2 Hyperbolic problems - dimensions

Hyperbolic problems often come from a small-amplitude linearization, or as the non-diffusion (or inviscid) limit of a more general problem.

As a result of these simplification, these problems may experience **shocks** (i.e. discontinuity in the solution, where the differential equations stop to hold, and integral equations and jump conditions are required). **todo** *classification of discontinuities on the massflow across the surface*

The very nature of these problem also suggest methods for the solution or the analysis of these equations, like **characteristic method**.

53.2.1 Scalar linear

1-dimensional

$$\partial_t u(x,t) + a\partial_x u(x,t) = f(x,t)$$

Caracteristic method. $U(t) = u(X(t),t)$, with the caracteristic curves $X(t)$ defined as those curves where the PDE becomes a ODE. Evaluating the time derivative of the function $u(X(t),t)$, the hyperbolic equation can be recast as

$$\frac{dU}{dt} + \left[a(X(t),t) - \frac{dX}{dt} \right] \partial_x u = f(X(t),t) \, .$$

The equation of characteristic lines is

$$\frac{dX}{dt} = a(X(t),t) \, ,$$

and the PDE on characteristic line becomes the ODE

$$\frac{dU}{dt}(X(t),t) = f(X(t),t) \, .$$

53.2.2 Scalar non-linear

53.2.3 System linear

1-dimensional

$$\mathbf{u}(x,t)$$

$$\partial_t \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} = \mathbf{f}$$

Method of characteristics

Characteristics. $\mathbf{U}(t) = \mathbf{u}(X(t), t)$

$$\frac{d\mathbf{U}}{dt} - \frac{dX}{dt}\partial_x \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} = \mathbf{f}$$

In order to get the equations of characteristic lines where PDE turns into ODEs, the eigenproblem

$$\mathbf{A}\partial_x \mathbf{u} = \frac{dX}{dt}\partial_x \mathbf{u} \ ,$$

holds. This problem has non trivial solution if $\frac{dX}{dt}$ and $\partial_x \mathbf{u}$ are pairs of eigenvalues and (right) eigenvectors of the array $\mathbf{A}$.

Diagonalization.

$$\mathbf{A} = \mathbf{R}\mathbf{\Lambda}\mathbf{L}$$

$$\mathbf{L}\left[\partial_t \mathbf{u} + \mathbf{R}\mathbf{\Lambda}\mathbf{L}\partial_x \mathbf{u}\right] = \mathbf{L}\mathbf{f}$$

Since $\mathbf{L} = \mathbf{R}^{-1}$, and defining the **characteristic variables** by $d\mathbf{q} = \mathbf{L}d\mathbf{u}$ - in linear problems matrix $\mathbf{A}$ is constant, and so its spectral decomoposition, and thus $\mathbf{q} = \mathbf{L}\mathbf{u}$ - , it's possible to recast the original problem in diagonal form

$$\partial_t \mathbf{q} + \mathbf{\Lambda}\partial_x \mathbf{q} = \mathbf{L}\mathbf{f}$$

$$\partial_t q_i + \Lambda_i \partial_x q_i = \sum_k L_{ik}\, f_k =: F_i \ .$$

Thus, on the ith family of characteristic lines, $\dfrac{dX}{dt} = \lambda_i$, $Q_i(t) = q_i(x(t), t)$ evolves as

$$\frac{dQ_i}{dt} = F_i \ .$$

If $F_i = \left[\mathbf{L}\mathbf{f}\right]_i = 0$, the characteristic variable q_i is constant along the characteristic lines. Once the characteristic variables are determined, the conservative variables are evalauted as $\mathbf{u}(x, t) = \mathbf{R}\mathbf{q}(x, t)$.

Domain of influence and domain of dependence

Riemann problem

A Riemann problem is defined as the evolution of the initial state

$$\mathbf{u}(x, t_0) = \begin{cases} \mathbf{u}_a \ , & x < x_0 \\ \mathbf{u}_b \ , & x > x_0 \end{cases}$$

This problem is quite useful in quite a wide range of numerical methods for hyperbolic problems - Godunov schemes in Finite Volume Methods -, to evaluate the **boundary state** to be used numerical flux.

For linear problems, the matrix $\mathbf{A}$ is constant ad so it is its spectral decomposition, $\mathbf{A} = \mathbf{R}\mathbf{\Lambda}\mathbf{L}$, and the solution of a Riemann problem of an homogeneous linear hyperbolic system can be easily determined analytically with the method of characteristics,

Let's change the origin of space and time, so that the initial state is in $t = 0$, and the jump in the initial condiiton in $x = 0$. Each charactersitic variable $q_k(x, t)$ is constant on its family of characteristic lines, $x = X_k(t) = x_{0,k} + \lambda_k t$.

$$q_k(x, t) = q_k(x_{0,k} + \lambda_k t, t) = q_k(x_{0,k}, 0) = q_k(x - \lambda_k t, 0) = L_{ki} u_j(x - \lambda_k t, 0) \ .$$

Thus, the solution in conservative variables $\mathbf{u}(x,t)$ in x at time t reads

$$\begin{aligned}\mathbf{u}(x,t) &= \mathbf{R}\mathbf{q}(x,t)\\ u_i(x,t) &= R_{ik}q_k(x,t) = R_{ik}q_k(x-\lambda_k t,0) = R_{ik}L_{kj}u_j(x-\lambda_k t,0)\end{aligned}$$

In a Riemann problem for a N-dimensional linear system the solution shows $N+1$ homogeneous regions (at most, in general the same number as the number of the non-coincident eigenvalues $+1$), delimited by the characteristic lines with origin in the discontinuity. Sorting the eigenvalues in increasing order

$$\lambda_1 > \lambda_2 > \cdots > \lambda_N \,,$$

and defining the homogeneous regions

$$\begin{aligned}S_0 &: \frac{x}{t} \in (-\infty, \lambda_1)\\ S_1 &: \frac{x}{t} \in (\lambda_1, \lambda_2)\\ &\cdots\\ S_i &: \frac{x}{t} \in (\lambda_i, \lambda_{i+1})\\ &\cdots\\ S_{N-1} &: \frac{x}{t} \in (\lambda_{N-1}, \lambda_N)\\ S_N &: \frac{x}{t} \in (\lambda_N, +\infty)\end{aligned}$$

the solution is in the S_i region is

$$u_i(x,t) = \sum_{\lambda_k > \frac{x}{t}} R_{ik}q_{a,k} + \sum_{\lambda_k < \frac{x}{t}} R_{ik}q_{b,k}$$

Example 53.2.1 (Linear(ized) P-system)

The linear(ized) P-system around a uniform reference state $\overline{\rho}$, $\overline{u}$ in convective form reads

$$\partial_t \begin{bmatrix}\rho\\ u\end{bmatrix} + \begin{bmatrix}\overline{u} & \overline{\rho}\\ \frac{a^2}{\overline{\rho}} & \overline{u}\end{bmatrix} \partial_x \begin{bmatrix}\rho\\ u\end{bmatrix} = \mathbf{0}\,.$$

Spectral decomposition.

$$0 = |-\lambda\mathbf{I} + \mathbf{A}| = (\overline{u}-\lambda)^2 - a^2$$

$$\lambda_{12} = \overline{u} \mp a \qquad , \qquad \mathbf{r}_{12} = \begin{bmatrix}\overline{\rho}\\ \mp a\end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix}\overline{\rho} & \overline{\rho}\\ -a & a\end{bmatrix}$$

$$\mathbf{L} = \mathbf{R}^{-1} = \frac{1}{2\overline{\rho}a}\begin{bmatrix}a & -\overline{\rho}\\ a & \overline{\rho}\end{bmatrix}$$

Reference state.

$$|u| \;:\; \begin{cases} = 0 & \text{at rest}\\ < a & \text{subsonic flow}\\ > a & \text{supersonic flow to the left/right}\end{cases}$$

Subsonic: the two families of characteristic lines have opposite direction; supersonic: the two families of characteristic lines have the same direction.

Example 53.2.2 (Linearized shallow water equations)

Example 53.2.3 (Linearized Euler equations (acoustics))

Example 53.2.4 (Wave equation)

A wave equation arises in many different fields of science. As an example, 1-dimensional wave equation descrives the axial dynamics of a truss

$$m\partial_{tt}u - EA\partial_{xx}u = f \ ,$$

that can be recast in the general expression of wave equation

$$\partial_{tt}u - c^2\partial_{xx}u = F$$

The 2nd order differential operator appearing in 1-dimensional wave equation can be factored as the "product" of 2 1st order differentail operators,

$$\left(\partial_{tt} - c^2\partial_{xx}\right)u = \left(\partial_t - c\partial_x\right)\left(\partial_t + c\partial_x\right)u$$

and thus a wave equation can be written as

$$\begin{cases} \partial_t u + c\partial_x u - w = 0 \\ \partial_t w - c\partial_x w = F \end{cases}$$

In the regime of small displacement, the velocity field is the partial time derivative of the dispalcement field, $v = \partial_t u$, and the axial force reads $N = EA\partial_x u$. Exploiting Schwartz's theorem about mixed partial derivatives to write $\partial_t N = EA\partial_x v$, it's possible to write the wave function as the following system of hyperbolic equations in the physical unknowns v, N

$$\begin{cases} \partial_t N - EA\partial_x v = 0 \\ \partial_t v - \frac{1}{m}\partial_x N = f \end{cases}$$

P-system and wave equation - reference state at rest, $\overline{u} = 0$.

$$\begin{cases} \partial_t \rho + \overline{\rho}\partial_x u = 0 \\ \partial_t u + \frac{a^2}{\overline{\rho}}\partial_x \rho = 0 \end{cases}$$

Taking time partial derivative of the first and space partial derivative of the second equation times $\overline{\rho}$, and evaluating their difference, a wave equation for rho appears

$$\partial_{tt}\rho - a^2\partial_{xx}\rho = 0 \ .$$

Analogously, taking space derivative of the first and time derivative of the second, a wave equation for the velocity field appears

$$\partial_{tt}u - a^2\partial_{xx}u = 0 \ .$$

53.2.4 System non-linear

1-dimensional space

$$\mathbf{u}(x,t)$$

$$\begin{aligned}\partial_t\mathbf{u}+\partial_x\mathbf{F}(\mathbf{u})&=\mathbf{f} \qquad \text{(conservative form)}\\ \partial_t\mathbf{u}+\partial_{\mathbf{u}}\mathbf{F}(\mathbf{u})\partial_x\mathbf{u}&=\mathbf{f} \qquad \text{(convective form)}\end{aligned}$$

53.2.5 n-dimensional space

$$\mathbf{u}(\vec{r},t)$$

$$\begin{aligned}\partial_t\mathbf{u}+\nabla\cdot\mathbf{F}(\mathbf{u})&=\mathbf{f} \qquad \text{(conservative form)}\\ \partial_t\mathbf{u}+\nabla\mathbf{u}\cdot\partial_{\mathbf{u}}\mathbf{F}(\mathbf{u})&=\mathbf{f} \qquad \text{(convective form)}\end{aligned}$$

Different descritpions of integral problem,

$$\begin{aligned}\frac{d}{dt}\int_V\mathbf{u}+\oint_{\partial V}\hat{n}\cdot\mathbf{F}(\mathbf{u})&=\int_V\mathbf{f} \qquad \text{(Eulerian)}\\ \frac{d}{dt}\int_{V_t}\mathbf{u}-\oint_{\partial V_t}\mathbf{u}\vec{u}\cdot\hat{n}+\oint_{\partial V_t}\hat{n}\cdot\mathbf{F}(\mathbf{u})&=\int_{V_t}\mathbf{f} \qquad \text{(Lagrangian)}\\ \frac{d}{dt}\int_{v_t}\mathbf{u}-\oint_{\partial v_t}\mathbf{u}\vec{u}_b\cdot\hat{n}+\oint_{\partial v_t}\hat{n}\cdot\mathbf{F}(\mathbf{u})&=\int_{v_t}\mathbf{f} \qquad \text{(arbitrary)}\end{aligned}$$

todo *in coordinates*

$$\begin{aligned}f_i&=\partial_t u_i+\partial_{x_k}F_{ki}(u_l)=\\ &=\partial_t u_i+\partial_{x_k}u_m\partial_{u_m}F_{ki}(u_l)=\end{aligned}$$

Example 53.2.5 (P-system in 1-dimensional domain)

$$\begin{cases}\partial_t\rho+u\partial_x\rho+\rho\partial_x u=0\\ \rho\partial_t u+\rho u\partial_x u+\partial_x P=0\end{cases}$$

with $\partial_x P=a^2\partial_x\rho$,

Convective form

$$\partial_t\begin{bmatrix}\rho\\ u\end{bmatrix}+\begin{bmatrix}u & \rho\\ \frac{a^2}{\rho} & u\end{bmatrix}\partial_x\begin{bmatrix}\rho\\ u\end{bmatrix}=\underline{0}\,.$$

Conservative form

$$\partial_t\begin{bmatrix}\rho\\ \rho u\end{bmatrix}+\partial_x\begin{bmatrix}\rho u\\ \rho u^2+\rho a^2\end{bmatrix}=\underline{0}\,.$$

Spectral decomposition of $\mathbf{A}(\mathbf{u})$ gives

$$0=\left|\begin{bmatrix}u-s & \rho\\ \frac{a^2}{\rho} & u-s\end{bmatrix}\right|=(u-s)^2-a^2$$

$$
\begin{aligned}
s_{1,2} &= u \mp a \\
\mathbf{R} &= \begin{bmatrix} \rho & \rho \\ a & -a \end{bmatrix} \\
\mathbf{L} &= \frac{1}{2\rho a} \begin{bmatrix} a & \rho \\ a & -\rho \end{bmatrix}
\end{aligned}
$$

Example 53.2.6 (Euler equations in 1-dimensional domain)

Conservative form

$$
\partial_t \begin{bmatrix} \rho \\ \rho u \\ \rho e^t \end{bmatrix} + \partial_x \begin{bmatrix} \rho u \\ \rho u^2 + P \\ \rho u h^t \end{bmatrix} = \underline{0} \, ,
$$

with $h^t = e^t + \frac{P}{\rho}$ and $e^t = e + \frac{u^2}{2}$, and the pressure field can be written as a function of the other thermodynamic variables. As an example, using conservative variables $(\rho, m, E^t) = (\rho, \rho u, \rho e^t) = \left(\rho, \rho u, \rho \left(e + \frac{u^2}{2}\right)\right)$

$$
P(\rho, e) = P\left(\rho, \frac{E^t}{\rho} - \frac{m^2}{2\rho^2}\right) = \Pi\left(\rho, m, E^t\right)
$$

so that

$$
\begin{aligned}
\partial_\rho \Pi &= \partial_\rho P\big|_e + \partial_e P\big|_\rho \left(-\frac{E^t}{\rho^2} + \frac{m^2}{\rho^3}\right) = \\
&= \partial_\rho P\big|_e + \partial_e P\big|_\rho \left(-\frac{e^t}{\rho} + \frac{u^2}{\rho}\right) \\
&= c^2 - \frac{P}{\rho^2} \partial_e P\big|_\rho + \partial_e P\big|_\rho \left(-\frac{e^t}{\rho} + \frac{u^2}{\rho}\right) \\
&= c^2 + \partial_e P\big|_\rho \left(-\frac{h^t}{\rho} + \frac{u^2}{\rho}\right) \\
\partial_m \Pi &= \partial_e P\big|_\rho \left(-\frac{m}{\rho^2}\right) \\
\partial_{E^t} \Pi &= \partial_e P\big|_\rho \left(\frac{1}{\rho}\right)
\end{aligned}
$$

The speed of sound reads

$$
\begin{aligned}
c^2 &= \partial_\rho P\big|_s = \\
&= \partial_\rho P\big|_e + \partial_e P\big|_\rho \, \partial_\rho e\big|_s = \\
&= \partial_\rho P\big|_e + \frac{P}{\rho^2} \partial_e P\big|_\rho \, ,
\end{aligned}
$$

Conservative form in conservative variables.

$$
\partial_t \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} + \partial_x \begin{bmatrix} m \\ \frac{m^2}{\rho} + \Pi(\rho, m, E^t) \\ \frac{m}{\rho}\left(E^t + \Pi(\rho, m, E^t)\right) \end{bmatrix} = \underline{0} \, ,
$$

Convective form in conservative variables.

$$
\partial_t \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ -\frac{m^2}{\rho^2} + \partial_\rho \Pi & \frac{2m}{\rho} + \partial_m \Pi & \partial_{E^t} \Pi \\ -\frac{m}{\rho^2}(E^t + \Pi) + \frac{m}{\rho} \partial_\rho \Pi & \frac{1}{\rho}(E^t + \Pi) + \frac{m}{\rho} \partial_m \Pi & \frac{m}{\rho}\left(1 + \partial_{E^t} \Pi\right) \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} = \underline{0} \, ,
$$

Spectral decomposition of $\mathbf{A}(\mathbf{u})$

$$
\begin{aligned}
0 = &\left|\begin{bmatrix} -s & 1 & 0 \\ -u^2+\partial_\rho \Pi & 2u+\partial_m \Pi - s & \partial_{E^t}\Pi \\ -u\left(e^t+\frac{P}{\rho}\right)+u\partial_\rho \Pi & e^t+\frac{P}{\rho}+u\partial_m \Pi & u\left(1+\partial_{E^t}\Pi\right)-s \end{bmatrix}\right| = \\
&= -s\left[(2u+\partial_m\Pi - s)\left(u\left(1+\partial_{E^t}\Pi\right)-s\right) - \partial_{E^t}\Pi\left(h^t+u\partial_m\Pi\right)\right] + \\
&\quad - uh^t\partial_{E^t}\Pi + u\partial_\rho\Pi\partial_{E^t}\Pi + \\
&\quad + \left(u^2-\partial_\rho\Pi\right)\left(u(1+\partial_{E^t}\Pi)-s\right) = \\
&= -s^3 + \\
&\quad + s^2\left(2u+\partial_m\Pi+u+u\partial_{E^t}\Pi\right) + \\
&\quad + s\left(-2u^2-2u^2\partial_{E^t}\Pi - u\partial_m\Pi - u\partial_m\Pi\,\partial_{E^t}\Pi + \partial_{E^t}\Pi\,h^t + u\partial_{E^t}\Pi\,\partial_m\Pi - u^2 + \partial_\rho\Pi\right) + \\
&\quad + \left(-uh^t\partial_{E^t}\Pi + u\partial_\rho\Pi\,\partial_{E^t}\Pi + u^3 + u^3\partial_{E^t}\Pi - u\partial_\rho\Pi - u\partial_\rho\Pi\,\partial_{E^t}\Pi\right) + \\
&= -s^3 + \\
&\quad + s^2\left(3u+\partial_m\Pi+u\partial_{E^t}\Pi\right) + \\
&\quad + s\left(-3u^2-2u^2\partial_{E^t}\Pi - u\partial_m\Pi + \partial_{E^t}\Pi\,h^t + \partial_\rho\Pi\right) + \\
&\quad + \left(u^3 - uh^t\partial_{E^t}\Pi + u^3\partial_{E^t}\Pi - u\partial_\rho\Pi\right) = \\
&= -(s-u)^3 + (s-u)c^2 = \\
&= (s-u)\left[-(s-u)^2+c^2\right]
\end{aligned}
$$

being

$$
\begin{aligned}
\partial_m\Pi + u\partial_{E^t}\Pi &= \left(-\frac{u}{\rho}+\frac{u}{\rho}\right)\partial_\rho P\big|_e = 0 \\
-2u^2\partial_{E^t}\Pi - u\partial_m\Pi + \partial_{E^t}\Pi\,h^t + \partial_\rho\Pi &= -\frac{u^2}{\rho}\partial_\rho P\big|_e + \frac{1}{\rho}\partial_e P\big|_\rho\, h^t + \partial_\rho P\big|_e + \partial_e P\big|_\rho\left(-\frac{e^t}{\rho}+\frac{u^2}{\rho}\right) = \\
&= \partial_e P\big|_\rho \frac{P}{\rho^2} + \partial_\rho P\big|_e = \\
&= c^2 \\
-u\partial_\rho\Pi + u^3\partial_{E^t}\Pi - uh^t\partial_{E^t}\Pi &= u\left(-\partial_\rho P\big|_e - \partial_e P\big|_\rho\left(-\frac{e^t}{\rho}+\frac{u^2}{\rho}\right) + \frac{u^2}{\rho}\partial_e P\big|_\rho - \frac{h^t}{\rho}\partial_e P\big|_\rho\right) \\
&= u\left(-\partial_\rho P\big|_e - \frac{P}{\rho^2}\partial_e P\big|_\rho\right) = \\
&= -uc^2\ .
\end{aligned}
$$

Thus,

$$
s_{1,3} = u \mp c \quad , \quad s_2 = u
$$

$$
\mathbf{r}_{1,3} = \begin{bmatrix} 1 \\ u\mp c \\ \dots \end{bmatrix}\hat{\rho} \quad , \quad \mathbf{r}_2 = \begin{bmatrix} \dots \\ 0 \\ \dots \end{bmatrix}\hat{\rho}
$$

being

$$
\begin{aligned}
\hat{E}^t\partial_{E^t}\Pi &= \left[u^2 - \partial_\rho\Pi + (-u\pm c - \partial_m\Pi)(u\mp c)\right]\hat{\rho} = \\
&= \left[u^2 - \partial_\rho\Pi - u^2 + \pm 2uc - c^2 - \partial_m\Pi(u\mp c)\right]\hat{\rho} = \\
&= \left[-c^2 - \partial_e P\big|_\rho\left(-\frac{h^t}{\rho}+\frac{u^2}{\rho}\right) \pm 2uc - c^2 + \frac{u}{\rho}\partial_e P\big|_\rho(u\mp c)\right]\hat{\rho} = \\
&= \left[-2c^2 + \partial_e P\big|_\rho\frac{h^t}{\rho} \pm 2uc \mp \frac{uc}{\rho}\partial_e P\big|_\rho\right]\hat{\rho} =
\end{aligned}
$$

$$\mathbf{R} = ...$$

$$\mathbf{L} = ...$$

Example 53.2.7 (Shallow water equation in 1-dimensional domain)

Let $b(x)$..., $h(x)$ the height of the free surface, $\eta(x) = h(x) - b(x)$ the depth.

Derivative of integrals with non-constant extremes

$$\partial_x \int_{z=0}^{\eta(x,t)} \rho u \, dz = \int_{z=0}^{\eta(x,t)} \partial_x(\rho u) \, dz + \rho u(x, \eta(x,t), t) \partial_x \eta(x,t) \, .$$

Continuity equation reads

$$\partial_t \rho + \partial_x(\rho u) + \partial_z(\rho w) = 0 \, ,$$

for fluids with constant and uniform density

$$\begin{aligned} 0 &= \int_{z=0}^{\eta(x,t)} \left(\partial_t \rho + \partial_x(\rho u) + \partial_z(\rho w)\right) \, dz = \\ &= \partial_x \int_{z=0}^{\eta(x,t)} (\rho u) \, dz - \rho u(x, \eta, t) \partial_x \eta + \rho w(x, \eta(x,t), t) = \\ &= \partial_x \int_{z=0}^{\eta(x,t)} (\rho u) \, dz + \rho \partial_t \eta = \qquad\qquad \simeq \partial_x \left(\rho \eta u\right) + \partial_t \left(\rho \eta\right) \, . \end{aligned}$$

having linked the velocity to the material derivative of the position, whose vertical component reads

$$w(x, \eta(x,t), t) = \frac{D\eta}{Dt} = \partial_t \eta(x,t) + u(x, \eta(x,t), t) \partial_x \eta \, .$$

Assuming hydrostatic pressure distribution, $p = P_a + \rho g z$ at depth z under the level of local free surface,

Momentum equation reads

$$0 = \partial_t(\rho u) + \partial_x(\rho u^2) + \partial_z(\rho u w) + \partial_x P \, .$$

and integration in z-direction **todo** Explicitly treat the z term

$$\begin{aligned} 0 &= \partial_t(\rho \eta u) + \partial_x(\rho u^2 \eta) + \partial_x \int_{z=0}^{\eta(x)} (P_a + \rho g z) \, dz = \\ &= \partial_t(\rho \eta u) + \partial_x \left(\rho u^2 \eta + \frac{1}{2} \rho g \eta^2\right) \, . \end{aligned}$$

Conservative form of the equations.

$$\begin{cases} \partial_t(\eta) + \partial_x m = 0 \\ \partial_t m + \partial_x \left(\frac{m^2}{\eta} + \frac{g\eta^2}{2}\right) = 0 \end{cases}$$

Convective form of the equations.

$$\partial_t \begin{bmatrix} \eta \\ m \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -\frac{m^2}{\eta^2} + g\eta & 2\frac{m}{\eta} \end{bmatrix} \partial_x \begin{bmatrix} \eta \\ m \end{bmatrix} = \underline{0}$$

Spectrum of matrix $\mathbf{A}(\mathbf{u})$.

$$0 = |\mathbf{A}(\mathbf{u}) - s^2\mathbf{I}| = -s\,(2u - s) + u^2 - g\eta = (s-u)^2 - g\eta\ .$$

Example 53.2.8 (P-system in n-dimensional domain)

- Conservative variables: $(\rho, \vec{m})$
- Physical variables: e.g. $(\rho, \vec{u})$, $(P, \vec{u})$,…

$$\begin{cases} \partial_t \rho + \nabla \cdot \vec{m} = 0 \\ \partial_t \vec{m} + \nabla \cdot \left[\frac{\vec{m} \otimes \vec{m}}{\rho} + \rho a^2 \mathbb{I} \right] = 0 \end{cases}$$

Example 53.2.9 (Euler system in n-dimensional domain)

- Conservative variables: $(\rho, \vec{m}, E^t)$
- Physical variables: e.g. $(\rho, \vec{u}, e)$,…

$$\begin{cases} \partial_t \rho + \nabla \cdot \vec{m} = 0 \\ \partial_t \vec{m} + \nabla \cdot \left[\frac{\vec{m} \otimes \vec{m}}{\rho} + \Pi\, \mathbb{I} \right] = \vec{0} \\ \partial_t E^t + \nabla \cdot \left[\frac{\vec{m}(E^t + \Pi)}{\rho} \right] = 0 \end{cases}$$

where Π represents the pressure field as a function of the conservative varaibles,

$$\Pi\,(\rho, \vec{m}, E^t) = P\,(\rho, e) = P\left(\rho, \frac{E^t}{\rho} - \frac{|\vec{m}|^2}{\rho^3}\right)\ ,$$

and P the pressure field expressed by the **equation of state of the fluid** as a function of density and internal energy per unit mass as the pair of independent variables determining the thermodynamic state.

Example 53.2.10 (Shallow water equations in 2-dimensional domain)

$$\begin{cases} \partial_t(\rho\eta) + \nabla \cdot (\rho\eta\vec{u}) = 0 \\ \partial_t(\rho\eta\vec{u}) + \nabla \cdot \left(\rho\eta\vec{u}\vec{u} + \frac{1}{2}\rho g \eta^2 \mathbb{I}\right) = 0 \end{cases}$$

53.3 Scalar linear equation

Differential equation.

$$\begin{aligned} r &= \partial_t u + a\partial_x u && \text{convective form} \\ &= \partial_t u + \partial_x(au) && \text{conservative form} \end{aligned}$$

Integral equation. On a control volume V at rest

$$0 = \frac{d}{dt}\int_V u + \oint_{\partial V} n_x a u\ .$$

Integral equation in an arbitrary domain v_t can be derived using *Reynolds' transport theorem*.

Method of characteristics. Let $U(t) = u(X(t), t)$. From derivative of composite functions,

$$d_t U(t) = \partial_t u|_{X(t),t} + \dot{X}(t)\partial_x u|_{X(t),t} \ ,$$

and thus the PDE can be recast as a system of two ODEs

$$\begin{cases} \dot{X}(t) = a \\ \dot{U}(t) = r \end{cases}$$

…

53.4 Linear vector equation

Differential equation.

$$\begin{aligned} \mathbf{r} &= \partial_t \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} \qquad \text{convective form} \\ &= \partial_t \mathbf{u} + \partial_x (\mathbf{A}\mathbf{u}) \quad \text{conservative form} \end{aligned}$$

Integral eqaution. On a control volume V at rest

$$\int_V \mathbf{r} = \frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} n_x \mathbf{A}\mathbf{u} \ .$$

Integral equation in an arbitrary domain v_t can be derived using *Reynolds' transport theorem*.

Method of characteristics. Let $\mathbf{A}$ be diagonalizable,

$$\begin{aligned} \mathbf{AR} &= \mathbf{RS} \\ \mathbf{A} &= \mathbf{RSL} \end{aligned}$$

Using spectral decomposition and multiplying the PDE by $\mathbf{L}$, the PDE is transformed in a set of scalar PDE equations,

$$\begin{aligned} \mathbf{Lr} &= \mathbf{L}\left(\partial_t \mathbf{u} + \mathbf{RSL}\partial_x \mathbf{u}\right) = \\ &= \mathbf{L}\partial_t \mathbf{u} + \mathbf{SL}\partial_x \mathbf{u} = \\ &= \partial_t \mathbf{v} + \mathbf{S}\partial_x \mathbf{v} \ , \end{aligned}$$

or

$$\partial_t v_i + s_i \partial_x v_i = L_{ik} r_k \ ,$$

having introduced the characteristic variables $\mathbf{v}(x,t) := \mathbf{L}\mathbf{u}(x,t)$. If the source is not function of the unknown function $\mathbf{u}(x,t)$, this is a system of decoupled PDEs. The solution of the vector PDE follows from the solution of n scalar PDE linear equations. Using methods of characteristics, the solution of the ith scalar PDE equation can be recast as the solution of pairs of ODEs, after writing $V_i(t) = v_i(X_i(t), t)$, being $X_i(t)$ the equation of a curve belonging to the ith familiy of characteristics,

$$\begin{cases} \dot{X}_i = s_i \\ \dot{V}_i = L_{ik} r_k \ . \end{cases}$$

As the spectral decomposition of a constant matrix is constant, the characteristic lines are straight lines $X_i(t) = s_i t + c$. In general, the solution of the dynamical equations of the characteristic variables V_i is a full system of n ODEs.

Particular case: no source. As a particular case, if the source term is identically zero, $\mathbf{r}(x,t) = \mathbf{0}$, characteristic variable $V_i(t)$ is constant along each characteristic line $X_i(t; \dots)$ of the ith family of characteristics.

…

$$u_k(x,t) = R_{ki} v_i(x,t) = R_{ri} v_i(x_{0,i} + s_i(t - t_{0,i}), t) \ .$$

53.5 Scalar non-linear equation

todo list

- When does a jump occur? I.e. when is the physical solution discontinuous? Have non physical jumps already been discussed?
- Boundary conditions for hyperbolic problems
- …

53.5.1 Examples

Burgers's equation

Differential equation and integral balance

Burgers' differential equation reads

$$\begin{aligned} 0 = \partial_t u + u\partial_x u = \quad & \text{(convective form)} \\ = \partial_t u + \partial_x \frac{u^2}{2} = \quad & \text{(conservative form)} \\ = \partial_t u + \partial_x F(u)\ , & \end{aligned}$$

being the flux $F(u) = \frac{u^2}{2}$, and it can be derived as the local counterpart of the integral balance equation,

$$0 = \frac{d}{dt}\int_V u + \oint_{\partial V} n_x \frac{u^2}{2} = \frac{d}{dt}\int_V u + F(u(x))\big|_a^b\ ,$$

for a control volume at rest $V = [a, b]$. For a control volume with arbitrary motion $v_t = [a_t, b_t]$ the integral balance equation reads

$$\begin{aligned} 0 &= \frac{d}{dt}\int_{v_t} u - \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{v}_b\, u + \oint_{\partial v_t} n_x F(u) = \\ &= \frac{d}{dt}\int_{v_t} u - \oint_{\partial v_t} n_x v_{b,x}\, u + \oint_{\partial v_t} n_x F(u) = \\ &= \frac{d}{dt}\int_{v_t} u + \left[F(u(x)) - v_{b,x} u(x)\right]\Big|_a^b\ , \end{aligned}$$

having used Reynolds' transport theorem[1], and the unit normal vector in the only space direction $n_x(x = a) = -1$ on the left extreme and $n_x(x = b) = 1$ on the right extreme of the domain $V = [a, b]$.

[1] For a generic volume v_t, $d_t \int_{v_t} f = \int_{v_t} \partial_t f + \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{v}_b\, f$. Thus, for a control volume at rest V, $d_t \int_V f = \int_V \partial_t f$. Comparing the two expressions, it immediately follows that $d_t \int_{v_t} f = d_t \int_{V \equiv v_t} f + \oint_{\partial v_t} \hat{\mathbf{n}} \cdot \mathbf{v}_b\, f$.

Jump condition

Jump condition immediately follows from the integral form of the balance equations for the arbitrary volume v_t.

$$[F] = v_{b,x}[u] \, .$$

For Burgers' equation, $F(u) = \frac{u^2}{2}$, and thus the jump relation becomes

$$\frac{u_2^2}{2} - \frac{u_1^2}{2} = v_{b,x} \left(u_2 - u_1 \right) \, ,$$

and thus, except for the trivial condition $u_1 = u_2$ where no jump occurs,

$$v_{b,x} = \frac{1}{2} \left(u_2 + u_1 \right) \, ,$$

the velocity of the point where the discontinuity appears - the shock - is the average of the velocity u on its sides.

todo *When does a jump occur? I.e. when is the physical solution discontinuous? Have non physical jumps already been discussed?*

Characteristics

Let $U(t) = u(X(t), t)$. Burgers' equation can be recast as[2]

$$0 = d_t U - \dot{X} \partial_x u + u \partial_x u \, .$$

A characteristic line is defined as a curve for which the coefficient of the spatial partial differential equation is zero, i.e. $\dot{X}(t) = U(X(t), t)$. On these lines, the PDE becomes a ODE, so that it's somehow equivalent **todo** *be more precies* to the system of ODEs

$$\begin{cases} \dot{X}(t) = U(X(t), t) \\ \dot{U}(t) = 0 \, . \end{cases}$$

These equations need initial and boundary conditions.

todo *Boundary conditions for hyperbolic problems*

Let x_0, t_0 a point in space and time where the solution is known, $u_0 = u(x_0, t_0)$. On the characteristic line originating from x_0, t_0, the velocity is constant in time as $\dot{U}(t) = 0$, and thus $U(t)_{X(t;x_0,t_0)} = u_0$. The characteristic line satisfying the first equation and the initial conditions is the line with equation

$$X(t; t_0, x_0, u_0) = u_0 (t - t_0) + x_0 \, .$$

Riemann problem

Riemann problem is defined as the PDE with a discontinuity in a piecewise constant initial conditons. For an unbounded domain $V = (-\infty, +\infty)$[3]

$$u(x, 0) = \begin{cases} u_1 & , \quad x < 0 \\ u_2 & , \quad x > 0 \end{cases}$$

Two cases must be discussed here: 1) $u_1 < u_2$, 2) $u_1 > u_2$.

[2] Using derivation of composite functions, $d_t U(t) = d_t u(X(t), t) = \partial_t u + d_t X \partial_x u = \partial_t u|_{(x,t)=(X(t),t)} + \dot{X}(t) \partial_x u|_{(x,t)=(X(t),t)}$.

[3] In an unbounded domain, there's no need to treat boundary conditions…**todo** *add link to boundary conditions for hyperbolic problems.*

Expansion fan, $u_1 < u_2$**.** A self-similar solution exists between lines $r_1 : x = u_1 t$ and $r_2 : x = u_2 t$. Defining $\xi := \frac{x}{t}$, the solution becomes $u(x,t) = U(\xi(x,t))$, so that

$$\partial_t U = \partial_t \xi U'(\xi) = -\frac{x}{t^2} U'(\xi)$$
$$\partial_x U = \partial_x \xi U'(\xi) = \frac{1}{t} U'(\xi) ,$$

so that Burgers' equation in the rarefaction fan becomes

$$0 = \partial_t u + u \partial_x u = \frac{1}{t} \left[-\xi + U(\xi) \right] U'(\xi) .$$

Looking for a continuous solution blending the two states, the condition $U'(\xi) = 0$ is not feasible. Then,

$$U(\xi) = \xi ,$$

or

$$u(x,t) = \frac{x}{t} \quad , \quad \frac{x}{t} \in [u_1, u_2] .$$

The solution of the Riemann problem reads

$$u(x,t) = \begin{cases} u_1 & , \quad \frac{x}{t} \leq u_1 \\ \frac{x}{t} & , \quad \frac{x}{t} \in [u_1, u_2] \\ u_2 & , \quad \frac{x}{t} \geq u_2 \end{cases}$$

Shock, $u_1 > u_2$**.** Characteristic method fails at a shock originating from the origin. Characteristic lines originating from $x < 0$ have a slope $\dot{X}(t) = u_1$ larger than the slope of the characteristic lines originating from $x > 0$: characteristic lines collapses into the shock, that has velocity

$$\dot{s} = \frac{u_1 + u_2}{2} .$$

The solution of the Riemann problem reads

$$u(x,t) = \begin{cases} u_1 & , \quad \frac{x}{t} < \dot{s} \\ u_2 & , \quad \frac{x}{t} > \dot{s} \end{cases}$$

Boundary conditions

In order to have a well-defined hyperbolic problem, boundary conditions are **required** on the regions of the boundary **where the information enters the domain**. For a scalar problem, the local direction of the information is determined by the velocity of the local characteristic lines. For Burgers' equation, the direction of the local intormation is just the sign of the velocity of the characteristics, that is equal to the local velocity. Thus, boundary conditions are strictly needed on the left boundary $x = a$ if $u(a,t) > 0$ and on the right boundary $x = b$ if $u(b,t) < 0$.

53.6 Method of characteristics

This page is an old version of the material. It contains some useful and (incredibly?) interesting things, but it will be used to write new pages first and eventually deleted. Check all the **cross-references**

53.6.1 Scalar multi-dimensional problem

$$\mathbf{a}^T(u(\mathbf{x}),\mathbf{x})\,\nabla u(\mathbf{x}) = c(u(\mathbf{x}),\mathbf{x}) \quad , \quad \mathbf{x}\in D$$
$$a_k(u(\mathbf{x}),\mathbf{x})\frac{\partial}{\partial x_k}u(\mathbf{x}) = c(u(\mathbf{x}),\mathbf{x})$$

The method of characteristics aims at transforming the PDE over the domain D, in a set of ODEs over lines - the characteristic lines. Both the coordinates of these lines and the unknown function on these lines are written in parametric form as

$$\mathbf{x} = \mathbf{X}(s)$$
$$u(\mathbf{X}(s)) = U(s)\ .$$

Now, taking the ordinary derivative of $U'(s)$,

$$U'(s) = \frac{dX_k}{ds}(s)\frac{\partial u}{\partial x_k}\left(\mathbf{X}(s)\right)\ .$$

If the characteristic lines are defined by the differential equation $X_k'(s) = a_k(U(s),X(s))$, the set of ODEs become

$$U'(s) = c\left(U(s),\mathbf{X}(s)\right)$$
$$\mathbf{X}'(s) = \mathbf{a}\left(U(s),\mathbf{X}(s)\right)\ .$$

53.6.2 Vector multi-dimensional problem

todo *Characteristic method fails…characteristic - eikonal - equation. Uncomment*

Let

$$\mathbf{u}(t,\mathbf{r}) : \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}^n$$
$$\mathbf{F}(\mathbf{u}) : \mathbb{R}^n \to \mathbb{R}^d\times\mathbb{R}^n$$
$$\mathbf{s}(t,\mathbf{r}) : \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}^n$$

the conservative form of a hyperbolic problem reads

$$\partial_t\mathbf{u} + \nabla\cdot\mathbf{F}(\mathbf{u}) = \mathbf{s}$$
$$\partial_t u_i + \partial_j F_{ji} = s_i\ .$$

Expanding the divergence of the flux, the convective (quasi-linear? It makes little to no sense to me…) form follows

$$s_i = \partial_t u_i + \sum_{j=1}^{d}\partial_j F_{ji} =$$
$$= \partial_t u_i + \sum_{j=1}^{d}\sum_{k=1}^{n}\partial_j u_k \partial_{u_k} F_{ji} =$$
$$= \partial_t u_i + \sum_{j=1}^{d}\sum_{k=1}^{n} A_{ik}^j(\mathbf{u})\,\partial_j u_k\ ,$$

for every component $i = 1 : n$, or

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_{j=1}^{d} \mathbf{A}^j \partial_j \mathbf{u} \, .$$

This problem can be made a little more general with by defining $\mathbf{x} = (t, \mathbf{r}) = (x_0, \mathbf{r})$, as

$$\sum_{j=0}^{d} \mathbf{A}^j \, \partial_j \mathbf{u} = \mathbf{s} \, .$$

The former expression of the hyperbolic problem immediately follows if $\mathbf{A}_0 = \mathbf{I}_n$, $A^0_{ik} = \delta_{ik}$, and

$$\widetilde{\nabla} \cdot \tilde{\mathbf{F}} = \partial_0 \mathbf{F}_0 + \sum_{j=1}^{d} \partial_j \mathbf{F}_j = \widetilde{\nabla} \cdot [\, \mathbf{u} \mid \mathbf{F}] \, .$$

as, for $i = 1 : n$,

$$\delta_{ik} = \partial_{u_k} F_{0i} \quad \rightarrow \quad F_{0i} = u_k \delta_{ik} = u_i \, .$$

Taylor expansion of a solution

Starting from the solution on a manifold determined by the equation $S : f(\mathbf{x}) = 0, \mathbf{x}_0 \in S$, if the solution is differentiable, the solution in a point $\mathbf{x} = \mathbf{x}_0 + \Delta\mathbf{x}$ reads

$$\mathbf{u}(\mathbf{x}) \sim \mathbf{u}(\mathbf{x}_0) + \Delta\mathbf{x} \cdot \nabla \mathbf{u}(\mathbf{x}_0)$$
$$u_i(\mathbf{x}) \sim u_i(\mathbf{x}_0) + \Delta_j \frac{\partial u_i}{\partial x_j} \, .$$

Now, with a change of coordinates from $\mathbf{x}$ to $\xi = (n, \mathbf{t})$, i.e. to local normal and tangential direction,

$$\nabla \mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial}{\partial x_i} \mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial x_i} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial \xi_k} \frac{\partial \xi_k}{\partial x_i} = \hat{\xi}_k \frac{\partial \mathbf{u}}{\partial \xi_k}$$

Inserting in the hyperbolic equation

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \frac{\partial u_k}{\partial x_j} = \\ &= \sum_{j=0}^{d} A^j_{ik} \sum_{\ell=0}^{d} \frac{\partial u_k}{\partial \xi_\ell} \frac{\partial \xi_\ell}{\partial x_j} = \\ &= \sum_{\ell=0}^{d} \sum_{j=0}^{d} A^j_{ik} \frac{\partial \xi_\ell}{\partial x_j} \frac{\partial u_k}{\partial \xi_\ell} = \\ &= \sum_{\ell=0}^{d} \sum_{j=0}^{d} A^j_{ik} \xi^\ell_j \frac{\partial u_k}{\partial \xi_\ell} \, , \end{aligned}$$

and separating the normal $\ell = 0$ from the tangential $\ell = 1 : d$ coordinates,

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \xi^0_j \frac{\partial u_k}{\partial \xi_0} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} A^j_{ik} \xi^t_j \frac{\partial u_k}{\partial \xi_t} = \\ &= \sum_{j=0}^{d} A^j_{ik} n_j \frac{\partial u_k}{\partial n} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} A^j_{ik} t^\ell_j \frac{\partial u_k}{\partial t_\ell} = \\ &= \sum_{j=0}^{d} n_j \mathbf{A}^j \partial_n \mathbf{u} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} t^\ell_j \mathbf{A}^j \partial_{t_\ell} \mathbf{u} \, . \end{aligned}$$

On a smooth surface where the solution is known, all the tangential derivatives of the solution are known as well. In order to evaluate the normal derivative $\partial_n \mathbf{u}$ from the PDE, the (formal) inversion of the matrix

$$\mathbf{A}_{\hat{\mathbf{n}}} := \sum_{j=0}^{d} n_j \, \mathbf{A}^j \ ,$$

must be invertible, to formally get

$$\partial_n \mathbf{u} = \mathbf{A}_{\hat{\mathbf{n}}}^{-1} \left(\mathbf{s} - \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{t}}_\ell} \partial_{t_\ell} \mathbf{u} \right) \ .$$

This expression of the normal derivative of the solution - whenever it exists - can be used to find the approximation of the solution in normal direction w.r.t. the surface S, i.e. with $\Delta \mathbf{x} = \Delta \ell \hat{\mathbf{n}}$ as

$$\begin{aligned} \mathbf{u}(\mathbf{x}) \sim \mathbf{u}(\mathbf{x}_0) + \Delta \ell \, \hat{\mathbf{n}} \cdot \nabla \mathbf{u}(\mathbf{x}_0) = \\ = \mathbf{u}(\mathbf{x}_0) + \Delta \ell \, \partial_n \mathbf{u}(\mathbf{x}_0) \ . \end{aligned}$$

If the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ is diagonalizable, it's invertible if it has no zero eigenvalue. Let $\hat{\mathbf{n}}_i$ be a unit vector so that the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ has an eigenvalue equal to zero $s_0 = 0$, with right and left eigenvectors $\mathbf{r}_0$, $\mathbf{l}_0$. Recalling the expression of the PDE in normal and tangential components

$$\mathbf{s} = \mathbf{A}_{\hat{\mathbf{n}}} \partial_n \mathbf{u} + \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{d}}_\ell} \partial_{t_\ell} \mathbf{u} \ ,$$

left-multiplying by $\mathbf{l}$ gives

$$\mathbf{l}_0^T \mathbf{s} = \mathbf{l}_0^T \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{d}}_\ell} \partial_{t_\ell} \mathbf{u} = \qquad \ldots \qquad = \mathbf{l}_0^T \sum_{k=0}^{d} \mathbf{A}^k \partial_k \mathbf{u} \ .$$

todo

- *compatibility conditions*
- *eikonal equation*
- *explicitly treating steady vs. unsteady problems*

Examples

Isothermal compressible flow - 2d

- Conservative variables: $(\rho, \vec{m})$
- Physical variables: e.g. $(\rho, \vec{u})$, $(P, \vec{u})$,…

In isothermal compressible flow model, pressure is a function of density only and

$$dp = a^2 d\rho \ ,$$

with constant a, speed of sound.

Conservative form reads

$$\begin{cases} \partial_t \rho + \nabla \cdot \mathbf{m} = 0 \\ \partial_t \mathbf{m} + \nabla \cdot \left[\frac{\mathbf{m} \otimes \mathbf{m}}{\rho} + \rho a^2 \mathbf{I} \right] = 0 \end{cases}$$

Convective form in a 2-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & 1 & \cdot \\ a^2 - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} & \cdot \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & 1 \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \\ a^2 - \frac{m_y m_y}{\rho^2} & \cdot & \frac{2m_y}{\rho} \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} = \mathbf{0}$$

Some algebra

$$
\begin{aligned}
0 &= \rho_{/t} + m_{x/x} + m_{y/y} \\
0 &= m_{i/t} + \partial_j \left(\frac{m_j m_i}{\rho} + \rho a^2 \delta_{ij} \right) .
\end{aligned}
$$

Thus,

$$
\begin{aligned}
\mathbf{A}_{\hat{\mathbf{n}}} &= \begin{bmatrix} \cdot & n_x & n_y \\ a^2 n_x - u_n u_x & u_x n_x + u_n & u_x n_y \\ a^2 n_y - u_n u_y & u_y n_x & u_n + u_y n_y \end{bmatrix} = \\
&= \begin{bmatrix} 0 & \mathbf{n}^T \\ a^2 \mathbf{n} - (\mathbf{n}^T \mathbf{u}) \mathbf{u} & \mathbf{n}^T \mathbf{u} \mathbf{I} + \mathbf{u} \mathbf{n}^T \end{bmatrix} = \\
&= \begin{bmatrix} 0 & \hat{\mathbf{n}}^T \\ a^2 \mathbf{n} - (\mathbf{n} \cdot \mathbf{u}) \mathbf{u} & (\mathbf{n} \cdot \mathbf{u}) \mathbb{I} + \mathbf{u} \otimes \mathbf{n} \end{bmatrix} .
\end{aligned}
$$

Eigenvalues. Details

The eigenvalue decomposition of the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ follows from

$$
\begin{aligned}
0 = |\mathbf{A}_{\hat{\mathbf{n}}} - s\mathbf{I}| &= \left| \begin{bmatrix} -s & \mathbf{n}^T \\ a^2 \mathbf{n} - u_n \mathbf{u} & (u_n - s)\mathbf{I} + \mathbf{u} \mathbf{n}^T \end{bmatrix} \right| = \\
&= -s(u_n + u_x n_x - s)(u_n + u_y n_y - s) + n_x u_x n_y (a^2 n_y - u_n u_y) + n_y u_y n_x (a^2 n_x - u_n u_x) + \\
&- (a^2 n_y - u_n u_y)(u_x n_x + u_n - s) n_y - (a^2 n_x - u_n u_x)(u_y n_y + u_n - s) n_x - s u_x u_y n_x n_y = \\
&= s^3 (-1) + \\
&\quad + s^2 (u_n + u_y n_y + u_n + u_x n_x) + \\
&\quad + s(-(u_n + u_x n_x)(u_n + u_y n_y) + n_y (a^2 n_y - u_n u_y) + n_x (a^2 n_x - u_n u_x) - u_x n_x u_y n_y) + \\
&\quad + 1 \cdot (n_x u_x n_y (a^2 n_y - u_n u_y) + n_y u_y n_x (a^2 n_x - u_n u_x) - n_y (a^2 n_y - u_n u_y)(u_x n_x + u_n) - n_x (a^2 n_x - u_n u_x)(u_y n_y + u_n)) = \\
&= -s^3 + 3 u_n s^2 + s(-u_n^2 - u_n (u_x n_x + u_y n_y) - u_x n_x u_y n_y - u_n u_y n_y - u_n u_x n_x + a^2 - u_x n_x u_y n_y) + \\
&\quad + (a^2 \left(n_y^2 u_x n_x + n_x^2 n_y u_y - n_y^2 u_x n_x - n_y^2 u_n - n_x^2 u_y n_y - u_n n_x^2 \right) - u_n u_x n_x u_y n_y - u_n u_x n_x u_y n_y + u_n u_x n_x u_y n_y + u_n^2 u_y n_y + \\
&= -s^3 + 3 u_n s^2 + s(a^2 - 3 u_n^2) - u_n (a^2 - u_n^2) = \\
&= -(s - u_n)^3 + (s - u_n) a^2 = \\
&= -(s - u_n)((s - u_n)^2 - a^2) .
\end{aligned}
$$

Thus the eigenvalues of the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ are

$$
s_{1,3} = u_n \mp a \quad , \quad s_2 = u_n ,
$$

and the right and left eigenvectors follow

$$
\begin{aligned}
\mathbf{R} &= \\
\mathbf{L} &=
\end{aligned}
$$

Right and left eigenvectors. Details

For $s_{1,3} = u_n \mp a$,

$$\mathbf{0} = \begin{bmatrix} -u_n \pm a & \mathbf{n}^T \\ a^2\mathbf{n} - u_n\mathbf{u} & \pm a\mathbf{I} + \mathbf{u}\mathbf{n}^T \end{bmatrix} \begin{bmatrix} \hat{\rho} \\ \hat{\mathbf{m}} \end{bmatrix}$$

From the first equation it follows $\hat{m}_n = (u_n \mp a)\hat{\rho}$, and thus the momentum equation gives

$$\begin{aligned} \mathbf{0} &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) \pm a\hat{\mathbf{m}} + \mathbf{u}\hat{m}_n = \\ &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) \pm a\hat{\mathbf{m}} + \mathbf{u}(u_n \mp a)\hat{\rho} = \\ &= \hat{\rho}(a^2\mathbf{n} \mp a\mathbf{u}) \pm a\hat{\mathbf{m}} \,, \end{aligned}$$

and thus

$$\hat{\mathbf{m}}_{1,3} = \hat{\rho}_{1,3}(\mathbf{u} \mp a\mathbf{n}) \,.$$

For $s_2 = u_n$,

$$\mathbf{0} = \begin{bmatrix} -u_n & \mathbf{n}^T \\ a^2\mathbf{n} - u_n\mathbf{u} & \mathbf{u}\mathbf{n}^T \end{bmatrix} \begin{bmatrix} \hat{\rho} \\ \hat{\mathbf{m}} \end{bmatrix}$$

From the first equation it follows $\hat{m}_n = u_n\hat{\rho}$, and thus the momentum equation gives

$$\begin{aligned} \mathbf{0} &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) + \mathbf{u}\,\hat{m}_n = \\ &= \hat{\rho}(a^2\mathbf{n} - u_n\mathbf{u}) + \mathbf{u}\,u_n\,\hat{\rho} = \\ &= \hat{\rho}a^2\mathbf{n} \,, \end{aligned}$$

and thus $\hat{\rho}_2 = 0$. The components of the momentum readily follows from mass equation that is equivalent to the condition

$$\mathbf{n}^T\hat{\mathbf{m}}_2 = 0 \,,$$

i.e. as an example $\hat{\mathbf{m}}_2 = \begin{bmatrix} n_y \\ -n_x \end{bmatrix}$.

Assuming positive density, it's possible to write all the eigenvectors with the proper physical dimensions, and collect them in the matrix of right eigenvectors,

$$\mathbf{R} = \begin{bmatrix} \rho & 0 & \rho \\ \rho(u - an_x) & \rho a n_y & \rho(u + an_x) \\ \rho(v - an_y) & -\rho a n_x & \rho(v + an_y) \end{bmatrix} \,.$$

The determinant reads

$$\frac{1}{\rho^3 a^2}|\mathbf{R}| = n_y(v + n_y) - (u - n_x)n_x - n_y(v - n_y) + n_x(u + n_x) = 2 \,.$$

The inverse matrix reads

$$\begin{aligned} \mathbf{L} &= \frac{1}{|\mathbf{R}|}\,\rho^2 \begin{bmatrix} a(a + u_n) & -an_x & -an_y \\ 2a(n_x v - n_y u) & 2an_y & -2an_x \\ a(a - u_n) & an_x & an_y \end{bmatrix} = \\ &= \frac{1}{2\rho a^2} \begin{bmatrix} a(a + u_n) & -an_x & -an_y \\ 2a(n_x v - n_y u) & 2an_y & -2an_x \\ a(a - u_n) & an_x & an_y \end{bmatrix} \,. \end{aligned}$$

as

$$
\begin{aligned}
L_{11} &\propto an_y(v+an_y)+an_x(u+an_x) = au_n + a^2 \\
-L_{21} &\propto (u-an_x)(v+an_y)-(v-an_y)(u+an_x) = 2aun_y - 2an_xv \\
L_{31} &\propto -an_x(u-an_x)-an_y(v-an_y) = -au_n + a^2 \\
-L_{12} &\propto an_x \\
L_{22} &\propto (v+an_y)-(v-an_y) = 2an_y \\
-L_{32} &\propto -an_x \\
L_{13} &\propto -an_y \\
-L_{23} &\propto (u+an_x)-(u-an_x) = 2an_x \\
L_{33} &\propto an_y
\end{aligned}
$$

Some check

$$
\begin{aligned}
\mathbf{LR} &= \frac{1}{2\rho a^2}\begin{bmatrix} a(a+u_n) & -a\mathbf{n}^T \\ 2a(n_xv-n_yu) & 2a\mathbf{t}^T \\ a(a-u_n) & a\mathbf{n}^T \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ \mathbf{u}-a\mathbf{n} & a\mathbf{t} & \mathbf{u}+a\mathbf{n} \end{bmatrix}\rho = \\
&= \frac{1}{2a^2}\begin{bmatrix} a^2+au_n-au_n+a^2 & -a\mathbf{n}^T\mathbf{t} & a^2+au_n-au_n-a^2 \\ 2a(n_xv-n_yu)+2a(n_yu-n_xv) & 2a^2 & 2a(n_xv-n_yu)+2a(n_yu-n_xv) \\ a^2-au_n+au_n-a^2 & a\mathbf{n}^T\mathbf{t} & a^2-au_n+au_n+a^2 \end{bmatrix} = \mathbf{I}_3 \,.
\end{aligned}
$$

Normal vectors of the characteristic surfaces. For **locally supersonic flows**, $|\mathbf{u}| > a$, there are three directions, i.e. three unit vectors $\hat{\mathbf{n}}_i$ that make the eigenvalues equal to zero,

$$
\hat{\mathbf{n}}_{1,3}\cdot\mathbf{u} = \pm a \quad , \quad \hat{\mathbf{n}}_2\cdot\mathbf{u} = 0 \,,
$$

or equivalently

$$
\hat{\mathbf{n}}_{1,3}\cdot\left(\mathbf{u}\mp a\hat{\mathbf{n}}_{1,3}\right) = 0 \quad , \quad \hat{\mathbf{n}}_2\cdot\mathbf{u} = 0 \,.
$$

Let $\hat{\mathbf{u}}$ the unit vector along the local velocity field, it follows

$$
u\hat{\mathbf{u}}\cdot\hat{\mathbf{n}}_{1,3} = \pm a \,,
$$

i.e.

$$
\cos\theta_{1,3} = \pm\frac{a}{u} = \pm\frac{1}{M} \,.
$$

For **locally subsonic flows**, $|\mathbf{u}| < a$, the eigenvalues $s_{1,3}$ are always negative and positive respectively, while the eigenvalue s_2 becomes equal to zero for unit vectors that are orthogonal w.r.t. the local velocity.

Right and left eigenvectors

Right eigenvector problem reads

$$
\mathbf{AR} = \mathbf{RS} \,,
$$

s.t. $\mathbf{A}\mathbf{r}_i = s_i\mathbf{r}_i$, with $\mathbf{r}_i$ the ith column of matrix $\mathbf{R}$.

Left eigenvector problem reads

$$
\mathbf{LA} = \mathbf{SL} \,,
$$

s.t. $\mathbf{l}_i^T\mathbf{A} = s_i\mathbf{l}_i^T\mathbf{A}$, with $\mathbf{l}_i^T$ the ith row of matrix $\mathbf{L}$.

Compatibility equations. For **locally supersonic flows**, for $s_{1,3} = u_n \mp a$, $u_n = \pm a$ to get $s_{1,3} = 0$, and thus the left eigenvectors for those choices of unit vector become

$$\begin{aligned}\mathbf{l}_{0;1,3}^T &= \frac{1}{2\rho a^2}\left[\, a(a \pm u_n) \mid \mp a\mathbf{n}_{1,3}^T\right] = \\ &= \frac{1}{2\rho a}\left[\, 2a \mid \mp \mathbf{n}_{1,3}^T\right] \ .\end{aligned}$$

The quasi linear form of the equations becomes

todo *Check algebra, and UNCOMMENT!*

$$0 = \frac{1}{2\rho a}\left[\, 2a \mid \mp \mathbf{n}_{1,3}^T \,\right]\left(\partial_t \begin{bmatrix}\rho\\ m_x\\ m_y\end{bmatrix} + \begin{bmatrix}\cdot & 1 & \cdot\\ a^2 - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} & \cdot\\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho}\end{bmatrix}\partial_x \begin{bmatrix}\rho\\ m_x\\ m_y\end{bmatrix} + \begin{bmatrix}\cdot & \cdot & 1\\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho}\\ a^2 - \frac{m_y m_y}{\rho^2} & \cdot & \frac{2m_y}{\rho}\end{bmatrix}\partial_y \begin{bmatrix}\rho\\ m_x\\ m_y\end{bmatrix} = \mathbf{0}\right) =$$

$$\begin{aligned}0 = &\dots \partial_t \cdots + \\ &+ \left[\mp a^2 n_x \pm u_n u_x \mid 2a \mp u_x n_x \mp u_n \mid \mp n_y u_x \,\right]\partial_x \begin{bmatrix}\rho\\ \mathbf{m}\end{bmatrix} + \\ &+ \left[\mp a^2 n_y \pm u_n u_y \mid \mp n_x u_y \mid 2a \mp u_y n_y \mp u_n \,\right]\partial_y \begin{bmatrix}\rho\\ \mathbf{m}\end{bmatrix} = \\ = &\left[\left(\mp a^2 n_x \pm u_n u_x\right)\partial_x + \left(\mp a^2 n_y \pm u_n u_y\right)\partial_y\right]\rho + \\ &+ \left[\left(2a \mp u_x n_x \mp u_n\right)\partial_x + \left(\mp n_x u_y\right)\partial_y\right] m_x + \\ &+ \left[\left(\mp n_y u_x\right)\partial_x + \left(2a \mp u_y n_y \mp u_n\right)\partial_y\right] m_y =\end{aligned}$$

Comparing directional derivative of ρ and m_x

By direct comparison of the coefficients of the differential operator applied to physical quantities, it follows that these differential operators are proportional to the same directional derivative. Here, as an example, the ccoefficients of the differential operators acting on ρ and m_x, $X^\rho\partial_x + Y^\rho\partial_y$ and $X^{m_x}\partial_x + Y^{m_x}\partial_y$, are compared with the goal of proving that

$$\frac{X^\rho}{Y^\rho} = \frac{X^{m_x}}{Y^{m_x}} \ ,$$

or, avoiding singularities, $X^\rho Y^{m_x} = Y^\rho X^{m_x}$. With the explicit expressions of the coefficients,

$$\begin{aligned}(\mp a^2 n_x \pm u_n u_x)(\mp n_x u_y) &= ? = (\mp a^2 n_y \pm u_n u_y)(2a \mp u_x n_x \mp u_n)\\ 0 \ ? &= a^2 n_x^2 u_y - u_n u_x n_x u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n u_x u_y n_x + u_n^2 u_y\\ 0 \ ? &= a^2 n_x^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n^2 u_y\\ 0 \ ? &= a^2 u_y - a^2 n_y^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n^2 u_y\end{aligned}$$

and if $u_n = \pm a$,

$$\begin{aligned}0 \ ? &= a^2 u_y - a^2 n_y^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y - a^2 u_n n_y \mp 2a u_n u_y + u_n^2 u_y\\ 0 \ ? &= a^2 u_y - a^2 n_y^2 u_y \pm 2a^3 n_y - a^2 u_x n_x n_y \mp a^3 n_y - 2a^2 u_y + a^2 u_y\\ 0 \ ? &= -a^2 n_y(n_y u_y + n_x u_x) \pm 2a^3 n_y \mp a^3 n_y\\ 0 \ ? &= \mp a^3 n_y \pm 2a^3 n_y \mp a^3 n_y = 0 \ .\end{aligned}$$

Ratio of the coefficients of the differential operators

With $a = \pm u_n$, the comparison of coefficients in differential operator acting on ρ and m_x

$$\begin{aligned}
\frac{X^\rho}{X^{m_x}} &= \frac{\mp a^2 n_x \pm u_n u_x}{2a \mp u n_x \mp u_n} = \\
&= \frac{\mp a^2 n_x + a u}{2a \mp u n_x - a} = \\
&= a \frac{\mp a n_x + u}{a \mp u n_x} = \\
&= a \frac{-u_n n_x + u}{\pm u_n \mp u n_x} = \\
&= \pm a \frac{-(u n_x + v n_y) n_x + u}{u n_x + v n_y - u n_x} = \\
&= \pm a \frac{u n_y^2 - v n_y n_x}{+v n_y} = \\
&= \pm a \frac{u n_y - v n_x}{v} .
\end{aligned}$$

$$\begin{aligned}
\frac{Y^\rho}{Y^{m_x}} &= \frac{\mp a^2 n_y \pm u_n v}{\mp n_x v} = \\
&= \frac{\mp a \cdot a n_y \pm u_n v}{\mp n_x v} = \\
&= \frac{-a u_n n_y + a v}{\mp n_x v} = \\
&= \pm a \frac{u_n n_y - v}{n_x v} = \\
&= \pm a \frac{(u n_x + v n_y) n_y - v}{n_x v} = \\
&= \pm a \frac{u n_x n_y - v n_x^2}{n_x v} = \\
&= \pm a \frac{u n_y - v n_x}{v} .
\end{aligned}$$

The comparison of coefficients in differential operator acting on ρ and m_y

$$\begin{aligned}
\frac{X^\rho}{X^{m_y}} &= \frac{\mp a^2 n_x + a u}{\mp n_y u} = \\
&= \frac{\mp a \cdot a n_x \pm u_n u}{\mp n_y u} = \\
&= \frac{-a u_n n_x + a u}{\mp n_y u} = \\
&= \pm a \frac{u_n n_x - u}{n_y u} = \\
&= \pm a \frac{(u n_x + v n_y) n_x - u}{n_y u} = \\
&= \pm a \frac{-u n_y + v n_x}{u} = \\
&= \mp a \frac{u n_y - v n_x}{u} .
\end{aligned}$$

$$\frac{Y^{\rho}}{Y^{m_y}} = \quad =$$

Thus it follows that on the characteristic lines,

$$0 = \pm a(un_y - vn_x)d\rho + vdm_x - udm_y \, ,$$

or, recalling that $m_x = \rho u$, $m_y = \rho v$

$$\begin{aligned} 0 &= \pm a(un_y - vn_x)d\rho + vdm_x - udm_y = \\ &= \pm a(un_y - vn_x)d\rho + \rho vdu + uvd\rho - \rho udv - uvd\rho = \\ &= \pm a(un_y - vn_x)d\rho + \rho vdu - \rho udv = \end{aligned}$$

or evaluating the ratio of these coefficients

$$\frac{\rho v}{\pm a(un_y - vn_x)}$$

todo *Find the general relation.*

Prandtl-Meyer relation

$$0 = \pm a(un_y - vn_x)d\rho + \rho vdu - \rho udv \, .$$

Defining θ as the flow angle w.r.t. the x-axis, s.t. and V the absolute value of the local velocity, it follows

$$\begin{aligned} u &= V\cos\theta \\ v &= V\sin\theta \, . \end{aligned}$$

From the relation $\mathbf{u} \cdot \hat{\mathbf{n}} = \pm a$ on Mach lines, being $\hat{\mathbf{n}}$ the unit vector orthogonal to the Mach lines.,

$$u_n = \pm a \quad , \quad V\cos\mu = \pm a \, ,$$

with μ the angle between the Mach line and the flow. The components of the unit normal vector read

$$\pm a = n_x u + n_y v = V\left(n_x\cos\theta + n_y \sin theta\right)$$

$$\begin{aligned} \pm\sin\mu &= n_x\cos\theta + n_y\sin\theta = \\ &= \pm\sin(\mu - \theta + \theta) = \\ &= \pm\left[\sin(\mu-\theta)\cos\theta + \cos(\mu-\theta)\sin\theta\right] = \end{aligned}$$

so that …

todo *Treat properly $\mp$ and $\pm$ for the 2 families of Mach lines*

$$\begin{aligned} n_x &= \sin(\theta \mp \mu) \\ n_y &= -\cos(\theta \mp \mu) \, . \end{aligned}$$

Now

$$\begin{aligned} un_y - vn_x &= V\left(-\cos\theta\cos(\theta\mp\mu) - \sin\theta\sin(\theta\mp\mu)\right) = \\ &= -V\cos(\theta - \theta \pm \mu) = -V\cos\mu \, . \end{aligned}$$

The differentials of the Cartesian components of the velocity in polar coordinates become

$$\begin{aligned} du &= d(V\cos\theta) = dV\cos\theta - V\sin\theta d\theta \\ dv &= d(V\sin\theta) = dV\sin\theta + V\cos\theta d\theta \end{aligned}$$

So that, putting everything together,

$$\begin{aligned}0 &= \pm a(un_y - vn_x)d\rho + \rho v du - \rho u dv = \\ &= \mp aV\cos\mu d\rho + \rho V\left(\sin\theta\cos\theta - \cos\theta\sin\theta\right)dV + \rho V^2\left(-\sin\theta\sin\theta - \cos\theta\cos\theta\right)d\theta = \\ &= \mp aV\cos\mu d\rho - \rho V^2 d\theta\ ,\end{aligned}$$

i.e.

$$0 = \pm a\cos\mu\, d\rho + \rho V\, d\theta\ .$$

Now, as $\sin\mu = \frac{a}{V} = \frac{1}{M}$, $\cos\mu = \sqrt{1-\sin^2\mu} = \sqrt{1-\frac{1}{M^2}}$, Prandtl-Meyer relation can be recast as

$$\begin{aligned}0 &= \pm\frac{1}{M}\sqrt{1-\frac{1}{M^2}}d\rho + \rho d\theta \\ 0 &= \pm\sqrt{M^2-1}d\rho + M^2\rho d\theta\ .\end{aligned}$$

For an isothermal and irrotational flow, momentum equation reads

$$\begin{aligned}\mathbf{0} &= \mathbf{u}\cdot\nabla\mathbf{u} + a^2\frac{\nabla\rho}{\rho} = \\ &= \nabla\frac{|\mathbf{u}|^2}{2} + \omega\times\mathbf{u} + a^2\frac{\nabla\rho}{\rho} = \\ &= \nabla\frac{|\mathbf{u}|^2}{2} + a^2\frac{\nabla\rho}{\rho}\ ,\end{aligned}$$

and thus, as $\frac{d\rho}{\rho} = -\frac{1}{a^2}d\frac{V^2}{2} = -\frac{1}{a^2}VdV$, Prandtl-Meyer relation becomes

$$\begin{aligned}0 &= \mp\sqrt{M^2-1}\frac{1}{a^2}VdV + M^2\, d\theta \\ 0 &= \mp\sqrt{M^2-1}\,\underbrace{\frac{V^2}{a^2}}_{=M^2}\,\frac{VdV}{V^2} + M^2\, d\theta \\ 0 &= \mp\sqrt{M^2-1}\frac{dV}{V} + d\theta \\ 0 &= \mp\sqrt{M^2-1}\frac{dM}{M} + d\theta\ ,\end{aligned}$$

with the last equation containing only non-dimensional quantities.

…

For $s_2 = u_n$, $u_n = 0$ to get $s_2 = 0$, and thus the corresponding left eigenvector reads

$$\mathbf{l}^T_{0;2} = \frac{1}{\rho a}\left[-\mathbf{t}^T\mathbf{u}\mid\mathbf{t}^T\right]\ ,$$

with $\mathbf{t} = \begin{bmatrix} n_y \\ -n_x \end{bmatrix}$, so that $\mathbf{t}^T\mathbf{n} = 0$. The quasi linear form of the equations becomes

$$0 = \frac{1}{\rho a}\left[\, -\mathbf{t}^T\mathbf{u} \mid \mathbf{t}^T \,\right]\left(\partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & 1 & \cdot \\ a^2 - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} & \cdot \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \end{bmatrix}\partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & 1 \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \\ a^2 - \frac{m_y m_y}{\rho^2} & \cdot & \frac{2m_y}{\rho} \end{bmatrix}\partial_y \begin{bmatrix} \rho \\ m_x \\ m_y \end{bmatrix} = \mathbf{0}\right) =$$

$$0 = \left[\, -\mathbf{t}^T\mathbf{u} \mid \mathbf{t}^T \,\right]\partial_t \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} +$$

$$+ \left[\, t_x a^2 - \mathbf{t}^T\mathbf{u}u_x \mid -\mathbf{t}^T\mathbf{u} + t_x u_x + \mathbf{t}^T\mathbf{u} \mid t_y u_x \,\right]\partial_x \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} +$$

$$+ \left[\, t_y a^2 - \mathbf{t}^T\mathbf{u}u_y \mid t_x u_y \mid -\mathbf{t}^T\mathbf{u} + t_y u_y + \mathbf{t}^T\mathbf{u} \,\right]\partial_y \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} =$$

$$0 = \left[\, -\mathbf{t}^T\mathbf{u} \mid \mathbf{t}^T \,\right]\partial_t \begin{bmatrix} \rho \\ \mathbf{m} \end{bmatrix} +$$

$$+ a^2\left(t_x\partial_x + t_y\partial_y\right)\rho - u_t\left(u_x\partial_x + u_y\partial_y\right)\rho + t_x\left(u_x\partial_x + u_y\partial_y\right)m_x + t_y\left(u_x\partial_x + u_y\partial_y\right)m_y =$$

$$=$$

or choosing $\hat{\mathbf{t}} = \hat{\mathbf{u}}$, s.t. $\mathbf{u} = u\hat{\mathbf{t}}$,

$$\begin{aligned} 0 &= \ldots\partial_t \cdots + a^2\partial_u\rho - u^2\partial_u\rho + ut_x\partial_u m_x + ut_y\partial_u m_y = \\ &= \ldots\partial_t \cdots + a^2\partial_u\rho - u^2\partial_u\rho + \rho\underbrace{ut_x}_{u_x}\partial_u u_x + \rho\underbrace{ut_y}_{u_y}\partial_u u_y + \underbrace{uu_x t_x\partial_u\rho + uu_y t_y\partial_u\rho}_{=u^2\partial_u\rho} = \\ &= \ldots\partial_t \cdots + a^2\,\partial_u\rho + \rho\partial_u\frac{|\mathbf{u}|^2}{2}\,. \end{aligned}$$

In **steady conditions**, dividing by $\rho \neq 0$

$$0 = a^2\,\frac{\partial_u\rho}{\rho} + \partial_u\frac{u^2}{2} = \partial_u\left(a^2\ln\rho + \frac{u^2}{2}\right)\,.$$

This is the isothermal version of the Bernoulli's theorem along a streamline for compressible flows, in absence of volume force. This can be derived from momentum equation in steady conditions, after scalar multiplication by the velocity field

$$\begin{aligned} 0 &= \mathbf{u}\cdot\left[\rho\mathbf{u}\cdot\nabla\mathbf{u} + \nabla p\right] = \\ &= \mathbf{u}\cdot\left[\rho\nabla\frac{|\mathbf{u}|^2}{2} + \omega\times\mathbf{u} + \nabla p\right] = \\ &= \rho\mathbf{u}\cdot\left[\nabla\frac{|\mathbf{u}|^2}{2} + a^2\frac{\nabla\rho}{\rho}\right] = \\ &= \rho\mathbf{u}\cdot\nabla\left(\frac{|\mathbf{u}|^2}{2} + a^2\ln\rho\right)\,. \end{aligned}$$

Euler equations for homoentropic inviscid compressible flows - 2d

Homoentropic flows have constant and uniform entropy, $s(\mathbf{r}, t) = \overline{s}$. This condition immediately follows from the entropy equation under the assumptions of:

1. negligible viscous effects
2. negligible heat conductivity
3. uniform entropy at the boundaries

so that, there's no source in the governing equation of the entropy of material particles (because of 1. and 2., $D_t s = 0$) and the value of the entropy of material particles entering the domain is constant and uniform (because of 3., $s = \overline{s}$).

todo *Check if anything changes from the very beginning, e.g. transforming $\nabla p = a^2 \nabla \rho$ and collecting in the flux term $\nabla \cdot (\rho a^2 \mathbf{I})$.*

Under these conditions, all the computation done for the isothermal flow model are valid except for the ones assuming constant speed of sound a. In general the speed of sound is a function of the thermodynamic state. As an example, using ρ, s as the pair of thermodynamic independent variables, the speed of sound reads

$$a^2(\rho, s) = \left(\frac{\partial p}{\partial \rho} \right)_s .$$

Here a is a function of the density only, as the constant value of entropy can be treated as a constant parameter.

Euler equations for inviscid compressible flows - 2d

1. Using conservative variables.

Direct computation with conservative variables

Conservative form reads

$$\begin{cases} \partial_t \rho + \nabla \cdot \mathbf{m} = 0 \\ \partial_t \mathbf{m} + \nabla \cdot \left[\frac{\mathbf{m} \otimes \mathbf{m}}{\rho} + p \mathbf{I} \right] = 0 \\ \partial_t E^t + \nabla \cdot \left(\frac{E^t + p}{\rho} \mathbf{m} \right) = 0 \, . \end{cases}$$

Convective form in a 2-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \begin{bmatrix} \cdot & 1 & \cdot & \cdot \\ \partial_\rho p - \frac{m_x m_x}{\rho^2} & \frac{2m_x}{\rho} + \partial_{m_x} P & \partial_{m_y} P & \partial_{E^t} p \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} & \cdot \\ \frac{-h^t + \partial_\rho p}{\rho} m_x + \frac{\partial_\rho P}{\rho} & h^t + \frac{\partial_{m_x} P}{\rho} m_x & \frac{\partial_{m_y} P}{\rho} m_x & \frac{1 + \partial_{E^t} P}{\rho} m_x \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & 1 \\ -\frac{m_x m_y}{\rho^2} & \frac{m_y}{\rho} & \frac{m_x}{\rho} \\ \partial_\rho p - \frac{m_y m_y}{\rho^2} & \partial_{m_x} P & \frac{2m_y}{\rho} + \partial_{m_y} P \\ \frac{-h^t + \partial_\rho p}{\rho} m_y & \frac{\partial_{m_x} P}{\rho} m_y & h^t + \frac{\partial_{m_y} P}{\rho} m_y \end{bmatrix}$$

Thus,

$$\mathbf{A}_n = \begin{bmatrix} 0 & \mathbf{n}^T & 0 \\ \partial_\rho p \mathbf{n} - u_n \mathbf{u} & u_n \mathbf{I} + \mathbf{u} \mathbf{n}^T + \mathbf{n} \partial_{\mathbf{m}} P & \partial_{E^t} p \mathbf{n} \\ (-h^t + \partial_\rho p) u_n & h^t \mathbf{n}^T + u_n \partial_{\mathbf{m}} P & (1 + \partial_{E^t} P) \, u_n \end{bmatrix} =$$

$$= \begin{bmatrix} 0 & n_x & n_y & 0 \\ \partial_\rho p \, n_x - u_n u_x & u_n + n_x u + n_x \partial_{m_x} P & n_y u + n_x \partial_{m_y} P & \partial_{E^t} p \, n_x \\ \partial_\rho p \, n_y - u_n u_y & n_x v + n_y \partial_{m_x} P & u_n + n_y v + n_y \partial_{m_y} P & \partial_{E^t} p \, n_y \\ (-h^t + \partial_\rho P) u_n & h^t n_x + u_n \partial_{m_x} P & h^t n_y + u_n \partial_{m_y} P & (1 + \partial_{E^t} P) \, u_n \end{bmatrix} .$$

with

$$\partial_{m_i} P = \partial_e P|_\rho \left(-\frac{m_i}{\rho^2} \right)$$

$$\partial_{\mathbf{m}} P = -\frac{\mathbf{m}}{\rho^2} \, \partial_e P|_\rho = -\frac{\mathbf{u}}{\rho} \partial_e P|_\rho$$

and thus

$$\mathbf{A}_n = \begin{bmatrix} 0 & n_x & n_y & 0 \\ \partial_\rho P\, n_x - u_n u_x & u_n + n_x u(1-\partial_{E^t} P) & n_y u - n_x v \partial_{E^t} P & \partial_{E^t} P\, n_x \\ \partial_\rho P\, n_y - u_n u_y & n_x v - n_y u \partial_{E^t} P & u_n + n_y v(1-\partial_{E^t} P) & \partial_{E^t} P\, n_y \\ (-h^t + \partial_\rho P) u_n & h^t n_x + u_n \partial_{m_x} P & h^t n_y + u_n \partial_{m_y} P & (1+\partial_{E^t} P)\, u_n \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \mathbf{n}^T & 0 \\ \partial_\rho P \mathbf{n} - u_n \mathbf{u} & u_n \mathbf{I} + \mathbf{u}\mathbf{n}^T - \mathbf{n}\mathbf{u}^T \partial_{E^t} P & \partial_{E^t} P \mathbf{n} \\ (-h^t + \partial_\rho P) u_n & h^t \mathbf{n}^T - u_n \mathbf{u}^T \partial_{E^t} P & (1+\partial_{E^t} P)\, u_n \end{bmatrix} .$$

Derivative of the pressure function $P(\rho, \mathbf{m}, E^t)$

As $E^t = \rho\left(e + \frac{|\mathbf{u}|^2}{2}\right) = \rho e^t + \frac{|\mathbf{m}|^2}{2\rho}$, $e^t = \frac{E^t}{\rho} - \frac{|\mathbf{m}|^2}{2\rho^2}$

$$P(\rho, \mathbf{m}, E^t)$$

$$P(\rho, e) = P\left(\rho, e\left(\rho, \mathbf{m}, E^t\right)\right) ,$$

$$\begin{aligned} \partial_\rho P\big|_{\mathbf{m},E^t} &= \partial_\rho P\big|_e + \partial_e P|_\rho\, \partial_\rho e\big|_{\mathbf{m},E^t} &&= \partial_\rho P\big|_e + \partial_e P|_\rho \left(-\frac{E^t}{\rho^2} + \frac{|\mathbf{m}|^2}{\rho^3}\right) \\ \partial_{m_i} P\Big|_{\rho,E^t} &= \partial_e P|_\rho\, \partial_{m_i} e\Big|_{\rho,E^t} &&= \partial_e P|_\rho \left(-\frac{m_i}{\rho^2}\right) \\ \partial_{E^t} P|_{\rho,\mathbf{m}} &= \partial_e P|_\rho\, \partial_{E^t} e|_{\rho,\mathbf{m}} &&= \partial_e P|_\rho \left(\frac{1}{\rho}\right) \end{aligned}$$

and by direct comparison,

$$\begin{aligned} \partial_\rho P\big|_{\mathbf{m},E^t} &= \partial_\rho P\big|_e + \partial_{E^t} P|_{\rho,\mathbf{m}} \left(-e^t + |\mathbf{u}|^2\right) \\ \partial_{\mathbf{m}} P|_{\rho,E^t} &= -\mathbf{u}\, \partial_{E^t} P|_{\rho,\mathbf{m}} \end{aligned}$$

Speed of sound

…link to an updated version of Thermodynamics:potentials:Maxwell's relations

$$a^2(\rho, s) = \frac{\partial p}{\partial \rho}\Big|_s$$

Independent variables $f(\rho, e)$

$$a^2(\rho, e) = \ ?$$

$$\begin{aligned} a^2(\rho, e(\rho, s)) &= \frac{\partial p}{\partial \rho}\Big|_s = \\ &= \frac{\partial p}{\partial \rho}\Big|_e + \frac{\partial p}{\partial e}\Big|_\rho \frac{\partial e}{\partial \rho}\Big|_s = \\ &= \frac{\partial p}{\partial \rho}\Big|_e + \frac{\partial p}{\partial e}\Big|_\rho \frac{p}{\rho^2} . \end{aligned}$$

Thermodynamic potentials, partial derivatives and Maxwell's relations

$$de = T ds + \frac{p}{\rho^2} d\rho \quad , \quad T = \frac{\partial p}{\partial s}\Big|_\rho \quad , \quad \frac{p}{\rho^2} = \frac{\partial e}{\partial \rho}\Big|_s$$

Independent variables $f(\rho, \mathbf{m}, E^t)$ With

$$E^t = \rho\left(\frac{|\mathbf{u}|^2}{2} + e\right) = \frac{|\mathbf{m}|^2}{2\rho} + \rho e \,.$$

$$a^2(\rho, \mathbf{m}, E^t(\rho, \mathbf{m}, s))$$

$$\begin{aligned} a^2(\rho, \mathbf{m}, E^t) &= \left.\frac{\partial p}{\partial \rho}\right|_s (\rho, \mathbf{m}, E^t(\rho, \mathbf{m}, e(\rho, s))) = \\ &= \left.\frac{\partial p}{\partial \rho}\right|_{\mathbf{m}, E^t} + \left.\frac{\partial p}{\partial E^t}\right|_{\rho, \mathbf{m}} \left[\left.\frac{\partial E^t}{\partial \rho}\right|_{\mathbf{m}, e} + \left.\frac{\partial E^t}{\partial e}\right|_{\rho, \mathbf{m}} \left.\frac{\partial e}{\partial \rho}\right|_s\right] = \\ &= \left.\frac{\partial p}{\partial \rho}\right|_{\mathbf{m}, E^t} + \left.\frac{\partial p}{\partial E^t}\right|_{\rho, \mathbf{m}} \left[\left(-\frac{|\mathbf{m}|^2}{2\rho^2} + e\right) + \rho \left.\frac{\partial e}{\partial \rho}\right|_s\right] = \end{aligned}$$

Eigenvalues

$$0 = |\mathbf{A}_n - s\mathbf{I}| = \begin{bmatrix} -s & n_x & n_y & 0 \\ \partial_\rho p\, n_x - u_n u_x & -s + u_n + n_x u & n_y u & \partial_{E^t} p\, n_x \\ \partial_\rho p\, n_y - u_n u_y & n_x v & -s + u_n + n_y v & \partial_{E^t} p\, n_y \\ (-h^t + \partial_\rho P) u_n & h^t n_x + u_n \partial_{m_x} P & h^t n_y + u_n \partial_{m_y} P & -s + (1 + \partial_{E^t} P)\, u_n \end{bmatrix} .$$

Using physical variables $(\rho, \mathbf{u}, s)$ and variable transformation

Convection matrix

Let $\mathbf{u} = (\rho, \mathbf{m}, E^t)$ the vector of conservative variables, and $\mathbf{v} = (\rho, s\mathbf{u}, e)$ the vector of physical variables. As proved in the following section, the relation between the convection matrices reads

$$\mathbf{A}^{\mathbf{u}}_{\mathbf{n}} = \mathbf{U}_{\mathbf{v}} \mathbf{A}^{\mathbf{v}}_{\mathbf{n}} \mathbf{V}_{\mathbf{u}} \,,$$

with

$$\mathbf{A}^{\mathbf{v}}_{\mathbf{n}} = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{1}{\rho}\partial_\rho p|_e \mathbf{n} & u_n \mathbf{I} & \frac{1}{\rho}\partial_e p|_\rho \mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix} ,$$

and

$$\mathbf{U}_{\mathbf{v}} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ u & \rho & \cdot & \cdot \\ v & \cdot & \rho & \cdot \\ e + \frac{|\mathbf{u}|^2}{2} & \rho u & \rho v & \rho \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ \mathbf{u} & \mathbf{I}_2 & \cdot \\ e^t & \rho \mathbf{u}^T & \rho \end{bmatrix}$$

$$\mathbf{V}_{\mathbf{u}} = \mathbf{U}_{\mathbf{v}}^{-1} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ -\frac{u}{\rho} & \frac{1}{\rho} & \cdot & \cdot \\ -\frac{v}{\rho} & \cdot & \frac{1}{\rho} & \cdot \\ \frac{1}{\rho}\left(\frac{|\mathbf{u}|^2}{2} - e\right) & -\frac{u}{\rho} & -\frac{v}{\rho} & \frac{1}{\rho} \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ -\frac{\mathbf{u}}{\rho} & \frac{1}{\rho}\mathbf{I}_2 & \cdot \\ \frac{1}{\rho}\left(|\mathbf{u}|^2 - e^t\right) & -\frac{\mathbf{u}^T}{\rho} & \frac{1}{\rho} \end{bmatrix}$$

the gradient and its inverse (easy to evaluate for the triangular structure of the $\mathbf{U}_{\mathbf{v}}$ matrix) of the change of variables

$$\mathbf{u} = \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho\left(e + \frac{|\mathbf{u}|^2}{2}\right) \end{bmatrix}$$

Right and left eigenvectors

With the right and left eigenvectors using physical variables $(\rho, \mathbf{u}, e)$,

$$\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -an_x & 0 & an_y & an_x \\ -an_y & 0 & -an_x & an_y \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -a\mathbf{n} & \mathbf{0} & a\mathbf{t} & a\mathbf{n} \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix}$$

$$\mathbf{L}^{\mathbf{v}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} & -\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} & \mathbf{0}^T & -\frac{P_e}{\rho a^2} \\ 0 & \frac{\mathbf{t}^T}{a} & 0 \\ \frac{P_\rho}{2\rho a^2} & +\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \end{bmatrix}$$

Right eigenvectors w.r.t. conservative variables are

$$\mathbf{R}^{\mathbf{u}} = \mathbf{U}_{\mathbf{v}}\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho e^t - \rho u_n a + p & \rho e^t - \rho^2\frac{P_\rho}{P_e} & \rho a u_t & \rho e^t + \rho u_n a + p \end{bmatrix} =$$

$$= \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho h^t - \rho u_n a & \rho e^t - \rho^2\frac{P_\rho}{P_e} & \rho a u_t & \rho h^t + \rho u_n a \end{bmatrix},$$

with $h^t = e^t + \frac{|\mathbf{u}|^2}{2}$, and

$$\rho e^t - \rho^2\frac{P_\rho}{P_e} = \rho e^t + P - P - \rho^2\frac{P_\rho}{P_e} = \rho h^t - \frac{\rho^2}{P_e}\left(\frac{P}{\rho^2}P_e + P_\rho\right) = \rho h^t - \frac{\rho^2}{P_e}a^2 .$$

Or using the derivatives of $\Pi(\rho, \mathbf{m}, E^t)$,

$$E^t - \rho^2\frac{P_\rho}{P_e} = E^t - \rho^2\frac{\partial_\rho\Pi - \partial_{E^t}\Pi(-e^t + |\mathbf{u}|^2)}{\rho\partial_{E^t}\Pi} =$$

$$= E^t - \rho\frac{\partial_\rho\Pi}{\partial_{E^t}\Pi} - E^t + \rho|\mathbf{u}|^2 =$$

$$= \rho|\mathbf{u}|^2 - \rho\frac{\partial_\rho\Pi}{\partial_{E^t}\Pi} .$$

Left eigenvectors w.r.t. conservative variables are

$$\mathbf{L}^{\mathbf{u}} = \mathbf{L}^{\mathbf{v}}\mathbf{V}_{\mathbf{u}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} + \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & -\frac{1}{2\rho a}n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & -\frac{1}{2\rho a}n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} - \frac{P_e}{\rho a^2}\frac{|\mathbf{u}|^2 - e^t}{\rho} & \frac{P_e}{\rho a^2}\frac{u}{\rho} & \frac{P_e}{\rho a^2}\frac{v}{\rho} & -\frac{P_e}{\rho^2 a^2} \\ -\frac{u_t}{\rho a} & \frac{t_x}{\rho a} & \frac{t_y}{\rho a} & 0 \\ \frac{P_\rho}{2\rho a^2} - \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & \frac{1}{2\rho a}n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & \frac{1}{2\rho a}n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \end{bmatrix} =$$

$$= \frac{1}{2\rho a^2}\begin{bmatrix} \Pi_\rho + a u_n & -a n_x + \Pi_{m_x} & -a n_y + \Pi_{m_y} & \Pi_{E^t} \\ 2a^2 - 2\Pi_\rho & -2\Pi_{m_x} & 2 - \Pi_{m_y} & -2\Pi_{E^t} \\ -2a u_t & 2a t_x & 2a t_y & 0 \\ \Pi_\rho - a u_n & a n_x + \Pi_{m_x} & a n_y + \Pi_{m_y} & \Pi_{E^t} \end{bmatrix}$$

being

$$\Pi_\rho = P_\rho + \frac{P_e}{\rho}\left(-e^t + |\mathbf{u}|^2\right)$$
$$\Pi_{\mathbf{m}} = -\frac{\mathbf{u}}{\rho}P_e$$
$$\Pi_{E^t} = \frac{1}{\rho}P_e$$

$$2\frac{P}{\rho^2}P_e - 2P_e\frac{|\mathbf{u}|^2 - e^t}{\rho} = 2\frac{P}{\rho^2}P_e + 2P_\rho - 2\Pi_\rho = \\ = 2a^2 - 2\Pi_\rho$$

and the speed of sound

$$a^2(\rho, s) = \partial_\rho P|_s$$

with $P(\rho, e(\rho, s))$

$$a^2 = \partial_\rho P|_e + \partial_\rho e|_s\, \partial_e P|_\rho = \\ = \partial_\rho P|_e + \frac{P}{\rho^2}\partial_e\, P|_\rho\,.$$

2. Using physical variables $(\rho, \mathbf{u}, s)$**.** So that speed of sound $a^2(\rho, s) = \left(\partial_\rho p\right)_s$ naturally appears.

Change of coordinates and convective form

Starting from a set of variables **u** and the convective form of the equations

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u}\,,$$

using another set of variables **v**, the equations become

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u}$$
$$\mathbf{s} = \mathbf{U}_\mathbf{v}(\mathbf{v})\partial_t \mathbf{v} + \sum_k \mathbf{A}^k \mathbf{U}_\mathbf{v}(\mathbf{v})\partial_k \mathbf{v}\,,$$

being $d\mathbf{u} = \mathbf{U}_\mathbf{v} d\mathbf{v}$. If the transformation is non-singular, and $\mathbf{U}_\mathbf{v}^{-1} = \mathbf{V}_\mathbf{u}(\mathbf{v})$, then the differential equations can be recast as

$$\partial_t \mathbf{v} + \sum_k \mathbf{V}_\mathbf{u}\mathbf{A}^k\mathbf{U}_\mathbf{v}\partial_k \mathbf{v} = \mathbf{V}_\mathbf{u}\mathbf{s}$$

With a coordinate transformation, the normal and tangential matrices become

$$\mathbf{A}^\mathbf{v}_\mathbf{n} = \sum_k n_k \mathbf{V}_\mathbf{u}\mathbf{A}^k\mathbf{U}_\mathbf{v} = \\ = \mathbf{V}_\mathbf{u} \underbrace{\sum_k n_k \mathbf{A}^k}_{\mathbf{A}^\mathbf{u}_\mathbf{n}} \mathbf{U}_\mathbf{v} = \mathbf{V}_\mathbf{u}\mathbf{A}^\mathbf{u}_\mathbf{n}\mathbf{U}_\mathbf{v}\,,$$

$$\mathbf{A}^\mathbf{v}_{\mathbf{t}_i} = \cdots = \mathbf{V}_\mathbf{u}\mathbf{A}^\mathbf{u}_{\mathbf{t}_i}\mathbf{U}_\mathbf{v}\,.$$

If the change of variables is non-singular, the Jacobian matrices $\mathbf{V_u}$, $\mathbf{U_v}$ are non singular and reciprocally inverse. Thus, the **determinant** of the matrix $\mathbf{A_n}$ is independent from the set of chosen variables, as

$$\begin{aligned}\det\left(\mathbf{A}_\mathbf{n}^\mathbf{v}\right) &= \det\left(\mathbf{V_u}\right)\det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right)\det\left(\mathbf{U_v}\right) = \\ &= \det\left(\mathbf{V_u}\right)\det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right)\det\left(\mathbf{V_u}\right)^{-1} = \\ &= \det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right)\ .\end{aligned}$$

Thus, the **eigenvalues** are independent from the set of variables as well

$$\begin{aligned}0 &= \det\left(s\mathbf{I}-\mathbf{A}_\mathbf{n}^\mathbf{v}\right) = \\ &= \det\left(\mathbf{V_u}\right)\det\left(s\mathbf{I}-\mathbf{A}_\mathbf{n}^\mathbf{u}\right)\det\left(\mathbf{U_v}\right) = \\ &= \det\left(s\mathbf{I}-\mathbf{A}_\mathbf{n}^\mathbf{u}\right)\ .\end{aligned}$$

The relation between **right eigenvectors** immediately follows

$$\begin{aligned}\mathbf{A}_\mathbf{n}^\mathbf{v}\mathbf{R}^\mathbf{v} &= \mathbf{R}^\mathbf{v}\mathbf{S} \\ \mathbf{V_u}\mathbf{A}_\mathbf{n}^\mathbf{u}\mathbf{U_v}\mathbf{R}^\mathbf{v} &= \mathbf{R}^\mathbf{v}\mathbf{S} \\ \mathbf{A}_\mathbf{n}^\mathbf{u}\underbrace{\mathbf{U_v}\mathbf{R}^\mathbf{v}}_{\mathbf{R}^\mathbf{u}} &= \underbrace{\mathbf{U_v}\mathbf{R}^\mathbf{v}}_{\mathbf{R}^\mathbf{u}}\mathbf{S} \\ \mathbf{A}_\mathbf{n}^\mathbf{u}\mathbf{R}^\mathbf{u} &= \mathbf{R}^\mathbf{u}\mathbf{S}\ ,\end{aligned}$$

with $\mathbf{R}^\mathbf{u} = \mathbf{U_v}\mathbf{R}^\mathbf{v}$, or for any individual eigenvector $\mathbf{r}_i^\mathbf{u} = \mathbf{U_v}\mathbf{r}_i^\mathbf{v}$.

The relation between **left eigenvectors** analogously follows

$$\begin{aligned}\mathbf{L}^\mathbf{v}\mathbf{A}_\mathbf{n}^\mathbf{v} &= \mathbf{S}\mathbf{L}^\mathbf{v} \\ \mathbf{L}^\mathbf{v}\mathbf{V_u}\mathbf{A}_\mathbf{n}^\mathbf{u}\mathbf{U_v} &= \mathbf{S}\mathbf{L}^\mathbf{v} \\ \underbrace{\mathbf{L}^\mathbf{v}\mathbf{V_u}}_{\mathbf{L}^\mathbf{u}}\mathbf{A}_\mathbf{n}^\mathbf{u} &= \mathbf{S}\underbrace{\mathbf{L}^\mathbf{v}\mathbf{V_u}}_{\mathbf{L}^\mathbf{u}} \\ \mathbf{L}^\mathbf{u}\mathbf{A}_\mathbf{n}^\mathbf{u} &= \mathbf{S}\mathbf{L}^\mathbf{u}\ ,\end{aligned}$$

with $\mathbf{L}^\mathbf{u} = \mathbf{L}^\mathbf{v}\mathbf{U_v}$, or for any individual eigenvector $\left(\mathbf{l}_i^\mathbf{u}\right)^T = \left(\mathbf{l}_i^\mathbf{v}\right)^T\mathbf{U_v}$, or $\mathbf{l}_i^\mathbf{u} = \mathbf{U}_\mathbf{v}^T\mathbf{l}_i^\mathbf{v}$.

Convective form in a 2-dimensional domain reads

$$\begin{aligned}D_t\rho + \rho\nabla\cdot\mathbf{u} &= 0 \\ D_t\mathbf{u} + \frac{1}{\rho}\nabla p &= \mathbf{0} \\ D_t e + \frac{p}{\rho}\nabla\cdot\mathbf{u} &= 0\ .\end{aligned}$$

$$\begin{aligned}0 &= \partial_t\rho + u\partial_x\rho + \rho\partial_x u + v\partial_y\rho + \rho\partial_y v \\ 0 &= \partial_t u + u\partial_x u + v\partial_y u + \frac{1}{\rho}\partial_x p = \\ &= \partial_t u + u\partial_x u + v\partial_y u + \frac{1}{\rho}\partial_\rho p|_e\partial_x\rho + \frac{1}{\rho}\partial_e p|_\rho\partial_x e \\ 0 &= \partial_t v + u\partial_x v + v\partial_y v + \frac{1}{\rho}\partial_y p = \\ &= \partial_t v + u\partial_x v + v\partial_y v + \frac{1}{\rho}\partial_\rho p|_e\partial_y\rho + \frac{1}{\rho}\partial_e p|_\rho\partial_y e \\ 0 &= \partial_t e + u\partial_x e + v\partial_y e + \frac{p}{\rho}\left(\partial_x u + \partial_y v\right)\ .\end{aligned}$$

Convective form in a 2-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} u & \rho & \cdot & \cdot \\ \frac{1}{\rho}\partial_\rho p & u & \cdot & \frac{1}{\rho}\partial_e p \\ \cdot & \cdot & u & \cdot \\ \cdot & \frac{p}{\rho} & \cdot & u \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} v & \cdot & \rho & \cdot \\ \cdot & v & \cdot & \cdot \\ \frac{1}{\rho}\partial_\rho p & \cdot & v & \frac{1}{\rho}\partial_e p \\ \cdot & \cdot & \frac{p}{\rho} & v \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} = \mathbf{0}$$

Thus,

$$\mathbf{A}_n = \begin{bmatrix} u_n & \rho\mathbf{n}^T & \cdot \\ \frac{1}{\rho}\partial_\rho p\,\mathbf{n} & u_n\mathbf{I} & \frac{1}{\rho}\partial_e p\,\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix} = \\ = \dots$$

Eigenvalues of the $\mathbf{A_n}$ matrix using variables $(\rho, \mathbf{u}, e)$.

Let $\mathbf{M} = \mathbf{A}_n - \lambda\mathbf{I}$. We expand $\det(\mathbf{M})$ along the **first row**:

$$\det(\mathbf{M}) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

Identification of Minors Substituting the elements from the first row $[\chi, \rho n_x, \rho n_y, 0]$:

$$\det(\mathbf{M}) = \chi \det \underbrace{\begin{bmatrix} \chi & 0 & \beta n_x \\ 0 & \chi & \beta n_y \\ \eta n_x & \eta n_y & \chi \end{bmatrix}}_{M_{11}} - \rho n_x \det \underbrace{\begin{bmatrix} \alpha n_x & 0 & \beta n_x \\ \alpha n_y & \chi & \beta n_y \\ 0 & \eta n_y & \chi \end{bmatrix}}_{M_{12}} + \rho n_y \det \underbrace{\begin{bmatrix} \alpha n_x & \chi & \beta n_x \\ \alpha n_y & 0 & \beta n_y \\ 0 & \eta n_x & \chi \end{bmatrix}}_{M_{13}} - 0$$

Evaluating Minor M_{11} Expanding M_{11} along its first row:

$$\det(M_{11}) = \chi(\chi^2 - \beta\eta n_y^2) - 0 + \beta n_x(0 - \chi\eta n_x) = \chi^3 - \chi\beta\eta n_y^2 - \chi\beta\eta n_x^2$$

Since $n_x^2 + n_y^2 = 1$:

$$\det(M_{11}) = \chi(\chi^2 - \beta\eta)$$

Evaluating Minor M_{12} Expanding M_{12} along its first row:

$$\det(M_{12}) = \alpha n_x(\chi^2 - \beta\eta n_y^2) - 0 + \beta n_x(\alpha n_x \eta n_y - 0) = \alpha\chi^2 n_x$$

Note. Cross terms often cancel in symmetric projections)}

Note. In the final summation, the ρn_x multiplier simplifies this to $\alpha\rho n_x^2\chi^2$.

Evaluating Minor M_{13} Expanding M_{13} along its first row:

$$\det(M_{13}) = \alpha n_x(0 - \beta\eta n_x) - \chi(\alpha n_y\chi - 0) + \beta n_x(\alpha n_y\eta n_x - 0) = -\alpha\chi^2 n_y$$

Total Characteristic Polynomial Combining the terms:

$$0 = \chi[\chi(\chi^2 - \beta\eta)] - \rho n_x[\alpha n_x\chi^2] + \rho n_y[-\alpha n_y\chi^2] = \\ = \chi^2(\chi^2 - \beta\eta) - \alpha\rho\chi^2(n_x^2 + n_y^2) = \\ = \chi^2(\chi^2 - \beta\eta - \alpha\rho)$$

Final Substitution Recall $\alpha\rho = \partial_\rho p$ and $\beta\eta = \frac{p}{\rho^2}\partial_e p$. The speed of sound is $a^2 = \alpha\rho + \beta\eta$:

$$\chi^2(\chi^2 - a^2) = 0$$

Thus the solutions $\chi = u_n - s$ are:

- $\chi_{1,2} = \pm a \implies s_{1,2} = u_n \mp a$
- $\chi_{3,4} = 0 \implies s_{3,4} = u_n$.

Right eigenvectors

The right eigenvectors $\mathbf{R}$ satisfy the equation $(\mathbf{A}_n - \lambda\mathbf{I})\mathbf{R} = 0$. Using the substituted variable $\chi = u_n - \lambda$, the system for any eigenvector $\mathbf{R} = [R_\rho, R_u, R_v, R_e]^T$ is:

1. $\chi R_\rho + \rho n_x R_u + \rho n_y R_v = 0$
2. $\alpha n_x R_\rho + \chi R_u + \beta n_x R_e = 0$
3. $\alpha n_y R_\rho + \chi R_v + \beta n_y R_e = 0$
4. $\eta n_x R_u + \eta n_y R_v + \chi R_e = 0$

Convective Eigenvectors ($\lambda_{1,2} = u_n \implies \chi = 0$) When $\chi = 0$, the system collapses to:

- From (1) and (4): $n_x R_u + n_y R_v = 0$ (Velocity change is purely tangential).
- From (2) and (3): $\alpha R_\rho + \beta R_e = 0 \implies R_\rho = -\frac{\beta}{\alpha} R_e$.

We choose two linearly independent vectors representing the non-acoustic transport:

Entropy Wave ($\mathbf{R}_1$): Only density changes; pressure (and thus velocity/energy) remains constant.

$$\mathbf{R}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -\alpha/\beta \end{bmatrix}$$

Vorticity/Shear Wave ($\mathbf{R}_2$): Only the tangential velocity changes; no change in density or pressure.

$$\mathbf{R}_2 = \begin{bmatrix} 0 \\ n_y \\ -n_x \\ 0 \end{bmatrix}$$

Acoustic Eigenvectors ($\lambda_{3,4} = u_n \pm a \implies \chi = \mp a$) Substituting $\chi = \mp a$ into the system:

1. $\mp a R_\rho + \rho(n_x R_u + n_y R_v) = 0 \implies n_x R_u + n_y R_v = \pm \frac{a}{\rho} R_\rho$
2. From (2) and (3): The velocity changes R_u, R_v must be proportional to n_x, n_y (the wave is longitudinal). Let $R_u = k n_x$ and $R_v = k n_y$.
3. Then $k = \pm \frac{a}{\rho} R_\rho$.

For the energy component from (4): $\eta k(n_x^2 + n_y^2) \mp a R_e = 0 \implies R_e = \pm \frac{\eta k}{a} = \frac{\eta}{\rho} R_\rho$.

Setting $R_\rho = \rho$ as a normalization:

Right-Running Acoustic Wave (C^+):

$$\mathbf{R}_3 = \begin{bmatrix} \rho \\ a n_x \\ a n_y \\ p/\rho \end{bmatrix}$$

Left-Running Acoustic Wave (C^-):

$$\mathbf{R}_4 = \begin{bmatrix} \rho \\ -a n_x \\ -a n_y \\ p/\rho \end{bmatrix}$$

Left eigenvectors

The left eigenvectors $\mathbf{L}$ are found by solving $\mathbf{L}(\mathbf{A}_n - \lambda\mathbf{I}) = 0$. Using $\chi = u_n - \lambda$, $\alpha = \frac{1}{\rho}\partial_\rho p$, and $\beta = \frac{1}{\rho}\partial_e p$:

Convective Eigenvectors ($\chi = 0$)

- $\mathbf{L}_1$ **(Entropy):** Projects onto the state where pressure is constant.

$$\mathbf{L}_1 = [1, 0, 0, -\alpha/\beta]$$

- $\mathbf{L}_2$ **(Vorticity):** Projects onto the tangential velocity change.

$$\mathbf{L}_2 = [0, n_y, -n_x, 0]$$

Acoustic Eigenvectors ($\chi = \mp a$) For the acoustic waves, the weights for u and v align with the normal $\mathbf{n}$.

- $\mathbf{L}_3$ **($u_n + a$):**

$$\mathbf{L}_3 = \left[\frac{\alpha}{2a^2}, \frac{n_x}{2a}, \frac{n_y}{2a}, \frac{\beta}{2a^2}\right]$$

- $\mathbf{L}_4$ **($u_n - a$):**

$$\mathbf{L}_4 = \left[\frac{\alpha}{2a^2}, -\frac{n_x}{2a}, -\frac{n_y}{2a}, \frac{\beta}{2a^2}\right]$$

Compatibility Equations Multiplying the primitive variable differential vector $d\mathbf{V} = [d\rho, du, dv, de]^T$ by these eigenvectors yields:

$$\frac{\alpha}{a^2}d\rho + \frac{\beta}{a^2}de \pm \frac{1}{a}(n_x du + n_y dv) = 0$$

Using $a^2 d\rho_{total} = \alpha\rho d\rho + \beta\rho de$, this simplifies to the physical form:

$$\frac{dp}{\rho a} \pm du_n = 0$$

Shallow water - 2d

53.7 Hyperbolic problems in multi-dimensional domains

Examples

- ***draft*** *dummy page with the scheme used for all the systems*
- ***P-system***
- ***Euler equations***
- ***Shallow water***

53.7.1 General hyperbolic problem

Integral equations

For a volume V at rest

$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \hat{n}\cdot\mathbf{F} = \int_V \mathbf{s}$$

Integral equations for a control volume V at rest

Integral equation for a control volume at rest V

$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \mathbf{u}\vec{u}\cdot\hat{n} = \int_{\partial V} \hat{n}\cdot\mathbf{f} + \int_V \mathbf{s}$$
$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \hat{n}\cdot\mathbf{F} = \int_V \mathbf{s}$$

being $\mathbf{u}$ the vector of the conservative variables, $\mathbf{u} \in \mathbb{R}^n$, and $\vec{u}$ the velocity field in the d-dimensional domain, and $\mathbf{F} := -\mathbf{f} + \vec{v}\mathbf{u}$.

Integral equations for an arbitary domain v_t

Integral equation for an arbitrary domain v_t, with *Reynolds' transport theorem*

$$\frac{d}{dt}\int_{v_t} \mathbf{u} + \oint_{\partial v_t} \mathbf{u}(\vec{u}-\vec{v}_b)\cdot\hat{n} = \int_{\partial v_t} \hat{n}\cdot\mathbf{f} + \int_{v_t} \mathbf{s}$$
$$\frac{d}{dt}\int_{v_t} \mathbf{u} - \oint_{\partial v_t} \mathbf{u}\vec{v}_b\cdot\hat{n} + \int_{\partial v_t} \hat{n}\cdot\mathbf{F} = \int_{v_t} \mathbf{s}$$

Jump conditions

$$\hat{n}\cdot\vec{v}_b\,[\,\mathbf{u}\,] = \hat{n}\cdot[\,\mathbf{F}\,] \tag{53.1}$$

From integral equations on v_t to jump relations

Collapsing v_t on a surface only the boundary terms on the two sides of the surface persist and the balance equations becomes

$$\begin{aligned}
\mathbf{0} &= \oint_{\partial v_t} (-\hat{n}\cdot\vec{v}_b\mathbf{u} + \hat{n}\cdot\mathbf{F}) = \\
&= \int_{S_-} (-\hat{n}_-\cdot\vec{v}_b\mathbf{u}_- + \hat{n}_-\cdot\mathbf{F}_-) + \int_{S_+} (-\hat{n}_+\cdot\vec{v}_b\mathbf{u}_+ + \hat{n}_+\cdot\mathbf{F}_+) = \\
&= \int_S (-\hat{n}_-\cdot\vec{v}_b(\mathbf{u}_- - \mathbf{u}_+) + \hat{n}_-\cdot(\mathbf{F}_- - \mathbf{F}_+))
\end{aligned}$$

being $\hat{n}_- = -\hat{n}_+$. From the arbitrariness of the surface S, jump relations follow

$$\hat{n}\cdot\vec{v}_b\,[\,\mathbf{u}\,] = [\,\mathbf{F}\,]\ ,$$

being $[\,\cdot\,] = (\cdot)_+ - (\cdot)_-$ the jump across the surface.

Differential equations

In regions where the fields are smooth, differential equations follow from integral equations using *theorems of differential calculus*.

Conservative form

Conservative form of the differential equations reads

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F}(\mathbf{u}) = \mathbf{s} \, . \tag{53.2}$$

From integral equations on V to conservative form of differential equations

For a volume V at rest

$$\frac{d}{dt} \int_V \mathbf{u} + \oint_{\partial V} \hat{n} \cdot \mathbf{F} = \int_V \mathbf{s}$$
$$\int_V \partial_t \mathbf{u} + \oint_{\partial V} \hat{n} \cdot \mathbf{F} = \int_V \mathbf{s}$$

or using components

$$\int_V \partial_t u_i + \oint_{\partial V} n_k F_{ki} = \int_V s_i \, ,$$

and using [divergence theorem] under the assumption of sufficient regularity of the flux

$$\begin{aligned} 0 = \int_V \partial_t u_i + \oint_{\partial V} n_k F_{ki} - \int_V s_i = \\ = \int_V \partial_t u_i + \int_V \partial_k F_{ki} - \int_V s_i = \\ = \int_V \{ \partial_t u_i + \partial_k F_{ki} - s_i \} \, , \end{aligned}$$

and from the aribitrariness of the domain V,

$$\partial_t u_i + \partial_k F_{ki} = s_i \, ,$$

or using abstract vector notation

$$\partial_t \mathbf{u} + \nabla \cdot \mathbf{F} = \mathbf{s} \, .$$

$$\partial_t \mathbf{u} + \sum_{k=1}^{d} \mathbf{A}^{(k)}(\mathbf{u}) \, \partial_k \mathbf{u} = \mathbf{s} \, ,$$

with the convection matrices

$$\left\{ \mathbf{A}^{(k)} \right\}_{i\ell} = \partial_{u_\ell} F_{ki} \, .$$

From conservative form to convective form of differential equations

Starting from the component expression {eq}`` of the convective differential equations

$$
\begin{aligned}
s_i &= \partial_t u_i + \partial_k F_{ki}(u_\ell) = \\
&= \partial_t u_i + \partial_{u_\ell} F_{ki}(u_\ell) \partial_k u_\ell = \\
&= \partial_t u_i + A^{(k)}_{i\ell}(u_\ell) \partial_k u_\ell = \\
&= \partial_t u_i + \sum_{k=1}^{d} A^{(k)}_{i\ell}(u_\ell) \partial_k u_\ell \ ,
\end{aligned}
$$

having defined $A^{(k)}_{i\ell} = \partial_{u_\ell} F_{ki}$, or using vector notation

$$
\partial_t \mathbf{u} + \sum_{k=1}^{d} \mathbf{A}^{(k)}(\mathbf{u}) \partial_k \mathbf{u} = \mathbf{s} \ .
$$

Method of characteristics

Spectral decomposition

Summary of method of characteristics in multi-dimensional domains.

Let

$$
\begin{aligned}
\mathbf{u}(t, \vec{r}) &: \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^n \\
\mathbf{F}(\mathbf{u}) &: \mathbb{R}^n \to \mathbb{R}^d \times \mathbb{R}^n \\
\mathbf{s}(t, \vec{r}) &: \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^n
\end{aligned}
$$

the conservative form of a hyperbolic problem reads

$$
\begin{aligned}
\partial_t \mathbf{u} + \nabla \cdot \mathbf{F}(\mathbf{u}) &= \mathbf{s} \\
\partial_t u_i + \partial_j F_{ji} &= s_i \ .
\end{aligned}
$$

Expanding the divergence of the flux, the convective (quasi-linear? It makes little to no sense to me…) form follows

$$
\begin{aligned}
s_i &= \partial_t u_i + \sum_{j=1}^{d} \partial_j F_{ji} = \\
&= \partial_t u_i + \sum_{j=1}^{d} \sum_{k=1}^{n} \partial_j u_k \partial_{u_k} F_{ji} = \\
&= \partial_t u_i + \sum_{j=1}^{d} \sum_{k=1}^{n} A^j_{ik}(\mathbf{u}) \, \partial_j u_k \ ,
\end{aligned}
$$

for every component $i = 1 : n$, or

$$
\mathbf{s} = \partial_t \mathbf{u} + \sum_{j=1}^{d} \mathbf{A}^j \partial_j \mathbf{u} \ .
$$

This problem can be made a little more general with by defining $\mathbf{x} = (t, \mathbf{r}) = (x_0, \vec{r})$, as

$$
\sum_{j=0}^{d} \mathbf{A}^j \, \partial_j \mathbf{u} = \mathbf{s} \ .
$$

The former expression of the hyperbolic problem immediately follows if $\mathbf{A}_0 = \mathbf{I}_n$, $A^0_{ik} = \delta_{ik}$, and

$$\widetilde{\nabla} \cdot \tilde{\mathbf{F}} = \partial_0 \mathbf{F}_0 + \sum_{j=1}^{d} \partial_j \mathbf{F}_j = \widetilde{\nabla} \cdot [\, \mathbf{u} \mid \mathbf{F}] \ .$$

as, for $i = 1 : n$,

$$\delta_{ik} = \partial_{u_k} F_{0i} \quad \rightarrow \quad F_{0i} = u_k \delta_{ik} = u_i \ .$$

Taylor expansion of a solution. Starting from the solution on a manifold determined by the equation $S : f(\mathbf{x}) = 0$, $\mathbf{x}_0 \in S$, if the solution is differentiable, the solution in a point $\mathbf{x} = \mathbf{x}_0 + \Delta\mathbf{x}$ reads

$$\mathbf{u}(\mathbf{x}) \sim \mathbf{u}(\mathbf{x}_0) + \Delta\mathbf{x} \cdot \nabla \mathbf{u}(\mathbf{x}_0)$$
$$u_i(\mathbf{x}) \sim u_i(\mathbf{x}_0) + \Delta x_j \frac{\partial u_i}{\partial x_j}(\mathbf{x}_0) \ .$$

Now, with a change of coordinates from $\mathbf{x}$ to $\xi = (n, \mathbf{t})$, i.e. to local normal and tangential directions,

$$\nabla \mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial}{\partial x_i} \mathbf{u} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial x_i} = \hat{\mathbf{x}}_i \frac{\partial \mathbf{u}}{\partial \xi_k} \frac{\partial \xi_k}{\partial x_i} = \hat{\xi}_k \frac{\partial \mathbf{u}}{\partial \xi_k}$$

Inserting in the hyperbolic equation

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \frac{\partial u_k}{\partial x_j} = \\ &= \sum_{j=0}^{d} A^j_{ik} \sum_{\ell=0}^{d} \frac{\partial u_k}{\partial \xi_\ell} \frac{\partial \xi_\ell}{\partial x_j} = \\ &= \sum_{\ell=0}^{d} \sum_{j=0}^{d} A^j_{ik} \frac{\partial \xi_\ell}{\partial x_j} \frac{\partial u_k}{\partial \xi_\ell} = \\ &= \sum_{\ell=0}^{d} \sum_{j=0}^{d} A^j_{ik} \xi^\ell_j \frac{\partial u_k}{\partial \xi_\ell} \ , \end{aligned}$$

and separating the normal $\ell = 0$ from the tangential $\ell = 1 : d$ coordinates,

$$\begin{aligned} s_i &= \sum_{j=0}^{d} A^j_{ik} \xi^0_j \frac{\partial u_k}{\partial \xi_0} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} A^j_{ik} \xi^t_j \frac{\partial u_k}{\partial \xi_t} = \\ &= \sum_{j=0}^{d} A^j_{ik} n_j \frac{\partial u_k}{\partial n} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} A^j_{ik} t^\ell_j \frac{\partial u_k}{\partial t_\ell} = \\ &= \sum_{j=0}^{d} n_j \mathbf{A}^j \partial_n \mathbf{u} + \sum_{\ell=1}^{d} \sum_{j=0}^{d} t^\ell_j \mathbf{A}^j \partial_{t_\ell} \mathbf{u} \ . \end{aligned}$$

On a smooth surface where the solution is known, all the tangential derivatives of the solution are known as well. In order to evaluate the normal derivative $\partial_n \mathbf{u}$ from the PDE, the (formal) inversion of the matrix

$$\mathbf{A}_{\hat{\mathbf{n}}} := \sum_{j=0}^{d} n_j \, \mathbf{A}^j \ ,$$

must be feasible, and thus $\mathbf{A}_{\hat{\mathbf{n}}}$ must be invertible, to formally get

$$\partial_n \mathbf{u} = \mathbf{A}^{-1}_{\hat{\mathbf{n}}} \left(\mathbf{s} - \sum_{\ell=1}^{d} \mathbf{A}_{\hat{\mathbf{t}}_\ell} \partial_{t_\ell} \mathbf{u} \right) \ .$$

This expression of the normal derivative of the solution - whenever it exists - can be used to find the approximation of the solution in normal direction w.r.t. the surface S, i.e. with $\Delta\mathbf{x} = \Delta\ell\hat{\mathbf{n}}$ as

$$\begin{aligned}\mathbf{u}(\mathbf{x}) &\sim \mathbf{u}(\mathbf{x}_0) + \Delta\ell\,\hat{\mathbf{n}}\cdot\nabla\mathbf{u}(\mathbf{x}_0) = \\ &= \mathbf{u}(\mathbf{x}_0) + \Delta\ell\,\partial_n\mathbf{u}(\mathbf{x}_0)\ .\end{aligned}$$

If the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ is diagonalizable, it's invertible if it has no zero eigenvalue. Let $\hat{\mathbf{n}}_i$ be a unit vector so that the matrix $\mathbf{A}_{\hat{\mathbf{n}}}$ has an eigenvalue equal to zero $s_0 = 0$, with right and left eigenvectors $\mathbf{r}_0$, $\mathbf{l}_0$. Recalling the expression of the PDE in normal and tangential components

$$\mathbf{s} = \mathbf{A}_{\hat{\mathbf{n}}}\partial_n\mathbf{u} + \sum_{\ell=1}^{d}\mathbf{A}_{\hat{\mathbf{d}}_\ell}\partial_{t_\ell}\mathbf{u}\ ,$$

left-multiplying by $\mathbf{l}$ gives

$$\begin{aligned}\mathbf{l}_0^T\mathbf{s} &= \mathbf{l}_0^T\left(\mathbf{A}_{\hat{\mathbf{n}}}\partial_{\hat{n}}\mathbf{u} + \sum_{\ell=1}^{d}\mathbf{A}_{\hat{\mathbf{d}}_\ell}\partial_{t_\ell}\mathbf{u}\right) = \\ &= \mathbf{l}_0^T\sum_{\ell=1}^{d}\mathbf{A}_{\hat{\mathbf{d}}_\ell}\partial_{t_\ell}\mathbf{u} = \\ &= \mathbf{l}_0^T\sum_{k=0}^{d}\mathbf{A}^k\partial_k\mathbf{u}\ ,\end{aligned}$$

i.e. the **compatibility condition** on the surface with unit normal $\hat{\mathbf{n}}$ that makes $s_0 = 0$,

$$\mathbf{l}_0^T\mathbf{s} = \mathbf{l}_0^T\sum_{k=0}^{d}\mathbf{A}^k\partial_k\mathbf{u} = \mathbf{l}_0^T\widetilde{\nabla}\cdot\tilde{\mathbf{F}}\ .$$

Surfaces with unit normal vectors $\hat{\mathbf{n}}$ making eigenvalues of the matrix $\mathbf{A}_{\mathbf{n}}$ equal to zero are defined as **characteristic surfaces**. On each characteristic surface, the corresponding compatibility condition holds.

Change of variables

Let $\mathbf{u}$ the set of conservative variables[1] and let $\mathbf{v}$ another set of variables. Let the transfomration of variables $\mathbf{v}(\mathbf{u})$ be invertible and let $\mathbf{u}(\mathbf{v})$ be the inverse transformation. Let $\mathbf{U}_\mathbf{v}$ the gradient of the transformation $\mathbf{u}(\mathbf{v})$ w.r.t. variables $\mathbf{v}$ and $\mathbf{V}_\mathbf{u}$ the gradient of the inverse transformation $\mathbf{v}(\mathbf{u})$, so that the differentials of the variables read

$$d\mathbf{u} = \mathbf{U}_\mathbf{v}d\mathbf{v} \quad , \quad d\mathbf{v} = \mathbf{V}_\mathbf{u}d\mathbf{u}\ .$$

These gradients are mutually inverse, $\mathbf{U}_\mathbf{v} = \mathbf{V}_\mathbf{u}^{-1}$. The convective form of the equations using $\mathbf{u}$ or $\mathbf{v}$ as primary variables are respectively

$$\partial_t\mathbf{u} + \sum_{k=1}^{d}\mathbf{A}^{\mathbf{u},k}\partial_k\mathbf{u} = \mathbf{s}^\mathbf{u}$$

$$\partial_t\mathbf{v} + \sum_{k=1}^{d}\mathbf{A}^{\mathbf{v},k}\partial_k\mathbf{v} = \mathbf{s}^\mathbf{v}$$

with

$$\begin{aligned}\mathbf{A}^{\mathbf{v},k} &= \mathbf{V}_\mathbf{u}\mathbf{A}^{\mathbf{u},k}\mathbf{U}_\mathbf{v} \\ \mathbf{s}^\mathbf{v} &= \mathbf{V}_\mathbf{u}\mathbf{s}^\mathbf{u}\ .\end{aligned}$$

The equations using $\mathbf{v}$ immediately follows after introducing the relation between the differentials and pre-multiplication of the equations by $\mathbf{V}_\mathbf{u}$,

[1] Once the problem is written in convective form, this set of variable could be an arbitrary set of variables and not only the conservative variables.

Change of variables and spectral decomposition of the matrix $\mathbf{A}_{\hat{n}}$.

Starting from a set of variables $\mathbf{u}$ and the convective form of the equations

$$\mathbf{s} = \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u} \ ,$$

using another set of variables $\mathbf{v}$, the equations become

$$\begin{aligned} \mathbf{s} &= \partial_t \mathbf{u} + \sum_k \mathbf{A}^k \partial_k \mathbf{u} \\ \mathbf{s} &= \mathbf{U}_\mathbf{v}(\mathbf{v}) \partial_t \mathbf{v} + \sum_k \mathbf{A}^k \mathbf{U}_\mathbf{v}(\mathbf{v}) \partial_k \mathbf{v} \ , \end{aligned}$$

being $d\mathbf{u} = \mathbf{U}_\mathbf{v} d\mathbf{v}$. If the transformation is non-singular, and $\mathbf{U}_\mathbf{v}^{-1} = \mathbf{V}_\mathbf{u}(\mathbf{v})$, then the differential equations can be recast as

$$\partial_t \mathbf{v} + \sum_k \mathbf{V}_\mathbf{u} \mathbf{A}^k \mathbf{U}_\mathbf{v} \partial_k \mathbf{v} = \mathbf{V}_\mathbf{u} \mathbf{s}$$

With a coordinate transformation, the normal and tangential matrices become

$$\begin{aligned} \mathbf{A}_\mathbf{n}^\mathbf{v} &= \sum_k n_k \mathbf{V}_\mathbf{u} \mathbf{A}^k \mathbf{U}_\mathbf{v} = \\ &= \mathbf{V}_\mathbf{u} \underbrace{\sum_k n_k \mathbf{A}^k}_{\mathbf{A}_\mathbf{n}^\mathbf{u}} \mathbf{U}_\mathbf{v} = \mathbf{V}_\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{U}_\mathbf{v} \ , \\ \mathbf{A}_{\mathbf{t}_i}^\mathbf{v} &= \cdots = \mathbf{V}_\mathbf{u} \mathbf{A}_{\mathbf{t}_i}^\mathbf{u} \mathbf{U}_\mathbf{v} \ . \end{aligned}$$

If the change of variables is non-singular, the Jacobian matrices $\mathbf{V}_\mathbf{u}$, $\mathbf{U}_\mathbf{v}$ are non singular and reciprocally inverse. Thus, the **determinant** of the matrix $\mathbf{A}_\mathbf{n}$ is independent from the set of chosen variables, as

$$\begin{aligned} \det\left(\mathbf{A}_\mathbf{n}^\mathbf{v}\right) &= \det\left(\mathbf{V}_\mathbf{u}\right) \det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right) \det\left(\mathbf{U}_\mathbf{v}\right) = \\ &= \det\left(\mathbf{V}_\mathbf{u}\right) \det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right) \det\left(\mathbf{V}_\mathbf{u}\right)^{-1} = \\ &= \det\left(\mathbf{A}_\mathbf{n}^\mathbf{u}\right) \ . \end{aligned}$$

Thus, the **eigenvalues** are independent from the set of variables as well

$$\begin{aligned} 0 &= \det\left(s\mathbf{I} - \mathbf{A}_\mathbf{n}^\mathbf{v}\right) = \\ &= \det\left(\mathbf{V}_\mathbf{u}\right) \det\left(s\mathbf{I} - \mathbf{A}_\mathbf{n}^\mathbf{u}\right) \det\left(\mathbf{U}_\mathbf{v}\right) = \\ &= \det\left(s\mathbf{I} - \mathbf{A}_\mathbf{n}^\mathbf{u}\right) \ . \end{aligned}$$

The relation between **right eigenvectors** immediately follows

$$\begin{aligned} \mathbf{A}_\mathbf{n}^\mathbf{v} \mathbf{R}^\mathbf{v} &= \mathbf{R}^\mathbf{v} \mathbf{S} \\ \mathbf{V}_\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v} &= \mathbf{R}^\mathbf{v} \mathbf{S} \\ \mathbf{A}_\mathbf{n}^\mathbf{u} \underbrace{\mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v}}_{\mathbf{R}^\mathbf{u}} &= \underbrace{\mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v}}_{\mathbf{R}^\mathbf{u}} \mathbf{S} \\ \mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{R}^\mathbf{u} &= \mathbf{R}^\mathbf{u} \mathbf{S} \ , \end{aligned}$$

with $\mathbf{R}^\mathbf{u} = \mathbf{U}_\mathbf{v} \mathbf{R}^\mathbf{v}$, or for any individual eigenvector $\mathbf{r}_i^\mathbf{u} = \mathbf{U}_\mathbf{v} \mathbf{r}_i^\mathbf{v}$.

The relation between **left eigenvectors** analogously follows

$$\begin{aligned} \mathbf{L}^\mathbf{v} \mathbf{A}_\mathbf{n}^\mathbf{v} &= \mathbf{S} \mathbf{L}^\mathbf{v} \\ \mathbf{L}^\mathbf{v} \mathbf{V}_\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} \mathbf{U}_\mathbf{v} &= \mathbf{S} \mathbf{L}^\mathbf{v} \\ \underbrace{\mathbf{L}^\mathbf{v} \mathbf{V}_\mathbf{u}}_{\mathbf{L}^\mathbf{u}} \mathbf{A}_\mathbf{n}^\mathbf{u} &= \mathbf{S} \underbrace{\mathbf{L}^\mathbf{v} \mathbf{V}_\mathbf{u}}_{\mathbf{L}^\mathbf{u}} \\ \mathbf{L}^\mathbf{u} \mathbf{A}_\mathbf{n}^\mathbf{u} &= \mathbf{S} \mathbf{L}^\mathbf{u} \ , \end{aligned}$$

with $\mathbf{L}^\mathbf{u} = \mathbf{L}^\mathbf{v} \mathbf{U}_\mathbf{v}$, or for any individual eigenvector $\left(\mathbf{l}_i^\mathbf{u}\right)^T = \left(\mathbf{l}_i^\mathbf{v}\right)^T \mathbf{U}_\mathbf{v}$, or $\mathbf{l}_i^\mathbf{u} = \mathbf{U}_\mathbf{v}^T \mathbf{l}_i^\mathbf{v}$.

- Eigenvalues, right and left eigenvectors
- Characteristic lines/surfaces, and compatibility conditions

Riemann problem

- Useful in numerical schemes in finite volume methods, using Godunov flux

Linearization - Roe intermediate state

- Local linearization of the problem, to reduce the computational cost of solving non-linear Riemann problems at all the interfaces in a grid in FVM

Boundary conditions

- characteristic-based
- wall
- …

53.7.2 P-system

53.7.3 Euler equations

Thermodynamics and constitutive equations

todo *Reference to Physics: Thermodynamics bbook*

Thermodynamics

Principles of thermodynamics and differential of the internal energy

The differential of the internal energy as a function of s, ρ as independent thermodynamic variables, $e(\rho, s)$ reads

$$de = T\, ds + \frac{p}{\rho^2} d\rho \,, \tag{53.3}$$

from the principles of thermodynamics, and the definition of the entropy s in classical thermodynamics. Following Gibbs' mathematical foundation of classical thermodynamics, the partial derivatives of the internal energy can be related to temperature, pressure and density

$$T = \left(\frac{\partial e}{\partial s}\right)_\rho \quad , \quad \frac{p}{\rho^2} = \left(\frac{\partial e}{\partial \rho}\right)_s . \tag{53.4}$$

Speed of sound, $a(\rho, s)$.

As it should be clear from the study of Euler equations as an hyperbolic system the speed of sound a can be defined as

$$a^2(\rho, s) = \left(\frac{\partial p}{\partial \rho}\right)_s . \tag{53.5}$$

As an example, the definition of the speed of sound appears in two of the eigenvalues of the system, as shown in (53.16) using the convective form (53.15) of the equations using $(\rho, \vec{u}, s)$ as primary variables.

Speed of sound, and change of independent thermodynamic variables

All the relations in this box mainly exploit **rule of derivatives of composite functions** and basic thermodynamic relations.

As an example, if the pressure is defined as a function of (ρ, e), and the relation $e(\rho, s)$ is known, with pressure can be written as $p(\rho, e) = p(\rho, e(\rho, s))$. With the derivation of the composite function, an alternative expression of the speed of sound follows

$$\begin{aligned} a^2 = \left(\frac{\partial p}{\partial \rho}\right)_s = \\ = \left(\frac{\partial p}{\partial \rho}\right)_e + \left(\frac{\partial e}{\partial \rho}\right)_s \left(\frac{\partial p}{\partial e}\right)_\rho = \\ = \left(\frac{\partial p}{\partial \rho}\right)_e + \frac{p}{\rho^2}\left(\frac{\partial p}{\partial e}\right)_\rho . \end{aligned} \tag{53.6}$$

Conservative variables of Euler equations are $(\rho, \vec{m}, E^t) = \left(\rho, \rho\vec{u}, \rho\left(e + \frac{|\vec{u}^2}{2}\right)\right)$. Pressure as a function of conservative variable reads

$$p = \Pi\left(\rho, \vec{m}, E^t\right) = \Pi\left(\rho, \rho\vec{u}, \rho e(\rho, s) + \rho\frac{|\vec{u}|^2}{2}\right)$$

Constitutive equations for an ideal Newtonian fluid

- Stress tensor $\mathbb{T} = -p\mathbb{I}$
- Stress vector $\vec{t}_{\hat{n}} = \hat{n} \cdot \mathbb{T} = -p\hat{n}$
- Heat conduction flux $\vec{q} = \vec{0}$

Integral equations

Conservative variables and flux are respectively

$$\mathbf{u} = \begin{bmatrix} \rho \\ \vec{m} \\ E^t \end{bmatrix} \tag{53.7}$$

$$\mathbf{F} = \begin{bmatrix} \vec{m} \\ \frac{\vec{m}\otimes\vec{m}}{\rho} + p\mathbb{I} \\ H^t\frac{\vec{m}}{\rho} \end{bmatrix} \tag{53.8}$$

Integral equations for a control volume V at rest

Integral equation for a control volume at rest V immediately follows from the three principles of classial mechanics: conservation of mass, 2-nd principle of Newtonian dynamics, 1-st principle of thermodynamics

$$\begin{aligned}
\frac{d}{dt}\int_V \rho + \oint_{\partial V} \rho\vec{u}\cdot\hat{n} &= 0 \\
\frac{d}{dt}\int_V \rho\vec{u} + \oint_{\partial V} \rho\vec{u}\vec{u}\cdot\hat{n} &= -\oint_{\partial V} p\hat{n} \\
\frac{d}{dt}\int_V \rho e^t + \oint_{\partial V} \rho e^t\vec{u}\cdot\hat{n} &= -\oint_{\partial V} p\vec{u}\cdot\hat{n}
\end{aligned}$$

with the need for an equation of state with the expression of pressure p as a function of the dynamical variables $p = \Pi(\rho, \vec{u}, e^t)$. Using conservative variables $(\rho, \vec{m}, E^t) = (\rho, \rho\vec{u}, \rho e^t)$

$$\begin{aligned}
\frac{d}{dt}\int_V \rho + \oint_{\partial V} \vec{m}\cdot\hat{n} &= 0 \\
\frac{d}{dt}\int_V \vec{m} + \oint_{\partial V} \left(\frac{\vec{m}\otimes\vec{m}}{\rho} + p\mathbb{I}\right)\cdot\hat{n} &= 0 \\
\frac{d}{dt}\int_V E^t + \oint_{\partial V} H^t\frac{\vec{m}}{\rho}\cdot\hat{n} &= 0\,.
\end{aligned}$$

having defined the total enthalpy per unit volume

$$H^t = \rho h^t = \rho\left(e^t + \frac{p}{\rho}\right) = \rho\left(e + \frac{p}{\rho} + \frac{|\vec{u}|^2}{2}\right) = \rho\left(h + \frac{|\vec{v}|^2}{2}\right)\,.$$

Integral equations for an arbitrary domain v_t

Integral equation for an arbitrary domain v_t, with *Reynolds' transport theorem*

$$\begin{aligned}
\frac{d}{dt}\int_{v_t} \rho + \oint_{\partial v_t} \rho(\vec{u}-\vec{v}_b)\cdot\hat{n} &= \\
\frac{d}{dt}\int_{v_t} \rho\vec{u} + \oint_{\partial v_t} \rho\vec{u}(\vec{u}-\vec{v}_b)\cdot\hat{n} &= \\
\frac{d}{dt}\int_{v_t} \rho e^t + \oint_{\partial v_t} \rho e^t(\vec{u}-\vec{v}_b)\cdot\hat{n} &=
\end{aligned}$$

Jump conditions

Jump conditions (53.1) for Euler equations read

$$\vec{v}_b\cdot\hat{n}\left[\begin{pmatrix}\rho\\ \vec{m}\\ E^t\end{pmatrix}\right] = \hat{n}\cdot\left[\begin{pmatrix}\vec{m}\\ \frac{\vec{m}\vec{m}}{\rho}+p\\ H^t\frac{\vec{m}}{\rho}\end{pmatrix}\right] \tag{53.9}$$

Differential equations

In regions where the fields are smooth, from integral to differential equations using *theorems of differential calculus*

Conservative form

Conservative form of the differential equations immediately follows introducing the expressions (53.7) of the conservative variables and (53.8) of the flux in the general expression of the conservative form of hyperbolic problems (53.2)

$$\partial_t \begin{bmatrix} \rho \\ \vec{m} \\ E^t \end{bmatrix} + \nabla \cdot \begin{bmatrix} \vec{m} \\ \frac{\vec{m} \otimes \vec{m}}{\rho} + p\mathbb{I} \\ H^t \frac{\vec{m}}{\rho} \end{bmatrix} = \mathbf{0} \,. \tag{53.10}$$

Convective forms

Convective form with conservative variables

Using Cartesian coordinates for a 2-dimensional domain,

$$\begin{cases} \partial_t \rho + \partial_x m_x + \partial_y m_y = 0 \\ \partial_t m_x + \partial_x \left(\frac{m_x^2}{\rho} + \Pi \right) + \partial_y \left(\frac{m_x m_y}{\rho} \right) = 0 \\ \partial_t m_y + \partial_x \left(\frac{m_x m_y}{\rho} \right) + \partial_y \left(\frac{m_y^2}{\rho} + \Pi \right) = 0 \\ \partial_t E^t + \partial_x \left((E^t + \Pi) \frac{m_x}{\rho} \right) + \partial_y \left((E^t + \Pi) \frac{m_y}{\rho} \right) = 0 \end{cases}$$

$$\begin{cases} \partial_t \rho + \partial_x m_x + \partial_y m_y = 0 \\ \partial_t m_x - \frac{m_x^2}{\rho^2} \partial_x \rho + 2 \frac{m_x}{\rho} \partial_x m_x + \partial_x \Pi - \frac{m_x m_y}{\rho^2} \partial_y \rho + \frac{m_y}{\rho} \partial_y m_x + \frac{m_x}{\rho} \partial_y m_y = 0 \\ \partial_t m_x - \frac{m_x m_y}{\rho^2} \partial_x \rho + \frac{m_y}{\rho} \partial_x m_x + \frac{m_x}{\rho} \partial_x m_y - \frac{m_y^2}{\rho^2} \partial_y \rho + 2 \frac{m_y}{\rho} \partial_y m_y + \partial_y \Pi = 0 \\ \partial_t E^t - u h^t \partial_x \rho + h^t \partial_x m_x + u \partial_x E^t + u \partial_x \Pi - v h^t \partial_y \rho + h^t \partial_y m_y + v \partial_y E^t + v \partial_y \Pi = 0 \end{cases}$$

and using the matrix formalism

$$\begin{aligned} \partial_t \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 & 0 \\ -u^2 + \Pi_\rho & 2u + \Pi_u & \Pi_v & \Pi_{E^t} \\ -uv & v & u & \cdot \\ (-h^t + \Pi_\rho) u & h^t + u \Pi_{m_x} & u \Pi_{m_y} & u(1 + \Pi_{E^t}) \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} + \\ + \begin{bmatrix} 0 & 0 & 1 & 0 \\ -uv & v & u & \cdot \\ -v^2 + \Pi_\rho & \Pi_u & 2v + \Pi_v & \Pi_{E^t} \\ (-h^t + \Pi_\rho) v & v \Pi_{m_x} & h^t + v \Pi_{m_y} & v(1 + \Pi_{E^t}) \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ m_x \\ m_y \\ E^t \end{bmatrix} = \mathbf{0} \end{aligned} \tag{53.11}$$

Convective form with primary variables $(\rho, \vec{u}, e)$.

$$\begin{cases} \partial_t \rho + \rho \nabla \cdot \vec{u} + \vec{u} \cdot \nabla \rho = 0 \\ \partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{0} \\ \partial_t e + \vec{u} \cdot \nabla e + \frac{1}{\rho} p \nabla \cdot \vec{u} = 0 \,. \end{cases} \tag{53.12}$$

or using Cartesian coordinates and matrix formalism

$$\partial_t \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} u & \rho & \cdot & \cdot \\ \frac{p_\rho|_e}{\rho} & u & \cdot & \frac{p_e|_\rho}{\rho} \\ \cdot & \cdot & u & \cdot \\ \cdot & \frac{p}{\rho} & \cdot & u \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} + \begin{bmatrix} v & \cdot & \rho & \cdot \\ \cdot & v & \cdot & \cdot \\ \frac{p_\rho|_e}{\rho} & \cdot & v & \frac{p_e|_\rho}{\rho} \\ \cdot & \cdot & \frac{p}{\rho} & v \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ u \\ v \\ e \end{bmatrix} = \mathbf{0} \tag{53.13}$$

Convective form with primary variables $(\rho, \vec{u}, s)$.

Using the thermodynamic relation $de = Tds + \frac{p}{\rho^2} d\rho$ and combining mass and internal energy equations, the differential equation $D_t s = 0$ follows for the entropy, having introduced the material derivative $D_{t_} = \partial_{t_} + \vec{u} \cdot \nabla__$. The convective form of the equations becomes

$$\begin{cases} \partial_t \rho + \rho \nabla \cdot \vec{u} + \vec{u} \cdot \nabla \rho = 0 \\ \partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{0} \\ \partial_t s + \vec{u} \cdot \nabla s = 0 \, . \end{cases} \tag{53.14}$$

or using Cartesian coordinates and matrix formalism in a 2-dimensional domain

$$\partial_t \begin{bmatrix} \rho \\ u \\ v \\ s \end{bmatrix} + \begin{bmatrix} u & \rho & \cdot & \cdot \\ \frac{p_\rho|_s}{\rho} & u & \cdot & \frac{p_s|_\rho}{\rho} \\ \cdot & \cdot & u & \cdot \\ \cdot & \cdot & \cdot & u \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ u \\ v \\ s \end{bmatrix} + \begin{bmatrix} v & \cdot & \rho & \cdot \\ \cdot & v & \cdot & \cdot \\ \frac{p_\rho|_s}{\rho} & \cdot & v & \frac{p_s|_\rho}{\rho} \\ \cdot & \cdot & \cdot & v \end{bmatrix} \partial_y \begin{bmatrix} \rho \\ u \\ v \\ s \end{bmatrix} = \mathbf{0} \tag{53.15}$$

Using this expression of the equations, it's pretty easy to find the eigenvalues of the system, evaluating the eigenvalues of the matrix

$$\mathbf{A}^{\mathbf{v}}_{\hat{n}} = n_x \mathbf{A}^{\mathbf{v}}_x + n_y \mathbf{A}^{\mathbf{v}}_y = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{p_\rho|_s}{\rho} \mathbf{n} & u_n \mathbf{I}_2 & \frac{p_s|_\rho}{\rho} \mathbf{n} \\ \cdot & \cdot & u_n \end{bmatrix}$$

as the zeros of the following equation,

$$0 = |\mathbf{A}^{\mathbf{v}}_{\hat{n}} - s\mathbf{I}_4| = (u_n - s)^2 \left((u_n - s)^2 - \left. \frac{\partial p}{\partial \rho} \right|_s \right) = (u_n - s)^2 \left((u_n - s)^2 - a^2 \right) \, ,$$

having used the definition of the spped of sound (53.5). Thus the **eigenvalues of the system** are

$$\begin{aligned} s_{1,4} &= u_n \mp a \\ s_{2,3} &= u_n \, . \end{aligned} \tag{53.16}$$

Spectral decomposition

It may be convenient to evaluate the spectrum using physical variables $\mathbf{v} = (\rho, \vec{u}, e)$ and then use the rules for the spectral decomposition after a *change of variables* to get the spectrum in terms of the conservative variables.

Transformation of variables. The tranformation between physical variables $\mathbf{v} = (\rho, \vec{u}, e)$ and the conservative variables $\mathbf{u} = (\rho, \vec{m}, E^t)$ reads

$$\mathbf{v}(\mathbf{u}) = \begin{bmatrix} \rho \\ \vec{u} \\ e \end{bmatrix} = \begin{bmatrix} \rho \\ \frac{\vec{m}}{\rho} \\ \frac{E^t}{\rho} - \frac{|\vec{m}|^2}{2\rho^2} \end{bmatrix} \qquad , \qquad \mathbf{u}(\mathbf{v}) = \begin{bmatrix} \rho \\ \vec{m} \\ E^t \end{bmatrix} = \begin{bmatrix} \rho \\ \rho \vec{u} \\ \rho \left(e + \frac{|\vec{u}|^2}{2} \right) \end{bmatrix} \, .$$

Gradients of the transformations.

$$\mathbf{U}_{\mathbf{v}} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ u & \rho & \cdot & \cdot \\ v & \cdot & \rho & \cdot \\ e + \frac{|\mathbf{u}|^2}{2} & \rho u & \rho v & \rho \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ \mathbf{u} & \mathbf{I}_2 & \cdot \\ e^t & \rho \mathbf{u}^T & \rho \end{bmatrix}$$

$$\mathbf{V}_{\mathbf{u}} = \mathbf{U}_{\mathbf{v}}^{-1} = \begin{bmatrix} 1 & \cdot & \cdot & \cdot \\ -\frac{u}{\rho} & \frac{1}{\rho} & \cdot & \cdot \\ -\frac{v}{\rho} & \cdot & \frac{1}{\rho} & \cdot \\ \frac{1}{\rho}\left(\frac{|\mathbf{u}|^2}{2} - e\right) & -\frac{u}{\rho} & -\frac{v}{\rho} & \frac{1}{\rho} \end{bmatrix} = \begin{bmatrix} 1 & \cdot & \cdot \\ -\frac{\mathbf{u}}{\rho} & \frac{1}{\rho}\mathbf{I}_2 & \cdot \\ \frac{1}{\rho}\left(|\mathbf{u}|^2 - e^t\right) & -\frac{\mathbf{u}^T}{\rho} & \frac{1}{\rho} \end{bmatrix}$$

Spectral decomposition of $\mathbf{A}^{\mathbf{v}}_{\hat{n}}$, using $\mathbf{v} = (\rho, \vec{u}, e)$. The normal convective matrix using physical variables $\mathbf{v}$ reads

$$\mathbf{A}^{\mathbf{v}}_{\hat{n}} = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{1}{\rho}\partial_\rho p|_e \mathbf{n} & u_n \mathbf{I} & \frac{1}{\rho}\partial_e p|_\rho \mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix} ,$$

Eigenvalues

In a 2-dimensional domain, the system of equations is 4-dimensional. As shown with the convective form (53.15) using $(\rho, \vec{u}, s)$ as primary variables, the eigenvalues of the system are

$$\{u_n - a, u_n, u_n, u_n + a\} .$$

In a 3-dimensional domain, the system of equations is 5-dimensional, and its eigenvalues are

$$\{u_n - a, u_n, u_n, u_n, u_n + a\} .$$

Right eigenvectors

In a 2-dimensional domain

$$\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -an_x & 0 & an_y & an_x \\ -an_y & 0 & -an_x & an_y \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ -a\mathbf{n} & \mathbf{0} & a\mathbf{t} & a\mathbf{n} \\ \frac{p}{\rho} & -\rho\frac{\partial_\rho p|_e}{\partial_e p|_\rho} & 0 & \frac{p}{\rho} \end{bmatrix}$$

Details for d-dimensional domain

Given the eigenvalues of the system in a d-dimensional ($d = 2, 3$), the right eigenvectors of the matrix $\mathbf{A}^{\mathbf{v}}_{\hat{n}}$,

$$\mathbf{A}^{\mathbf{v}}_{\hat{n}} = n_i \mathbf{A}^{\mathbf{v}}_i = \begin{bmatrix} u_n & \rho \mathbf{n}^T & \cdot \\ \frac{p_\rho|_s}{\rho}\mathbf{n} & u_n \mathbf{I}_d & \frac{p_s|_\rho}{\rho}\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & u_n \end{bmatrix}$$

are evaluated. For $s_{1,d+2} = u_n \mp a$, the singular system

$$\mathbf{0} = \begin{bmatrix} \pm a & \rho \mathbf{n}^T & \cdot \\ \frac{p_\rho|_e}{\rho}\mathbf{n} & \pm a \mathbf{I}_3 & \frac{p_e|_\rho}{\rho}\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & \pm a \end{bmatrix} \begin{bmatrix} \tilde{\rho} \\ \tilde{\mathbf{u}} \\ \tilde{e} \end{bmatrix} ,$$

has the non-trivial solution - it should be easy to prove it recalling the expression (53.6) of the speed of sound

$$\begin{bmatrix} \tilde{\rho}_{1,d+2} \\ \tilde{\vec{u}}_{1,d+2} \\ \tilde{e}_{1,d+2} \end{bmatrix} = \begin{bmatrix} \rho \\ \mp a\mathbf{n} \\ \frac{p}{\rho} \end{bmatrix} .$$

For $s_{2:d+1} = u_n$, the solution of singular system

$$\mathbf{0} = \begin{bmatrix} \cdot & \rho\mathbf{n}^T & \cdot \\ \frac{p_\rho|_e}{\rho}\mathbf{n} & \mathbf{0}_d & \frac{p_e|_\rho}{\rho}\mathbf{n} \\ \cdot & \frac{p}{\rho}\mathbf{n}^T & \cdot \end{bmatrix} \begin{bmatrix} \tilde{\rho} \\ \tilde{\mathbf{u}} \\ \tilde{e} \end{bmatrix} ,$$

is a linear combination of the basis of the kernel of the matrix. It shouldn't be hard to prove that the kernel is spanned by the right eigenvectors

$$\mathbf{r}_2 = \begin{bmatrix} \rho \\ \mathbf{0} \\ -\rho\frac{p_\rho|_e}{p_e|_\rho} \end{bmatrix} ,$$

$$\mathbf{r}_{2+i} = \begin{bmatrix} \cdot \\ a\mathbf{t}_i \\ \cdot \end{bmatrix} \quad , \quad i = 1 : d \, ,$$

with $\mathbf{t}_i$ $d-1$ independent "tangent vectors", orthogonal w.r.t. the unit "normal vector" $\mathbf{n}$.

Left eigenvectors

In a 2-dimensional domain

$$\mathbf{L}^{\mathbf{v}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} & -\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} & \mathbf{0}^T & -\frac{P_e}{\rho a^2} \\ 0 & \frac{\mathbf{t}^T}{a} & 0 \\ \frac{P_\rho}{2\rho a^2} & +\frac{1}{2a}\mathbf{n}^T & \frac{P_e}{2\rho a^2} \end{bmatrix}$$

Spectral decomposition of $\mathbf{A}^{\mathbf{u}}_{\hat{n}}$, using $\mathbf{u} = (\rho, \vec{m}, E^t)$.

Right eigenvectors

In a 2-dimensional domain

$$\mathbf{R}^{\mathbf{u}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ \vec{m} - \rho a\hat{n} & \vec{m} & \rho a\hat{t} & \vec{m} + \rho a\hat{n} \\ H^t - m_n a & \frac{|\vec{m}|^2}{\rho} - \rho\frac{\Pi_\rho}{\Pi_{E^t}} & am_t & H^t + m_n a \end{bmatrix}$$

Left eigenvectors

In a 2-dimensional domain

$$\mathbf{L}^{\mathbf{u}} = \frac{1}{2\rho a^2}\begin{bmatrix} \Pi_\rho + au_n & -an_x + \Pi_{m_x} & -an_y + \Pi_{m_y} & \Pi_{E^t} \\ 2a^2 - 2\Pi_\rho & -2\Pi_{m_x} & 2 - \Pi_{m_y} & -2\Pi_{E^t} \\ -2au_t & 2at_x & 2at_y & 0 \\ \Pi_\rho - au_n & an_x + \Pi_{m_x} & an_y + \Pi_{m_y} & \Pi_{E^t} \end{bmatrix}$$

Some algebra

Right eigenvectors w.r.t. conservative variables are

$$\mathbf{R}^{\mathbf{u}} = \mathbf{U}_{\mathbf{v}}\mathbf{R}^{\mathbf{v}} = \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho e^t - \rho u_n a + p & \rho e^t - \rho^2 \frac{P_\rho}{P_e} & \rho a u_t & \rho e^t + \rho u_n a + p \end{bmatrix} =$$

$$= \begin{bmatrix} \rho & \rho & 0 & \rho \\ \rho u - \rho a n_x & \rho u & \rho a n_y & \rho u + \rho a n_x \\ \rho u - \rho a n_y & \rho v & -\rho a n_x & \rho u + \rho a n_y \\ \rho h^t - \rho u_n a & \rho e^t - \rho^2 \frac{P_\rho}{P_e} & \rho a u_t & \rho h^t + \rho u_n a \end{bmatrix},$$

with $h^t = e^t + \frac{|\mathbf{u}|^2}{2}$, and

$$\rho e^t - \rho^2 \frac{P_\rho}{P_e} = \rho e^t + P - P - \rho^2 \frac{P_\rho}{P_e} = \rho h^t - \frac{\rho^2}{P_e}\left(\frac{P}{\rho^2} P_e + P_\rho\right) = \rho h^t - \frac{\rho^2}{P_e} a^2 .$$

Or using the derivatives of $\Pi(\rho, \mathbf{m}, E^t)$,

$$E^t - \rho^2 \frac{P_\rho}{P_e} = E^t - \rho^2 \frac{\partial_\rho \Pi - \partial_{E^t}\Pi(-e^t + |\mathbf{u}|^2)}{\rho \partial_{E^t}\Pi} =$$

$$= E^t - \rho \frac{\partial_\rho \Pi}{\partial_{E^t}\Pi} - E^t + \rho |\mathbf{u}|^2 =$$

$$= \rho |\mathbf{u}|^2 - \rho \frac{\partial_\rho \Pi}{\partial_{E^t}\Pi} .$$

Left eigenvectors w.r.t. conservative variables are

$$\mathbf{L}^{\mathbf{u}} = \mathbf{L}^{\mathbf{v}}\mathbf{V}_{\mathbf{u}} = \begin{bmatrix} \frac{P_\rho}{2\rho a^2} + \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & -\frac{1}{2\rho a} n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & -\frac{1}{2\rho a} n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \\ \frac{P}{\rho^2}\frac{P_e}{\rho a^2} - \frac{P_e}{\rho a^2}\frac{|\mathbf{u}|^2 - e^t}{\rho} & \frac{P_e}{\rho a^2}\frac{u}{\rho} & \frac{P_e}{\rho a^2}\frac{v}{\rho} & -\frac{P_e}{\rho^2 a^2} \\ -\frac{u_t}{\rho a} & \frac{t_x}{\rho a} & \frac{t_y}{\rho a} & 0 \\ \frac{P_\rho}{2\rho a^2} - \frac{u_n}{2\rho a} + \frac{1}{2\rho a^2}\frac{P_e}{\rho}(|\mathbf{u}|^2 - e^t) & \frac{1}{2\rho a} n_x - \frac{u}{2\rho a^2}\frac{P_e}{\rho} & \frac{1}{2\rho a} n_y - \frac{v}{2\rho a^2}\frac{P_e}{\rho} & \frac{P_e}{2\rho^2 a^2} \end{bmatrix} =$$

$$= \frac{1}{2\rho a^2}\begin{bmatrix} \Pi_\rho + a u_n & -a n_x + \Pi_{m_x} & -a n_y + \Pi_{m_y} & \Pi_{E^t} \\ 2a^2 - 2\Pi_\rho & -2\Pi_{m_x} & 2 - \Pi_{m_y} & -2\Pi_{E^t} \\ -2a u_t & 2a t_x & 2a t_y & 0 \\ \Pi_\rho - a u_n & a n_x + \Pi_{m_x} & a n_y + \Pi_{m_y} & \Pi_{E^t} \end{bmatrix}$$

being

$$\Pi_\rho = P_\rho + \frac{P_e}{\rho}\left(-e^t + |\mathbf{u}|^2\right)$$

$$\Pi_{\mathbf{m}} = -\frac{\mathbf{u}}{\rho} P_e$$

$$\Pi_{E^t} = \frac{1}{\rho} P_e$$

$$2\frac{P}{\rho^2} P_e - 2P_e \frac{|\mathbf{u}|^2 - e^t}{\rho} = 2\frac{P}{\rho^2} P_e + 2P_\rho - 2\Pi_\rho =$$

$$= 2a^2 - 2\Pi_\rho$$

and the speed of sound

$$a^2(\rho, s) = \partial_\rho P|_s$$

with $P(\rho, e(\rho, s))$

$$a^2 = \partial_\rho P|_e + \partial_\rho e|_s \, \partial_e P|_\rho =$$
$$= \partial_\rho P|_e + \frac{P}{\rho^2} \partial_e \, P|_\rho \, .$$

- Eigenvalues, right and left eigenvectors
- Characteristic lines/surfaces, and compatibility conditions

Riemann problem

- Useful in numerical schemes in finite volume methods, using Godunov flux

Linearization - Roe intermediate state

- Local linearization of the problem, to reduce the computational cost of solving non-linear Riemann problems at all the interfaces in a grid in FVM

Boundary conditions

- characteristic-based
- wall
- …

53.7.4 Shallow water

Integral equations

- Intgral equation for a control volume at rest V
- Intgral equation for an arbitrary domain v_t, with Reynolds' transport theorem

Jump conditions

- From the integral equation for an arbitrary domain v_t, collapsed on a surface

Differential equations

- In regions where the fields are smooth, from integral to differential equations with "divergence theorem"

Spectral decomposition

- Eigenvalues, right and left eigenvectors
- Characteristic lines/surfaces, and compatibility conditions

Riemann problem

- Useful in numerical schemes in finite volume methods, using Godunov flux

Linearization - Roe intermediate state

- Local linearization of the problem, to reduce the computational cost of solving non-linear Riemann problems at all the interfaces in a grid in FVM

Boundary conditions

- characteristic-based
- wall
- …

53.8 Riemann problems

Riemann problems in compressible fluid mechanics

53.9 Physical solution in hyperbolic problems

Topics

This section deals with:

- existence of infinite number of solution of hyperbolic problems, and existence of one physical solution of the problem as the inviscid limit of the solution of the diffusive problem
- definition of an entropy function, $\eta(u)$, and its flux, $q(\eta)$
- properties of the entropy functions, and fluxes: **convex**, and non-convex fluxes
- discussion of the general case, and examples (Euler equations, P-sys, shallow water)

53.9.1 Scalar non-linear equation in 1-dimensional domain

The hyperbolic equation for $u(x,t)$, $\partial_t u + \partial_x F(u) = 0$, is often the inviscid limit of the 2-nd order equation

$$\partial_t u + \partial_x F(u) - \partial_x \left(\varepsilon \partial_x u\right) = 0 \, .$$

Term $\varepsilon \partial_x u$ represents a diffusive flux, and the inviscid limit is defined by the condition $\varepsilon \to 0$.

Definition of an entropy function $\eta(u)$, and its flux $q(u)$. Let $\eta(u)$ be a function of the field u. Evaluating its time derivative using the viscous equation,

$$\begin{aligned} \partial_t \eta(u) &= \eta'(u) \partial_t u = \\ &= \eta'(u) \left[-\partial_x F(u) - \partial_x (\varepsilon \partial_x u) \right] = \\ &= -\eta'(u) F'(u) \partial_x u - \eta'(u) (\varepsilon u_{/x})_{/x} \, . \end{aligned}$$

If a function $q(u)$ satisfies the condition

$$q'(u) = \eta'(u) F'(u) \, ,$$

function $\eta(u)$ can be defined as an **entropy function** and $q(u)$ as the corresponding **entropy flux**.

Differential inequality. Let these functions exist. Then

$$\partial_t \eta + \partial_x q = -\eta' \left(\varepsilon u_{/x} \right)_{/x} \, ,$$

is a differential balance equation for the entropy. The last term can be manipulated to find

$$\begin{aligned} 0 &= \partial_t \eta + \partial_x q - \eta' \left(\varepsilon u_{/x} \right)_{/x} = \\ &= \partial_t \eta + \partial_x q - \left(\varepsilon \eta'(u) u_{/x} \right)_{/x} + \eta''(u) \varepsilon \left(u_{/x} \right)^2 \, , \end{aligned}$$

For $\varepsilon \geq 0$, for **convex entropy function** $\eta''(u)$, a differential inequality follows

$$\partial_t \eta + \partial_x q + \left(\varepsilon \eta'(u) u_{/x} \right)_{/x} \leq 0 \, .$$

Integral inequality and jump conditions. Differential inequality is the local counterpart of the integral equation for a volume V at rest

$$\frac{d}{dt} \int_V \eta + \oint_{\partial V} n_x \, q + \oint_{\partial V} \varepsilon q'(u) u_{/x} n_x \leq 0 \, ,$$

or for a generic volume v_t,

$$\frac{d}{dt} \int_{v_t} \eta - \oint_{\partial v_t} n_x v_b \eta + \oint_{\partial v_t} n_x \, q + \oint_{\partial v_t} \varepsilon q'(u) u_{/x} n_x \leq 0 \, ,$$

being v_b the velocity of the points of the boundary of the domain ∂v_t. Jump condition in the inviscid limit $\varepsilon \to 0$ follows from the collapse of the volume on a surface ($\frac{|v_t|}{|\partial v_t|} \to 0$), and from the arbitrariness of this surface

$$v_b [\eta] \geq [q] \, .$$

Weak shocks. Expanding in Taylor series jump condition for weak shocks provides an alternative version of the entropy condition, as a relation between the eigenvalues of the system and the shock speed. The shock speed reads $\dot{s} = \frac{[f]}{[u]} =$

$\frac{f_R - f_L}{u_R - u_L}$. Taylor expansion of jump condition relies on Taylor expansion of the jumps there in and of the speed of the shock

$$
\begin{aligned}
f_R - f_L &= f'(L)\Delta u + f''(L)\frac{\Delta u^2}{2} + f'''(L)\frac{\Delta u^3}{6} + o(\Delta u^3) \\
\dot{s} &= f'(L) + f''(L)\frac{\Delta u}{2} + f'''(L)\frac{\Delta u^2}{6} + o(\Delta u^2) \\
\eta_R - \eta_L &= \eta'(L)\Delta u + \eta''(L)\frac{\Delta u^2}{2} + \eta'''(L)\frac{\Delta u^3}{6} + o(\Delta u^3) \\
q_R - q_L &= q'(L)\Delta u + q''(L)\frac{\Delta u^2}{2} + q'''(L)\frac{\Delta u^3}{6} + o(\Delta u^3)
\end{aligned}
$$

and comparing terms of the same order

$$
0 \leq \Delta u(\eta'_L f'_L - q'_L) + \Delta u^2 \left(\frac{1}{2} f'_L \eta''_L + \frac{1}{2} f''_L \eta'_L - \frac{1}{2} q''_L\right) + \Delta u^3 \left(\frac{1}{6}\eta'''_L f'_L + \frac{1}{4}\eta''_L f''_L + \frac{1}{6}\eta'_L f'''_L - \frac{1}{6} q'''_L\right) + o(\Delta u^3) \, .
$$

Exploiting the relation $q' = \eta' f'$, and evaluating higher-order derivatives

$$
\begin{aligned}
q'' &= \eta'' f' + \eta' f'' \\
q''' &= \eta''' f' + 2\eta'' f'' + \eta' f'''
\end{aligned}
$$

it follows that the first two terms of the jump condition inequality are identically zero, so that it can be written as

$$
0 \leq -\frac{1}{12}\Delta u^3 \eta''_L f''_L \, .
$$

For a system with **convex entropy function** $\eta''(u) \geq 0$, and **convex flux** $f''(u) \geq 0$, it follows that $\Delta u \leq 0$, i.e. $u_R < u_L$. Using the Taylor expansion (either centered in the left or right state) of the speed of the shock,

$$
\begin{aligned}
\dot{s} &= f'_L + f''_L \frac{\Delta u}{2} \leq f'_L \\
\dot{s} &= f'_R - f''_R \frac{\Delta u}{2} \geq f'_R
\end{aligned}
$$

the relation between the eigenvalues of the systems in the left $\lambda_L = f'(u_L)$ and right state $\lambda_R = f'(u_R)$, and the speed of the shock immediately follows

$$
\lambda_L \geq \dot{s} \geq \lambda_R \, .
$$

Burgers' equation

53.9.2 Non-linear system of equations in 1-dimensional domain

$$
\partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \mathbf{0} \, .
$$

If a shock originates from the ith characteristic family, the condition $\lambda_{i,L} \geq \dot{s}_i \geq \lambda_{i,R}$ holds.

Details

Let $\mathbf{A}(\mathbf{u}) = \nabla_{\mathbf{u}} \mathbf{F}(\mathbf{u})$ the convection matrix, $\mathbf{A} = \mathbf{R}\Lambda\mathbf{L}$ its spectral decomposition, and $\mathbf{v}$ the characteristic variables s.t. $d\mathbf{v} = \mathbf{L}d\mathbf{u}$, or $d\mathbf{u} = \mathbf{R}d\mathbf{v}$.

The convective form of the equation reads

$$
\partial_t \mathbf{u} + \mathbf{A}\partial_x \mathbf{u} = \mathbf{0} \, ,
$$

or in diagonal form using characteristic variables,

$$\partial_t \mathbf{v} + \Lambda \partial_x \mathbf{v} = \mathbf{0} \, .$$

A shock originating from the ith family of characteristics comes from the scalar equation

$$\partial_t v_i + \lambda_i \partial_x v_i = 0 \, .$$

The speed of the shock satisfies $\dot{s}_i [\mathbf{u}]_i = [\mathbf{F}(\mathbf{u})]_i$, with $[\mathbf{u}]_i = \mathbf{u}_i^+ - \mathbf{u}_i^-$, and

$$\begin{aligned} \mathbf{u}_i^- &= \mathbf{u}(\mathbf{v}_{\not\in i}, v_i^-) \\ \mathbf{u}_i^+ &= \mathbf{u}(\mathbf{v}_{\not\in i}, v_i^+) \end{aligned}$$

Taylor expansion of the relation gives (no sum on i, as Δv_i is the only non-zero component of $\Delta \mathbf{v}$ across a shock originating from the ith family of characteristics)

$$\begin{aligned} \dot{s}_i \Delta_i u_k = \Delta_i F_k = \\ &= \partial_{u_\ell} F_k \Delta_i u_\ell + \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m = \\ &= A_{k\ell} R_{\ell i} \Delta v_i + \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m \end{aligned}$$

Exploiting spectral decomposition of the convection matrix

$$\left(\dot{s}_i R_{ki} - R_{km} \Lambda_{mn} \underbrace{L_{n\ell} R_{\ell i}}_{\delta_{ni}} \right) \Delta v_i = \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m$$

$$\left(\dot{s}_i - \lambda_i \right) R_{ki} \Delta v_i = \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m$$

$$\left(\dot{s}_i - \lambda_i \right) \Delta_i u_k = \frac{1}{2} \partial_{u_\ell u_m} F_k \Delta_i u_\ell \Delta_i u_m \, .$$

If the flux is convex (here the flux is a vector quantity. Is this the requirement for each of its components F_k?), i.e. the Hessian is (semi)definite positive, the right-hand side is always non-negative.

The following relations have been used above

$$du_l = R_{lm} dv_m \, .$$

$$\Delta_i u_l = \partial_{v_m} u_l \underbrace{\Delta_i v_m}_{= \delta_{im} \Delta v_i} = \partial_{v_i} u_l \Delta v_i = R_{li} \Delta v_i \, .$$

Entropy function for systems

Let $\eta(\mathbf{u})$ the entropy function of the system $\partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \mathbf{0}$. It follows

$$\begin{aligned} \partial_t \eta &= \partial_{u_i} \eta \, \partial_t u_i \\ &= -\partial_{u_i} \eta \, \partial_x F_i = \\ &= -\partial_{u_i} \eta \, \partial_{u_j} F_i \, \partial_x u_j = \\ &= -\partial_{u_j} q \, \partial_x u_j = \\ &= -\partial_x q \, , \end{aligned}$$

i.e. the compatibility equation between the entropy function $\eta(\mathbf{u})$ and the entropy flux $q(\mathbf{u})$ reads

$$\partial_{u_j} q = \partial_{u_j} F_i \, \partial_{u_i} \eta = A_{ij} \partial_{u_i} \eta \, ,$$

or using vector formalism

$$\nabla_{\mathbf{u}} q = \nabla_{\mathbf{u}} \mathbf{F} \cdot \nabla_{\mathbf{u}} \eta = \mathbf{A} \cdot \nabla_{\mathbf{u}} \eta \, .$$

Entropy function in diffusive equations

Let $\partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \partial_x(\mathbf{D}\partial_x \mathbf{u})$ the diffusive equation. The entropy equation becomes

$$\begin{aligned}\partial_t \eta &= \partial_{u_i}\eta\, \partial_t u_i \\ &= -\partial_{u_i}\eta\, \partial_x F_i + \partial_{u_i}\eta \partial_x \left(D_{ik}\partial_x u_k\right) = \\ &= -\partial_{u_i}\eta\, \partial_{u_j} F_i\, \partial_x u_j + \partial_x \left(\partial_{u_i}\eta D_{ik}\partial_x u_k\right) - \partial_x \partial_{u_i}\eta D_{ik}\partial_x u_k = \\ &= -\partial_{u_j} q\, \partial_x u_j + \partial_x \left(\partial_{u_i}\eta D_{ik}\partial_x u_k\right) - \partial_x u_l \partial_{u_l u_i}\eta D_{ik}\partial_x u_k \,.\end{aligned}$$

By definition, if the **diffusion term** is **compatible** with the entropy function, the matrix (is this a matrix, a tensor or a matrix of tensors?) $\nabla_{\mathbf{uu}} \cdot \mathbf{D}$ is semi-definite positive. If the diffusion term is compatible with the entropy function, in the inviscid limit $D_{ik} \to 0$, the entropy inequality becomes

$$\partial_t \eta - \partial_x q \leq 0 \,.$$

Entropy variables

Now, let the **entropy variables** be $\mathbf{w} := \nabla_{\mathbf{u}}\eta$, $w_i = \partial_{u_i}\eta$. The last term becomes

$$\begin{aligned}\frac{\partial u_l}{\partial x}\frac{\partial u_k}{\partial x}\frac{\partial}{\partial u_l}\frac{\partial}{\partial u_i}\eta D_{ik} &= \frac{\partial u_l}{\partial x}\frac{\partial u_k}{\partial x}\frac{\partial}{\partial u_l} w_i D_{ik} = \\ &= \frac{\partial w_i}{\partial x} D_{ik}\frac{\partial u_k}{\partial x} = \\ &= \partial_x \mathbf{w} \cdot \mathbf{D} \cdot \left(\nabla_{\mathbf{uu}}\eta\right)^{-1} \cdot \partial_x \mathbf{w} \,,\end{aligned}$$

being

$$\partial_x \mathbf{u} = \partial_x \mathbf{w} \cdot \nabla_{\mathbf{w}} \mathbf{u} = \partial_x \mathbf{w} \cdot \left(\nabla_{\mathbf{u}} \mathbf{w}\right)^{-1} = \partial_x \mathbf{w} \cdot \left(\nabla_{\mathbf{uu}}\eta\right)^{-1}$$

or, by symmetry of the Hessian, $\partial_x \mathbf{u} = \left(\nabla_{\mathbf{uu}}\eta\right)^{-1} \cdot \partial_x \mathbf{w}$.

Compatibility between the diffusion and the entropy function needs $\mathbf{D} \cdot \left(\nabla_{\mathbf{uu}}\eta\right)^{-1}$ to be semi-definite positive.

$$\partial_x u_k = \partial_{w_j} u_k \partial_x w_j = \left[\partial_{u_k} w_j\right]^{-1} \partial_x w_j \,,$$

$$\delta_{ij} = \partial_{u_i} w_k \partial_{w_k} u_j \,.$$

P-system

A common choice of *entropy function* in a P-system is the mechanical energy (the "reversible" part of the energy).

$$\eta = \rho\left(\frac{1}{2}u^2 + a^2 \ln\frac{\rho}{\rho_0}\right) \,.$$

This quantity decreases across shocks. Corresponding flux reads

$$q = \rho\left(\frac{1}{2}u^2 + a^2 \ln\frac{\rho}{\rho_0} + a^2\right) u \,.$$

Details

Taking the time derivative of the entropy function,

$$
\begin{aligned}
\eta_{/t} &= \rho_{/t}\frac{u^2}{2} + \rho u u_{/t} + a^2\rho_{/t}\ln\frac{\rho}{\rho_0} + a^2\rho_{/t} = \\
&= \left(\frac{u^2}{2} + a^2\ln\frac{\rho}{\rho_0} + a^2\right)\rho_{/t} + u\rho u_{/t} = \\
&= -\left(\frac{u^2}{2} + a^2\ln\frac{\rho}{\rho_0} + a^2\right)(\rho u)_{/x} - u\left(\rho u u_{/x} + a^2\rho_{/x}\right) = \\
&= -\left(\rho\frac{u^2}{2}u\right)_{/x} \underbrace{- a^2\ln\frac{\rho}{\rho_0}\rho u_{/x} - a^2\ln\frac{\rho}{\rho_0}u\rho_{/x} - a^2\rho_{/x}u}_{=-\left(a^2\rho\ln\frac{\rho}{\rho_0}u\right)_{/x}} \underbrace{- a^2\rho u_{/x} - a^2\rho_{/x}u}_{=-(a^2\rho u)_{/x}} = \\
&= -\left[\left(\rho\frac{u^2}{2} + a^2\rho\ln\frac{\rho}{\rho_0} + \rho a^2\right)u\right]_{/x}.
\end{aligned}
$$

Shallow water

Euler equations

53.10 Convexity in hyperbolic problems

Convexity of the entropy function and the flux are assumption used discussing *physical solutions in hyperbolic problems*.

Convexity of the entropy function, $\eta''(u) \geq 0$, and the flux F, F"(u)$$, is a requirement for the entropy inequality.

Next section discusses problems with non-convex fluxes, and the **Olenik-criterion**.

53.10.1 Examples

Burgers' equation

- Flux: $F(u) = \frac{u^2}{2}$
- Entropy: $\eta(u) = \frac{u^2}{2}$
- Entropy flux: $q(u) = \frac{u^3}{3}$

so that the compatibility condition $q'(u) = F'(u)\eta'(u)$ is satisfied.

The flux is convex, $F''(u) = 1 > 0$; the entropy function is convex as well, $\eta''(u) = 1 > 0$.

P-system

$$\partial_t \begin{bmatrix} \rho \\ m \end{bmatrix} + \partial_x \begin{bmatrix} m \\ \frac{m^2}{\rho} + \rho a^2 \end{bmatrix} = \partial_x \begin{bmatrix} 0 \\ \mu \partial_x \left(\frac{m}{\rho} \right) \end{bmatrix}$$

- Flux: $F(\mathbf{u}) = \begin{bmatrix} m \\ \frac{m^2}{\rho} + \rho a^2 \end{bmatrix}$
- Entropy: $\eta(\mathbf{u}) = \frac{m^2}{2\rho} + a^2 \rho \ln \frac{\rho}{\rho_0}$
- Diffusion flux: $\begin{bmatrix} 0 & 0 \\ -\mu \frac{m}{\rho^2} & \mu \frac{1}{\rho} \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m \end{bmatrix}$

Flux. First component

$$\nabla_{\mathbf{u}} F_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_1 = \mathbf{0}_2 \, .$$

Second component

$$\nabla_{\mathbf{u}} F_2 = \begin{bmatrix} a^2 - \frac{m^2}{\rho^2} \\ 2\frac{m}{\rho} \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_2 = \begin{bmatrix} 2\frac{m^2}{\rho^3} & -\frac{2m}{\rho^2} \\ -\frac{2m}{\rho^2} & \frac{2}{\rho} \end{bmatrix} \, .$$

Entropy. Gradient

$$\nabla_{\mathbf{u}} \eta = \begin{bmatrix} -\frac{m^2}{2\rho^2} + a^2 \left(1 + \ln \frac{\rho}{\rho_0} \right) \\ \frac{m}{\rho} \end{bmatrix}$$

and Hessian

$$\nabla_{\mathbf{uu}} \eta = \begin{bmatrix} \frac{m^2}{\rho^3} + \frac{a^2}{\rho} & -\frac{m}{\rho^2} \\ -\frac{m}{\rho^2} & \frac{1}{\rho} \end{bmatrix}$$

Compatibility of diffusion and entropy.

$$\nabla_{\mathbf{uu}} \eta \cdot \mathbf{D} = \mu \begin{bmatrix} \frac{m^2}{\rho^4} & -\frac{m}{\rho^3} \\ -\frac{m}{\rho^3} & \frac{1}{\rho^2} \end{bmatrix}$$

This matrix is semi-definite positive with one zero eigenvalue (associated with exact conservation and no dissipation in mass equation), and the other eigenvalue

Euler equations for compressible flows

Navier-Stokes equations

$$\begin{aligned} &\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0 \\ &\rho D_t \mathbf{u} = \nabla \cdot \mathbb{T} \\ &\rho D_t e^t = \nabla \cdot (\mathbb{T} \cdot \mathbf{u}) - \nabla \cdot \mathbf{q} \, , \end{aligned}$$

with $\mathbb{T} = -p\mathbb{I} + 2\mu\mathbb{D} + \lambda(\nabla \cdot \mathbf{u})\mathbb{I}$ the constitutive equation for Newtonian fluids, and $\mathbf{q} = -k\nabla T$ if Fourier law holds.

The conservative form of the equations in 1-dimensional domain reads

$$\partial_t \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix} + \partial_x \begin{bmatrix} m \\ \frac{m^2}{\rho} + p \\ \frac{m}{\rho}\left(E^t + p\right) \end{bmatrix} = \partial_x \begin{bmatrix} 0 \\ (\mu+\lambda)\partial_x\left(\frac{m}{\rho}\right) \\ (\mu+\lambda)\frac{m}{\rho}\partial_x\left(\frac{m}{\rho}\right) - q \end{bmatrix}$$

The convective form using (ρ, u, e) as primary variables reads

...

Entropy function is chosen as the thermodynamic entropy volume density $\eta = \rho s$. Entropy equation naturally follows from the thermodynamic relation $ds = \frac{1}{T}\left(de - \frac{p}{\rho^2}d\rho\right)$,

$$\rho\partial_t s + \rho\mathbf{u}\cdot\nabla s = \frac{2\mu|\mathbb{D}|^2 + \lambda(\nabla\cdot\mathbf{u})^2}{T} + \frac{k|\nabla T|^2}{T^2} - \nabla\cdot\left(\frac{\mathbf{q}}{T}\right)$$

or in conservative form (using mass equation)

$$\partial_t\left(\rho s\right) + \nabla\cdot\left(\rho s\mathbf{u}\right) = \frac{2\mu|\mathbb{D}|^2 + \lambda(\nabla\cdot\mathbf{u})^2}{T} + \frac{k|\nabla T|^2}{T^2} - \nabla\cdot\left(\frac{\mathbf{q}}{T}\right)$$

- Flux: $F(\mathbf{u}) = \begin{bmatrix} m \\ \frac{m^2}{\rho} + p \\ \frac{m}{\rho}(E^t + p) \end{bmatrix}$
- Entropy: $\eta(\mathbf{u}) = \rho s\left(\rho, \mathbf{m}, E^t\right)$
- Diffusion flux: $\begin{bmatrix} 0 & 0 & 0 \\ -(\mu+\lambda)\frac{m}{\rho^2} & (\mu+\lambda)\frac{1}{\rho} & 0 \\ \dots & \dots & \dots \end{bmatrix} \partial_x \begin{bmatrix} \rho \\ m \\ E^t \end{bmatrix}$

Flux. First component

$$\nabla_{\mathbf{u}} F_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_1 = \mathbf{0}_3 \,.$$

Second component

$$\nabla_{\mathbf{u}} F_2 = \begin{bmatrix} -\frac{m^2}{\rho^2} + \Pi_{/\rho} \\ 2\frac{m}{\rho} + \Pi_{/m} \\ \Pi_{E^t} \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_2 = \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix} \,.$$

Third component

$$\nabla_{\mathbf{u}} F_2 = \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix}$$

$$\nabla_{\mathbf{uu}} F_2 = \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix} \,.$$

Entropy. Gradient

$$\nabla_{\mathbf{u}}\eta = \begin{bmatrix} s + \rho s_{/\rho} \\ \rho s_{/m} \\ \rho s_{/E^t} \end{bmatrix}$$

and Hessian

$$\nabla_{\mathbf{uu}}\eta = \begin{bmatrix} 2s_{/\rho} + \rho s_{/\rho\rho} & s_{/m} + \rho s_{\rho m} & s_{/E^t} + \rho s_{\rho E^t} \\ ... & \rho s_{mm} & \rho s_{mE^t} \\ ... & ... & \rho s_{E^t E^t} \end{bmatrix}$$

Compatibility of diffusion and entropy.

$$\nabla_{\mathbf{uu}}\eta \cdot \mathbf{D} = \mu \left[... \right]$$

This matrix is semi-definite positive with one zero eigenvalue (associated with exact conservation and no dissipation in mass equation), and the other eigenvalue

53.11 Non-convex hyperbolic problems

In this section, non-convex hyperbolic problems are briefly discussed using a non-linear scalar equation as a toy problem to show their characteristics. The main topic is the solution of a Riemann problem. *Depending on the initial conditions*, the solution of the Rieamnn problem for a non-convex hyperbolic scalar equation may involve not only simple waves (like shock or rarefaction), but also **composite waves**, combining both a shock and a rarefaction.

First, a *numerical simulation* of a Riemann problem the viscous counterpart of the hyperbolic equation is performed and the solution is discussed. Then, the solution is discussed introducing the ***Olenik criterion*** to find the physical solution.

53.11.1 Toy problem

The non-linear scalar equation

$$\partial_t u + \partial_x F(u) = 0 \, ,$$

with flux $F(u) = u^3$ is a hyperbolic equation with non-convex flux, as

$$F''(u) = 6u \begin{cases} > 0 & \text{for } u > 0 \\ \leq 0 & \text{for } u \leq 0 \end{cases}$$

Numerical simulation

The numerical simulation of two Riemann problem of the viscous counterpart of the hyperbolic equation,

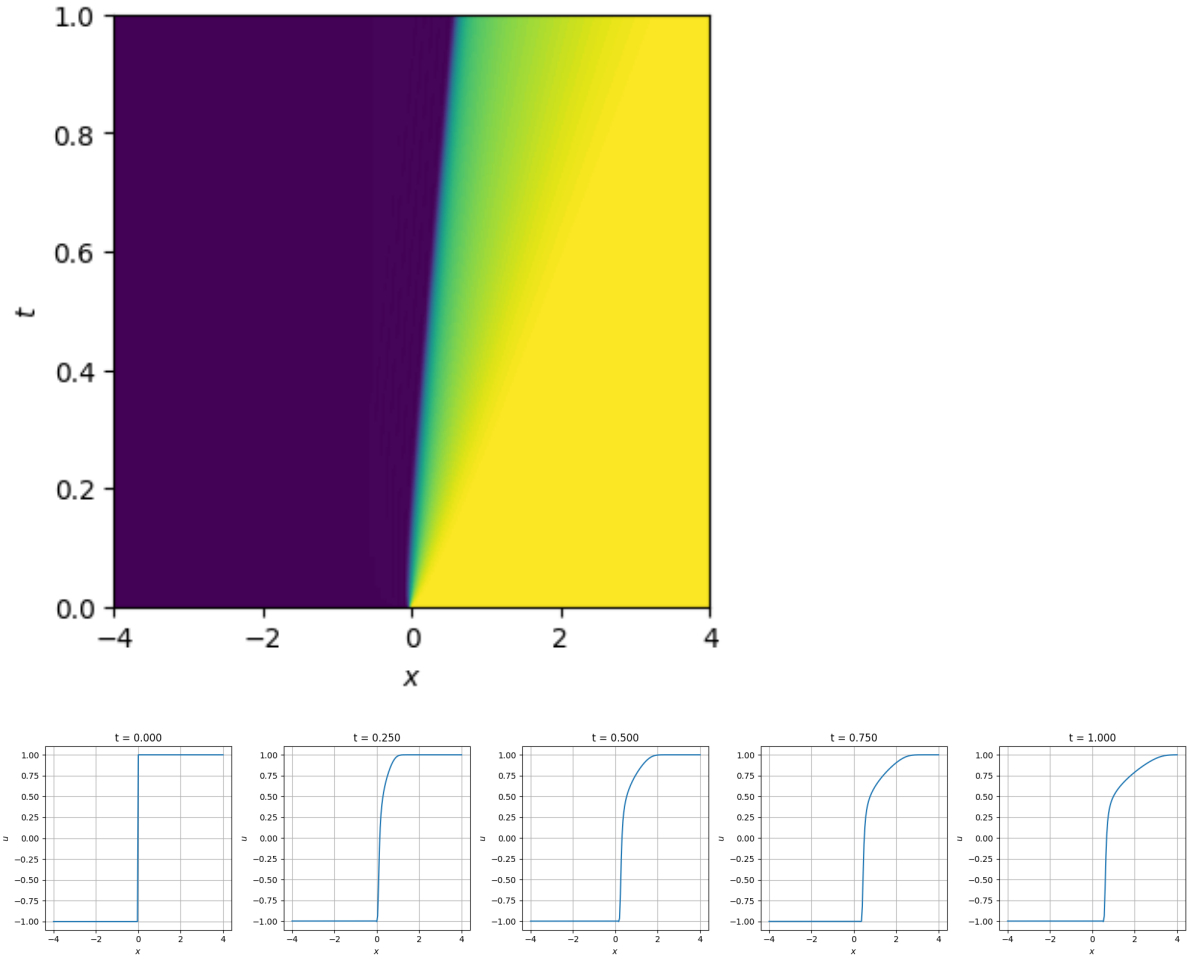
$$\partial_t u + \partial_x F(u) - \partial_x(\nu \partial_x u) = 0 \, ,$$

is performed here with a time-explicit finite-difference method for two different initial conditions: 1) $(u_L, u_R) = (-1, 1)$, 2) $(u_L, u_R) = (1, -1)$.

Here, the value of the diffusion coefficient is choosen as $\nu = .05$ as a compromise between thickness of the "shock" and the stability requirement of the numerical method.

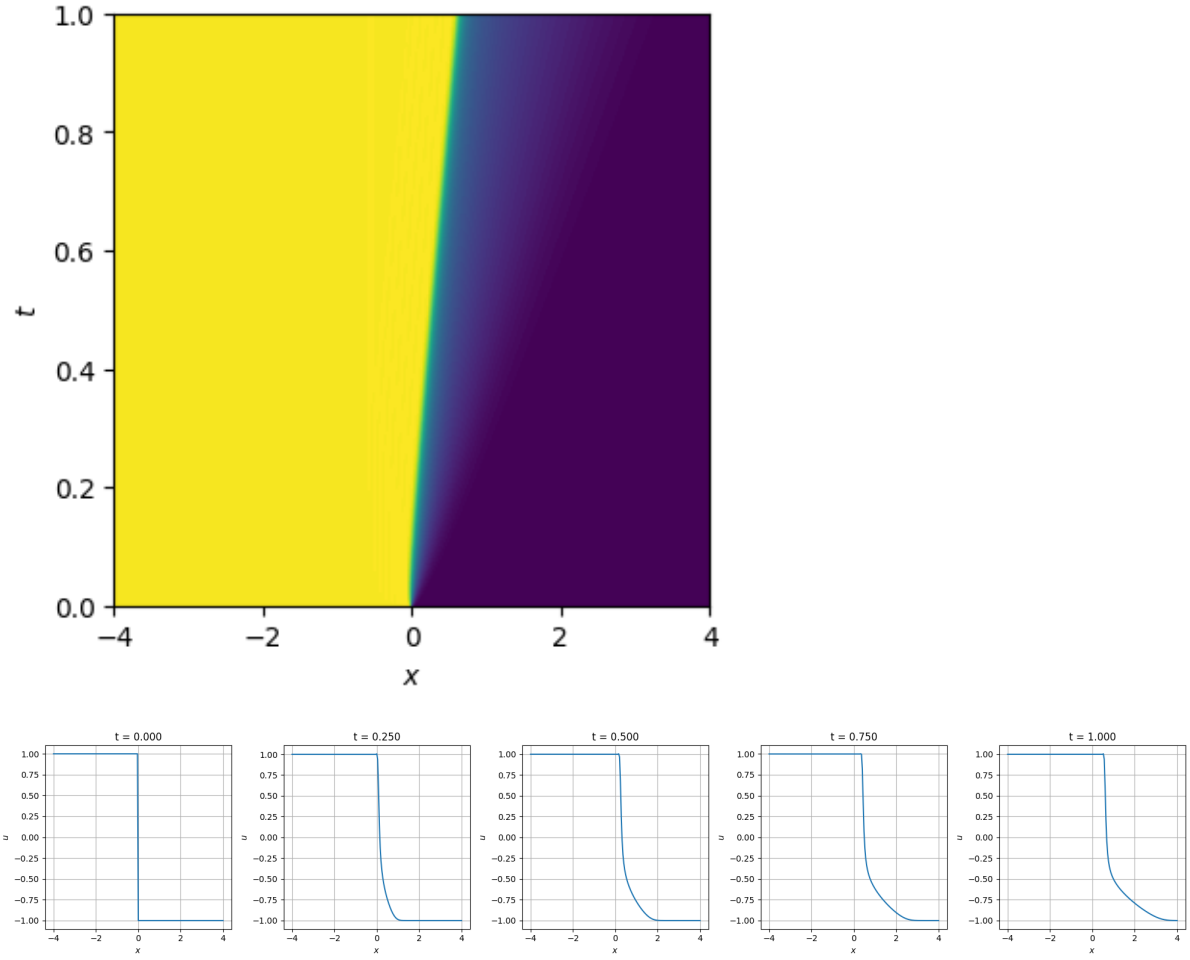
Simulation with $(u_L, u_R) = (-1, 1)$

The solution is the combination of a left shock between state $u_L = -1$ and an intermediate state $u^* = \frac{1}{2}$, and a right fan between u^* and $u_R = 1$.



Simulation with $(u_L, u_R) = (1, -1)$

The solution is the combination of a left shock between state $u_L = 1$ and an intermediate state $u^* = \frac{1}{2}$, and a right fan between u^* and $u_R = -1$.



Olenik's criterion

In a Riemann problem with two values u_L, u_R, with $F''(u_L)F''(u_R) < 0$ - so that non-convexity manifests itself, and assuming that $F''(u)$ changes sign only once (see the comment below for more complex flux functions -, a composite wave occurs. This composite wave is composed of a rarefaction wave and a shock wave.

The intermediate state u^* is found with the condition that the speed of the shock equals the characteristic speed of the fan at the shock, i.e.

$$s(u_L, u^*) = F'(u^*) \, ,$$

or

$$\frac{F(u^*) - F(u_L)}{u^* - u_L} = F'(u^*) \, .$$

i.e. the slope $F'(u^*)$ of the tangent line at $F(u)$ in u^* equals the slope of the line connecting the state $(\cdot)^*$ with $(\cdot)_L$. This condition prevents characteristic lines from originating from the shock.

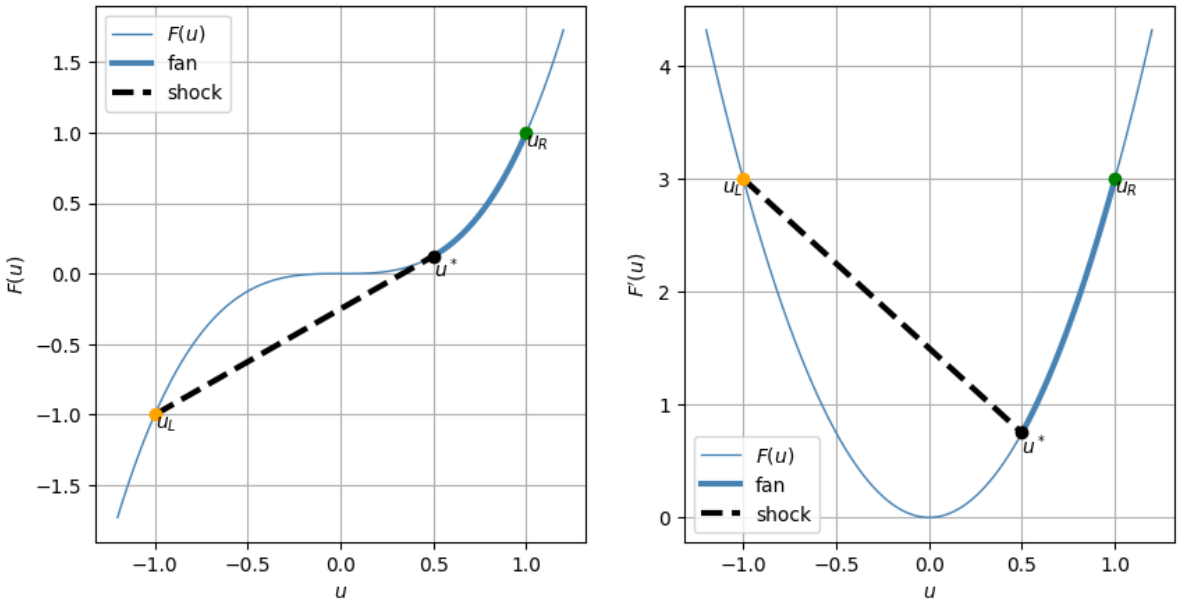
An alternative formulation of Olenik's criterion reads: for a shock to be stable, the shock speed s must be the slowest possible. For $u_L > u_R$,

$$\frac{f(u) - f(u_L)}{u - u_L} \geq s(u) \geq \frac{f(u) - f(u_R)}{u - u_R} \qquad s \in [u_L, u_R] \, .$$

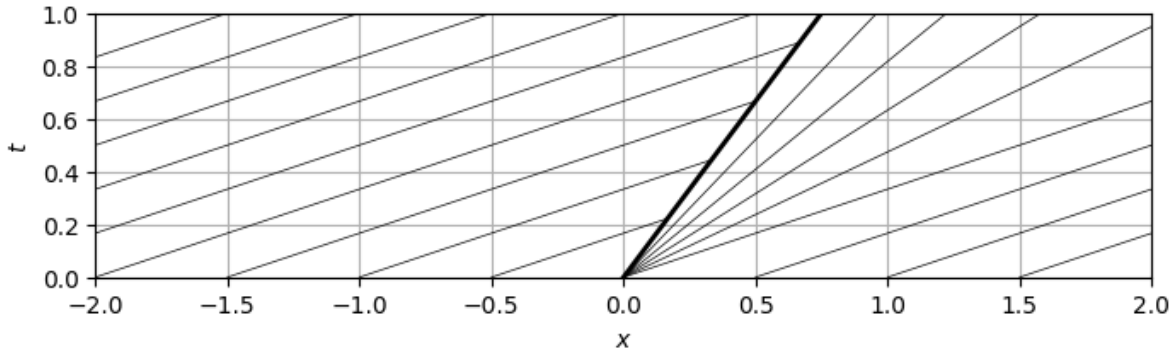
todo

- With more "complex" flux functions $F(u)$ (that have several ranges of u-values where $F''(u)$ changes sign), how many waves in the composite wave? Show how Olenik's criterion applies.

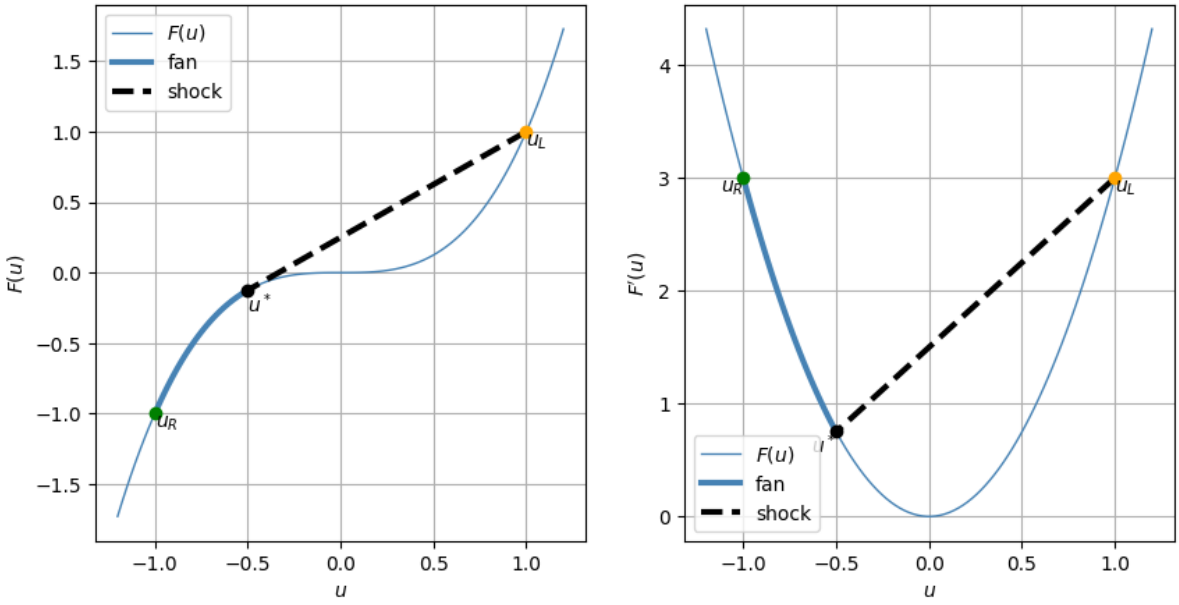
$(u_L, u_R) = (-1, 1)$



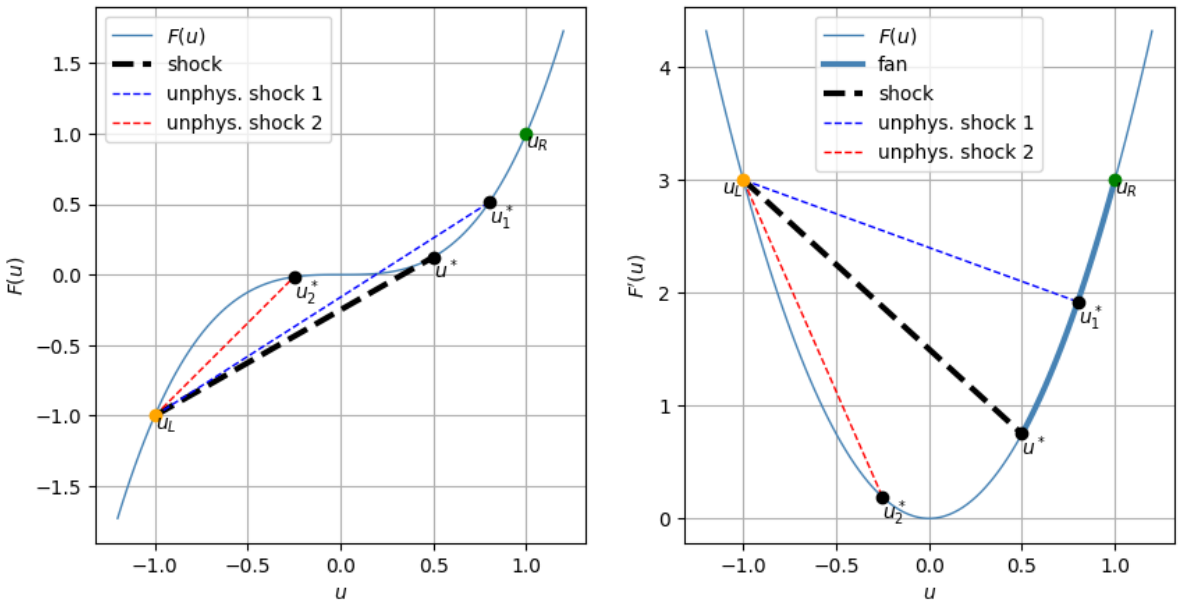
Characteristic lines in the $x - t$ plane

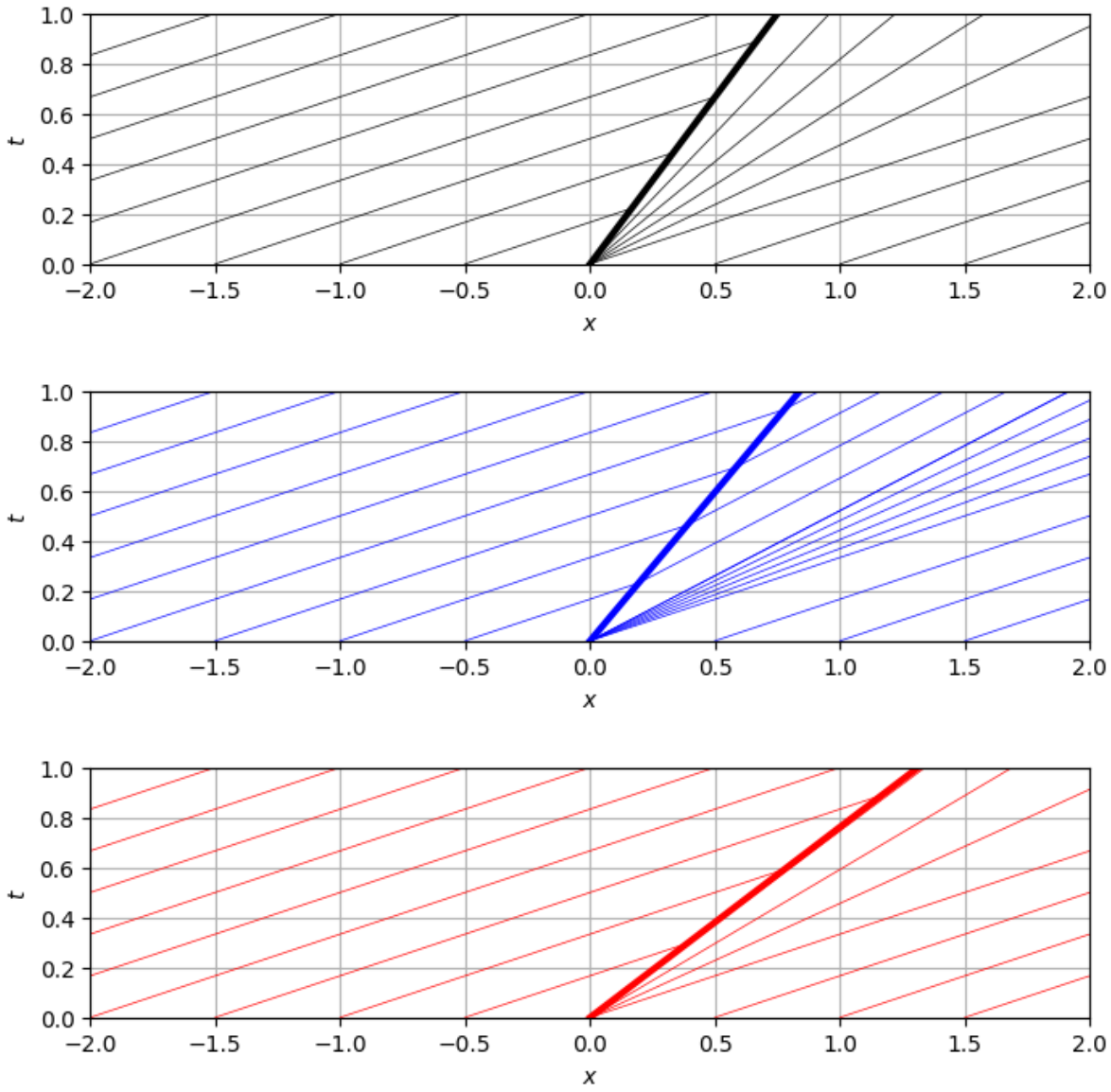


$(u_L, u_R) = (1, -1)$



Non-physical solutions for $(u_L, u_R) = (-1, 1)$





todo

- The blue unphysical solution looks unphysical for the characteristic lines orginating from the shock. But why is the red solution unphysical?
- Discussion about local stability
- Discussion with local production of entropy $\Sigma(u) = s(u)[\eta(u)] - [q(u)]$, and extreme points at the physical solution u^*? $\Sigma'(u^*) = 0$?

CHAPTER

FIFTYFOUR

NAVIER-CAUCHY EQUATIONS

Navier-Cauchy equations are the differential balance equation of the momentum of an elastic isotropic medium in the regime of small strain and displacement,

$$\rho_0 \partial_{tt} \vec{s} = \rho_0 \vec{g} + \nabla \cdot \boldsymbol{\sigma} \ .$$

Stress tensor for an isotropic medium reads

$$\begin{aligned} \boldsymbol{\sigma} &= 2\mu\boldsymbol{\varepsilon} + \lambda \operatorname{tr}(\boldsymbol{\varepsilon})\, \mathbb{I} = \\ &= \left(2\mu\boldsymbol{\varepsilon} - \frac{2}{3}\mu \operatorname{tr}(\boldsymbol{\varepsilon})\mathbb{I}\right) + \left(\lambda + \frac{2}{3}\mu\right) \operatorname{tr}(\boldsymbol{\varepsilon})\, \mathbb{I} \ , \end{aligned}$$

with the small strain tensor

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla \vec{s} + \nabla^T \vec{s}\right) \ .$$

Essential, natural and Robin boundary conditions read

$\vec{s} = \bar{\vec{s}}$	$\vec{r} \in S_D$	esserntial - Dirichlet b.c.
$\hat{n} \cdot \boldsymbol{\sigma} = \bar{\vec{t}}_n$	$\vec{r} \in S_N$	natural - Neumann b.c.
$a\vec{s} + \hat{n} \cdot \boldsymbol{\sigma} = \vec{b}$	$\vec{r} \in S_R$	Robin b.c.

54.1 Weak formulation

For $\forall \vec{w} \in ...$

$$\begin{aligned} 0 &= -\int_V \rho \vec{w} \cdot \partial_{tt} \vec{s} + \int_V \rho_0 \vec{w} \cdot \vec{g} + \int_V \vec{w} \cdot \nabla \cdot \boldsymbol{\sigma} = \\ &= -\int_V \rho \vec{w} \cdot \partial_{tt} \vec{s} + \int_V \rho_0 \vec{w} \cdot \vec{g} + \int_{\partial V} \hat{n} \cdot \boldsymbol{\sigma} \cdot \vec{w} - \int_V \nabla \vec{w} : \boldsymbol{\sigma} \end{aligned}$$

The volume integral containing the stress tensor can be written either as

$$\begin{aligned} \int_V \nabla \vec{w} : \boldsymbol{\sigma} &= \int_V w_{i/j} \left[\mu\left(s_{i/j} + s_{j/i}\right) + \lambda s_{k/k} \delta_{ij}\right] = \\ &= \int_V \mu w_{i/j}\left(s_{i/j} + s_{j/i}\right) + \int_V \lambda w_{j/j} s_{k/k} \end{aligned}$$

or

$$\begin{aligned} \int_V \frac{1}{2}\left(\nabla \vec{w} + \nabla^T \vec{w}\right) : \boldsymbol{\sigma} &= \int_V \frac{1}{2}\left(w_{i/j} + w_{j/i}\right)\left[\mu\left(s_{i/j} + s_{j/i}\right) + \lambda s_{k/k}\delta_{ij}\right] = \\ &= \int_V \frac{\mu}{2}\left(w_{i/j} + w_{j/i}\right)\left(s_{i/j} + s_{j/i}\right) + \int_V \lambda w_{j/j} s_{k/k} \end{aligned}$$

The weak formulation of the Navier-Cauchy equations reads

$$\int_V \rho_0 \vec{w} \cdot \partial_{tt} \vec{s} + \int_V 2\mu \frac{\nabla \vec{w} + \nabla^T \vec{w}}{2} : \frac{\nabla \vec{s} + \nabla^T \vec{s}}{2} + \int_V \lambda \nabla \cdot \vec{w} \nabla \cdot \vec{s} + \int_{S_R} \vec{w} \cdot a \vec{s} = \int_V \rho_0 \vec{w} \cdot \vec{g} + \int_{S_N} \vec{w} \cdot \bar{\vec{t}}_n + \int_{S_R} \vec{w} \cdot \vec{b} \,,$$

for $\forall \vec{w} \in ...$, and with $\vec{s} = \bar{\vec{s}}$ for $\vec{r} \in S_D$.

CHAPTER

FIFTYFIVE

NAVIER-STOKES EQUATIONS

Incompressible Navier-Stokes equations read

$$\begin{cases} \rho\partial_t\vec{u} + \rho\left(\vec{u}\cdot\nabla\right)\vec{u} - \mu\nabla^2\vec{u} + \nabla P = \rho\vec{g} \\ \nabla\cdot\vec{u} = 0\ . \end{cases}$$

Mass balance equation is replaced by the incompressiblity kinematic constraint, $\nabla\cdot\vec{u}=0$: this constraint is not dynamic, as time derivative of density does not appear in the equation. With the incompressibility constraint, mass equation tells us that material particles keep their density constant,

$$0 = \underbrace{\partial_t\rho + \vec{u}\cdot\nabla\rho}_{=\frac{D\rho}{Dt}} + \rho\underbrace{\nabla\cdot\vec{u}}_{=0} = \frac{D\rho}{Dt}\ ,$$

whose solution can be written using material coordinates $\vec{r}_0$ as $\rho(\vec{r}(\vec{r}_0,t),t) = \rho_0(\vec{r}_0,t)$.

55.1 Incompressibility constraint

Incompressibility constraint makes thermodynamic fade, while pressure field is replaced by/contains the contribution of a Lagrangian multiplier related to the incompressiblity constraint.

55.1.1 Wave-vector transformed space

Transforming the fields from physical space to the wave-vector space $\tilde{u}(\vec{\kappa},t) = \mathcal{F}\left\{\vec{u}(\vec{r},t)\right\}$, Navier-Stokes equations for incompressible fluids with uniform and constant density $\rho(\vec{r},t)=\rho$ becomes

$$\begin{cases} \rho\partial_t\tilde{u} + \mathcal{F}\left\{\left(\vec{u}\cdot\nabla\right)\vec{u}\right\} + \mu|\vec{\kappa}|^2\tilde{u} + i\vec{k}\widetilde{P} = \rho\tilde{g} \\ i\vec{\kappa}\cdot\tilde{u} = 0\ . \end{cases}$$

Taking the divergence of the momentum balance equation, i.e. taking the scalar product with $i\vec{\kappa}$ in the transformed space, and using the incompressibility constraint to set $i\vec{\kappa}\cdot\tilde{u}=0$,

$$i\vec{\kappa}\cdot\mathcal{F}\left\{(\vec{u}\cdot\nabla)\vec{u}\right\} - |\vec{\kappa}|^2\widetilde{P} = i\vec{\kappa}\cdot\rho\tilde{g}\ ,$$

so that the transformed pressure field becomes

$$\widetilde{P} = \frac{i\vec{\kappa}}{|\vec{\kappa}|^2}\cdot\mathcal{F}\left\{\left(\vec{u}\cdot\nabla\right)\vec{u} - \rho\vec{g}\right\}\ ,$$

Replacing this expression in the transformed Navier-Stokes equations, the meaning of the pressure field as a Lagrange multiplier associated with incompressibility constraint becomes clear,

$$\rho\partial_t\tilde{u} + \mu|\vec{\kappa}|^2\tilde{u} = \left[1 - \frac{\vec{\kappa}\vec{\kappa}}{|\vec{\kappa}|^2}\right]\cdot\mathcal{F}\left\{-(\vec{u}\cdot\nabla)\vec{u} + \rho\tilde{g}\right\}$$

as the orthogonal projector $\left[1-\frac{\vec{\kappa}\vec{\kappa}}{|\vec{\kappa}|^2}\right]$ onto the space of divergence-free functions acts on the non-linear and forcing terms.

55.2 Weak formulation of the problem

$$
\begin{aligned}
0 &= \int_V \vec{w}\cdot[\rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\vec{u} - 2\mu\nabla\cdot\mathbb{D}(\vec{u}) + \nabla P - \rho\vec{g}] - \int_V v\nabla\cdot\vec{u} = \\
&= \int_V \vec{w}\cdot[\rho\partial_t\vec{u} + \rho(\vec{u}\cdot\nabla)\vec{u}] + \int_V 2\mu\nabla\vec{w} : \mathbb{D} - \int_V \nabla\cdot\vec{w}P - \int_V \vec{w}\cdot\rho\vec{g} - \int_V v\nabla\cdot\vec{u} - \int_{\partial V}\hat{n}\cdot(\mathbb{S} - P\mathbb{I})\cdot\vec{w}\,,
\end{aligned}
$$

55.2.1 Weak formulation and incompressibility constraint

$$
\begin{aligned}
\vec{r}(\vec{r}_0, t) &= \vec{r}(q(t), t) \\
\vec{u} &= \frac{D\vec{r}}{Dt} = \dot{q}\frac{\partial\vec{r}}{\partial q} + \frac{\partial\vec{r}}{\partial t}
\end{aligned}
$$

In the weak formulation, using $\vec{w} = \frac{\partial\vec{r}}{\partial q} = \frac{\partial\vec{u}}{\partial\dot{q}}$

$$
0 = \int_V \vec{w}\cdot\rho\frac{D\vec{u}}{Dt} + \int_V 2\mu\nabla\vec{w} : \mathbb{D} - \int_V \nabla\cdot\vec{w}P - \int_V \rho\vec{w}\cdot\vec{g} - \int_V v\nabla\cdot\vec{u} - \int_{\partial V}\vec{t}_{\hat{n}}\cdot\vec{w}\,,
$$

$$
\begin{aligned}
\int_V \vec{w}\cdot\rho\frac{D\vec{u}}{Dt}\,dV &= \int_{V_0}\rho_0\frac{\partial\vec{u}}{\partial\dot{q}}\cdot\frac{D\vec{u}}{Dt} = \\
&= \int_{V_0}\rho_0\frac{D}{Dt}\left(\frac{\partial\vec{u}}{\partial\dot{q}}\cdot\vec{u}\right)dV_0 - \int_{V_0}\rho_0\frac{D}{Dt}\left(\frac{\partial\vec{r}}{\partial\dot{q}}\right)\cdot\vec{u}\,dV_0 = \\
&= \int_{V_0}\rho_0\frac{D}{Dt}\left(\frac{\partial}{\partial\dot{q}}\frac{|\vec{u}|^2}{2}\right)dV_0 - \int_{V_0}\rho_0\frac{\partial}{\partial q}\frac{|\vec{u}|^2}{2}\,dV_0 = \\
&\dots
\end{aligned}
$$

55.3 Non-linear term

Different ways to treat the non-linear term:

- Semi-linear approximation of the non-linear term

  $$(\vec{u}(\vec{r}, t^n)\cdot\nabla)\vec{u}(\vec{r}, t^n) \sim (\vec{u}^*(\vec{r}, t^n)\cdot\nabla)\vec{u}(\vec{r}, t^n)\,,$$

  with $\vec{u}^*(\vec{r}, t^n)$ an approximation of $\vec{u}(\vec{r}, t^n)$ involving values of the velocity field at previous time-steps, as an example

  $$\vec{u}^*(\vec{r}, t^n) = \begin{cases} \vec{u}(\vec{r}, t^{n-1}) & 1^{st}\text{-order} \\ 2\vec{u}(\vec{r}, t^{n-1}) - \vec{u}(\vec{r}, t^{n-2}) & 2^{nd}\text{-order} \end{cases}$$

CHAPTER

FIFTYSIX

ARBITRARY LAGRANGIAN-EULERIAN DESCRIPTION

Reynold's transport theorem allows for the formulation of intergal equations, and grid-based methods like FVM, on moving grids and changing domains. Rules for derivatives of composite functions provide the relations between time derivatives in a Lagrangian, Eulerian, or arbitrary description,

$$\left.\frac{\partial f}{\partial t}\right|_{\vec{r}_0} = \left.\frac{\partial f}{\partial t}\right|_{\vec{r}} + \vec{u}\cdot\nabla f$$
$$\left.\frac{\partial f}{\partial t}\right|_{\vec{r}_b} = \left.\frac{\partial f}{\partial t}\right|_{\vec{r}} + \vec{u}_b\cdot\nabla f$$

Equations governing the motion of the grid are usually required as well. E.g.:

- known and prescribed motion of the grid;
- boundary conditions only without changing grids (for small displacements)
- pseudo-elastic deformation (usually good for small strain and displacement;
- for large displacements of/or models with complex geometry, sliding and/or overlapping grids could an option for grid-based methods.

56.1 Integral problem

Application of Reynolds theorem to the balance equation of the quantity **u** for a material volume V_t

$$\frac{d}{dt}\int_{V_t}\rho\mathbf{u} = \int_{V_t}\rho\mathbf{f} + \oint_{\partial V_t}\hat{n}\cdot\mathbf{T}\,.$$

provides the expression of the balance equation for a geometrical volume v_t in arbitrary motion,

$$\frac{d}{dt}\int_{v_t}\rho\mathbf{u} + \oint_{\partial v_t}\rho\mathbf{u}\,(\vec{u}-\vec{u}_b)\cdot\hat{n} = \int_{v_t}\rho\mathbf{f} + \oint_{\partial v_t}\hat{n}\cdot\mathbf{T}\,.$$

Here, the integral forulation of the problem will be applied to each element of the grid in arbitrary motion, for domains with variable geometry.

56.2 Differential problem

Rules for derivatives of composite functions allows to write the differential w.r.t. the variables associated with the points of a moving grid. A balance equation in convective form can be written as

$$\rho \frac{D\mathbf{u}}{Dt} = \rho\mathbf{f} + \nabla \cdot \mathbf{T}$$

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + \vec{u} \cdot \nabla \mathbf{u} \right] =$$

$$\rho \left[\left. \frac{\partial \mathbf{u}}{\partial t} \right|_{\vec{r}_b} + (\vec{u} - \vec{u}_b) \cdot \nabla \mathbf{u} \right] =$$

Part XIII

Numerical Methods for PDEs

CHAPTER

FIFTYSEVEN

INTRODUCTION TO NUMERICAL METHODS FOR PDES

Different numerical methods for PDEs rely on the discretization of different formulations of the continunuos problems. As an example,

- **FDM, Finite difference methods** rely on the approximation of derivatives of the **strong formulation** of the problem
- ***FEM, Finite element methods*** rely on a finite dimensional approximation of the **weak formulation** of the problem; usually the finite dimensional approximation can be interpreted as a projection of a infinite dimensional continuous problem onto a finite dimensional space, the space of the choosen finite elements
- ***FVM, Finite volume methods*** rely on an approximation of the **integral formulation** of the problem
- **BEM, Boundary elemement methods** rely on an approximation of a **boundary integral formulation** of the problem, when it's feasible and convenient
- *…spectral methods*, *spectral element methods*,…

Characteristics.

- grid: domain-grid-based, buondary-grid-based, grid-free methods
- range of interaction: short in physical space for FDV, FEM, FVM; long-range for boundary element methods, even though clustering techniques are available, like FMM; (usually) over the whole domain in space, short-range interaction in wave-number space for spectral methods;

Pros and cons. *More suited methods for each problems…; domain, order,…*

Let's take Poisson equations for a scalar function $u(\vec{r})$ to show all the possible approaches above. Here we start from the most general version of the equation, namely the integral form. As it's shown in Continuum Mechanics:Governing Equations, the most general form of balance equations is the integral form, while the differential form can be seamlessly derived only when the quantities involved are regular enough, to apply Stokes' theorem and for the derivatives appearing to exist.

The problem of interest can be interpreted as the problem of finding the temperature field $u(\vec{r})$ during steady heat conduction in the domain Ω, with distributed volume heat source $f(\vec{r})$. Fourier's law assume that condcution heat flux is proportional to the gradient of the temperature, $\vec{q} = -k\nabla u$. Temperature is prescribed, $u = g$ on the region of the boundary S_D, while the heat flux is prescribed $\hat{n} \cdot \vec{q} = h$ on the region of the boundary S_N.

The **integral formulation** of the problem for all the boundary reads

$$0 = -\oint_{\partial\Omega} \hat{n} \cdot \vec{q} + \int_{\Omega} f \, . \tag{57.1}$$

Example 57.1 (Finite volume methods)

Finite volume methods rely on a tassellation of the domain $\Omega = \cup_k \Omega_k$, to write and solve the integral balance equation (57.1) for all the elemantary domain Ω_k.

$$0 = -\oint_{\partial\Omega_k} \hat{n} \cdot \vec{q} + \int_{\Omega_k} f \,. \tag{57.2}$$

evaluating volume integrals with the internal variables of the cell Ω_k, and boundary flux with the variables of the cell Ω_k and the neighbouring cells $\Omega_i \in B_k$. Introducing the definition of numerical flux $F_{ik}(u_i, u_k)$ at the interface between a generic discrete version of the balance equation (57.2) for the kth element becomes

$$0 = -\sum_{\Omega_i \in B_k} F_{ik}(u_i, u_k) + S_k(u_k) \,.$$

Different numerical methods differs in the evaluation of fluxes and sources. FVM is **conservative** as it evaluate flux at interfaces and then distribute it to neighoring cells, see *Property 59.1*. A rough elementwise-uniform variable with jump at interfaces and diffusive flux,

$$F_{ik} = -A_{ik}\, k\, \frac{u_i - u_k}{d_{ik}} \,,$$

with d_{ik} equal to the distance of the centers of elements i and k, leads to the balance equation for the internal volume Ω_k,

$$0 = \sum_i A_{ik}\, k\, \frac{u_i - u_k}{d_{ik}} + V_k f_k \,.$$

Volumes at the boundary are influenced by fluxes at the boundaries and boundary conditions in general.

For regular cubic mesh, $A_{ik} = \Delta x$, $d_{ik} = \Delta x$, $V_k = \Delta x$, and thus the equations for internal elements become

$$0 = k \sum_{i \in \Omega_k} \frac{u_i - u_k}{\Delta x^2} + f_k \,,$$

where the summation is exactly equal to the center-difference stencil for the Laplacian, used in finite difference method *Example 57.2*.

If fields are regular enough to apply Stokes' theorem, it's possible to derive differential problem from the integral formulation,

$$0 = -\oint_{\partial V} \hat{n} \cdot \vec{q} + \int_V f = -\int_V \nabla \cdot \vec{q} + \int_V f = \int_V (-\nabla \cdot \vec{q} + f)$$

Exploiting the arbitrariety of the volume V, the **strong form** of the differential problem is the Poisson equation with suitable boundary conditions,

$$\begin{cases} -\nabla \cdot (k \nabla u) = f(\vec{r}) \,, & \vec{r} \in V \\ u = g(\vec{r}) \,, & \vec{r} \in S_D \\ \hat{n} \cdot \nabla u = h(\vec{r}) \,, & \vec{r} \in S_N \end{cases}$$

only holds in regions of the domain where the functions involved are continuous and differentiable, i.e. everything written in the problem is not meaningful, i.e. at least it exists.

Example 57.2 (Finite difference methods)

Finite different method approximates the strong form of the problem, building a stencil to evaluate an approximation of

the Laplacian. On a cubic regular grid ($\Delta x = \Delta y = \Delta z$), the Laplacian van be evaluated with a 7-point stencil

$$\begin{aligned}\nabla^2 u &= \partial_{xx} u + \partial_{yy} u + \partial_{zz} u = \\ &= \frac{u_{i+1,j,k} - 2u_{i,j,k} + u_{i-1,j,k}}{\Delta x^2} + \frac{u_{i,j+1,k} - 2u_{i,j,k} + u_{i,j-1,k}}{\Delta y^2} + \frac{u_{i,j,k+1} - 2u_{i,j,k} + u_{i,j,k-1}}{\Delta z^2} = \\ &= \frac{1}{\Delta x^2}\left[-6u_{i,j,k} + u_{i+1,j,k} + u_{i-1,j,k} + u_{i,j+1,k} + u_{i,j-1,k} + u_{i,j,k+1} + u_{i,j,k-1}\right] = \\ &= \frac{1}{\Delta x^2}\sum_{i \in B_k}(u_i - u_k)\end{aligned}$$

and it's equal to the same expression already provided for the rough finite volume method with regular grid, in *Example 57.1*

Starting from strong formulation of the differential problem, the **weak formulation** of the problem is derived:

- multiplying the strong problem by an arbitrary test function $w(\vec{r})$ compatible with the essential constraints
- integrating over the whole domain

$$\begin{aligned}0 &= \int_V w(\vec{r}) \cdot [-\nabla \cdot (k\nabla u) - f(\vec{r})] = \\ &= \int_V \{k\nabla w(\vec{r}) \cdot \nabla u(\vec{r}) - w(\vec{r}) f(\vec{r})\} - \oint_{\partial V} w(\vec{r})\, \hat{n} \cdot \nabla u \qquad \forall w(\vec{r})\ .\end{aligned}$$

Using a test function $w(\vec{r})$ equal to zero on the boundary where essential boundary conditions are prescribed $w|_{S_D} = 0$, and introducing the natural boundary conditions of the Neumann boundary S_N, $\hat{n} \cdot \nabla u|_{S_N} = h$,

$$\int_V k\nabla w \cdot \nabla u = \int_V wf + \int_{S_N} wh\ , \quad \forall w$$

compatible with essential boundary conditions.

Example 57.3 (Finite element methods)

Finite element methods build an N-dimensional systems:

- approximating the solution as a linear combination of N base functions, $u(\vec{r}) = \sum_j \phi_j(\vec{r}) u_j$
- testing the equation over N independent test functions $\psi_i(\vec{r})$

i.e. the ith equation becomes

$$\int_V k\nabla\psi_i(\vec{r}) \cdot \nabla\phi_j(\vec{r})\, u_j = \int_V \psi_i(\vec{r}) f + \int_{S_N} \psi_i(\vec{r}) h\ ,$$

or briefly

$$K_{ij} u_j = f_i \qquad , \qquad \mathbf{Ku} = \mathbf{f}\ .$$

A common choice uses the same functions both as test and base functions, $\psi_i(\vec{r}) = \phi_i(\vec{r})$.

The differential problem in strong form can be recast as a boundary element problem exploiting the properties of Green's functions $G(\vec{r}; \vec{r}_0)$. The expression of *Green's function* $G(\vec{r}, \vec{r}_0)$ is known for some problems of interest like Poisson

equation, Helmholtz equation or wave equation. Exploiting the properties of Green's function and integration by parts, the integral buondary problem is derived as follow

$$\begin{aligned} E(\vec{r}_0)u(\vec{r}_0) &= \int_{\vec{r}\in V} u(\vec{r})\delta(\vec{r}-\vec{r}_0)\,dV = \\ &= -\int_{\vec{r}\in V} u(\vec{r})\nabla^2 G(\vec{r};\vec{r}_0)\,dV = \\ &= -\oint_{\vec{r}\in\partial V} u(\vec{r})\hat{n}(\vec{r})\cdot\nabla G(\vec{r};\vec{r}_0)\,dS + \oint_{\vec{r}\in\partial V} G(\vec{r};\vec{r}_0)\hat{n}(\vec{r})\cdot\nabla u(\vec{r})\,dS - \int_{\vec{r}\in V} G(\vec{r};\vec{r}_0)\nabla^2 u(\vec{r})\,dV = \\ &= -\oint_{\vec{r}\in\partial V} u(\vec{r})\hat{n}(\vec{r})\cdot\nabla G(\vec{r};\vec{r}_0)\,dS + \oint_{\vec{r}\in\partial V} G(\vec{r};\vec{r}_0)\hat{n}(\vec{r})\cdot\nabla u(\vec{r})\,dS + \int_{\vec{r}\in V} G(\vec{r};\vec{r}_0)f(\vec{r})\,dV\ . \end{aligned}$$

It must be paid attention that the integrals may be singular and may produce discontintuities in the fields across the surface ∂V (*so what's the value of a field in presence of a discontinuity?…virtual singularities, regularization close to the singularities…one way to correctly interpret them is via Cauchy principal value… evaluating some integrals in a check-point just inside or outside the domain may make the value of the function* $E(\vec{r}_0)$ *change, but the integral involving* $\hat{n}\cdot\nabla G$ *- related to the solid angle seen by the check-point of the surface - changes accordingly and keep the sum of these two terms constant=*.

The value of the unknown function or the flux $\hat{n}\cdot\nabla u$ are known on the Dirichlet and Neumann regions of the boundary respectively, and so the integro-differential problem becomes

$$E(\vec{r}_0)u(\vec{r}_0) + \int_{S_N} u\hat{n}\cdot\nabla G\,dS - \int_{S_D} G\,\hat{n}\cdot\nabla u\,dS = \int_{S_D} g\hat{n}\cdot\nabla G\,dS - \int_{S_N} Gh\,dS + \int_{\vec{r}\in V} Gf\,dV$$

The unknown function is approximated on the boundary of the domain ∂V as an N-dimensional approximation, as an example

$$u(\vec{r}) = \sum_j \phi_j(\vec{r})u_j \qquad , \qquad \vec{r}\in\partial V\ ,$$

and the integro-differential equation is evaluated in N different points $\vec{r}_{0,i}$ in order to get a N, N linear equation in the amplitudes u_j of the base functions,

$$[\mathbf{E} + \mathbf{D}_N + \mathbf{S}_D]\,\mathbf{u} = \mathbf{D}_D\mathbf{g} + \mathbf{S}_N\mathbf{h} + \mathbf{S}_V\mathbf{f}\ .$$

CHAPTER

FIFTYEIGHT

FINITE ELEMENT METHOD

Examples

- Structural mechanics of linear beam structures
- *Poisson equation, 1D*

58.1 1-dimensional Poisson equation

58.1.1 Strong form of the problem

$$-\left(\nu u'(x)\right)' = f(x) \quad , \qquad x \in [a,b]$$

Supplied with proper boundary conditions. As an example,

- Dirichlet boundary conditions: $u(\overline{x}) = \overline{u}$
- Neumann boundary conditions: $\nu u'(\overline{x}) = \overline{q}$
- Robin boundary conditions: $au(\overline{x}) + bu'(\overline{x}) = c$

58.1.2 Weak form of the problem

For $\forall w(x) \in \mathcal{H}^1([a,b])$[1] ,

$$\begin{aligned} 0 &= \int_{x=a}^{b} w(x) \left\{ -\left(\nu u'(x)\right)' - f(x) \right\} = \\ &= \int_{x=a}^{b} w'(x)\nu u'(x)\,dx - \int_{x=a}^{b} w(x) f(x)\,dx - \left[w(x)\nu u(x) \right]\Big|_{x=a}^{b} \end{aligned}$$

Boundary conditions…

[1] Which functional space? It would be nice to be as more precise as possible, without introducing too many "unnecessary" complications. Without going into details, 1) everything appears in the weak form should exists for the functions of that space: here, the functions must have L^2-integrable first-order derivative; 2) some results exist when test and unkwown functions belong to the same space: so, some "special" treatment of essential boundary conditions (on Dirichlet boundaries) is required.

Poisson equation with Dirichlet boundary conditions

By definition, test functions are identically zero on Dirichlet boundary conditions $w|_{S_D} = 0$.

Thus the weak form becomes

$$\int_{x=a}^{b} w'(x)\nu u'(x)\,dx = \int_{x=a}^{b} w(x)f(x)\,dx\ ,$$

along with the essential boundary conditions $u(a) = u_a$, $u(b) = u_b$.

Finite element method

Discretization of the domain. The domain $[a, b] =: \Omega$, is divided into elements Ω_i s.t.

$$\cup_i \Omega_i = \Omega$$
$$\Omega_i \cap \Omega_j = \emptyset \quad i \neq j$$

Exploiting additive property of integrals on the union of domains, integrals of the weak form become summation of integrals on the individual elements

$$\int_{\Omega} \dots = \sum_i \int_{\Omega_i} \dots$$

Choice of the base functions. *Lagrangian base functions* (1) $\phi_i(x_j) = \delta_{ij}$, 2) $\phi_i(x) = 0$, if $x \notin B(x_i)$; 3) $\sum_i \phi_i(x) = 1$, $\forall x \in \Omega$)

$$u(x) = \sum_i \phi_i(x) u_i\ .$$

Definition on a reference element $\xi \in [-1, 1]$ and then change of coordinates between "physical" and reference space to evaluate integrals.

Here, as the maximum order of the derivatives involved in the problem is 1, test functions must be piecewise-*linear* (or better, first degree polynomial) at least, in order to avoid numerically setting to zero some terms in the equation only due to poor approximation (here the term $\int_{x=a}^{b} w'(x)\nu u'(x)dx$ would be zero with piecewise-constant test functions).

First degree-polynomial Lagrangian base functions on the reference elements are

$$\widetilde{\varphi}_1(\xi) = \frac{1}{2}(1 - \xi)$$
$$\widetilde{\varphi}_2(\xi) = \frac{1}{2}(1 + \xi)$$

Evaluation of integrals. Polynomials functions are exactly evaluated using Gaussian integration, i.e. with a linear combination of the values of the function on a set of Gaussian nodes *x_g*

…

Analytical exact integration for first degree-polynomial. The coordinate transformation[2]

$$x = a_i \frac{1 - \xi}{2} + b_i \frac{1 + \xi}{2} = \frac{a_i + b_i}{2} + \xi \frac{\ell_i}{2}\ ,$$

transforms the reference element into physical elements: e.g. $x(\xi = -1) = a_i$, $x(\xi = 1) = b_i$. The Jacobian of this transformation is constant over the element,

$$\frac{dx}{d\xi} = \frac{\ell_i}{2}\ .$$

[2] Here, the transformation between physical and reference space is iso-parametric, as it uses the same base functions as those used for function approximations, $x(\xi) = a_i\widetilde{\phi}_1(\xi) + b_i\widetilde{\phi}_2(\xi)$.

On the element i,

$$
\begin{aligned}
K_{i,11} &= \int_{x=a_i}^{b_i} \frac{d\phi_1}{dx}(x)\frac{d\phi_1}{dx}(x)\,dx = \\
&= \int_{\xi=-1}^{\xi=1} \frac{d\xi}{dx}\frac{d\tilde{\phi}_1}{d\xi}\frac{d\xi}{dx}\frac{d\tilde{\phi}_1}{d\xi}\frac{dx}{d\xi}d\xi = \\
&= \int_{\xi=-1}^{\xi=1} \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right)d\xi\frac{2}{\ell_i} = \frac{1}{\ell_i}\,.
\end{aligned}
$$

Analogously,

$$
\begin{aligned}
K_{i,11} &= K_{i,22} = \frac{1}{\ell_i} \\
K_{i,12} &= K_{i,21} = -\frac{1}{\ell_i}
\end{aligned}
$$

$$
\begin{aligned}
M_{i,11} &= \int_{x=a_i}^{b_i} \phi_1(x)\phi_1(x)dx = \\
&= \int_{\xi=-1}^{1} \varphi_1(\xi)\varphi_1(\xi)\frac{dx}{d\xi}d\xi = \\
&= \int_{\xi=-1}^{1} \left[\frac{1}{2}(1-\xi)\right]^2 d\xi\frac{\ell_i}{2} = \\
&= \frac{\ell_i}{8}\left(2+\frac{2}{3}\right) = \frac{1}{3}\ell_i\,.
\end{aligned}
$$

and analogously

$$
\begin{aligned}
M_{i,11} &= M_{i,22} = \frac{\ell_i}{3} \\
M_{i,12} &= M_{i,21} = \frac{\ell_i}{6}
\end{aligned}
$$

```
import numpy as np
import scipy as sp

import matplotlib.pyplot as plt

#> Domain
a, b = 0, 1
l = b - a

#> Physical properties
nu = .01                        # Diffusion coefficient
f = 1.

#> Discretization
nel = 20              # n.of elements
nnodes = nel + 1
#> Dirichlet boundaries
i_dir = np.array([ 0, nel ])    # Global idx of nodes on Dirichlet boundaries
u_dir = np.array([ .0, .0 ])    # Value of the solution at nodes on Dir. boundaries

#> Volume forcing contribution
```

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```
f_nodal = np.ones(nel+1) * f   # Nodal values f(x_i) of volume forcing, here uniform␣
↪f(x) = f

#> Node coordinates array, rr, and element-node connectivity array, ee
rr = np.linspace(a,b, nel+1)
ee = np.array([[i, i+1] for i in np.arange(nel)])
#> Size of the elements
e_vol = np.array([ rr[ee[i,1]] - rr[ee[i,0]] for i in np.arange(nel) ])
```

```
#> Assembling stiffenss matrix, mass matrix, and forcing term
#> Stiffness matrix
ki = np.concatenate([[ee[i,0], ee[i,0], ee[i,1], ee[i,1]] for i in np.arange(nel)])
kj = np.concatenate([[ee[i,0], ee[i,1], ee[i,0], ee[i,1]] for i in np.arange(nel)])
ke = np.concatenate([nu * np.array([1., -1., -1., 1.,])/e_vol[i] for i in np.
↪arange(nel)])
K = sp.sparse.coo_array((ke, (ki, kj)),)

#> Mass matrix, here used for integration of the volume force
mi = np.concatenate([[ee[i,0], ee[i,0], ee[i,1], ee[i,1]] for i in np.arange(nel)])
mj = np.concatenate([[ee[i,0], ee[i,1], ee[i,0], ee[i,1]] for i in np.arange(nel)])
me = np.concatenate([np.array([2., 1., 1., 2.,])*e_vol[i]/6. for i in np.arange(nel)])
M = sp.sparse.coo_array((me, (mi, mj)),)

F = M @ f_nodal   # Volume force
```

Slicing the linear system

```
#> Apply essential boundary conditions, solve the linear system and retrieve the␣
↪solution
#> Method 1. Slicing matrices
# K u = f
# [ Kuu  Kud ] [ uu ] = [ fu ]
# [ Kdu  Kdd ] [ ud ]   [ fd ]
# with known: ud, fu; unknown: uu, fd found solving the problem
# Kuu * uu = fuu - Kud * ud  -> uu = ...
# fd = Kdu * uu + Kdd * ud
iD = i_dir.copy()
iU = np.array(list(set(np.arange(nnodes)) - set(iD)))
ud = u_dir.copy()

#> Slice stiffness matrix
Kuu = (K.tocsc()[:,iU]).tocsr()[iU,:]
Kud = (K.tocsc()[:,iD]).tocsr()[iU,:]
Kdu = (K.tocsc()[:,iU]).tocsr()[iD,:]
Kdd = (K.tocsc()[:,iD]).tocsr()[iD,:]

#> Slice forcing
fu = F[iU]

#> Solve the linear system
uu = sp.sparse.linalg.spsolve(Kuu, fu - Kud @ ud )
fd = Kdu @ uu + Kdd @ ud
```

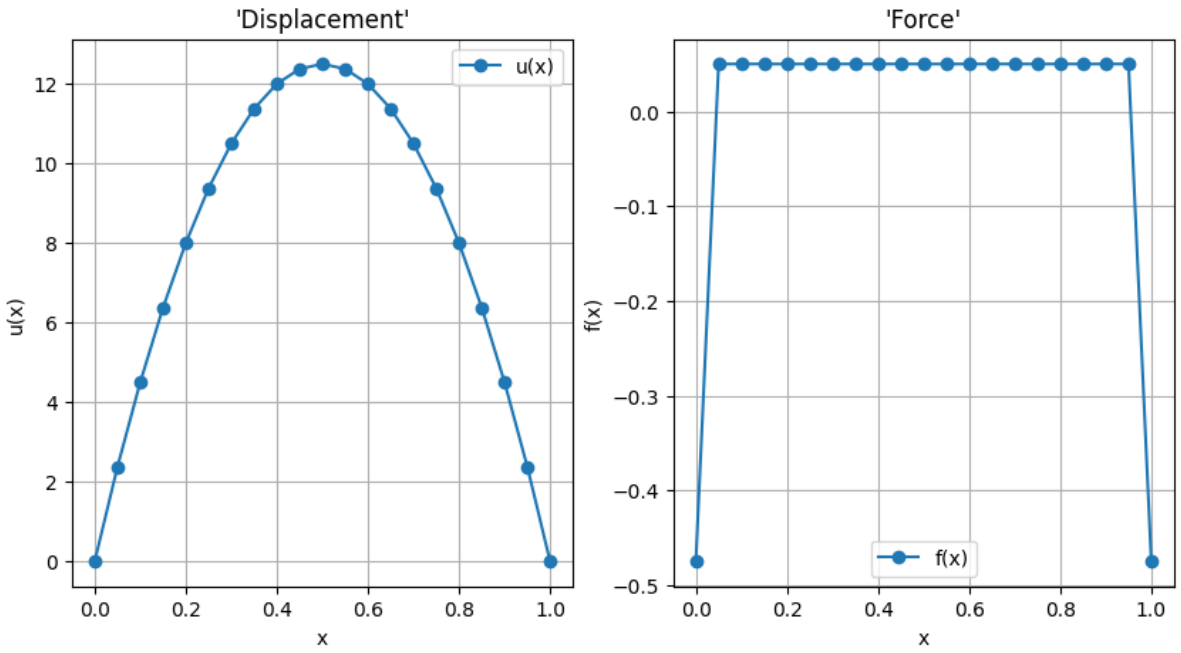
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```
#> Re-assemble the solution
u = np.zeros(nnodes);  u[iU] = uu;  u[iD] = ud
f = np.zeros(nnodes);  f[iU] = fu;  f[iD] = fd
# print("uu: \n", uu)
# print("Fd: \n", fd)

#> Plot some results
fig, ax = plt.subplots(1,2, figsize=(10, 5))
ax[0].plot(rr, u, '-o', label='u(x)')
ax[0].legend()
ax[0].grid()
ax[0].set_title('\'Displacement\'')
ax[0].set_xlabel('x')
ax[0].set_ylabel('u(x)')
ax[1].plot(rr, f, '-o', label='f(x)')
ax[1].legend()
ax[1].grid()
ax[1].set_title('\'Force\'')
ax[1].set_xlabel('x')
ax[1].set_ylabel('f(x)')
```

```
Text(0, 0.5, 'f(x)')
```




```
#> Method 2. Augmented system
# ...
```

Augmenting the linear system

See discussion at the end of the *script implementing a basic finite volume method solution of 1-dimensional Poisson equation*.

Some comments and todos.

- **Convergence analysis.** It's possible to evaluate both convergence with the dimension of the elements and/or the degree of the polynomial base functions. The *exact solution* can be easily computed analytically via direct integration for simple distribution of load $f(x)$, and uniform ν
- **Result discussion.** "Force" contribution on Dirichlet boundary conditions involves both the node contribution to distributed load and the "constraint reaction". As an example, with a domain of size ℓ and uniform force distribution f, with $u(0) = u(\ell) = 0$, by *symmetry* and *global equilibrium* the force at boundaries are $-\frac{1}{2}q\ell = -\frac{1}{2}$. With the modelling choices done, the contribution of the uniform distributed load on an element on any of its nodes is $\frac{1}{2}q\ell_i = \frac{1}{2}q\frac{\ell}{n_{el}}$. Here with $n_{el} = 20$, this elementary contribution is $\frac{1}{40} = .025$: thus, the nodal contribution of internal nodes is twice (contribution from two neighboring elements) this value .05, and the force on each Dirichlet boundary is $-\frac{1}{2} + \frac{1}{40} = -0.475$.
- Try to set different values of Dirichlet boundary conditions, u_dir

CHAPTER

FIFTYNINE

FINITE VOLUME METHOD

Property 59.1

Evaluate flux on interfaces between cells, and distribute between neighboring cells.

59.1 1-dimensional Poisson equation

59.1.1 Integral form of the problem

The integral form of a Poisson problem reads, **for every** $V \in \Omega$,

$$-\oint_{\partial V} \hat{\mathbf{n}} \cdot (\nu \nabla u) = \int_V f \, .$$

supplied with proper boundary conditions. As an example,

- Dirichlet boundary conditions: $u(\overline{x}) = \overline{u}$
- Neumann boundary conditions: $\nu \hat{\mathbf{n}} \cdot \nabla u(\overline{x}) = \overline{q}$
- Robin boundary conditions: $au(\overline{x}) + bu'(\overline{x}) = c$

The multi-dimensional set may be useful also for 1-dimensional domains, to avoid confusions with signs

59.1.2 Domain, volume forcing and boundary conditions

```
import numpy as np
import scipy as sp

import matplotlib.pyplot as plt

#> Domain
a, b = 0, 1
l = b - a

#> Physical properties
nu = .01                        # Diffusion coefficient
f = 1.

#> Discretization
```

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```
nel = 10             # n.of elements
nnodes = nel + 1     # n.of nodes
nvol = nnodes        # n.finite volumes (node-centered FVM)

#> Dirichlet boundaries
i_dir = np.array([ 0, nvol-1])    # Global idx of nodes on Dirichlet boundaries
u_dir = np.array([ .0, .0 ])    # Value of the solution at nodes on Dir. boundaries

#> Volume forcing contribution
f_nodal = np.ones(nel+1) * f    # Nodal values f(x_i) of volume forcing, here uniform␣
 ↪f(x) = f

#> Node centered FVM
rr_n = np.linspace(a,b, nel+1)        # node coordinates (center of the node-centered␣
 ↪FVs)
rr_i = .5 * ( rr_n[:-1] + rr_n[1:] ) # coords of the inner cell interface
rr_b = np.array([ 2*rr_n[0]-rr_n[1], 2*rr_n[-1]-rr_n[-2] ])  # coords of the␣
 ↪boundaries

vols = np.array(
     [ np.abs(rr_i[0]-rr_n[0]) ] + \
       list( np.abs(rr_i[1:] - rr_i[:-1]) ) + \
     [ np.abs(rr_i[-1]-rr_n[-1]) ]
)

ni = len(rr_i)
nb = len(rr_b)

#> Interface-volume connectivity (ni, 2) (normal dir: first element -> second element)
iee = np.array([ [i, i+1] for i in np.arange(ni) ])
#> Boundary-volume connectivity (nb) (convention, normal dir: elem -> outside)
bee = np.array([0, nvol-1])
```

59.1.3 Discrete matheamtical problem

```
#> Evaluating boundary contributions
# Here assmbling a stiffness matrix. In many FVM applications, there's explicit time
# dependence and explicit time integration schemes are used, so there's no need to
# assemble any matrix while the algorithm just rely on accumulation of flux␣
 ↪contributions
# -> Add a link to a script implementing time-dependent "solver" for FVM, e.g.␣
 ↪hyperbolic problems

i_flux, j_flux, e_flux = [], [], []

#> Loop over internal interface
# accumulating and keep changing the dimension! May be very inefficient and slow
for i in np.arange(ni):
    flux_u = nu / np.abs( rr_n[ iee[i,1] ] - rr_n[ iee[i,0] ] )
    i_flux += [ iee[i,0], iee[i,0], iee[i,1], iee[i,1] ]
    j_flux += [ iee[i,0], iee[i,1], iee[i,0], iee[i,1] ]
    e_flux += [ -flux_u, flux_u, flux_u, -flux_u ]

e_flux = -np.array(e_flux)
```

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```
K = sp.sparse.coo_array((e_flux, (i_flux, j_flux)),)

#> Volume forcing
F = vols * f

#> Loop over boundary surfaces
# ...
# - Neumann: add a forcing term for finite volumes with sides on Neumann boundaries
# - Dirichlet: prescribe node value (e.g. 1) matrix slicing, 2) augmented system,...)
# - Robin: add a forcing term depending on the unknown value of the function ->␣
 ↪modify K matrix

# ...
```

59.1.4 Applying essential boundary conditions

The discrete counterpart of the continuous problem is a linear system that can be formally written as $\mathbf{Au} = \mathbf{f}$. So far, the essential boundary conditions on Dirichlet boundary S_D have not been prescribed yet. Essential boundary conditions can be prescribed in (at least) two ways: 1) slicing the linear system; 2) augmenting the linear system.

Method 1. Slicing the linear system. The set of nodes $(\cdot)_d$ lying on Dirichlet boundaries can be distinguished from all the remaining nodes $(\cdot)_u$, and the linear system sliced accordingly

$$\begin{bmatrix} \mathbf{A}_{uu} & \mathbf{A}_{ud} \\ \mathbf{A}_{du} & \mathbf{A}_{dd} \end{bmatrix} \begin{bmatrix} \mathbf{u}_u \\ \mathbf{u}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_u \\ \mathbf{f}_d \end{bmatrix} .$$

On u-nodes, forcing $\mathbf{f}_u$ is known, while the value of the function $u(\mathbf{x})$ is unknown, and thus $\mathbf{u}_n$ is unknown; on the other hand, on d-nodes the value of the function $u(\mathbf{x})$ is known, and so $\mathbf{u}_d = \mathbf{u}_D$ is known, while the forcing $\mathbf{f}_d$ is not known (as it contains both a volume contribution and a "boundary contribution" required to prescribe essential boundary conditions, as it will be more clear later, hopefully).

After slicing, the problem is solved first finding $\mathbf{u}_u$ and then retrieving the value of $\mathbf{f}_d$,

$$\begin{aligned} \mathbf{A}_{uu}\mathbf{u} = \mathbf{f}_u - \mathbf{A}_{ud}\mathbf{u}_D \qquad &\rightarrow \quad \mathbf{u}_u = \mathbf{A}_{uu}^{-1}\left(\mathbf{f}_u - \mathbf{A}_{ud}\mathbf{u}_D\right) \\ \mathbf{f}_d = \mathbf{A}_{du}\mathbf{u}_u + \mathbf{A}_{dd}\mathbf{u}_D \qquad &\rightarrow \quad \mathbf{f}_d = \dots \end{aligned}$$

Method 2. Augmenting the linear system. The linear system is augmented, a) explicityl adding equations $\mathbf{u}_d = \mathbf{u}_D$ for the essential boudary conditions, and splitting the forcing contribution on Dirichlet nodes as the sum of a known volume contribution and an unknown boundary contribution,

$$\mathbf{f} = \begin{bmatrix} \mathbf{f}_u^{vol} \\ \mathbf{f}_d^{vol} + \mathbf{f}_d^D \end{bmatrix} = \begin{bmatrix} \mathbf{f}_u^{vol} \\ \mathbf{f}_d^{vol} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix} \mathbf{f}_d^D .$$

Moving all the unkowns on the LHS, the augmented linear system reads

$$\begin{bmatrix} \mathbf{A}_{uu} & \mathbf{A}_{ud} & \mathbf{0} \\ \mathbf{A}_{du} & \mathbf{A}_{dd} & -\mathbf{I} \\ \mathbf{0} & -\mathbf{I} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u}_u \\ \mathbf{u}_d \\ \mathbf{f}_d^D \end{bmatrix} = \begin{bmatrix} \mathbf{f}_u^{vol} \\ \mathbf{f}_d^{vol} \\ -\mathbf{u}_D \end{bmatrix} .$$

The (arbitrary) choice of the sign in the last block of equations (the ones prescribing the value of the unknown function on Dirichlet boundary) is usually chosen to preserve the symmetry of the original problem, if any.

todo *Discuss the reason why the system becomes determinate, after the application of the essential b.c.s*

Slicing the linear system

Slice the linear system to prescribe the essential boundary conditions, solve the linear system, retrieve and plot the results.

```
#> Apply essential boundary conditions, solve the linear system and retrieve the␣
 ↪solution
#> Method 1. Slicing matrices
# K u = f
# [ Kuu  Kud ] [ uu ] = [ fu ]
# [ Kdu  Kdd ] [ ud ]   [ fd ]
# with known: ud, fu; unknown: uu, fd found solving the problem
# Kuu * uu = fuu - Kud * ud  -> uu = ...
# fd = Kdu * uu + Kdd * ud
iD = i_dir.copy()
iU = np.array(list(set(np.arange(nnodes)) - set(iD)))
ud = u_dir.copy()

print(iD)
print(iU)

#> Slice stiffness matrix
Kuu = (K.tocsc()[:,iU]).tocsr()[iU,:]
Kud = (K.tocsc()[:,iD]).tocsr()[iU,:]
Kdu = (K.tocsc()[:,iU]).tocsr()[iD,:]
Kdd = (K.tocsc()[:,iD]).tocsr()[iD,:]

#> Slice forcing
fu = F[iU]

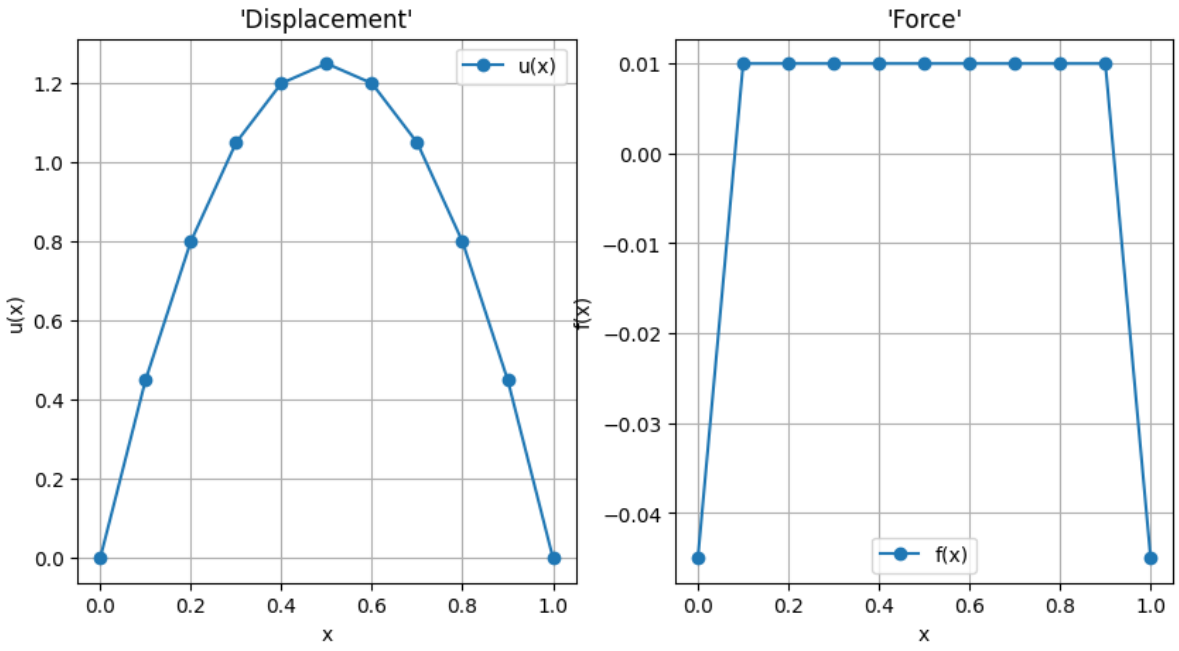
#> Solve the linear system
uu = sp.sparse.linalg.spsolve(Kuu, fu - Kud @ ud )
fd = Kdu @ uu + Kdd @ ud

#> Re-assemble the solution
u = np.zeros(nnodes);  u[iU] = uu;  u[iD] = ud
f = np.zeros(nnodes);  f[iU] = fu;  f[iD] = fd
# print("uu: \n", uu)
# print("Fd: \n", fd)

#> Plot some results
fig, ax = plt.subplots(1,2, figsize=(10, 5))
ax[0].plot(rr_n, u, '-o', label='u(x)')
ax[0].legend()
ax[0].grid()
ax[0].set_title('\'Displacement\'')
ax[0].set_xlabel('x')
ax[0].set_ylabel('u(x)')
ax[1].plot(rr_n, f, '-o', label='f(x)')
ax[1].legend()
ax[1].grid()
ax[1].set_title('\'Force\'')
ax[1].set_xlabel('x')
ax[1].set_ylabel('f(x)')
```

```
[ 0 10]
[1 2 3 4 5 6 7 8 9]
```

```
Text(0, 0.5, 'f(x)')
```



Augmenting the linear system

Augmenting the linear system to prescribe the essential boundary conditions, solve the linear system, retrieve and plot the results.

todo

```
#> Method 2. Augmented system
# ...
```

Some comments and todos.

- **Convergence analysis.** It's possible to evaluate both convergence with the dimension of the elements and/or the degree of the polynomial base functions. The *exact solution* can be easily computed analytically via direct integration for simple distribution of load $f(x)$, and uniform ν
- **Result discussion.** "Force" contribution on Dirichlet boundary conditions involves both the node contribution to distributed load and the "constraint reaction". As an example, with a domain of size ℓ and uniform force distribution f, with $u(0) = u(\ell) = 0$, by *symmetry* and *global equilibrium* the force at boundaries are $-\frac{1}{2}q\ell = -\frac{1}{2}$. With the modelling choices done, the contribution of the uniform distributed load on an element on any of its nodes is $\frac{1}{2}q\ell_i = \frac{1}{2}q\frac{\ell}{n_{el}}$. Here with $n_{el} = 20$, this elementary contribution is $\frac{1}{40} = .025$: thus, the nodal contribution of internal nodes is twice (contribution from two neighboring elements) this value .05, and the force on each Dirichlet boundary is $-\frac{1}{2} + \frac{1}{40} = -0.475$.
- Try to set different values of Dirichlet boundary conditions, u_dir

59.2 FVM for hyperbolic problems

Approach

- Integral equations, often from balance or conservation laws
- First, problems in 1-dimensional domains are treated. Most of the features of a finite volume method are treated for 1-dimensional problems
  - flux: characteristics and common schemes
  - implementation: evaluation of flux at interfaces and distribution on cells
  - boundary conditions
  - …

  then everything is naturally generalized to multi-dimensional domain: each interface of a cell in a multi-dimensional domain is locally treated as a 1-dimensional domain.
- *Numerical flux*
  - features:
    - must be conservative…
    - must satisfy **entropy condition**, i.e. produce the physical solution (the limit of viscous problem for negligible viscosity) among the infinite possible solutions
  - schemes:
    - low-order
      - *Godunov*: the numerical flux is evaluated after solving a *Riemann problem* at each cell interface, taking the $x = 0$ state $\mathbf{u}(x = 0, t)$ to evaluate the flux $\mathbf{F}(\mathbf{u})$ at that cell boundary
      - *Roe* (~ linearized Godunov) + *entropy fix* required for the numerical method to select the physical solution
    - high-order schemes:
      - improve accuracy of differentiable solutions (no shock, no discontinuity) on regular grids
      - fail to represent the physical solution
    - high-resolution, (TVD schemes with flux limiters, MUSCL,…):
      - high-order methods where the solution is differentiable
      - low order method where the solution has discontinuities
      - high-resolution achieved as a linear combination of a low and high order flux

        $$F(u_{i+1/2}) = f^{low}_{i+1/2} + \phi(r_{i-1,i}) \left(f^{high}_{i+1/2} - f^{low}_{i+1/2} \right) ,$$

        being $\phi(r_{i,i+1})$ the flux limiter, i.e. a function of the gradient r_i of the solution on the grid
- Theorems about numerics:
  - **CFL (Courant-Friedrichs-Lewy) condition** for explicit-time schemes (it could hold locally…),

    $$\Delta t \leq \frac{\Delta x}{|\lambda|_{\max}}$$

  - *…results about fluxes…*

59.2.1 Numerical fluxes

Godunov flux

The numerical flux in Godunov scheme is the flux $\mathbf{F}$ evaluated with the state at the interface of two neighboring cells, resulting from the solution of the *Riemann problem* between the two cells.

For a piecewise constant approximation of the fields (like in standard finite volume methods), the evolution at each cell interface can be treated as a Riemann problem

In order to calculate the numerical flux at the interface i, separating cells A, B

1. The Riemann problem is solved at interface i separating the two cells with uniform fields $\mathbf{u}_A(t)$, $\mathbf{u}_B(t)$. Let $\mathbf{u}_i(x, t+\Delta t)$ be the solution, being x a local coordinate (orthogonal to the interface for multi-dimensional domains) with $x = 0$ at the interface, and $\Delta t > 0$. Riemann problem is a **non-linear problem**, so a non-linear solver is required. In order to avoid (expensive) non-linear methods, *Roe scheme* is introduced in the next section.
2. The state at interface $\mathbf{u}_i(x = 0, t + \Delta t)$ is retrieved from the solution of the Riemann problem
3. The numerical flux is evaluated with the state at the interface

$$\mathbf{F}_i(t) = \mathbf{F}(\mathbf{u}_i(0, t)) \ .$$

Roe flux

Instead of solving the original non-linear problem,

$$\begin{aligned} \mathbf{0} &= \partial_t \mathbf{u} + \partial_x \mathbf{F}(\mathbf{u}) = \\ &= \partial_t \mathbf{u} + \mathbf{A}(\mathbf{u}) \partial_x \mathbf{u} \ . \end{aligned}$$

a Roe solver aims at solving a linearized problem at each interface, namely

$$\mathbf{0} = \partial_t \mathbf{u} + \hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R) \partial_x \mathbf{u} \ ,$$

being $\hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R)$ an approximation of the convection matrix $\mathbf{A}(\mathbf{u})$ built with the states in the cells sharing the interface of interest: in time-explicit schemes, states in the cells $\mathbf{u}_L$, $\mathbf{u}_R$ are known, and so it is the linearized matrix $\hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R)$.

Intermediate state, $\hat{\mathbf{u}}$. The linearized matrix can be written as the original matrix evaluated for an *intermediate state* $\hat{\mathbf{u}}(\mathbf{u}_L, \mathbf{u}_R)$,

$$\hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R) = \mathbf{A}(\hat{\mathbf{u}}(\mathbf{u}_L, \mathbf{u}_R)) \ .$$

Roe flux. Roe flux looks similar to upwind scheme,

$$\begin{aligned} \mathbf{F}^{Roe} &= \frac{1}{2} \left(\mathbf{F}(\mathbf{u}_L) + \mathbf{F}(\mathbf{u}_R) - \left| \hat{\mathbf{A}}(\mathbf{u}_L, \mathbf{u}_R) \right| (\mathbf{u}_L - \mathbf{u}_R) \right) = \\ &= \frac{1}{2} \left(\mathbf{F}(\mathbf{u}_L) + \mathbf{F}(\mathbf{u}_R) - \hat{\mathbf{R}} \left| \hat{\Lambda} \right| \hat{\mathbf{L}} (\mathbf{u}_L - \mathbf{u}_R) \right) \ . \end{aligned}$$

Properties. Roe matrix needs to satisfy the following properties:

1. **diagonalization**: $\hat{\mathbf{A}}$ is diagonalizable
2. **consistency**: if $\mathbf{u}_L, \mathbf{u}_R \to \mathbf{u}$, then $\hat{\mathbf{u}} \to \mathbf{u}$
3. **conservation**: $\mathbf{A}(\hat{\mathbf{u}}) (\mathbf{u}_L - \mathbf{u}_R) = \mathbf{F}(\mathbf{u}_L) - \mathbf{F}(\mathbf{u}_R)$

Example 59.2.1 (P-sys. Intermediate state)

Intermediate state. For each value of $\hat{\rho}$,

$$\hat{u} = \frac{\sqrt{\rho_L}u_L + \sqrt{\rho_R}u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} ,$$

and thus

$$\hat{m} = \hat{\rho}\hat{u} .$$

Details

3. **Conservation**

$$\mathbf{A}(\hat{\mathbf{u}})(\mathbf{u}_L - \mathbf{u}_R) = \mathbf{F}_L - \mathbf{F}_R .$$

explicitly, using conservative variables

$$\begin{bmatrix} 0 & 1 \\ a^2 - \frac{\hat{m}^2}{\hat{\rho}^2} & 2\frac{\hat{m}}{\hat{\rho}} \end{bmatrix} \begin{bmatrix} \rho_L - \rho_R \\ m_L - m_R \end{bmatrix} = \begin{bmatrix} \rho_L - \rho_R \\ \frac{m_L^2}{\rho_L} + \rho_L a^2 - \frac{m_R^2}{\rho_R} - \rho_R a^2 \end{bmatrix} .$$

The first equation is identically satisfied. The second equation in $\hat{u}$ reads

$$\begin{aligned} 0 &= (a^2 - \hat{u}^2)(\rho_L - \rho_R) + 2\hat{u}\,(\rho_L u_L - \rho_R u_R) - \rho_L u_L^2 - \rho_L a^2 + \rho_R u_R^2 + \rho_R a^2 = \\ &= -\hat{u}^2(\rho_L - \rho_R) + 2\hat{u}\,(\rho_L u_L - \rho_R u_R) - \rho_L u_L^2 + \rho_R u_R^2 = \end{aligned}$$

and collecting terms

$$0 = \rho_L(\hat{u} - u_L)^2 - \rho_R(\hat{u} - u_R)^2$$

or, with $\rho > 0$,

$$\sqrt{\rho_L}(\hat{u} - u_L) = \mp\sqrt{\rho_R}(\hat{u} - u_R) ,$$

so that the solution that avoids singularities for $\rho_L = \rho_R$ (the one with the minus sign above) reads

$$\hat{u} = \frac{\sqrt{\rho_L}u_L + \sqrt{\rho_R}u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} ,$$

2. **Consistency.** From this choice of sign, consistency of the velocity follows immediately,

$$\lim_{\mathbf{u}_L, \mathbf{u}_R \to \mathbf{u}} \hat{\mathbf{u}}(\mathbf{u}_L, \mathbf{u}_R) = \mathbf{u} .$$

Arbitrary intermediate value of the density $\hat{\rho}$ needs to satisfy consistency, as well. Just as an example, if $\hat{\rho}$ is chosen to be the average of the densities, $\hat{\rho} = \frac{1}{2}(\rho_L + \rho_R)$, this is consistent choice.

Example 59.2.2 (Euler equations. Intermediate state)

Intermediate state. For each value of $\hat{\rho}$,

$$\begin{aligned} \hat{u} &= \frac{\sqrt{\rho_L}u_L + \sqrt{\rho_R}u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} \\ \hat{h}^t = \langle h^t \rangle_{\sqrt{\rho}} &= \frac{\sqrt{\rho_L}h_L^t + \sqrt{\rho_R}h_R^t}{\sqrt{\rho_L} + \sqrt{\rho_R}} \end{aligned}$$

while $\hat{\rho}$ can be found (if not trivial) solving an additional condition

$$\Delta P = \Delta\rho\,\partial_\rho \Pi(\hat{\rho}, \hat{\rho}\hat{u}, \hat{E}^t(\hat{\rho}, \hat{u}, \hat{h}^t)) + \Delta m\,\partial_m \Pi(\hat{\rho}, \hat{\rho}\hat{u}, \hat{E}^t(\hat{\rho}, \hat{u}, \hat{h}^t)) + \Delta E^t\,\partial_{E^t} \Pi(\hat{\rho}, \hat{\rho}\hat{u}, \hat{E}^t(\hat{\rho}, \hat{u}, \hat{h}^t)) .$$

If this condition is an identity - as it is for a perfect ideal gas - any value of $\hat{\rho}$ satisfying consistency is allowed.

Details

3. **Conservation**

$$\mathbf{A}(\hat{\mathbf{u}})(\mathbf{u}_L - \mathbf{u}_R) = \mathbf{F}_L - \mathbf{F}_R \, .$$

explicitly, using conservative variables

$$\begin{bmatrix} 0 & 1 & 0 \\ -\frac{m^2}{\rho^2} + \partial_\rho \Pi & \frac{2m}{\rho} + \partial_m \Pi & \partial_{E^t} \Pi \\ -\frac{m}{\rho^2}(E^t + \Pi) + \frac{m}{\rho}\partial_\rho \Pi & \frac{1}{\rho}(E^t + \Pi) + \frac{m}{\rho}\partial_m \Pi & \frac{m}{\rho}\left(1 + \partial_{E^t}\Pi\right) \end{bmatrix} \begin{bmatrix} \rho_L - \rho_R \\ m_L - m_R \\ E^t_L - E^t_R \end{bmatrix} = \left| \begin{bmatrix} \rho \\ \frac{m^2}{\rho} + \Pi(\rho, m, E^t) \\ \frac{m}{\rho}\left(E^t + \Pi(\rho, m, E^t)\right) \end{bmatrix} \right|_R^L$$

or

$$\begin{bmatrix} 0 & 1 & 0 \\ -\hat{u}^2 + \partial_\rho \Pi & 2\hat{u} + \partial_m \Pi & \partial_{E^t} \Pi \\ -\hat{u}\hat{h}^t + \hat{u}\partial_\rho \Pi & \hat{h}^t + \hat{u}\partial_m \Pi & \hat{u}\left(1 + \partial_{E^t}\Pi\right) \end{bmatrix} \begin{bmatrix} \rho_L - \rho_R \\ \rho_L u_L - \rho_R u_R \\ E^t_L - E^t_R \end{bmatrix} = \left| \begin{bmatrix} \rho \\ \rho u^2 + \Pi(\rho, m, E^t) \\ \rho u h^t \end{bmatrix} \right|_R^L$$

- The first equation is an identity
- The second equation in $\hat{u}$ gives

$$0 = -\left[\rho(\hat{u} - u)^2\right]\Big|_R^L + \left(\rho\partial_\rho \hat{\Pi} + \rho u \partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi} - \Pi\right)\Big|_R^L \, ,$$

- The third equation in $\hat{h}^t$ gives

$$\begin{aligned} 0 = & \left[\rho(u - \hat{u})\hat{h}^t\right]\Big|_R^L + \left[\hat{u}\rho\partial_\rho \hat{\Pi} + \hat{u} m \partial_m \hat{\Pi} + \hat{u} E^t \partial_{E^t}\hat{\Pi} + \hat{u} \underbrace{E^t}_{=\rho h^t - p} - \rho u h^t\right]\Big|_L^R = \\ = & \left[\rho(u - \hat{u})\hat{h}^t + \rho h^t(\hat{u} - u)\right]\Big|_R^L + \hat{u}\left[\rho\partial_\rho \hat{\Pi} + m\partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi} - \Pi\right]\Big|_L^R = \\ = & \left[\rho(u - \hat{u})(\hat{h}^t - h^t)\right]\Big|_R^L + \hat{u}\left[\rho\partial_\rho \hat{\Pi} + m\partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi} - \Pi\right]\Big|_L^R \, . \end{aligned}$$

Adding the condition $\Delta P = \Delta\rho\partial_\rho \hat{\Pi} + m\partial_m \hat{\Pi} + E^t \partial_{E^t}\hat{\Pi}$, the second equation can be solved for $\hat{u}$ (choosing the solution providing consistency - see P-sys example for a brief discussion)

$$\hat{u} = \langle u \rangle_{\sqrt{\rho}} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}$$

and the third equation can be solved for $\hat{h}^t$ (using $\sqrt{\rho_L}(u_L - \hat{u}) = -\sqrt{\rho_R}(u_R - \hat{u})$ from the expression of $\hat{u}$ to get the equation $\sqrt{\rho}(\hat{h}^t - h^t_L) = -\sqrt{\rho_R}(\hat{h}^t - h^t_R)$),

$$\hat{h}^t = \langle h^t \rangle_{\sqrt{\rho}} = \frac{\sqrt{\rho_L} h^t_L + \sqrt{\rho_R} h^t_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}$$

For a medium with generic equations of state, the condition about Δp introduced above can be eventually solved to find the value of $\hat{\rho}$. For a perfect ideal gas (PIG), this condition is an identity providing no additional information on the value of $\hat{\rho}$

2. **Consistency.** …

Entropy fix

Entropy fix is required to exploit numerical dissipation in choosing the physical solution, when some eigenvalue is close to zero. As an example, a threshold is set as a ratio of the local speed of sound a, e.g. $\delta = 0.1 \div 0.2$ and the Roe matrix is modified as

$$\hat{\mathbf{A}}^{s\,fix} = \hat{\mathbf{R}} \left| \hat{\Lambda}^{s\,fix} \right| \hat{\mathbf{L}} \,,$$

with

$$\left| \hat{\Lambda}_i^{s\,fix} \right| = \begin{cases} |\lambda_i| & |\lambda_i| \geq \delta a \\ \dfrac{|\lambda_i|^2}{2\delta a} + \dfrac{\delta a}{2} & |\lambda_i| < \delta a \end{cases}$$

todo *Add discussion about* **numerical dissipation** *and some numerical examples...*

59.3 Boundary conditions in hyperbolic problems

59.3.1 Wall

Domain at rest

Example 59.3.1 (Wall for Roe flux in 1-dimensional P-sys)

Roe intermediate state for a P-sys is

$$\begin{bmatrix} \hat{\rho} \\ \hat{m} \end{bmatrix} = \begin{bmatrix} \hat{\rho} \\ \hat{\rho}\hat{u} \end{bmatrix} \,,$$

with $\hat{\rho}$ satisfying consistency (as an example $\hat{\rho} = \sqrt{\rho_L \rho_R}$ or $\hat{\rho} = \frac{1}{2}(\rho_L + \rho_R)$), and

$$\hat{u} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}} \,.$$

Wall boundary conditions for a P-sys implies no mass flux through the wall. The state of the ghost cell outside the domain is prescribed as

$$\mathbf{u}_g = \begin{bmatrix} \rho_g \\ \rho_g u_g \end{bmatrix} \,,$$

with $\rho_g = \rho_\partial$, $u_g = -u_\partial$. Roe intermediate state across the wall boundary interface has velocity

$$\hat{u}_\partial = \frac{\sqrt{\rho_g} u_g + \sqrt{\rho_\partial} u_\partial}{\sqrt{\rho_g} + \sqrt{\rho_\partial}} = 0 \,.$$

Upwind matrix in Roe flux. Using the spectral decomposition of the matrix

$$\mathbf{A} = \begin{bmatrix} u & \rho \\ \frac{a^2}{\rho} & u \end{bmatrix} \,,$$

i.e.

$$\mathbf{R} = \begin{bmatrix} \rho & \rho \\ a & -a \end{bmatrix} \quad , \quad \Lambda = \begin{bmatrix} u-a & 0 \\ 0 & u+a \end{bmatrix} \quad , \quad \mathbf{L} = \begin{bmatrix} \frac{1}{2\rho} & \frac{1}{2a} \\ \frac{1}{2\rho} & -\frac{1}{2a} \end{bmatrix}$$

the upwind matrix evaluated on the Roe intermediate state (no need for entropy fix; with entropy fix, the result doesn't change anyway, as there's no eigenvalue close to zero) reads

$$
\begin{aligned}
|\hat{\mathbf{A}}| = \hat{\mathbf{R}} \left|\hat{\Lambda}\right| \hat{\mathbf{L}} &= \begin{bmatrix} \rho & \rho \\ a & -a \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} \frac{1}{2\rho} & \frac{1}{2a} \\ \frac{1}{2\rho} & -\frac{1}{2a} \end{bmatrix} = \\
&= \begin{bmatrix} \rho a & \rho a \\ a^2 & -a^2 \end{bmatrix} \begin{bmatrix} \frac{1}{2\rho} & \frac{1}{2a} \\ \frac{1}{2\rho} & -\frac{1}{2a} \end{bmatrix} = \\
&= \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} .
\end{aligned}
$$

Roe flux. Putting everything together, Roe flux at wall reads

$$
\begin{aligned}
\mathbf{F}_\partial &= \frac{1}{2} \left[\mathbf{F}_g + \mathbf{F}_\partial - \left|\hat{\mathbf{A}}\right| (\mathbf{u}_g - \mathbf{u}_\partial) \right] = \\
&= \frac{1}{2} \left[\begin{bmatrix} -m_\partial \\ \rho_\partial \left(a^2 + u_\partial^2\right) \end{bmatrix} + \begin{bmatrix} m_\partial \\ \rho_g \left(a^2 + u_\partial^2\right) \end{bmatrix} - \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \left(\begin{bmatrix} \rho_\partial \\ -m_\partial \end{bmatrix} - \begin{bmatrix} \rho_\partial \\ m_\partial \end{bmatrix} \right) \right] = \\
&= \begin{bmatrix} 0 \\ \rho_\partial (a + u_\partial)^2 \end{bmatrix} ,
\end{aligned}
$$

i.e. there's **no mass flux** across the wall, as the first component of the vector $\mathbf{F}_\partial$ is zero.

Example 59.3.2 (Wall for Roe flux in 1-dimensional Euler equations)

Moving domain

59.3.2 Characteristic-based boundary conditions

59.4 Examples of FVM for hyperbolic problems

1-dimensional P-sys

1-dimensional Euler equations for perfect ideal gas

Quasi 1-dimensional Euler equations for perfect ideal gas

1-dimensional Euler equations for perfect ideal gas on moving domains (ALE)

59.4.1 1-dimensional P-system

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

P-system class

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

Boundary conditions

Simulation parameters

Time loop

Post-processing

59.4.2 1-dimensional Euler equations for Perfect Ideal Gas

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

Euler equations for perfect ideal gas class

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

Boundary conditions

Initial conditions

Simulation parameters

Time loop

Post-processing

59.4.3 Quasi 1-dimensional Euler equations for Perfect Ideal Gas

Unsteady Euler equations for quasi-1 dimensional flows read

$$\partial_t \begin{bmatrix} \rho A \\ \rho u A \\ \rho e^t A \end{bmatrix} + \partial_z \begin{bmatrix} \rho u A \\ \frac{(\rho u A)^2}{\rho A} + pA \\ \frac{(\rho e^t A + pA)(\rho u A)}{\rho A} \end{bmatrix} = \begin{bmatrix} 0 \\ p \partial_z A \\ 0 \end{bmatrix}$$

Method 1. The same conservative variables as the 1-dimensional flow solver are used (ρ, m, E^t), and non-uniform

- appears as a source in momentum equation
- appears in discretization for cells A_i and as a factor multiplying fluxes $A_{i+1/2} F_{i+1/2}$

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^n - \frac{\Delta t}{\Delta A_i} \left(\mathbf{F}_{i+1/2} A_{i+1/2} - \mathbf{F}_{i-1/2} A_{i-1/2} \right) + \Delta t \, \mathbf{s}_i$$

Method 2. Another approach would be using $(\rho A, mA, E^t A)$ as conservative variables.

todo

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

Euler equations for perfect ideal gas class

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

Boundary conditions

Initial conditions

Simulation parameters

Time loop

Post-processing

59.4.4 1-dimensional Euler equations for Perfect Ideal Gas on moving mesh (ALE)

Integral equations for general domain v_t. Starting from the balance equation for a control volume V at rest,

$$\frac{d}{dt}\int_V \mathbf{u} + \oint_{\partial V} \mathbf{F}(\mathbf{u})\cdot\hat{\mathbf{n}} = \int_V \mathbf{s}\,,$$

using Reynolds' transport theorem, the balance equation for an arbitrary volume v_t follows

$$\frac{d}{dt}\int_{v_t} \mathbf{u} - \oint_{\partial v_t} \mathbf{u}\mathbf{v}_b\cdot\hat{\mathbf{n}} + \oint_{\partial v_t} \mathbf{F}(\mathbf{u})\cdot\hat{\mathbf{n}} = \int_{v_t} \mathbf{s}\,,$$

or collecting all the boundary terms

$$\frac{d}{dt}\int_{v_t} \mathbf{u} + \oint_{\partial v_t} [\mathbf{F}(\mathbf{u}) - \mathbf{u}\mathbf{v}_b]\cdot\hat{\mathbf{n}} = \int_{v_t} \mathbf{s}\,.$$

Explicit Euler scheme

$$(\mathbf{u}_i^{n+1}V_i^{n+1}) = (\mathbf{u}_i^n V_i^n) - \Delta t \sum_{j\in B_i} \mathbf{F}_j^n\,,$$

Roe flux …

Boundary conditions

Useful classes for numerical methods

Basic libraries

Parent class

`HyperbolicSystem1d()` class with common functions (or just their templates) required to implement a finite volume method solver.

Euler equations for perfect ideal gas class

Time dependent mesh

Time dependent mesh class, producing a mesh with uniform spacing, given the law of motion of the extreme points of the domain.

Numerical simulation

System of equations

Initialized the system of equations to be solved

Domain

VV of the moving mesh

Boundary conditions

Initial conditions

Simulation parameters

Time loop

Post-processing

CHAPTER

SIXTY

BOUNDARY ELEMENT METHOD

Part XIV

Boundary Methods for PDEs

CHAPTER

SIXTYONE

GREEN'S FUNCTION METHOD

61.1 Poisson equation

General Poisson's problem

$$\begin{cases} -\nabla^2 \mathbf{u}(\mathbf{r}, t) = \mathbf{f}(\mathbf{r}, t) \\ + \text{b.c.} \end{cases}$$

with common boundary conditions

$$\begin{cases} \mathbf{u} = \mathbf{g} & \text{on } S_D \\ \hat{\mathbf{n}} \cdot \nabla \mathbf{u} = \mathbf{h} & \text{on } S_N \end{cases}$$

over Dirichlet and Neumann regions of the boundary.

Poisson's problem for Green's function, in infinite domain

$$-\nabla^2_{\mathbf{r}} G(\mathbf{r}; \mathbf{r}_0) = \delta(\mathbf{r} - \mathbf{r}_0)$$

Green's function method

$$\begin{aligned} E(\mathbf{r}_0, t) u_i(\mathbf{r}_0, t) &= \int_{\mathbf{r} \in \Omega} u_i(\mathbf{r}, t) \delta(\mathbf{r} - \mathbf{r}_0) = \\ &= -\int_{\mathbf{r} \in \Omega} u_i(\mathbf{r}, t) \nabla^2_{\mathbf{r}} G(\mathbf{r} - \mathbf{r}_0) = \\ &= -\int_{\mathbf{r} \in \Omega} \nabla_{\mathbf{r}} \cdot (u_i \nabla_{\mathbf{r}} G - G \nabla_{\mathbf{r}} u_i) - \int_{\mathbf{r} \in \Omega} G \nabla^2 u_i = \\ &= -\oint_{\mathbf{r} \in \partial\Omega} \hat{\mathbf{n}} \cdot (u_i \nabla_{\mathbf{r}} G - G \nabla_{\mathbf{r}} u_i) + \int_{\mathbf{r} \in \Omega} G(\mathbf{r} - \mathbf{r}_0) f_i(\mathbf{r}, t). \end{aligned}$$

An integro-differential boundary problem can be written using boundary conditions. As an example, using Dirichlet and Neumann boundary conditions, the integro-differential problem reads

$$\begin{aligned} &E(\mathbf{r}_0, t) \mathbf{u}(\mathbf{r}_0, t) + \int_{\mathbf{r} \in S_N} \mathbf{u}(\mathbf{r}, t)\, \hat{\mathbf{n}} \cdot \nabla_{\mathbf{r}} G(\mathbf{r} - \mathbf{r}_0) - \int_{\mathbf{r} \in S_D} G(\mathbf{r} - \mathbf{r}_0)\, \hat{\mathbf{n}} \cdot \nabla_{\mathbf{r}} \mathbf{u}(\mathbf{r}, t) = \\ &= -\int_{\mathbf{r} \in S_D} \mathbf{g}(\mathbf{r}, t)\, \hat{\mathbf{n}} \cdot \nabla_{\mathbf{r}} G(\mathbf{r} - \mathbf{r}_0) + \int_{\mathbf{r} \in S_N} G(\mathbf{r} - \mathbf{r}_0)\, \mathbf{h}(\mathbf{r}, t) + \int_{\mathbf{r} \in \Omega} G(\mathbf{r} - \mathbf{r}_0)\, \mathbf{f}(\mathbf{r}, t). \end{aligned}$$

Green's function of the Poisson-Laplace equation reads

$$G(\mathbf{r}; \mathbf{r}_0) = \frac{1}{4\pi} \frac{1}{|\mathbf{r} - \mathbf{r}_0|}\,.$$

Green's function of the Laplace equation

$$-\nabla^2 G = 0 \qquad \text{for } \mathbf{r} \neq \mathbf{r}_0$$

Solutions with spherical symmetry,

$$0 = \nabla^2 G = \frac{1}{r^2}\left(r^2 G'\right)' \quad \rightarrow \quad G'(r) = \frac{A}{r^2} \quad \rightarrow \quad G(r) = -\frac{A}{r} + B$$

Choosing $B = 0$ s.t. $G(r) \rightarrow 0$ as $r \rightarrow \infty$, and integrating over a sphere centered in $r = 0$ to get $A = -\frac{1}{4\pi}$,

$$1 = \int_V \delta(r) = -\int_V \nabla^2 G = -\oint_{\partial V} \hat{\mathbf{n}} \cdot \nabla G = -\oint_{\partial V} \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} \frac{A}{r^2} = -4\pi A$$

61.2 Helmholtz equation

todo from Fourier to Laplace trasnform in the first lines of this section

A Helmholtz's equation can be thouoght as the time Fourier transform of a wave equation,

$$\begin{cases} \frac{1}{c^2}\partial_{tt}\mathbf{u}(\mathbf{r},t) - \nabla^2 \mathbf{u}(\mathbf{r},t) = \mathbf{f}(\mathbf{r},t) \\ + \text{b.c.} \\ + \text{i.c.} \, , \end{cases}$$

Fourier transform in time of field $\mathbf{u}(\mathbf{r},t)$ reads

$$\tilde{\mathbf{u}}(\mathbf{r},\omega) = \mathcal{F}\{\mathbf{u}(\mathbf{r},t)\} = \int_{t=-\infty}^{+\infty} \mathbf{u}(\mathbf{r},t) e^{-i\omega t} \, d\omega$$

and, if $\mathbf{u}(\mathbf{r},t)$ is compact in time, Fourier transform of its time partial derivatives read

$$\begin{aligned} \mathcal{F}\{\dot{\mathbf{u}}(\mathbf{r},t)\} &= \int_{t=-\infty}^{+\infty} \dot{\mathbf{u}}(\mathbf{r},t) e^{-i\omega t} \, d\omega = \\ &= \mathbf{u}(\mathbf{r},t) e^{-i\omega t}\Big|_{t=-\infty}^{+\infty} + i\omega \int_{t=-\infty}^{+\infty} \mathbf{u}(\mathbf{r},t) e^{-i\omega t} \, d\omega = \\ &= i\omega \mathcal{F}\{\mathbf{u}(\mathbf{r},t)\} \end{aligned}$$

$$\mathcal{F}\{\partial_t^n \mathbf{u}(\mathbf{r},t)\} = (i\omega)^n \tilde{\mathbf{u}} \, .$$

The differential problem in the transformed domain thus reads

$$-\frac{\omega^2}{c^2}\tilde{\mathbf{u}} - \nabla^2 \tilde{\mathbf{u}} = \tilde{\mathbf{f}}$$

Green's function of Helmholtz'e equation reads

$$G(\mathbf{r},s) = \alpha^+ \frac{e^{\frac{s|\mathbf{r}-\mathbf{r}_0|}{c}}}{|\mathbf{r}-\mathbf{r}_0|} + \alpha^- \frac{e^{-\frac{s|\mathbf{r}-\mathbf{r}_0|}{c}}}{|\mathbf{r}-\mathbf{r}_0|}$$

with $\alpha^+ + \alpha^- = \frac{1}{4\pi}$.

Being the Laplace transform,

$$\mathcal{L}\{f(t)\} = \int_{t=0^-}^{+\infty} f(t) e^{-st} dt \, ,$$

the Laplace transform of a causal function with time delay $\tau \geq 0$ reads

$$\mathcal{L}\{f(t-\tau)\} = \int_{t=0^-}^{+\infty} f(t-\tau)e^{-st}dt = \int_{z=-\tau}^{+\infty} f(z)e^{-s(z+\tau)}\,dz = e^{-s\tau}\int_{z=0}^{+\infty} f(z)e^{-sz}\,dz = e^{-s\tau}\,\mathcal{L}\{f(t)\}$$

having used causality $f(t) = 0$ for $t < 0$. Laplace transform of Dirac's delta $\delta(t)$ reads

$$\mathcal{L}\{\delta(t)\} = \int_{t=0^-}^{+\infty} \delta(t)\,dt = 1\,,$$

so that $e^{-s\tau} = e^{-s\tau}\,1 = \mathcal{L}\{\delta(t-\tau)\}$.

Thus, Green's function for the wave equation reads

$$G(\mathbf{r},t;\mathbf{r}_0,t_0) = \alpha^+\frac{\delta\left(t-t_0+\frac{|\mathbf{r}-\mathbf{r}_0|}{c}\right)}{|\mathbf{r}-\mathbf{r}_0|} + \alpha^-\frac{\delta\left(t-t_0-\frac{|\mathbf{r}-\mathbf{r}_0|}{c}\right)}{|\mathbf{r}-\mathbf{r}_0|}$$

If $t \geq t_0$, and $G(\mathbf{r},t;\mathbf{r}_0,t_0)$ connects the past t_0 with the future t, the first term is not causal, and thus $\alpha^+ = 0$ and

$$G(\mathbf{r},t;\mathbf{r}_0,t_0) = \frac{1}{4\pi}\frac{\delta\left(t-t_0-\frac{|\mathbf{r}-\mathbf{r}_0|}{c}\right)}{|\mathbf{r}-\mathbf{r}_0|}\,.$$

Green's function of Helmholtz's equation

$$\frac{s^2}{c^2}G - \nabla^2 G = \delta(r)$$

$$G(r) = \frac{\alpha e^{kr} + \beta e^{-kr}}{r}$$

Proof:

- Gradient

$$\nabla G(r) = \hat{\mathbf{r}}\partial_r G = \hat{\mathbf{r}}\frac{\alpha(kr-1)e^{kr} + \beta(-kr-1)e^{-kr}}{r^2}$$

- Laplacian

$$\begin{aligned}\nabla^2 G(r) &= \frac{1}{r^2}\left(r^2 G'(r)\right)' = \\ &= \frac{1}{r^2}\left(\alpha(kr-1)e^{kr} + \beta(-kr-1)e^{-kr}\right)' = \\ &= \frac{1}{r^2}\left(\alpha k e^{kr} + \alpha k^2 r e^{kr} - \alpha k e^{kr} - \beta k e^{-kr} + \beta k^2 r e^{-kr} + \beta k e^{-kr}\right) = \\ &= \frac{1}{r}\left(\alpha e^{kr} + \beta e^{-kr}\right)k^2 = k^2 G(r)\,.\end{aligned}$$

  and thus $k^2 G(r) - \nabla^2 G = 0$, for $r \neq 0$;

- Unity

$$1 = \int_V \delta(r) = \int_V (k^2 G - \nabla^2 G) = \int_V k^2 G - \oint_{\partial V} \hat{\mathbf{n}}\cdot\nabla G$$

  the second term is the sum of two contributions of the form

$$\oint_{\partial V} \hat{\mathbf{n}}\cdot\nabla G^{\pm} = \oint_{\partial V} \frac{\alpha^{\pm}(\pm kr-1)e^{\pm kr}}{r^2} = 4\pi\alpha^{\pm}(\pm kr-1)e^{\pm kr}$$

the first term is the sum of two contributions of the form

$$k^2 \int_V G(r) = k^2 \int_V \frac{\alpha^{\pm} e^{\pm kr}}{r} =$$
$$= k^2 \alpha^{\pm} \int_{R=0}^{r} \int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} \frac{e^{\pm kR}}{R} R^2 \sin\phi \, dR \, d\phi \, d\theta =$$
$$= k^2 \alpha^{\pm} \, 4\pi \int_{R=0}^{r} R \, e^{\pm kR} \, dR \, .$$

the last integral can be evaluated with integration by parts

$$\int_{R=0}^{r} R \, e^{\pm kR} \, dR = \left[\frac{1}{\pm k} e^{\pm kR} R \right] \Big|_{R=0}^{r} \mp \frac{1}{k} \int_{R=0}^{r} e^{\pm kR} \, dR =$$
$$= \frac{1}{\pm k} e^{\pm kr} r - \frac{1}{k^2} e^{\pm kR} + \frac{1}{k^2} =$$

Thus summing everything together,

$$1 = \alpha^{+} \left[4\pi k^2 \left(\frac{r}{k} e^{kr} - \frac{1}{k^2} e^{kr} + \frac{1}{k^2} \right) - 4\pi \left(kr - 1 \right) e^{kr} \right] + \alpha^{-} \left[\dots \right] =$$
$$= 4\pi \left(\alpha^{+} + \alpha^{-} \right) \, .$$

61.3 Wave equation

Wave equation general problem

$$\begin{cases} \frac{1}{c^2} \partial_{tt} \mathbf{u}(\mathbf{r}, t) - \nabla^2 \mathbf{u}(\mathbf{r}, t) = \mathbf{f}(\mathbf{r}, t) \\ + \text{b.c.} \\ + \text{i.c.} \end{cases}$$

Green's problem of the wave equation

$$\frac{1}{c^2} \partial_{tt} G(\mathbf{r}, t; \mathbf{r}_0, t_0) - \nabla^2_{\mathbf{r}} G(\mathbf{r}, t; \mathbf{r}_0, t_0) = \delta(\mathbf{r} - \mathbf{r}_0) \delta(t - t_0)$$

Integration by parts

$$E(\mathbf{r}_\alpha, t_\alpha) \mathbf{u}(\mathbf{r}_\alpha, t_\alpha) = \int_{t \in T} \int_{\mathbf{r} \in V} \delta(t - t_\alpha) \delta(\mathbf{r} - \mathbf{r}_\alpha) \mathbf{u}(\mathbf{r}, t) =$$
$$= \int_{t \in T} \int_{\mathbf{r} \in V} \left\{ \frac{1}{c^2} \partial_{tt} G - \nabla^2_{\mathbf{r}} G \right\} \mathbf{u} =$$
$$= \int_{t \in T} \int_{\mathbf{r} \in V} \left\{ \frac{1}{c^2} \left[\partial_t \left(\mathbf{u} \partial_t G - G \partial_t \mathbf{u} \right) + G \partial_{tt} \mathbf{u} \right] - \nabla_{\mathbf{r}} \cdot \left(\nabla_{\mathbf{r}} G \, \mathbf{u} - G \nabla_{\mathbf{r}} \mathbf{u} \right) - G \, \nabla^2_{\mathbf{r}} \mathbf{u} \right\} =$$
$$= \int_{\mathbf{r} \in V} \frac{1}{c^2} \left[\mathbf{u}(\mathbf{r}, t) \partial_t G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) - G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \partial_t \mathbf{u}(\mathbf{r}, t) \right] \Big|_{t_0}^{t_1} +$$
$$+ \int_{t \in T} \oint_{\mathbf{r} \in \partial V} \left\{ -\hat{\mathbf{n}}(\mathbf{r}, t) \cdot \nabla_{\mathbf{r}} G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \, \mathbf{u}(\mathbf{r}, t) + G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \, \hat{\mathbf{n}}(\mathbf{r}, t) \cdot \nabla_{\mathbf{r}} \mathbf{u}(\mathbf{r}, t) \right\} +$$
$$+ \int_{t \in T} \int_{\mathbf{r} \in V} G(\mathbf{r}, t; \mathbf{r}_\alpha, t_\alpha) \underbrace{\left\{ \frac{1}{c^2} \partial_{tt} \mathbf{u}(\mathbf{r}, t) - \nabla^2_{\mathbf{r}} \mathbf{u}(\mathbf{r}, t) \right\}}_{= \mathbf{f}(\mathbf{r}, t)}$$

$$\int_{t \in T} \int_{\mathbf{r} \in V} \frac{1}{4\pi} \frac{\delta\left(t - t_\alpha + \frac{|\mathbf{r} - \mathbf{r}_\alpha|}{c} \right)}{|\mathbf{r} - \mathbf{r}_\alpha|} \mathbf{f}(\mathbf{r}, t) = \int_{\mathbf{r} \in V \cap B_{|\mathbf{r} - \mathbf{r}_\alpha| \le c(t_\alpha - t)}} \frac{1}{4\pi |\mathbf{r} - \mathbf{r}_\alpha|} \mathbf{f}\left(\mathbf{r}, t_\alpha - \frac{|\mathbf{r} - \mathbf{r}_\alpha|}{c} \right)$$

Part XV

Optimization

CHAPTER

SIXTYTWO

OPTIMIZATION

- **Gradient-based methods**, and Hessian based, for continuos functions
- Free and constrained optimization
- …

62.1 Unconstrained optimization

Given $f(\mathbf{x})$, with $\mathbf{x} \in \mathbb{R}^n$, a **candidate local** extreme $\mathbf{x}^*$ is a point where the *gradient* $\partial_{\mathbf{x}} f$ of function f is zero

$$\partial_{\mathbf{x}} f(\mathbf{x}^*) = \mathbf{0} \, .$$

Among these candidates, local extreme may be:

- maximum: if the Hessian evalauted in $\mathbf{x}^*$ is definite negative,

$$\begin{cases} \partial_{\mathbf{x}} f(\mathbf{x}^*) = \mathbf{0} \\ \partial_{\mathbf{xx}} f(\mathbf{x}^*) > 0 \end{cases}$$

- minimum: if the Hessian evalauted in $\mathbf{x}^*$ is definite positive,

$$\begin{cases} \partial_{\mathbf{x}} f(\mathbf{x}^*) = \mathbf{0} \\ \partial_{\mathbf{xx}} f(\mathbf{x}^*) < 0 \end{cases}$$

Here the formalism $\partial_{\mathbf{xx}} f(\mathbf{x}^*) < 0$ indicates that the Hessian matrix is negative definite, see *Example 62.1.1*.

Example 62.1.1 (Positive/negative definite Hessian)

Local polynomial expansion of a differentiable function $\mathbf{x}$ around $\mathbf{x}^*$ reads

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \Delta\mathbf{x}^T \partial_{\mathbf{x}} f(\mathbf{x}^*) + \frac{1}{2} \Delta\mathbf{x}^T \, \partial_{\mathbf{xx}} f(\mathbf{x}^*) \, \Delta\mathbf{x} + o(|\Delta\mathbf{x}|^2) \, ,$$

with $\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}^*$.

Candidate extreme points are thos points $\mathbf{x}^*$ where the gradient vanishes $\partial_{\mathbf{x}} f\mathbf{x}^*) = 0$. If the Hessian $H(\mathbf{x}^*) := \partial_{\mathbf{xx}} f(\mathbf{x}^*)$ is positive definite, by definition,

$$\mathbf{v}^T \, H(\mathbf{x}^*) \mathbf{v} > 0 \, , \qquad \forall \mathbf{v} \neq \mathbf{0} \, ,$$

then

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \frac{1}{2} \Delta\mathbf{x}^T H(\mathbf{x}^*) \, \Delta\mathbf{x} + o(|\Delta\mathbf{x}|^2) > f(\mathbf{x}^*) \, ,$$

as $|\Delta\mathbf{x}| \to 0$, and thus $\mathbf{x}^*$ is a point of minimum, as $f(\mathbf{x}) > f(\mathbf{x}^*)$ for all the neighoring points $\mathbf{x}$.

62.2 Constrained optimization

$$f(\mathbf{x}) \,, \qquad \mathbf{x} \in D \subseteq \mathbb{R}^n$$

with constraints, such as

$$\begin{aligned} \mathbf{g}(\mathbf{x}) &= \mathbf{0} \\ \mathbf{h}(\mathbf{x}) &\geq \mathbf{0} \end{aligned}$$

62.2.1 Method of Lagrange multipliers for equality constraints

$$\tilde{f}(\mathbf{x}; \lambda) = f(\mathbf{x}) - \lambda^T \mathbf{g}(\mathbf{x}) \,.$$

Part XVI

Reinforcement Learning

CHAPTER

SIXTYTHREE

INTRODUCTION TO REINFORCEMENT LEARNING

CHAPTER

SIXTYFOUR

MARKOV PROCESSES

Definition 64.1 (Stochastic process)

A stochastic process is as collection of random variables $X(t)$ defined on a common probability space $(\Omega, \mathcal{F}, P)$, indexed by some set T, with outcomes in a measurable space (S, Σ),

$$\{X(t) : t \in T\} \ .$$

Markov process is a memomry-less stochastic process (see def *Definition 64.1*). For discrete-time Markov processes, state at time n only depends on state at time $n-1$

$$\langle S, P, \mu \rangle$$

with

- S (finite) set of states
- **P** state transition probability matrix, $P_{ss'} = P(s'|s)$
- μ a set of initial probability, $\mu_i = P(X_0 = i), \forall i$

Properties of probability matrix.

A **stationary MP** is a process described by a constant state transition probability.

1-step transition.

Notation: probability vectors as row vectors

In treating stochastic processes, probability vectors are usually treated as row vectors. Probability distribution over states s' at time n is the sum of probabilities or reaching s' starting from s at time $n-1$

$$p_n(s') = \sum_s P(s'|s) p_{n-1}(s) = \sum_s p_{n-1}(s) P_{ss'} \ ,$$

or using matrix formalism

$$\mathbf{p}_n = \mathbf{p}_{n-1}\mathbf{P} \ .$$

Properties of probability matrix.

Since $P_{ss'} = P(s'|s)$ represents the probability of arriving in s' starting from s (i.e. the *conditional probability* of arriving in s', starting from s), the entries of a matrix representing a time-discrete process are

$$0 \leq P_{ss'} \leq 1 \ .$$

The sum over s' is 1 (sum over all the possibilities),

$$\sum_{s'} P(s'|s) = \sum_{s'} P_{ss'} = 1 \,, \quad \forall s \qquad \text{or} \qquad \mathbf{P1} = \mathbf{1} \,.$$

Thus, a probability matrix $\mathbf{P}$ has a eigenvalue $\lambda = 1$ with the uniform vector $\mathbf{1}$ as eigenvector.

Is this unique? *No, in general.*

Existence of a stationary distribution

Spectral radius of a probability matrix is $\rho(\mathbf{P}) = 1$.

Proof

By *Gershgorin circle theorem*,

$$|\lambda - P_{ii}| \leq \sum_{j \neq i} |P_{ij}| \,,$$

for i s.t. the components of the (right) eigenvalue $\mathbf{v}$ satisfy $|x_i| \geq |x_j|, \forall j$. As $P_{ij} \geq 0$ and $\sum_j P_{ij} = P_{ii} + \sum_{j \neq i} P_{ij} = 1$, it follows

$$|\lambda - P_{ii}| \leq \sum_{j \neq i} P_{ij} = \sum_j P_{ij} - P_{ii} = 1 - P_{ii} \,,$$

meaning that any eigenvalue λ is contained in a circle centered in P_{ii} with radius $1 - P_{ii}$. It's not hard to prove that such a circle is contained in a circle centered in 0 with radius 1[1]. Thus, for all the eigenvalues $|\lambda| \leq 1$ holds, and at least one eigenvalue with $|\lambda| = 1$, namely the $\lambda = 1$ eigenvalue shown above, exists. Thus, the spetral radius of a probability matrix is $\rho(\mathbf{P}) = 1$.

n-step transition. n-step transition in a stationary MP reads

$$\mathbf{p}_{m+n} = \mathbf{p}_{m+n-1}\mathbf{P} = \mathbf{p}_{m+n-2}\mathbf{PP} = \cdots = \mathbf{p}_m\mathbf{P}^n \,.$$

First passage. …; recurrent states

Classification of states: accessible, communicating (and dividing states into partitions, disjoint classes); positive recurrent, periodic, ergodic; absorbing or transient

Classification of MP: absorbing, ergodic, regular

Stationary distribution, a probability vector $\mathbf{p}$ so that probability doesn't change in the next timestep,

$$\mathbf{p} = \mathbf{pP} \,.$$

Fundamental matrix

Other definitions: mixing rate, spectral gap

64.1 Markov Reward Processes

$$\langle S, P, R, \gamma, \mu \rangle$$

with

[1] $1 - P_{ii} \geq |\lambda - P_{ii}| = |\lambda - 0 + 0 - P_i i| \geq ||\lambda| - P_i i|$; if $|\lambda| \geq P_{ii}$ it follows $1 \geq |\lambda|$; if $|\lambda| \leq P_{ii}$…**todo**

- R **reward function**; it could be a stochastic variable, depending on a state with probability $p(r|s)$ *(or on the transition $p(r|s, s')$?)*
- γ **discount factor**, $\gamma \in [0, 1]$; it's used to represent the present value of future rewards ($\gamma \sim 0$ short-sighted, $\gamma \sim 1$ far-sighted)

Return

- Time horizion: finite, indefinite (until stopping criteria, or absorbing state), infinite
- Possible choices of cumulative reward:
  - total reward, $V = \sum_i r_i$
  - average reward, $V = \frac{1}{n} \sum_{i=1}^{n} r_i$
  - discounted reward $V = \sum_i \gamma^{i-1} r_i$

Return v_t is defined as the total discount reward from timestep t

$$v_t = \sum_{k=0}^{+\infty} \gamma^k r_{t+k+1} = r_{t+1} + \gamma r_{t+2} + ...$$

Value function Expected value of the return v_t from state $s_t = s$

$$V(s) := \mathbb{E}\left[v_t | s_t = s\right]$$

64.1.1 Bellman Equation for $V(s)$

$$\begin{aligned}
V(s) &= \mathbb{E}\left[v_t | s_t = s\right] = \\
&= \mathbb{E}\left[r_{t+1} + \gamma \sum_{k=0}^{+\infty} \gamma^k r_{t+2+k} \middle| s_t = s \right] = \\
&= \mathbb{E}\left[r_{t+1} + \gamma v_{t+1} |\, s_t = s\right] = \\
&= \mathbb{E}\left[r_{t+1} | s_t = s\right] + \gamma \mathbb{E}\left[v_{t+1} | s_t = s\right] = \\
&= R(s) + \gamma \sum_{s'} P(s'|s) \mathbb{E}\left[v_{t+1} | s_{t+1} = s'\right] = \\
&= R(s) + \gamma \sum_{s'} P(s'|s)\, V(s') \,.
\end{aligned}$$

or in matrix form

$$\mathbf{V} = \mathbf{R} + \gamma \mathbf{P} \mathbf{V} \,.$$

Solution of Bellman equation. Bellman equation for value function is linear. It can be solved directly, solving the linear problem

$$(\mathbf{I} - \gamma \mathbf{P})\, \mathbf{V} = \mathbf{R} \,.$$

Solutions:

- direct solution of the linear system (feasible for small-dimensional MRP only?)
- iterative methods for large MRPs: DP (dynamic programming), MC (Monte-Carlo evaluation), TD (Temporal-Difference learning)

64.2 Markov Decision Processes

$$\langle S, A, P, R, \gamma, \mu \rangle$$

with

- A (finite) set of actions
- **P** state transition probability matrix, $P(s'|s, a)$
- R reward function, $R(s, a) = \mathbb{E}[r|s, a]$

64.2.1 Policies

A policy is a distribution over actions, given the initial state

$$\pi(a|s) = P(a|s)$$

Given a policy $\pi(a|s)$, a MDP becomes the MRP with

- transition probability

$$P^{\pi}(s'|s) = \sum_{a \in A} \pi(a|s) P(s'|s, a)$$

- reward (**todo** *Prove it!*)

$$R^{\pi}(s) = \sum_{a \in A} \pi(a|s) R(s, a)$$

Value functions.

- State value function, $V^{\pi}(s)$, expected return starting from s and following policy π

$$V^{\pi}(s) := \mathbb{E}_{\pi}[v_t|s_t = s]$$

- Action-value functon $Q^{\pi}(s, a)$, expected return starting from s, taking action a first and then following policy π

$$Q^{\pi}(s, a) := \mathbb{E}_{\pi}[v_t|s_t = s, a_t = a]$$

64.2.2 Bellman Equations

Bellman expectation equations. Bellman expectation equation for the state-value function $V^\pi(s)$ reads

$$\begin{aligned}
V^\pi(s) &:= \mathbb{E}_\pi\left[v_t | s_t = s\right] = \\
&= \mathbb{E}_\pi\left[r_{t+1} + \gamma \sum_{k=0}^{+\infty} \gamma^k r_{t+2+k} \middle| s_t = s \right] = \\
&= \sum_{a\in A} \pi(a|s) \mathbb{E}_\pi\left[r_{t+1} + \gamma \sum_{k=0}^{+\infty} \gamma^k r_{t+2+k} \middle| s_t = s, a_t = a \right] = \\
&= \sum_{a\in A} \pi(a|s) \mathbb{E}\left[r_{t+1} | s_t = s, a_t = a\right] + \gamma \sum_{a\in A} \sum_{s'\in S} \pi(a|s) P(s'|s,a) \mathbb{E}_\pi\left[v_{t+1} | s_{t+1} = s'\right] = \\
&= \sum_{a\in A} \pi(a|s) R(s,a) + \gamma \sum_{s'\in S} P^\pi(s'|s) \mathbb{E}_\pi\left[v_{t+1} | s_{t+1} = s'\right] = \\
&= \sum_{a\in A} R(s,a) \left\{ \pi(a|s) + \gamma \sum_{s'\in S} P(s'|s,a) V^\pi(s') \right\} = \\
&= R^\pi(s) + \gamma \sum_{s'\in S} P^\pi(s'|s) V^\pi(s') =
\end{aligned} \tag{64.1}$$

Bellman expectation equation for the action-value function $Q^\pi(s)$ reads

$$\begin{aligned}
Q^\pi(s,a) &:= \mathbb{E}_\pi\left[v_t | s_t = s, a_t = a\right] = \\
&= \mathbb{E}\left[r_{t+1} | s_t = s, a_t = a\right] + \gamma \mathbb{E}_\pi\left[v_{t+1} | s_t = s, a_t = a\right] = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) \mathbb{E}_\pi\left[v_{t+1} | s_{t+1} = s'\right] = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) V^\pi(s') = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) \sum_{a'\in A} \pi(a'|s') \mathbb{E}_\pi[v_{t+1} | s_{t+1} = s', a_{t+1} = a'] = \\
&= R(s,a) + \gamma \sum_{s'\in S} P(s'|s,a) \sum_{a'\in A} \pi(a'|s') Q^\pi(s',a')
\end{aligned} \tag{64.2}$$

Bellman expectation operators todo

Optimal value functions, and optimal policies

Definition 64.2.1 (Optimal value functions)

Optimal state-value function $V^*(s)$ is defined as

$$V^*(s) = \max_\pi V^\pi(s) \, .$$

Optimal action-value function $Q^*(s,a)$ is defined as

$$Q^*(s,a) = \max_\pi Q^\pi(s,a) \, .$$

Usually, the goal of a problem involving MDP is the maximization of value functions, as the performance of the MDP.

Valus functions define a partial ordering of policies,

$$\pi \geq \pi' \text{ if } V^\pi(s) \geq V^{\pi'} \; \forall s \in S \, .$$

Theorem 64.2.1 (Optimal policies)

For a MDP,

- an optimal policy π^* exists, s.t. $\pi^* \geq \pi$, $\forall \pi$
- all optimal policies achieve the same value of optimal functions, namely

$$V^{\pi^*}(s) = V^*(s) \quad , \quad Q^{\pi^*}(s,a) = Q^*(s,a) \; .$$

- a deterministic optimal policy exists, and it's found optimizing action-value function over actions a,

$$\pi^*(a|s) = \begin{cases} 1 & , \quad \text{if } a = \text{argmax}_{a \in A} Q^*(s,a) \\ 0 & , \quad \text{otherwise} \end{cases}$$

Proof

todo

Property 64.2.1 (Property: optimal state-value and action-value functions)

$$V^*(s) = \max_a Q^*(s,a) \tag{64.3}$$

todo *Prove it!*

Bellman optimality equations. Using *Property 64.2.1* and the expression of action-value function in (64.2)

$$\begin{aligned}
V^*(s) &= \max_a Q^*(s,a) = \\
&= \max_a \max_\pi Q^\pi(s,a) = \\
&= \max_a \max_\pi \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^\pi(s') \right\} = \\
&= \max_a \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^*(s') \right\} \\
Q^*(s,a) &= \max_\pi Q^\pi(s,a) = \\
&= \max_\pi \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^\pi(s') \right\} = \\
&= R(s,a) + \max_\pi \left\{ \gamma \sum_{s' \in S} P(s'|s,a) V^\pi(s') \right\} = \\
&= R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V^*(s') = \\
&= R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) \max_{a'} Q^*(s',a')
\end{aligned}$$

Bellman optimality operators

Properties of Bellman operators

Solving Bellman optimality equations

- non-linear
- no closed form solution
- iterative solutions:
  - Dynamic Programming (DP): Value Iteration (VI), Policy Iteration (PI)
  - Linear Programming (LP)
  - Reinforcement Learning (RL): Q-learning (off-policy), SARSA (on-policy),…

CHAPTER

SIXTYFIVE

METHODS OF SOLUTION OF MPD: DP AND LP

65.1 Brute force: policy search

65.2 Dynamic programming

Introduction to DP

- sequential or temporal approach to optimization
- solving complex problems breaking them down into sub-problems:
  - solve sub-problems
  - combine solutions of sub-problems
- for problems with 2 properties:
  - optimal substructure: optimal sol can be decomposed in sub-pbs
  - overlapping sub-problems: sub-pbs recur, sols can be cached and reused
- DP assumes full knowledge of the MDP
- used for planning (???)
- prediction: given MDP and policy, evaluate value function
- control: given MDP, evaluate optimal value function and policy

65.2.1 Policy iteration

Policy iteration alternates *policy evaluation* and *policy improvement* steps. When value function converges, iteration stops at **Bellman optimality condition**, as shown in the *algorithm* below.

Policy Evaluation

Bellman equation for a given policy π. Solution of Bellman equation (64.1),

$$V^{\pi}(s) = \sum_{a \in A} R(s,a) \left\{ \pi(a|s) + \gamma \sum_{s' \in S} P(s'|s,a) V^{pi}(s') \right\}$$

- direct solution. Complexity: …
- full policy-evaluation back-up

$$V_{k+1}(s) \leftarrow \sum_{a \in A} R(s,a) \left\{ \pi(a|s) + \gamma \sum_{s' \in S} P(s'|s,a) V_k^{\pi}(s') \right\}$$

  synchronous, asynchronous update; **todo** *convergence?*

Policy Improvement

Greedy update. Let π be a deterministic policy. Greedy update acts maximizing the action-value function for all state over all the actions,

$$\pi'(s) := \text{argmax}_{a \in A} Q^{\pi}(s,a) \,,$$

and it "improves the value function" (which value function?) of any state,

$$Q^{\pi}(s, \pi'(s)) := \max_{a \in A} Q^{\pi}(s,a) \geq Q^{\pi}(s, \pi(s)) = V^{\pi}(s) \,. \tag{65.1}$$

In stochastic optimization, usually *ε-greedy improvement* is introduced to improve **exploration** and try to avoid converging to local optimum.

Theorem 65.2.1 (Policy improvement theorem)

Let π and π' a pair of deterministic policites s.t.

$$Q^{\pi}(s, \pi'(s)) \geq V^{\pi}(s) \,, \quad \forall s \in S$$

Then $\pi' \geq \pi$, i.e. $V^{\pi'}(s) \geq V^{\pi}(s)$, $\forall s \in S$.

Proof.

$$\begin{aligned}
V^{\pi}(s) &\leq Q^{\pi}(s, \pi'(s)) = \\
&=: \mathbb{E}_{\pi} \left[v_t | \, s_t = s, a_t = \pi'(s) \right] = \\
&= \mathbb{E}_{\pi} \left[r_{t+1} + \gamma v_{t+1} | \, s_t = s, a_t = \pi'(s) \right] = \\
(1) &= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma \mathbb{E}_{\pi} \left[v_{t+1} | \, s_{t+1} = s', a_{t+1} = \pi(s) \right] = \\
&= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma V^{\pi}(s) = \\
&\leq \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma Q^{\pi}(s, \pi'(s)) = \\
&= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma \mathbb{E}_{\pi} \left[r_{t+2} + \gamma v_{t+2} | \, s_{t+1} = s, a_{t+1} = \pi'(s) \right] = \\
(2) &= \mathbb{E}_{\pi'} \left[r_{t+1} | s_t = s \right] + \gamma \mathbb{E}_{\pi'} \left[r_{t+2} | s_{t+1} = s \right] + \gamma^2 \mathbb{E}_{\pi} \left[v_{t+2} | \, s_{t+2} = s'', a_{t+2} = \pi(s) \right] = \\
&= \mathbb{E}_{\pi'} \left[r_{t+1} + \gamma r_{t+2} | s_t = s \right] + \gamma^2 V^{\pi}(s) = \\
&\leq \mathbb{E}_{\pi'} \left[r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \ldots | s_t = s \right] = \\
&\leq V^{\pi'}(s) \,.
\end{aligned}$$

with in (1) the first action a_t is drawn from $\pi'(s)$, s' results from the transition

$$P^{\pi}(s'|s) = P(s'|s,a)\pi(a|s) \ ,$$

and the following are drawn from π. The same notation is used in (2), and in all the following manipulations.

Policy Iteration algorithm

Policy iteration algorithm

$$\pi_0 \to V^{\pi_0} \to \pi_1 \to V^{\pi_1} \to ... \pi^* \to V^* \to \pi^*$$

Convergence is reached when $\pi_{n+1}(s) = \pi_n(s)$. Greedy update (65.1) gives

$$Q^{\pi_n}(s, \pi_{n+1}(s)) = Q^{\pi_n}(s, \pi_n(s)) = V^{\pi_n}(s) \ ,$$

namely, when the optimality condition (64.3) is reached.

65.2.2 Value Iteration

Unlike policy iteration, value iteration algorithm doesn't provide any explicit policy, but only optimal value of the value-functions.

An optimal policy for a MDP is built with an optimal action a^* first, followed by actions drawn from the optimal policy. Thus, it must satisfy

$$V^*(s) = \max_{a \in A} R(s,a) + V^*(s') \ ,$$

with $s_n = s$ and $s_{n+1} = s'$, resulting from transition $s'|s,a$. Subdividing the problem in sub-problems, **deterministic value iteration** proceeds from s' backwards to s,

$$V^*(s) \leftarrow \max_{a \in A} R(s,a) + V^*(s') \ .$$

or

$$V^*(s) \leftarrow \max_{a \in A} \left\{ R(s,a) + \gamma \sum_{s' \in S} P(s'|s,a) V(s') \right\} .$$

65.2.3 Dynamic programming: extensions

- synchronous, or in-place
- prioritized sweeping/update
- …

65.3 Linear programming

…

CHAPTER

SIXTYSIX

METHODS OF SOLUTION OF MPD: RL

Introduction

- model-free / model-based
- on-policy / off-policy
- online / offline
- tabular / function approximation
- value-based / policy-based / actor-critic

Two main goals of methods in RL:

- *prediction or evaluation*: estimate the performance of a given policy
- *control*: find the best (or a good policy) to get the best performance

These two tasks usually rely on two processes: *evaluation* and *improvement*. Evaluation stage provides the perfomance of a given policy usually in terms of value functions. Improvement aims at finding a new policy with better performance.

Prediction involves only the evaluation step, while control usually involves an iteration over alternate evaluation and improvement processes.

66.1 Evaluation

66.1.1 Monte-Carlo RL

- model-free, learn from experience of complete episodes ((-) no bootstrapping)
- uses empirical mean return, instead of expected value
- first-visit MC (unbiased estimator) / every-visit MC (biased estimator)

First-visit MC

Initialize: π policy to be evaluated; V arbitrary state-value function, $R(s)$ empty list of returns

Loop until convergence or stopping criterion

- generate an episode using π
  - for each state s in the episode:
    - evalaute and append the return $r(s)$ following the first occurrence of s to the list of $R(s)$
    - update $V(s) = \text{average}(R(s))$

Every-visit MC

Initialize: π policy to be evaluated; V arbitrary state-value function, $R(s)$ empty list of returns

Loop until convergence or stopping criterion

- generate an episode using π
  - for each state s in the episode:
    - for each occurrence of state s in the episode:
      - evaluate and append the return $r(s)$ following the occurrence of s to the list of $R(s)$
      - update $V(s) = \text{average}(R(s))$

Using incremental mean update

$$\begin{aligned}\mu_{N+1} &= \frac{x_1 + \cdots + x_N + x_{N+1}}{N+1} = \\ &= \frac{x_1 + \cdots + x_N}{N}\frac{N}{N+1} + \frac{x_{N+1}}{N+1} = \\ &= \frac{N}{N+1}\mu_N + \frac{1}{N+1}x_{N+1} = \\ &= \mu_N + \frac{1}{N+1}\left(x_{N+1} - \mu_N\right)\end{aligned}$$

Incremental update of the average of the value function reads

$$V_k(s) \leftarrow V_{k-1}(s) + \frac{1}{k}\left(v_k - V_{k-1}(s)\right) \ ,$$

with v_k the return of state s_k.

A generalized update may follow, to introduce a free hyperparameter as a weight of older estimation,

$$\begin{aligned}V_k(s) &\leftarrow V_{k-1}(s) + \alpha_k\left(v_k - V_{k-1}(s)\right) \\ &= (1-\alpha_k)V_{k-1}(s) + \alpha_k v_k\end{aligned}$$

66.1.2 Temporal Difference (TD) Learning

- model-free, learn from experience of incomplete episodes (bootstrapping)
- …

66.1.3 TD prediction

- Starting from every visit MC

$$V(s_t) \leftarrow V(s_t) + \alpha(v_t - V(s_t))$$

- Update towards an **estimation of the return**
  - the simplest TD algorithm is $TD(0)$.
    - estimated return (called **TD target**) reads

$$v_t \sim r_{t+1} + \gamma V(s_{t+1})$$

* **TD error** error δ_t is defined as

$$\delta_t := r_{t+1} + \gamma V(s_{t+1}) - V(s_t)$$

* update step

$$V(s) \leftarrow V(s) + \alpha_k \left(r_{t+1} + \gamma V(s_{t+1}) - V(s_{t+1})\right) = \\ = V(s) + \alpha_k \, \delta_t$$

– $TD(n)$ algorithm uses n-step prediction

$$v_t^{(n)} = r_{t+1} + \gamma r_{t+2} + ... \gamma r_{t+n} + \gamma^{n+1} V(s_{t+n})$$

– λ-return v_t^{λ} combines all n-step returns $v_t^{(n)}$ as

$$v_t^{\lambda} := (1-\lambda) \sum_{n=1}^{+\infty} \lambda^{n-1} v_t^{(n)}$$

weights s.t. sum of weights $= 1$ and decay as $\lambda (< 1)$

– …eligibility traces, forward and backward TD,…

66.2 Model control

66.2.1 On- and Off-Policy Learning

- On-policy: learn policy π from experience sampled from π
- Off-policy: learn policy π from experience sampled from $\tilde{\pi}$

Greedy policy improvement

- over $V(s)$ requires model of MDP,

$$\pi'(s) = \text{argmax}_{a \in A} \{R(s,a) + P(s'|s,a) V(s')\}$$

- **over $Q(s,a)$ is model-free**

$$\pi'(s) = \text{argmax}_{a \in A} Q(s,a) \, .$$

66.2.2 MC control

Note: Generalized policy iteration with MC evaluation

MC evaluation is model-free. To keep a model-free control task, a model-free policy improvement is required, like greedy policy improvement over $Q(s,a)$.

66.2.3 ϵ-greedy policy improvement

$$\pi(a|s) = \begin{cases} 1 - \varepsilon\left(1 - \frac{1}{N}\right) & , \quad \text{if } a = \text{argmax}_{a \in A} Q^{\pi}(s,a) \\ \frac{\varepsilon}{N} & , \quad \text{otherwise} \end{cases}$$

Theorem 66.2.1 (ε-greedy Policy Improvement)

For …, the ε-greedy policy π' w.r.t. Q^{π} is an improvement, $V^{\pi'}(s) \geq V^{\pi}(s)$.

Proof

$$\begin{aligned} Q^{\pi}(s, \pi'(s)) &= \sum_{a \in A} \pi'(a|s) Q^{\pi}(s,a) = \\ &= \sum_{a \in A, a \neq a^*} \frac{\varepsilon}{m} Q^{\pi}(s,a) + \left(1 - \varepsilon + \frac{\varepsilon}{m}\right) Q^{\pi}(s,a^*) = \\ &= \sum_{a \in A} \frac{\varepsilon}{m} Q^{\pi}(s,a) + (1-\varepsilon)\, Q^{\pi}(s,a^*) = \\ (1) &\geq \sum_{a \in A} \frac{\varepsilon}{m} Q^{\pi}(s,a) + (1-\varepsilon) \sum_{a \in A} \frac{\pi(a|s) - \frac{\varepsilon}{m}}{1-\varepsilon} Q^{\pi}(s,a) = \\ &= \sum_{a \in A} \pi(a|s) Q^{\pi}(s,a) = V^{\pi}(s) \end{aligned}$$

having used in (1) … And for the policy improvement theorem *Theorem 65.2.1*, $V^{\pi'}(s) \geq V(s)$. **todo** *Check it!*

GLIE (Greedy in the Limit of Infinite Exploration) …

GLIE MC Control

- k^{th} episode using \pi$$
- for each state s_t, and action a_t in the episode, update number of (s,a) pair occurence and action-value function

$$\begin{aligned} N(s_t, a_t) &\leftarrow N(s_t, a_t) + 1 \\ Q(s_t, a_t) &\leftarrow Q(s_t, a_t) + \frac{1}{N(s_t, a_t)} (v_t - Q(s_t, a_t)) \end{aligned}$$

- improve policy with ε-greedy method over action-value function Q,

$$\begin{aligned} \varepsilon &\leftarrow \frac{1}{k} \\ \pi &\leftarrow \varepsilon - \text{greedy}(Q) \end{aligned}$$

66.2.4 TD control

- (+): online, from incomplete episodes, lower variance
- use TD instead of MC for policy evaluation, **SARSA**:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left(r + \gamma Q(s',a') - Q(s,a)\right)$$

…

Variations. SARSA(λ),…

Note: SARSA is on-policy

66.2.5 Off-policy learning

Q-learning, is off-policy as a' is not taken from π

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left(r + \gamma \max_{a'} Q(s',a') - Q(s,a)\right)$$

66.3 Actor-Critic

todo

66.4 Learning and planning

- Model-free RL: no model; **learn** policy and/or value function from real experience
- Model-based RL: learn a model from real experience; **plan** policy and/or value function from simulated experience
- Dyna: leanr a model from real experience; **learn and plan** a policy and/or value function from real and simulated experience

CHAPTER

SIXTYSEVEN

LARGE OR CONTINUOUS MDPS

For small-dimensional problems, value functions can be efficiently represented by arrays or look-up tables,

$$\begin{aligned} V(s) \quad &\rightarrow \quad \mathbf{V} \in \mathbb{R}^{|S|} \\ Q(s,a) \quad &\rightarrow \quad \mathbf{Q} \in \mathbb{R}^{|S|\times|A|} \end{aligned}$$

For large problems with large number of discrete states $\sim 10^N$ ($N = 20$ for backgammon, $= 10^{170}$ for Go), or continuous state space, RL usually relies on **function approximation** of the value function,

$$\begin{aligned} V^\pi(s) &\sim V(s;\theta) \\ Q^\pi(s,a) &\sim Q(s,a;\theta) \end{aligned}$$

Parameters θ are updated during learning. Examples of function approximators, the most suitable depend on the problem itself (ANN, Fourier/wavelets basis, polynomial, piecewise-polynomial, coarse coding,…)

Optimizing/minimizing mean-squared error between $V(s,\theta)$ and $V^\pi(s)$

$$L(\theta) := \mathbb{E}_\pi \left[\left(V^\pi(s) - V(s;\theta) \right)^2 | s_t = s \right]$$

Gradient descent,

$$\begin{aligned} \Delta\theta = -\frac{1}{2}\alpha\nabla_\theta L(\theta) = \\ = -\frac{1}{2}\alpha\nabla_\theta \, \mathbb{E}_\pi \left[\left(V^\pi(s) - V(s,\theta) \right)^2 | s_t = s \right] = \\ = \alpha \, \mathbb{E}_\pi \big[\left(V^\pi(s) - V(s,\theta) \right) \nabla_\theta V(s,\theta) \mid s_t = s \big] \, , \end{aligned}$$

and the stochastic gradient decent update reads

$$\Delta\theta = \alpha \, \left(V^\pi(s) - V(s,\theta) \right) \, \nabla_\theta V(s,\theta) \, .$$

State representation with **features** $\phi(s)$,

$$\phi(s) = \begin{bmatrix} \phi_1(s) \\ ... \\ \phi_n(s) \end{bmatrix}$$

State-value function as a linear combination of features

$$V(s;\theta) = \phi^T(s) \, \theta \, ,$$

and the gradient of the objective function directly follows from

$$\nabla_\theta V(s;\theta) = \phi(s) \, .$$

…

Action-value function as a linear combination of features

$$Q(s,a;\theta) = \phi^T(s,a) \, \theta \, ,$$

…

67.1 Value and Policy-Based RL

- value based
- policy based
- actor-critic

…

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